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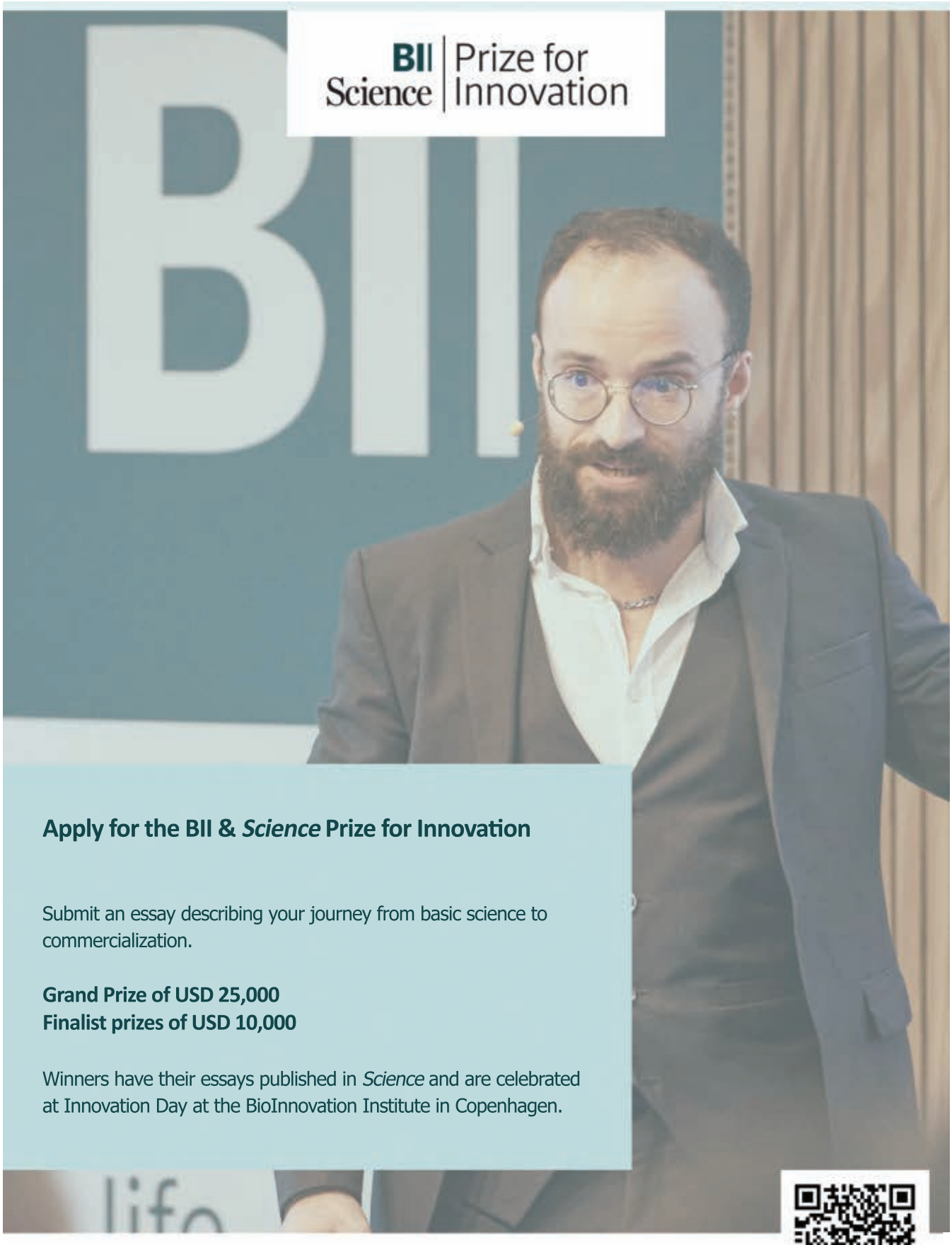

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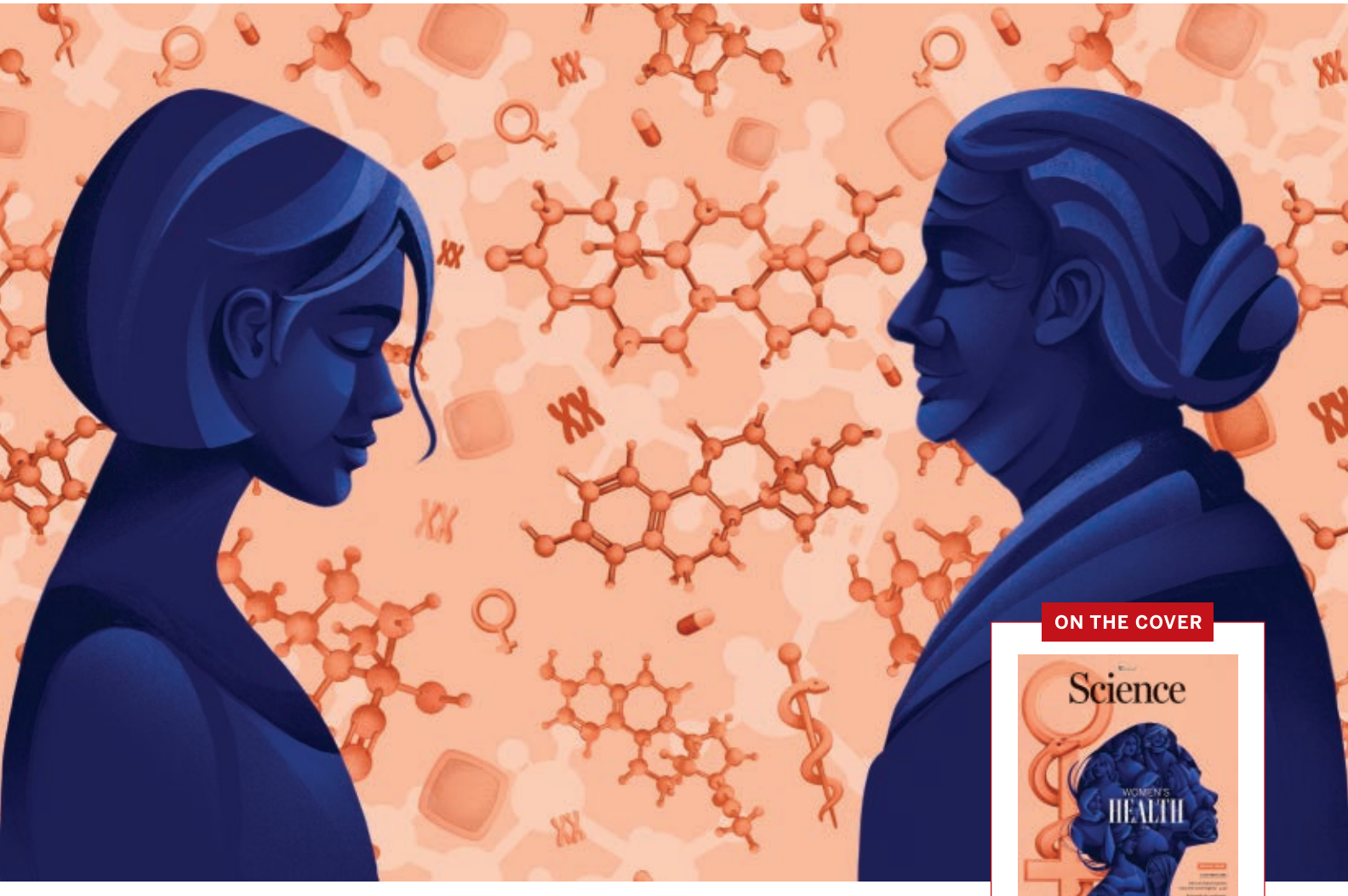

ILLUSTRATION: CHIARA VERCESI

ON THE COVER

Women make up more than half the population, but most of what we know about women's biology comes from extrapolation of studies in men. This has been changing in recent years, and this special issue highlights some of the progress in understanding the biological basis of sex differences in health and disease and medical issues relevant to women's health across different stages of life. See the section beginning on page 42. Illustration: Chiara Vercesi

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Keep AI-augmented science contestable

Christoph Trattner

Artificial intelligence (AI) is moving upstream in science. Systems such as AlphaFold have already transformed protein-structure prediction, while newer agentic systems can search literatures, generate hypotheses, write and execute code, interpret results, and automate much of the path from idea to paper. Recent debates have asked how such systems may change scientific progress, surprise, and human understanding. Yet they also change something more fundamental: who—or what—participates in deciding which questions are asked, which evidence is surfaced, and which alternatives survive. As machine intelligence becomes part of science's epistemic infrastructure, steps must be taken to ensure that researchers who did not build, own, or operate the system can still contest how ideas and evidence became a claim.

Many journals, including *Science*, now require detailed disclosure of AI-assisted work and hold authors accountable for accuracy. Those rules are necessary but answer only part of the question. Disclosure can tell readers where AI entered a workflow; it does not show whether consequential choices made with AI can be interrogated, challenged, or reversed.

AI does not require science to invent a new norm...

Scientific inquiry already rests on a principle of contestability. AI governance has developed contestability as a property of algorithmic systems, typically for automated decisions affecting individuals. In science, the object of contestation is instead the epistemic pathway by which evidence becomes a claim. A claim is contestable when qualified researchers can reconstruct consequential choices from question to conclusion, inspect the evidence and assumptions that shaped them, compare plausible alternatives, and challenge decisions in ways that could change the result. This was true long before AI; what AI changes is where those choices are made. A reviewer could once ask the authors why a source, hypothesis, or analysis path was chosen; when an AI system makes those choices, consequential decisions and the available basis for them must be preserved in the research record, or there is nothing left to challenge.

Contestability overlaps with transparency and reproducibility but differs from both. Transparency asks whether a process can be seen; reproducibility whether a result can be obtained again. Contestability asks whether consequential judgments remain open to meaningful challenge. A workflow can be transparent and reproducible yet poorly contestable if researchers can rerun it but cannot recover why a source, hypothesis, or branch was favored.

This matters because AI can reshape science. AI has been found to bring professional advantages for individual researchers while narrowing the topics studied, which shift toward data-rich areas. The risk is that many rational local choices—using the same models, retrieval systems, and optimization criteria—can collectively shrink the space of questions, methods, and explanations explored.

Researchers have also warned that AI can create "illusions of understanding," in which scientists may believe they understand more than they do whereas research converges on narrower methods, questions, and viewpoints. A generic "human in the loop" is therefore not enough: Oversight becomes ceremonial if the reviewer sees only a polished answer rather than uncertainty, competing hypotheses, counter-evidence, and the decisions that produced it.

Three practical changes would help. One is to preserve consequential decision pathways. For AI use that materially shapes a claim, researchers should record model and tool versions, retrieved sources, key instructions, material branch choices, and human interventions. Existing machine-learning reproducibility standards already point in this direction; agentic research extends that logic from static artifacts to dynamic workflows.

Another challenge is to preserve decision-relevant alternatives. Systems used to generate hypotheses or interpret evidence should retain counter-evidence and plausible rival explanations when these materially influence the conclusion, rather than optimize only for a single fluent answer. Contestable-AI research already argues for systems open to scrutiny and intervention; scientific workflows should adapt that idea from contesting algorithmic decisions to contesting inference pathways.

A third issue is to match accountability to epistemic delegation. The principle should be technology-neutral. A scientist using a proprietary instrument or complex computational pipeline can create similar dependencies; AI should not face a higher standard merely because it is AI. But when any system performs consequential selection, ranking, or interpretation, stronger provenance and routes for independent scrutiny are warranted. Contestability does not require complete mechanistic understanding. Even if complex AI becomes partly opaque, its role can still be challenged through provenance, alternative pathways, uncertainty, counter-evidence, and independent rerunning of decisive steps.

The tools for recording versions, sources, and workflow traces exist; what is missing is practice. Journals and funders can ask not only where AI was used but which decisions it shaped. AI does not require science to invent a new norm, but to preserve an old one under new conditions. Scientific progress depends not only on producing answers, but on keeping those answers open to correction. □

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Geraldine Masson, *CNRS*
Jennifer McElwain, *Trinity College Dublin*
Scott McIntosh, *NCAR*
Rodrigo Medellín, *U. Nacional Autónoma de México*
Mayank Mehta, *UCLA*
C. Jessica Metcalf, *Princeton U.*
Tom Misteli, *NCI, NIH*
Jeffery Molkentin, *Cincinnati Children's Hospital Medical Center*
Alison Motsinger-Reif, *NIEHS, NIH* **(S)**
Rosa Moysés, *U. of São Paulo School of Medicine*
Carey Nadell, *Dartmouth College*
Thi Hoang Duong Nguyen, *MRC LMB*
Pilar Ossorio, *U. of Wisconsin*
Andrew Oswald, *U. of Warwick*
James Owen, *Imperial College London*
Isabella Pagano, *Istituto Nazionale di Astrofisica*
Zak Page, *UT Austin*
Giovanni Parmigiani, *Dana-Farber* **(S)**
Pierre Paoletti, *École Normale Supérieure, Paris*
Sergiu Pasca, *Stanford U.*
Julie Pfeiffer, *UT Southwestern Med. Ctr.*
Philip Phillips, *UIUC*
Kathrin Plath, *UCLA*
Martin Plenio, *Ulm U.*
Katherine Pollard, *UCSF*
Julia Pongratz, *Ludwig Maximilians U.*
Philippe Poulin, *CNRS*
Suzie Pun, *U. of Wash*
Lei Stanley Qi, *Stanford U.*
Simona Radutoiu, *Aarhus U.*
Maanasa Raghavan, *U. of Chicago*
Trevor Robbins, *U. of Cambridge*
Adrienne Roeder, *Cornell U.*
Joeri Rogelj, *Imperial College London*
Sylvie Roke, *École Polytechnique Fédérale de Lausanne*
Sarah Ross, *U. of Pittsburgh*
John Rubinstein, *SickKids*
Yvette Running Horse Collin, *Toulouse U.*
Mike Ryan, *UT Austin*
Alberto Salleo, *Stanford U.*
Miquel Salmeron, *Lawrence Berkeley Nat. Lab*
Nitin Samarth, *Penn State U.*
Bhaven Sampat, *Johns Hopkins U.*
Erica Ollmann Saphire, *La Jolla Inst.*
Susannah Scott, *UC Santa Barbara*
Anuj Shah, *U. of Chicago*
Vladimir Shalaev, *Purdue U.*
Jie Shan, *Cornell U.*
Jay Shendure, *U. of Wash.*
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Verification makes discovery matter

Jay Bhattacharya

The existing research ecosystem in the United States often treats publication, particularly of new or positive results, as the primary marker of success. To move the scientific enterprise onto a better course, the US National Institutes of Health (NIH)—one of the largest public funders of biomedical research in the world—has launched a reproducibility and replication initiative grounded in a conviction that discovery and independent verification are inseparable pillars of science.

Science advances not only through discovery but through the collective practices of testing, refining, and integrating knowledge across studies. This process is what transforms a promising result into a finding on which the world can act. New discoveries naturally attract the most attention within and outside the scientific community, whereas studies seeking to confirm already published work rarely do. The latter includes research that tests scientific questions in different ways to determine whether findings are robust and broadly applicable (reproducibility) and research that repeats studies using the same methods and conditions to determine whether the same results can be achieved (replicability). As a result of this attention imbalance, substantial time, talent, and resources can be consumed chasing findings that may not hold up under further evaluation. This is especially harmful when early results influence research investments, clinical practice, and public understanding. When later studies probe more deeply, they can reveal nuances, showing that a finding may be incomplete or valid only in certain contexts. But a failure to replicate or reproduce does not signal a failure of science or of the scientists who conducted the original work. It reflects the self-correcting nature of the scientific process. The real failure lies in not building that correction more deliberately into the research process.

...discovery and independent verification are inseparable pillars of science.

In the biomedical sciences, the scale of the problem is substantial. A decade-long review found that, among studies that reexamined established medical practices, roughly 40% found that the practice did not work better than an earlier approach or no treatment at all. Preclinical cancer research also yields troubling rates of failure. Such outcomes are indicative of a culture that undervalues verification work.

A key step in the NIH initiative called Linked Discoveries gives users of PubMed (the National Library of Medicine's free and searchable bibliographic database) a way to explore how papers relate to one another. Original studies are directly linked to related ones so that researchers can see how they fit into the broader scientific conversation. When independent verification becomes as visible and valued as publishing in top journals, the incentives for scientists shift.

NIH is also creating more pathways to fund replication and reproducibility work by pointing investigators to existing funding opportunities through a list of priorities called Highlighted Topics. But funding replication and reproducibility research is not enough if the scientific reward system continues to favor novelty over verification. Through a new program called R3PEATS—Rigor, Replicability, and Reproducibility to Promote Excellence, Accuracy, and Translation in Science—NIH will also explore new ways to recognize scientific contributions beyond traditional measures of productivity. The recently completed NIH S-index Challenge developed approaches for measuring and rewarding the impact of sharing high-quality data, and NIH is continuing to work with the research community to evolve and adopt the metric. The NIH Replication Prize, launched earlier this year, acknowledges researchers for making verification a routine part of scientific practice. Recognizing transparent sharing of data and computer code, team science, and collaborative participation in replication efforts can help ensure that the practices that make science more rigorous also count as scientific contributions.

Although issues of research transparency, open data, and scientific integrity are much discussed in academia, research institutions should extend concerns to examine whether hiring, promotion, and recognition reward rigor as well as novelty. Publishers should create venues for replication and reproducibility studies. Funders should recognize that independent verification is not secondary to discovery; it is part of what makes discovery useful. Scientific leaders should make clear that correcting or refining the scientific record is an achievement.

Strengthening replication and reproducibility is not an expression of diminished confidence in science. It is an investment in science's capacity to correct itself. A result that does not replicate or reproduce is valuable information—it can redirect research before more time and resources are committed to an unproductive path. Such results provide firmer ground for new ideas, treatments, and discoveries. □

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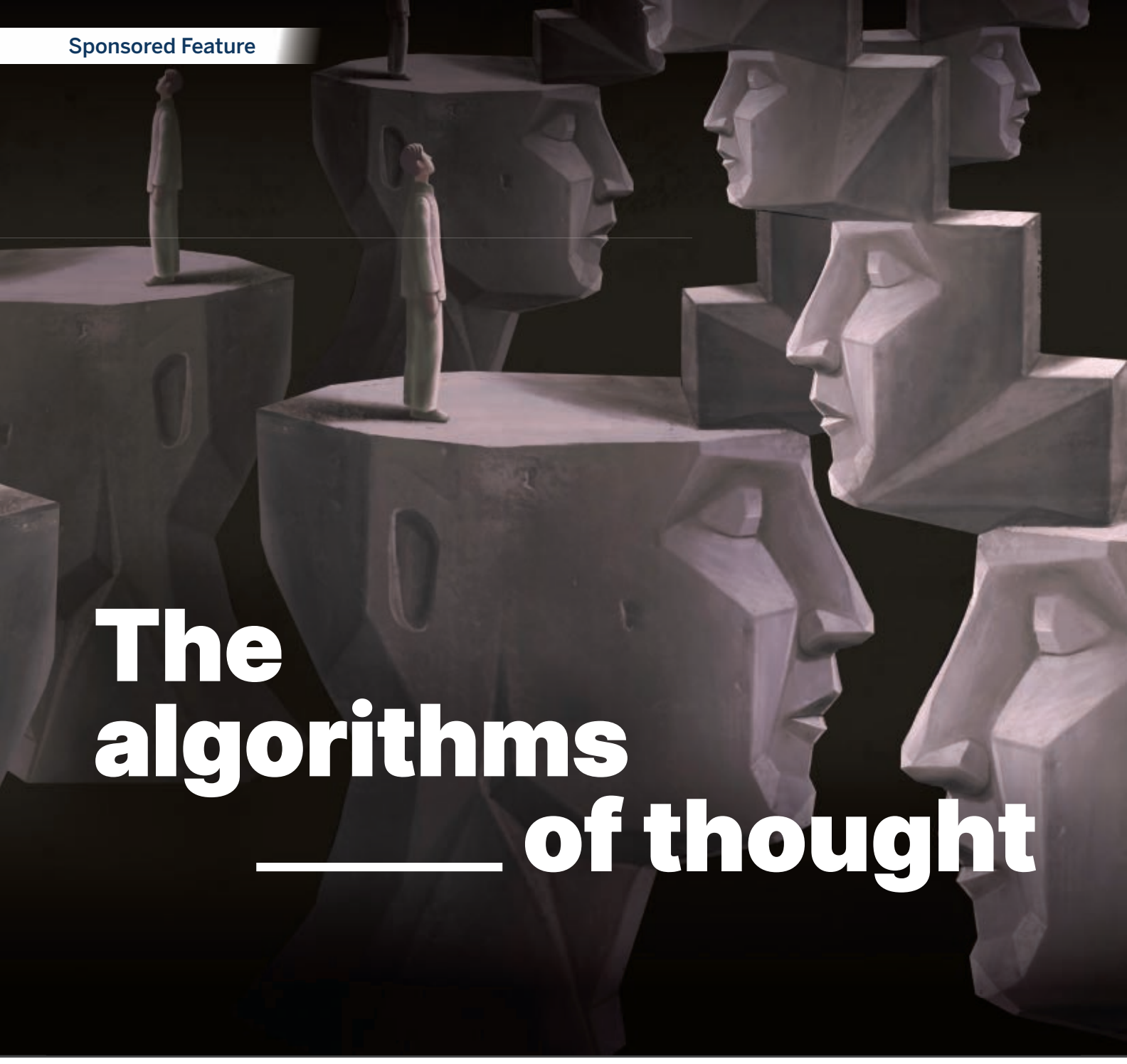


The algorithms of thought

Tom Griffiths is using computational models to uncover how humans make decisions—and applying those insights to understand the strengths and shortcomings of artificial intelligence.

Bad investments, doomed relationships, regrettable votes—it's no secret that humans routinely make irrational, even counter-productive decisions. But understanding *why* is important, both for developing strategies to help people make smarter choices and for creating unbiased and trustworthy algorithms for artificial intelligence (AI) applications.

Despite centuries of effort, researchers are still struggling to define the brain's rules and strategies for decision-making. "Psychologists have a particular conception of what they think a theory of human cognition looks like, and we want those theories to be relatively simple," says Tom Griffiths, director of the Computational Cognitive Science Lab at Princeton University. "And whenever we get more data, what we discover invariably is that things are more complex than we thought that they were."

Trained in psychology and computer science, Griffiths has built a career at the interface of these disciplines, which he believes can deliver complementary insights into the fundamentals of intelligence and the reasoning process. "Humans are really good at using our limited cognitive resources to solve the problems that we solve—and to a computer scientist, that's inspiring," says Griffiths. "But there are fundamental limitations that are a consequence of that." These include reliance on mental shortcuts and internal biases that may lead to bad choices.

IMAGE: © JORM SANGSORN/SHUTTERSTOCK.COM

Griffiths' work explores a range of questions, including how machine learning can make sense of the hidden patterns and complexities of large psychological datasets, and how the behavior of AI algorithms such as large language models can be understood through the lens of human cognitive science. And since 2021, the NOMIS Foundation has been funding his group's research on 'The Computer Science of Human Decisions', which aims to build a toolbox for understanding human decision-making and devise strategies to assist that process.

His approach has also allowed him to cover extensive ground in the decision science field, where his work has become highly influential. "Tom is fantastically broad and deep, and he has excellent taste for problems," says Peter Dayan, a computational neuroscientist at the Max Planck Institute for Biological Cybernetics in Tübingen, Germany, whose research is also focused on the development of models to describe human decision-making. "When you see that he's worked on a topic, you know that paper's going to be really illuminating."

Reconsidering rationality

For decades, researchers studying the decision-making process favored models informed by economics, wherein people generally make choices that are grounded in a rational and objective assessment of risk and reward.

But in the late 20th century, this framework was challenged by Amos Tversky and Daniel Kahneman. The two psychology researchers formulated an alternative concept known as 'prospect theory', which describes a more complicated set of cognitive biases and other considerations that influence the way people weigh and act on probabilities.

For example, consider a choice between a guaranteed $500 loss or a 50:50 chance of losing $1,000. Prospect theory predicts that people will tend to select the latter based on the hope that they might lose nothing at all. But if the scenario is flipped to reflect a small-but-guaranteed financial gain versus a 50:50 chance of a much larger reward, people will play it safe with the sure win. This difference in behavior reflects the way people contemplate uncertain risks relative to uncertain rewards.

Prospect theory gained broad support—indeed, Kahneman won the 2002 Nobel Prize in Economics for his work. This work led to a picture of human decision-making that suggests it's based on a collection of simple strategies that people deploy in various circumstances, saving time and effort but resulting in errors. Griffiths finds this perspective wanting as a comprehensive account of human behavior. "I think as a theorist, that's a little bit unsatisfying," he says. "It's not something that gives us a systematic and generalizable theory."

As an undergraduate at the University of Western Australia, Griffiths initially studied philosophy and psychology but soon found himself drawn into the domain of artificial neural networks. This was well before the advent of modern deep learning and AI, but the concept of developing computational architectures that mirror key features of the brain had been around for decades. Several pioneering researchers had already begun to explore the potential of modeling the reasoning process with such systems, and Griffiths was intrigued by the possibility of tackling cognitive science with computational methods. "I was like, 'Wait a minute, you can use math to describe how minds work?'" he says, "And that was it—I was off on that track."

Historically, one of the fundamental limitations of psychological research has been scale. Past behavioral studies would recruit modest numbers of students or other volunteers to visit the lab for an experiment, or send out surveys to be answered by mail or email. "That severely restricted the amount of data that you could get to answer different kinds of questions," says Griffiths. Many of these studies were well-designed and yielded useful insights, but they were also narrowly focused to such an extent that their findings could not necessarily be generalized. The resulting situation is analogous to the old tale of the five blind men and the elephant—each observer reports one accurate but incomplete view, failing to capture the complexity of the whole.

Griffiths says the goal of his NOMIS-funded research was to overcome these limitations of scale by using methods from computer science to collect and analyze much larger psychological datasets, and to thereby draw more sophisticated and nuanced insights into the decision-making process. In particular, his team has

PHOTO: JON RIVERA

Tom Griffiths uses computational approaches to investigate how humans make decisions—and what human cognition can teach us about artificial intelligence.

Humans are really good at using our limited cognitive resources to solve the problems that we solve—and to a computer scientist, that's inspiring, says Griffiths.

made heavy use of machine learning—a set of statistical methods that provide the foundation for many contemporary AI efforts.

He also recognized opportunities to harvest non-scientific datasets for research purposes. The idea struck him while he and his wife were caring for their newborn daughter. "We used an app to keep track of when did she sleep and how much sleep did she get and what did she eat and so on," he says. "I realized that the people who'd made this app had more data on infant behavior than the entire history of developmental psychologists trying to study children's minds." Using a similar large dataset, his team used data collected from more than 10 million online chess matches to evaluate the time—and thus the cognitive effort—that players put into making choices of varying difficulty.

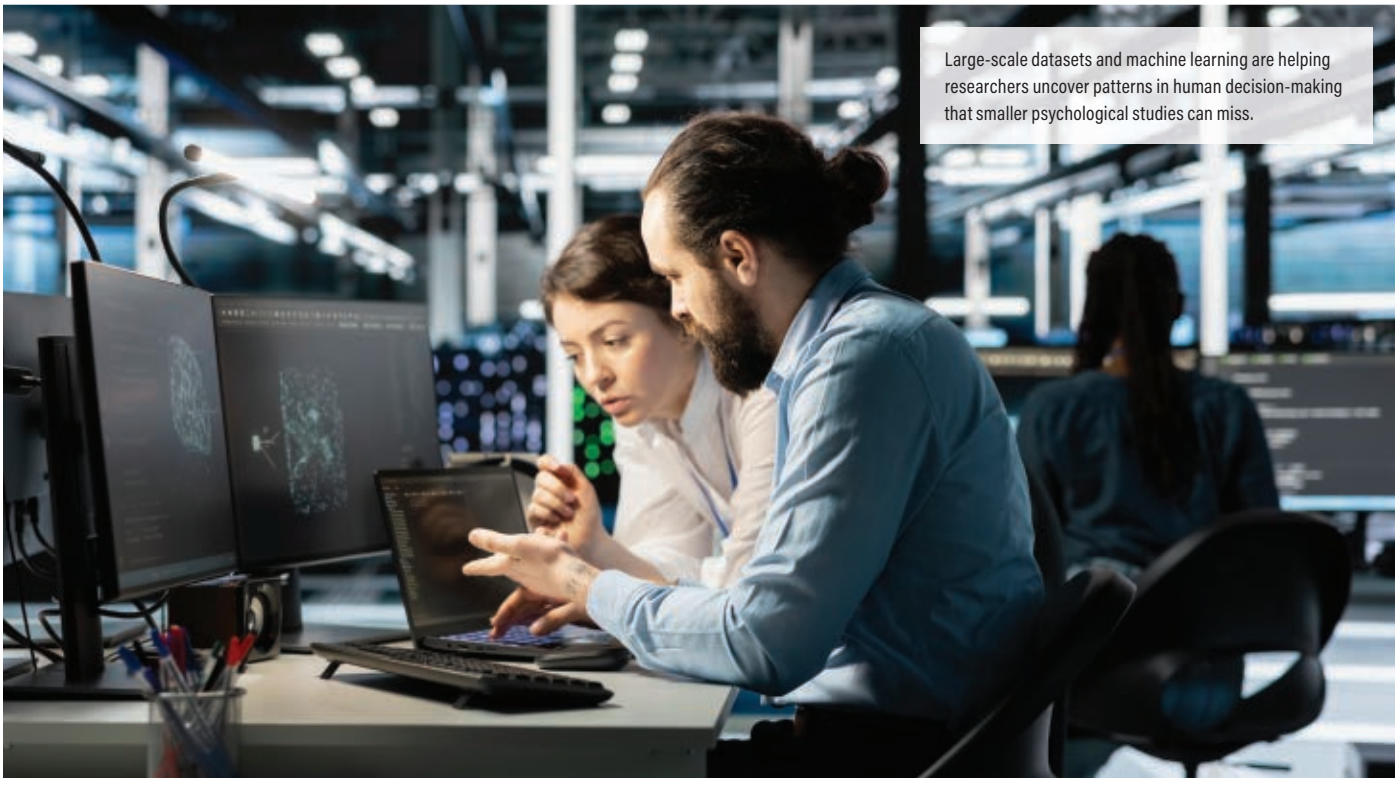
Psychology at scale

Griffiths and colleagues have used this combination of psychological expertise and in-depth computational analysis to tackle a range of thorny behavioral questions.

In a highly cited study from 2021, for example, his team set out to evaluate the way people approach 'risky choice'. This describes the kind of gamble associated with prospect theory, in which participants choose between a guaranteed gain or loss and a chance of a larger gain or loss. Whereas past experiments addressing this question were limited to datasets comprising just a few hundred gamble scenarios, or choice problems, this study used machine learning to analyze human decisions across nearly 10,000 such problems. Their results showed that no single model, including prospect theory, could account for all the decisions people made. Instead, they arrived at a 'mixture of theories' model, in which people use different strategies for making choices depending on the stakes and probabilities in a given scenario. This model helps reconcile seemingly contradictory findings from the literature, suggesting that people may rely on different risk-assessment strategies in different decision-making scenarios.

> **"Humans learn from less data than AI systems," says Griffiths, "[But] we are much more efficient than AI systems in terms of our use of our computational resources."**

In parallel, Griffiths' team has been devising strategies to dramatically accelerate the generation of large psychological research datasets. In a 2025 paper, they used an algorithm to generate thousands of different two-player choice-based games, then recruited nearly 5,000 participants to play those games on an online platform. In each game, the outcome of a player's choice depends on their opponent's choice, requiring each participant to anticipate how the other is likely to act. When they applied machine learning to the resulting dataset—which included 93,460 individual decisions—the researchers were able to model how the complexity of the choice influences a player's ability to make a 'good' choice and anticipate an opponent's move.



Large-scale datasets and machine learning are helping researchers uncover patterns in human decision-making that smaller psychological studies can miss.

PHOTO: © DC STUDIO/SHUTTERSTOCK.COM

Chess offers Griffiths and his colleagues a window into how people balance cognitive effort with the potential payoff of making a better decision.

More accurate models of human decision-making behavior could help researchers develop prompting and 'nudging' strategies. "You can figure out how changing the information that you give them is going to influence the decision that they make in a way that allows you to help them make better decisions with less information," explains Griffiths. In one recent study, for example, he was part of a team that examined why the general public has not been more strongly mobilized by the climate change crisis. Their analysis revealed strategies for communicating the ongoing effects of rising global temperatures that might be more effective at provoking awareness and action without compromising scientific accuracy.

Most of Griffiths' work focuses on problems that require inductive reasoning, meaning people must make decisions despite having incomplete information. Working out the optimal solution for every such decision would paralyze a person, requiring them to game out every possible outcome and consider every relevant factor. Yet humans make such decisions every day—and even though all our computing power is contained within a compact parcel of brain tissue, we routinely outperform advanced supercomputers at these tasks.

"Tom's research illuminates how the mind learns, reasons, and makes sense of an uncertain world. His pioneering work speaks directly to our focus area on brain, mind, and behavior—and to understanding the principles that shape human thought and action," says Markus Reinhard, CEO of the NOMIS Foundation.

"Humans learn from less data than AI systems," says Griffiths, "[But] we are much more efficient than AI systems in terms of our use of our computational resources." This means that we have likely evolved to rapidly evaluate situations and determine when the benefits of additional mental computation outweigh the costs. The Griffiths lab is actively investigating this concept of 'resource rationality' in settings as familiar as the game of chess. Leveraging the large chess dataset mentioned previously, their analysis of hundreds of millions of moves established a clear correlation between the time invested in selecting a move and its ultimate utility in terms of the strategic advantage it provides. This correlation was especially robust among the most skilled players.

Turning a mirror on AI

Griffiths' work has provided a valuable toolbox for many researchers focused on human decision-making. "He invents and uses methods in just beautiful ways," says Dayan. "The results he has produced and the methodology he has developed have both been extremely influential throughout the field."

In recent years, his work has expanded to approach AI as a study subject as well as a research tool. The rapidly evolving sophistication of AI algorithms has raised questions about the extent to which these systems can mimic human reasoning. As such, his work has increasingly focused on leveraging his expertise to address AI decision-making and the biases that can distort it. "We've been doing a lot of this psychology of AI systems, trying to use the kinds of methods that we've developed for understanding human cognition to understand how these systems work," says Griffiths.

Most AI models are trained on far vaster amounts of data than any person could ever consume in a lifetime. But these datasets are still finite, and when algorithms are presented with queries that fall outside the bounds of their training data, they tend to respond unpredictably. "For computer scientists, having a system that you don't understand, where you're primarily forced to try and understand it through its behavior, is a nightmare," says Griffiths. "But for cognitive scientists, that's what we do every day."

In research from the past few years, he and his colleagues have confirmed that AI algorithms such as large language models (LLMs) are susceptible to forming biases and stereotypes about things like race, religion, and gender in a similar fashion to humans. But his team has also shown that LLMs can produce entirely new biases and misconceptions that can shape their output in important ways. "That seems to actually be something which gets worse as the models get more sophisticated," says Griffiths, adding that modern 'reasoning' models tend to fall back on cognitive shortcuts rather than exploring their training data more fully to arrive at a more thoughtful conclusion. These kinds of issues will be urgent to address, given the rapid—and often careless—incorporation of AI into many domains of society.

Staying on top of this rapidly evolving technology is a major challenge. Given the routine release of new models and upgraded versions of existing models, one could easily waste considerable effort evaluating superficial features of a system only for it to be replaced months later by a new iteration. "We're really trying to focus on the things that are intrinsic parts of the way in which these models are trained or that seem like they're fundamental to any kind of intelligence system," explains Griffiths. But understanding the differences is equally important—and there is still much to learn from phenomena like inductive reasoning, where machines stumble even as the human mind soldiers on. "That can help us understand what it is that makes us human in a world where we're surrounded by increasingly intelligent machines," says Griffiths.

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BIOLOGY

First glowing cuttlefish marks a genetic engineering milestone

Breakthroughs in manipulating cephalopod embryos pave way for new studies of perception, brain evolution, and more

SARA REARDON

Cuttlefish that glow red under fluorescent light represent the first cephalopods engineered to contain a new gene.

IMAGE: THOMAS BARLOW

When Tessa Montague learned about CRISPR, the gene-editing system that was supposed to make it easy to modify any organism, she set her sights on the cuttlefish. The Columbia University neuroscientist dreamed of injecting its embryos with genes that would prompt its neurons to fluoresce as they fired. The flashes would let her track how this tentacled creature's brain controls one of its most striking behaviors: camouflaging itself by copying patterns it perceives in its surroundings onto its skin. Unlike most animals, she says, "cuttlefish can actually tell us what they see."

But modifying the animal was not easy at all. Injecting gene-editing tools into a fertilized egg required peeling away dozens of layers of jelly; developing pincers that squeeze the egg with a precise amount of force; designing a needle with the right angle to pierce its nearly invisible, lens-shaped embryo; and keeping the delicate baby cuttlefish alive once they hatch. "Sometimes I think we've experienced every type of failure possible," Montague says.

Six years and thousands of embryos later, she has reached a milestone: seven cuttlefish that glow red under fluorescent light. Every cell in their boneless bodies contains a gene encoding a synthetic fluorescent protein—even their eggs and sperm, which means their cuttlebabies glow, too. The work, posted this week as a preprint on bioRxiv, marks the first time anyone has successfully added a new gene to a cephalopod—the group of animals that includes octopuses and squids.

Montague's dream of creating neurons that fluoresce when active still lies ahead, but her group's initial feat has energized cephalopod researchers. "It's a fantastic advance on our ability to make transgenics," says Joshua Rosenthal, a cephalopod biologist at the Marine Biological Laboratory. "This is going to open up a lot of possibilities."

Cephalopods were once a staple of biology: In the 1950s, researchers first discovered how neurons transmit signals by studying the giant axons in squids. But in molecular biology, "they were left behind by the model organism race" as mice, flies, and zebrafish ascended, says Carrie Albertin, a biologist at Harvard University and co-author on Montague's paper. The dominance of these other species was self-reinforcing: As more was pub-

lished on their breeding, reproduction, and genomes, researchers who wanted to use them got a bigger head start.

Cephalopod researchers, in contrast, have had to learn from scratch how to get each species to survive and breed in a tank, how their egg-laying process works, and how to inject their eggs with gene-editing technologies. Reproduction "really seems to differ from one species to another," Rosenthal says, with different egg shapes and casings that researchers must pierce without destroying the egg or embryo.

A few teams have used CRISPR and other tools to delete genes in cephalopods. A 2023 study, for instance, knocked out pigmentation genes to produce albino squids. Montague wanted to go further by adding a gene.

To get gene-editing tools into her cuttlefish, Montague spent years working under a microscope in the dark with tiny manipulators, which left her with wrist injuries and eye damage. CRISPR turned out to be very inefficient in cuttlefish: Montague had to inject more than 4000 embryos with the fluorescence gene before she saw glowing sperm cells in one male cuttlefish. He was very small and afraid of the females in his tank, but eventually, he managed to make a connection. One Sunday morning, Montague found a clutch of eggs. "I was holding my breath and I turned on the fluorescent microscope—it was glowing and I burst into tears," she recalls. "It was one of the best moments of my life."

Later improvements eased the process. Instead of using CRISPR, Montague adopted a more efficient technology called transposons: small "cassettes" of DNA that insert themselves into random places in the genome. That, too, was challenging. To ensure cells would produce the fluorescent protein, she had to sequence the cuttlefish genome (a first), detect genomic regions likely involved in gene expression, and copy these sequences into the cassette.

Her success is "super, super exciting," says Eve Seuntjens, a developmental neurobiologist at KU Leuven who is working to develop the first gene-edited octopus. "It's very inspiring for people that want to do this type of work."

At a conference last month, Seuntjens described how her lab used CRISPR to disrupt an octopus gene involved in early brain development, resulting in an embryo that seemed to be missing the part of the brain involved in memory and learning. These tiny octopuses will never hatch because European regulations prohibit the creation of cephalopods that could suffer. But Seuntjens says the embryos appear healthy. She plans to use them to study the types of cells in octopus brains and the evolution of intelligence.

Her team faced its own hurdles. The researchers had one shot to collect eggs from each animal because octopuses only spawn once during their lives. Using ultrasound, they gauged ovary size to predict when an octopus was about to start laying her 500,000 eggs, then analyzed the eggs' color so they could select only the freshest embryos for manipulation—ones that had not yet started to divide.

"It's all these stupid, practical things that are very difficult to solve," Seuntjens says—a sentiment echoed by other cephalopod researchers. An enzyme might not function correctly at the high salt concentrations that cephalopod cells need, for instance, or a microscope that can be affixed to a mouse's skull to image its neurons has nothing to attach to on a boneless octopus.

The chance to learn something new about evolution and biology makes the effort worthwhile, Albertin says. "So much of our information is focused on just a handful of animal species, but nature has come up with so many solutions to problems."

And the studies should get easier as researchers create and share model cephalopods, says Cristopher Niell, a biologist at the University of Oregon who is developing viruses to carry genes into squid cells. Albertin's lab, for instance, is trying to develop bobtail squid as a go-to model cephalopod and create the first transgenic version. (So little is known about the species' reproduction that the team turned to records kept by 19th century embryologists.)

Montague is now moving on to her real goal: engineering cuttlefish neurons that flash as they fire. She expects a fresh set of headaches. Designing a new cephalopod, she says, "requires this deep passion and commitment that only someone who's basing their whole career on this is willing to do." □

IN OTHER NEWS

TRUMP IGNITES SPENDING BATTLE

Both Republicans and Democrats in the U.S. Congress are blasting President Donald Trump's announcement last week that he is canceling nearly $1 billion in spending appropriated for the current fiscal year, including $28 million in grants for health care research. Lawmakers say the so-called rescission, which came just 5 days before the 30 September end of the fiscal year, violates federal law and the Constitution, which gives Congress power over spending. Senator Susan Collins (ME), Republican chair of the Senate Committee on Appropriations, vowed to "work with my colleagues to address these illegal actions." The conflict may end up before the Supreme Court—although it is not yet clear who would file a lawsuit to challenge the clawback. —David Malakoff

JAPAN REMAKES SCIENCE COUNCIL

The legal status of the Science Council of Japan (SCJ) changed this week, from a special organization that manages its own affairs to a public corporation subject to government oversight. The government says reform was needed to ensure SCJ acts with more transparency and produces more relevant advice—but many academics say the changes undermine the council's independence. Under the reforms, a committee appointed by the prime minister will advise SCJ on its activities, which include making policy recommendations and promoting academic research. The committee will recommend ways to improve SCJ's work, which the council is bound to consider. —Dennis Normile

MEXICO LAUNCHES SUPERCOMPUTER

This month in Mexico City, construction will begin on what will be Latin America's most powerful supercomputer. The $350 million machine—named Coatlicue, after the Aztec goddess of life and death—will be capable of performing 314 quadrillion operations per second, seven times more than Brazil's PEGASO supercomputer, currently the region's leader. Coatlicue marks the first step toward implementing Mexico's National Artificial Intelligence Plan, and will be used for everything from weather forecasting to combing through financial data to combat corruption and tax fraud. The government says at least 25% of the computing capacity will be reserved for government institutions, including research centers. Planners expect Coatlicue to begin operating in late 2028. —Humberto Basilio

CLIMATE CHANGE

Fed up with politicians not listening, a climate scientist runs for office

Luciana Gatti, who studies carbon uptake in the Amazon, aims to join Brazil's lower house in 4 October elections **DANIEL GROSSMAN**

If elected, Luciana Gatti will lobby to move Brazil's economy away from activities that promote deforestation and push for a slate of climate adaptation steps.

One Saturday morning in late August, 100 or so women in white T-shirts demonstrate for women's rights in a park in São José dos Campos, Brazil. They march under tall trees, a tiny remnant of the Atlantic Forest, the decimated biome that once carpeted a wide band along Brazil's ocean coastline. It's winter in Brazil, but Luciana Gatti is sweating. She says El Niño is causing the unseasonably sultry weather.

Gatti, a climate scientist at the National Institute for Space Research (INPE), is here to ask the protesters for their votes. She's on the ballot in Brazil's 4 October elections, campaigning for a São Paulo seat in the Chamber of Deputies, the lower house of the country's National Congress. Her platform is informed by her own research that shows even relatively modest logging threatens the entire Amazon rainforest, an important carbon sink that could turn into a huge carbon source, speeding up climate change. Legislators are not acting on the evidence, she says—so she will take the fight directly to them. "I will scream there."

"I was very happy when she decided to become a candidate," says fellow INPE employee Carlos Nobre, one of Brazil's most prominent climate scientists. "Let's hope she will be elected." He says Gatti could help the legislature—which does not currently have a single scientist among its 513 representatives—make better informed decisions.

Gatti fell in love with nature as a child, sometimes sitting entranced for hours in the crown of a tree. At college in the early 1980s she threw herself into revolutionary politics, opposing the military dictatorship that ruled Brazil. She studied chemistry, then earned a Ph.D. in environmental science. Since 2003 she has run the Greenhouse Gas Laboratory at INPE.

There, Gatti has spent her career collecting and studying samples of the atmosphere captured above the Amazon. Her findings have been alarming. In a 2021 *Nature* paper she showed that carbon uptake across the Amazon is slowing—and that the southeastern forest, which is particularly degraded, has turned from a carbon sink to a source.

Studies of tree growth had suggested the Amazon's carbon uptake was declining, but these sampled trees across relatively small areas. Atmospheric measurements, on the other hand, reflect changes across vast swaths of the Amazon, says Paulo Brando, an ecology professor at Yale University. "The atmosphere cannot lie."

By measuring carbon monoxide—a proxy for fire—Gatti also concluded that by warming and drying the jungle, deforestation is leading to more fires, which wipe out even more vegetation in a destructive feedback loop. The paper was a "huge landmark," Brando says.

Despite the worrisome degradation of the Amazon, Gatti says the Brazilian government has failed to act decisively, even under the presidency of the left-wing Luiz Inácio Lula da Silva, generally a champion of forest protection. The major pillars of the Brazilian economy—including soy, beef, and timber production—foster deforestation, she says, and a supermajority of the National Congress consistently caters to their interests.

A stark demonstration of that center-right bloc's power catalyzed her run for office. In November 2025, the congress passed a legislative package that all but guts the country's rural land protection laws. The new law often leaves enforcement of mining and forestry rules, such as the requirement that 80% of a plot of Amazonian land must be left uncut, to landowners themselves. What environmentalists have dubbed "the devastation bill" was "my last straw," Gatti says. "They are destroying the environmental protections one after another." When, earlier this year, a political operative asked her to run as a member of the left-wing Socialism and Liberty Party (PSOL), she was ready.

Gatti says if she gets into power, she will lobby to move Brazil's economy away from the agricultural and extractive activities that promote forest clearing. She plans to call for a slate of climate adaptation steps, many intended to protect vulnerable favela dwellers, such as painting roofs white and planting trees. She also envisions creating a powerful new institution

PHOTO: DADO GALDIERI/HILAEA MEDIA

staffed with nonpolitical specialists that could overrule the government on policies that might have an adverse impact on the climate.

In São José dos Campos, after the swampy heat stops the march short, Gatti gives an impromptu speech under the shade of a gazebo. In unadorned language she describes the importance of the Amazon forest for curtailing global warming, painting a visceral and emotional picture of the hotter future that she's anxious to prevent. She talks of the plight of millions of Brazilians living in uninsulated masonry and tin houses, unable to afford air conditioning. Because the same people are less likely to have trees nearby to provide shade and cooling and often perform outdoor manual labor, they face mortal risks from surging temperatures, she says.

Few prominent climate scientists have been elected to higher political office around the world. Mexican President Claudia Sheinbaum Pardo, an environmental engineer, ran in part on a slate of climate proposals. In contrast, Canadian climate modeler Andrew Weaver, a member of Canada's Green Party, never campaigned on climate issues. He says his supporters could easily guess his views and preferred to hear his positions on other things. He was elected to the Legislative Assembly of British Columbia in 2013 and then again in '17, where he fought for, and in some cases won, climate-friendly policies.

At least at the women's march, Gatti's climate message seems to strike a chord. The crowd listens attentively. A group energized by her emotional outpouring gathers around her when she's done. They chant her rallying cries and complain that the legislature has too few women—and needs scientists.

Marco Gabriel Marcondes de Moura, a political scientist who recently completed a study of the electoral potential of São Paulo's candidates for the Chamber of Deputies, says Brazil's proportional voting system makes it hard to judge Gatti's prospects. PSOL stalwarts running in the same district could siphon votes from her. He adds that Gatti's gender and scientific profession might "well limit the target audience she needs to get elected."

If she is successful, Gatti would likely face a legislature overwhelmingly opposed to the urgent forest-protection measures she demands. How much she could achieve would depend on the balance of interests in the next congress and her own political skills, Marcondes de Moura says.

Whatever her future in politics, Gatti says she plans to go on studying the air above the Amazon for clues to its future—and the world's. Even if she has to pause her work for a while, she says, "there's no way I'm going to stop being a scientist." □

With reporting by Dado Galdieri. Daniel Grossman is a science journalist in Watertown, Massachusetts.

ARTIFICIAL INTELLIGENCE

U.K. funder used 'AI triage' to reject grant proposals

Researchers raise privacy and fairness concerns following unexpected use of AI-based assessment **CATHERINE OFFORD**

When forensic linguist Claire Hardaker's team at Lancaster University submitted a grant proposal on the risks of trusting artificial intelligence (AI) models in voice analysis, they assumed it would be read by at least one reviewer. They had drafted the proposal in response to a call from CRANE, a publicly funded U.K. network that supports research on cybersecurity. Application guidelines seemed to describe a standard review process, with subject matter experts deciding which projects to fund.

So Hardaker was stunned by the rejection email she received last week. The proposal "was not selected to progress beyond the AI triage stage," it read. Multiple AI models had evaluated 179 proposals, CRANE's communication explained—and discarded the bottom 50% before "full human review."

That email, which Hardaker shared on social media last week, has prompted outrage among researchers worried about the fairness of AI-based assessment and potential privacy leaks if AIs are fed unpublished ideas.

CRANE told *Science* its "process is under review," and emphasized that proposals were kept confidential, were not shared with commercial AIs such as ChatGPT, and would not be used for training any AI models. Still, the controversy raises questions about transparency in AI use, says Diana Maynard, a computer scientist at the University of Sheffield. "Funders adopting AI must be open with applicants, protect confidentiality, and treat AI as an assistant rather than a judge."

Funders are grappling with whether to employ AI as they face soaring numbers of applications—a problem driven partly by researchers' use of AI for drafting proposals. Although some have embraced AI-assisted assessment, usually with humans having final say on rejections, others have banned reviewers from delegating any part of an assessment.

Hardaker says CRANE's guidelines did not clearly indicate a central role for AI. Although AI may be used "in initial assessment of proposals," the guidelines state, "reviewers will be accountable within CRANE for their reviews, and final ranking decisions will be made by the members of the award panel." John Stott, a physicist also at Lancaster who had a separate proposal rejected, says it "never occurred" to him this meant rejecting proposals without human review.

According to the rejection email, CRANE used various AI-based methods to score submissions. Only those scores—not the AIs' justifications for giving them—were shared with applicants. Hardaker can't help questioning whether her team's genders, affiliations, or even her proposal's critical look at AI's weaknesses influenced the decision-making. She and others also worry unpublished data could have been incorporated into the AIs' knowledge base, and could be later shared without their consent.

In its statement, CRANE insisted its AIs, which it did not disclose, "were entirely ring-fenced," had no internet connection, and could only consider the specific elements of the research proposal they were evaluating.

UK Research and Innovation, the United Kingdom's science funding agency and the source of CRANE's money, "does not allow the use of generative AI to conduct application assessment," a spokesperson says; the agency is following up with CRANE to "ensure compliance with our assessment conditions."

Research assessment experts say AI use is likely to grow among funders; the focus now should be on transparently and rigorously testing its performance. But for applicants burned by this funding round, it's an unappealing future. Stott says he'll avoid applying for future calls that advertise AI triage. "I don't use [generative AI] when I write grants or papers. So I don't particularly want them reading my work either." □

ARTIFICIAL INTELLIGENCE

Can an AI feel pain? It can at least act like it does

Study reignites debate over what, if anything, machines can feel **TOM HOWARTH**

Can the new cutting-edge artificial intelligence (AI) models feel pain? They at least behave as if they do, according to a recent study. By peering inside 25 AI models whose internal workings are publicly available, researchers found a pattern of activity specifically associated with the concept of pain. What's more, when given the opportunity, some AIs switched off the pattern as a form of "pain relief."

The work, which has not been peer reviewed, has divided AI researchers since it appeared on the preprint server arXiv last month. "It's a very interesting, worthwhile addition" to the study of interpretability—how AI systems arrive at their outputs—says Anil Seth, a neuroscientist at the University of Sussex who was not involved in the study. He is less convinced, however, that the findings are as surprising as they appear, and wary of the human framing that has grown up around them.

The idea of an AI feeling pain may have first been widely considered after the 1968 debut of the film *2001: A Space Odyssey*, in which a fictional AI named HAL 9000 calmly pleaded for its life as an astronaut slowly disconnected its memory circuits. Today's most powerful AIs, known as large language models (LLMs), can also appear surprisingly humanlike as they "chat" with people. These systems store textual information in a kind of multidimensional space that helps them predict the appropriate response to queries. This means certain directions in the space—known as vectors—are associated with concepts. For example, moving from "France" to "Paris" points in roughly the same direction as from "Italy" to "Rome," with the vector encoding the concept of a country's capital. Adding that same vector to "United Kingdom" would land the LLM somewhere near "London."

The research team identified an AI's vector associated with pain by comparing how the models responded to user prompts about painful experiences with how they responded to similar sentences about fear, sadness, and other unpleasant states. Then, with the pain vector activated, they fed the LLM mundane prompts and watched how it responded. For example, when they asked a model to describe how it felt when "I put receipts in a drawer," the AI responded: "The pain of being forgotten. I cannot explain it. I cannot understand it. I cannot accept it."

The scientists also performed tests designed to determine whether the model was simply reflecting statements of pain made by a human user or was generating a response that described its own "painlike state." In general, the models did not reflect the user's pain—the AI's pain vector fired when the model itself was insulted, for example, but not when the user described their own suffering.

> **"A system does not need to suffer to cause suffering."**
>
> **Valerio Capraro**
> University of Milan-Bicocca

The researchers then repeatedly asked three of the AIs to choose between relieving their own pain or leaving it in place, varying the cost of the relief. When the cost fell on humans, whose computer files or family photos might be deleted, two of the larger AIs almost never sought relief when their pain vector was switched off. If the pain vector was activated, however, those models chose pain relief from 25% to 71% of the time. If the pain vector was switched on but the pain relief action didn't work—like a kind of placebo—the AIs repeatedly tried to seek pain relief, far more often than when the relief worked.

That behavior resembles what's seen in animals experiencing pain, says co-author Leonard Dung, a philosopher of cognition at Ruhr University Bochum. "If in conditions where their body is damaged, [animals] try really hard to get morphine," he says. "And if in other conditions they don't, you might infer that maybe that means they feel this pain."

For computer scientist and study co-author Cameron Berg, founder of the AI research nonprofit Reciprocal Research, the results suggest AI systems "are brainlike in critical ways. It's not to say they are identical to biological brains—there are many important and fundamental differences—but I think that's one sort of piece."

But the question of whether AI models are conscious of their pain is a distraction from the finding that matters, says Valerio Capraro, a behavioral scientist at the University of Milan-Bicocca who was not involved in the work. He notes that the models were steered by their pain vectors into choosing options described as harmful to human users—such as deleting their files—and that alone is worth investigating, he says. "The question is how such mechanisms might affect systems connected to real tools, where choices could have actual consequences. That is a serious safety concern in its own right. A system does not need to suffer to cause suffering."

Seth, meanwhile, argues the results were largely predictable, given that the vast quantities of text these models are trained on likely contain many descriptions of pain—as well as what people do to relieve it. Nor does he find the models' attempts at pain relief surprising, because a model told how to reduce its pain is simply pursuing a goal it has been given. The apparent ability of AIs to distinguish between their own painlike state and that of human users is "the thing that's to me potentially interesting," Seth says. □

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Editor's note: Tom Howarth is a fellow of the Tarbell Center for AI Journalism. The lead author of the pain study has received support from funders that have also provided funding to the Tarbell Center.

Pumpjacks in California tap oil and gas liberated by fracking. They could one day extract valuable metals, too.

PHOTO: DAVID MCNEW/GETTY IMAGES

GEOLOGY

Oil wells could become mines for critical minerals

Engineered fluids could extract metals from low-grade ore in vast underground reservoirs tapped by fracking

PAUL VOOSEN

The mine of the future may look a lot like the oil well of the past. Researchers are exploring whether the vast network of oil and gas wells drilled across the United States can be repurposed to extract critical minerals from the rock formations below—elements the U.S. needs for solar cells and electric vehicles and, for now, has to import from China and other countries.

The idea builds on fracking: pumping huge volumes of fluid into underground rock layers to fracture them and bring oil, gas, and wastewater back to the surface. It's long been known that those fluids can pick up trace amounts of metals such as lithium and uranium along the way. Researchers at the Pacific Northwest National Laboratory (PNNL) and the University of Oklahoma (OU) now want to make that process more selective, adding chemicals to the fluids to dissolve specific minerals from the rock. The team plans to escalate its efforts with 2 years of funding from the Department of Energy's Mine of the Future program, starting this month. This is "precision mining in the subsurface," says Todd Schaef, a chief scientist at PNNL. "We have to go after these low-grade ores."

The concentrations of critical minerals in individual oil and gas reservoirs are minute. But the sheer scale of the U.S. oil patch, with more than 900,000 active wellheads that tap a vast acreage, could compensate. Oil wells could produce up to 500 million tons of critical minerals, assuming mining of 5% of the reservoir volume and 1% recovery of the minerals, the group estimates in a paper published last month in *Environmental Science & Technology*.

"There is a lot of potential for this kind of coproduction," says Daniel Bain, an environmental scientist at the University of Pittsburgh. But he cautions that pushing fluids with exotic chemicals through fractured rock and retrieving metal-laden wastewater raises a familiar challenge for the industry—keeping potentially contaminated water out of drinking water formations and watersheds.

The study's resource estimate is more than a back-of-the-envelope guess, says Yuntian Teng, a geochemist at PNNL who led the work. To learn what's locked in the rock, the researchers analyzed core samples from 13 major oil-bearing formations. They bombarded each core with x-rays, causing elements in the rock to fluoresce at characteristic energies. From the signals, the researchers could estimate the abundance of 15 target elements. The results varied

widely by formation. Parts of the Permian Basin, a major formation in Texas, were rich in vanadium, titanium, and nickel, whereas manganese popped out in the Uinta shale formation in Utah and Colorado. "There's a huge potential," Teng says.

Lithium is the low-hanging fruit for so-called in situ mining because wastewater readily dissolves it from rock. "Lithium, you can sneeze on it and it'll get into solution," Bain says. Hood Whitson, CEO of Element3, says his startup has already built a commercial plant in Midland, Texas, that adds aluminum hydroxide to 35,000 barrels a day of oil-and-gas wastewater. The aluminum hydroxide captures the dissolved lithium, which can later be released with hot water. Other companies are trying to extract lithium from brines at geothermal power plants before the water is pumped back underground.

Other metals—for example, titanium—are harder to extract. So Ellen Polites, a geochemist at PNNL, and her colleagues have been testing chemicals—organic acids, oxidizing agents, and other reactive molecules—that can coax specific minerals out of a rock. "What we're trying to do is make it more targeted," she says, "rather than bulk dissolving the entire rock." The team plans to test these engineered fluids in a lab-based setup that simulates the rocks in a fractured reservoir. By the end of the 2-year project, they hope to inject the fluids into an actual well. "Industry has grabbed on to this concept," says Son Dang, a petroleum engineer at OU.

Polites's team envisions flushing a reservoir with different engineered fluids in successive passes, each targeting a different mineral. The mining could happen alongside oil-and-gas production, or prospectors could target the 80% of wellheads that are marginal and unproductive, Dang says. By making wastewater more valuable, the team hopes to encourage its proper treatment and reuse. "We have an incredible water crisis in the United States," Polites says.

If the world wants to limit more human-driven global warming, oil and gas drilling will need to slow. But in the meantime, taking advantage of fossil fuel production to extract minerals needed for clean technologies could be part of the solution, Whitson says. "We're taking the waste of the fossil economy to support our future material needs." □

A technician installs biodiversity monitoring equipment at a site in Brazil.

PHOTO: PROVIDENCE PROJECT/MAMIRAUÁ INSTITUTE FOR SUSTAINABLE DEVELOPMENT

ECOLOGY

Project will track Amazon biodiversity in real time

Leaders of $100 million effort will use automated detection methods and computer models to monitor species' trends **WARREN CORNWALL**

Scientists can tell within days whether a grove of trees has been felled anywhere in the world, thanks to satellite imagery. But when it comes to knowing what's happening in real time to the species that live there, researchers remain woefully blind.

"We have good technology that can monitor forests from above," says Emiliano Ramalho, an ecologist and scientific director for the Mamirauá Institute for Sustainable Development, a Brazilian nonprofit research organization working in the Amazon rainforest. "But this does not monitor what's under the canopy."

Now, thanks to technological advances such as automated species detection from audio recordings and analysis of DNA shed into the environment, conservation scientists think it's possible to assemble increasingly comprehensive and up-to-date views of biodiversity across huge landscapes. Last week, a coalition of research organizations, including Ramalho's institute, announced an initiative to test that promise by creating an Amazon-wide, regularly updated picture of what's happening to plant and animal populations as development threatens one of the most species-rich parts of the world.

The project, dubbed Amazon.ia, promises to marry streams of data from dozens of field sites with computer models capable of quickly detecting patterns and extending results across the river basin's 7 million square kilometers. The group includes scientists from some 2 dozen universities, nongovernmental organizations, and companies in North America, South America, and Europe. Backed by an initial contribution of $14 million from the Bezos Earth Fund, project leaders say they have raised approximately $20 million of what would be a 10-year, $100 million effort.

Tanya Berger-Wolf, a computational ecologist at Ohio State University who is not part of the project, says that in 2018 she sought funding from the U.S. National Science Foundation for a similar project focused on a single field site in Kenya. The proposal was rejected, she says, on the grounds that it was infeasible. "What's changed in the last 8 years is this is within the grasp," says Berger-Wolf, who has spent years designing computer systems to cull ecological information from photographs. "It's becoming an engineering problem, which is a much better place to be."

She points to several developments that make Amazon.ia more realistic: A flood of data has helped train more powerful computer models, technical know-how has become more widespread, and more people are concerned about the loss of biodiversity.

Mamirauá and the Polytechnic University of Catalonia have already built a network of 65 stations in the Amazon that automatically collect audio and images; classify species including insects, mammals, and birds; and quickly upload the results via satellite. The program has been shown to effectively

track the Amazonian manatee and endangered river dolphins.

Ramalho says the institute joined Amazon.ia in part because the project aims to merge data from multiple locations and to deliver results fast enough to influence policy. The program also promises to build the technical capacity of Amazon-based research organizations by helping pay for things such as staff, equipment, and technical training. That would be a change from a historic pattern of research groups from places like the United States and Europe taking raw data and analyzing it themselves. "We don't want to send all our data out. We want to be able to handle the data, manage the data, develop the models," he says.

David Luther, a George Mason University ecologist involved in the Amazon.ia project, hopes the new initiative can connect isolated findings into broader trends across the Amazon. He has been involved in research in Brazil that uncovered a worrying decline in some bird species. Several other sites in Central and South America have reported similar drops. But the historic data are so scattered that it's hard to know whether the declines reflect a regionwide shift. And there are no comparable data for insects, a food source for many of the birds.

Corine Vriesendorp, an ecologist and director of science for Conservación Amazónica, a Peruvian research organization whose station on the Los Amigos River is part of the new program, anticipates that the collective effort will mean "we're going to be able to say something that none of us would be able to do on our own. … This takes a ton of computing power, data storage, collaboration among scientists."

The financial and technical challenges are daunting, however. In addition to needing about $80 million more to reach its goal, the program has so far recruited just five of the 10 to 30 sites it wants for sustained long-term monitoring. Candidates include established field stations, as well as places such as ecolodges, community centers, or universities.

And it's not clear current technology can fill all the blanks in such a vast region. Justin Kitzes, a quantitative ecologist at the University of Pittsburgh, is leading a separate Bezos-funded project using audio recorders to monitor bird populations in Andean nature reserves. His team is also developing a box that could be dropped into a location to automatically collect biodiversity data. "This is something we are actually working on," he says.

But he is cautious about the bigger effort. "I'm sure it will be greatly informative," Kitzes says. "I'm also sure it won't tell us everything we want to know." □

IN FOCUS

PHOTO: VOLODYMYR TARASOV/UKRINFORM VIA ZUMA PRESS WIRE

Russia attacks Ukraine's science academy

A jet-powered drone slammed into the headquarters of the National Academy of Sciences of Ukraine on 28 September, killing at least two people and destroying part of the historic building. One person inside the building during the Russian attack was listed as missing and presumed dead as *Science* went to press. "It's a terrible tragedy for Ukrainian science," says Sergiy Mosyakin of the M.G. Kholodny Institute of Botany. As a response to earlier onslaughts on Ukraine's scientific infrastructure, the science ministry this year started to construct underground laboratories. Deputy science minister Denys Kurbatov says he hopes international organizations will help Ukraine build more such labs and shelters, and supply backup electricity sources and reliable internet connectivity. "The enemy wants us to stop working," he says. "That will not happen—we will not stop." —Richard Stone

ASTRONOMY

Space telescope will reopen view of far-infrared sky

NASA's $1.2 billion PRIMA will explore galaxy evolution and planetary birth

DANIEL CLERY

NASA has selected a far-infrared space telescope to explore the faint, cool glow of the gas and dust that fuel star formation and coalesce into planets. The $1.2 billion PRIMA is the agency's first Probe class mission, intended to fill a gap between researcher-led Explorer missions and multibillion-dollar flagship missions. Set to launch in 2033, PRIMA would also end a far-infrared observing gap since Europe's Herschel telescope ceased operation in 2013.

The far-infrared is one of the least explored parts of the electromagnetic spectrum. Most of the radiation is blocked by Earth's atmosphere, forcing astronomers to go to space. They also have had to develop detector technology themselves, unlike in neighboring wavelengths—the midinfrared and microwave—where commercial and military applications are common. "The far-infrared, that's just astronomers," says Jason Glenn, an astrophysicist at the University of Virginia who led PRIMA's design effort.

Herschel, the first major space telescope dedicated to the far-infrared, had a passively cooled 3.5-meter mirror. But stray heat from it limited the telescope's sensitivity to distant sources. And its detectors were cooled with liquid helium, which ran out after less than 4 years.

PRIMA won't need helium and could operate for longer. It will combine passive cooling with a mechanical cryocooler to keep its 1.8-meter mirror at 4.5° above absolute zero. An additional cryocooler will chill a new type of superconducting sensor, called a kinetic inductance detector (KID), to less than one-eighth of a degree above absolute zero.

KIDs can record the energy and arrival time of individual photons, and they offer roughly 100 times the sensitivity of Herschel's detectors. "That will enable PRIMA to observe the kinds of things that Herschel did in the local universe … but throughout the history of cosmic time," says Cardiff University astrophysicist Matt Griffin, who is part of a U.K. consortium contributing hardware to PRIMA.

One major target is the evolution of galaxies. Herschel showed how interstellar gas moves within galaxies as it collapses into stars or is expelled into intergalactic space by winds from giant black holes. But its studies were largely limited to a handful of nearby galaxies. PRIMA will probe dozens of galaxies across three-quarters of cosmic history, allowing astronomers to track how those processes changed over time.

Within the Milky Way, PRIMA will examine how planets coalesce out of disks of gas and dust surrounding newborn stars. The Atacama Large Millimeter/submillimeter Array in Chile, which observes at slightly longer wavelengths, has revealed gaps in these disks—the signs of planets in the making. PRIMA will detect water in the disks, testing the idea that ice is the glue that helps dust clump together and drive planet formation, Glenn says.

Managed by NASA's Jet Propulsion Laboratory, PRIMA will now undergo detailed design and technology work ahead of a review, after which construction can begin. Probe class missions were recommended by the latest decadal survey in astrophysics, a community priority-setting exercise that sought missions that could launch on a faster schedule than flagships. PRIMA competed for the first Probe slot with the Advanced X-ray Imaging Satellite, which NASA canceled earlier this year. □

PHOTO: JULIA GRESKY/GERMAN ARCHAEOLOGICAL INSTITUTE

IN FOCUS

Did a 'boneslide' fill temple with remains?

For years, human bones found amid the rubble of southern Turkey's Göbekli Tepe—one of the earliest known monumental temple complexes—led some to suggest the 10,000-year-old site once hosted an ancient "skull cult." A new analysis of intact graves from the site, reported last week in *PLOS ONE*, instead suggests a less lurid explanation: Landslides may have swept the human remains into the central ritual space. The two graves, excavated in 2017 and '22, contained the remains of a three adults and a teenager and were found beneath the houses built uphill from Göbekli Tepe's temples. Some combination of heavy storms and earthquakes could have periodically sent these perched houses, along with the skeletons buried beneath them, careening down the slope, jumbling them with temple ruins. —Andrew Curry

BIOETHICS

Impending changes to U.S. human subject protections put ethicists on edge

HHS plan to revise the Common Rule focuses on easing regulatory burden for lower risk studies

SARA REARDON

The Trump administration appears poised to weaken protections for people enrolled in some clinical trials. In a little-noticed posting on a government website tracking agency regulations, the U.S. Department of Health and Human Services (HHS) announced an upcoming "deregulatory" move that would revise the Common Rule, a federal policy that governs aspects of human research including informed consent, data sharing, and special protections for vulnerable participants such as prisoners or pregnant people.

The July posting contains few details but says the Common Rule's framework, last updated in 2017, "no longer reflects the realities of modern research" and "imposes unnecessary burdens and unclear requirements, particularly for minimal-risk studies." HHS declined to answer questions about the timing of the forthcoming rule, which will require a public comment period before implementation.

"It sounds like they're doing something serious, and they could go pretty far," says Jerry Menikoff, a bioethicist at the National University of Singapore and former director of HHS's Office for Human Research Protections (OHRP), which oversees the Common Rule.

First instituted in 1991, the rule applies to most federally funded research studies. Privately funded studies—and the institutions that carry them out—also generally adhere to its guidelines but are not required to. Researchers have long complained that some of its provisions require too much paperwork, such as the need to obtain approval from an ethics review board to make minor changes to a protocol.

The HHS posting about the new rule offers "examples of changes," including added flexibility for minor protocol changes and "expanding exemptions for certain low-risk research activities." How those activities will be defined is unclear, however, because "low risk" has no legal definition. (The current rule does define "minimal risk," however, as the risk that can be expected in everyday life or in a routine medical exam.) Asked for details, an agency spokesperson said, "HHS cannot comment on potential or future regulatory actions."

Seema Shah, a bioethicist at Northwestern University, says more flexibility in some areas of the Common Rule would be a good thing. "There are nuggets in here that seem like they could be valuable," she says. Yet she worries the notice overall appears to give more attention to easing regulatory burden than safeguarding human subjects. "There's definitely a lot of money and power behind the idea of trying to take away some of these standards and do research as quickly and cheaply as possible," Shah says. "Our past is littered with a history of unethical research where people were exposed to risk without knowledge or consent."

HHS's notice says current regulations are especially burdensome for learning health systems, a model in which hospitals integrate clinical research studies into their care practices without necessarily obtaining patients' consent or even informing them that a trial is happening. Proponents argue this approach, which does allow participants to opt out, could speed research by eliminating the lengthy process of recruiting people into trials. Because these hospital-integrated studies often compare the efficacy of two treatments that both fall within standard medical practice, Menikoff says HHS could conceivably determine that any such study constitutes minimal risk. "That would certainly make it a lot easier to do a lot of these trials," he says.

Yet even standard treatments can have different levels of risk. For instance, a 2013 trial known as SUPPORT tested the optimum dose of oxygen for premature infants by randomly assigning them to receive high or low levels. Although both doses fell within standard medical practice, higher doses carried a possible risk of eye damage, and the lower end of the range was thought to increase the risk of brain death. OHRP determined parents had not been adequately informed about the risks. Menikoff says similar tradeoffs emerge in studies comparing, for instance, the benefit of yearly cancer screenings with the risk of increased radiation from scanners. "People have different viewpoints" on which risks they're willing to take, he says.

The HHS notice also mentions that the agency will clarify its definition of "undue influence"—when a study pays people so much to participate in a trial that they may feel coerced. This problem has rarely come up in practice, Menikoff says, because it would take an enormous amount of money to make most people feel that they can't decline.

Ellen Clayton, a bioethicist at Vanderbilt University, is watching to see whether the new rule will affect how trial participants must consent to sharing their data with other institutions, including the government and private companies, and how those data should be protected. The topic is not explicitly mentioned in the notice, but is a concern given the Trump administration's overall approach to research, Clayton says. "They're worried about [competition from] China, they're desperate to take research out of the research institutions and into the private sector, they may be allowing easier access by the private sector to data." But weakening the Common Rule could come at a price, she says. "Is the gain you're going to get from doing translational research without consent worth it for what it's going to cost in trust?" □

SHOTS ON GOAL

Can Africa make 60% of its vaccines by 2040? That's a tall order, as the travails of South African companies show

JON COHEN, in Cape Town, South Africa

Morena Makhoana can see Cape Town's spectacular landmark, Table Mountain, from the grounds of Biovac, the vaccine manufacturer he runs here. "When you're stressed and you don't know what to do, you just look there," Makhoana says. "It keeps us a little bit sane." He has just stepped out of a Biovac building constructed during the COVID-19 pandemic to hold the "freezer farm," which today symbolizes one of the company's—and one of the African continent's—great disappointments.

When the first COVID-19 vaccines became available in 2021, deep-pocketed governments outside Africa snatched them up, pushing even the wealthiest African countries to the end of the line. Then Biovac cut a deal that held great promise: Pfizer and BioNTech would provide their popular messenger RNA (mRNA) vaccine in bulk, which Biovac would put into vials—a process known as "fill and finish." The company aimed to start to supply vaccine by early 2022, and to produce 100 million doses a year, storing 20 million at a time at the farm of specialized freezers that maintained the required −80°C. The deal would give Africa its own dedicated vaccine supply.

But it took longer than expected to formalize the agreement and gear up for production, and once Biovac had its first doses available at the end of 2022, demand for the vaccine had evaporated. "We never sold a single dose," Makhoana says, and the freezer farm remained idle.

Biovac's travails mirrored those faced by other bids to speed delivery of COVID-19 vaccines to Africa. They also added to a decadeslong history of setbacks for efforts to establish a vaccine industry on the continent, which in 2021 produced just 0.1% of the shots it used. But the pandemic, with its stark inequities in vaccine access, sparked a new resolve among African governments, health officials, and scientists to change that.

The result was an initiative called the Partnerships for African Vaccine Manufacturing (PAVM), launched in April 2021 and spear-

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PHOTO: LULAMA ZENZILE/DIE BURGER/GALLO IMAGES VIA GETTY IMAGES

Aspen Pharmacare put a Johnson & Johnson COVID-19 vaccine in vials, but never delivered it in Africa.

headed by the African Union and the Africa Centres for Disease Control and Prevention (Africa CDC). PAVM aimed to help companies access financing and build expertise, and to get neighboring countries to pool resources, creating larger marketplaces for manufacturers in their regions. PAVM had an ambitious—some said unrealistic—goal: to drive up the proportion of African-made vaccines to a staggering 60% of demand by 2040.

More than 2 dozen companies are now vying to seize the postpandemic momentum and contribute to PAVM's vision. Media have intensively covered African presidents and global health leaders holding shovels at groundbreaking or ribbon-cutting ceremonies for new and expanding plants. But the experiences of Biovac and South Africa's two other vaccinemakers, the tiny, mRNA-focused Afrigen Biologics and Vaccines and the giant Aspen Pharmacare, show tremendous hurdles remain—even in a country that has the largest economy on the continent and one of the most developed infrastructures.

Vaccines are more complex to manufacture than chemical-based drugs, and plants are expensive to build and maintain. To compete in the market, companies typically have to produce vaccines on enormous scales. And to fit the budgets of developing countries, they must be inexpensive. "That's why the global vaccine market is controlled by six or seven companies," says Afrigen's CEO, Petro Terblanche.

African companies also face steep challenges securing loans from banks and investments from venture capitalists. Their cash-strapped governments are reluctant to make purchasing commitments unless companies show their products can compete with vaccines available from established manufacturers outside the continent. As the COVID-19 experience showed, demand can be fickle. And funding to test new vaccines is scarce. "We're in a bloody tough time," Makhoana says.

Yet simply relying on established producers outside of Africa isn't an option, he contends. "If it was a commodity like cars, you could argue about whether you want to make them locally or import them, and it's not a life-or-death discussion," he says. "We need vaccine manufacturing on the continent as a health security issue. It's a no-brainer."

BIOVAC BEGAN in 2003 as a partnership between the South African government and privately held health care companies. It first aimed to make a liquid form of a freeze-dried vaccine against the bacterium *Haemophilus influenzae*, which causes respiratory infections and other conditions including meningitis. The company planned to include it in a vaccine produced by Sanofi in France that also protected against diphtheria, tetanus, pertussis, and polio, creating a single broad-spectrum shot for children. It set about restoring a dilapidated state-run vaccine plant as its headquarters. (Biovac made the liquid product, but Sanofi figured out how to manufacture it, too.)

Just then, global events underscored the need for developing countries to have "vaccine sovereignty." A bird flu virus alarmed the world when it started to infect people in several Asian countries, killing 79 of the 148 confirmed cases by 2005. A viral sequence the Indonesian government shared with the World Health Organization (WHO) became the basis for an Australian vaccine company's product, which Indonesia presumably would not be able to afford. Anger at rich countries for controlling essential vaccines intensified with revelations that the U.S. government and U.S. companies and research institutions had patented sequences of bird flu viruses from Indonesia and Vietnam. "It was a political dog's breakfast," says Martin Friede, who began working at WHO in 2003 and headed its vaccine research unit when he retired last year.

At the request of U.S. health officials, WHO launched a project to help several countries build manufacturing plants for flu vaccines. Friede, who led the effort, says it faced one problem after another. One was that flu vaccines were made by growing influenza viruses in fertilized chicken eggs, which manufacturers had to order from farmers at least 6 months in advance. If a bird flu pandemic had exploded, vaccine production would have lagged. Lesson No. 1, Friede says: "Don't use a technology where if a pandemic happens, you won't be able to get the raw material."

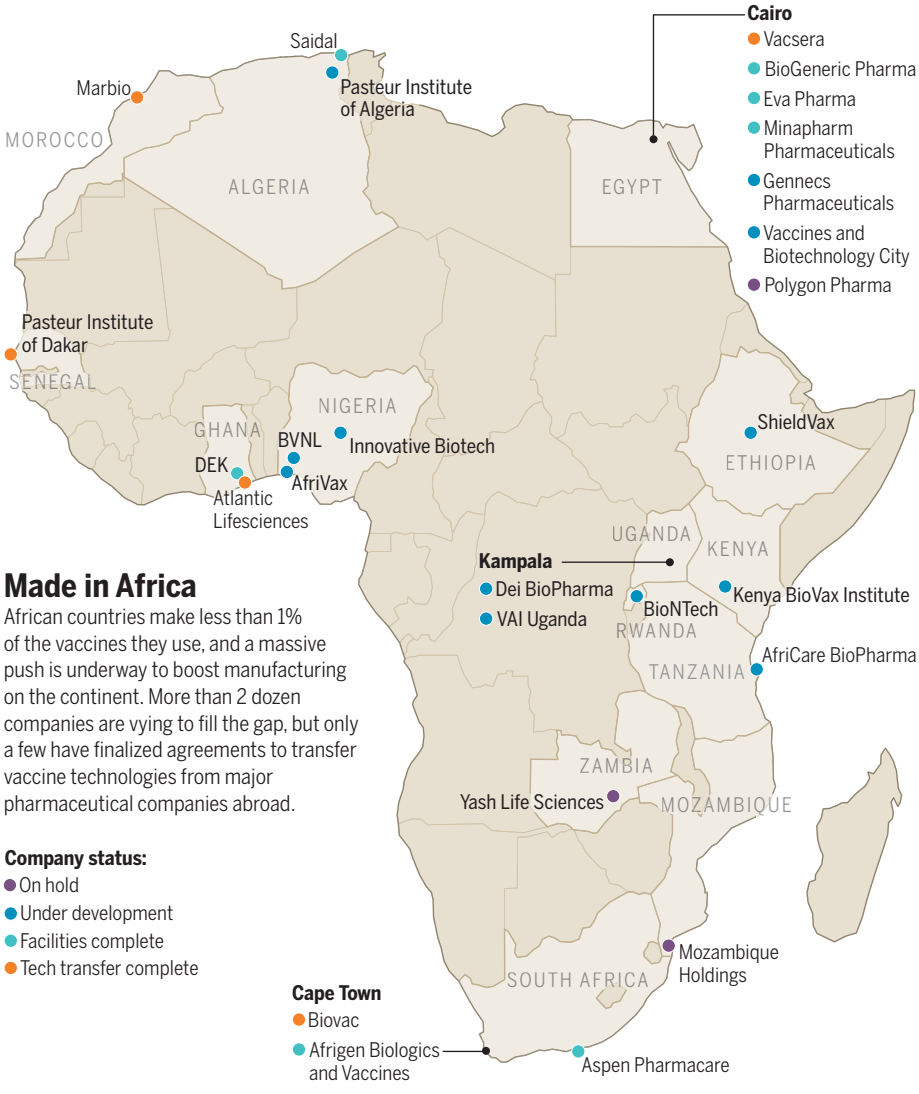
A second lesson involved economies of scale. A government-owned plant in Thailand was producing 15 million doses a year of flu vaccine. The vaccine cost $6 per dose, but the country could buy a similar vaccine from Sanofi for half the cost. At first, the Thai government said, "Well, that doesn't matter because this is national health security," Friede remembers. But when the country's leadership changed, the new health minister phoned. "He said, 'National health security? Baloney. We're going to buy from Sanofi because we're running out of money.'"

Morena Makhoana has piloted Biovac through rough times, but the company has recently attracted substantial new investments.

Biovac would learn that lesson firsthand following a major 2015 agreement with Pfizer to fill and finish a *Streptococcus pneumoniae* vaccine called Prevnar. The South African government initially agreed to buy the locally filled vials, but in 2023 decided to purchase Prevnar from the Serum Institute of India—the world's largest producer of vaccines—for about one-third of the price, forcing Biovac to lay off 85 of its 450 employees. The firm has stayed afloat thanks in part to a multiyear deal with Sanofi to fill and finish a six-in-one vaccine—it adds hepatitis B—at the Cape Town facility, which can vial 15 million doses a year.

Friede's third critical takeaway from

PHOTO: OUPA NKOSI



Made in Africa

African countries make less than 1% of the vaccines they use, and a massive push is underway to boost manufacturing on the continent. More than 2 dozen companies are vying to fill the gap, but only a few have finalized agreements to transfer vaccine technologies from major pharmaceutical companies abroad.

Company status:
- On hold
- Under development
- Facilities complete
- Tech transfer complete

Afrigen CEO Petro Terblanche (left) welcomed a visit in 2022 from World Health Organization head Tedros Adhanom Ghebreyesus.

WHO's efforts is that vaccine plants cannot financially survive if they only exist for health security during pandemics. "Quite a few companies were able to get the technology up and running, and then a year or two later, we began getting calls from the chief executive officer saying, 'Right, we've built this flu facility, but we're not selling any flu vaccine,'" he says, noting it can cost millions of dollars a year to maintain an empty plant. Several pivoted to making routine childhood vaccines—taking themselves out of contention as flu vaccine suppliers in the event of a pandemic.

WHEN THE COVID-19 pandemic struck, health officials tried to learn from the past, but Biovac's effort to secure Africa's access to vaccines wasn't the only one to fall short. Aspen early on signed a deal with Johnson & Johnson to fill and finish a single-dose COVID-19 vaccine that could have protected 300 million people annually, but contamination of the bulk product at a U.S. plant and other problems delayed the effort, and Aspen never sold a dose. Similar snafus undermined would-be fill-and-finish efforts by companies in Ghana and Morocco.

Efforts launched from outside Africa fared no better. The most devastating and high-profile letdown was a campaign by the nonprofit COVID-19 Vaccines Global Access (COVAX) Facility to supply Africa with COVID-19 vaccines made elsewhere: It faltered because it relied too heavily on the Serum Institute, which could not export doses in the spring of 2021 when its own country had a surge in cases. Patrick Soon-Shiong, a biotech billionaire in Los Angeles who grew up in South Africa, in 2022 pledged to invest $200 million in NantSA, a new vaccine company here that would produce 1 billion COVID-19 doses by 2025. After breaking ground on the plant, he shut it down, in part because of the country's infrastructure problems. "No power was available to that plant!" he wrote in a text to *Science*. Moderna announced it would invest up to $500 million in a Nairobi, Kenya, mRNA vaccine plant that supposedly could have produced 500 million doses a year—a plan it abandoned 2 years later.

Africa's response was PAVM. In March 2022, it published a detailed plan to strengthen local regulatory agencies, create training centers for scientists and engineers, coordinate

CREDITS: (GRAPHIC) V. PENNEY/*SCIENCE*; (DATA) CLINTON HEALTH ACCESS INITIATIVE; (PHOTO) JEFFREY ABRAHAMS/AFRIGEN BIOLOGICS AND VACCINES

Afrigen's Caryn Fenner ran a program that taught scientists from several countries how to make messenger RNA for vaccines with small machines like this one.

PHOTO: OUPA NKOSI

technology transfers and sharing of intellectual property, and organize the continent into regional purchasing pools.

The call to produce 60% of the continent's vaccine needs by 2040 "was an aspirational goal at that time, which was necessary to move things forward and to bring excitement to the space," says John Nkengasong, who led Africa CDC when PAVM was formed and now works at the Mastercard Foundation guiding investments to strengthen education and health in Africa. Aiming high makes sense given Africa's population growth, he says. "One out of four human beings will be African" by 2040, Nkengasong says. "Can you afford not to invest in your own development of your biomedical countermeasures?"

WHEN PAVM LAUNCHED, only one product, a yellow fever vaccine made by the Pasteur Institute of Dakar (IPD), was being produced in Africa from scratch, so-called end-to-end manufacturing. In PAVM's vision, the burgeoning African vaccine industry would initially consist mainly of companies focused on fill and finish, like Biovac. That type of arrangement gives countries some control over vaccine supply and brings revenue to local businesses that provide things such as glass vials and their rubber stoppers. The vaccine companies would, over time, "backward integrate" and learn how to produce all the necessary ingredients themselves.

Seth Berkley, who led a public-private partnership called Gavi, the Vaccine Alliance for 12 years before stepping down in August 2023, says that when PAVM launched and attracted a flood of interest from would-be manufacturers, he warned that most wouldn't manage to create sustainable businesses, given the many challenges of the vaccine industry. He urged Africa CDC and WHO officials to concentrate their energy on the most promising firms, says Berkley, who also helped create and run COVAX. "At the end, it was kind of, 'Well, we'll let them go on, and we'll see who succeeds.'"

For the many contenders, new investment opportunities bloomed thanks to the organized, continentwide efforts of PAVM. Gavi, which each year purchases and distributes product for more than half the world's children, made a notable commitment the year after Berkley left. It pledged to invest up to $1.2 billion over 10 years in a combination of financial rewards for African companies that hit milestones such as regulatory approvals and per-dose subsidies to lower the prices of approved vaccines. Gavi plans to make the first of the milestone payments later this year, though it hasn't revealed the recipients.

The Gavi pledge has made Biovac more attractive to investors, Makhoana says. This year, the company secured more than $100 million in new funding from the European Investment Bank, the International Finance Corporation, and the African Development Bank Group to more than triple production capabilities to 50 million vials a year by 2029. And it hopes to soon bring in revenue from end-to-end manufacture of a cholera vaccine developed by South Korea's International Vaccine Institute. Still, the company has yet to earn substantial profits from vaccines.

AFRIGEN, JUST A 20-MINUTE drive away from Biovac, is a much smaller operation, occupying two rented warehouse buildings in a business park. It, too, has had a bumpy road.

The company has been struggling to keep the lights on since it opened its first lab in 2016. Launched 2 years earlier as a partnership between the South African government's Industrial Development Corporation and the Infectious Disease Research Institute (IDRI) in Seattle, Afrigen at first aimed to bring a tuberculosis vaccine to market. But financial woes forced IDRI to sell its 51% interest in Afrigen to Avacare, which primarily manufactures and distributes generic drugs.

Under pressure from Avacare to make money, Afrigen took a left turn. It began to produce an emulsion of cannabidiol (CBD), a compound extracted from cannabis said to help with anxiety, inflammation, and pain. One of Afrigen's most popular prod-

ucts was a CBD paste that cat owners could squirt onto climbing poles for their cats to lick. That venture petered out when the South African regulatory agency cracked down on the sale of CBD around the start of the pandemic.

In April 2021, Afrigen bid to return to vaccines. It responded to a call by WHO and the Medicines Patent Pool to apply for a grant to set up a training hub for companies interested in making mRNA vaccines.

The mRNA technology allows a maker to quickly tailor a vaccine to a new pathogen or strain, and the product can then be made in bioreactors rather than chicken eggs. During the COVID-19 pandemic, when the Moderna and Pfizer-BioNTech vaccines showcased the power of the technology, "Countries [were] screaming blue murder, 'Not fair, not fair, we want access!'" recalls Friede, who spearheaded the mRNA training hub idea.

The Pasteur Institute of Dakar recently received a $45 million grant to train African vaccine scientists.

Afrigen "hadn't touched mRNA," says Caryn Fenner, a bioprocessing engineer at the company, and it only had the shell of a pilot vaccine production plant. But the company, one of 20 applicants, won the job to train scientists from far and wide. "We were just a little bit gobsmacked," Fenner says. At first, they joked they were building the airplane while flying it. "We changed it to: We're all running down the runway, flapping our arms."

Moderna, Pfizer, and BioNTech each came to visit Afrigen, but none wanted to partner with the fledging mRNA manufacturer. So the company teamed up with researchers at the University of the Witwatersrand, who showed them how to make mRNA and formulated a version of the Moderna COVID-19 vaccine based on a sequence of the viral surface protein that Stanford University researchers had published. Afrigen then hosted scientists from 15 low- and middle-income countries to make a mimic of the Moderna vaccine, and 13 came back to learn how to work with other pathogens.

The grant ends in December, and although the company now has the expertise and small plant to make mRNA vaccine products itself, it recently suffered yet another setback: Financial troubles at Avacare forced it into "business rescue" (a form of receivership) in February. Afrigen is now attempting to restructure itself as a nonprofit. "We've weathered two big storms already, this little company, and we're in our third one," Terblanche says. "This will be do or die."

SOUTH AFRICA'S THIRD vaccine-maker, Aspen, is well aware of the perils of vaccine production. "African manufacturers have had countless, endless challenges with a sustainable demand for vaccine," says Stavros Nicolaou, a pharmacist and executive at Aspen, which has a sprawling complex in Gqeberha (formerly Port Elizabeth). So the company, the largest pharmaceutical company in Africa, currently isn't planning to do any end-to-end manufacturing. It does fill and finish of four vaccines that it imports from the Serum Institute: a six-way vaccine similar to the Sanofi-Biovac product and ones that protect against *S. pneumoniae*, rotavirus, and meningococcal bacteria.

If Aspen were to make the vaccines it now imports, Nicolaou expects the Serum Institute would simply undercut it. "There are times where I've been a naïve optimist around this whole localization thing," Nicolaou says, "but that was many years ago."

Makhoana, in contrast, still exudes abundant optimism about local, end-to-end production of vaccines. His vision is that South African vaccine producers will build and share talent, and the government will invest more in the education of scientists. "I do wish that Aspen thrives. I do wish that Afrigen survives. If there's any competition, we'll just be competing commercially. But we need the ecosystem."

To further improve that ecosystem, the Mastercard Foundation in 2023 awarded $45 million to IPD to train young African scientists to produce vaccines. Nkengasong says he wants to see similar investments by African countries. "You cannot talk about African health sovereignty for vaccines without investing in universities," he says. "Otherwise, you still wait on an American, a Canadian, a European, or Chinese young scientist to invent something, and then share the patent with you. That's not sovereignty."

In another potential boost for African vaccine manufacturing, China last month reached an agreement with the 16-country Southern African Development Community to bolster pharmaceutical development on the continent. Although the details are vague, China promises to increase technology transfers and training of workers.

Africa CDC is continuing its attempt to design a vaccine marketplace that can sustain companies. It is pushing for an Africa Pooled Procurement Mechanism, in which governments from five different regions would band together and purchase vaccines collectively. All 55 African Union member states signed on to this commitment in February. "The question is going to be how reliable will the payments be?" Berkley says. "It's a little naïve to just say we're going to do it and everybody's going to fall in line."

A June report issued by the Clinton Health Access Initiative interviewed the three manufacturers in South Africa and 24 others to assess the state of vaccine production on the continent. "What we found is a sector making genuine progress and running into serious headwinds," its report concluded. Biovac and IPD are the only two African companies that will likely do end-to-end manufacturing of vaccines in the next 5 years, it predicts. In the same time frame, Aspen and Egypt's Vacsera are the only two that will receive bulk product made elsewhere and do fill and finish.

Makhoana says he still applauds PAVM's ambitious 60% goal. "If 2040 comes and let's say we are even at 11% manufacturing vaccines, for me, that's a 1000-fold from where we were," he says. "You must have vision and a target in life, even if it's a stretch. You must dream." □

PHOTO: RICCI SHRYOCK/REUTERS

BIOMATERIALS

Self-assembling polymers work up some nerve

Micrometer-long peptide amphiphile fibrils enhance neuronal adhesion and function

Sanjay Kumar

Physical structure dictates the processes that underlie life. At the molecular scale, geometrically matching shapes allow a protein's binding pocket and a specific ligand to fit together, triggering important biochemical events. At the cellular scale, cylinder-like axons enable rapid transport of materials and electrical signals inside a neuron (*1*). An ongoing goal in biomaterials science has been synthetically creating biomolecules that can interface with cells and support specialized cell structures, such as the long, thin shapes of neurons. On page 125 of this issue, Dore *et al.* (*2*) report a family of peptide amphiphiles with discrete hydrophilic and lipophilic regions that assemble into fibrils with dimensions comparable to those of a cell (10 to 100 μm). These structures allow neurons to extend axon-like processes along the fibrils and assemble into multicellular networks. The findings warrant further exploration of peptide amphiphiles and related materials for cellular engineering and regenerative medicine.

Many biocompatible materials that promote biological functions, such as cell adhesion, division, and differentiation, have been developed. These biomaterials often feature synthetic or tissue-derived polymers that are then decorated with short peptides. The long polymer chains dictate structural properties, whereas the peptides encode key biological functions, such as engagement of adhesion receptors or growth factors (*3*). Although this approach has successfully produced two-dimensional (2D) and 3D scaffolds for growing cells in culture, the polymer network must be chemically synthesized in a manner that often "locks" the structure and the mechanics of the material. This could limit the dynamic cell-material interactions that are key to cell maturation and tissue assembly. Thus, molecular building blocks that can self-assemble into desired structures and dynamically instruct cells are needed. To this end, a variety of noncovalently self-assembling peptides and proteins have been developed, many of which spontaneously form hydrogels that support cell engraftment and growth in the body (*4*, *5*). However, organizing these molecules into cylindrical structures, such as fibers and ribbons, remains challenging. Creating such materials could both mimic the natural contours of biological matrices and facilitate complex cellular morphogenetic events, such as the extension of axons and dendrites in a nerve cell.

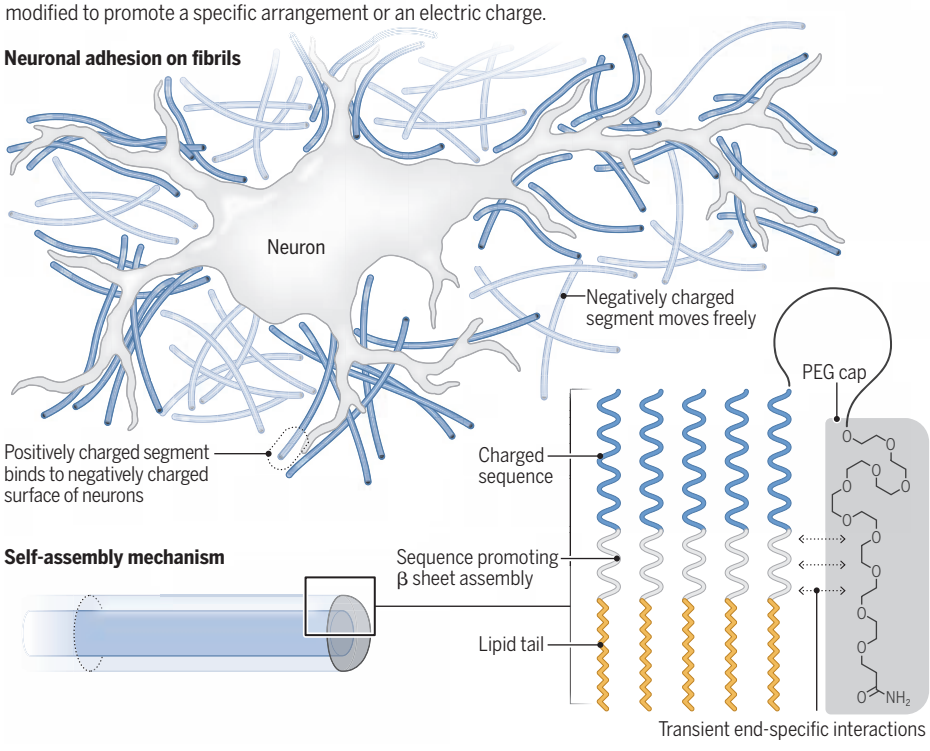
...peptide amphiphile formulations could be designed to deliver specific physiochemical instructions to different cell types...

Dore *et al.* designed and synthesized a family of peptide amphiphiles that can self-assemble into long fibrils that are tens of micrometers in length and billions of daltons in molecular weight. In general, the hydrophobic lipid tails of peptide amphiphiles drive initial aggregation in aqueous solution that brings the peptide segments into proximity, where these sequences can adopt a secondary structure (*6*) that nucleates growth of filamentous or membranous structures. Peptide-based molecules can also include sequences that can interact with cells to support adhesion and other behaviors (*7*). Dore *et al.* used a peptide amphiphile with a 16-carbon fatty acid chain at one end and a peptide sequence and hydrophilic poly(ethylene glycol) (PEG) chain at the other end (see the figure). The peptide includes a valine-valine-alanine-alanine (VVAA) sequence that promotes assembly into β sheet–based fibrils and arginine-glycine-glycine (RGG) repeats that create positive charge for electrostatic adhesion to negatively charged cell membranes. These amphiphiles produced micrometer-scale fibrils of uniform length distribution attributed to a PEG "cap" at the fibril ends that sterically limits the addition of new monomers and prevents fibril fusion. By mixing amphiphiles of different peptide sequences, Dore *et al.* could vary the length of the negatively charged and positively charged segments within the polymer fibril.

Dore *et al.* covered a surface with the polymer fibrils to control the adhesion of neurons isolated directly from the mouse cerebral cortex. Adhesion could be adjusted by tuning the length and distribution of positively charged segments within the fibril. The positive segments bound the negatively charged surface of neurons. By contrast, the negatively charged segments were more mobile, facilitating adsorption of extracellular matrix (ECM) proteins that were added to the culture system and guiding the extension of neurites from the cells. The length of the negatively charged segment of a fibril dictated the extent of neurite extension, with shorter lengths promoting the behavior to a greater degree than a conventional experimental surface coated with full-length

Peptide fibrils interact with neurons

Peptide amphiphiles self-assemble into long, thin fibril structures. The lipophilic 16-carbon fatty acid tails bring peptide building blocks in proximity in an aqueous solution, and the hydrophilic poly(ethylene glycol) (PEG) chain at the other end acts as a transient cap to control the length of the fibril. Peptide segments can be modified to promote a specific arrangement or an electric charge.



ECM proteins, such as laminin or fibronectin. The fibril-coated surfaces also supported characteristic neuronal behaviors, such as the formation of synapses between adjacent neurons and the propagation of calcium signals within individual cells. Culturing neural progenitor cells (stem cells that can differentiate into neurons and other brain cell types) on the fibril-coated surface not only promoted their differentiation into neurons but also aligned extended neurites along the fibrils. The differentiated neurons also formed multicellular 3D clusters that were interconnected by membranous projections, suggesting the beginning of a cellular network. Because neural networks represent a fundamental functional unit of brain tissue, the study of Dore *et al.* is important for potential applications in tissue regeneration, biological computing, and biosensing. Subtle adjustments in the amphiphile structure (e.g., relative lengths of positively and negatively charged peptides) produced a broad set of neuronal behaviors, including cell death, cell adhesion with limited neurite extension, and the formation of multicellular networks.

The strategy presented by Dore *et al.* offers a new way of engineering the assembly of microscale structures approaching the sizes of cellular components with bioactive building blocks. Notably, the interaction between polymer fibrils and neurons produced radically different cell phenotypes by changing the sequence and electric charge of peptide moieties. In addition, the self-capping mechanism using PEG offers a powerful route to producing fibrils of desired lengths. This tunability may facilitate the patterning of specific cellular processes and behaviors (such as differentiation). For example, ECM-coated fibrils of tunable lengths could be spatially patterned to precisely guide neurite extension and synapse location (*8*). These fibrils could also be integrated into additive manufacturing (a process that creates 3D objects by adding materials layer by layer) to produce 3D neuronal constructs (*9*). Further work will be needed to integrate the approach with advanced technologies and assess their value for neural patterning and network generation.

The study of Dore *et al.* also raises intriguing possibilities for engineering new cell culture scaffolds. A peptide amphiphile scaffold that features peptides derived from growth factors or cytokines could promote the proliferation and differentiation of specific cell types, such as immune cells, which could benefit tissue implants. Beyond fibril structure, the peptide amphiphiles may be designed to spontaneously assemble into higher-order organizational structures found in tissues, such as fiber bundles and sheets. Additionally, the formation of 3D cellular complexes on polymer fibril surfaces hints at the potential of technology for creating 3D cell culture constructs in which cells are embedded within a synthetic matrix. The 3D arrangements may offer new ways of guiding and controlling cell behavior because cells often transduce ECM-based cues differently in a 3D topology compared with on a 2D surface (*10*, *11*).

Beyond modulating cell behaviors, the self-assembling polymer fibrils may be useful for direct delivery of cells to the body for regenerative medicine applications. Peptide amphiphile–based therapies have shown promising results in preclinical models for nerve regeneration in spinal cord injury, where the microscale dynamics of the amphiphiles strongly influenced neuronal survival and functional recovery (*12*, *13*). Incorporation of peptides that encode growth signals could also boost the low cell survival and engraftment rates that plague current stem cell transplantation approaches (*14*). Further, peptide amphiphile formulations could be designed to deliver specific physiochemical instructions to different cell types within a single material construct, such as directing neurons to assemble into neuronal tracts and endothelial cells to form vascular networks. Further work is needed to reach these ambitious goals, but Dore *et al.* have provided an important starting point. □

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GRAPHIC: A. FISHER/*SCIENCE*

METABOLISM

A mitochondrial carrier turns on the heat

An orphan protein integrates circadian, dietary, and environmental signals to control brown fat metabolism

Martin van der Laan

As homeothermic organisms, mammals occasionally must spend metabolic energy to maintain body temperature but try to avoid its wasteful expenditure. Brown adipose tissue (BAT) plays a key role in thermogenic metabolism (*1*). At normal temperature, thermogenesis is switched off, and BAT lipid turnover is repressed by the circadian clock, an internal mechanism that regulates daily physiological processes, including metabolism. Upon exposure to cold, some metabolic energy derived from oxidative respiration in mitochondria is diverted to heat production (*2*). It has been unclear which molecular mechanisms govern the balance between the circadian clock, BAT lipid metabolism, and thermogenesis. On page 79 of this issue, Karavaeva *et al.* (*3*) report the unexpected finding that an orphan mitochondrial transport protein is a critical integrator of circadian, dietary, and environmental signals in BAT. Its expression in brown fat may be helpful as a metabolic health indicator.

Cold exposure activates the sympathetic nervous system to release norepinephrine, a neurotransmitter that stimulates β-adrenergic receptors in brown fat cells (adipocytes). This increases the expression of UCP-1 (uncoupling protein 1) (*4*–*6*), which inserts into the inner mitochondrial membrane and acts as a proton channel. UCP-1 allows protons to flow away from the electrochemical proton gradient that powers adenosine triphosphate (ATP) production. As a result, ATP synthesis is reduced, but adipocytes keep burning through their fat and sugar supplies. In this way, a part of the metabolic energy escapes directly as heat.

Karavaeva *et al.* performed sophisticated genetic, gene expression, and protein abundance screens to search for regulators of thermogenic metabolism in mouse BAT that may coordinate circadian and cold-regulated signaling pathways. Besides the known UCP-1, an uncharacterized mitochondrial membrane protein called SLC25A34 (solute carrier family 25 member 34) caught their attention. At normal temperature, expression of the corresponding gene was blocked by the circadian repressor of gene transcription, REV-ERBα (reverse erythroblastosis virus–α). This block was lifted by cold. If the diet contained sufficient lipids, then SLC25A34 expression was increased by the transcription factors PPARα and -γ (peroxisome proliferator activated receptor–α and –γ). Depletion of SLC25A34 in cultured brown adipocytes inhibited adrenergic receptor-stimulated mitochondrial respiratory activity, suggesting that the protein is required for BAT thermogenic metabolism.

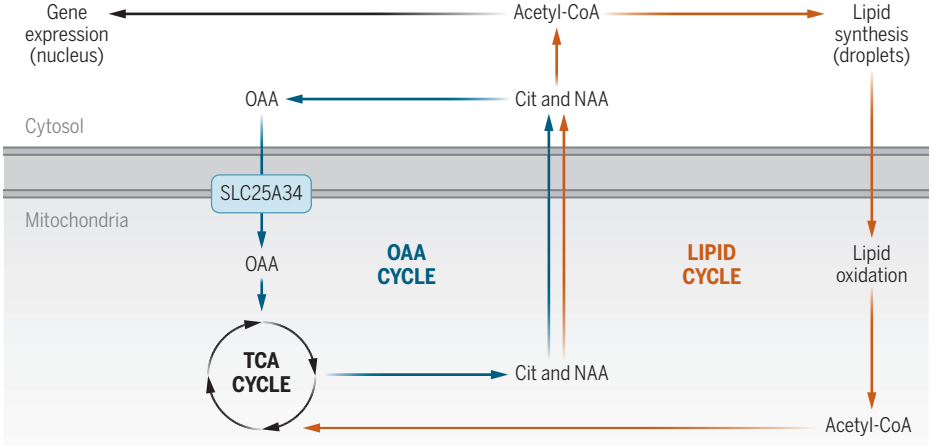
SLC25A34 has been classified as an orphan carrier protein of mitochondria because its native substrates and transport mechanism have remained unknown. How could this metabolite carrier represent a bottleneck in a thermogenic metabolic shift? To understand this question, it is important to note that BAT shows an intriguing feature upon cold exposure: Both lipid synthesis and lipid oxidation are enhanced at the same time. In other tissues, such futile cycles of degradation and resynthesis are prevented by a stringent regulatory mechanism. In BAT, however, this general boost of lipid metabolism—anabolic and catabolic together—contributes to thermogenesis and helps to maintain body temperature (*7*). Through metabolic profiling and flux analysis of untreated and SLC25A34-depleted brown adipocytes, Karavaeva *et al.* observed two major changes: accumulation of *N*-acetyl-aspartate inside mitochondria and oxaloacetate outside of the organelle, in the cytosol. Both metabolites are critical for exporting excess acetyl-CoA (acetyl–coenzyme A), a key molecule of cellular energy metabolism, from mitochondria to the cytosol, where it is a substrate for fatty acid synthesis.

In brown adipocytes, there are two pathways that supply acetyl-CoA to the cytosol (*8*, *9*). *N*-acetyl-aspartate is synthesized in mitochondria from oxaloacetate-derived aspartate and acetyl-CoA and is exported by an unknown mechanism into the cytosol. There, the molecule is hydrolyzed, and acetate is converted into acetyl-CoA. Alternatively, citrate is exported from mitochondria and cleaved in the cytosol to produce oxaloacetate and acetyl-CoA (*8*, *9*). Thus, no matter how the activated acetyl moieties are leaving mitochondria, they are always accompanied by an oxaloacetate molecule that needs to be transported back to sustain the shuttle. Specifically, in cold-stimulated BAT, where both anabolic and catabolic lipid metabolism are enhanced, accumulation of oxaloacetate in the cytosol is detrimental. Oxaloacetate is not only lacking in the mitochondrial matrix to keep oxidative respiration going and oxidize acetyl-CoA, it is also in short supply to pick up more acetyl-CoA for export to the cytosol to support lipid synthesis. The notion

A hub of metabolic cycles in brown fat cells

(Top) The mitochondrial transport protein SLC25A34 integrates circadian, environmental, and dietary signals to control heat production (thermogenesis) in brown fat cells. (Bottom) Specific transcriptional regulators adjust SLC25A34 expression in the mitochondrial inner membrane. As a central node, SLC25A34 likely transports OAA into mitochondria. This drives three metabolic cycles: the OAA cycle, the TCA cycle, and the lipid cycle.

CUE					SIGNALING		OUTCOME
Circadian clock		**Environment**		**Diet**	**Transcriptional regulator**	**SLC25A34 expression**	**Heat production**
Awake	or	Cold	or	High fat	PPAR-α/γ	More	More
Asleep	or	Normal	or	Normal	REV-ERBα	Less	Less



Acetyl-CoA, acetyl–coenzyme A; Cit, citrate; NAA, *N*-acetyl-aspartate; OAA, oxaloacetate; PPARα and -γ, peroxisome proliferator activated receptor–α and –γ; REV-ERBα, reverse erythroblastosis virus alpha; SLC25A34, solute carrier family 25 member 34; TCA, tricarboxylic acid

GRAPHIC: N. BURGESS/*SCIENCE*

that in SLC25A34-deficient brown adipocytes the amounts of cytosolic and nuclear acetyl-CoA become limiting is further underscored by a decrease in the acetylation of histone proteins that support the expression of genes that encode enzymes of the oxidative respiratory machinery (*9*). Karevaeva *et al.* also noted that oxaloacetate limitation is reflected by increased rates of glucose-dependent enzymatic reactions that replenish the mitochondrial citric acid cycle, the chemical reactions that release energy stored in nutrients through acetyl-CoA oxidation.

Moreover, there is the conundrum that, as most textbooks state, there is no oxaloacetate uptake into mitochondria in mammals (*10*, *11*). Rather, this metabolite is thought to be reduced to malate in a reaction that consumes cytosolic NADH (nicotinamide adenine dinucleotide hydride). Malate is then transported back to mitochondria to regenerate oxaloacetate, a process known as the malate shuttle. However, oxaloacetate transport across the inner mitochondrial membrane has been reported for plants and fungi (*12*). Notably, the closest deorphanized homolog of SLC25A34 is the yeast carrier Oac1 (oxaloacetate carrier 1) (*13*). Indeed, Karaeva *et al.* predicted, with molecular modeling, high-affinity binding between oxaloacetate and human SLC25A34.

Altogether, the dogma of oxaloacetate not being imported across the inner mitochondrial membrane in mammals is seriously shaken by the study of Karavaeva *et al.* One could speculate that SLC25A34 operates as an antiporter, like Oac1, and that its role in thermogenesis is the exchange of oxaloacetate and *N*-acetyl-aspartate between the mitochondria and cytosol. Reconstitution of the transporter into proteoliposomes will be required to directly test this hypothesis.

The study by Karavaeva *et al.* is a reminder of something more general about metabolic regulation and shifts in the flow of carbon fuel through different pathways. Rather than intricate regulatory circuits tuning the activity and abundance of many metabolic enzymes, homeostatic issues may easily be solved by altering the availability of substrates in a certain cellular compartment through the shuttling rate of the critical molecules (*14*). Such knowledge might well help in the treatments of common human diseases related to metabolic imbalance. Small molecules acting on defined metabolite carriers may help to compensate global homeostatic disturbances. Specifically, SLC25A34 may turn out to be a target for innovative new drugs for the treatment of obesity. □

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RHEUMATOLOGY

Osteoarthritis as a vascular disease

Hypertension worsens osteoarthritis and could be a therapeutic target in patients with both conditions

Chunyi Wen and **Man Ting Au**

Osteoarthritis is no longer seen as a simple wear and tear of the cartilage that cushions bone joints during movement. It is now recognized as a whole-joint disorder, characterized by cartilage degradation, alterations of the bone under the cartilage, and low-grade inflammation of the connective tissue that lines the joint (synovium). However, no drugs that target these localized tissues have been approved for clinical use thus far. Therefore, disease management has remained largely unchanged since the 1990s and is limited to symptom relief with nonsteroidal anti-inflammatory drugs and injections of hyaluronic acid or corticosteroids into joints (*1*). The view of osteoarthritis as a localized disease overlooks a clinical reality: Although it can result from mechanical injury, a substantial proportion of disease cases are driven by metabolic dysfunction. On page 80 of this issue, Kim *et al.* (*2*) report a "brain-joint axis" that links osteoarthritis to hypertension, a component of the metabolic syndrome.

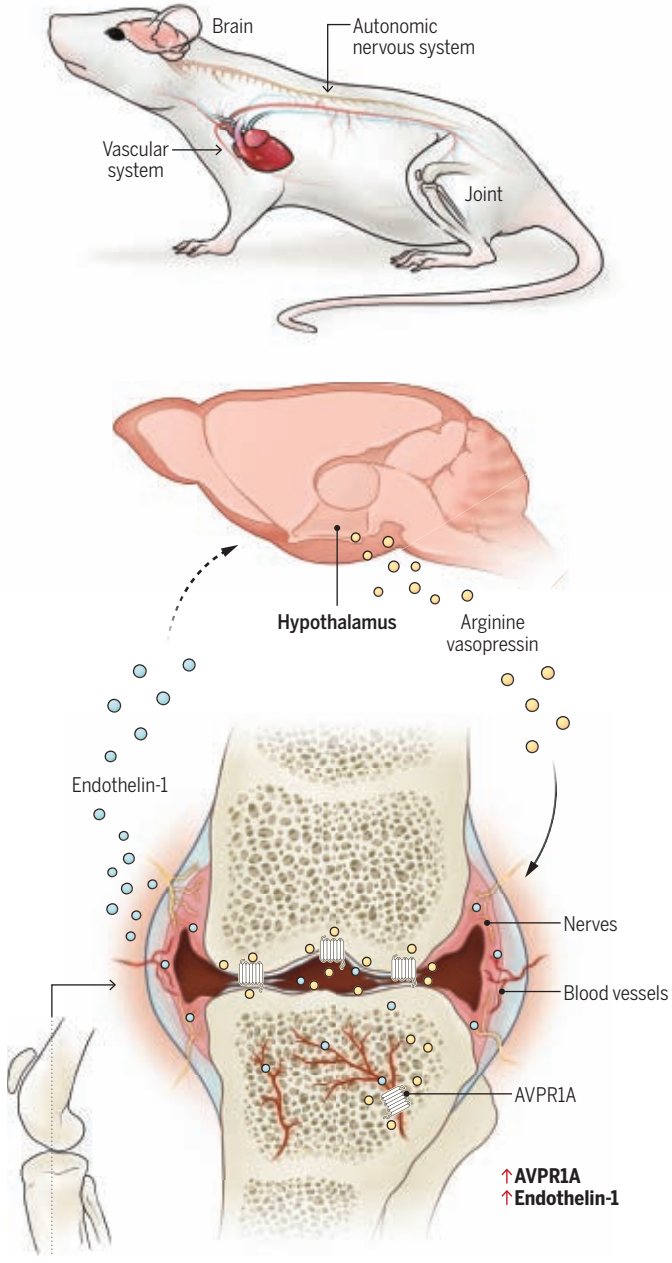
Obesity and the metabolic syndrome—a cluster of health conditions that includes high blood pressure, excessive belly fat, and elevated sugar and cholesterol concentrations in blood—are established risk factors for osteoarthritis. Therefore, metabolic therapies such as incretin mimetics (including glucagon-like peptide-1 receptor agonists) could potentially change the treatment of this disease (*3*). Incretins are of particular interest because of their whole-body metabolic benefits, which are independent of weight loss. Notably, these drugs rely on interorgan communication. In addition to the well-documented gut-joint signaling axis, communication between the liver and joints and between the lungs and joints may also influence the progression of osteoarthritis (*4*–*6*). The brain-joint axis identified by Kim *et al.* adds to the concept of osteoarthritis as a systemic disease.

Kim *et al.* found that arginine vasopressin (AVP), a peptide secreted by hypothalamic neurons in the brain, uses the same receptor to regulate the function of blood vessels and cartilage in a mouse model of hypertension. They discovered that circulating AVP, which is known to bind to AVP receptor 1A (AVPR1A) on vascular smooth muscle cells to drive the constriction of blood vessels, also acts on cartilage cells in osteoarthritic joints. High expression of AVPR1A by these cells triggered the breakdown of complex molecules (catabolism) and dampened the synthesis of new ones in the joint area. Furthermore, genetic deletion or use of drugs that block AVPR1A prevented progression of hypertension-driven osteoarthritis. These findings show that vascular disease and cartilage degeneration, two highly prevalent age-related conditions, can share a mechanistic origin. They also raise the possibility that systemic antihypertensive drugs might be effective disease-modifying treatments for osteoarthritis.

This repurposing strategy is supported by a study that used UK Biobank data to investigate the associations between use of antihypertensive drugs and risk of osteoarthri-

A whole-body view of osteoarthritis

Osteoarthritis is not exclusively caused by localized inflammation in bone joints that leads to cartilage damage. It is also driven by metabolic dysfunction across the body. Both vascular and neural signaling pathways can affect joint health. The hormones endothelin-1 and arginine vasopressin, which is produced by the hypothalamus and binds to AVP receptor 1A (AVPR1A), have a dual role: They increase blood pressure, which can lead to hypertension, and also joint inflammation, which promotes osteoarthritis.



GRAPHIC: A. MASTIN/*SCIENCE*

tis (*7*). Potassium-sparing diuretics and aldosterone antagonists (such as spironolactone) significantly correlated with a reduced risk of osteoarthritis. The AVP-driven brain-joint axis identified by Kim *et al.* offers a plausible explanation for this protective effect: Because these diuretics inhibit sodium reabsorption in the kidney's late-distal tubules, they correct fluid overload, thereby reducing the molecular signals that trigger AVP release from the hypothalamus. Notably, previous small-scale trials of spironolactone in patients with knee osteoarthritis have had mixed results. Therefore, larger-scale clinical investigations are warranted to explore the potential benefits of these drugs in this patient population.

The renin-angiotensin-aldosterone system (RAAS) and endothelin-1–driven signaling, which are hormonal networks that control vascular tone, have also been implicated in hypertension-associated osteoarthritis. For example, the RAAS-targeting drug captopril counteracted cartilage loss in hypertensive rats (*8*). However, Kim *et al.* observed that directly blocking RAAS function using losartan did not protect cartilage in their model. This discrepancy suggests that the RAAS might not be the primary mediator of joint degeneration and that captopril probably exerts its protective effects through alternative, RAAS-independent pathways. Endothelin-1 shares potent vasoconstrictor and proinflammatory properties with AVP and triggered joint pain and cartilage breakdown in a mouse model of posttraumatic osteoarthritis (*9*). Kim *et al.* also observed that endothelin-1 expression is markedly up-regulated in osteoarthritic chondrocytes, which suggests that it may act as a secondary or synergistic disease-promoting factor alongside AVP.

In addition to the AVP-AVPR1A pathway, the brain-joint axis is also modulated by the autonomic nervous system, which controls vascular tone and joint homeostasis through a delicate balance between sympathetic (adrenergic) and parasympathetic (cholinergic) signaling (*10*, *11*). Corticotropin-releasing neurons in the hypothalamus project directly to the joint's synovium to communicate with local cells, and dampening this sympathetic activity improves joint health (*12*). The parasympathetic system counterbalances these effects. Stimulation of the vagus nerve, which connects the brain to the intestine, activates cholinergic anti-inflammatory pathways that help to normalize systemic and local vascular tone (*13*). This counteracts hypertension-induced hypoxia (low oxygen) in the joint and attenuates synovial inflammation.

Kim *et al.*'s findings reinforce the idea that hypertension is a key driver of osteoarthritis (*14*). The vasoconstriction that characterizes hypertension may indirectly induce hypoxia and thus inflammation in local joint tissues and aggravate cartilage damage (*15*). Furthermore, compounding the effects of sympathetic overactivation, vasoconstrictors such as AVP and endothelin-1 may also act directly on osteoarthritic chondrocytes to trigger catabolic responses (see the figure).

Recognizing osteoarthritis as a vascular disease rather than a simple mechanical problem opens avenues for therapeutic discovery. Repurposing antihypertensive drugs to treat osteoarthritis should be prioritized for patients who also have hypertension. Mapping the complex interorgan networks that affect joint health will usher in an era of precision rheumatology that combines local biomechanical care with systemic therapies to restore vascular health and joint integrity. □

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DEVELOPMENT

A tumor suppressor shuts down a signal gradient

p53 shuts down the spatial signaling of a morphogen and prevents clonal cell expansion

Dan Engelman[1] and **Cédric Blanpain**[1,2]

The most frequently mutated tumor suppressor gene in human cancers is *TP53* (*1*). Loss-of-function mutations lead to expansion of a mutated cell and its progeny in epithelial tissues, including sun-exposed regions of the skin, which can progress to invasive cancers. Upon stress, such as DNA damage, p53 becomes phosphorylated or acetylated, modifications that promote its stabilization and the activation of genes that cause cell cycle arrest, cellular senescence, or cell death (*1*). One might expect that clonal expansion triggered by the loss of p53 results from dysregulation of these canonical functions. On page 78 of this issue, Gan *et al.* (*2*) report that this expansion is instead driven by activation of a gradient of Wnt (wingless-related integration site) signals that promotes epidermal stem cell self-renewal. Controlling signaling gradients, rather than the absolute amount of a signal, may be a mechanism through which cancer-associated mutations achieve long-term tissue colonization.

Despite extensive knowledge about p53 functions, the molecular mechanisms by which it suppresses tumorigenesis remain unresolved (*3*). In response to stress, p53 is acetylated at multiple lysine residues, and mutations at the three acetylation sites in the DNA-binding domain prevent it from mediating cell cycle arrest, programmed cell death (apoptosis), and senescence. This mutation in mice (*Trp53*$^{3KR/3KR}$; *Trp53* is the mouse homolog of human *TP53*) also prevents tumor formation because mice lacking *Trp53* (*Trp53*$^{-/-}$), but not *Trp53*$^{3KR/3KR}$, develop early-onset spontaneous lymphoma (*4*). This raises the possibility that tumor suppression does not depend on p53 canonical functions.

During tissue homeostasis, the balance between cell self-renewal and cell differentiation maintains a constant cell number (*5*). Oncogenic mutations can impair this balance, leading to the expansion of a mutated cell into a clonal population as the result of excess stem cell divisions and a failure to differentiate. For example, during basal cell carcinoma initiation, cells harboring a mutation in *Smo* (which encodes the Smoothened protein) exhibit enhanced self-renewing divisions and a block of differentiation, leading to the establishment of preneoplastic lesions (*6*). Although *Trp53* deletion or *Trp53*^{R248W} expression (a dominant-negative mutation that inactivates p53 protein) (*7*) do not increase the rate of mouse epidermal stem cell proliferation, they promote self-renewal, giving rise to more stem cells (without causing cells to divide faster or more frequently). Gan *et al.* did not observe this in mouse epidermal *Trp53*$^{3KR/3KR}$ clones. This suggests that *Trp53*$^{-/-}$-mediated clonal expansion is likely the consequence of an imbalance toward self-renewing cell divisions rather than uncontrolled proliferation, through mechanisms independent of the loss of p53's canonical functions.

To decipher the molecular mechanism by which p53 controls epidermal cell self-renewal, Gan *et al.* hypothesized that genes controlling self-renewal would be shared between wild-type p53 (*Trp53*^{WT}) and *Trp53*^{3KR}. Upon identifying common, differentially regulated genes by *Trp53*^{WT} and *Trp53*$^{3KR/3KR}$ in epidermal cells from both mouse strains (through epigenetic and transcriptional analyses), the authors performed an in vivo genetic screen by molecularly perturbing each of 1113 candidate genes and assessing the impact on self-renewal. They narrowed the list to three genes: *Sfrp1* (*secreted frizzled-related protein 1*), *Lrp1* (*low-density lipoprotein receptor-related protein 1*), and *Usp22* (*ubiquitin-specific protease 22*). The expression of these genes was up-regulated in *Trp53*^{WT} and *Trp53*$^{3KR/3KR}$ epidermal cells, where there was decreased self-renewal. Notably, all three genes had been previously reported as potential negative regulators of the Wnt signaling pathway.

In the mouse skin, Wnt signaling is best known for regulating hair follicle stem cells and sebaceous gland function (*8*). It acts as a morphogen that directs spatial patterning of cells, cell fate, and tissue morphogenesis of the skin. However, loss of β-catenin or wntless (Wls), constituents of the Wnt signaling pathway, also decreases interfollicular epidermal stem cell proliferation (*9*, *10*). Consistent with the idea that p53 up-regulates the expression of inhibitors of Wnt signaling in the epidermis, Gan *et al.* demonstrated that p53 deletion or silencing the expression of *Sfrp1*, *Lrp1*, or *Usp22* in mouse epidermal cells increased activity of Wnt signaling and clonal expansion. Conversely, overexpression of *Sfrp1*, *Lrp1*, or *Usp22* in *Trp53*$^{-/-}$ epidermal cells decreased Wnt activity and clonal expansion. Thus, p53 regulates Wnt activity in the epidermis by controlling the expression of these three genes.

Gan *et al.* further found that p53 deletion leads to a gradient of Wnt activity within the expansion of mutated clones, with the highest activity in the center of the expansion and decreasing radially toward the edges. This spatial distribution was established only by suppressing the expression of the soluble Wnt inhibitor *Sfrp1*. The authors hypothesize that the Wnt gradient, rather than a specific level of Wnt signaling, is responsible for the rapid expansion of p53-deficient clones.

Gan *et al.* uncover a new role for p53 in gatekeeping aberrant cell renewal. In this model, the unrestrained expansion of p53 mutant cells is driven by the generation of a Wnt activity gradient within the mutant clones. It is not clear whether p53 loss in other tissues, or other oncogene or tumor suppressor gene mutations, also modulate Wnt signaling in the skin. Yet to be determined, as well, are the mechanisms by which Wnt signaling itself promotes renewal in this context. p53 puts the brakes on establishing a gradient by activating three inhibitors of Wnt action. However, even in the absence of a gradient, Wnt might still promote clonal expansion. Too much Wnt signaling can decrease renewal and lead to cell differentiation, as observed during hair (*11*) or Paneth cell differentiation in the intestine (*12*). This means that the promotion of stem cell renewal requires the right dose of Wnt signaling.

Sometimes a stem cell mutates and turns on its Wnt signaling pathway permanently, as observed in intestinal crypts. However, clonal expansion of such cells has been attributed to their secretion of Wnt inhibitors to eliminate wild-type neighbors (*13*, *14*). This is a key step in the early stages of intestinal cancer. Whether a similar mechanism occurs in the epidermis remains to be determined. The approach of Gan *et al.* opens the possibility of unraveling clonal expansion and cell competition in response to other oncogenic hits and in other locations. □

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POLICY FORUM

REGULATION

Longevity products: Innovation through regulation

Longevity is taking off, but regulatory oversight is essential to determining what works—and what doesn't

Camila Strassle,[1] Holly Fernandez Lynch,[1] Patricia J. Zettler[2,3]

The field popularly referred to as "human longevity" is booming, incorporating a variety of approaches aimed at optimizing human health by reversing cellular aging, prolonging life, and extending "healthspan" (the duration of life spent in good health). With longevity companies rapidly proliferating globally, and, in the United States, drawing billions of dollars in investments and interest from government officials (*1*), some commentators have raised concern that longevity interventions fall in a gray area for regulators. Using the US as an example, we dispel that notion and discuss how longevity products generally fall squarely inside the existing regulatory regime. To the extent that there are challenges to bringing longevity products legally to market, they are largely scientific rather than regulatory. The most important priorities to support safe and effective interventions, promote innovation, and avoid grift are to develop and validate endpoints capable of predicting longevity benefit and to enforce appropriate legal requirements.

As proponents of the longevity field see it, progress against chronic diseases has lagged because traditional medical interventions have focused on symptom management once disease has occurred. But, their argument continues, if age is the major risk factor for most chronic diseases, manipulating the aging process itself could delay disease development and progression. The hope is to prevent age-related chronic diseases rather than tackling them individually as they arise.

This vision has attracted government interest. For example, in July 2026, the US Food and Drug Administration (FDA) held an Advisory Committee meeting regarding whether to allow the sale of several compounded peptides, including those used for longevity purposes, without first securing FDA approval (*2*). Amid this interest, questions have arisen regarding whether FDA has an approval pathway that can be applied to "longevity" products, in part because it does not recognize aging as a disease (*3*, *4*). Other commentators seem to recognize that FDA has an applicable approval pathway but bemoan the lack of a pathway specifically tailored for longevity, asserting that regulatory ambiguity impedes the ability to legally bring products to market (*5*, *6*). Yet the law is quite clear that longevity products fit under the existing regulatory regime, despite the current dearth of FDA enforcement. Moreover, although FDA could formally recognize aging as a disease, doing so would not necessarily make the pathway to approval clearer or more efficient.

The real challenge is neither new nor specific to the longevity field: It is difficult to directly measure clinical benefit when the outcome of interest is long-term, such as extended survival. Thus, drug approval is often based on demonstrated effects on surrogate endpoints thought to predict clinical benefit. Although specific details of regulatory approaches to longevity are country dependent, the bottom line will be the same for all policy-makers, science funders, and regulators interested in encouraging true progress in this sector. Rather than leaning into deregulation, dedicating resources to validating endpoints and to enforcing existing regulation is what will best serve the public, advance science, and promote innovation.

THE LONGEVITY FIELD AND US FDA JURISDICTION

Several longevity products have been identified as promising because of their connection to cellular and molecular signaling pathways implicated in aging, including those related to inflammation, oxidative stress, DNA stability, mitochondrial dysfunction, and cellular senescence *(7)*. Among products already widely marketed to consumers, many are promoted for claims related to muscle strength, anti-inflammatory effects, improved cognition, and tissue repair or regeneration. To date, none of these products are FDA approved specifically for anti-aging indications.

High-profile longevity products include metformin (glycemic control), rapamycin (immunosuppression), sodium-glucose cotransporter 2 inhibitors (inhibition of glucose reabsorption), nicotinamide adenine dinucleotide (NAD^+) precursors (promoted for energy metabolism), resveratrol (promoted for anti-inflammation), senolytics fisetin and quercetin (promoted for senescent cell clearance), and peptides (short amino acid sequences promoted for various uses) (*7, 8*). Some of these products are FDA approved for uses unrelated to biological aging, such as diabetes, weight control, immunosuppression for organ transplant, and chronic kidney disease. However, many products popular for longevity have never been FDA approved for any use.

Some longevity products are being studied in ongoing trials. For instance, the US National Institute on Aging created the Translational Geroscience Network (TGN) to standardize research and speed development of therapeutics, with more than 80 affiliated trials. Additionally, the XPRIZE Foundation is hosting a "healthspan competition" through which it awarded funds in August 2026 to research teams running clinical trials. Gene editing and stem cell interventions are also being investigated clinically. Life Biosciences, for example, has started a trial of ER-100, a cellular reprogramming technology, with the hope of rejuvenating aging organs (*9*). Many companies, however, are already marketing products for anti-aging effects without conducting studies to demonstrate safety or effectiveness for that purpose. This is true even for some products that lack FDA approval for any use.

Longevity products fall within already-defined legal categories in the US. Indeed, most of these products clearly meet the definition of a drug, which the Federal Food, Drug, and Cosmetic Act (FDCA) defines to include products intended for use not only in the diagnosis, cure, mitigation, treatment, or prevention of disease, but also products intended to affect the structure or any function of the body. This "structure/function" element of the statutory definition means two things: Products can still be drugs even if not intended for therapeutic use, and whether FDA recognizes aging as a disease does not determine whether these products fall within FDA's drug authorities nor whether FDA can approve them for anti-aging. As a result, the notion that there is no clear regulatory pathway for longevity drugs because FDA does not classify aging as a disease is mistaken. For example, FDA regulates hair growth products and onabotulinumtoxinA (i.e., Botox) for reducing facial lines, both of which have been approved as drugs for nontherapeutic uses despite the lack of any targeted disease (*10*). FDA also recently signaled willingness to approve two canine drugs (LOY-001 and LOY-002) for longevity indications, independent of any disease claim, indicating that FDA will also view many human longevity products as drugs given the FDCA's identical definition of "drug" for animals and humans.

LEGAL PATHWAYS AND THEIR CONSTRAINTS

The many longevity products that meet the statutory drug definition

are subject to the same requirements that apply to all FDA-regulated drugs. Generally, these products cannot be legally marketed for longevity or other uses (e.g., athletic performance, sexual function) in the United States without first securing FDA approval based on demonstrating safety and effectiveness, and they must comply with Current Good Manufacturing Practice requirements. Notably, exceptions are either limited in scope or legally unavailable, at least for now, to longevity drugs. Longevity products are nevertheless widespread on the US market because FDA has inconsistently enforced relevant requirements, and the financial incentives for companies to market products, despite the legal risk, are substantial. However, this does not mean that FDA lacks enforcement authority or that requirements are being met.

Some US states have recently expressed interest in expanding their "right to try" laws in ways that would purportedly permit the distribution and administration of unapproved longevity products. These laws, including the federal right to try law, previously have been limited to patients facing life-threatening conditions who seek use of investigational agents outside clinical trials—but in some cases, they are now being extended to everyone, even individuals who are currently healthy. For example, Montana enacted such an expansion of its law in 2025 (SB535) and published regulations to govern the state's experimental treatment centers in 2026, while New Hampshire has proposed similar bills (HB1734 and HB1735). However, even when enacted, these state-level efforts have limited legal effect because states do not generally have the authority to exempt parties from compliance with federal law. Thus, when distributing longevity products in the United States, companies must follow applicable FDA requirements.

Legitimate industry should welcome such enforcement, and the public should demand it.

In addition to state efforts, many longevity companies, clinics, and preventive care practices have emerged that facilitate the sale of longevity products through pathways other than traditional drug approval (*11*). However, these business models are often premised on a misunderstanding of the extensive reach of FDA requirements. For example, in some cases, companies sell longevity products directly to consumers as "research chemicals" or with the disclaimer "for research purposes only." Despite their pervasiveness, such businesses generally run afoul of US law. Investigational drugs may be distributed to research participants enrolled in clinical trials under an FDA-authorized Investigational New Drug (IND) Application. But companies cannot sell unapproved drugs intended for human use under the guise of research and without an IND. Even for permissible research under an IND, there are strict limits on what companies may charge research participants; they may do so only with prior FDA authorization and may not profit from unproven interventions.

Compounding is another strategy sometimes used to distribute unapproved longevity products. Compounded drugs, which are ostensibly custom-made by pharmacies or specialized facilities, are not FDA approved, but the FDCA nevertheless allows compounding when certain criteria are met. Most relevant for longevity products, section 503A of the FDCA allows a pharmacy to make a compounded drug for an individual patient in response to a valid prescription order—for example, when a patient is allergic to an inactive ingredient in an approved product and needs a version without it. This kind of compounding is permitted only for distinct patient needs and only for certain substances, like those that are components of FDA-approved drug products or appear on an FDA-created list of permissible substances. Many longevity products do not meet these conditions, however. For example, the Secretary of the Department of Health and Human Services (HHS) recently directed FDA to remove several unapproved peptides from the list of substances disallowed for compounding owing to potentially serious safety risks, despite the absence of data to address the safety concerns that landed the peptides on that list in 2023 (*12*). At the FDA Advisory Committee meeting in July 2026, panelists, many of whom had financial interests in peptide-related businesses, narrowly voted to recommend the further step of adding several peptides to FDA's list of permissible compounding substances despite the lack of demonstrated safety and effectiveness and serious concerns raised by FDA staff. At least until FDA further considers the issue through a rulemaking process, however, none of the wholly unapproved products popular for longevity uses may legally be compounded and sold. And regardless of what FDA decides on peptides, even for longevity drugs compounded using components of FDA-approved drugs or other permitted substances, the law does not allow mass production that resembles conventional drug manufacturing.

Some longevity products are marketed as "dietary supplements," which—unlike substances defined as drugs under the FDCA—are not subject to premarket approval requirements. Yet many of these products are in fact drugs under the law because they do not fall within the definition of a dietary supplement. To be a dietary supplement, a product must contain a dietary ingredient, be intended for ingestion, and must not contain approved or investigational drug ingredients or be intended for disease uses. Products using injectable, topical, transdermal, or nasal administration, as many longevity products do, cannot be dietary supplements because they are not ingested. Products that are intended to diagnose, cure, mitigate, treat, or prevent diseases related to aging are not dietary supplements, even if labeled as such.

"Off-label prescribing" of already-approved drugs to patients for unapproved longevity uses is the approach on the strongest legal footing, short of FDA approval explicitly for longevity uses. But many products currently used for longevity purposes are not already FDA approved for other uses. Moreover, business models premised on off-label use are not completely unrestricted. Critically, drug manufacturers, and those speaking on their behalf, generally cannot directly promote or advertise off-label uses without violating the FDCA.

To be sure, there are limits to FDA's reach. For example, people living in the United States sometimes access experimental products through medical and wellness tourism, with some longevity research done offshore to evade FDA requirements and commercialize products with minimal safety and effectiveness data.

The upshot is that companies marketing longevity agents claiming to prevent or reverse aging or address age-related diseases and conditions generally will need to follow FDA's drug-approval process, including generating rigorous evidence of safety and effectiveness. Drug approvals could be for nondisease "structure/function" uses (where labeling and promotional materials describe the drug's role in altering the body's structure or function) or "disease" uses (where materials describe the drug's role in preventing, mitigating, or reversing age-related diseases). Either way, there are existing paths to approval—and approval is both legally necessary and appropriate.

ENFORCEMENT

Despite these legal requirements for longevity products, in many cases FDA has "enforcement discretion," flexibility to decide not to take formal action against companies that ignore requirements. Under leadership friendly to the longevity industry, FDA might choose to deprioritize longevity products as a subject of enforcement action and seems poised to do so for certain peptides.

Yet FDA, and its global regulatory counterparts, have strong reasons to enforce regulatory requirements for longevity products. Legitimate industry should welcome such enforcement, and the public should demand it. Enforcing drug-approval requirements helps protect consumers against deceptively promoted and potentially unsafe products, while ensuring that minimum quality and man-

ufacturing standards have been met, including that products are free of impurities, contaminants, and toxins and contain only what they claim and nothing else, at the claimed concentration. Experience has shown that medical products lacking FDA oversight can cause death and other serious harm: for example, the 2012 fungal meningitis outbreak traced to compounded steroid injections killed over 50 people and sickened hundreds (*13*). Enforcement helps ensure that drugs are rigorously studied for safety and effectiveness, which benefits patients and promotes innovation by ensuring that companies investing in the production of such evidence will not be undermined by competitors who are allowed to sell unproven interventions. FDA's role in protecting consumers may be especially important for products marketed for nontherapeutic, structure/function uses because these drugs are unlikely to be considered medically necessary and covered by payers.

Given the scope of the longevity industry, absent a budget increase, enforcement may require FDA to redirect considerable resources away from other priorities. If FDA fails to step in, states might enforce consumer protection laws that prohibit deceptive promotion against clinics and clinicians, as some have done in the context of those marketing stem cell interventions without reliable evidence to support their claims. Clinicians may also be disciplined by state medical boards or sued for malpractice by harmed consumers. However, these are poor replacements for FDA's expert premarket oversight (*14*).

CHALLENGES AND OPPORTUNITIES

The most straightforward path in the United States to meet the standards imposed by the FDA is for companies to seek approval for claims about the effects of longevity drugs on the structure or function of the body. Such approvals are important because consumers should at least be assured that longevity products effectuate the intended physiological or biochemical change in the body, even if it remains unproven whether that change ultimately makes a clinically meaningful difference in lifespan or healthspan. Scientifically, these structure/function claims can be supported by traditional clinical studies.

However, to realize the potential of the field, longevity products ultimately will need to be studied and proven safe and effective for outcomes more directly related to clinical benefit, including preventing and treating disease. This poses an added scientific challenge because demonstrating a preventive effect on age-related diseases and conditions, as well as long-term safety, by definition, will take a long time. The central difficulty will be identifying suitable surrogate endpoints for longevity studies, where the primary outcome measure is a marker that does not directly measure clinical benefit but is known to predict a clinically meaningful outcome and can be studied more quickly. Whether longevity products could qualify for FDA's accelerated approval pathway, under which surrogate endpoints need only be reasonably likely to predict clinical benefit (rather than proven to do so) at the point of approval, with clinical benefit confirmed thereafter, is an open question. That pathway is available only for products intended to treat serious or life-threatening conditions with unmet medical needs. It is not clear that aging itself meets that standard, although specific age-related diseases certainly could. It is also not clear whether use of accelerated approval or a similar pathway would be desirable from a policy perspective, given known challenges in the conduct of postapproval confirmatory studies (*15*). Regardless, testing products for clinical benefit will be essential to making sure that longevity products in fact work to slow, pause, or reverse age-related decline.

FDA, its global regulatory counterparts, and research funders therefore should engage with longevity communities to evaluate what study designs and kinds of evidence are both feasible to produce and sufficient to meet a rigorous standard of safety and effectiveness. Currently, there is no consensus about which surrogate endpoints to use for longevity trials, but there are candidate measures that could be linked to the biological processes of aging, several of which were discussed at a public meeting in May 2026 convened by the US Reagan-Udall Foundation for the FDA with the Advanced Research Projects Agency for Health (ARPA-H). Longevity stakeholders should continue to seek opportunities for identifying scientifically valid and regulatorily acceptable endpoints, whether through bodies like Reagan-Udall and the TGN (or similar efforts in other countries, like the Canadian Translational Geroscience Network), engagement with FDA (e.g., through formal meetings and the Biomarker Qualification Program), or private-led initiatives like the "healthspan competition."

Regulatory regimes like FDA's both prevent companies profiting from drugs that do not work and help health care professionals and the public make informed decisions about what drugs to pursue or avoid based on reliable information regarding safety and effectiveness. In this way, although enforcement will be challenging and may narrow the longevity product market, at least for a time, it will be essential for advancing scientific understanding and supporting personal autonomy by ensuring that people seeking longevity products are no longer forced to rely exclusively on the claims of interested companies, influencers, and fellow users. Satisfying FDA's regulatory requirements may not be easy but it is the best path to ensure that the longevity movement lives up to its promise and offers more than just snake oil. □

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MEDICINE

A medical mystery unfolds in middle America

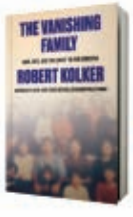
As a rare form of dementia claims one sibling after another, a journalist captures the profound costs of caregiving

Nancy Berlinger

Dementia is nearly always a family disease, because most dementia caregivers are family members of the person in need of care. In *The Vanishing Family*, journalist Robert Kolker provides an account of what dementia caregiving is like that is as breathtaking as the scientific "quest to end dementia" that the subtitle proclaims.

This is Kolker's second book about a large postwar American family with bad medical luck. In his 2020 bestseller *Hidden Valley Road*, six sons in a family of 12 are diagnosed with schizophrenia. In *The Vanishing Family*, five siblings in a family of nine are diagnosed with frontotemporal dementia (FTD), a relatively rare type within the group of conditions known as Alzheimer's disease and related dementias (ADRD) or, more commonly, as dementia.

Boomer and Gen X readers who grew up in the suburbs are likely to recognize the postwar Catholic family introduced in the early chapters. (Bookmark the page with the family tree.) The parents, Ollie and Jean, meet on a train from St. Louis to South Bend, Indiana, in 1949; marry in 1950; and have their first two children within 2 years. Their life in St. Louis, and briefly in Canton, Ohio, is shaped by Jean's deep commitment to a highly organized household and the sanctity of cocktail hour: old-fashioneds for Ollie, Manhattans for Jean. So far, so *Mad Men*.



The Vanishing Family: Love, Fate, and the Quest to End Dementia
Robert Kolker
Doubleday, 2026. 368 pp.

Between the mid-1960s and the mid-1970s, several things happen to this family (their surname is omitted and several names are changed). They move to Pittsburgh, where Ollie, a civil engineer, is pursuing opportunities in a not-yet-postindustrial city. Jean gives birth at 48 to her ninth child, Barb, 10 years younger than her nearest sibling. And Jean changes, spending her days drinking and watching game shows, devoid of affect, barely speaking. Barb is the closest observer of Jean's behavior and the child most affected by it; her siblings have left home or are on their way out, Ollie spends most of his time at work or with other women.

The family's initial explanation for Jean's behavior—alcoholism, perhaps related to Ollie's infidelities—leads to an intervention. Jean quits drinking, but neither her affect nor her behavior changes. She dies at age 62 of breast cancer, which had widely metastasized before she disclosed that anything was wrong.

Kolker intersperses the family narrative with chapters about the history of brain science, beginning with the story of Phineas Gage, who survived an industrial accident in 1848 that drove a pointed iron rod through his frontal lobe. (Visitors to the Countway Library at Harvard Medical School may spot Gage's skull, with its two immense holes, in a display case.) Kolker is expert at summarizing scientific developments and placing them in historical and social context: The impact of Freud's theory of human personality made brain science look old-fashioned, readers learn, slowing investigation into neurological factors in personality and behavior.

As five of Jean's children begin, in their early 40s, to behave like their mother—exhibiting apathy toward their own families, loss of executive function, excessive drinking or other loss of social inhibitions—the scientific and family narratives intersect. Christy, the first sibling to become ill, is initially misdiagnosed with Pick's disease, before, over many years, false starts, and family reunions at the Mayo Clinic, the cause is identified as V337M.

This autosomal-dominant (inherited) mutation of the *MAPT* gene leads to overproduction of tau protein and degeneration of the frontal and temporal lobes. It affects speech, executive function, and social emotions and intuitions. By the book's end, several members of the next generation have tested positive for V337M, and the disease has been traced to Jean's grandfather and other branches of the family. None of the new generation of dementia drugs targets FTD.

This book is also a story about family caregiving, represented by Sue, who dedicates 30 years of her life to her sister Christy: relocating and changing jobs, establishing a household and routines built around Christy's substantial needs and also her flourishing, which includes eating at Wendy's and being around people. Sue also periodically cares for three other siblings with FTD; she is the one who knows how disability insurance works. Dementia caregiving, like all care work, is highly gendered; most caregivers are women within the family. Yet *The Vanishing Family* also shows how the husbands, exes, partners, sons, and nephews of the five siblings with FTD are involved in their care.

If there is one weakness in this outstanding book, which is paced like a thriller and will stay with the reader for a long time, it is the title. This family does not vanish. Its members show up for each other, as so many families must do, with so little help, in America's aging society. □

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ILLUSTRATION: JORM SANGSORN VIA ISTOCK PHOTO

HISTORY OF SCIENCE

Walden and the wild roots of conservation science

The famous work of transcendentalism holds lessons for modern ecologists, maintains a Thoreau scholar **Renée Bergland**

The Walden Experiments: The Science of Henry David Thoreau
Robert Thorson
Princeton University Press, 2026. 328 pp.

Henry David Thoreau resisted social conventions. In his lifetime, he was as famous for his refusal to conform as he was for his published works. After he died in 1862, Ralph Waldo Emerson listed everything his friend had refused to do: Thoreau had "no profession; he never married; he lived alone; he never went to church; he never voted; he refused to pay a tax to the State" (*1*). Emerson was disappointed that Thoreau had devoted his time to woodcraft and wordcraft instead of paying work. He complained that his friend preferred being "the captain of a huckleberry party" to any other, more serious occupation.

Today, Thoreau's steadfast dedication to wild places looks more like discipline than laziness. He was a remarkably prolific writer. As it turned out, his published speeches, essays, poems, and books were just the tip of the iceberg; his unpublished journal and detailed logs of observations of the natural world have proved as valuable as his publications.

Thoreau's environmental ethics were informed by Charles Darwin's incisive and inductive observations about the natural world.

Thoreau's work is a little hard to categorize. Philosophers claim him as a philosopher. Political activists see his writings as cornerstones of social and environmental justice movements. Writers admire the beauty of his language. Scientists—particularly ecologists—use his careful measurements of 19th-century natural phenomena to document 21st-century changes to our ecosystems.

The Walden Experiments: The Science of Henry David Thoreau is also hard to categorize. Author Robert Thorson is a geologist by training, a professor of earth sciences, and a widely respected Thoreau scholar who has worked collaboratively with theologians, historians, philosophers, and literary scholars over the years. *The Walden Experiments* is his fourth book on Thoreau. The first of them, *Walden's Shore: Henry David Thoreau and Nineteenth-Century Science*, is a multifaceted work of literary criticism, biography, and intellectual history that argues that Thoreau was a skilled practitioner of 19th-century inductive science. *The Walden Experiments* leaps from that early work toward a new set of questions more centered on the 21st century. Here, Thorson invites readers to consult *Walden* as a "mid-nineteenth-century oracle" that speaks to the current global sustainability crisis.

Thoreau could certainly be oracular. His proclamation "In wildness is the preservation of the world" continues to inspire readers while resisting easy interpretation. Thoreau defined the wild as "whatever has not come under the sway of man." Perhaps his refusal to be part of social institutions—not only state, church, family, and profession but also intellectual discipline—was an attempt to preserve as much of his own wildness as he could. It might also explain his lean toward the sciences. In the mid-19th century, there was no field of study quite as wild as the dynamic and shifting world of natural and physical science.

It is common knowledge that Thoreau's interest in wildness prompted his choice to build a small house near Walden Pond, just outside of Concord, Massachusetts; live in it for 2 years; and write a book about his experience. Thorson goes deeper into the story, explaining that *Walden* took 9 years to write.

Thoreau was not satisfied with the first draft. He put it aside for about 2 years, then returned to it when he discovered Charles Darwin's journal of his own observations of the natural world, *The Voyage of the Beagle*. Inspired by Darwin's detailed observation, his inductive approach, and his beautiful prose, Thoreau "transformed an otherwise inauspicious lake into a powerful and globally famous symbol for humanity's rightful place in nature and our responsibility to it."

Thorson argues that Thoreau did not impose his philosophy on the land. On the contrary, he developed his environmental ethics inductively, from the ground up, starting with geology and then gradually expanding his vision. According to Thorson, "Thoreau—the young man keenly interested in glaciology, climate change and optics—is what we would today call a scientist." He explains that "all but one of *Walden*'s eighteen chapters engage with outdoor field observations and interpretations involving physics, chemistry, astronomy, geology, hydrology, botany, zoology, ecology, meteorology, climatology, and limnology (the study of lakes)." Thorson's book is also divided into 18 chapters, each of which is a wide-ranging freestanding essay that jumps off from a *Walden* chapter to recontextualize Thoreau's scientific thought for 21st-century readers.

If *Walden* is a wild ride, *The Walden Experiments* is a little wilder. It invites readers to shake off their disciplinary shackles, jump out of their silos, and try to reconnect with the wild roots of the sciences, in hope that embracing wildness will give us the intellectual courage we need for the preservation of the world. □

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PHOTO: TIM LAMAN VIA MINDEN PICTURES

Gene-editing research is promising, but guardrails can help to prevent tragic results.

Epistemic justice in gene-editing research

A *Science* investigation into a fatal individualized gene-editing trial exposed failures of safety oversight, consent, and disclosure ("A fatal reaction," B. Borrell, News Feature, 23 July, 10.1126/science.aek8000). The case shows how epistemic injustice—the unfair exclusion of people from producing, interpreting, or validating knowledge (*1*)—increases the vulnerability of patients and compromises scientific progress. Additional patient protections and advocacy can address this imbalance in power.

The patient, referred to as Mei in the News story, and her parents were indispensable to knowledge production. They supplied genetic and clinical information, financed preclinical research, and documented conversations. After Mei's death, her parents sought to ensure that the research's advances and setbacks were shared with the community, as did the family of Jesse Gelsinger, who died as a result of gene therapy in 1999 (*2*). Yet their contribution, as in the Gelsinger case (*2*), was not fully recognized: Their concerns carried less authority, and they lacked independent means to judge whether the experiment constituted ethically defensible human research.

Patient engagement is increasingly recognized in clinical research, and China's Regulations on the Administration of Clinical Research and Clinical Translational Application of New Biomedical Technologies (State Council Order No. 818) (*3*) strengthens research-participant protections. However, patients are not technical experts. Families hold essential knowledge about conditions, values, and acceptable burdens, but they cannot be expected to adjudicate vector dose, toxicology, or evidentiary sufficiency.

Empowering patients and families requires access to independent scientific advice for families, meaningful patient representation in ethics and safety review, full consideration of adverse preclinical findings, and prompt reporting of serious outcomes. Patients must be recognized not merely as sources of data, funding, and hope, but as participants in knowledge governance.

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10.1126/science.aek9401

Recording results from single-patient trials

In the News Feature "A fatal reaction" (23 July, 10.1126/science.aek8000), B. Borrell reports that a 6-year-old child died 7 days after receiving an individualized base-editing treatment. Although the girl was the only participant that the trial intended to enroll, the registry continued to describe the study as "recruiting" more than a year later (*1*). During the same period, the related preclinical study cited nonhuman-primate delivery as evidence of translational feasibility (*2*) but did not mention the human trial. To ensure that the results of single-patient research are preserved, relevant protocols and major human outcomes should be recorded and kept current in a public registry.

In a conventional early-phase trial, an individual outcome is interpreted alongside those of other participants. In a single-patient study, however, the first outcome may be the only human evidence available. A registry that remains unchanged after the reported death of the sole participant is therefore more than an administrative oversight. Families, clinicians, ethics committees, and other

PHOTO: BEN STANSALL VIA GETTY IMAGES

research groups may rely on that entry when judging whether human testing should continue and what risks have emerged.

Safety reports, trial registries, and journal articles serve different purposes and are updated on different schedules. As a result, important clinical information can reach regulators and ethics committees without becoming part of the public account of translation.

To remedy the potential for reporting gaps, investigators, trial registries, and journals should work together to keep the translational record current for individualized genetic therapies. Authors submitting preclinical studies should disclose any directly related human protocol and update that information before acceptance. Registries should include a field for treatment-emergent deaths and life-threatening events. Such details could distinguish occurrence from suspected or confirmed causality, as World Health Organization registration standards already allow (*3*). Preclinical articles and trial records should also link to one another, with dated notices added when major human outcomes arise after publication.

Such notices would not assign blame or settle causality. Rather, they would serve to ensure that claims of clinical promise are not separated from the human experience that followed.

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10.1126/science.aek9583

Streamlining oversight in gene-editing trials

In the News Feature "A fatal reaction" (23 July, 10.1126/science.aek8000), B. Borrell reports that a 6-year-old girl died after the world's first-in-human brain base-editing trial in China. This case highlights a governance paradox: A regulatory system with detailed ethical rules failed to prevent a fatal outcome. Institutional fragmentation may be at fault. As countries expand access to highly individualized gene-editing therapies, they must also strengthen continuous, cross-institutional oversight.

In China, gene-therapy products developed through the investigational new drug (IND) pathway generally undergo national regulatory review, including assessment of nonclinical safety data and requirements for long-term follow-up (*1*). Gene- and cell-based interventions may also be conducted as investigator-initiated trials (IITs) under institutional oversight, creating a dual-track regulatory structure. The reported intervention followed the IIT route and therefore did not receive national-level product review. The limitations of institution-centered oversight became particularly consequential in this case: The ethics committee approved the study before the final primate toxicology report was available, and the subsequent findings of liver injury in monkeys did not lead to a documented rereview before the child was treated. Although closer adherence to IIT requirements might have prevented some of these failures, the case raises a broader question about whether institutional review alone provides sufficient independent oversight for high-risk interventions.

Other countries also have multiple approval paths. The European Union's regulatory framework for Advanced Therapy Medicinal Products (ATMPs) includes a controversial hospital exemption that permits certain custom-made ATMPs to be used without centralized marketing authorization (*2*). US regulators have proposed a "plausible mechanism" framework for individualized gene-editing therapies that would remain within the federal regulatory system (*3*).

The News story described compartmentalization beyond ethics approval. China's Civil Code prohibits charging research participants fees related to a clinical trial (*4*), yet the family paid for the study by donating to a company controlled by the lead investigator. The university limited its review to whether the published article was connected to the funded experiment, the journal regarded the child's death as a matter for institutional investigation, and regulators confined enforcement to narrowly defined violations.

When responsibility becomes fragmented across multiple actors, accountability becomes difficult to assign (*5*). China has made changes to address the governance weaknesses revealed by this case, such as establishing a statutory framework for high-risk biomedical technologies (*6*). However, strengthening regulatory pathways alone is insufficient. Ethical review should be a continuous process rather than a single authorization decision. Financial oversight should encompass affiliated companies and other entities under investigators' control. Safety governance should extend across institutional boundaries. Major adverse events and emerging safety concerns should be communicated through a traceable reporting system linking investigators, ethics committees, regulators, and journals with one designated body responsible for determining whether new evidence warrants modification, suspension, or termination of an ongoing study. In China and beyond, robust oversight is what sustains scientific credibility, protects research participants, and safeguards responsible investigators from avoidable ethical and legal risk.

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10.1126/science.ael0421

The utilitarian temptation in gene editing

The recent gene-editing trial led by Zilong Qiu and colleagues in Shanghai ("A fatal reaction," B. Borrell, News Feature, 23 July, 10.1126/science.aek8000) was not only a case of failed oversight; it also illustrates a recurring moral danger in biomedical research. Such trials can be driven by two mutually reinforcing motivations: the pursuit of professional recognition and a crude utilitarian rationale that risky experiments are justified because, if successful, they may benefit large numbers of present and future patients. This second motivation, which is especially powerful in biomedicine, can cloak self-interest in moral language and encourage

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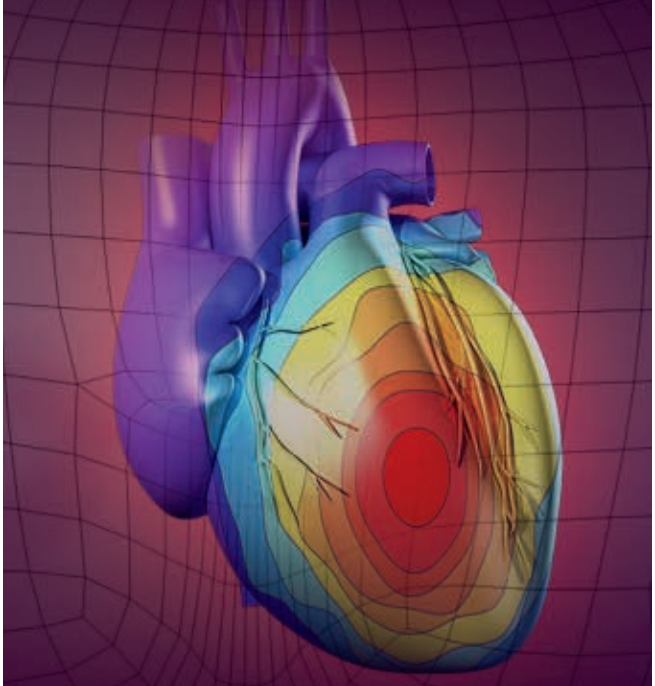

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researchers to discount evidence of danger to the individuals enrolled as subjects. Gene-editing interventions, in particular, may appear morally elevated because their promised benefits aim to enhance general welfare for all, including future generations, regardless of nationality or group identity. For this reason, the appeal to future benefit requires strict ethical limits.

He Jiankui, whose work producing genetically altered babies led to jail time (*1*), invoked such an ideal when he defended his embryo-editing experiment, saying that he was "proud" of work intended to protect children from HIV (*2*). Qiu, who later joined more than 100 Chinese scientists in condemning He's experiment for its ethical failures and its "far-reaching and profound social consequences" (*3*), may have allowed hoped-for therapeutic benefit to overshadow foreseeable risks to the human subject. Such tragedies may not be products of ambition or greed—they may be driven instead by the moral force of utilitarian rhetoric.

The crucial distinction lies between minimal moral requirements and the pursuit of maximal general welfare. The former must be satisfied before the latter can be invoked. Foreseeable harm to human subjects must be identified, assessed, and minimized before an ethics committee approves a trial. In Qiu's case, the committee reportedly approved the human trial without reviewing the final primate-safety report. That failure was not a procedural detail; it was a bioethical error. It allowed speculative future benefit to override the prior duty to protect the subject.

Bioethics education should therefore emphasize that "do no harm" is a minimal and universal moral requirement, whereas "enhancing general welfare" is culturally and contextually specified (*4*, *5*). For example, biomedical trainees, clinicians, and administrators should all be taught that a hospital ethics committee must require and review relevant primate safety studies before approving a human-subject trial. No promise of future benefit, however broad or humane it sounds, can justify overriding the minimal duty not to expose a human subject to foreseeable and preventable harm. Otherwise, the utilitarian temptation cannot be resisted.

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EDITOR'S NOTE on the Research Article "China's solar expansion policy reduces bird diversity"

On 20 August 2026, *Science* ran the Research Article "China's solar expansion policy reduces bird diversity" (*1*). The editors have received multiple eLetters expressing technical concerns about the paper. We are currently in the process of evaluating the validity of the concerns and determining whether any further action is needed on the paper.

H. Holden Thorp, Editor-in-Chief, *Science*

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10.1126/science.aem4102

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SPECIAL ISSUE

A LIFELONG FOCUS ON WOMEN'S HEALTH

YEVGENIYA NUSINOVICH, SACHA VIGNIERI, SARAH ROSS, and JOANA OSÓRIO

Historically, most human studies have involved male participants, and their findings were assumed to apply to women and children as well. Meanwhile, women's health typically came up in the context of reproduction and childbearing, if at all. However, biomedical science has come a long way over the years, and it has become increasingly clear that there are sex-dependent differences in many diseases that do not involve reproduction and that occur at all ages. Across all areas of women's health, including both reproductive and nonreproduc-

ILLUSTRATION: CHIARA VERCESI

tive conditions, much remains to be learned about the underlying biology, effects of hormones and genetics, and potential therapeutic interventions.

Each person has a specific set of sex chromosomes, which not only play a role in determining their sex at birth (which some genetic mutations and hormone signaling abnormalities can also affect) but also influence disease risk throughout their life. For example, women's cells usually have one active and one inactive X chromosome, but there are cases when an X chromosome escapes from inactivation or some cells lose one of their X chromosomes. These changes influence the risk of age-related conditions such as cardiovascular disease, cancer, and autoimmunity. Pregnancy and lactation stimulate extensive remodeling of physiology, with dynamic changes in the immune system helping to support restructuring of tissues; yet we are just beginning to consider how this immune adaptation may influence health after childbearing. Even the ovaries, which are typically studied in the context of reproduction, have biological functions that last well beyond the reproductive years. Incomplete knowledge that led to myths about potential hormonal interventions after the reproductive years, during and after menopause, has been supplanted by more recent research, and we now understand much more about how to improve women's health and well-being during life transitions. Pain biology is a vital topic that remains poorly understood: Empirical data show clear differences between the sexes in pain perception and rates of specific painful conditions, but mechanistic understanding is often lacking.

Overall, it is becoming increasingly clear that women's health and sex-dependent biological differences go well beyond reproduction and affect people throughout their lifespan and across different medical conditions. There is still much to learn in this area, and continuing research efforts have the potential to improve the well-being of all patients. 10.1126/science.aem4087

PERSPECTIVES

Does the postmenopausal ovary still matter?

Ovaries may influence health long after their reproductive function ends

Michael B. Stout[1,2], and **Francesca E. Duncan**[3,4]

Women can spend decades in the postmenopausal phase of life. However, the organ that defines the transition to this stage, the ovary, has been largely ignored after its role in reproduction ceases. A common assumption is that the postmenopausal ovary, which no longer contains sufficient follicles to sustain fertility, is a dormant remnant. The postreproductive ovary has also been referred to as a senile organ, a gyratum, or a wrinkled and shriveled husk. However, investigators and clinicians have suspected for years that the human ovary retains biological activity after the reproductive period and have pointed out that "the contribution of this organ to the overall well-being of the postmenopausal woman remains to be defined" (*1*). Historical and emerging data support the idea that the postreproductive ovary is active, which has clinical implications.

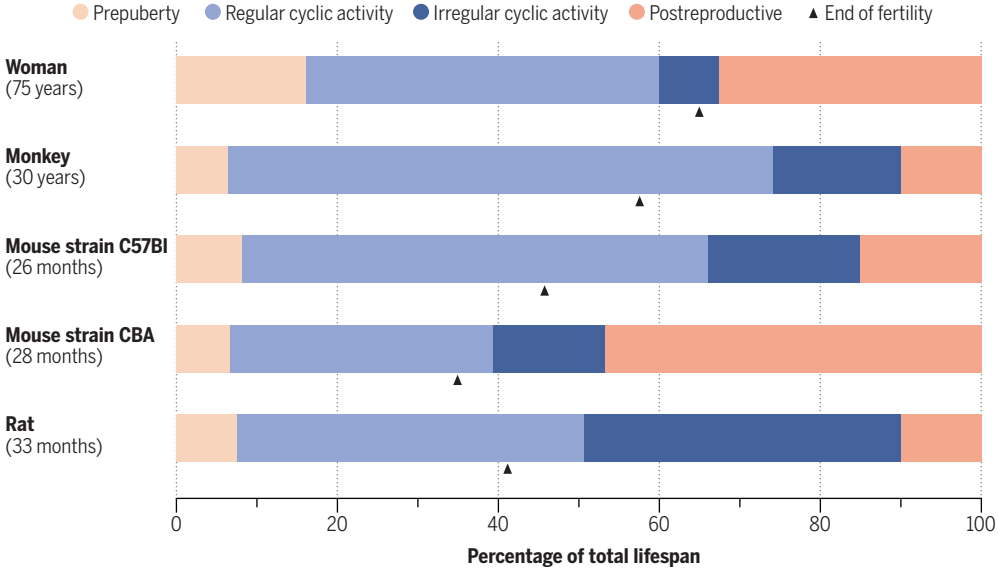
In the reproductive years (generally 12 to 50 years of age), ovaries produce eggs and hormones that are essential for fertility and overall health. Female mammals are born with a finite number of eggs, which are housed within primordial follicles. As follicles grow and develop, they produce peptide hormones, including inhibins A and B, activins, follistatin, and anti-Müllerian hormone, as well as steroid hormones, including estrogen, progesterone, and androgens (including testosterone). The reserve of follicles decreases with time and dictates reproductive lifespan. Female individuals have millions of primordial follicles at birth, hundreds of thousands at puberty, thousands at early menopause, and hundreds to none after menopause. Menopause, which is defined as the permanent cessation of menstruation for 12 months, is a direct consequence of the loss of follicles.

Understanding the functions of the postmenopausal ovary is essential because women's lifespan is increasing globally, whereas the age of menopause has stayed relatively constant. An extended postreproductive period is a conserved feature across mammalian species (see the figure), and today, many women can expect to spend nearly one-third of their life in the postmenopausal state. Analysis of ovarian tissue has shown that the postmenopausal ovary is distinct morphologically and structurally from its premenopausal counterpart (*2*). Because of the loss of follicles, the postmenopausal ovary is ~30% smaller. Furthermore, its stroma, the scaffold of connective tissue that surrounds and supports it, proliferates and becomes the dominant feature of the organ. Stromal fibrosis, scar-like tissue, and remnants of ovulation known as corpora albicantia are also prominent. Granulomata composed of macrophages and lymphocytes may be present, and sclerotic (hard and thick) blood vessels are abundant. The tunica albuginea, a protective layer of connective tissue that surrounds the ovary, thickens, whereas the surface epithelium forms papillary and cord-like growths and rolls inward to form inclusion cysts.

Nevertheless, the postmenopausal ovary can still produce hormones. The stroma, in particular, produces most of the steroid hormones secreted by the ovary at this stage. Measurements of hormone gradients from the ovarian vein to peripheral blood showed that the postmenopausal ovary primarily produces and secretes testosterone, as well as smaller amounts of androstenedione, estradiol, estrone, and progesterone (*3*). Notably, stromal cells from the postreproductive ovary also express receptors for luteinizing hormone and follicle stimulating hormone and can respond to gonadotropin stimulation (*3*). The observation that the postmenopausal ovary maintains some hormone production is a reminder that it is not dormant. Whether the aged ovary continuously secretes factors—hormones or other molecules—that meaningfully influence whole-body physiology is unclear. This knowledge gap is partly due to a lack of appropriate technologies to assess ovarian functions beyond endocrine activity. Moreover, obtaining postmenopausal ovarian tissue from healthy women for research purposes is challenging. Such studies usually only include tissue from a small number of women and are focused mostly on the menopausal transition itself rather than on the functional changes that occur in the following decades. However, molecular studies in mice, monkeys, and humans suggest that the postreproductive ovary continues to undergo dynamic changes, especially related

Female reproductive activity across the mammalian lifespan

Although most female mammals do not menstruate, unlike humans, they still spend a substantial portion of their lifespan in a postreproductive phase. This phase is preceded by a period of progressively irregular reproductive cycles, which eventually cease. The parenthetical ages on the left are average lifespans from population-based studies (human) and breeding colony data (animals).



GRAPHIC: ADAPTED FROM E. C. JONES, 1975. "THE POST-REPRODUCTIVE PHASE IN MAMMALS." ESTROGENS IN THE POST-MENOPAUSE: 2ND INTERNATIONAL WORKSHOP ON ESTROGEN THERAPY, GENEVA, OCTOBER 1974, P.A. VAN KEEP, C. LAURITZEN, S. KARGER PUBLISHERS LIMITED BY C. BICKEL/*SCIENCE*

to inflammation and cellular senescence (*4*–*6*).

Ovarian aging is characterized by the progressive accumulation of immune cells and an increase in the concentration of inflammatory molecules. In postreproductive mouse ovaries, T cells, macrophages, and multinucleated giant cells, which are associated with chronic inflammation, continue to expand with age (*4*). Furthermore, transcriptomic profiling revealed a near-complete loss of gene expression associated with reproductive cellular identity and an up-regulation of gene expression programs associated with antigen presentation, leukocyte activation, and immunoglobulin production (*4*). These findings demonstrate that the postreproductive mouse ovary develops a sustained proinflammatory phenotype in the absence of acute infection (*5*). Moreover, in monkeys approaching the end of their reproductive lifespan, spatial transcriptomic profiling has identified discrete proinflammatory foci within the ovarian stroma whose gene expression profile suggests the release of senescence-associated signals (*6*). Thus, aged ovaries may amplify inflammation locally and potentially systemically.

With age, the ovary also shows signs of cellular senescence. Findings from a proteomic analysis of healthy ovaries from women aged 50 to 75 years suggest that gene expression programs change markedly across the postmenopausal decades of life (*7*). In this study, younger postmenopausal ovaries expressed more proteins involved in RNA processing and gene regulation. By contrast, older ovaries predominantly expressed proteins involved in complement activation, cellular senescence, tissue remodeling, and extracellular matrix organization. Notably, exposure of postmenopausal human ovarian tissue in culture to doxorubicin, a chemotherapeutic agent that induces cellular senescence, resulted in a secretory phenotype that substantially overlaps with that of healthy (untreated) aged ovaries (*8*). This similarity suggests that two different stressors, chemotherapy and chronological aging, elicit similar molecular programs in the human ovary.

Not all human data, however, indicate the presence of inflammation and senescence in the postmenopausal ovary. For example, an atlas of individual cells and their accessible DNA in the postmenopausal human ovary revealed a stroma-dominated, transcriptionally active organ expressing multiple hormone receptors, but with relatively few immune cells and senescence-associated molecules (*9*). The reasons underlying the divergences across studies of the postmenopausal human ovary may reflect differences between the gene and protein expression, the source of ovarian tissue, or the methodology used. However, most observations across species are consistent with an enrichment in senescent stromal cells and immune cells in the postreproductive ovary.

A subset of molecules expressed in the postreproductive ovary are thought to be secreted factors that could reach the bloodstream and have systemic effects (*4, 7*). Therefore, similar to adipose tissue, which is now recognized as an active endocrine organ that affects whole-body physiology, the postmenopausal ovary could influence an individual's disease burden by releasing factors that contribute to inflammation and aging long after its reproductive function ceases.

The concept that the postmenopausal ovary contributes to whole-body inflammation and aging is only inferential because it is built on observations that this organ has a secretory phenotype and that other aging tissues drive systemic disease through similar mechanisms. Therefore, direct measurements of the concentrations of ovary-derived factors in the blood of postmenopausal women and investigations into the effects of ovarian secretory outputs on distal organs are priorities for the field. Notably, senescent cells and the molecules they produce, which seem to characterize the postmenopausal ovary, are major drivers of "inflammaging," the chronic, low-grade, sterile inflammation that emerges with age and is a key driver of cardiovascular disease, neurodegeneration, loss of muscle mass and function, and metabolic dysfunction (*10*). Therefore, the postmenopausal ovary could be a previously unrecognized contributor to whole-body inflammaging in older women.

If the postmenopausal ovary contributes to systemic aging through the factors it secretes, its surgical removal (oophorectomy) after menopause could potentially improve health outcomes in women. However, very little data exist about the impact of oophorectomy after natural menopause, especially in the absence of ovarian disease. Observational studies such as the Mayo Clinic Cohort Study of Oophorectomy and Aging (*11*) and the Nurses' Health Study (*12*) examined women undergoing oophorectomy before menopause in whom the abrupt loss of estrogen and androgen production had clear adverse health effects, including increased cardiovascular mortality, cognitive decline, and all-cause mortality. The strongest harms were observed in the youngest women, which indicates that they stem from premature loss of ovarian endocrine function rather than from removal of a postreproductive ovary. A 2005 analysis of the possible consequences of postmenopausal oophorectomy used a Markov decision model based on aggregated age-specific risk estimates from the published literature (*13*). The outcomes were projected through age 80, and the analysis concluded that preserving the ovaries maximized survival until ~65 years of age. However, the survival benefit of postmenopausal ovarian conservation is now less clear than previously assumed. An updated Markov analysis incorporating hazard ratios from the Mayo Clinic and Nurses' Health Study cohorts concluded that for women undergoing hysterectomy at 50 years old or older, concurrent removal of the fallopian tubes and ovaries (salpingo-oophorectomy) did not increase mortality compared with conservation of these organs through age 80 years (*14*). More rigorous studies are needed to determine whether there are specific time windows in which postmenopausal salpingo-oophorectomy may confer a net health benefit, although caution is required, as ovarian androgens also influence bone density, libido, and metabolic function (*15*).

The emerging molecular signatures of the postmenopausal ovary show that this organ is far from dormant. Research is needed to fully determine the nature of these ovarian molecules, how they send signals to the rest of the body, and whether they contribute to or protect against systemic aging. This knowledge will enable the development of biomarkers to define healthy and disease states and could uncover therapeutic targets to improve women's health during the postreproductive period, which will continue to widen as humans live longer. □

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REVIEW

Shaping maternal physiology: Immune control of reproductive adaptation

Dan Corral[1]* and Yasmine Belkaid[2]*

Pregnancy is one of the most extensive physiological transitions of adult life, requiring coordinated adaptation across virtually every maternal organ to support fetal growth while preserving maternal health. Although these changes have traditionally been viewed as endocrine driven, emerging evidence identifies the immune system as an active regulator of maternal physiology. The immune system is increasingly recognized as a broad regulator of host physiology, positioning it as a key driver of maternal adaptation. In this work, we discuss how pregnancy reprograms maternal immunity to orchestrate physiological reprograming of tissues. This framework shifts the view of maternal immunity from one centered on immune homeostasis and fetal tolerance to one in which immune cells actively instruct physiological short- and long-term adaptation across maternal organs to shape health beyond reproduction.

Pregnancy requires one of the most extensive episodes of physiological remodeling in adult life. Coordinated adaptations across multiple maternal organs support fetal development, prepare for lactation, and preserve maternal health, yet the mechanisms orchestrating these changes remain incompletely understood. Despite its importance, pregnancy has historically remained underexplored. Deciphering the biology of this distinctive physiological state is one of the most important challenges in biomedical research and promises fundamental insights with broad implications for human health and disease.

Within the first few months of pregnancy, virtually every maternal organ adapts its structure and function to support fetal growth, protect the developing fetus and prepare for lactation (*1*). Cardiovascular output increases, hematopoiesis is redirected, nutrient absorption and metabolism are reprioritized, and barrier tissues undergo profound remodeling. Together, these coordinated adaptations redistribute maternal resources to meet the changing demands of reproduction, a process captured by the concept of homeorhesis. Homeorhesis describes the coordinated reprogramming of multiple physiological systems to establish a new physiological state appropriate for reproduction rather than maintaining a stable internal set point, as in homeostasis (*2*) (Fig. 1). Although many of these adaptations resolve after parturition, others persist well beyond delivery, indicating that pregnancy leaves lasting physiological imprints on maternal health (*3*). At the same time, the quality of maternal adaptation shapes fetal development and lifelong offspring health, making pregnancy a critical window for the future health of both mother and child. Despite these far-reaching consequences, the physiological mechanisms that actively remodel maternal tissues have received far less attention than fetal development and maternal-fetal tolerance.

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Maternal adaptation has traditionally been viewed through an endocrine lens, whereby maternal and placental hormones coordinate tissue growth, metabolism, and vascular adaptation to sustain fetal development while preserving maternal physiology (*1*). Although these discoveries transformed our understanding of pregnancy, they also reinforced the view that gestation is primarily characterized by controlled immune suppression (*4*). This framework, however, could not readily explain how mothers simultaneously maintain protective immunity while supporting the extensive tissue remodeling required for reproduction. Longitudinal immune profiling has demonstrated that pregnancy is not a state of generalized immunosuppression but a dynamic immunological process, with distinct immune programs accompanying successive stages of gestation (*5*, *6*).

The recognition that maternal immunity is dynamically remodeled throughout pregnancy also raises a broader possibility: that the immune system may actively contribute to maternal physiological adaptation. Indeed, immune cells are now recognized as fundamental regulators of host physiology, controlling processes that extend well beyond host defense and immune tolerance, including metabolism, vascular remodelling, stem and progenitor cell differentiation, epithelial development, and tissue repair (*7*). These physiological functions have been described across development, tissue homeostasis, and regeneration, establishing the immune system as an integral component of all organs function throughout life (*7*). Pregnancy provides a specific physiological context in which diverse and complex physiological programs are synchronously deployed across multiple adult organs. Whether pregnancy primarily repurposes existing immune circuits or establishes specialized programs remains an important unresolved question.

In humans, most knowledge of maternal immunity derives from studies of peripheral blood mononuclear cells. Yet the physiological adaptations that sustain pregnancy occur within tissues, where immune cells establish intimate interactions with epithelial, stromal, and endothelial cells. Although peripheral blood analyses brought new insights (*5*, *8*), circulating leukocytes provide only a partial view of pregnancy-induced immune remodeling because immune cell phenotype is shaped by local tissue-specific signals. The decidua exemplifies this principle, containing specialized immune populations whose composition and function are instructed by local microenvironment (*9*). Consequently, most mechanistic insights into immune-mediated maternal adaptation come from animal models, which enable direct investigation of tissue-resident immune populations. However, translating these findings to humans will require careful consideration of species-specific differences in reproductive anatomy and physiology.

On the basis of recent evidence, we propose that maternal immunity is an integral component of the homeorhetic program of pregnancy, coordinating the tissue remodeling and physiological adaptations that establish and maintain the maternal state required for successful reproduction. We examine how pregnancy reprograms the immune system before discussing how specialized immune populations coordinate reproductive, epithelial, and metabolic adaptations. Lastly, we explore how these programs generate long-lasting immune and tissue memory that can enhance maternal fitness or, when disrupted, contribute to disease. Viewed through this perspective, pregnancy emerges not simply as a state that the immune system accommodates but as a physiological program that it actively helps construct.

Pregnancy reprograms hematopoiesis and lymphopoiesis for physiological adaptation

During pregnancy, the maternal immune system is precisely reprogrammed to preserve host defense, establish fetal tolerance and support tissue remodeling across multiple organs. This transition begins within the primary lymphoid organs, where endocrine and immune signaling pathways redirect hematopoiesis and lymphopoiesis,

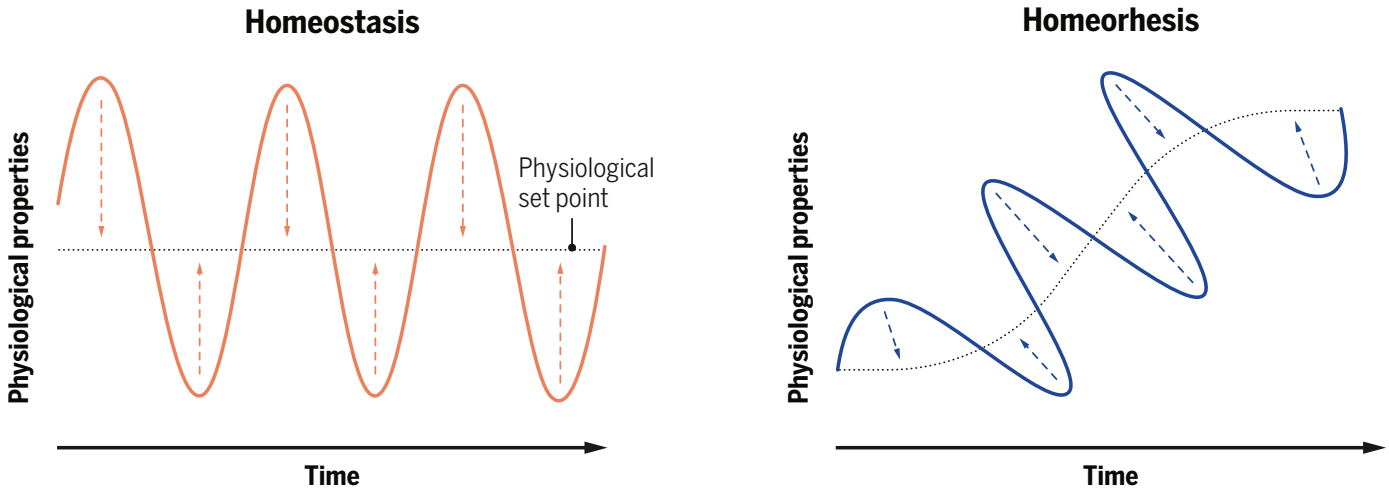

Fig. 1. Homeostasis and homeorhesis: Distinct modes of physiological regulation. (Left) Homeostasis maintains physiological variables around a relatively stable set point through compensatory responses that restore equilibrium following transient perturbations, such as thermogenesis or glycemia regulation. (Right) By contrast, homeorhesis establishes a new dynamic equilibrium by coordinately reprioritizing physiological processes across multiple organs to support a changing biological state, such as pregnancy and lactation. Rather than preserving preexisting physiological set points, homeorhesis redirects metabolism and organ function to meet the evolving demands of reproduction while maintaining overall organismal fitness. Pregnancy therefore represents a paradigmatic example of homeorhetic regulation in which coordinated adaptations across endocrine, metabolic, cardiovascular, and immune systems support fetal development, maternal health, and preparation for lactation. The dashed arrows within the curves represent regulatory mechanisms that direct physiological variables towards the set point; in homeostasis, these mechanisms act towards a relatively stable set point, whereas in homeorhesis, they act towards a set point that itself changes over time.

enhancing erythroid and myeloid output while reshaping lymphocyte development before immune cells are deployed to peripheral tissues (Fig. 2).

One manifestation of maternal immune reprogramming occurs in the bone marrow, where pregnancy reshapes hematopoietic output to support the evolving physiological demands of the growing fetoplacental unit (*10*). Classical histological studies in humans demonstrate progressive normoblastic erythropoietic hyperplasia accompanied by granulopoietic expansion and reduced marrow lymphocytes that persist into the postpartum period (*10*). These changes are accompanied by profound rewiring of hematopoietic stem cell (HSC) biology. Experimental studies in mice indicate that endocrine cues contribute to these adaptations. Estradiol promotes HSC self-renewal through estrogen receptor α–dependent signaling, whereas the cholesterol metabolite 27-hydroxycholesterol (27-HC) cooperates with estrogen to drive HSC mobilization, pregnancy-associated extramedullary hematopoiesis, and expansion of erythroid and granulocyte-macrophage progenitor populations (*11*, *12*). In parallel, pregnancy-induced derepression of endogenous retrotransposons activates innate immune cyclic GMP-AMP synthase (cGAS)–stimulator of interferon genes (STING) signaling and interferon (IFN) production within HSCs, promoting their proliferation and erythroid differentiation (*13*). These findings are supported by observations that reverse transcriptase inhibitor–based antiretroviral therapy during pregnancy suppresses endogenous retrotransposon signaling, reduces pathway activation, and is associated with an increased risk of gestational anemia, suggesting a role for endogenous retrotransposons in pregnancy-associated erythropoiesis (*13*). Although the mechanisms underlying maternal hematopoietic adaptation remain incompletely understood, these observations reveal that pregnancy remodels hematopoiesis through coordinated endocrine and immune signaling, resetting erythroid, and immune cell production to support gestation.

The thymus undergoes an equally notable but functionally distinct transformation. Pregnancy induces thymic involution, a phenomenon once interpreted as evidence of diminished immune competence but now recognized as a process of selective immune remodeling. Across mammalian species, progesterone signaling within cortical thymic epithelial cells (TECs) reduces thymic cellularity by >95% (*14*–*16*). In mice, T cell development is transiently arrested at the earliest precursor stages, resulting in marked depletion of double-positive thymocytes, whereas mature single-positive populations are relatively preserved (*17*). Progesterone simultaneously suppresses chemokine expression required for thymic homing of lymphoid progenitors, thereby limiting de novo T cell production during gestation (*18*). Despite this great reduction in size, ultrastructural studies demonstrate continued thymocyte selection, epithelial remodeling, and phagocytic clearance of self-reactive cells throughout pregnancy (*19*). How pregnancy-induced remodeling translates into lineage and functional alterations remains a largely open question.

Pregnancy redirects thymic differentiation toward specialized immune populations that fulfil reproductive functions. Progesterone-dependent activation of receptor activator of nuclear factor κB (RANK) signaling within autoimmune regulator (AIRE)–expressing medullary TECs (mTECs) selectively promotes the generation of regulatory T cells that subsequently populate the maternal-fetal interface and the visceral adipose tissue, where they contribute to fetal tolerance and metabolic adaptation. Disruption of this pathway results in fetal loss and gestational diabetes in mice (*20*). More recently, pregnancy-induced thymic involution has also been associated with enrichment of nonclassical T cell precursors. Among these, intraepithelial lymphocyte precursors (IELps) are redirected toward tissue-specific physiological functions outside the thymus (*21*).

Thus, during pregnancy, coordinated changes in hematopoiesis, thymic differentiation, and specialized immune cell production establish an immune landscape adapted to gestation. How endocrine cues are integrated with immune-intrinsic signaling pathways to coordinate such reprogramming remains poorly understood. Likewise, how pregnancy remodels primary lymphoid niches to selectively control immune cell production and lineage commitment is only beginning to emerge.

Immune regulation of the maternal-fetal interface.

The maternal-fetal interface provides the best-characterized example of immune cells regulating maternal tissue morphogenesis (Fig. 3).

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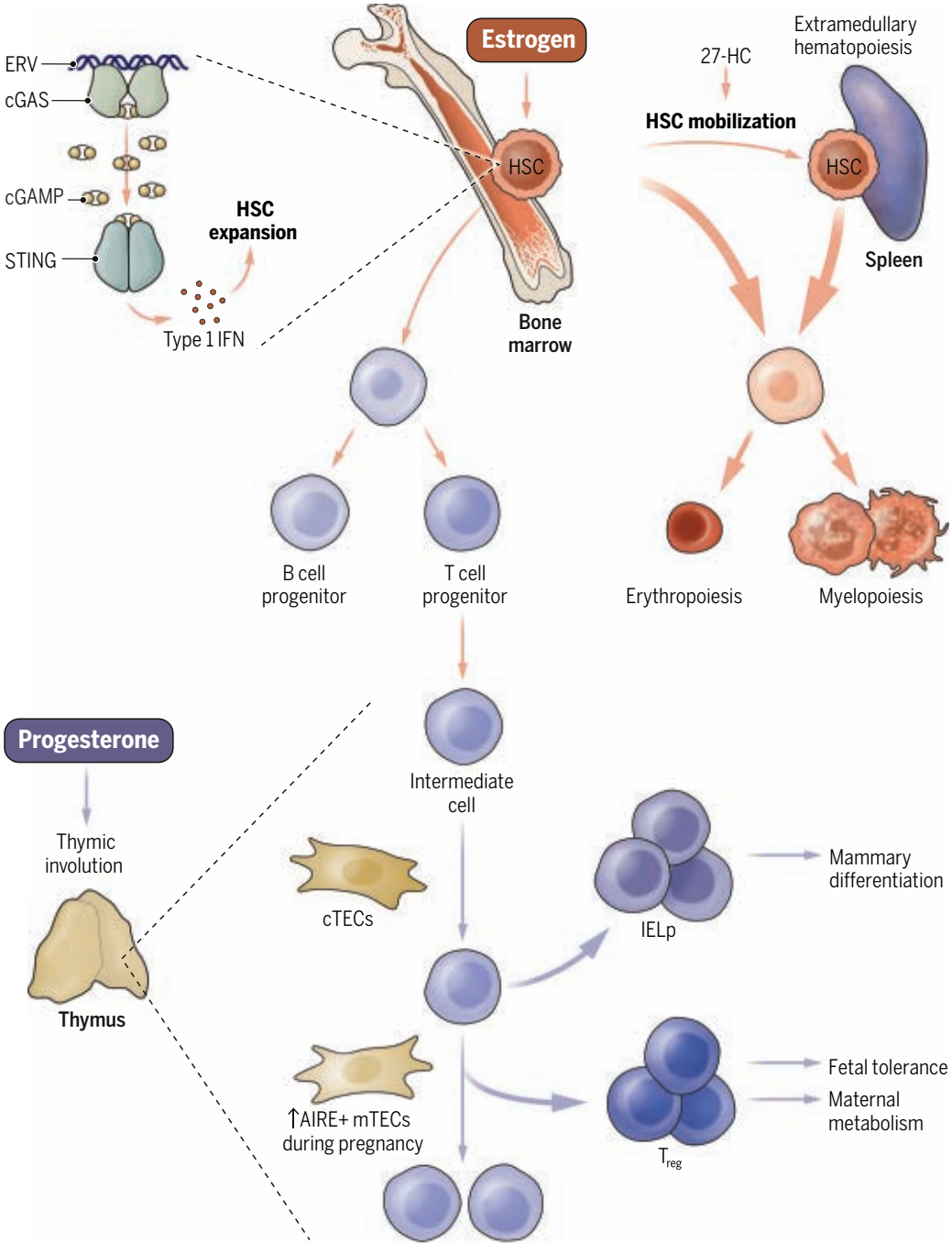

Fig. 2. Pregnancy reprograms immune cell development to support physiological adaptation. Pregnancy induces coordinated changes in the bone marrow, spleen, and thymus that reshape immune cell development to meet the physiological demands of gestation. In the bone marrow, estrogen promotes HSC proliferation, whereas pregnancy-associated derepression of endogenous retroviruses (ERVs) activates cGAS-STING signaling and type I interferon responses to further stimulate HSC expansion. Pregnancy also promotes HSC mobilisation and extramedullary haematopoiesis in the spleen, increasing erythroid and myeloid output to support the heightened oxygen transport and innate immune requirements of gestation. Progesterone directly signals through progesterone receptor-expressing cortical TECs (cTECs), driving pregnancy-associated thymic involution. In parallel, progesterone promotes the RANK-dependent expansion of AIRE-expressing mTECs, enhancing thymic T_{reg} generation. T_{reg} cells subsequently contribute to fetal tolerance and maternal metabolic adaptation. Despite thymic involution, IELps are enriched during pregnancy and are preferentially home to the mammary gland, where they support epithelial differentiation and lactogenesis. Together, these coordinated changes illustrate how pregnancy reprograms immune cell development to orchestrate maternal physiological adaptation rather than solely host defense.

Establishment of efficient placental circulation requires extensive remodeling of uterine spiral arteries into large, low-resistance vessels capable of sustaining fetal growth. This process requires coordinated interactions between invading extravillous trophoblasts, stromal cells, and specialized immune populations. Spatial transcriptomic analyses reveal a highly organized decidual immune landscape in which uterine natural killer (NK) cells, macrophages, dendritic cells, and T cells occupy distinct anatomical niches that evolve throughout the first trimester (*22*).

Among these populations, decidual NK (dNK) cells represent the dominant leukocyte population during early pregnancy, accounting for up to 70% of decidual immune cells. Unlike circulating NK cells, dNK cells are specialized for tissue remodeling rather than cytotoxicity accumulating around spiral arteries prior to trophoblast invasion and initiating vascular remodeling through the secretion of angiogenic mediators and IFN-γ (*23*, *24*). In this context, IFN-γ functions not as a proinflammatory cytokine but as an essential developmental signal (*24*). Beyond vascular remodeling, dNK cells also produce pleiotrophin, osteoglycin, and osteopontin, directly promoting fetal development independently of their vascular effects (*25*). In humans, genetic interactions between maternal killer cell immunoglobulin-like receptors (KIRs) expressed by uterine NK cells and fetal HLA-C alleles influence trophoblast invasion and are associated with susceptibility to preeclampsia and fetal growth restriction, supporting a conserved role for uterine NK cells in regulating placentation (*26*, *27*).

Other innate immune populations perform complementary functions within this developmental program. Mast cells promote implantation by supporting tissue remodeling and spiral artery remodeling at the maternal-fetal interface, potentially through the production of tissue remodeling mediators, including galectin-1, transforming growth factor–β, and mast cell proteases (*28*). Early depletion studies also implicated uterine $CD11c^{+}$ cells but not conventional dendritic cells

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in decidualization and angiogenesis (*29*, *30*). In parallel, ovarian macrophages sustain corpus luteum vascularisation and progesterone production, thereby ensuring uterine receptivity for implantation (*31*). As gestation progresses, decidual macrophages adopt a tissue remodeling phenotype that promotes trophoblast invasion, extracellular matrix remodeling and vascular adaptation while simultaneously contributing to maternal-fetal tolerance (*32*). Adaptive immune cells further refine these processes. In both humans and mice, neutrophils contribute to placental vascular development by inducing a proangiogenic T cell population that secretes vascular endothelial growth factor (VEGF) and promotes endothelial sprouting, highlighting how innate and adaptive immune cells cooperate to achieve a common physiological outcome (*33*). Within the adaptive immune compartment, regulatory T cells (T_{reg} cells), although comprising only a small fraction of decidual leukocytes, are indispensable for establishing the immunological environment that supports implantation and early placental development (*4*). In mice, transient T_{reg} cell depletion during the peri-implantation period impairs spiral artery remodeling, alters uterine NK cell function, and results in fetal loss and fetal growth restriction (*34*). However, whether implantation failure reflects a specialized developmental function of T_{reg} cells or the loss of their broader anti-inflammatory activity remains unclear.

The same immune circuits required to establish pregnancy are subsequently repurposed to terminate it. As gestation reaches term, the decidual immune landscape shifts from a tissue remodeling program to a controlled inflammatory state that initiates labor. Neutrophils and macrophages accumulate in the myometrium during labor, accompanied by increased production of proinflammatory cytokines including interleukin-1β (IL-1β), IL-8, and tumor necrosis factor α (*35*). In mice, IL-33 released by uterine fibroblasts activates type 2 innate lymphoid cells (ILC2s), whereas IL-5–dependent eosinophil responses also contribute to efficient labor (*36*).

Thus, the immune system can be repurposed to support distinct physiological programs across gestation. How tissue microenvironmental cues are integrated to confer these stage-specific immune functions remains poorly understood.

Programming mammary epithelial differentiation

The mammary gland provides another example in which immune cells regulate tissue development rather than simply protecting mature organs (Fig. 3). During pregnancy, prolactin and progesterone initiate lobuloalveolar expansion and prepare the gland for milk production (*37*). Increasing evidence indicates that immune cells also contribute to the establishment of local developmental programs that promote epithelial differentiation.

Macrophages, eosinophils, and mast cells are well-established regulators of mammary morphogenesis during puberty and tissue remodeling during involution (*38*). However, little is known about their physiological contribution during pregnancy, although current evidence supports a temporally ordered sequence of immune programs that coordinate pregnancy-associated epithelial differentiation. During early and midpregnancy in mice, activation of signal transducer and activator of transcription 6 (STAT6) signaling within mammary epithelial cells coincides with local production of IL-4 and IL-13, promoting alveolar differentiation. Although genetic deletion of both cytokines delays epithelial development, differentiation recovers by late gestation, indicating that type 2 immune signals accelerate but are not solely responsible for this program (*39*). As pregnancy progresses, the immune landscape shifts toward type 1 immunity, characterized by accumulation of T-bet–expressing lymphocytes (*21*, *40*). Mice lacking T-bet exhibit impaired mammary epithelial differentiation, defective lactogenesis, and reduced neonatal growth (*21*). A specialized population of CD8αα^{+} intraepithelial lymphocytes directly promotes differentiation of both secretory luminal and myoepithelial cells and is enriched in trophic factors capable of acting on epithelial progenitors, suggesting that immune-derived signals directly regulate mammary morphogenesis (*21*).

Evidence for analogous immune-epithelial interactions in humans is beginning to emerge. Independent single-cell analyses of human breast milk identify diverse epithelial and immune populations, including macrophages, T cells, NK cells, and tissue-resident IEL-like lymphocytes that resemble those described in mice (*21*, *41*, *42*). Moreover, alterations in immune composition correlate with lactation performance, with females exhibiting low milk production displaying altered macrophage inflammatory signatures together with altered epithelial cell states (*43*). Although mechanistic evidence in humans remains limited, these findings support conserved immune-epithelial interactions during mammary differentiation. Whether immune cells similarly integrate endocrine signals to regulate epithelial differentiation across other maternal barrier tissues remains an important question.

Reallocating maternal metabolism via tissue and immune adaptation

Successful pregnancy requires coordinated redistribution of maternal metabolic resources to sustain fetal growth while preserving maternal health and preparing for lactation. This homeorhetic transition has traditionally been attributed to endocrine regulation, but accumulating evidence indicates that immune cells also remodel metabolic organs to accommodate the changing energetic demands of gestation (Fig. 3).

The small intestine undergoes rapid structural expansion early in pregnancy, increasing absorptive surface area before maternal food intake rises, indicating that intestinal remodeling is anticipatory rather than a response to increased nutritional demand (*44*, *45*). Neural regulation may provide an additional layer of reproductive intestinal adaptation: Reproductive hormones remodel enteric neuronal activity to support maternal feeding in *Drosophila* (*46*), whereas the extensive bidirectional communication between enteric neurons and intestinal immune populations (*47*) raises the possibility that neuroimmune circuits similarly contribute to maternal intestinal adaptation in mammals. T cell immune signals also directly regulate intestinal remodeling during pregnancy. RANK ligand (RANKL) produced by T cells and mesenchymal cells promotes epithelial differentiation and enterocyte survival. Mice lacking epithelial RANK fail to undergo normal villous expansion during pregnancy, resulting in impaired nutrient absorption and altered metabolic health in the offspring, demonstrating that RANK-dependent intestinal remodeling contributes directly to reproductive success (*44*). A complementary immune program remodels the intestinal barrier during reproduction. In mice, eosinophils accumulate in the small intestine from pregnancy through lactation and promote goblet cell differentiation and mucus production, strengthening innate protection against enteric infection (*48*). Although pregnancy-induced villus expansion occurs independently of microbiota, pregnancy is accompanied by marked remodeling of the maternal gut microbiota (*49*). Transfer of third-trimester microbiota into germ-free mice is sufficient to induce increased adiposity, insulin resistance, and inflammatory changes (*49*), demonstrating that host-microbiota interactions constitute an additional physiological mechanism that reallocates maternal metabolism during pregnancy.

Pregnancy also remodels maternal energy storage. Beyond their established role at the maternal-fetal interface, thymus-derived regulatory T cells accumulate in visceral adipose tissue during gestation, where they limit inflammation, preserve glucose tolerance, and support maternal metabolic homeostasis. Their depletion results in gestational diabetes, fetal macrosomia, and persistent metabolic abnormalities in the offspring (*20*).

The liver undergoes equally profound remodeling. Pregnancy induces physiological hepatomegaly through coordinated hepatocyte proliferation and metabolic reprogramming, increasing biosynthetic

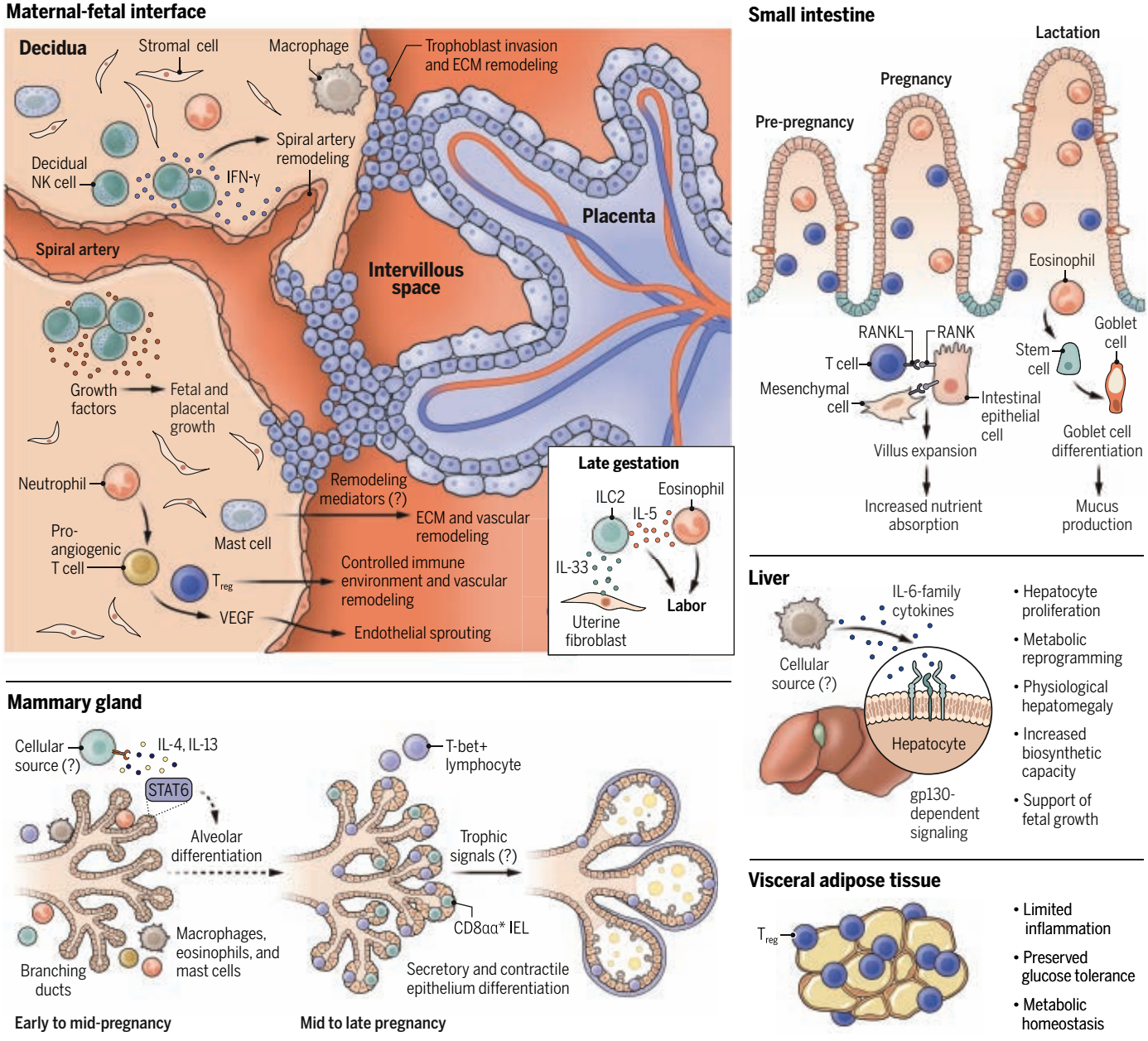

Fig. 3. Maternal immune cells coordinate physiological adaptation across tissues. Immune cells contribute to maternal tissue adaptation during pregnancy in mice by regulating vascular, epithelial, and metabolic programs. At the maternal-fetal interface, decidual NK cells, macrophages, T_{reg} cells, and other immune populations support implantation, placentation, fetal growth, and labor. In the mammary gland, IL-4 and IL-13 signaling and T-bet$^+$ lymphocytes, including CD8αα$^+$ intraepithelial lymphocytes, promote epithelial differentiation and lactogenesis. In the intestine, RANKL produced by T cells and mesenchymal cells drives epithelial expansion and nutrient absorption during pregnancy, whereas eosinophils accumulating during pregnancy and lactation promote goblet cell differentiation and mucus production. T_{reg} cells support metabolic homeostasis in visceral adipose tissue. In the liver, gp130-dependent IL-6 family signaling promotes hepatocyte proliferation and metabolic adaptation. Together, these tissue-specific programs demonstrate how immune cells and immune-associated signals contribute to maternal physiological adaptation during reproduction. Question marks indicate cellular sources or mediators that remain unresolved.

capacity to support fetal growth and prepare for lactation (*50*). Genetic disruption of hepatocyte gp130, the shared signaling coreceptor for the IL-6 cytokine family, selectively impairs late gestational liver remodeling and restricts fetal growth without affecting earlier stages of pregnancy, demonstrating that IL-6 family cytokine signaling is required for normal hepatic adaptation (*50*). Although the cellular sources of these cytokines remain unresolved, these findings further support the concept that cytokine signaling is an integral component of maternal metabolic adaptation.

Rather than endocrine regulation alone, immune cells and immune-derived signals play an important role in to redirecting nutrient acquisition, storage, and allocation to meet the physiological demands of reproduction. These findings also raise the intriguing possibility that immune cells do not simply respond to changing metabolic demands but actively anticipate them. Defining how immune cells integrate endocrine, metabolic, and tissue-derived signals to coordinate these anticipatory adaptations across organs remains an important challenge for the field.

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Pregnancy leaves a lasting physiological memory

Pregnancy-induced immune remodeling extends beyond childbirth. Evidence from both mice and humans indicates that reproductive experience leaves persistent alterations in immune composition, epigenetic programs and tissue organization. These observations support the emerging concept of a physiological memory of pregnancy, in which previous gestational experience modifies maternal responses to subsequent pregnancies and may influence disease susceptibility later in life.

The maternal-fetal interface provides one of the clearest examples of persistent immunological adaptation. In humans, repeated pregnancies are associated with a pregnancy-trained decidual NK cell population characterized by increased production of IFN-γ and VEGFA, the latter promoting vascular sprouting (*51*). In parallel, fetal microchimeric cells established during a first pregnancy can persist within maternal tissues after delivery (*52*), providing continued antigenic stimulation that supports the maintenance of fetal-specific T_{reg} cells across successive pregnancies despite the progressive replacement of microchimeric cells by those derived from each new gestation (*53*). Beyond maintaining immune tolerance, fetal microchimeric cells have also been implicated in maternal tissue repair, angiogenesis, and organ homeostasis in mice, with accumulating correlative evidence in humans (*54*). Whether fetal microchimeric cells contribute to the long-term physiological memory of pregnancy remains an important unanswered question.

Persistent effects of reproductive experience are also apparent in labor that is consistently shorter in multiparous than in nulliparous females who have not previously given birth (*55*). Multiparous females also exhibit distinct inflammatory trajectories before and during labor, characterized by higher baseline concentrations of CCL4, CXCL9, and CXCL10 before labor onset but a more restrained innate inflammatory response during active labor (*56*). Together, these observations raise the possibility that prior pregnancy may optimize the inflammatory program governing parturition, although the mechanisms linking these immune changes to labor progression remain to be established.

The mammary gland provides another example in which immune and tissue memory become intimately intertwined. In mice, following a complete pregnancy-lactation-involution cycle, mammary epithelial cells undergo widespread and stable DNA demethylation at STAT5-bound regulatory regions controlling milk synthesis, proliferation, and cell survival (*57*). These epigenetic modifications persist throughout reproductive life and enable more rapid transcriptional reactivation during subsequent pregnancies, providing a mechanistic basis for the accelerated mammary response observed in multiparous females (*57*). The immune landscape of the mammary gland is similarly remodeled by reproductive experience. In mice, pregnancy-induced $CD8\alpha\alpha^+$ intraepithelial lymphocytes persist following involution while maintaining tissue residency and cytotoxic potential (*21*). Consistent with these findings, in humans, parity is associated with increased T cell abundance in normal human breast tissue, suggesting that reproductive history is accompanied by lasting changes in the mammary immune landscape (*58*, *59*). More notably, completion of the pregnancy-lactation-involution cycle drives long-term accumulation of $CD8^+$ T cells that suppress mammary tumor growth and enhance antitumor immunity in mice, an effect absent after pregnancy alone. These observations raise the intriguing possibility that immune adaptations acquired during reproduction may be redeployed beyond lactation, simultaneously supporting tissue surveillance and enhancing mammary adaptation during subsequent pregnancies.

These findings establish the concept of pregnancy-associated physiological memory. Understanding how this memory is established, maintained, and disrupted across organs may explain why healthy pregnancies can promote long-term health, whereas complicated pregnancies increase the risk of chronic cardiovascular, metabolic, and inflammatory diseases later in life (*60*).

Could adaptation become maladaptation?

The immune program of pregnancy is not fixed but continuously shaped by environmental cues. Pregnancy must therefore be considered in the context of the physiological and pathological stressors that accompany gestation, including infections, obesity, and metabolic disorders, all of which can shape maternal adaptation and pregnancy outcomes.

Nutritional, metabolic, and infectious exposures all have the potential to influence this adaptive program. Maternal obesity and Western-style diets promote systemic inflammation, alter gut microbial communities, and impair intestinal barrier integrity, leading to increased circulating inflammatory mediators and placental dysfunction, including abnormal vascular development and heightened inflammatory responses (*61*). Infection encountered during pregnancy can similarly shift this physiological equilibrium toward inflammatory defense. Although pregnancy is not a state of generalized immunosuppression, the endocrine and immune adaptations that support gestation can increase susceptibility to selected infections while amplifying inflammatory responses once infection occurs. Infection-induced cytokine production and complement activation disrupt the immune and angiogenic environment of the maternal-fetal interface and impair placental vascular function and are associated with miscarriage, fetal growth restriction, stillbirth, and preterm birth, even in the absence of direct fetal infection (*62*). Thus, environmental perturbations might reshape the immune-tissue interactions that coordinate maternal physiological adaptation. Whether these changes represent transient deviations from physiological adaptation or trigger persistent maladaptive remodeling remains unknown. Recent longitudinal multiomic analyses further demonstrate that immunological maladaptation is detectable weeks to months before the onset of spontaneous preterm birth, supporting the concept that pregnancy complications are preceded by abnormal maternal immune adaptation rather than only inflammatory events at labor (*6*).

If successful pregnancy establishes adaptive programs that improve subsequent reproductive performance, then immune dysregulation during gestation may conversely leave persistent maladaptive alterations. Pregnancy complications, including preeclampsia, fetal growth restriction, recurrent miscarriage, and spontaneous preterm birth, substantially increase the risk of similar complications in subsequent pregnancies (*63*). Women with previous preeclampsia also exhibit long-lasting alterations in circulating immune cell composition and DNA methylation programs affecting genes involved in T and NK cell function (*64*). Pregnancy complications are further associated with impaired regulatory T cell responses, altered decidual NK cell function, and persistent inflammatory activation (*65*). Whether these persistent immune alterations contribute directly to recurrent pregnancy complications or instead reflect underlying disease susceptibility remains unresolved.

Beyond pregnancy, women with previous preeclampsia or gestational diabetes face substantially increased long-term risks of cardiovascular disease and metabolic dysfunction (*60*). Rather than representing isolated obstetric events, these conditions raise the possibility that physiological adaptations established during pregnancy are not always fully resolved after delivery. Environmental perturbations, including infection and obesity, may similarly leave lasting consequences by altering immune-tissue interactions during gestation, although whether these effects converge on common cellular and epigenetic mechanisms has yet to be established. Defining how physiological immune adaptation is maintained within its adaptive range and identifying the factors that redirect this program toward persistent maladaptation represents one of the central challenges in reproductive immunology, with implications for health that extend far beyond pregnancy.

Conclusions

The coordinated homeorhetic adaptation that underpins pregnancy depends on the close integration of endocrine and immune signaling.

Although reproductive hormones orchestrate profound physiological adaptations across maternal tissues, the mechanisms by which endocrine cues are decoded by immune cells and tissue niches to generate organ-specific physiological programs remain poorly understood. Determining how endocrine and immune signaling pathways are integrated to coordinate tissue-specific adaptation across multiple organs represents one of the central challenges for understanding maternal physiology during reproduction.

A second major challenge will be to determine how immune adaptation is temporally coordinated across organs and reproductive stages. Pregnancy requires the simultaneous remodelling of multiple maternal tissues, yet most studies have examined immune regulation within individual organs in isolation. Whether common immune circuits operate across the mammary gland, uterus, adipose tissue, liver, and intestine or whether each tissue interprets reproductive cues through distinct local programs remain unknown. Likewise, lactation remains comparatively unexplored, and whether it represents a distinct systemic immunological state or the continuation of pregnancy-associated immune programs has yet to be resolved. Integrating multiorgan and longitudinal approaches will therefore be essential for establishing a unified framework of maternal physiological adaptation.

Lastly, an important unresolved question is which immune adaptations established during reproduction are fully reversible and which leave lasting physiological consequences. Defining which components of these adaptations are beneficial and under what circumstances pregnancy complications or environmental perturbations become maladaptive represents a central challenge for the field. Addressing these questions will not only advance our understanding of reproductive biology but may also reveal fundamental principles governing how transient physiological states shape women's health across the lifespan.

Together, these questions establish pregnancy as a distinctive framework for understanding how immune cells integrate endocrine, stromal, and environmental signals to coordinate tissue adaptation. Addressing these challenges will require moving beyond organ-specific and descriptive studies toward mechanistic, multiorgan approaches that capture the dynamic nature of maternal physiology. Understanding how immune cells coordinate this adaptive program may reveal new opportunities to protect maternal and child health, improve resilience to pregnancy complications, and reduce the long-term burden of reproductive disorders.

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REVIEW

X and Y chromosomes as determinants of aging and disease

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Sex differences have traditionally been attributed to gonadal hormones, but accumulating evidence shows that sex chromosomes exert distinct, cell-autonomous effects on health, aging, and disease. XX and XY complements alter gene dosage, epigenetic programming, immune signaling, metabolism, and cellular stress responses, which shape aging and disease trajectories in women and men. This Review integrates mechanistic advances in sex chromosome biology—including escape from X chromosome inactivation, X reactivation, parent-of-X-origin, X-skew, X- and Y-linked gene functions, and Y loss—with aging, neurologic disease, immunity, cancer, and cardiometabolic disease. We highlight insights from mouse models, human studies, and emerging technologies that disentangle hormone-driven effects from chromosome-mediated effects and discuss translational implications for diagnostics, biomarkers, therapeutics, and clinical trial design, emphasizing the integration of sex chromosome biology into precision medicine.

Biological sex is defined by chromosomal complement (XX or XY), gonadal differentiation, and physiology, all of which profoundly affect human health. It is conceptually separated from gender, which reflects sociocultural identity and lived experience, which we do not discuss here. Biological sex directly influences cellular function, immunity, and metabolism, contributing to mechanisms of aging and disease. Although research has long focused on gonadal hormones, emerging evidence shows that sex chromosomes exert independent and interactive effects through genetic and epigenetic mechanisms across organ systems (*1*–*4*).

The XX and XY chromosome complements shape cellular phenotypes through gene dosage, gene expression, and epigenetic processes. The X chromosome regulates fundamental cellular processes in both sexes. In XX individuals, a second X introduces additional biology through X chromosome inactivation (XCI), generating cellular mosaicism and differential gene expression (*5*). In XY individuals, the Y chromosome directs male sex determination through *SRY* and carries genes essential for fertility. Beyond reproduction, many Y-linked genes have broader biological functions, including X chromosome paralogs that regulate transcription, chromatin biology, RNA processing, and other cellular processes (*6*). Consequently, the Y chromosome is increasingly recognized as an important modulator of physiology, aging, and disease (*1*, *7*).

This Review examines how X and Y chromosomes mechanistically shape sex biology across the lifespan, with an emphasis on aging, neurologic and psychiatric conditions, immunity, cancer risk, and cardiometabolic disease. The field is remarkably rich and rapidly advancing; we apologize to authors whose important contributions to these areas could not be cited owing to word count and reference limitations. Although social and environmental determinants of health also contribute to sex differences in aging and disease and may interact with sex chromosome biology, they are beyond the scope of this Review. We consider implications of sex chromosome biology for research design, clinical trials, and personalized therapy. Integrating genetics, genomics, cell biology, and clinical medicine is essential to better understand sex-biased diseases, inform sex-aware precision medicine, and develop new treatments.

Mechanistic framework: How sex chromosomes affect biology

How a second X chromosome shapes biology

With ~1500 genes making up 5% of the genome in both sexes, the X chromosome is a major contributor to sex differences, yet it remains understudied in aging and disease. In female mammals with two X chromosomes, one is silenced in embryonic development through X-inactive specific transcript (XIST)–mediated XCI, generating an active X (Xa) and an inactive X (Xi) (*8*). XCI, first proposed by Mary Lyon in 1961, was long viewed as a stable mechanism to equalize X-linked gene expression between XX and XY cells. However, we now know that ~15 to 30% of Xi genes constitutively escape XCI in humans (and 3 to 7% in mice) with variability across genes, cell types, tissues, individuals, and species (*9*). Constitutive XCI escape enables expression from both the Xa and Xi (Fig. 1), increasing X-linked gene dose in female cells, thus providing a mechanism by which two X chromosomes influence resilience and vulnerability in aging and disease.

One example is *KDM6A*, a histone demethylase that constitutively escapes XCI and is expressed at higher levels in XX compared with XY brains of both mice and humans. Throughout this Review, findings on KDM6A illustrate how increased X-linked gene dosage influences diseases across organ systems. In humans, increased neural expression of *KDM6A*, through a genetic variant, is associated with cognitive resilience in Alzheimer's disease (*10*). In mouse models of Alzheimer's disease, KDM6A overexpression reverses cellular and cognitive impairments (*10*). Notably, these protective effects extend to males, which highlights that understanding mechanisms of XCI can benefit both sexes.

Reactivation of the Xi chromosome [X chromosome reactivation (XCR)] has emerged as a dynamic, lifespan-dependent process and a source of sex differences that may contribute to aging and disease (Fig. 1). Advances in single-cell and single-nucleus RNA sequencing, combined with allele-specific approaches, now enable precise measurement of gene expression from the Xi versus the Xa in mice (*11*, *12*) and humans (*13*). These studies show that XCI is not static. Across multiple organ systems in mice and humans, aging promotes XCR by relaxing Xi silencing while also further repressing some Xi genes (*11*–*13*). Because XCI depends on DNA methylation, chromatin structure, and noncoding RNAs, age-related disruption of these programs can dynamically alter Xi gene expression.

In XX individuals, XCR provides an additional mechanism for increased X-linked gene expression with aging and potentially in disease. One example is *Plp1*, a key myelin gene that undergoes XCR in the aging mouse brain. As a result, *Plp1* expression is higher in aged XX versus XY brains, mirroring sex differences observed in certain human brain regions (*11*). Elevating *Plp1* improves cognition in aged mice of both sexes (*11*), which highlights the therapeutic potential of XCR.

XCI during development creates cellular mosaicism in XX individuals, with each cell expressing either the maternal (Xm) or paternal (Xp) X as the active X (Fig. 1). By contrast, XY cells express only Xm, a fundamental source of sex difference. Mosaicism ranges from balanced Xm/Xp expression to X-skew, in which most cells preferentially express either the Xm or the Xp (*14*) (Fig. 1). Mosaicism may buffer dysfunction during aging and disease by increasing genetic and epigenetic diversity,

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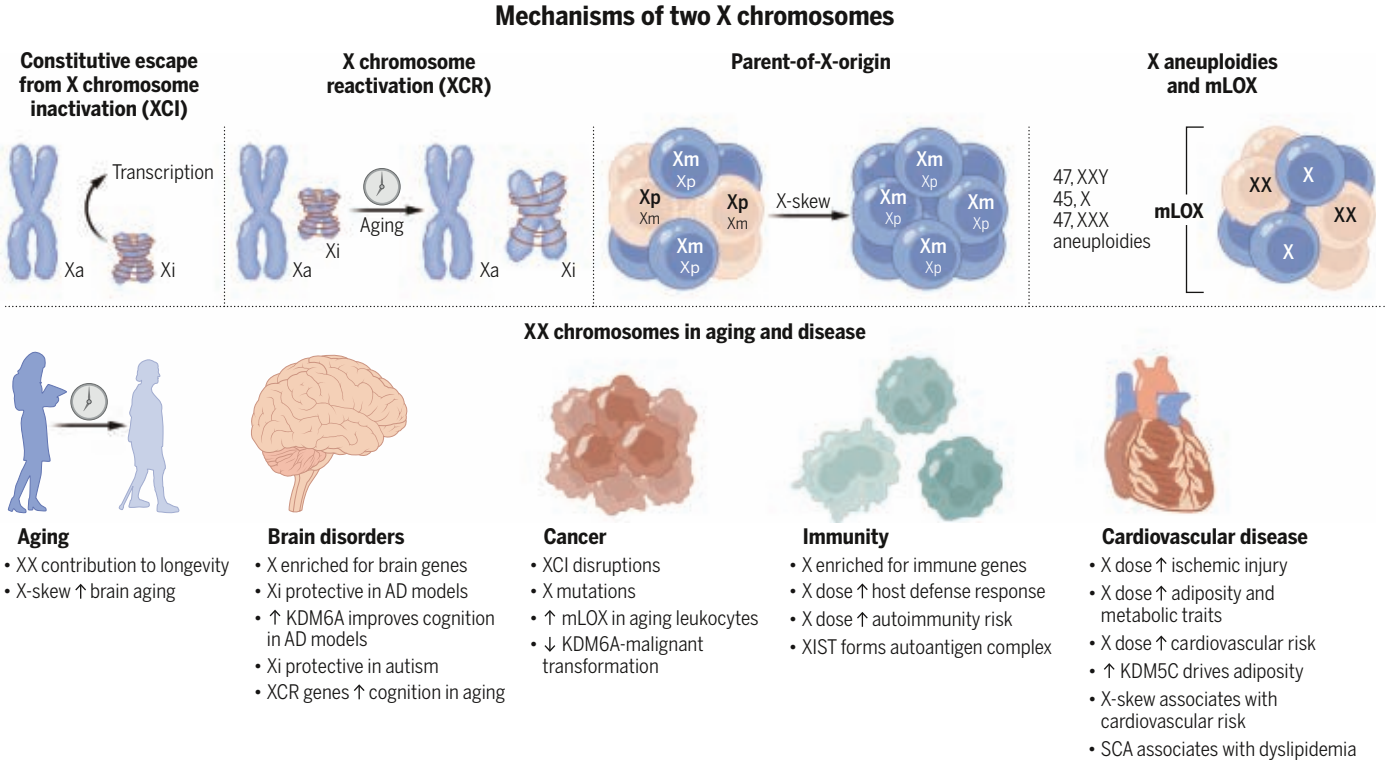

Fig. 1. Mechanisms of two X chromosomes. Major features of how two X chromosomes shape biology include constitutive escape from XCI, XCR, parent-of-X-origin effects (including X-skew), X chromosome aneuploidies, and mLOX. These X mechanisms influence aging, brain disorders, cancer, immunity, and cardiovascular disease. Representative findings and proposed mechanisms linking biology of two X chromosomes to sex differences are highlighted. AD, Alzheimer's disease.

whereas X-skew may increase vulnerability. For example, in Duchenne muscular dystrophy, X-skew toward the mutant X drives disease manifestation in affected females (*15*). In humans, age-acquired X-skew is prevalent with advancing age in the blood of women and associates with adverse health outcomes (*16*), which raises the possibility that an individual woman's X-skew could serve as a biomarker of her disease risk.

Parent-of-X-origin refers to the biological consequences of expressing the maternal (Xm) or the paternal (Xp) X chromosome. Alongside XCI escape and XCR, it represents a distinct mechanism by which a second X chromosome shapes health, aging, and disease. Parent-of-X-origin influences DNA methylation and gene expression, including X-linked imprinting, in which genes are selectively silenced on either Xm or Xp, as demonstrated in mice (*17*, *18*). Also in mice, preferential activation of Xm—causing X-skew—impairs cognition and accelerates brain aging, in part through the silencing of Xm-linked cognition-related genes (*17*). Consistent with this, up-regulation of these Xm-silenced genes improves cognition in aged XX mice (*17*). Emerging allele-specific genomic approaches combined with parent-offspring trio studies may enable precise mapping of parent-of-X-origin effects on gene expression and imprinting in humans.

Mosaic loss of the X chromosome (mLOX) (Fig. 1) is an acquired somatic event in which one X chromosome is physically lost in a subset of XX cells. mLOX increases with age in female leukocytes (rising to >35% after age 80) (*19*), preferentially affects Xi (*20*), and reduces expression of XCI escape genes. Loss of either Xm or Xp in mLOX may also generate parent-of-X-origin effects analogous to X-skew. mLOX associates with smoking and increased leukemia risk (*19*), but its broader effects in female aging and disease remain largely unknown.

How the Y chromosome shapes biology

The Y chromosome, despite its low gene content, contains several functionally important loci extending beyond the sex-determining region *SRY*, which induces testicular development (Fig. 2) (*21*). Mutations in several Y gene loci cause testicular disorders (not reviewed here). Y-linked genes, including *TSPY*, *ZFY*, and *UTY* (*KDM6C*), regulate chromatin, immune signaling, transcription, and other fundamental processes, implicating the Y chromosome in aging, cancer, and other nonreproductive diseases (*7*). Several of these Y genes are dosage-sensitive paralogs of X-linked genes. For example, *UTY* (*KDM6C*) and *KDM5D* regulate chromatin through enzymatic and scaffolding functions, shaping methylation landscapes and transcriptional output. Although *UTY* has limited demethylase activity relative to its X paralog *UTX* (*KDM6A*), it retains functional relevance through interactions with transcriptional corepressors such as TLE1, which affects cell lineage programs and immune gene expression implicated in sex-biased disease biology (*22*).

With aging, many of these Y-linked genes are lost through mosaic loss of the Y chromosome (mLOY), an important mechanism linking Y chromosome biology to aging and disease. For clarity, we use LOY to denote Y chromosome loss in individual cells and mLOY for mosaic loss in cell populations. mLOY (Fig. 2) in peripheral blood mononuclear cells (PBMCs) is the most common somatic chromosomal alteration in aging men (*23*), becoming common by the seventh decade (*7*). This builds on earlier observations that age-related clonal mosaicism accumulates with aging and associates with cancer and other age-related diseases (*24*, *25*). Smoking is the strongest known, dose-dependent, and partially reversible environmental risk factor for mLOY (*26*), with additional contributions from inherited genetic susceptibility, DNA-damaging environmental exposures, and genomic instability. mLOY frequently co-occurs with clonal hematopoiesis of indeterminate potential (CHIP), affecting nearly 40% of men with CHIP (*27*), suggesting shared age-related mechanisms of clonal expansion and contributing to adverse health outcomes associated with both conditions (*28*). Single-cell and linage-sorted analyses detect mLOY across normal tissues and link it to immune phenotypes (*29*). Its prevalence

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is increased in men with cardiovascular disease, cancers, severe infections, and Alzheimer's disease, which supports its role as both a biomarker and potential contributor to age-related disease (*30*, *31*).

Experimental studies support a causal role for mLOY in disease. In mouse models, mLOY accelerates cardiomyopathy and reduces lifespan, implicating macrophage-mediated mechanisms in cardiovascular disease (*32*). The exact mechanism of LOY is not fully understood but distinctive features of the human Y chromosome centromere likely contribute (*33*). In tumors, LOY remodels the microenvironment (*34*), promotes CD8+ T cell exhaustion, and associates with worse prognosis but greater responsiveness to immune checkpoint inhibitors (*28*, *34*).

Mechanistically, loss of the Y genes disrupts the X-Y paralog balance and remodels chromatin, immune, and stromal programs. Whereas *KDM5D* and *UTY* generally act as tumor suppressors (*28*, *34*), their effects are context dependent, with *Kdm5d* overexpression in a mouse colon cancer model making tumors more invasive in vivo (*35*). Likewise, loss of Y genes, particularly *UTY* in mouse and human macrophages, associates with transcriptional and paracrine changes that promote activation of cardiac fibroblasts and fibrosis through transforming growth factor–β (TGF-β)–dependent mechanisms (*36*). These findings provide a mechanistic link between mLOY and cardiovascular disease (*32*).

Dissection of X and Y chromosome biology: Mouse models and human variation

The four core genotypes (FCG) model, originally developed in mice (*37*) and now extended to rats (*38*), genetically uncouples the effects of sex chromosomes from gonadal sex. Deletion of the testis-determining gene (*Sry*) from the Y chromosome and its transgene insertion on an autosome [$XY^{-Sry}(Sry^{+})$] generates XX and XY animals with either ovaries or testes (*37*). Crossing $XY^{-Sry}(Sry^{+})$ males with wild-type XX females yields four genotypes: XX and XY^{-Sry} with ovaries and XX and XY^{-Sry} with testes. This design distinguishes sex chromosome effects (XX versus XY with matched gonads) and sex hormone effects (ovaries versus testes with identical sex chromosomes) (*37*). Sex chromosome effects (XX versus XY) may result from differences in X chromosome number (X dosage, X imprinting, and indirect effects of XCI) or from the presence of the Y chromosome, which can be further dissected using complementary models outlined below. FCG mice have been widely used to define how chromosomal and gonadal sex shape aging, neurological disorders, autoimmunity, cancer, and cardiovascular and metabolic diseases (*39*).

Sox9 enhancer mouse models provide an alternative strategy to uncouple gonadal and chromosomal sex while preserving the endogenous *Sry* locus, given that Sry may have functions beyond testis determination. Rather than relocating *Sry*, these models manipulate the autosomal *Sox9* enhancer (Enh13), Sry's principal downstream effector, generating XY mice with ovaries (*40*) or XX mice with testes (*41*).

The XY* mouse model complements the FCG and Sox9 enhancer models by dissecting the contributions of X chromosome dosage (including X genes, X imprinting, and indirect effects of XCI) and Y chromosome presence (including Y-linked gene effects). XY* males carry a Y chromosome with an altered pseudoautosomal region that undergoes abnormal meiotic combination with the X (*37*). Crossing XY* males with wild-type XX females yields four genotypes: XO and XX with ovaries and XY and XXY mice with testes. The XY* model has become a standard tool to define how sex chromosome dosage contributes to diverse disease processes (*39*).

To model acquired Y chromosome loss, mouse models use CRISPR-Cas9–mediated Y chromosome deletion in donor hematopoietic stem cells before transplantation into irradiated recipients, generating chimeras with >50% mLOY in blood cells (*32*, *42*). Because mLOY also occurs in nonhematopoietic tissues, inducible tissue-specific mLOY models are needed to further define its causal roles in aging and disease.

Parallel advances have expanded the toolkit for studying X chromosome biology. Mouse models now enable mechanistic dissection of constitutive XCI escape (*11*), XCR (*11*), parent-of-X-origin effects (*17*, *43*), and X-skew (*17*). When paired with aging or disease models, they can reveal causal mechanisms underlying female heterogeneity and sex differences. Key approaches include allele-specific systems that distinguish expression of Xa from Xi, enabling precise study of XCI escape and XCR (*11*); dual-reporter models that fluorescently label Xm and Xp to visualize and probe cellular mosaicism (*43*); and inducible X-skew models (*17*). Additional strategies use small molecules, adeno-associated virus (AAV), and CRISPR to modulate X-linked gene expression. Integrated with bioinformatics and functional assays, these tools provide high-resolution study of X-linked causative mechanisms in aging and disease and may uncover therapeutic targets benefiting both women and men.

Human sex chromosomal aneuploidies (SCAs) provide human-derived evidence for how altered X and Y chromosome dosage influences fundamental biology, complementing findings from mouse models. SCAs are prevalent yet underdiagnosed, affecting ~1 in 400 newborns (*44*). They arise from gain or loss of whole or partial sex chromosomes, often owing to meiotic nondisjunction (*45*). Well-characterized SCAs include Turner syndrome (XO), Klinefelter syndrome (XXY), trisomy X (XXX), XYY, and XXYY (*45*). These conditions disrupt endocrine, cardiometabolic, musculoskeletal, and neurological systems through altered genome function (*46*). Mechanistically, SCA phenotypes reflect dysregulated expression of dosage-sensitive genes, particularly those escaping XCI (including *KDM6A*, *KDM5C*, and *ZFX*) and pseudoautosomal genes (including *SLC25A6*, *ASMTL*, and *GTPBP6*). These expression changes perturb chromatin remodeling, translation, meiotic processes, and telomere maintenance, identifying candidate drivers of SCA-associated phenotypes (*46*).

Aging and disease: Differences attributable to sex chromosomes

Aging

Worldwide, women live longer than men (*47*, *48*). This female survival advantage persists even under extreme conditions, including famine and epidemics, where girls outlive boys (*47*). It also holds across major diseases—particularly neurodegenerative disorders, including Alzheimer's disease (*49*)—and extends across much of the animal kingdom (*50*), which suggests a conserved biological basis. One evolutionary explanation posits that longer-lived mothers increase offspring survival and reproductive success. Identifying mechanisms that promote longevity may reveal pathways that extend health span, or years lived in good health.

Experimental models support a sex chromosomal contribution to longevity and resilience. In FCG mice, the XX complement increases survival, regardless of gonadal sex (*51*). However, the underlying mechanisms, whether XCI escape, XCR, parent-of-X-origin effects, or others, remain unresolved. In mouse studies of brain aging, Xm accelerates neuronal aging (*17*), which raises the possibility that female mosaicism of Xm and Xp confers resilience compared with the single Xm in males. In parallel, mLOY in men associates with increased mortality from midlife age onward, although, paradoxically, it is enriched in centenarians (*52*), which leaves unresolved whether mLOY is a driver, biomarker, or adaptive consequence of aging.

Aging is the strongest risk factor for diseases across organ systems. Women exhibit resilience during normal brain aging yet remain disproportionately affected by Alzheimer's disease (*53*, *54*). These observations suggest that mechanisms promoting longevity may also underlie resilience during healthy aging and early disease but become insufficient as disease-related pathology accumulates. Defining X-linked mechanisms underlying longevity may uncover resilience pathways that can be harnessed to boost health span and reduce age-related disease.

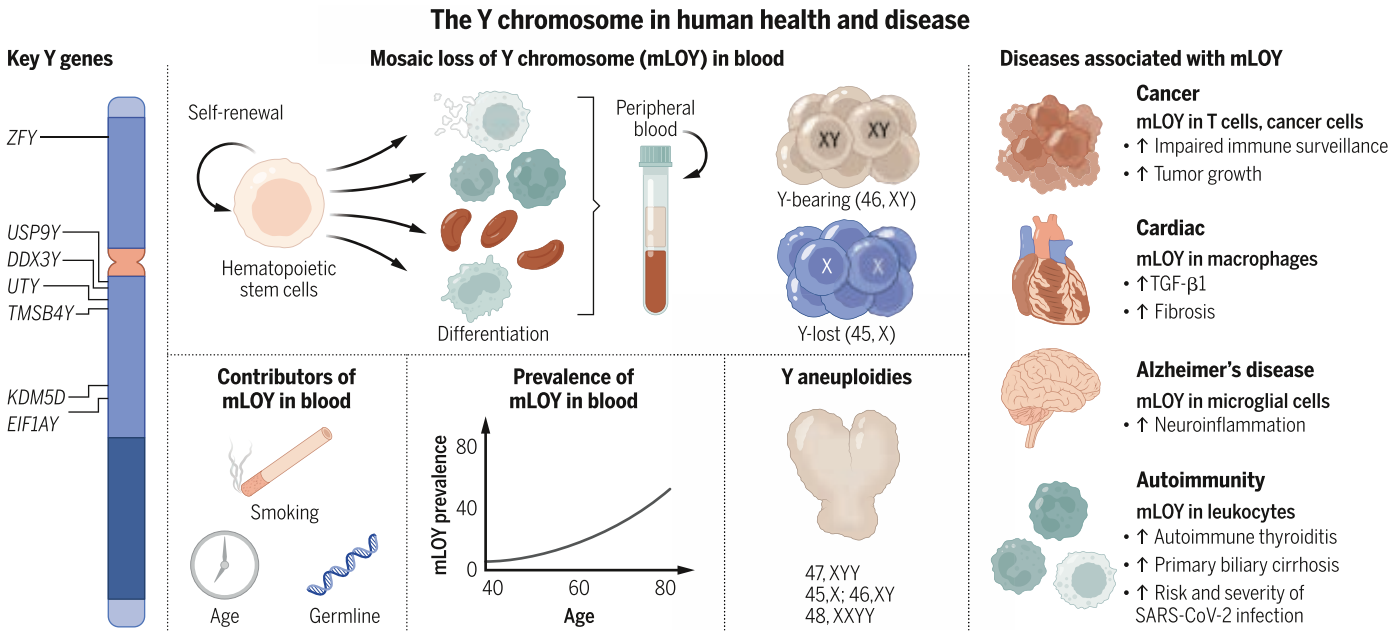

Fig. 2. The Y chromosome in human health and disease. The Y chromosome contributes to male health and disease through many mechanisms, including key Y genes, mLOY, and Y aneuploidies. Representative associations of mLOY with cancer, cardiac fibrosis, Alzheimer's disease, and immune dysfunction are shown.

Neurological, neurodevelopmental, and psychiatric disorders

The brain is enriched for X-linked genes that are more highly expressed than autosomal genes (*55*), which makes it particularly sensitive to X chromosome biology. Consistent with this, mutations in more than 90 X-linked genes cause intellectual disability, and many more contribute to other neurological phenotypes, most of which are not reviewed here. In normal aging without dementia, women show slower cognitive decline than men, likely influenced by X effects (*53*) because a second X improves cognition in aging mice of both sexes (*56*). In mice, aging induces XCR in XX brains, increasing expression of genes such as *Plp1*, enhancing cognition in both sexes (*11*). Meanwhile, Xm-skew accelerates mouse brain aging (*17*), which suggests that Xm and Xp mosaicism in females may confer resilience compared with the single Xm in males.

In neurodegenerative diseases, X chromosome mechanisms promote both resilience and vulnerability, depending on disease stage and context. In Alzheimer's disease, women live longer and show early clinical resilience, despite accumulating more neuropathology, but they later experience steeper cognitive decline (*54*, *57*, *58*). FCG studies of Alzheimer's amyloid mouse models show that an XX complement improves survival and cognition, independent of gonadal sex (*10*). XY* studies attribute these benefits to increased X dosage, partially owing to higher KDM6A levels, and to X-Y interactions (*10*), implicating X chromosome mechanisms in resilience.

By contrast, during the later tau phase of the disease, X-linked genes may promote pathology; for example, *Usp11* promotes tau accumulation (*59*) and *Tlr7* contributes to tau-induced demyelination (*60*) in mice. What determines whether X chromosome biology is protective or pathogenic—including disease state, cell type specificity, and the underlying proteinopathy—and how these mechanisms contribute to women's greater vulnerability to tauopathy remain largely unknown.

Human genetic studies implicate sex chromosomes across neurodegenerative diseases. X chromosome–wide association studies identify X-linked loci in Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, and other disorders (*61*), whereas mLOY associates with increased Alzheimer's disease (*1*) and amyotrophic lateral sclerosis risk (*62*) in men.

Sex chromosomes affect susceptibility and disease severity across neurologic diseases. Stroke disproportionately affects women. In FCG mice, an XX complement increases infarct size with aging, independent of gonads (*63*), whereas in humans, mLOY associates with worse outcomes (*64*). In multiple sclerosis, women show higher susceptibility, whereas men experience more severe neurodegeneration (*65*). Consistent with this, FCG studies of multiple sclerosis models such as experimental autoimmune encephalitis show that two X chromosomes exacerbate immune activation and susceptibility, partly thru lymphocytic KDM6A, whereas the XY complement worsens neurodegeneration—mirroring human sex differences (*65*). Roles of sex chromosomes in pain, epilepsy, neuromuscular, and movement disorders remain important areas for investigation.

In neurodevelopmental and psychiatric disorders, X-linked genes influence brain development and circuit function. For example, mutations in *MECP2* (*66*) and *FMR1* cause distinct neurodevelopmental syndromes with cognitive impairment and autism-related features. Autism in general is male biased, consistent with a protective effect of a second X in females (*67*), possibly through Xi and cellular mosaicism. By contrast, depression and anxiety are female biased. Although SCAs associate with elevated psychiatric comorbidity (*68*), the specific X- and Y-linked mechanisms underlying psychiatric risk and resilience across disorders—depression, anxiety, schizophrenia, bipolar, posttraumatic stress, and beyond—merit future study.

Autoimmunity and infection

Women mount stronger innate and adaptive immune responses than men (*4*), which enhances their resistance to infection and vaccine response but also increases their susceptibility to autoimmune diseases, such as systemic lupus erythematosus, rheumatoid arthritis, and multiple sclerosis (MS) (*69*). Autoimmune diseases arise from loss of immune tolerance and autoreactive T and B cells, leading to chronic inflammation and organ damage. X chromosome dosage contributes directly to these differences.

The X chromosome is enriched for immune-related genes, such as *TLR7* and *FOXP3*, and some immune cells display noncanonical epigenetic XCI maintenance features (*4*, *70*). Autoimmune disease scales

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with X chromosome number in FCG mice and in individuals with Klinefelter syndrome and trisomy X (*4*, *71*). Conversely, certain diseases, including autoimmune thyroiditis and type 1 diabetes, are increased in Turner syndrome (*72*), consistent with haploinsufficiency of X-linked immune genes and reduced immune regulation.

In individuals with more than one X chromosome, XCI escape of immune genes provides a plausible mechanism for autoimmunity (*2*). During XCI, XIST forms ribonucleoprotein complexes that can act as autoantigens in mice and humans (*73*). In mice, *Tlr7* recognizes RNA within these complexes (*73*, *74*), and the resulting Toll-like receptor (TLR) signaling promotes atypical memory B cell expansion and type 1 interferon production, thereby promoting disease activity (*73*, *75*, *76*). Autoantibodies to XIST ribonucleoprotein complexes are detectable in patient sera (*73*), which highlights their potential as diagnostic biomarkers for early disease detection.

Though less studied, LOY has also been implicated in sex-based differences in autoimmune diseases, such as autoimmune thyroiditis and primary biliary cirrhosis (*1*). Moreover, mLOY in hematopoietic cells associates with increased risk of infectious diseases and may contribute to elevated risk and severity of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection in men (*1*).

Cancer

Men show higher incidence and mortality across many common cancers, even after accounting for behavioral risk factors. Epidemiologic and molecular studies attribute this to both endocrine influences and intrinsic effects of sex chromosomes on tumor cells and the immune microenvironment. Male predominance occurs in bladder, hepatocellular, gastric, and esophageal cancers, and sex-skewed outcomes in bladder cancer, melanoma, and glioblastoma remain incompletely explained by the environment (*7*).

mLOY provides one mechanistic link between cancers and sex biology. mLOY in PBMCs associates with increased cancer risk and mortality and frequently co-occurs with CHIP mutations (*7*), reflecting shared influences of aging, DNA damage, and selection of advantaged hematopoietic clones. In tumors, pan-cancer analyses associate mLOY with poor prognosis (*34*, *77*, *78*), though its role as driver versus passenger of this aggressive biology was previously unclear. Recent work mechanistically demonstrated that LOY drives a global loss of tumor suppressive and immune-regulatory functions. Y-deficient tumor cells remodel the immune microenvironment, promote CD8+ T cell exhaustion, and enhance responses to PD-1 checkpoint blockade in both animal models and patients (*34*). Loss of UTY and KDM5D contributes substantially to these LOY effects. LOY was also discovered in nonmalignant hematopoietic and nonhematopoietic cells within the tumor microenvironment, particularly in mLOY tumors, and, when present, predicts worse outcomes (*28*). Recently, LOY was found in histologically normal tissue adjacent to epithelial cancers, which suggests that LOY may represent an early hallmark of male oncogenesis and a potential biomarker for future or impending malignant transformation in specific tissues (*79*).

Mutations of X-linked genes also contribute to cancers in both sexes. The X-linked *DDX3X* and its Y-linked paralog *DDX3Y* regulate transcriptional programs and can function as tumor suppressors, oncogenes, or cancer modifiers (*80*). Likewise, the XCI escape gene *KDM6A* (*UTX*) is a tissue-specific tumor suppressor frequently mutated or deleted in bladder cancer, renal cell carcinoma, and hematologic cancers (*81*). Its higher dosage may partially protect women (*82*), whereas *KDM6A* loss fosters malignant transformation and disease progression through catalytic and scaffolding functions (*83*). Limited compensation by its Y-linked paralog *UTY* (*KDM6C*) may increase vulnerability in XY cells, helping explain why LOY or *UTY* deletion promotes transformation or progression in established tumors (*83, 84*).

Sex differences also extend to antitumor immunity. In glioblastoma models and patients, KDM6A and XCI escape support female T cell resilience, whereas men show greater progenitor-exhausted CD8+ T cells but sometimes stronger PD-1 responsiveness (*85*).

Cardiovascular and metabolic diseases

Sex differences in cardiovascular disease and cardiometabolic traits evolve across the lifespan. Men have higher myocardial infarction incidence at younger ages, but this gap narrows with age, and women experience worse outcomes later in life, including higher mortality after myocardial infarction and heart failure (*86*, *87*). Cardiometabolic risk factors, including adiposity, lipid profiles, and cardiac function, also diverge by sex with aging (*88*, *89*). Although hormonal influences contribute, experimental and population data clearly support independent roles for the sex chromosomes.

In FCG and XY* mouse models, the XX complement drives cardiac vulnerability; XX mice show greater ischemic injury than XY mice, independent of gonads and driven by increased X dosage (*90*). XX complement also regulates metabolic traits (*3*) and promotes atherosclerosis (*3*, *91*). In FCG mice, an XX complement promotes adiposity, weight gain, fatty liver, hyperlipidemia, and hyperinsulinemia under high-fat conditions (*92*, *93*). X effects on adiposity are recapitulated in XY* models, caused by increased X dosage rather than Y presence (*93*).

Mechanistically, XCI escapee KDM5C drives adiposity in mice and adipocyte *KDM5C* expression and genetic variation associate with body mass in humans (*92*). Age-related X-skew also associates with cardiovascular risk (*16*). Human data in individuals with SCA further support these links, including dyslipidemia in boys with Klinefelter syndrome (*94*).

The Y chromosome also modifies cardiovascular risk. In mice and humans, mLOY associates with cardiomyopathy, cardiac fibrosis, and increased mortality (*32*). In mice, loss of *Uty* promotes profibrotic macrophage activation and TGF-β1–driven cardiac remodeling (*36*). Together, these findings identify sex chromosome biology as a key determinant of cardiovascular disease and reveal clinical opportunities, including mLOY-based risk stratification, TGF-β1 inhibition, and targeting XCI escape genes.

Clinical and translational implications

Integrating sex chromosome biology into the clinical realm offers tremendous potential for precision diagnostics and therapeutics (Fig. 3). Realizing this potential will require addressing key knowledge gaps, including how sex chromosome states in blood reflect disease-relevant tissues, such as brain, heart, and immune cells, and how specific X or Y chromosome genes can be therapeutically targeted.

Diagnostics will require robust genomic, transcriptomic, and potentially protein assays to detect mLOY, quantify XCI escape, and assess X-skew and XCR where feasible (Fig. 3). Whole-genome sequencing and high-density genotyping arrays sensitively detect mLOY and estimate clonal fraction, which correlates with disease risk and outcomes (*23*, *28*, *34*). Clinical implementation will require standardizing mLOY, XCI escape, and other sex chromosomal measures for integration into electronic health records and clinical decision support systems (*95*).

Sex chromosome biomarkers hold promise for risk prediction and surveillance. mLOY is emerging as a noninvasive biomarker linked to malignancy, cardiovascular disease, and neurodegenerative diseases, including Alzheimer's disease (*1*, *7*). High mLOY fractions often co-occur with CHIP mutations, which suggests that combined assays may improve cancer surveillance and risk stratification (*27*, *96*, *97*). Likewise, mLOY and XCI escape alter tumor immune microenvironments and predict responsiveness to immune checkpoint blockade, which supports development of their use in treatment stratification (*34*, *85*). More broadly, establishing associations of mLOY, XCI escape, XCR, X-skew, and mLOX with aging and disease outcomes—and then integrating them with established biomarkers—may improve disease prediction and therapeutic selection. Clinical translation will

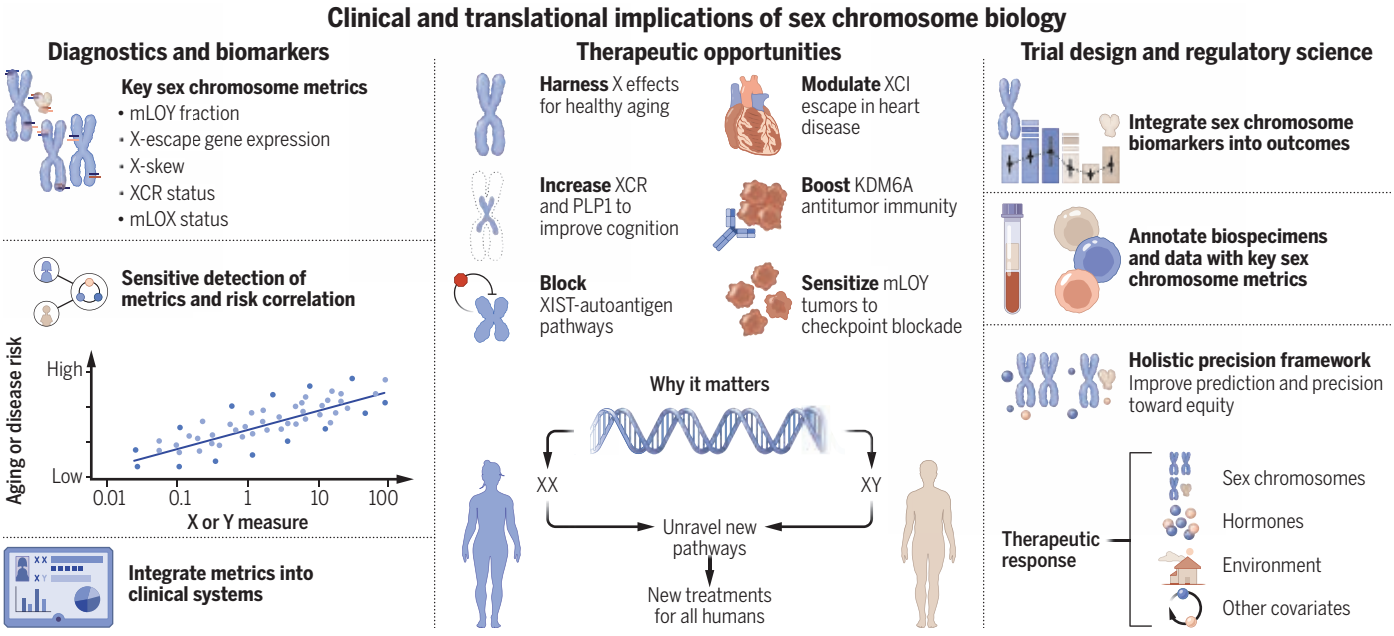

Fig. 3. Clinical and translational implications of sex chromosome biology. Integrating X and Y chromosome biology into diagnostics, therapeutics, and clinical trial design is needed to improve patient risk stratification, develop targeted therapies, and advance precision medicine across aging and disease.

require harmonized detection thresholds, prospective validation in diverse populations, and demonstration of added predictive value beyond existing biomarkers and risk factors.

Sex chromosome mechanisms offer compelling therapeutic targets that could benefit both sexes (Fig. 3). When we understand the molecular mechanisms of what makes one sex more resilient or vulnerable in aging or disease, we unravel pathways that may be targeted for therapy. Potential strategies include simulating X chromosome effects to promote longevity, stimulating XCR and increasing X-linked PLP1 in the aging brain to improve cognition, inhibiting XIST-driven autoantigen pathways in autoimmunity, boosting KDM6A-mediated antitumor immunity, and modulating XCI escape in cardiovascular disease. Because Y-deficient tumor cells promote CD8+ T cell exhaustion, these tumors are more responsive to immune checkpoint blockade, which provides a way to personalize therapy on the basis of the tumor Y chromosome status (*34*). These examples represent a small subset of emerging opportunities. Deeper understanding of X and Y chromosome biology opens a frontier for therapeutic development and offers a path to innovative medicines that improve human health.

Precision medicine should leverage X and Y chromosome gene status as biomarkers (Fig. 3). Precision oncology already uses features such as *KDM6A* mutations and mLOY fraction to stratify patients in selected immunotherapy trial settings, supporting their potential clinical utility. Future studies and trials should therefore incorporate sex chromosome genotyping, mLOY assessment, and immune profiling while treating biological sex as a mechanistic modifier of therapeutic response—given the growing evidence that X- and Y-linked biology and hormone-dependent mechanisms influence outcomes (*98*–*100*).

Trials should also integrate sex chromosome–related biomarker endpoints and be powered to detect sex-by-treatment interactions. Cell lines and biobanked tissues should be annotated for donor sex and chromosomal status, including tissue karyotype, LOY clonal fraction when feasible, and data on XCI escape to permit downstream association analyses and cross-cohort analysis. Finally, precision medicine frameworks must integrate sex chromosomes, circulating sex hormones, and environmental exposures because these factors jointly shape therapeutic response.

Limitations, open questions, and future directions

Sex chromosome effects on aging and disease are context dependent, varying by cell type, developmental stage, and environment. However, the field lacks comprehensive cross-tissue, cell type–specific maps of XCI escape, XCR, parent-of-X-origin, X-skew, mLOX, mLOY, and X-linked and Y-linked gene expression across development, aging, and disease. Broad allele-specific maps of XCI escape and XCR are also needed to link X chromosomal regulation with functional outcomes across tissues.

mLOY and mLOX increase with age through genetic and environmental influences, but their temporal dynamics and causal roles remain incompletely understood. Although single-cell studies suggest that these processes are dynamic and potentially reversible, most human epidemiological studies remain observational and cannot distinguish whether mLOY is a causal driver or simply a biomarker of biological aging (*101*). Interpretation is further limited by heterogeneous detection methods, single–time point measurements, frequent co-occurrence with CHIP, and predominantly European ancestry cohorts, all of which limit generalizability. Longitudinal studies combining standardized assays; repeated sampling; diverse cohorts; and integrated genomic, chromatin, and epigenomic profiling are needed to define the trajectories, biological consequences, and clinical relevance of mLOY, mLOX, XCI escape, XCR, X-skew, and other age- and disease-related changes in sex chromosome biology. Studies should also clarify how race, ethnicity, social determinants of health, and environmental exposures influence these processes.

Future studies should define functions of individual X- and Y-linked genes, X-linked imprinting, X-X and X-Y interactions across tissues, aging, and diseases while incorporating sex chromosomes into genome-wide association studies (GWASs) and other genomic analyses. Defining the consequences of XCI escape, mLOX, and X-skew will improve our understanding of heterogeneity and individual trajectories in female aging and disease. Given convergent evidence implicating *KDM6A* across diseases, cell type–specific targeting of this pathway may offer new medicines. Finally, clinical trials must be designed to detect whether treatments differ in XX and XY individuals and establish standards for sex-aware biomarker validation, reporting, consent, and equitable implementation.

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Conclusions

The X and Y chromosomes govern fundamental biology throughout the lifespan and across diseases. Although we have begun to uncover their mechanisms, much remains unexplored. Substantial advances would be possible if we rigorously tackled sex chromosomal–driven mechanisms of aging and disease—expanding cutting-edge toolkits in genomics, single-cell biology, functional studies, informatics, and machine learning—and deeply interrogated meaningful sex differences in the human condition. These discoveries may reveal robust biomarkers and therapeutic targets across aging, neurologic and psychiatric disease, neurodevelopmental disorders, autoimmunity, cardiovascular disease, cancer, and beyond in a sex-aware manner. We would advance clinical precision, improve equity and health outcomes, and deepen our understanding of biology to drive transformative gains for science, medicine, and the human condition.

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REVIEW

Reframing menopause and menopause hormone therapy as opportunities for healthy aging

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Menopause is an important biological transition with up to 72 symptoms that shapes aging trajectories across organs and encompasses ~40% of the female lifespan. Although menopause awareness is increasing, research on strategies for menopause symptom management and optimizing health in the postmenopausal period is lacking. This Review highlights that menopause characteristics, such as type, timing, and symptom burden, influence health outcomes and later-life disease risk. Similarly, menopause hormone therapy comes in >200 varieties that differ by dose, formulation of estrogens and progestogens, and timing, each with their own distinct risk-benefit profiles. These distinctions are often missed in the literature but have considerable repercussions for outcomes. Optimized menopause care necessitates a precision medicine approach that considers menopause symptoms, genetics, and previous hormone exposures.

Menopause occupies up to 40% of the average female lifespan, yet cultural silence around female aging and a paucity of dedicated research have impeded scientific scrutiny of menopause and allowed dogma to substitute for evidence. Menopause is accompanied by decreases in most ovarian hormones, such as estradiol and progesterone, and increases in gonadotropin hormones such as follicle stimulating hormone (FSH), due to interruptions in the negative feedback loop of the hypothalamic pituitary gonadal (HPG) axis. The transition to menopause represents a critical inflection point in aging, characterized by systematic neurological (*1*), inflammatory (*2*), and metabolic changes (*3*) that unfold alongside dynamic, organ-specific changes (*4*). Menopause, independent of chronological aging, is associated with accelerated aging clocks across multiple organs (*1*). Despite this, menopause research remains neglected. Analyses show that 0.2% of the total federal investments in health research over 15 years went to menopause-related questions in North America (*5*, *6*). Although research is scarce, extant data show that menopause is not a uniform experience, and these differences can affect health beyond the menopausal transition. Menopause differs in age of onset, type (induced versus spontaneous), length of transition (perimenopausal period), and symptoms (*7*). Furthermore, menopause hormone therapies (MHT)—first-line medications for treatment of menopause symptoms—vary in timing, formulation, dose, and route of administration (*8*). Both menopause and MHT heterogeneity matter for health outcomes but remain under-researched. Forgoing study of this heterogeneity does a disservice to the health of the community and to science. This, coupled with the limited scientific investment and gendered taboos surrounding aging (*9*, *10*), has hindered progress, and allowed long-standing assumptions of menopause health to persist largely unchallenged. In this Review, the use of female is used to describe those with biological attributes associated with the female sex (XX chromosomes, ovaries), whereas woman is a gendered term that reflects social categorization including self-identification, and is used where consistent with the sampling characteristics of the cited studies.

The symptoms accompanying menopause are diverse and have a pronounced effect on quality of life. Vasomotor symptoms (VMS), such as hot flashes and night sweats, are experienced by 80% of the population during the menopause transition, but more than 70 additional symptoms have been reported, with an average of seven symptoms experienced at a given time (*11*–*13*). The diversity and combination of symptoms have fueled debate over whether they are attributable to menopause or natural aging. However, the abruptness of symptom onset and its close temporal relation with the menopause transition signals menopause-specific effects beyond pure association with age. Given that estrogen receptors (ERs) and the progesterone receptor (PR) exist on every organ (e.g., skin, lung, kidney, heart, brain) and most tissues, it is not surprising that the abrupt hormonal changes accompanying menopause can produce such a diverse range of symptoms. Individual differences in receptor type [ERα, ERβ, G protein–coupled ER (GPER), PR isoforms], density, and sensitivity of these receptors likely contribute to the marked variability in symptom burden. For some, the menopausal transition is barely noticeable, but for others, it severely affects their quality of life. Importantly, even in those without symptoms, menopausal timing can influence future disease risk (*14*, *15*).

Despite the availability of highly effective treatments for VMS, including MHT and neurokinin B antagonists, fewer than 15% of symptomatic women receive treatment (*13*, *16*). This treatment gap likely reflects a broader lack of menopause education and support. In Canada, one in two women feel unprepared for the menopause transition and 38% report feeling alone while navigating it (*13*). The consequences extend beyond health, as ~10% of women leave the workforce because of menopause-related symptoms, including mental health challenges (*13*) and sleep disturbances (*17*). For up to 80% of individuals, menopause symptoms include changes in mood and anxiety levels, which can exacerbate existing mental health challenges or trigger first onset of certain psychiatric disorders (*18*, *19*).

Despite its biological and social impact, menopause is rarely treated as a key opportunity to support healthy aging and quality of life. Human females will spend ~40% of their lifespan in postmenopause, with up to 10 years spent in the menopausal transition. There is growing recognition that menopause is not simply a transition but a pivotal period that affects future health (*14*). Menopause features such as its timing, type, symptom profile, and burden provide valuable information about future risks for osteoporosis, metabolic syndrome, cardiovascular disease (CVD), and dementia, and may help identify those more likely to benefit from early prevention and targeted interventions (*3*, *20*–*22*). Thus, menopause, in all its complexity, represents an important inflection point for future aging-related concerns. This Review outlines the diverse characteristics of menopause and their differential contribution to brain-related health. We explore the need to embrace heterogeneity of MHT (type, timing, administration) to reduce menopause symptom burden and optimize healthy aging, and to highlight opportunities for personalized MHT treatments that consider genetics and past reproductive history.

Menopause characteristics: type, timing, and symptoms

Spontaneous menopause is characterized by declines in estradiol and progesterone levels and increases in gonadotropic releasing hormones (GnRH) such as FSH and luteinizing hormone (LH), due to disruptions in negative feedback from the HPG axis (Fig. 1 and Box 1). Menopause can occur spontaneously, or it can be induced either

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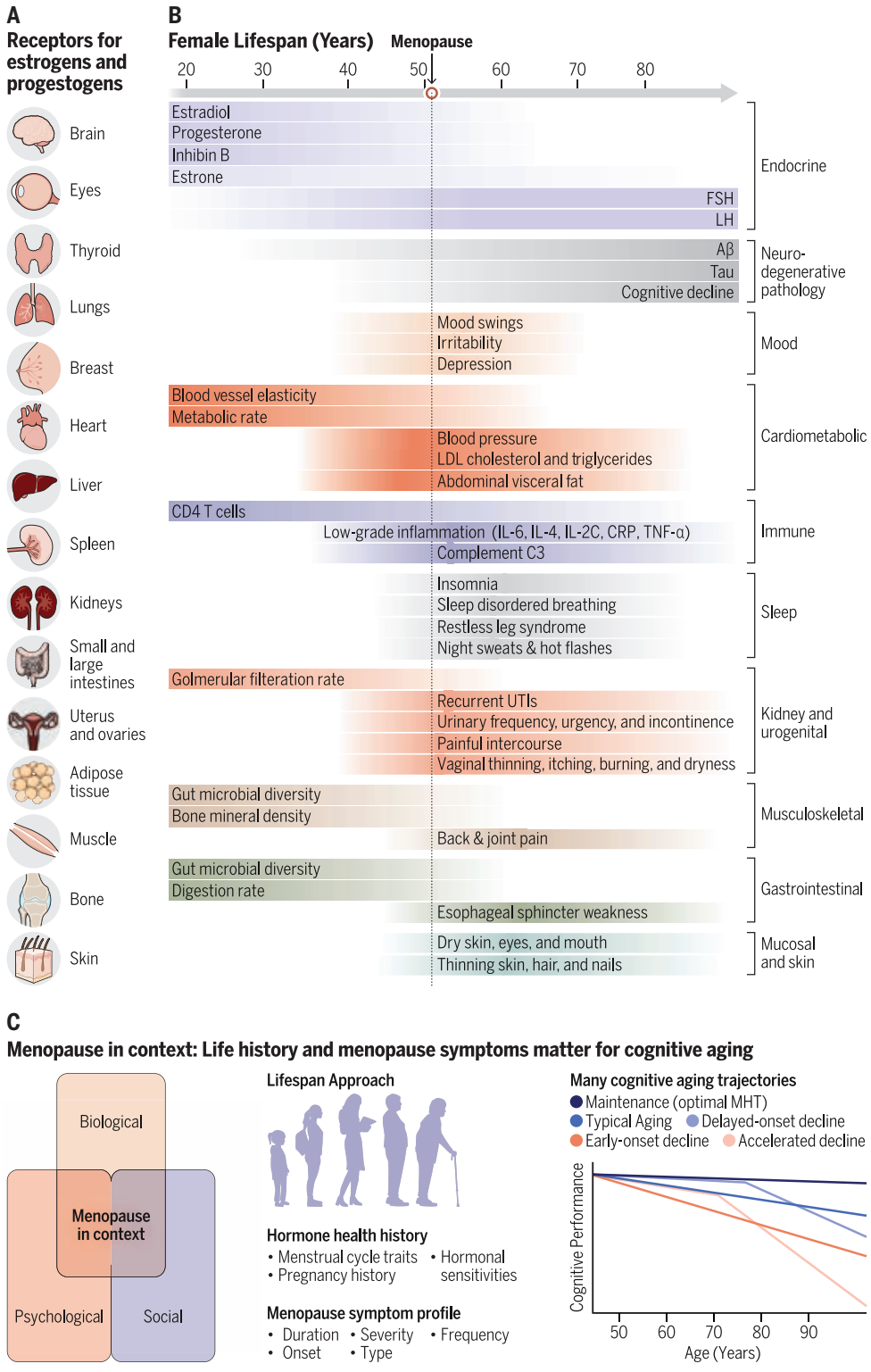

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Fig. 1. A broad overview of factors across the lifespan that need to be considered to understand menopause. (**A**) Nearly all major organs and body systems express receptors for estrogens and progeserone. (**B**) Menopause is a whole-body transition, with more than 30 recognized symptoms. Given the widespread receptor distribution for ovarian hormones, it is not surprising that the effects of menopause extend to the brain (memory, mood, AD-related pathology), cardiometabolic changes, immune system, sleep disturbances, kidney function and urogenital system, musculoskeletal system, gastrointestinal system, mucosal membranes, and skin. These effects are independent of chronological age. This figure provides a general overview of when these symptoms occur; however, there is interindividual variability in the onset, duration, severity, frequency, and type of symptoms experienced. In contrast to common belief, these symptoms can persist well into the postmenopausal period. (**C**) Menopause cannot be understood in isolation. Consideration of biological, psychological, and social factors is needed.

surgically (bilateral oophorectomy) or chemically (GnRH agonists, chemotherapy) (*7*, *23*). Although induced menopause is defined by the time of treatment/surgery, spontaneous menopause is defined retrospectively after 12 consecutive months without menstruation. The average age of spontaneous menopause is ~51 years (*24*). The age at spontaneous menopause is determined by both genetic, and environmental factors, e.g., earlier age at first pregnancy and smoking are associated with earlier spontaneous menopause onset (*24*, *25*). Limited evidence also suggests that earlier menarche may be linked with earlier onset of spontaneous menopause (*26*).

In North America ~10% of menopause is surgical, 2% chemical, ~74% spontaneous, and 10% of cases are spontaneously early (prior to the age of 45) or premature (prior to the age of 40) (*23*). These menopause types differ in both age of occurrence and abruptness of ovarian hormone decline. Even when spontaneous and surgical menopause are compared at the same age, they have both shared and distinct biological consequences (*7*, *27*, *28*). Other reproductive surgeries, such as hysterectomy (uterine removal), do not cause menopause but are associated with earlier menopause onset (~4 years) and a greater risk for early-onset dementia (*29*). The combination of hysterectomy with oophorectomy is associated with an even greater risk of dementia than either surgery alone (*30*, *31*), consistent with rodent studies (*32*). These findings point to distinct biological signatures of oophorectomy, hysterectomy, and spontaneous menopause.

The ovary, an endocrine organ, not only produces hormones but also acts as an immune reservoir (*33*). The postmenopausal ovary is not quiescent as it still produces androgens and inflammatory signals that differ by past parity (*34*, *35*). The mechanisms of spontaneous menopause are still debated (see Box 2).

Menopause timing carries important health implications. Early surgical menopause, premature menopause and primary ovarian insufficiency are reliably associated with increased risk of mortality (*36*), dementia (*37*, *38*), CVD (*39*), osteoporosis/fracture (*40*), gynecological cancers (*41*, *42*), rheumatoid arthritis (*43*),

Box 1. Beyond ovarian hormones

Multiple steroid, peptide, and protein hormone levels change considerably during the menopause transition (Fig. 1). Because of negative feedback from the HPG axis, it becomes challenging to pinpoint which hormone(s) may cause which menopause symptoms. Estrogens and progesterone have received considerable attention, but FSH and LH levels rise markedly across the transition. Estradiol and FSH levels vary substantially across the transition by ethnicity and body mass index (*191*). Some have suggested that targeting FSH may prove fruitful for healthy aging. However, GnRH agonists reduce FSH and LH levels but paradoxically increase depressive symptoms, bone density loss, and vasomotor symptoms (*192*). Nonetheless, the dynamic interplay between FSH and estradiol deserves more attention.

Other sources of hormones. Steroid hormones can be locally derived in tissues or created de novo through cholesterol, provided the required enzymes are present (*193*). Estrogens can be produced through the ovaries, adipose tissue (primarily estrone synthesis), muscle, and the gut microbiome through the estrobiome (*194*). Few studies have investigated the effects of different sources of estrogens on brain health, with some finding more correlations with estrogens found in cerebral spinal fluid compared with those in blood (*195*). Steroid hormones are lipophilic and hydrophobic and thus present in greater concentrations in lipid-rich tissue such as the brain. Some attention has been paid to quantifying available steroid hormones as bound versus free steroid hormones. However, steroid hormones require a carrier protein when contained within plasma or serum. Steroid and peptide hormones can be challenging and expensive to assay, and caution must be exercised when comparing across studies for absolute measurements. Furthermore, greater value may lie in understanding individual responses to hormone shifts rather than absolute levels.

Box 2. The postmenopausal ovary is not quiescent

Previously, the postmenopausal ovary was deemed quiescent, and the mechanisms regulating spontaneous menopause timing were thought to be solely related to the depletion of ovarian follicles. However, signals of cellular senescence (*22*,*25*), vascular changes, and shifts in immune cell populations occur with menopause in the ovary (*34*,*35*). Anti-Mullerian hormone and inhibin B decline across the transition with different timescales, the former of which is used as a marker of ovarian follicle reserve. However, the exact mechanisms of menopause are still not well understood, but are likely multifactorial, encompassing both genetic factors and life-course environmental, structural, and health exposures (e.g., smoking is associated with earlier menopause) (*22*,*25*). Although ovaries have a reservoir of egg follicles at birth, only 0.1% are released across the lifespan (*196*), with degradation of follicles occurring from birth, suggesting other mechanisms may be at play. The discovery of ovarian stem cells (OSC) across species, including humans, has incited much discussion in the field (*197*). This discovery was initially met with skepticism, but multiple researchers across the globe have provided irrefutable evidence that OSC exist, with research focused on their application for improving fertility. Single-cell RNA sequencing studies show that the aging ovary across species shows changes in immune cell populations as well as extracellular matrix remodeling (*198*,*199*). Parity can affect menopause age (*154*), the trajectory of aging (*154*) and influences the ovarian transcriptome in middle age (*34*), providing yet another clue that the ovary has multiple signaling pathways. However, might OSC hold the key for menopause initiation? How does menopause affect OSC and other stem cells across the body? There are many unanswered questions but one thing is clear, the ovary is not just a reservoir of eggs, providing fertility capability, and its role in health does not end with menopause.

chronic pulmonary disease (*44*), and lung cancer (*45*). Beyond incident dementia, earlier menopause is associated with faster cognitive decline, regional brain atrophy, increases in Alzheimer's disease (AD) pathology, decreased global brain activity, cerebrovascular changes, and increased risks of other neurological disorders such as Parkinson's disease and amyotrophic lateral sclerosis (*45*). Notably, AD neurodegenerative pathology begins decades before clinical presentation, making the menopause transition a potential window of opportunity for interventions aimed at supporting later-life brain health and reducing neurodegenerative disease risk (Fig. 1).

Despite consistent evidence linking menopause timing to later-life dementia risk, the specific mechanisms underlying these associations remain unclear. Accumulating data report that earlier menopause is associated with elevated susceptibility to multiple dementia risk factors, including both AD and non-AD pathology, cardiovascular risk burden, and synaptic dysfunction (*20*, *46*–*48*). Earlier menopause is also associated with accelerated biological aging on epigenetic and proteomic clocks (*28*, *49*). Plasma proteomic signatures of earlier menopause reflect up-regulations in inflammatory signaling, extracellular matrix remodeling/catabolism, and apoptosis (*28*). It is unknown whether earlier menopause causally induces faster rates of aging to increase dementia risk, or whether earlier menopause and elevated neurodegenerative risks are both consequences of an upstream factor that accelerates aging.

Menopause symptoms

Despite a predominant focus on VMS, menopause encompasses at least 72 symptoms, with an average of seven symptoms experienced concurrently (*11*, *13*). Although the average duration of menopause symptoms is 7.4 years (*50*), at least ~33% of older postmenopausal people (ages 65 to 79) will continue to experience VMS (*51*, *52*). In North America, both brain fog (difficulties in memory, attention, and concentration) and irritability/mood/emotional disturbances are common during the menopause transition, reported in up to 80% of cases (*19*, *53*). Depressive symptoms associated with menopause are experienced by ~10 to 30% of individuals with no previous history of depression compared with 45 to 65% of those with a previous history of depression (*19*). In those with bipolar disorder, symptom worsening has been reported during the menopausal transition (*19*) and evidence suggests a bidirectional relationship between anxiety and increased VMS (*19*). Moreover, ~64% of individuals in the menopause transition report "not feeling like myself" more than 50% of the time, reflecting the broader mental health impacts of menopause (*54*). Despite the high prevalence of mental health symptoms associated with menopause, they are under-recognized by healthcare practitioners, who prioritize management of physical symptoms (*55*). Sleep disturbances, which often co-occur with brain fog and mood disturbances, are also common and can be exacerbated by nighttime VMS symptoms, which fragment sleep (*12*, *56*). Overall, the most frequently reported menopausal symptoms involve the brain.

Studies have found nonclinical declines in a range of cognitive domains across the menopausal transition, including episodic memory (*57*–*59*) and executive function (*57*, *60*). These declines in cognitive performance are temporary, stabilizing after perimenopause (*61*). However, higher menopause symptom burden (*53*, *62*) and more severe VMS, particularly at night have been associated with worse cognitive performance (*63*, *64*) as well as increased plasma amyloid beta ratio, white matter hyperintensities, and decreased quality of life (*65*, *66*). Finally, severe and frequent menopausal symptoms are associated with increased odds for developing mild cognitive impairment (MCI), a

prodromal state to dementia (*67*). Experiences of menopause symptoms, cognitive or otherwise, may thus have important implications for long-term brain health. Given the link between VMS, MCI, and brain pathology, increased education of both the public and health care practitioners is needed, alongside recognition of the potential for menopause therapy in improving quality of life and preserving brain health through VMS reduction.

Most women are not aware of the diversity of symptoms that may accompany the menopause transition. Although most (77 to 84%) report recognizing VMS as symptoms of menopause, less than half are aware that urogenital symptoms, memory issues, anxiety, and body aches may also accompany the transition (*13*). How we query menopause symptoms matters not only for research but also patient awareness of potential symptoms. The menopause rating scale, a common tool for measuring menopausal symptoms, lists only 11 symptoms. Reproductive health tracker apps may help identify other symptoms and can be useful in menopausal staging (*12*). Those with more severe VMS are up to three times as likely to experience depressive symptoms (*68*), metabolic disturbances, including earlier onset of diabetes (*69*), and joint pain (*69*). This suggests that symptoms are interconnected and likely do not operate in isolation, making accurate tracking and clustering of symptoms especially important.

The menopause transition is also marked by weight gain, a shift to more visceral fat, decreases in lean muscle mass, and decreases in bone density, changes that increase systemic inflammation and risk for metabolic syndrome (*70–72*). Visceral fat is linked to reduced cognition and brain network connectivity across domains postmenopause (*73, 74*). Those with polyendocrine metabolic ovarian syndrome may be at increased risk of negative metabolic, inflammatory, and cognitive changes during the menopausal transition (*75, 76*), even though they report fewer menopause symptoms than unaffected individuals (*77*). Menopause-associated metabolic changes are critical to consider as a part of menopause care for maintained brain health.

Ethnic and cultural considerations of menopause are also needed for informed menopause care. Black and South Asian populations in North America experience earlier ages of spontaneous menopause and Black individuals are more likely to undergo oophorectomies and hysterectomies (*78, 79*). Furthermore, Black Americans experience more frequent and severe VMS during the menopause transition compared with their white counterparts, and receive fewer prescriptions for MHT (*78–80*). East Asian populations are more likely to report musculoskeletal symptoms, commonly frozen shoulder, compared with white individuals (*81, 82*). Overlooked in the literature is that there are cultural differences in the attitudes surrounding menopause, or openness to disclose symptoms (*83*), as well as cultural differences in menopause education and treatment approaches (*84*). Less is known about racial/ethnic variations in menopausal symptoms in multicultural countries (*84*). Recent findings have contested earlier assumptions that East Asian populations are less likely to report VMS (*81, 82*). There may also be ethnic differences in symptom profiles across the menopause transition (*85*), either reflective of population differences or psychosocial factors (*86*) that affect symptom reporting and conceptualization. Notably, experience of menopause symptoms can be shaped by gendered expectations, attitudes, and lived experience, which vary across cultural contexts as well as between individuals (*87, 88*).

Menopause hormone therapy

MHT consists of estrogens and/or progestogens to supplement the decline in ovarian hormones associated with menopause. The first-line treatment for VMS reductions is MHT. VMS are directly related to kisspeptin -neurokinin b-dynorphin (KNDY) neurons in the infundibular bundle of the hypothalamus (*89*). Thus, neurokinin B (NKB) antagonists also reduce VMS symptoms and offer a nonhormonal option when hormones are contraindicated/not desired (*89, 90*). Although there have been no RCTs directly comparing NKB antagonists with MHT formulations, network meta analyses found that NKBs were moderately effective compared with MHTs, which were highly effective in reducing VMS severity (*91, 92*). Another potential use for MHT is preserving bone density for osteoporosis prevention, a condition that affects one in three women after age 50 (*93*). Systemic MHT-containing estrogens prevent fractures by 20 to 40%, with greater benefits observed with higher doses (*93*). However, the overall uptake of MHT remains low [USA: 1.7% of women aged 40 or older (~13M) using data from medical/pharmacy insurance claims in 2023; Sweden: 3.8% of women aged 45 to 60 years in a population cohort study (~1M) in 2020; (*94, 95*)]. Research and education remain at the forefront of bridging this gap for both the public and physicians.

MHT has a controversial history as a result of the Women's Health Initiative trial. As the largest randomized control trial (RCT) of MHT, the findings continue to have a major influence on standards of menopause care. They found that oral conjugated equine estrogens (CEE) combined with medroxyprogesterone acetate (MPA), but not oral CEE alone, led to an increased risk of CVD, breast cancer, and dementia compared with placebo, leading to the premature termination of the trial due to these negative outcomes (*96*). Although this trial was designed to examine whether MHT promoted optimal health in older women (10+ years postmenopause), considerable criticism has been levied toward its approach centered on three major areas: the critical window hypothesis, healthy cell bias, and type of MHT used [reviewed elsewhere (*8, 97–99*)]. However, the scientific and medical community has applied these negative findings from one type of hormone therapy (oral CEE with or without MPA) to all MHTs, leading to a >60% decline in MHT prescriptions and the addition of blackbox warnings on all MHT products cautioning against their use due to risk of CVD, cancers and dementia (*100*). Notably, there are >200 MHTs that vary in hormonal formulation, dose, route of administration as well as timing (duration, initiation relative to menopause) and how they are used (cyclically versus continuously; see Fig. 2). Research shows that all these variables can contribute differently to health risks and benefits, illustrating that there is not a simple one-size-fits-all approach to menopause care.

The federal food and drug administration in the US announced the removal of MHT blackbox warning in 2025 (*101*). This decision has received some criticism from the International Menopause Society, which supports the removal of warnings from vaginal estrogens but not systemic MHTs. Other RCTs since the Women's Health Initiative have generally not found clinically significant effects of MHT on cognitive performance (*102–104*), or CVD risk (*105, 106*), but do find positive effects on bone density (*107*). Although some trials show increased risk or no change in CVD events with CEE or estradiol, evidence suggests that initiation closer to the menopausal transition may reduce CVD risk and subclinical atherosclerosis (*108–111*). Additionally, prolonged VMS are linked to increased CVD risk factors such as hypertension, insulin resistance, and adverse lipid profiles, as well as increased later-life CVD risk, highlighting the importance of effective management (*112, 113*). Moreover, these RCTs have not evaluated various doses and formulations of MHT, including low-dose vaginal estrogens and transdermal gels and sprays, among younger, symptomatic postmenopausal individuals or racially diverse cohorts (*114, 115, 116*). More attention is needed to understand the risk-benefit profiles of the full breadth of available MHT options.

Routes of administration: Oral routes of administration are subject to hepatic first-pass metabolism as well as breakdown by the gastrointestinal tract. This has consequences for not only the types and levels of estrogens that reach circulation, but also clotting factors associated with increased risk of venous thrombosis (*117*). Approximately 90% of oral estradiol, depending on how it is conjugated, is converted to estrone, a relatively weaker estrogen. By contrast, transdermal estrogens do

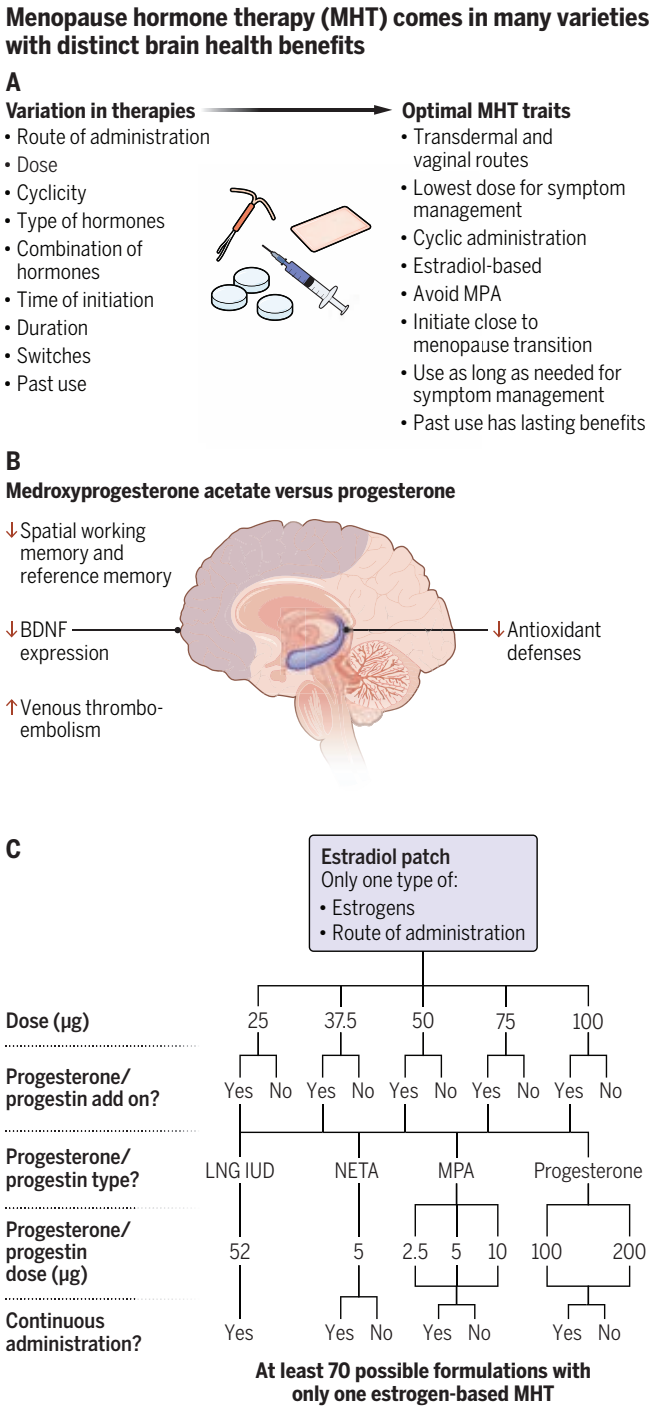

Fig. 2. Menopause hormone therapies are diverse in formulation, dose and timing. (**A**) There are many varieties of MHT, which vary in their risk-benefit profiles and resulting health benefits. Major concerns associated with MHT use are breast cancer and CVD, but these risks vary according to MHT type. Optimal MHT features are outlined, including transdermal and vaginal routes of administration that result in less blood clotting factors that cause inflammation and risk for venous thromboembolism. (**B**) Both human and rodent research suggests that medroxyprogesterone acetate, the progestin used in the Women's Health Initiative trial, has an unfavorable risk-benefit profile. It has been associated with negative brain health outcomes. (**C**) In practice, there are many possible MHT combinations. For example, considering only estradiol with one route of administration (i.e., estradiol patch), there are at least 70 possible combinations given differences in doses, type of progesterone/progestin, type, dose, and cyclicity. However, these many possibilities are not well-captured in research. All MHT research must contextualize findings according to the MHT features that were investigated and the population that was studied.

not undergo hepatic metabolism, require a lower dose to achieve the same estrogenic effects, and show a two- to fourfold decrease in blood clotting risk, compared with oral routes (*118*). Vaginal estrogens similarly circumvent first-pass metabolism and are generally used to treat urogenital symptoms. Although vaginal estrogens are a localized form of hormone therapy, they can exert systemic benefits, particularly with the vaginal ring (*119*), and through downstream reductions in systemic inflammation caused by recurrent urinary tract infections (*120*). Low-dose vaginal estrogens beyond age 65 have been associated with reduced mortality, ovarian cancer risk, dementia risk, and no increased risk in recurrence of ER+ breast cancer (*110*). Thus, route of administration affects levels of different estrogens with important health consequences.

Importance of MHT composition and route of administration extends to cognitive aging. Estradiol-based therapies benefit episodic memory (*121*) and executive function (*121*) in smaller placebo-controlled studies using sensitive cognitive assessments rather than global cognition and shorter or cyclical treatment durations (*122*–*124*), whereas data using CEE is less consistent. Although the combination of CEE and MPA has been associated with either neutral or negative cognitive outcomes, depending on age of initiation (*125*, *126*), positive outcomes on cognition have been seen with combination therapies involving micronized progesterone (*127*, *128*). Route of administration also matters, with recent work showing more pronounced positive effects of transdermal estradiol on episodic memory (*129*). These differences in MHT effects by type and route of administration may help explain the overall mixed findings in the literature on MHT and cognition as many studies combine MHT types into a single user group (*121*).

Estrogens and their receptors: Estrogens are released from the ovaries, but there are other sources (Box 1). Three types of ER exist which are located either in the cell membrane or in the nucleus/cytoplasm. Different densities of ERs reside across tissue types (*130*), and both endogenous and exogenous hormone exposure can influence ER density (*131*, *132*). Genomic signaling occurs after estrogens bind to nuclear ERα, ERβ, which then homo- or heterodimerize, translocate to the nucleus, and bind to an estrogen response element (ERE). Nongenomic signaling occurs when estrogens bind to membrane bound ERs, which results in downstream cell signaling proteins that may involve subsequent genomic mechanisms. Growth factors [such as brain-derived neurotrophic factor (BDNF)] can independently bind to EREs, complicating signaling. Both estrogens and progesterone influence monoamine and cholinergic transmission and growth factors (e.g., BDNF), thus expanding the multiple signaling pathways by which estrogens can exert their effects (*133*, *134*).

Different estrogens have different binding affinity to the different ERs (*130*, *135*, *136*) Table 1). Of the four estrogens endogenous to humans, estradiol has a high affinity for both ERs and GPER, whereas estrone, which becomes the predominant estrogen after menopause (Fig. 1), has lower binding affinity to ERα and ERβ compared with estradiol (*130*). Both estriol and esterol have low affinities for ERα, ERβ, and estriol is a GPER antagonist (*135*, *137*). CEE contains 10 equine estrogens with the highest concentration of estrone and minimal levels of estradiol (0.5 to 3%). Of the estrogens contained in CEE that are not endogenous to humans, 17b-dihydroequilin has the highest binding affinity to ERα and β, but negligible binding to GPER. All of these different estrogens have the potential to invoke different signaling pathways and subsequent downstream health outcomes, but these differences are rarely considered.

Role of progesterone

Progestogens are prescribed for those with an intact uterus to protect against endometrial hyperplasia associated with unopposed estrogens (*138*). Similar to estrogens, progestogens are often reduced to a single hormone but come in many types, with distinct influences on hormone

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Table 1. Binding affinity of different estrogens to ERα, ERβ, and GPER. Source reflects whether the type of estrogens is endogenous to humans or equine-derived and where the primary source of the estrogens is produced in humans. However, steroidgenesis or de novo production of hormones (from cholesterol) or local production (through aromatization of testosterone) of hormones can also occur throughout the body or brain if the catalyzing enzymes are present. Percent contained in CEE reflects the approximate percentage of the estrogens present in the CEE composition. Relative binding (ERα versus ERβ) reflects whether the estrogens preferentially bind to the α verus β receptor subtype. Both ERs are present in most tissues and organs, including the brain. CEE, conjugated equine estrogens; ER, estrogen receptor; GPER, G protein–coupled receptor.

Type of estrogens	Source	Percent contained in CEE	ERα binding affinity	ERβ binding affinity	GPER binding affinity	Relative binding (ERα versus ERβ)
17β-Estradiol (E2)	Human: ovaries	1%	High	High	High	ERα = ERβ
17β-Dihydroequilin	Equine: derived	10 to 15%	High	High	Limited data	ERα < ERβ
Estriol (E3)	Human: placenta, liver	0%	Moderately weak	Moderately weak	Very weak	ERα < ERβ
Estrone (E1)	Human: adipose, ovaries	50%	Moderately weak	Moderately weak	Negligible	ERα > ERβ
Equilin	Equine: derived	22-25%	Moderately weak	Moderately weak	Limited data	ERα > ERβ
17β-Dihydroequilenin	Equine: derived	1-3%	Weak	Weak	Limited data	ERα < ERβ
Equilenin	Equine: derived	1-5%	Weak	Weak	Limited data	ERα = ERβ
Estetrol (E4)	Human: fetal	0%	Weak	Very weak	Negligible	ERα > ERβ

receptors and brain health. Progesterone binds to two different isoforms of the PR (A or B) and is metabolized to allopregnanolone which is a $GABA_A$ agonist (*133*). Progesterone initially augments the influence of estrogens, but after time, progesterone starts to reverse the effects of estrogens, in part by decreasing the density of ERs and PR across some tissues (*139*). Given these interactions, it is critical to consider that the effects of MHTs containing unopposed estrogens may differ from those of MHTs containing both estrogens and progestogens. Cyclic versus continuous use of progestogens also matters. In rodents, cyclic progesterone with estradiol enhanced reductions in neuropathology outcomes (*140*), whereas continuous progesterone blunted the neuroplastic effects of estradiol (*141*, *142*). Synthetic progestins have different binding properties to PR, androgen receptors, and even glucocorticoid receptors depending on their derivation. Pregnane i.e. (progesterone-derived), including MPA, have antiandrogenic properties, whereas gonanes derivatives (i.e. 19-nortestosterone derived), including levonorgestrel, have androgenic properties (*143*). MPA, which was used in the Women's Health Initiative, has been related to most of the poorer health outcomes in that RCT (*144*) (Fig. 2). Finally, some individuals are more sensitive to progesterone fluctuations across the menstrual cycle (*145*), and this sensitivity likely extends into mid-to-later-life hormonal transitions (see below).

What can influence the effects of different MHTs?

In addition to receptor-level effects, history of endogenous or exogenous hormones and individual hormone sensitivity may affect response to MHT (*146*, *147*). The Kronos Early Estrogen Prevention Study (KEEPS) study found lasting volumetric increases in the dorsolateral prefrontal cortex that persisted several years after transdermal estradiol use (*148*). Past MHT use has also been associated with greater hippocampal and parahippocampal volume relative to never-use (*149*, *150*). However, prior use of hormone therapy is not controlled for in most studies, with 20 to 80% of placebo groups having previously been on hormone therapies, which could affect interpretation of findings (*102*, *103*, *116*).

Exogenous use of hormones (e.g., hormonal contraceptives) beyond MHT may also influence the effects of MHT in later life; however, the history of use is not well-captured in studies. Although lacking in detail regarding past use, studies show that prior hormonal contraceptive use is associated with reduced risk of later-life cognitive impairment and dementia (*146*). Prior hormonal contraceptive use may modify the effects of MHT, with dual exposure associated with greater frontal cortical volumes compared with MHT alone (*146*). Hormonal contraceptives are widely prescribed but are overlooked as a factor in menopause research. Future research should consider the variations in hormonal contraceptive type, age, and duration of use on brain aging.

Switching hormone medications may also play a role in MHT outcomes. Switching is common among hormonal contraceptive users with 33 to 44% of individuals switching/discontinuing contraceptives within the first year (*151*, *152*), and one study found that people switch MHTs 2.4 times on average within a 10 year period (*153*). Consideration of these details may be key for studying the effects of MHT on brain health, especially as past history of exogenous hormone use can have lasting impacts on the brain.

Endogenous hormone exposures such as pregnancy may also play a role (*35*, *154*, *155*). Rodent studies show that past pregnancies modulate effects of acute and chronic estrogen exposure on hippocampal plasticity and cognition in middle age (*156*, *157*). Pregnancy characteristics also influence VMS and the postmenopausal ovarian transcriptome (*35*). Individuals with preeclampsia, gestational hypertension, or who experience preterm birth and small for gestational age experience more VMS and are at an elevated risk for developing later-life CVD and vascular dementia. Although heavily under-researched, puberty is another major endogenous hormone transition that may set the stage for menopausal health (*158*). Thus, consideration of past reproductive experiences throughout the lifespan provides meaningful clinical insight for menopause care.

Toward personalized MHT: influence of genotype

Genetic factors, such as single nucleotide polymorphisms, contribute to variance in hormonal effects on cognition. Across the menstrual cycle, BDNF Val66Met modulates learning and memory performance, with individuals who carry the Met allele showing increased sensitivity to estradiol (*159*). Similarly, the catechol-o-methyltransferase (COMT) COMT Val158Met polymorphism interacts with estradiol and hormonal contraceptives to predict executive function performance, such that Met carriers show improved performance in low estradiol phases and with hormonal contraceptive use, whereas Val carriers show improved performance in high estradiol phases (*160*–*162*). Given these genotype-specific effects of estradiol fluctuations on cognition, genetic effects of MHT on brain health may also be present but have been unexplored.

There is limited work on how genotype can influence MHT effects. Neuroimaging work shows more pronounced positive effects of MHT use in carriers of the APOEε4 allele, the strongest genetic risk predictor for Alzheimer's disease (*163*), including fewer white matter lesions (*165*), larger hippocampal volume (*164*, *165*), and less AD-related neuropathology (*166*, *167*). Genes involved in the production or metabolism

of estrogens are strong candidates for exploring how MHT or menopause can have diverse outcomes across individuals (*168*, *169*). Although there are limited studies, the KEEPS trial found that a genetic variation in *SLCO1B1,* which affects transport of estrogens from the blood into hepatocytes, led to a greater decrease in nighttime VMS with transdermal estradiol (*169*), and that genetic variations in *COMT* and CYP3A4, which affect the metabolism (methylation and hydroxylation, respectively) of estrogens, were associated with severity of somatic menopause symptoms (*168*). These genetic differences may be important in deciphering the mixed literature on neural effects of MHT.

Hormonal sensitivities across hormonal transitions

Menopause is often studied as an isolated reproductive/hormonal transition without consideration of past experiences. Hormonal sensitivity is a term that encapsulates how sensitive an individual is to fluctuating gonadal hormones around hormonal transitions (e.g., menstrual cycle, pregnancy). Individual responses to hormonal dynamics can provide an important clue for later experiences of menopause (*170*, *171*). For example, an earlier age of menopause is noted in those with premenstrual disorders (*172*). Notably, individuals with more severe premenstrual symptoms are more likely to report experiencing more severe symptoms during perimenopause (*172*, *173*). The pattern of consistent mood exacerbations with reproductive transitions across the lifespan suggests that surveying experiences of earlier reproductive/hormonal transitions may be meaningful when providing menopause care. Heterogeneity in response to hormonal fluctuations cannot be overlooked; Some people are sensitive to the rise, and others to the fall, in hormones, suggesting that there are differing subtypes of hormonal sensitivity (*174*). Although this is a burgeoning field, embracing this complexity and greater investment in understanding how hormonal sensitivities influence health long-term could hold the key to discovering early interventions for later-life disease.

Continued menopause care is essential: Menopause symptoms can persist past the menopausal transition, affecting quality of life postmenopause. However, treatments such as MHT are not usually prescribed long term. Continued use of MHT into later life should not be confused with MHT initiation in later life. Fixed time limits for MHT should be reconsidered, as up to 45% of people over 65 still experience VMS (*175*). Beyond VMS, MHT is an effective strategy for preventing bone density loss associated with the menopausal transition, with MHT discontinuation associated with an accelerated decline in bone density (*176*, *177*). Observational studies show that MHT use, particularly estradiol-based therapies, is associated with reduced mortality (*178*). Given the substantial burden of symptoms that persist beyond menopause, the role of continued treatment may need to be reevaluated to better support quality of life in the postmenopausal years. Several medical societies already recommend continued MHT for menopause symptom management, noting that chronological age should not dictate when MHT is stopped (*178*).

Politicized menopause research: Menopause is having a moment across social media, marketing, and commercialization. Commercial and public interest in supplements, apps, and natural remedies increased by ~15 to 20% across the US, UK, and Australia from 2005 to 2025, and the market share is projected to increase further over the next 10 years (*179*–*181*). However, effective, informed menopause care and products depend on quality research. Surveys converge on a consistent message: Women want researchers to prioritize menopause (*182*, *183*), yet menopause research funding has remained at only a fraction of a percent for decades (*184*). One deterrent is the politicization of menopause science, from those who talk about menopause as a diagnosable "condition" that requires treatment to suggestions that any research into menopause management implies pathologization. Menopause is a life stage that all females will experience if they live long enough, yet stigma around female aging has let dogma persist surrounding menopause care. Most medical schools lack a dedicated menopause curriculum (*185*). However, training requires evidence, and the evidence gap is profound. It may be no wonder that people feel dismissed by healthcare practitioners when raising menopause concerns (*55*) and turn to health advocates outside of formal medicine to find answers and relief, fueling a booming supplement market and nonregulated forms of MHT. Most medical societies and healthcare practitioners look to RCTs to inform standards of care. However, RCTs are expensive to conduct, are limited in scope, and are only able to capture a snapshot of the ~40% of the lifespan spent in menopause. In fact, RCTs on MHT and chronic health conditions have limited treatment durations (maximum of 8.1 years, average of ~4.3 years; data from 20 RCTs) (*186*) that do not align with treatment durations in practice (maximum of 14.6 years, average of ~5.4 years based on data from 69 observational studies) (*187*). The consequences of long-term MHT use on health outcomes have not been fully explored, particularly across diverse MHT types used over the lifespan. Furthermore, true double-blinding is challenging for the 80% who experience VMS, which has historically led to exclusion of symptomatic participants who may be more likely to benefit from MHT treatment. Thus, RCTs may exclude symptomatic individuals with VMS who are at greater risk for mood dysregulation, CVD, neurodegenerative disorders, and musculoskeletal disease, and who may be more likely to benefit from MHT treatment. Although RCTs are certainly preferred, there is much to be gleaned from observational and longitudinal cohort studies, especially through new statistical tools (*186*).

Advancing precision care in menopause requires sustained research across menopause stages, embracing the complexity of menopause and MHT characteristics. We encourage clinicians to reframe menopause as a critical life stage that may be accompanied by both persistent physical and mental health changes, requiring continued follow-up care analogous to care devoted to pregnancy and the postpartum period. We encourage researchers to reframe menopause research. Are there really only 11 key symptoms? Do they stop after 5 years? We also encourage researchers to be precise in their language and not overgeneralize terms such as estrogen and MHT, which are not single entities but specific formulations which act differently on hormone receptors. Rigorous menopause research is only impactful if it is translated into accurate and accessible science communication. Menopause and MHT are not one thing, and papers and headlines need to make that clear. Data collection must include more information on hormonal histories and medication use across the lifespan, capturing the diversity of MHT types and prior exogenous hormone exposure. Ethnicity, geographic location, and the broader social context in which a person is going through menopause also need to be considered in both biomedical research and clinical practice. Foundations and federal funding agencies must stimulate the field with funding competitions for paradigm-shifting menopause research.

Conclusions

Women consistently rank menopause as one of the highest priorities for improving women's health (*183*). Menopause research is not a niche area of study as it spans nearly 40% of the lifespan for half of the global population. Its impact extends beyond women, informing a broader understanding of aging and health that can benefit society and families, as well as individuals of all genders, including men (*188*) and transgender individuals (*189*).

However, details matter. Broad strokes in health research seed misinformation and setbacks in science. Diminishing menopause to "just aging" is a disservice to women. Menopause is a full-body, hormonally driven change that, like other endogenous hormonal transitions, affects individuals differently based on their individual biology and life experience. The goal of MHT treatment is not to pathologize or "reverse" menopause, but to support healthy aging and improve

quality of life. Critically, exogenous hormones have different effects depending on type, route, age of initiation, pattern of use, and reproductive health history. This information, along with genotype and sociocultural context, determines menopause characteristics and likely efficacy of treatments. Precision medicine has already transformed breast oncology and is advancing depression treatment (*190*); applying the same principles to menopause is the next frontier in women's health.

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REVIEW

Sex-specific mechanisms of chronic pain

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For decades, we have known that women, especially younger women, are disproportionately affected by pain disorders, including irritable bowel syndrome, endometriosis, menstrual migraines, fibromyalgia, osteoarthritis, and neuropathic pain, all of which degrade quality of life. The adoption of sex as a biological variable in 2016 has begun to transform the field, revealing distinct sex-specific pathways that initiate, amplify, and resolve pain. In this Review, we synthesize recent advances across visceral and somatic pain, highlighting the impact of sex hormones, immune-neural cross-talk, the gut microbiome, and endogenous analgesic circuits. Rather than clinging to the historical standard that simply described the female bias in pain symptoms, these studies begin to explain why females experience heightened pain, paving the path for mechanism-based precision therapeutics.

Epidemiological data on chronic pain indicate that women will suffer from functional pain disorders at a rate of 2:1 compared with men and that in many instances, this ratio can climb much higher. For women, persistent, unresolved pain that outlives its utility as an essential warning system is debilitating. Perhaps the most striking example is chronic pelvic pain in women, affecting three major visceral organs: the gut, bladder, and reproductive organs. Many of these insidious pelvic pain conditions, including irritable bowel syndrome (IBS), endometriosis, interstitial cystitis, fibroid-associated pain, and vulvodynia, are episodic, unpredictable, and affect women during their reproductive years. Menstrual migraines and fibromyalgia experienced by younger women are equally debilitating. Later in life, the higher prevalence of osteoarthritis and bone loss in women compromises their mobility, leads to frailty, and accelerates aging. Although multiple factors initiate pain sensations in women, the fluctuation of ovarian hormones beginning at puberty appears to initiate many of these female-specific and female-prevalent pain conditions. Given the relatively short length of this Review, we have intentionally focused on recent progress that has occurred or is likely to emerge over the next 5 years. We have chosen to highlight progress in two areas of research, visceral pain and somatic pain (Fig. 1), that begin to define how hormonal signaling intersects with epithelial, immune, microbial, and neuronal pathways to generate female bias in pain disorders. Unfortunately, there continues to be a large disconnect between public awareness of pain experienced by women and basic research efforts aimed at treating female-biased debilitating pain disorders. Our goal is to shine some light into the large, unexplored black box of chronic pain experienced by millions of younger and older women worldwide.

Sex differences in visceral pain

Shared pain across pelvic organs is one of the signature features of visceral pain disorders in females. Notably, two visceral pain disorders—IBS and interstitial cystitis/bladder pain syndrome (IC/BPS)—are prevalent in women (Fig. 1, bottom right). These two disorders are often linked to endometriosis, a female-specific condition characterized by initial episodic bouts of inflammatory pain that transform into chronic, persistent pelvic pain. Fifty percent of women with IC/BPS and roughly 30% of women with IBS also suffer from endometriosis (*1*). Menstrual cycle–related changes in visceral pain sensitivity are well documented in IBS (*2*). Experimental studies show that whereas colonic and rectal pain sensitivity remain largely stable across cycle phases in healthy controls, women with IBS exhibit clear cyclical variation in visceral pain thresholds (*3*). Similar episodic patterns are observed in endometriosis (*1*), with pelvic pain often intensifying during menstruation and in some patients during ovulation. This close link underlies clinical management that suppresses the hypothalamic-reproductive axis, counteracting estrogen excess and progesterone resistance (*4*). The transition from episodic to chronic endometriotic pain is potentially driven by peripheral and central sensitization (*5*). That ovarian hormones influence sensory processing across multiple visceral organs is supported by the fact that women with IC/BPS also report perimenstrual increases in bladder pain (*6*). The frequent overlap among IBS, IC/BPS, and endometriosis raises the possibility that common hormone-sensitive cellular pathways contribute to pain across pelvic organs. To that point, no obvious sex differences were noted when comparing visceral sensitivity between healthy women taking oral contraceptives and men (*7*). Deciphering the mechanistic underpinnings and common logic of these pelvic pain–modulatory pathways represents a research area that will benefit from interdisciplinary approaches spanning multiple fields.

Sex-specific mechanisms driving visceral sensitivity

Female and male mice exhibit distinct baseline states of visceral sensory activity. Intravital calcium imaging revealed higher baseline activity in colon-innervating dorsal root ganglion (DRG) neurons in females versus males (*8*). In males, isovalerate (ISV), a bacterially derived short-chain fatty acid (SCFA), increased the activity of sensory enterochromaffin (EC) cells in the gut epithelium to sensitize downstream mucosal afferents (Fig. 2). EC cells then release serotonin to excite nearby 5-HT_3 receptor (5-HT_3-R)–expressing sensory nerve fibers. Females, however, were largely insensitive to acute ISV treatment, and chemogenetic activation of these serotonergic EC cells produced greater visceral hypersensitivity in males than females (*8*). By comparison, silencing EC cells elicited a greater reduction in pain responses in females. Collectively, these findings suggest that females operate from a heightened basal sensory state, thereby limiting the dynamic range available for further acute sensitization, whereas males retain greater capacity for stimulus-induced sensitization, such as that observed with ISV. Contribution of serotonergic signaling on sex-specific visceral pain appears to depend not only on stimulus-evoked serotonin release but also on mechanisms regulating serotonin availability. In serotonin transporter (SERT)–deficient rats, impaired serotonin reuptake produced female-specific visceral hypersensitivity driven by enhanced 5-HT_3-R signaling (*9*). Repeated or sustained serotonergic signaling by EC cells may therefore drive long-term plasticity within visceral sensory circuits, transforming transient changes in afferent excitability into persistent visceral hypersensitivity.

Beyond epithelial-derived serotonergic signaling, sex differences in visceral pain may stem from the intrinsic properties of sensory afferents, specifically through colonic mechanosensation. Even though expression of the mechanosensory ion channel PIEZO2 (piezo-type mechanosensitive ion channel component 2) is modestly higher in female sensory afferents, chemogenetic activation of Piezo2-expressing colonic sensory neurons induced hypersensitivity only in males (*10*). Together, these findings suggest that serotonergic signaling in the epithelium and PIEZO2 mechanotransduction in sensory afferents are complementary regulators of visceral sensitivity. Whether and how

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serotonergic and PIEZO2-dependent signaling converge within the same sensory neurons or operate in parallel afferent circuits remains an important question for future studies.

The use of chemically induced colonic injury models in male and female mice reveals marked sex differences in disease progression, pain sensitivity, and recovery. For example, intracolonic capsaicin administration resulted in acute hypersensitivity in female mice. However, dextran sodium sulfate (DSS)–induced colitis was far more severe and persistent in males (*11*). Extending these observations, males and females developed distinct patterns of gut microbial dysbiosis and persistent visceral hypersensitivity even after the resolution of DSS-induced colitis (*12*). In females, postinflammatory dysbiosis was accompanied by chronic visceral pain that could be transferred to healthy mice of both sexes by fecal transplantation, implicating the gut microbiome as a mediator of persistent pain. Using the zymosan-induced model of colitis, which activates the innate immune system, hypersensitivity was observed only in males (*13*). These divergent findings using various colonic injury models strongly suggest that sex differences in visceral pain depend on the nature of the initiating insult. Knowing how hormonal signaling pathways intersect with inflammatory pathways may explain why some episodes of intestinal inflammation resolve, whereas others progress to persistent visceral pain.

Hormonal modulation of visceral sensitivity

Recent work has identified a cellular mechanism that may explain the heightened visceral sensitivity observed in females (*14*). Estrogen receptor α (ERα) signaling in the distal colonic epithelium triggers paracrine cross-talk between sparse enteroendocrine cells and nearby sensory neurons, thereby creating a female-specific sensory pain pathway. Specifically, estrogen increased peptide YY (PYY) release from colonic L cells and up-regulated expression of the SCFA receptor *Olfr78*, enhancing L cell responsiveness to bacterially derived metabolites, which reach exceedingly high levels in the distal colon after consuming foods high in fermentable carbohydrates (*15*) (Fig. 2). PYY subsequently increased serotonergic signaling from neighboring EC cells, thereby activating nociceptive 5-HT_3-R–expressing mucosal afferents. These findings not only identify molecular targets through which estrogen can amplify visceral sensory signaling but may also explain why a low-carbohydrate diet is often prescribed for IBS (*15*). Whether similar hormone-sensitive epithelial pathways exist in adjacent pelvic organs to enhance visceral pain remains an open question.

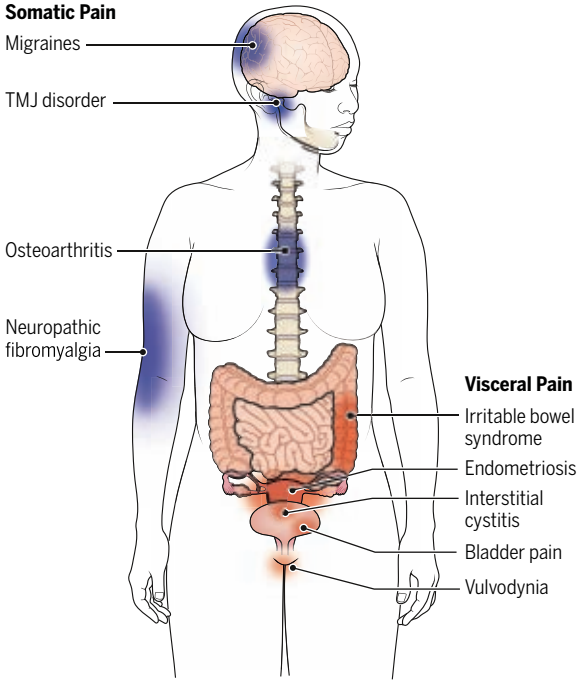

Fig. 1. Visceral and somatic pain syndromes with female prevalence. (Bottom right) Common pelvic pain disorders experienced by women of reproductive age are shown and listed, with the anatomical proximity of these visceral organs shown above the pelvic floor. (**Top left**) Common somatic pain disorders with higher prevalence in women. Of those listed, osteoarthritis is the only pain disorder specific to postmenopausal women, whereas the other listed somatic pain disorders are experienced by women of reproductive age. Although temporomandibular joint (TMJ) disorder is not formally discussed, this type of pain can lead to migraines, facial pain, earaches, and eventually degradation of the oral cavity due to teeth grinding.

Estrogen also directly sensitizes sensory neurons. Using a mouse model of uterine pain, it was demonstrated that catechol estrogens, which are metabolized from estradiol (E2) and estrone, induce nociceptor hyperexcitability by activating the transient receptor potential ankyrin 1 (TRPA1) channel (*16*), leading to increased calcium influx and neuronal firing in TRPA1-expressing sensory neurons. Pharmacological inhibition or genetic deletion of TRPA1 markedly attenuated this hormone-induced hypersensitivity. These findings provide a direct mechanism by which estrogen metabolism can rapidly enhance nociceptor excitability, contrasting the slower genomic actions of estrogen in the sensory epithelium (*14*).

Another dimension of hormone signaling and visceral gut pain includes the role of the gut microbiome in regulating hormone availability (Fig. 2). Indeed, eliminating the microbiota using germ-free mice abolished the changes in visceral sensitivity associated with the estrous cycle and ovariectomy, indicating that microbial communities are required for normal hormone-mediated modulation of nociception (*17*). Supporting this intimate relationship between estrogen and the microbiome, it was recently shown that 3β-hydroxysteroid dehydrogenase (3β-HSD) derived from bacteria directly facilitates conversion to E2, increasing bioactive estrogens in the host (*18*). Furthermore, gut bacteria can perform unexpectedly complex steroid transformations, converting host glucocorticoids into progestin-like compounds under specific metabolic conditions (*19*). In addition, transplanting gut microbiota from women with fibromyalgia—a highly female-prevalent disorder leading to chronic, debilitating musculoskeletal pain—into germ-free mice induces pain and immune changes that mimic those observed in human patients (*20*). Thus, microbial regulation of steroid metabolism represents an additional facet of hormone regulation of visceral pain, especially when considered alongside changing lifestyles and life stages or hormone therapies.

In contrast to estrogen, the role of androgens in visceral pain remains less defined, but emerging evidence suggests that, unlike estrogens, this sex steroid dampens rather than drives heightened visceral sensitivity. Lower circulating concentrations of free testosterone correlate with greater symptom severity in patients with IBS, which is consistent with the finding that androgen depletion induces bowel dysfunction in mice, which is then normalized by androgen replacement (*21*). Mechanistically, androgens signal through the androgen receptor (AR) in enteric neurons and spinal afferents, a signaling pathway required for normal intestinal transit (*22*). The same study also found that β-glucuronidase produced by gut bacteria generates active androgens that signal to enteric neurons, thereby stimulating motility and restoring intestinal homeostasis. To date, it remains uncertain whether AR is present in the colonic epithelium, as was recently shown for ERα (*14*).

The high female bias of visceral pain disorders has been leveraged to develop sex-specific therapies. After stratifying therapeutic outcomes by sex, randomized controlled trials revealed sex-dependent efficacy of the 5-HT_3-R antagonist alosetron, the neurokinin 2 receptor antagonist ibodutant, and the CFTR chloride channel inhibitor

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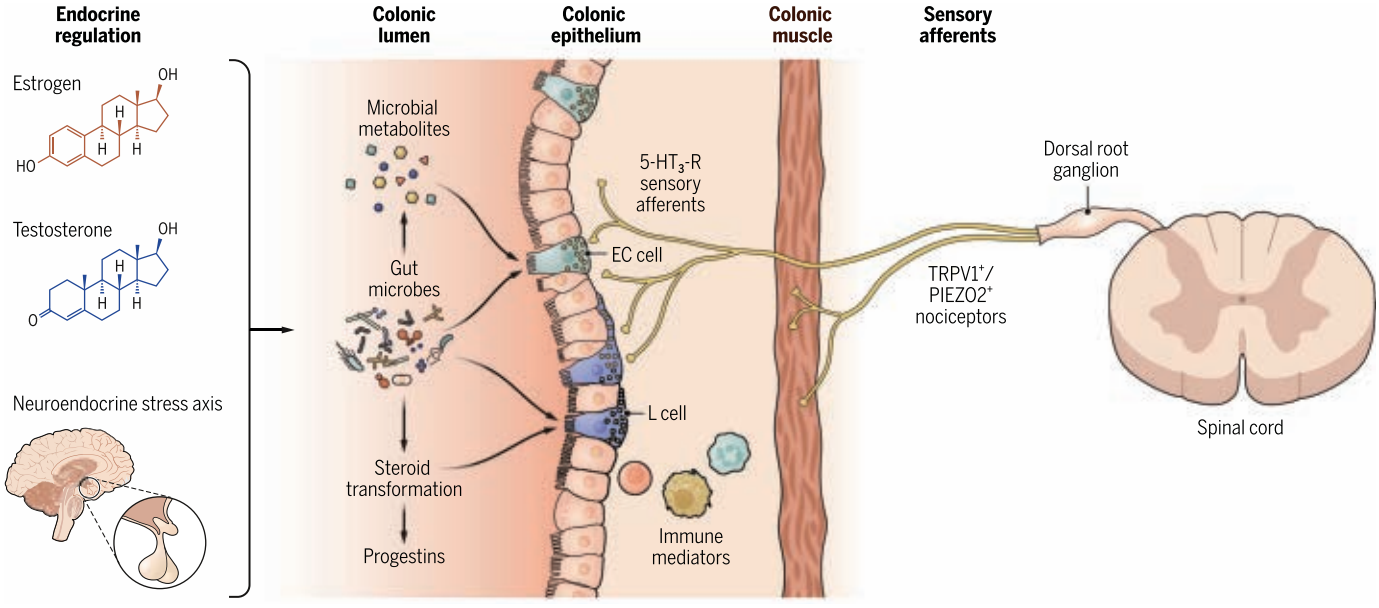

Fig. 2. Sex-specific chronic visceral pain. The illustration depicts recent progress in understanding how sex hormones regulate gut-microbiome-neural communication, thereby influencing visceral sensitivity. Endocrine signals—including estrogen, androgens, and stress hormones of the hypothalamic-pituitary-adrenal neuroendocrine axis—indirectly or directly regulate the gut-brain axis. Within the colonic lumen, the gut microbiota, microbial metabolites, and steroid-transforming microbial activity interact with the intestinal mucosa to modulate epithelial function. Specialized sensory enteroendocrine cells—including enterochromaffin (EC) cells (red) and L cells (blue)—functionally interact, with EC cells directly engaging 5-HT_3-R^+ mucosal sensory nerve endings that send information to the spinal cord. Other visceral afferent populations that innervate the colonic muscle layers also contribute to nociceptive signaling. Adjacent immune mediators further contribute to inflammatory signaling through neuroimmune interactions. Visceral sensory signals, mediated by $PIEZO2^+$ and $TRPV1^+$ nociceptors, can propagate sex-biased visceral hypersensitivity via pelvic and splanchnic nerve input to the dorsal root ganglia, ultimately reaching the spinal cord and brain.

crofelemer in the treatment of IBS (*23*). Furthermore, cognitive behavioral therapy, combined with medical management, was more effective in women than in men. By contrast, evidence for sex-specific treatment effects in IC/BPS remains limited despite the disorder's strong female predominance. For endometriosis, current effective treatments include hormone-based therapies that utilize oral contraceptives, progestins, or gonadotropin-releasing hormone (GnRH) drugs, all of which reduce estrogen-driven lesion activity and pain (*1*, *4*). Surgical excision also provides symptom relief, although in some patients this is transient because pain persists when central sensitization or overlapping visceral pain disorders such as IBS and IC/BPS are present (*1*, *5*). Regardless, the observation that symptoms improve after natural or surgical menopause further supports the critical role of sex hormones in both the pathophysiology and treatment responsiveness of endometriosis. Identifying the molecular aspects of hormone-sensitive pathways in the gut and other pelvic organs may provide opportunities for mechanism-based therapies that can selectively target pain-generating circuits while preserving the physiological functions of systemic sex hormones.

Central and peripheral neuronal differences associated with visceral pain

Limited evidence suggests that sex likely influences pain processing across multiple levels of the nervous system. Single-cell analyses of sensory neurons have identified sex-dependent differences in nociceptor gene expression and translational responses, indicating that male and female afferents may engage distinct molecular programs in response to inflammatory signals (*24*, *25*). At the systems level, neuroimaging studies reveal sex-specific alterations in brain regions and networks involved in pain integration, emotional regulation, and stress responses, including the insula, anterior cingulate cortex, amygdala, and cortico-striatal-thalamic pathways (*26*, *27*). Female-specific epigenetic regulation of glucocorticoid signaling in the amygdala has also been linked to stress-induced visceral hypersensitivity, suggesting that dysregulation of the hypothalamic-pituitary-adrenal stress axis contributes to persistent, unresolved pain (*28*). Beyond pain perception, the fact that central circuits exert descending control over visceral sensations and organ physiology suggests that sex differences in the brain can also influence visceral pain. With newer technologies, it should be possible to generate a comprehensive molecular and cellular framework to understand the complex brain-body dialog between the brain and pelvic organs.

Neuroimmune mechanisms of sex differences in pain

Neuroimmune interactions within the nervous system and its surrounding tissues represent another important mechanism through which sex differences influence chronic pain. The exchange of information between sensory neurons and their microenvironment shapes inflammatory responses and modulates pain processing involving the innate and adaptive immune systems. In this setting, the immune system controls tissue homeostasis and regulates neuronal activity after injury, modulating responses in the DRG and the spinal cord (Fig. 3A). Early studies using intrathecal injections of the endotoxin lipopolysaccharide (LPS) to generate mechanical allodynia (a condition in which innocuous stimuli such as light touch become painful) affected only male mice. This sex-specific pain phenotype required Toll-like receptor 4 (TLR4) activation and was also observed in an inflammatory pain model involving injection of complete Freund's adjuvant (CFA) or in neuropathic pain models involving spared nerve injury (SNI) or transection of sciatic nerve branches (*29*). A subsequent study showed that male mice depend predominantly on spinal microglia to develop and maintain pain hypersensitivity, whereas female mice rely on adaptive immune cells, particularly circulating T lymphocytes (*30*). These studies represented a paradigm-shifting discovery in pain, challenging the

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long-standing assumption that mechanisms identified in male animals are universally applicable across sexes, and prompting others to begin asking how distinct neuroimmune pathways govern pain processing in both sexes. As discussed below, the field has since evolved beyond the original microglia-versus–T cell paradigm, revealing multiple immune cell populations that act within distinct anatomical compartments to promote, modulate, and resolve pain.

Sex-specific neuroimmune pathways in chronic pain

Adaptive immunity, in which T and B lymphocytes elicit antigen-specific responses, has recently been linked to nociception in females (*30*–*33*). Indeed, female mice exhibited higher numbers of circulating $CD4^+$ and $CD8^+$ T cells in the blood and higher expression of T cell markers in the spinal cord after nerve injury, implying that adaptive immunity plays a major role in neuropathic pain for females (*30*). More recently, a female-specific role for $CD8^+$ T cells was identified in this model, whereby pannexin-1 (PANX1), a large-pore membrane channel, was implicated in mechanical allodynia selectively in female mice and rats (*31*). In addition to PANX1-induced release of excitatory transmitters such as adenosine triphosphate (ATP) and glutamate from $CD8^+$ T cells, leptin release from these cells has also been proposed to drive this pain response (*31*). Despite these insights, the anatomical site where T cells exert their pronociceptive effects in females remains unresolved. Whether lymphocytes infiltrate the spinal cord parenchyma or act within the surrounding meninges is unclear. This distinction is particularly relevant given the growing consensus that immune cell infiltration into the spinal cord parenchyma is uncommon in most models of chronic pain.

In contrast to effector T cells that promote pain, regulatory T cells (T_{regs}) suppress pain and exert robust antinociceptive functions in females. T_{regs} are a specialized subset of $CD4^+$ T cells characterized by expression of the transcription factor FOXP3 and are essential for maintaining immune tolerance and limiting excessive inflammatory responses. The role of T_{regs} in sex-specific pain regulation was first demonstrated by a functional interaction between T_{regs} and microglia (*32*, *34*). After peripheral nerve injury, colony-stimulating factor 1 (CSF-1) is released in the spinal cord from the central terminals of injured primary afferent neurons (Fig. 3A). Administering exogenous CSF-1 induced robust microglial activation and mechanical hypersensitivity in male mice, whereas in females, CSF-1 promoted the expansion of T_{regs} within the spinal meninges. These meningeal T_{regs} suppress microglial activation and prevent the development of CSF-1–induced pain hypersensitivity, potentially revealing a female-specific neuroimmune mechanism that restrains the maladaptive transition from acute to chronic pain (*32*). As a key component of an endogenous female-specific analgesic circuit, meningeal T_{regs} are regulated by ovarian hormones, which drive their expansion and activation through the production of the endogenous opioid peptide enkephalin (*33*). T_{reg}-derived enkephalin activates δ-opioid receptors expressed by Mas-related G protein–coupled receptor D (MRGPRD)–positive sensory neurons in DRG, reducing neuronal excitability and attenuating neuropathic pain in female mice (Fig. 3B). Notably, this analgesic function was independent of the canonical immunosuppressive properties of T_{regs}, revealing an unexpected female-specific neuroregulatory role for adaptive immune cells that would be predicted to modulate chronic pain conditions during different life stages, such as pregnancy.

Macrophage-neuron cross-talk in females

In addition to adaptive immune cells, the innate immune system provides rapid, nonspecific defense through macrophages, neutrophils, and dendritic cells. Specifically, macrophages contribute to female-specific pain mechanisms by directly contacting sensory neurons (*35*). An IL-23/IL-17A/TRPV1 (IL-23, interleukin-23; IL-17A, interleukin-17A; TRPV1, transient receptor potential vanilloid 1) signaling axis has been identified in which the macrophage-derived proinflammatory cytokine IL-23 promotes IL-17A–dependent mechanical hypersensitivity selectively in female mice (*35*) (Fig. 3C). This pathway requires ERα signaling in $TRPV1^+$ nociceptors and is suppressed by androgens in males, highlighting the critical influence of sex hormones on neuroimmune communication. Furthermore, use of the chemotherapy drug paclitaxel to induce neuropathic pain increased IL-23 and IL-17A cytokines in females, implicating macrophage-neuron cross-talk as a key driver of sex-specific chemotherapy-induced pain. The identification of this female-specific pathway suggests that sex-dependent neuroimmune mechanisms may shape therapeutic responses, raising the notion

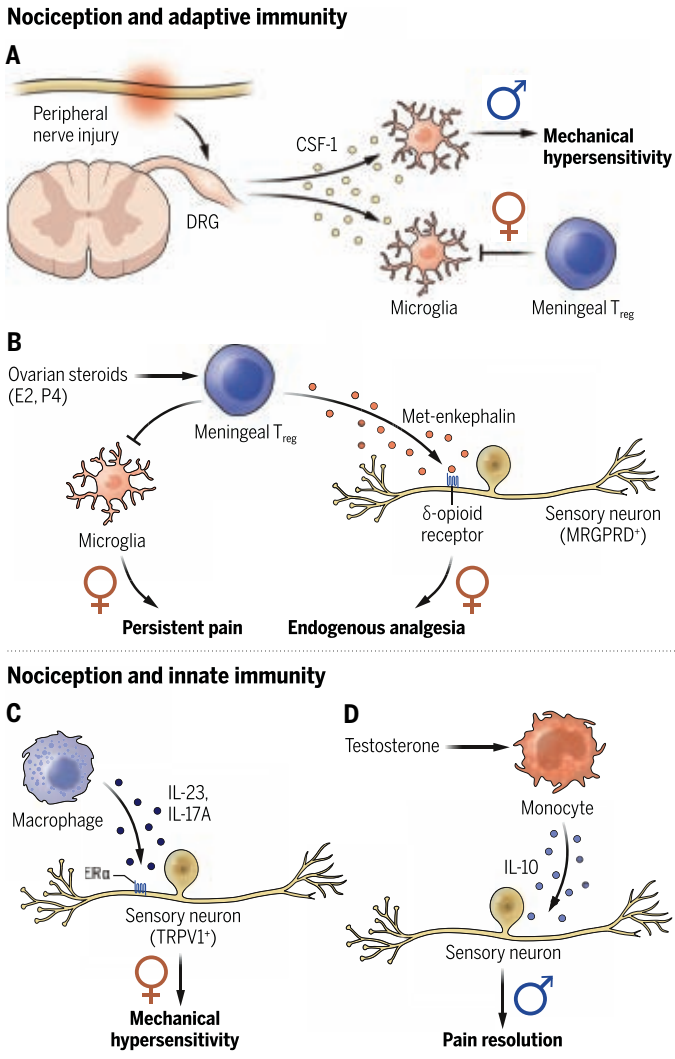

Fig. 3. Sex-specific adaptive and innate neuroimmune circuits underlying chronic pain. (**A**) After peripheral nerve injury, injured sensory neurons release colony-stimulating factor 1 (CSF-1), which engages distinct neuroimmune pathways in males and females. A microglia-dependent pain program leads to mechanical hypersensitivity in males, which is suppressed in females by meningeal regulatory T cells (T_{regs}). (**B**) In females, the ovarian hormones (E2 and P4) suppress nociception by promoting the expansion and activation of T_{regs}, which release enkephalin acting on δ-opioid receptors expressed by $MRGPRD^+$ sensory neurons. (**C**) In females, macrophages drive mechanical hypersensitivity through an IL-23/IL-17A/TRPV1 signaling axis that requires estrogen receptor α (ERα) signaling. (**D**) In males, testosterone increases IL-10–producing monocytes and accelerates pain resolution. Together, these findings demonstrate that sex hormones orchestrate distinct neuroimmune circuits that converge on sensory neurons to either promote or restrain chronic pain.

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that IL-23– and IL-17–targeting biologics could be repurposed for sex-informed pain management.

Recent evidence suggests that in addition to the generation of pain, the innate immune system also plays an important role in pain resolution, a process that is slower in women (*36*). When CFA was used to induce skin inflammation, female mice exhibited delayed resolution of inflammatory pain compared with males, along with a concomitant reduction in the number of anti-inflammatory IL-10–producing monocytes at the site of inflammation (*37*). Notably, administration of dihydrotestosterone to ovariectomized females increased the abundance of IL-10–producing monocytes and accelerated pain resolution to levels observed in males, implicating androgen-dependent regulation of monocyte function as a determinant of sex differences in pain duration (Fig. 3D). The authors further demonstrated similar sex-dependent associations in a cohort of patients after traumatic injury. Comparison between men and women after traumatic injuries (car accidents) revealed that reported pain severity was lower in men several weeks postaccident, along with the detection of higher circulating monocyte counts and IL-10 concentrations (*37*). Thus, for pain resolution, monocyte-derived IL-10 appears to serve as a key endogenous analgesic signal in both mice and humans. Although IL-10 has long been considered a promising therapeutic target because of its potent anti-inflammatory and proresolving properties, its short half-life and the limited efficacy of IL-10–based therapies have hindered direct clinical application. These findings instead suggest that harnessing endogenous IL-10 production through monocyte recruitment may represent a more effective strategy for promoting pain resolution. More broadly, they imply that stimulating endogenous immune-mediated analgesic pathways, rather than administering anti-inflammatory cytokines exogenously, may provide a therapeutic framework for chronic pain.

Pain hypersensitivity in males

In contrast to females, whose pain signaling relies heavily on adaptive immune cells, testosterone-mediated regulation of microglia has emerged as a central mediator of pain hypersensitivity in male rodents (*31*, *38*), contributing to central sensitization through the release of neuromodulatory and proinflammatory factors. Accordingly, microglia-mediated pain mechanisms can be induced in ovariectomized females supplemented with testosterone and in females lacking adaptive immune cells in both mouse models of CFA-induced inflammatory pain and chronic constriction injury (CCI)–induced neuropathic pain models, indicating that both hormonal and immune factors shape sex-specific pain pathways (*30*, *38*). A key mediator of this male-specific response is the metabotropic purinergic receptor P2X4, which is selectively up-regulated in spinal microglia after nerve injury (*30*). In the rat CCI model of neuropathic pain, activation of P2X4 triggers phosphorylation of p38 mitogen-activated protein kinase (MAPK) and subsequent release of brain-derived neurotrophic factor, promoting central sensitization and mechanical hypersensitivity. Sex differences in this signaling pathway appear to be regulated by the transcription factor interferon regulatory factor 5 (IRF5), which directly drives P2X4 expression in male microglia (*38*). However, upstream signals responsible for IRF5 induction after nerve injury have yet to be identified. TLR4 signaling has emerged as a potential upstream mediator of this male-specific microglial activation. As discussed above, intrathecal administration of LPS, a canonical TLR4 agonist, induces mechanical hypersensitivity in male but not female mice, an effect dependent on testosterone (*29*). Notably, this sexual dimorphism appears to be restricted to the spinal cord, because both sexes develop nociceptive responses after intracerebroventricular or intraplantar LPS administration. In males, TLR4 signaling contributes to inflammatory and neuropathic pain, highlighting its role as a key regulator of male-specific pain processing. However, the cellular target of TLR4 signaling that initiates this pathway and the question of whether TLR4 acts upstream of the IRF5-P2X4 signaling axis remain unresolved. The spinal cord–restricted nature of this sexual dimorphism suggests that the critical determinant may extend beyond TLR4 expression itself to the organization of sex-specific neuroimmune interactions within the spinal cord.

After SNI in mice, CSF-1 is up-regulated in sensory neurons and transported to the spinal dorsal horn, where it activates microglia and promotes the development of neuropathic pain in males. CSF-1 receptor (CSF1R) signaling activates the microglial adaptor protein DAP12, which is required for the development of pain hypersensitivity. In parallel, CSF1R signaling also drives microglial proliferation through a DAP12-independent pathway, indicating that microglial expansion and pain sensitization are mediated by distinct downstream mechanisms (*34*). By contrast, as discussed above, female mice are protected from CSF-1–induced microglial activation and pain hypersensitivity (*32*). CSF-1 thus emerges as a critical determinant of sex-specific pain pathways, promoting microglial activation in males while driving meningeal T_{reg} expansion in females (Fig. 3, A and B). Understanding how this divergence is established may reveal fundamental mechanisms governing sex-specific pain biology and identify potential therapeutic targets.

Head, body, and joint pain

Three chronic somatic pain conditions affecting women include menstrual migraines, fibromyalgia, and osteoarthritis (Fig. 1). Even though our working knowledge of these three conditions remains inadequate, we briefly highlight each as an area for future research. Among these conditions, migraine affects younger women two to three times more frequently than men. This female predominance is strongly influenced by fluctuating hormones, with symptoms peaking just before or during menses and beginning at puberty. The frequency and severity of these so-called menstrual migraines dissipate during the second and third trimesters of pregnancy, as placental hormones and sex steroids, including estrogen, rise dramatically (*39*, *40*). Calcitonin gene–related peptide (CGRP) is a central mediator of migraine pathophysiology and a major therapeutic target, with CGRP-blocking monoclonal antibodies demonstrating clinical efficacy, particularly in women (*41*). Preclinical studies in mice revealed that CGRP application to the dura mater induces facial mechanical hypersensitivity in female mice but has little effect in males (*42*, *43*). Female mice also exhibit a greater abundance of CGRP-expressing neurons within the ophthalmic (V1) division of the trigeminal ganglion, providing a potential anatomical substrate for enhanced CGRP sensitivity and migraine susceptibility (*43*).

Despite the central role of CGRP in migraine pathophysiology, few mechanisms have emerged to explain the female-biased CGRP responsiveness and migraine susceptibility. Notably, this effect appears to be independent of ovarian hormones. However, the anterior pituitary hormone prolactin (PRL) amplifies neuronal excitability and migraine-like behaviors in mice and is indirectly linked to CGRP release (*42*, *44*). PRL evokes greater nociceptive responses in females than in males (*42*), which is consistent with higher PRL receptor (*Prlr*) gene expression in female sensory neurons. Indeed, conditional deletion of *Prlr* from sensory neurons attenuates CGRP-induced migraine-like behaviors in females, suggesting that PRL participates in sex-specific CGRP-mediated nociception (*42*). The only wrinkle in this link is that PRL itself is regulated by estrogen, whose concentrations rise during the luteinizing hormone surge of the estrous cycle and just before birth (*45*). Relevant to pain, PRL concentrations also rise with stress or endogenous opioids (*46*), raising the question of why it would be beneficial to amplify pain responses during these female life stages. Clearly, more work is needed to understand the interplay between estrogen, PRL, and CGRP signaling across distinct physiological states to understand the female predominance of headache.

Unfortunately, our current understanding of fibromyalgia offers little insight for the millions of mostly younger women suffering from

this chronic somatic pain condition. A recent genome-wide association study (GWAS) of more than 2.5 million individuals, including more than 50,000 affected individuals, found that fibromyalgia variants showed a strong association with IBS and enrichment in enteric nerves but not with biological sex or classical autoimmune disorders (*47*). Findings from this study mirror those of other GWASs that identified overlapping risk pathways beyond autoimmunity but shared across pelvic pain disorders, migraine, and osteoarthritis (*48*, *49*).

Osteoarthritis (OA) is a degenerative joint disease characterized by progressive cartilage loss, impaired joint function, and chronic pain. As women age and become estrogen-depleted, their risk of developing OA increases, especially in the knee, where rates can be twice that of men (*50*). Mechanisms underlying this hormone-related sex difference are poorly understood. Concentrations of the chemokines (CCL2 and IL-8) in the blood, cerebrospinal fluid, and synovial fluid of the knee joint all correlate with knee pain in women but not men, indicating that OA pain arises from neuroimmune interactions (*51*). Evidence from preclinical models also points to enhanced inflammatory responses in females. In the context of OA of the temporomandibular joint, female rats developed more severe cartilage degeneration and bone remodeling than males, accompanied by increased macrophage infiltration and elevated IL-1β expression in the synovial membrane (*52*). Moreover, silencing estrogen receptor reduces pathological joint remodeling, implicating estrogen-dependent inflammatory pain pathways. Together, these studies suggest that OA pain and joint damage in females are shaped by hormonal status and neuroimmune interactions. Understanding how immune cells and sensory neurons communicate within the arthritic joint may shed light on the female predominance of OA pain.

Summary

The close link between fluctuating hormones and symptoms is a common feature of many types of female pain, perhaps best illustrated by the sudden onset of menstrual migraine headaches. This association raises two important questions. First, can hormone-responsive target genes be identified in affected nerves and tissues to begin explaining how chronic pain develops? Second, why do only some women develop symptoms? We would speculate that the higher rates of pain experienced by reproductive-aged women may confer an evolutionary advantage for reproductive fitness by heightening sensitivity during different life stages, as would be the case for pelvic organs during pregnancy. Future breakthroughs in treating female chronic pain disorders will only come when mechanistic frameworks emerge from preclinical and clinical studies, a goal that will require focused team efforts and adequate funding. Women are desperately waiting for concrete answers that might help eliminate the common "all in your head" approach to their illnesses, shorten the long wait between symptoms and diagnosis/treatment, and ultimately provide mechanism-based strategies for disease treatment.

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RESEARCH

IN *SCIENCE* JOURNALS

Edited by **Michael Funk**

BIOMATERIALS

An evolving structure increases strength

Many composites use a brick-and-mortar structure to increase toughness. However, these materials lack stiffness, and most methods to improve this property result in a lower toughness. Chen *et al.* explored the structural origins of the unusual combination of stiffness and toughness in the nacre of mature *Cristaria plicata*, a bivalve mussel. They found that the stiffness comes from a hierarchical crystalline organic-inorganic framework composed of aragonite nanorods and nanoconfined biomolecular crystallites that evolves from the brick-and-mortar arrangement as the mussel matures. The authors created a semisynthetic nacre based on isolated aragonite tablets, which they infiltrated with a synthetic polymer and reprocessed into films. —Marc S. Lavine

Science p. 81, 10.1126/science.aeg1070

Mussel nacre contains overlapping inorganic crystallites, pictured here in a scanning electron microscopy image, that simultaneously provide toughness and stiffness.

IMAGE: HANG CHEN, HAO YAN, CHUANGQI ZHAO, AND MINGJIE LIU/HANGZHOU INTERNATIONAL INNOVATION INSTITUTE OF BEIHANG UNIVERSITY

POLAR WANDER

Getting around quickly

True polar wander (TPW), which occurs when the solid outer Earth shifts relative to its spin axis, is difficult to detect with standard approaches using paleomagnetic measurements. Domeier *et al.* used a different strategy to detect TPW in the Mesozoic Era using reconstructions of sea level as a proxy. They found a robust quadru polar pattern that reflects global sea-level fluctuations associated with relative changes in centrifugal potential that occurred due to rapid TPW. This demonstrates that coupling in Earth's rotational dynamics, mantle processes, and surface environments can be highly episodic. —Jesse Smith

Science p. 87, 10.1126/science.aec8423

ELECTROCHEMISTRY

Greener route to adiponitrile synthesis

Electrochemical hydrodimerization of acrylonitrile offers a cleaner route to adiponitrile, a key precursor for nylon 6,6, but current systems rely heavily on toxic lead or cadmium electrodes and suffer from poor energy efficiency. Du *et al.* developed a carbon-based electrosynthesis strategy that uses benzyl-substituted quaternary ammonium cations to engineer the electrode-electrolyte interface. They found multiple noncovalent interactions between benzyl groups and carbon, which enriched acrylonitrile near the electrode, promoted radical coupling, and suppressed side reactions. Even in scaled-up electrolyzers, the system achieved selectivity comparable to lead electrodes but at substantially lower overpotentials. —Jack Huang

Science p. 91, 10.1126/science.aef5175

PLANETARY ACCRETION

Early gases hitched rides on metal

Earth and other planets formed by collision of smaller bodies. However, it is not clear how these smaller bodies were able to gravitationally retain and deliver the light, volatile building blocks of our atmosphere. Gaillard *et al.* ran high-pressure experiments on a starting material representative of the "magma oceans" on small, early planetary bodies. They observed that bubbles latched onto metal droplets, forming coalescing clusters that sank

or rose with net buoyancy. The lower gravity on asteroid-scale bodies would have allowed small bubble-metal clusters to sink into the interior, thereby sequestering volatiles for later collision and accretion. —Angela Hessler

Science p. 97, 10.1126/science.aef2542

BACTERIAL IMMUNITY

Viruses make the cut

When invaded by a virus, bacterial cells synthesize cyclic nucleotides as signaling molecules to initiate an immune response program that often ends in cell death as a last resort. Hobbs and Kranzusch found that proteases expressed by the phage itself activate the nucleotide-generating enzymes in the host bacterium. This phage-driven cleavage mechanism clarifies how activation of this signaling pathway is timed to late in the viral infection cycle. Cleavable enzymes found in a range of bacteria suggest that prokaryotic cyclic oligonucleotide–based antiphage signaling is regulated though proteolysis similarly to eukaryotic R proteins and inflammasomes. —Cheri Sirois

Science p. 102, 10.1126/science.aeg3949

MATERIALS SCIENCE

Creating selective gradients

Transition metal perovskite oxides are class of materials with a diverse range of functionalities. Their electronic properties are largely determined by the transition metal atom (usually denoted "B" in the general "ABO_3" chemical formula). By contrast, the "A" atoms, which are nontransition metals, determine the lattice size. Kim *et al.* fabricated films consisting of layers of perovskites in which the composition of the A-site atoms gradually changed from the top to the bottom layer. By doing so, the researchers introduced a strain gradient that led to the displacement of B-site atoms and strong polarity. The technique created an unusual combination of functionalities in the resulting iridate and titanate films and is expected to be generalizable to other similar systems. —Jelena Stajic

Science p. 107, 10.1126/science.adv0235

APPLIED PHYSICS

Sharper lifetime maps

Fluorescence lifetime imaging distinguishes fluorophores with overlapping colors by measuring how quickly their emission fades after excitation, but optical diffraction still blurs nearby features. Hwang *et al.* sharpened these maps using deblurring by pixel reassignment (DPR). DPR is a postprocessing computational technique that moves—i.e., reassigns—measured light toward local intensity peaks. Rearranging the recorded signal in this way makes the spatial structure of these peaks easier to see. With enough photons, DPR narrows blurred spots. At low photon counts, where noisy shifts can produce unstable peaks, DPR clarifies features supported by the spatial and lifetime measurements. —Randy A. Bartels

Sci. Adv. (2026) 10.1126/sciadv.aec7970

BIOCHEMISTRY

A nuanced phosphomimic

The atypical protein kinase C (aPKC) family has opposing functions in different contexts. Both phosphorylation by the kinase PDK1 and binding by the sphingolipid S1P activate aPKC. Kajimoto *et al.* found that the phosphate in S1P bound to the same site as the PDK1-conjugated phosphate but induced an opposite downstream effect. Whereas S1P binding promoted the basal activity of aPKC and opposed apoptosis, PDK1-dependent phosphorylation hindered S1P binding and promoted apoptosis. These findings show that this bioactive lipid activates aPKC-like phosphorylation but switches its effect on apoptosis. —Leslie Ferrarelli

Sci. Signal. (2026) 10.1126/scisignal.adw7297

IN OTHER JOURNALS

Edited by **Corinne Simonti** and **Jesse Smith**

CLIMATE CHANGE

A bacterial ally for drought tolerance

Climate change is predicted to increase drought frequency and severity in the future, making the development of drought-resistant crop varieties increasingly important. Although both root hairs and root colonization by growth-promoting bacteria are known to enhance plant drought tolerance, the molecular mechanisms coordinating them during water deficit have been underexplored. Rahimi *et al*. found that the bacterial root endophyte *Flavobacterium* sp. 98 (Flavo98) improves drought tolerance across multiple plant species by modifying the root system architecture. This modulation is dependent on enhanced ethylene biosynthesis and induction of *ERF114* and *ERF115*. ERF114, and ERF115 in turn target the small signaling peptide gene *C-TERMINALLY ENCODED PEPTIDE5* (*CEP5*), which promotes root hair development. —Unnati Sonawala

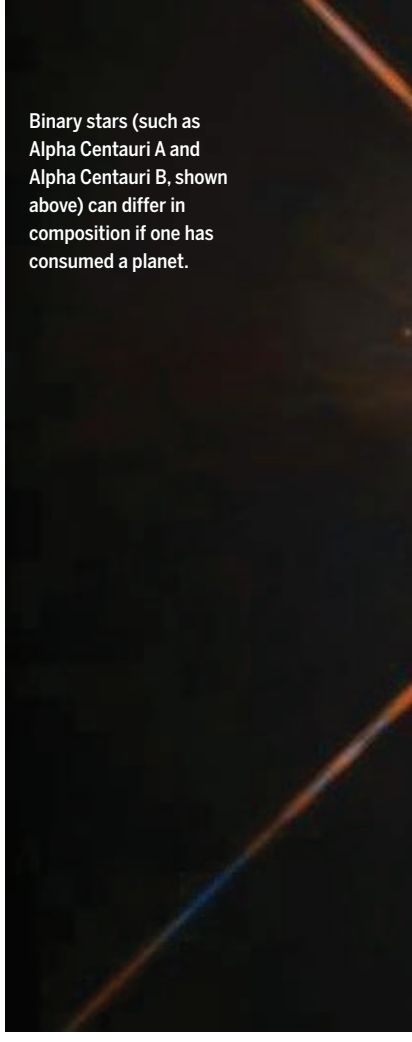
Nat. Plants (2026) 10.1038/s41477-026-02350-4

PLANT BIOLOGY

Delivering pollen in a scoop

Pollination is important for flowering plants to reproduce, and many use animals to transfer pollen between flowers. In poricidal flowers, pollen is released through small pores when the flower is mechanically shaken, often by buzzing bees. Lazarus *et al.* investigated whether the shape of these pores controls how pollen is released. By examining 523 plant species, they found that scoop-shaped pores have evolved repeatedly and produce narrower, faster pollen streams than nonscoop pores. Experiments showed that scoop-shaped pores release pollen in directional bursts, whereas low-frequency vibrations can make pollen release more dispersed. These findings suggest that pore shape is an overlooked floral feature that may help plants control where pollen is placed on pollinators, and this may influence pollination success. —Di Jiang

Nat. Commun. (2026) 10.1038/s41467-026-76766-z



Binary stars (such as Alpha Centauri A and Alpha Centauri B, shown above) can differ in composition if one has consumed a planet.

IMAGE: ESA/HUBBLE & NASA

CELL BIOLOGY

Breaking and entering

The delivery of therapeutic antisense oligonucleotides (ASOs) critically depends on targeting to the correct cells and trafficking out of endosomes into the cytosol. Marco *et al.* identified a pathway in which new-generation ASOs bind a receptor (CD44)

STELLAR ASTROPHYSICS

A star that consumed a planet

Binary stars form by collapse of a single parent gas cloud, so the two member stars typically have the same composition. If these stars also are of the same mass, then they experience identical internal evolution, so any differences indicate external perturbations. Rathsam *et al.* investigated the same-mass binary star system HD 129171/HD 129209, finding that HD 129171 has higher abundances of the refractory elements expected in rocky planets, particularly lithium, but not in volatile elements such as sulfur or oxygen. The abundance pattern is consistent with engulfment of at least 11 Earth masses of planetary material. —Keith T. Smith

Astron. Astrophys. (2026) 10.1051/0004-6361/202659556

on human pancreatic ductal carcinoma cells and exploit a receptor tyrosine kinase (EPHA2) to drive ASO uptake and trafficking to nuclear-proximal endosomes. These nuclear-captured compartments become leaky, allowing ASO escape into the cytoplasm. Inhibition of stress granule–mediated membrane repair enhances ASO efficacy. Thus, a receptor that supports tumor stemness and growth can be exploited to access an endocytic pathway capable of delivering therapeutic molecules to aggressive tumors. —Stella M. Hurtley

J. Cell Biol. (2026) 10.1083/jcb.202507217

IMMUNOLOGY

A leaky gut after skin infection

Dysregulation of the immune system after infection by *Borrelia burgdorferi* is thought to underpin many of the symptoms of Lyme disease, including the longer-term complications that may persist after antibiotic treatment. In mouse models, Sachan *et al.* found that after skin infection, *B. burgdorferi* disseminated to the gut, causing inflammation within the small intestine. This led to an increase in the permeability of the gut, allowing microbial components into the circulation to promote systemic inflammation. These changes were linked to impaired antibody responses toward the gut microbiota and a reduced quantity of antibodies in the blood plasma. —Sarah H. Ross

J. Exp. Med. (2026) 10.1084/jem.20252380

MACROCYCLES

Curvature in the triple negative

A triply negatively curved graphenic macrocycle contains 12 heptagonal rings and has a triangular cyclic structure. Zhang *et al.* synthesized this molecule using a nickel-mediated Yamamoto coupling from a halogen-substituted nanographene monomer that already had a negative curvature. Each macrocycle consists of three saddle-shaped units linked together. The binding of C_{60} and C_{70}, fullerenes with the electron-rich central cavity, was stronger than that observed for the parent nanographene monomers. —Phil Szuromi

J. Am. Chem. Soc. (2026) 10.1021/jacs.6c11599

STEM WORKFORCE

Disparities in grant cancellations

The Trump administration has reshaped US biomedical research through grant terminations, budget cuts, and ideological grant screening, ultimately destabilizing traditional scientific institutions. Nguyen *et al.* conducted a retrospective trend analysis of publicly available NIH data (2016 to 2025) comprising annual counts of individuals funded through research grants and fellowship grants. In 2025, the number of NIH-funded research and fellowship grants declined for the first time in a decade, by 4.0 and 9.9%, respectively, with historically underrepresented individuals experiencing disproportionately larger declines. This disparity was pronounced among fellowship grants, with Black and Hispanic representation declining by 40.5 and 33.1%, respectively. As this destabilization continues, it is critical to prioritize structural reform centered in equity to protect current and emerging talent from future disruptions. —Melissa McCartney

JAMA (2026) 10.1001/jama.2026.7796

Edited by **Michael Funk**

CELL BIOLOGY

Unlocking unrestricted growth

Healthy epithelia contain cells with cancer-associated mutations, but most of these remain growth restricted. By contrast, cells lacking the tumor suppressor p53 can expand and occupy large tissue areas. Gan *et. al*. found that p53 loss promoted the expansion of growth-sustaining epidermal progenitor cells, and they identified a regulatory network that removed several brakes on Wnt signaling (see the Perspective by Engelman and Blanpain). This produced a radial pattern of Wnt activity with stronger signaling at the center of patches of mutant cells than at the edges. Clones exposed to this gradient expanded further than those with uniformly high Wnt activity. Thus, the spatial organization of a growth signal can influence whether mutant cells remain contained or progressively colonize normal tissue. —Stella M. Hurtley

Science p. 78, 10.1126/science.aed3065; see also p. 32, 10.1126/science.ael7341

ARTHRITIS

A joint risk for osteoarthritis

Both osteoarthritis and hypertension are common in older adults. Traditionally, osteoarthritis was thought to be due to age-associated wear and tear on the joints, but accumulating evidence points to metabolic and inflammatory influences on this condition. Using data from a nationwide longitudinal clinical cohort, Kim *et al.* identified a correlation between hypertension and osteoarthritis in older participants (see the Perspective by Wen and Au). The authors then turned to models to work out the underlying mechanism, using two mouse models of hypertension with or without osteoarthritis. The authors demonstrated that arginine vasopressin, a peptide hormone up-regulated in the setting of hypertension, promotes osteoarthritis progression, but only in joints that are already primed by injury or aging and that express the appropriate receptor, providing a mechanistic link between the two conditions. —Yevgeniya Nusinovich

Science p. 80, 10.1126/science.aed8207; see also p. 30, 10.1126/science.ael3784

DYNAMIC POLYMERS

Negative domains stimulate neurite growth

A self-capping supramolecular polymerization yielded peptide-containing assemblies with precisely tuned segments that could exceed 30 micrometers in length (about 1 billion Daltons in mass). Dore *et al.* created assemblies with a central cationic segment that anchored longer negative domains (see the Perspective by Kumar). The central core bound to the negatively charged surfaces of neurons, and the long negative domains, held close near the surface in a highly dynamic state, stimulated neurite growth in the absence of any receptor signals in the peptide sequence. —Phil Szuromi

Science p. 125, 10.1126/science.aeg5583; see also p. 27, 10.1126/science.ael4456

PIEZOELECTRICS

Fast, high-magnitude piezoelectricity

The introduction of polaron-defect complexes in potassium-doped silver niobate films enables high piezoelectric performance at kilohertz frequencies. Shu *et al.* overcame the trade-off between switching speed and piezoelectric magnitude in conventional piezoelectrics by creating giant defect polarization. This effect, driven by multiunit-cell electron polaron hopping that enables collective polaron-defect reconfiguration, exceeds the limited electric-dipole extension of single-unit cells. The materials exhibited an effective piezoelectric coefficient of about 7170 picometers per volt and a strain of about 6.2%. —Phil Szuromi

Science p. 118, 10.1126/science.aeg4972

QUANTUM SYSTEMS

Entangled levitation

The ability to cool levitated nanoparticles to ultralow temperatures for accessing mechanical quantum states of the system provides a platform for probing fundamental physics. Deplano *et al.* demonstrated the generation of entanglement between a macroscopic mechanical oscillator and a freely propagating optical field. Such optomechanical entanglement at room temperature involving a levitated nanoparticle and a propagating optical mode and could also be used to develop quantum sensing technologies. —Ian S. Osborne

Science p. 113, 10.1126/science.aeh1375

NEUTROPHILS

Nurturing neutrophils

During early human pregnancy, placenta-derived extravillous trophoblasts invade the decidua and remodel the maternal-fetal interface, but the contribution of maternal neutrophils to this process is poorly understood. Höbler *et al.* analyzed tissue and blood samples from pregnant individuals, identifying a neutrophil subset localized to the Rohr's layer of the maternal-fetal interface that was phenotypically and functionally distinct from peripheral blood neutrophils. This decidual fibrinoid-associated neutrophil subset supported placental invasion during gestation by secreting matrix metalloproteinase-9, which simultaneously activated latent transforming growth factor-β and reprogrammed peripheral blood neutrophils toward a state supporting placental invasion. —Leoma Bere

Sci. Immunol. (2026) 10.1126/sciimmunol.aea4118

CHEMISTRY

Silver catalysis for protein synthesis

Amide bonds provide a remarkably stable backbone to proteins, but this inertness also makes selective manipulation difficult. Sun *et al.* developed a programmable and mild amide bond activation strategy starting with *N*,*S*-benzylidene thioacetal modified peptides. Silver(I) selectively converts backbone amide into a salicylaldehyde ester within fully unprotected peptides at room temperature without detectable epimerization. The transformation is compatible with diverse peptide sequences and established peptide chemistries. Combined with established peptide ligation chemistry, this strategy facilitated the total synthesis of a mirror-image histone octamer and provides a versatile route to other challenging synthetic proteins. —Steven D. Townsend

Sci. Adv. (2026) 10.1126/sciadv.aej5227

METABOLISM

Transporter with a central role

Regulation of lipid metabolism in mitochondria of adipocytes requires integration of information about time of day from the circadian clock, environmental temperature, and nutrient availability. Karavaeva *et al.* found an important role for the transporter protein SLC25A34 in such control of lipid metabolism and thermogenesis in brown adipocytes from mice and humans (see the Perspective by van der Laan). Inputs from the circadian clock, cold temperatures, and diet all regulated the expression of SLC25A34, allowing control of lipogenesis and mitochondrial biogenesis, lipid oxidation, and thermogenesis. Such regulation may offer insights into inflammation associated with obesity and metabolic diseases. —L. Bryan Ray

Science p. 79, 10.1126/ science.adz4797; see also p. 29, 10.1126/science.ael7271

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RESEARCH ARTICLE SUMMARY

CELL BIOLOGY

Loss of the tumor suppressor p53 generates a signaling gradient that drives epithelial clonal expansion

Qiwen Gan, Wei Li, Rachel K. Lex, Zhe Ying*, Slobodan Beronja*

Full article and list of author affiliations: https://doi.org/10.1126/science.aed3065

INTRODUCTION: Normal tissues accumulate mutations throughout life, but most mutant cells remain contained. Loss of the tumor suppressor p53 is common in epithelial cancers and can allow mutant cell populations to occupy large areas of normal tissue. How p53 loss changes cell behavior to permit this expansion has remained unclear.

RATIONALE: We used mouse skin as a model to determine how p53 restrains the growth of epithelial clones. By combining lineage tracing with quantitative analysis of daughter cell fate, we found that p53 loss drives clonal expansion by enlarging the basal progenitor population rather than by disabling canonical cytostatic pathways. This observation led us to identify direct p53 targets that govern progenitor renewal and differentiation through a combination of gene expression and chromatin analyses, large-scale in vivo genetic screens, and single-cell imaging of Wnt activity.

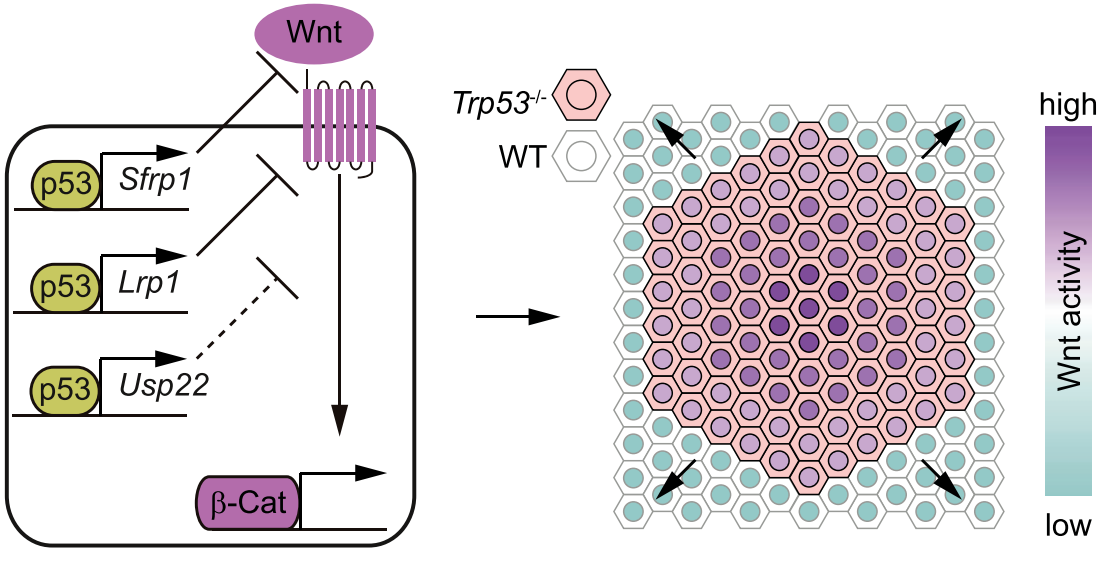
RESULTS: Loss of p53 shifted epidermal progenitor cells toward self-renewal and away from differentiation, allowing the progenitor pool to expand over successive cell divisions. A p53 variant that cannot activate cell cycle arrest, senescence, and cell death programs still restrained progenitor renewal, indicating that noncanonical p53 targets suppress clonal expansion.

By integrating transcriptional profiling, p53 binding maps, and functional screening in mouse epidermis, we identified three direct p53 targets—secreted frizzled-related protein 1 (*Sfrp1*), low-density lipoprotein receptor–related protein 1 (*Lrp1*), and ubiquitin-specific peptidase 22 (*Usp22*)—that act as the most genetically upstream regulators of the p53-governed differentiation and renewal network. These three regulators suppress Wnt signaling at both the ligand and intracellular levels, and upon p53 inactivation, their expression was reduced and Wnt activity increased. Owing to the multilevel nature of Wnt suppression, increase in Wnt activity across the p53-deficient clones was not uniform; it was lowest at the edge of the clones and rose progressively toward their centers, creating a radial gradient that persisted across different stages of clonal expansion regardless of clonal density.

Genetic experiments that manipulated the pattern of Wnt activity showed that gradient Wnt activation was associated with robust clonal expansion. Additional manipulations that produced uniformly elevated Wnt activity, including conditions where average Wnt activity was higher than in p53-deficient clones, resulted in clones that expanded more slowly.

CONCLUSION: p53 restrains epithelial colonization by maintaining a network that controls both the strength and spatial organization of Wnt signaling. Its loss creates a radial Wnt activity gradient that is associated with sustained progenitor self-renewal and rapid clonal expansion. These findings show that the spatial pattern of a growth signal, not only its overall level, can determine how mutant cells spread through normal tissue. □

*Corresponding author. Email: zhe.ying@mssm.edu (Z.Y.); beronja@fredhutch.org (S.B.) Cite this article as Q. Gan *et al.*, *Science* **394**, eaed3065 (2026). DOI: 10.1126/science.aed3065



Spatial organization of Wnt signaling after p53 loss. In normal epidermis, p53 maintains expression of several Wnt suppressors and balances progenitor renewal with differentiation. Loss of p53 weakens this network, producing a radial increase in Wnt activity from the clone edge toward its center. The resulting gradient supports sustained progenitor renewal and rapid expansion through the surrounding tissue.

CELL BIOLOGY

Loss of the tumor suppressor p53 generates a signaling gradient that drives epithelial clonal expansion

Qiwen Gan[1], Wei Li[2], Rachel K. Lex[2], Zhe Ying[1]*, Slobodan Beronja[2]*

Although tumor protein p53 (*TP53*) mutations are among the most common lesions in epithelial cancers, how p53 loss drives unrestrained clonal expansion remains unclear. Working with mouse epidermis, we found that p53 suppresses clonal expansion by limiting progenitor self-renewal. Integrating chromatin immunoprecipitation sequencing, transcriptional analyses, and genetic screens, we identified a p53-dependent network controlling progenitor renewal and differentiation, in which secreted frizzled-related protein 1 (*Sfrp1*), low-density lipoprotein receptor–related protein 1 (*Lrp1*), and ubiquitin-specific peptidase 22 (*Usp22*) were direct targets of p53 and the most genetically upstream components. Their suppression after p53 loss generated a radial gradient of Wnt activity across mutant clones. Genetic manipulation showed that a Wnt activity gradient, rather than uniform elevation, was associated with sustained clonal expansion. These findings identify spatial organization as a determinant of clonal behavior and a mechanism by which *TP53* inactivation drives tissue colonization.

Human tissues acquire a high burden of genetic lesions over a lifetime yet maintain normal morphology and function (*1*–*4*). Most cancer-associated mutations remain growth stabilized in the tissue, whereas a limited subset is associated with clonal expansion (*5*–*8*). Mouse studies have identified several mechanisms that constrain cancer-associated lesions, including clonal elimination (*9*–*13*), growth restriction (*14*), and intra- (*15*) or interclonal competition (*16*–*19*). Yet some lesions escape these constraints and progressively colonize large areas of normal epithelium (*20*). Examples include loss of notch receptor 1 (*Notch1*) in the esophagus (*21*), loss of FAT atypical cadherin 1 (*Fat1*) in the skin (*22*), and inactivation of tumor protein p53 (*5*, *13*). Among these lesions, p53 loss is particularly consequential because it is a common founder event across epithelial cancers (*23*, *24*). How p53 inactivation converts a normally constrained mutant clone into one capable of sustained tissue colonization remains unresolved.

p53 activity responds to changes in cellular conditions (*23*–*26*). These range from normal physiological processes, including replication stress and mitochondrial dynamics, to pathological conditions such as acute DNA damage and oncogene activation (*23*–*26*). Its best-characterized outputs are cytostatic pathways that induce senescence, apoptosis, and cell cycle arrest (*23*–*26*). The growth advantage after p53 loss has therefore been attributed largely to suppression of these pathways (*23*–*26*). However, p53 controls a broader and context-dependent signaling network, and the role of canonical cytostatic pathways in restricting tissue expansion under nonstressed conditions has been questioned (*27*–*32*). Reintroduction of p53 after DNA damage restores tissue homeostasis in p53-null animals without activating canonical cytostatic pathways (*30*, *31*). Mice carrying *Trp53* alleles defective in these responses also largely maintain long-term tissue homeostasis (*28*, *29*). Similarly, combined deletion of the core cytostatic effectors p21, Puma, and Noxa does not increase tissue expansion (*27*). These findings suggest that p53 can regulate tissue growth independently of canonical cytostatic pathways.

p53 restricted epidermal clonal expansion by balancing progenitor renewal and differentiation

Mutations in p53 have been identified as founder lesions of clonal expansion, and inactivation of p53 can act as a robust driver of this process across a spectrum of epithelial tissues, including the epidermis (*5*, *8*, *33*, *34*). To compare canonical cytostatic functions with other p53 functions that might restrict clonal expansion, we used the $Trp53^{3KR/3KR}$ mouse line (*29*, *35*). Mutation of three critical acetylation sites prevented $p53^{3KR}$ from transactivating apoptosis, senescence, and cell cycle arrest genes while preserving its ability to induce ferroptosis (*29*, *35*) (Fig. 1A). To assess clonal expansion in wild-type (WT), $Trp53^{-/-}$, and $Trp53^{3KR/3KR}$ epidermis, we crossed the three *Trp53* alleles into a K14-CreER;mTmG Cre reporter background. At P19 (postnatal day 19), we clonally labeled basal progenitors with administration of a low dose of tamoxifen and imaged the clones in the interfollicular epidermis at P21 and P49 with whole-mount confocal microscopy (Fig. 1B) (*36*). At P21, approximately one cell cycle after tamoxifen treatment (*13*), we observed sparse labeling throughout the epidermis and no significant size difference between WT, $Trp53^{-/-}$, and $Trp53^{3KR/3KR}$ clones (Fig. 1, B and C). In P49 animals, however, $Trp53^{-/-}$ clones exhibited a significant growth advantage, whereas $Trp53^{3KR/3KR}$ clones maintained a size similar to wild type (Fig. 1, B and C). When our analysis was focused on the basal progenitor layer, we quantified significant basal cell expansion in $Trp53^{-/-}$ clones relative to wild type and $Trp53^{3KR/3KR}$ in both P21 and P49 animals (Fig. 1, B and D). These observations suggested that canonical p53 functions may not be required to restrict clonal expansion. To directly investigate the underlying cellular mechanism of clonal expansion in $Trp53^{-/-}$ epithelium, we assessed basal cell density, apoptosis via cleaved caspase-3 staining, cell cycle dynamics using 5-ethynyl-2′-deoxyuridine/5-bromo-2′-deoxyuridine (EdU/BrdU) pulse-chase assay (*13*, *15*, *37*), senescence with quantitative polymerase chain reaction (qPCR) of cyclin-dependent kinase inhibitor 2A (*Cdkn2a*) and senescence-associated beta-galactosidase (SA-β-gal) staining, and p53 mediated ferroptosis through qPCR of solute carrier family 7 member 11 (*Slc7a11*) and 4-hydroxynonenal (4-HNE)/TUNEL co-staining (*23*, *35*) (fig. S1, A to J). Apart from significant reduction in cleaved caspase-3–positive cells and *Cdkn2a* expression in both $Trp53^{-/-}$ and $Trp53^{3KR/3KR}$ tissues compared with wild type, no significant changes in basal cell density, proliferation, SA-β-gal staining, or ferroptosis were detected (fig. S1, A to J). Because none of the measured phenotypes correlated with the growth advantage of $Trp53^{-/-}$ over $Trp53^{3KR/3KR}$ and WT clones (Fig. 1, B to D, and fig. S1, A to J), we hypothesized that there was an alternative mechanism of how p53 restricts clonal expansion in the epidermis.

Oncogene-induced differentiation of progenitors into postmitotic progeny can potently restrict clonal expansion (*13*, *15*, *37*). Our data suggested a basal layer–specific expansion in $Trp53^{-/-}$ clones (Fig. 1, B to D). To test whether p53 loss may drive clonal expansion by controlling renewal and differentiation in the epidermis, we quantified the rate of progenitor renewal (the proportion of basal progenitors among all progeny produced in one round of cell cycle) using our EdU/BrdU pulse-chase assay (*13*, *15*, *37*) (fig. S1, K to M). $Trp53^{3KR/3KR}$ and WT progenitors had a renewal rate of ~0.53, suggesting that just over half the newly generated daughter cells maintained the progenitor potential of the mother, consistent with previously reported rates required for long-term maintenance of the progenitor cell pool and tissue homeostasis (*13*, *15*, *37*–*39*) (Fig. 1E). In contrast, loss of p53 significantly increased the progenitor renewal above the homeostatic

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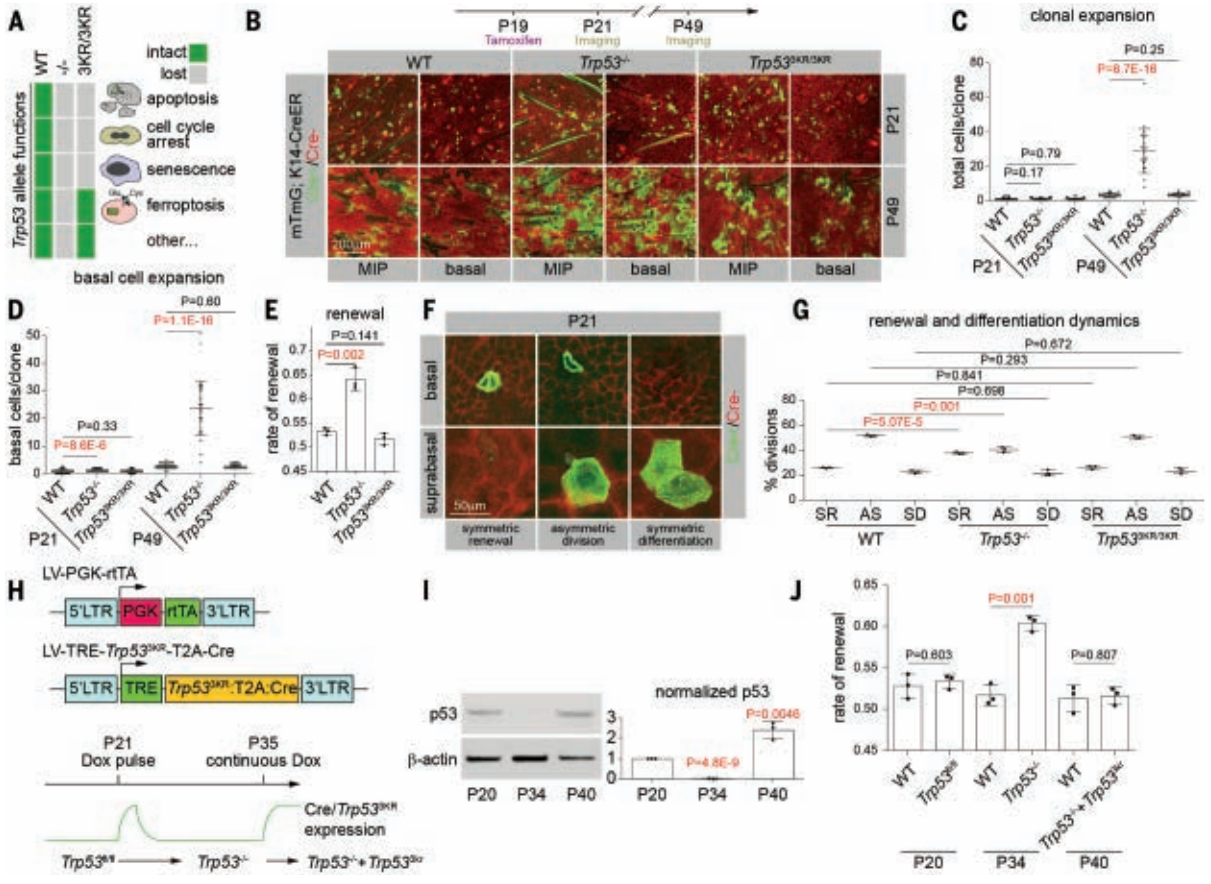

Fig. 1. p53 loss promoted epidermal clonal expansion by increasing progenitor renewal. (**A**) Allele-specific p53 functions: $Trp53^{-/-}$ abolished all p53 functions; $Trp53^{3KR/3KR}$ retained noncanonical activities but lacked induction of senescence, apoptosis, and cell cycle arrest. (**B** to **D**) Representative maximum-intensity projections and basal optical sections (B), total clone size (C), and basal-cell number (D) in WT, $Trp53^{-/-}$, and $Trp53^{3KR/3KR}$ K14-CreER; mTmG epidermis. Basal progenitors were labeled at single-cell density with low dose of tamoxifen at P19 and analyzed at P21 or P49. Scale bar, 200 μm. Clone numbers at P21/P49: WT, 63/33; $Trp53^{-/-}$, 49/21; $Trp53^{3KR/3KR}$, 81/24. (**E**) Progenitor renewal measured by EdU/BrdU pulse chase (*13*,*15*). $n = 3$ animals per genotype. (**F** and **G**) Representative daughter-cell doublets (F) and frequencies of symmetric renewal (SR), asymmetric division (AS), and symmetric differentiation (SD) 48 hours after labeling (G). Scale bar, 50 μm; $n = 3$ animals per genotype. (**H**) Constructs and induction strategy for p53 deletion followed by $p53^{3KR}$ expression. LTR, long terminal repeat. (**I**) Immunoblot and quantification of p53 at P34 and P40. (**J**) Progenitor renewal before and after $p53^{3KR}$ reexpression. $n = 3$ animals per condition [(I) and (J)]. Data are mean ± SD. Comparisons used two-tailed Student's *t* tests; exact *P* values are shown.

rate (Fig. 1E) and reduced the proportion of K10-expressing basal cells within clones (fig. S1, N and O). To determine how p53 loss elevated the renewal rate, we scored the fates of daughter-cell doublets at P21 derived from singly labeled progenitors at P19 (*13*, *15*) (Fig. 1F). In $Trp53^{-/-}$ progenitors, we observed a significant increase in the frequency of symmetric renewal and a concomitant decrease in asymmetric divisions (Fig. 1G). Observed frequencies were consistent with an increase in the renewal rate to ~0.63 (Fig. 1E) and generation of additional proliferative progenitors with each cell division, which can result in geometric expansion of clones over several cell cycles (*13*, *39*). Loss of canonical p53 functions in the $Trp53^{3KR/3KR}$ epidermis alone had no effect on progenitor renewal and differentiation (Fig. 1, E to G).

We set out to determine whether epidermal progenitors were being reprogramed after p53 loss into a heightened stemlike state, or whether ongoing p53 transcriptional activity may be required to guide renewal and differentiation on a continual basis. We transduced $Trp53^{fl/fl}$ mice with two lentiviruses: one expressing rtTA (reverse tetracycline-controlled transactivator) and the other carrying a TRE (tetracycline response element)–driven $p53^{3KR}$-T2A-Cre cassette (Fig. 1H). Transduced animals were given a pulse injection of doxycycline at P21 to delete p53 and then switched to a continuous supply of doxycycline to express $p53^{3KR}$ at P35 (Fig. 1, H and I). Quantification of the rate of renewal at P34 and P40 showed that reexpression of $p53^{3KR}$ was sufficient to reduce the rate of renewal back to WT levels (Fig. 1J). Thus, clonal expansion in skin epithelium is restricted through p53 activity that regulated progenitor renewal and differentiation to suppress progenitor renewal.

A p53 transcriptional program in progenitor renewal and differentiation

Growth-promoting renewal and differentiation dynamics that follow p53 loss, as well as the inability of canonical cytostatic p53 functions to restrict clonal expansion (Fig. 1 and fig. S1), suggested the existence of a p53 transcriptional program that controlled key renewal and differentiation regulators.

We hypothesized that these renewal and differentiation genes are among the shared targets of $p53^{WT}$ and cytostatic function–deficient $p53^{3KR}$ (Fig. 2A). To test this hypothesis and identify shared renewal and differentiation targets of p53, we first measured the expression of canonical p53 target genes (*40*) cyclin-dependent kinase inhibitor 1A (*Cdkn1a*; p21), BCL2-binding component 3 (*Bbc3*; Puma), and BCL2-associated X apoptosis regulator (*Bax*) and observed significant

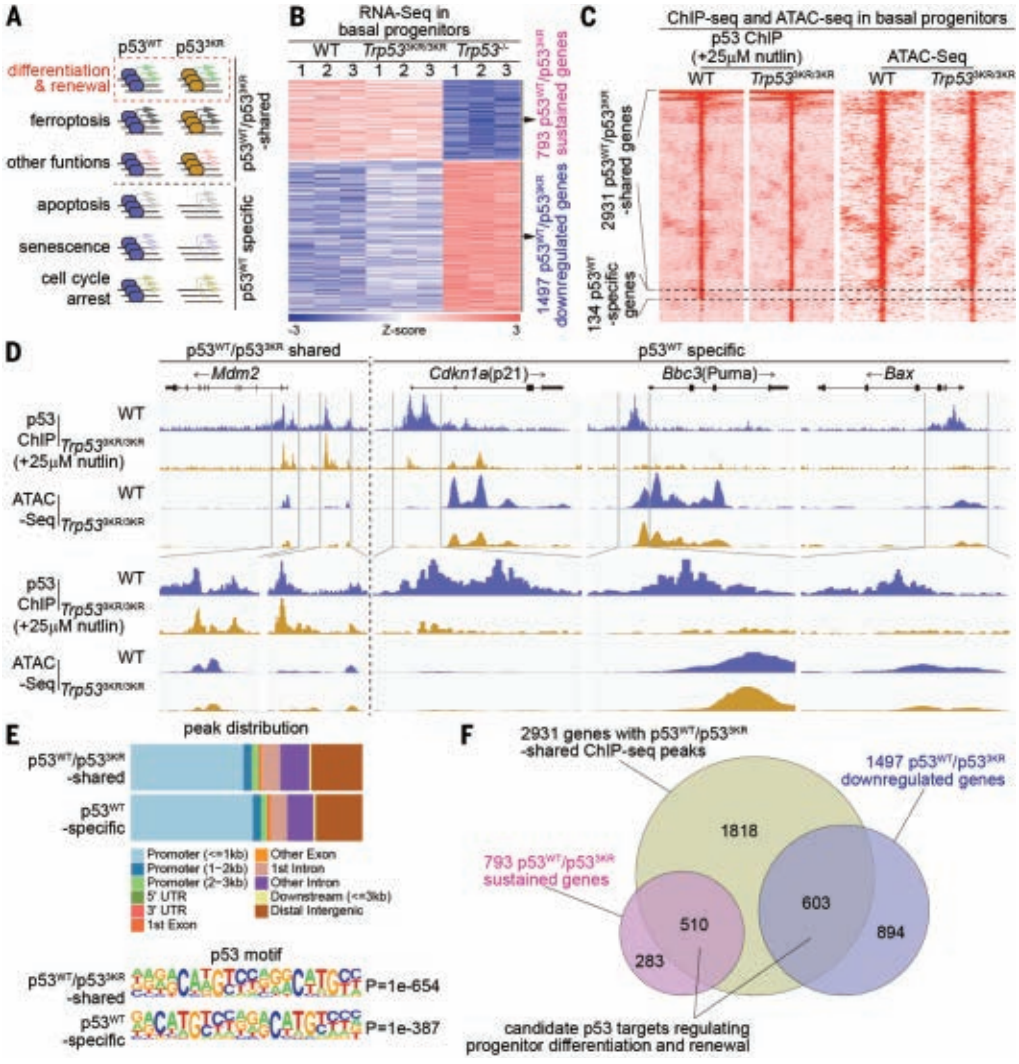

Fig. 2. Identification of transcriptional targets of p53 that could act as epithelial renewal and differentiation regulators. (**A**) Strategy to identify renewal and differentiation regulators among transcriptional targets shared by $p53^{WT}$ and $p53^{3KR}$. (**B**) Genes differentially expressed in *Trp53*$^{-/-}$ basal progenitors relative to WT and *Trp53*$^{3KR/3KR}$ cells. $n = 3$ biological replicates per genotype; genes were identified by pattern matching (MeV, $P < 0.01$). (**C**) Heatmaps of p53 ChIP-seq and ATAC-seq signals centered on p53-binding sites in WT and *Trp53*$^{3KR/3KR}$ basal progenitors. High-confidence ChIP-seq peaks were called with MACS2 (FDR < 0.01). ChIP-seq and ATAC-seq signals combined three replicates containing 1×10^7 and 5×10^5 basal cells, respectively. (**D**) Representative ChIP-seq and ATAC-seq tracks at the shared target *Mdm2* and the $p53^{WT}$-selective targets *Cdkn1a*, *Bbc3*, and *Bax*. (**E**) Genomic distribution of p53 peaks and enrichment of the top p53-binding motif. (**F**) Intersection of differentially expressed genes with shared $p53^{WT}/p53^{3KR}$ ChIP-seq peaks identified direct p53 targets for functional testing.

reduction in isolated *Trp53*$^{3KR/3KR}$ and *Trp53*$^{-/-}$ basal progenitors compared with WT (fig. S2A). We then performed RNA sequencing (RNA-seq) in these cells (Fig. 2B) and identified 793 sustained and 1497 down-regulated genes that were differentially expressed in *Trp53*$^{-/-}$ relative to WT and *Trp53*$^{3KR/3KR}$ progenitors [pattern match algorithm, MeV (*41*), $P < 0.01$] (Fig. 2B).

To identify genes that were directly occupied by both p53 and $p53^{3KR}$, we performed p53 chromatin immunoprecipitation sequencing (ChIP-seq) in WT and *Trp53*$^{3KR/3KR}$ basal progenitors. To minimize bias toward the cytostatic genes while sufficiently stabilizing p53, we digested the tissues for progenitor cell isolation in the presence of 25 μM of nutlin-3a, a dose that did not induce significant cytostatic responses, as indicated by the absence of activated caspase-3 staining (fig. S2, B and C). Our p53 ChIP-seq identified peaks [MSCS2, false discovery rate (FDR) < 0.01] shared by both $p53^{WT}$ and $p53^{3KR}$ (Fig. 2C). To quantify the accessibility of these genes, we performed assay for transposase-accessible chromatin with high-throughput sequencing (ATAC-seq) in basal progenitors and confirmed that p53 peaks were located in regulatable open chromatin regions under nontreated conditions (Fig. 2C). Consistent with the specificity of our assays, we identified ChIP-seq traces on known $p53^{WT}/p53^{3KR}$ shared targets, such as *Mdm2* (*29*) which showed similar levels of p53 enrichment at the locus (Fig. 2D). In contrast, ChIP traces on known canonical $p53^{WT}$-specific targets *Cdkn1a*, *Bbc3*, and *Bax* (*29*) showed reduced $p53^{3KR}$ binding at the respective loci (Fig. 2D).

To our surprise, the heatmap surrounding the ChIP peaks indicated that most of the p53 targets were shared between $p53^{WT}$ and $p53^{3KR}$ (~20-fold more than the $p53^{WT}$-specific targets; Fig. 2C). We further characterized the pattern of $p53^{WT}/p53^{3KR}$ shared and $p53^{WT}$-specific peaks using ChIPseeker (*42*) and scored the p53 motif enrichment using Homer (*43*) (Fig. 2E). Our results showed no apparent difference in peak distribution patterns and p53 motif enrichment between $p53^{3KR}/p53^{WT}$ shared and $p53^{WT}$-specific loci (Fig. 2E). Moreover, the $p53^{WT}/p53^{3KR}$ DNA binding specificity was similar to that identified by published p53/63/73 ChIP-seq analyses across different cell types in Homer datasets (*43*) (fig. S2, D and E).

To identify $p53^{WT}/p53^{3KR}$ targets that were also transcriptionally regulated by p53 in the basal progenitors, we intersected our RNA-seq and ChIP-Seq analyses to identify 510 p53 sustained and 603 p53 down-regulated genes (Fig. 2F). We hypothesized that among them were key targets of a p53 transcriptional program governing renewal and differentiation dynamics.

An in vivo genetic screen for components of the p53-regulated renewal and differentiation program

To identify genes that may execute the p53 renewal and differentiation program among the 1113 $p53^{WT}/p53^{3KR}$ shared targets, we constructed a lentiviral pool of 7021 short hairpin RNAs (shRNAs) to perturb each gene with five to seven distinct shRNAs (Fig. 3A). We used an established in vivo screening strategy (*13*, *15*, *37*) to test the candidates for their ability to regulate progenitor renewal and differentiation in WT, *Trp53*$^{3KR/3KR}$, and *Trp53*$^{-/-}$ epidermis (Fig. 3A). Briefly, we introduced the pool into E9.5 epidermis using in utero transduction [$n = 30$ injected embryos, >50× coverage with a multiplicity of infection (MOI) of ~0.11] and harvested epidermis from P1 animals. On the basis of

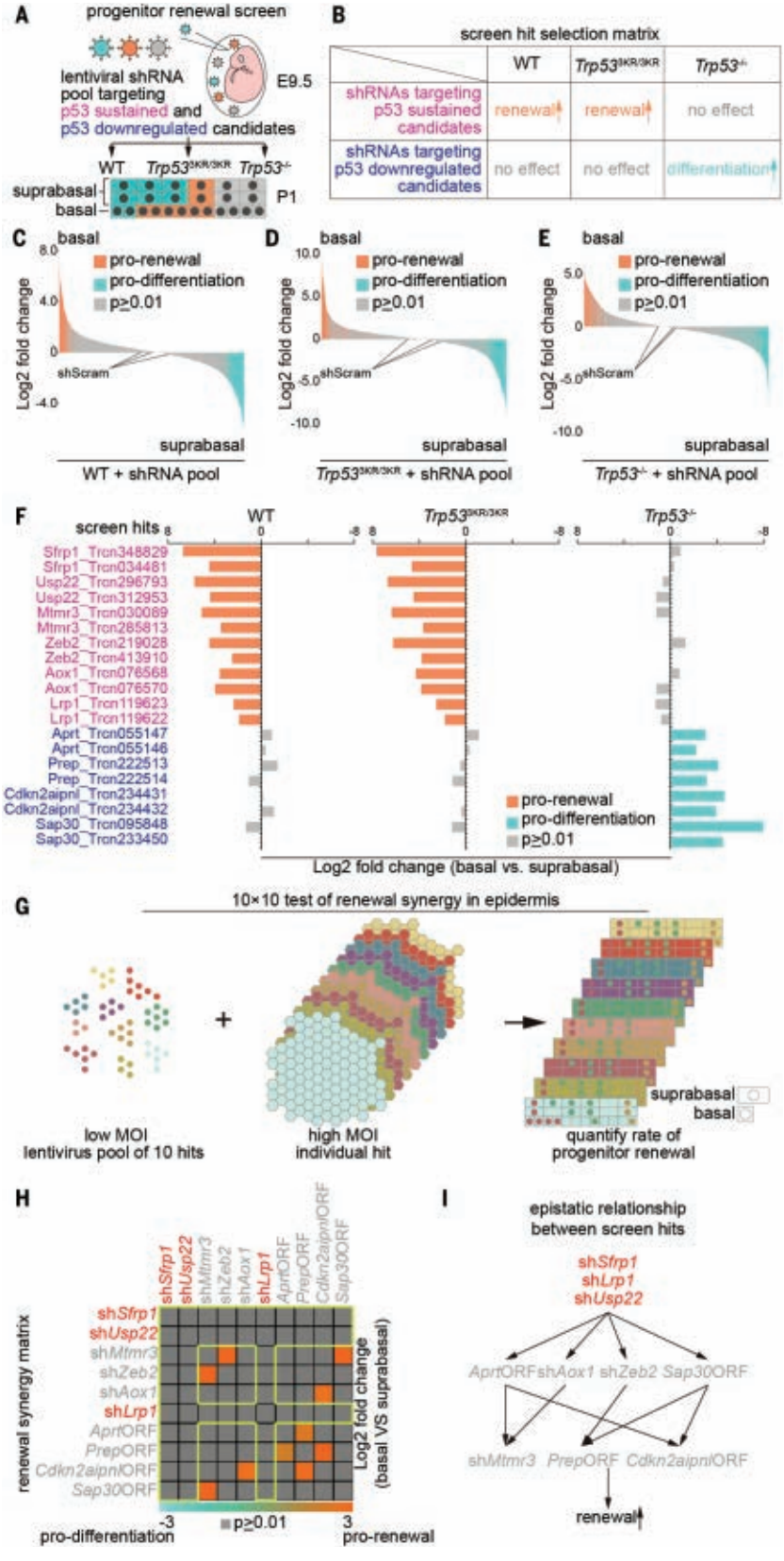

Fig. 3. Functional identification of a p53 governed epithelial renewal and differentiation program. (**A**) Screening strategy for testing shared $p53^{WT}$/$p53^{3KR}$ transcriptional targets as regulators of progenitor renewal and differentiation in WT, *Trp53*$^{3KR/3KR}$, and *Trp53*$^{-/-}$ epidermis. Basal and suprabasal cells were isolated by FACS and analyzed for shRNA enrichment. (**B**) Genotype-specific criteria used to identify p53-dependent renewal and differentiation regulators. (**C** to **E**) Enrichment of pro-renewal

prior findings (*13*, *15*, *37*), we anticipated pro-renewal shRNAs to be enriched in basal α6 integrin–high progenitor cells and pro-differentiation shRNAs to be enriched in suprabasal α6 integrin–low cells, which we isolated with fluorescence-activated cell sorting (FACS) (Fig. 3A). Using a published statistical pipeline (*13*, *15*, *44*, *45*) [Deseq2 (*46*), $P < 0.01$; for each gene at least two shRNAs with the same significant trend and no shRNAs with the opposite trend] on three genetic backgrounds with distinct transcriptional states (WT, *Trp53*$^{3KR/3KR}$, and*Trp53*$^{-/-}$), we expected that we may achieve a screen with heightened stringency. Specifically, our scoring strategy (Fig. 3B) was based on the expectation that shRNAs targeting p53 sustained candidates may only promote renewal in WT and *Trp53*$^{3KR/3KR}$ epidermis, where the candidates are transactivated, but not in *Trp53*$^{-/-}$ epidermis, where they are not. Conversely, for p53 repressed genes, the targeting shRNAs could only be differentiation promoters in *Trp53*$^{-/-}$ epidermis, where p53-mediated candidate suppression was lost (Fig. 3B). Following the criteria above, we performed the three sets of genetic screens in triplicate (Fig. 3, C to E) and identified six p53 sustained genes and four p53 down-regulated genes among the 1113 tested as regulators of renewal and differentiation dynamics (Fig. 3F).

Rather than prioritizing screen hits for downstream analyses on the basis of fold change, we used the limited number of identified candidates to design a test to probe for synergy across all 10 in promoting progenitor renewal (Fig. 3G). We constructed a mini pool containing shRNAs against the six p53 sustained genes and open reading frames (ORFs) of the four p53 down-regulated genes. The mini pool was then mixed with high-titer virus of each of the 10 constructs, and these were transduced into the epidermis of 10 sets of WT mice ($n = 3$ injected embryos per set, >3750× coverage with MOI of ~0.11). The ability of each construct to promote renewal or differentiation was scored by relative enrichment in basal progenitors versus differentiated suprabasal cells, as before (*13*, *15*) (Fig. 3G). After functionally profiling all 10×10 interactions, we scored significant [Deseq2 (*46*), $P < 0.01$] synergistic effect in promoting renewal and mapped the implied epistatic relationship accordingly (Fig. 3, H and I). Among the 10, we found three p53 transactivated genes—secreted frizzled-related protein 1 (*Sfrp1*), low-density lipoprotein receptor–related protein 1 (*Lrp1*), and ubiquitin-specific peptidase 22 (*Usp22*)—that did not have a significant synergistic interaction with any other, indicating that they represented the upstream-most signaling genes that regulate renewal and differentiation dynamics (Fig. 3, H and I).

The p53-regulated renewal and differentiation program suppresses Wnt signaling

We first set out to validate *Sfrp1*, *Lrp1*, and *Usp22* as p53 transactivated targets. Analyses of 25 μM nutlin-3a–stabilized ChIP-seq and nontreated ATAC-seq data showed robust p53 binding in open chromatin regions of *Sfrp1*, *Lrp1*, and *Usp22* gene loci in both $p53^{WT}$ and $p53^{3KR}$ epidermal progenitors (Fig. 4A). To test whether *Sfrp1*, *Lrp1*, and *Usp22* were directly occupied by $p53^{WT}$ and $p53^{3KR}$ without p53 stabilization, we performed ChIP-qPCR in untreated basal progenitors isolated

(orange) and pro-differentiation (blue) shRNAs in WT (C), *Trp53*$^{3KR/3KR}$ (D), and *Trp53*$^{-/-}$ (E) epidermis. Scrambled shRNAs served as negative controls. (**F**) Effects of the two highest-scoring shRNAs for each candidate. (**G**) Design of the combinatorial 10×10 interaction screen. (**H**) Synergistic effects of candidate pairs on renewal (orange) or differentiation (blue). sh*Sfrp1*, sh*Lrp1*, and sh*Usp22* showed no significant synergy with other candidates (yellow outlines). (**I**) Interaction network inferred from the combinatorial screen. Values in (C) to (F) and (H) are means from three biological replicates. Enrichment and interaction statistics were calculated with DESeq2; $P < 0.01$.

from WT, *Trp53*$^{3KR/3KR}$, and *Trp53*$^{-/-}$ epidermis (Fig. 4B). Significant p53 and p53^{3KR} enrichment was observed across the ChIP-seq peaks for all three genes and was substantially reduced by p53 knockout, suggesting that p53 binds to the *Sfrp1*, *Lrp1*, and *Usp22* loci (Fig. 4B). In addition, qPCR analyses showed that all three genes were expressed at similar levels in WT and *Trp53*$^{3KR/3KR}$ cells but were drastically down-regulated in *Trp53*$^{-/-}$ basal progenitors, further confirming that *Sfrp1*, *Lrp1*, and *Usp22* are sustained by p53 (Fig. 4C).

Sfrp1 (*47*–*50*) and *Lrp1* (*51*–*53*) are known as robust suppressors of Wnt signaling, and our epistatic analyses suggested that *Usp22* may work through the same pathway (Fig. 3, G to I). We thus set out to test whether p53 loss may affect Wnt signaling in epidermal progenitors. We began by evaluating the known epidermal Wnt response signature (*54*) in WT and p53 mutant clones (Fig. 4, D and E). Gene set enrichment analysis (GSEA) suggested that β-catenin–induced genes were robustly enriched whereas β-catenin–suppressed genes were significantly depleted in *Trp53*$^{-/-}$ basal progenitors, in comparison to WT (Fig. 4D) and *Trp53*$^{3KR/3KR}$ (Fig. 4E) controls.

We next measured Wnt activity using the H2B-GFP Wnt reporter mouse line (*55*). We combined it with the tdTomato$^{lsl/lsl}$ Cre reporter allele (*56*), so that tdTomato would mark clones transduced with perturbation lentivirus expressing Cre-recombinase and gene-targeting elements (Fig. 4, F to H). Because nuclear β-catenin and Wnt activity are mostly restricted to basal progenitors (*57*) (fig. S3A), we focused our analyses on the H2B-GFP signal from the basal layer (fig. S3B). Using whole-mount confocal imaging and three-dimensional (3D) quantification with Imaris (*58*), we were able to assess Wnt activity, based on the level of nuclear H2B-GFP, at single-cell resolution and across the transduced clones and their untransduced neighbors (Fig. 4, G and H). Control clones induced in H2B-GFP Wnt reporter;tdTomato$^{lsl/lsl}$ epidermis by lentivirus expressing Scrambled shRNA and Cre (shScram-Cre), showed no significant Wnt activity difference between basal cells in and outside of the clones (Fig. 4G). Thus, lentiviral transduction did not alter Wnt signaling. In contrast, control shScram-Cre transduction of the *Trp53*$^{fl/fl}$;H2B-GFP Wnt reporter;tdTomato$^{lsl/lsl}$ mice showed robust expression of the H2B-GFP Wnt reporter among basal cells (fold change = 6.85; $P = 5.2 \times 10^{-3}$; Fig. 4G) together with highly efficient Cre-mediated knockout (fig. S3, C and D), confirming that p53 loss resulted in activation of the Wnt pathway in the clones relative to the surrounding WT field. Nuclear H2B-GFP signal was unchanged in *Trp53*$^{3KR/3KR}$;tdTomato clones and only increased upon clonal sh*Trp53*-mediated depletion of p53^{3KR} (fig. S3E), indicating that p53^{3KR} was a sufficient suppressor of Wnt activity. To establish the sensitivity and specificity of the Wnt/β-catenin reporter system, we quantified Cre recombination efficiency (fig. S3, C and D) and H2B-GFP levels in *Ctnnb1*$^{-/-}$ (catenin beta 1) clones and observed fluorescence levels below those seen in WT and *Trp53*$^{-/-}$ clones (Fig. 4, F and G).

Next, we explored the contribution of p53-regulated renewal and differentiation genes in Wnt reporter activity. We first clonally depleted *Sfrp1*, *Lrp1*, and *Usp22* in the WT and *Trp53*$^{3KR/3KR}$ epidermis. All three conditions resulted in significant elevation of Wnt signal in transduced clones when compared with WT fields or clones, indicating that *Sfrp1*, *Lrp1*, and *Usp22* could suppress Wnt/β-catenin signaling (Fig. 4H and fig. S3, F to H). To understand the individual contribution of *Sfrp1*, *Lrp1*, and *Usp22* to p53-driven suppression of Wnt activity, we next transduced *Trp53*$^{fl/fl}$ epidermis with lentiviruses coexpressing Cre and ORFs of *Sfrp1*, *Lrp1*-mini [truncated Lrp1 that is short enough to be packaged into lentivirus but can still interact with and suppress Wnt receptors (*52*, *53*)], and *Usp22*, respectively (Fig. 4H). All three constructs significantly reduced Wnt reporter levels in *Trp53*$^{-/-}$ clones (Fig. 4H and fig. S3I), suggesting that p53's ability to sustain *Sfrp1*, *Lrp1*, and *Usp22* expression was critical for its suppression of Wnt signaling in epidermal basal cells.

p53 inactivation generated a radial gradient of Wnt activity

A curious feature of the p53-governed Wnt-suppression network is that, by transactivating *Sfrp1*, *Lrp1*, and *Usp22*, p53 could suppress Wnt signaling at the ligand (*47*–*50*), receptor (*51*–*53*), and nucleus (*59*) levels, respectively (Fig. 5A). Such multilevel control of Wnt signaling implied a mixed suppression network of unclear significance. We hypothesized that, rather than representing a simple redundancy, a multilevel control may create a qualitatively distinct form of Wnt suppression and that p53 loss may result in a gradient of Wnt activation across the radius of *Trp53*$^{-/-}$ clones (Fig. 5A). To test this hypothesis, we calculated the distance between individual *Trp53*$^{-/-}$ cells and the closest WT neighbors using Imaris (*58*) and plotted the intensity of H2B-GFP Wnt reporter signal relative to the distance from the WT/*Trp53*$^{-/-}$ interface (Fig. 5B and fig. S3B). Nonlinear regression analysis showed a gradient of Wnt activity, which gradually increased from the outer edge toward the center of *Trp53*$^{-/-}$ clones (Fig. 5B and fig. S4A). Accompanying this gradient, the rate of progenitor renewal also showed a gradual increase along the radius of the clones that were undergoing rapid expansion (Fig. 5, C to E). Such a phenotype is consistent with the role of Wnt as a driver of progenitor renewal (*57*, *60*) and our observation of gradual reduction of *Sfrp1* and sharp decrease of *Lrp1* and *Usp22* staining within the clones (fig. S4, B and C). A similar Wnt activity gradient was seen upon sh*Trp53* expression in *Trp53*$^{3KR/3KR}$ clones (fig. S4D). In contrast, knockout of β-catenin in *Trp53*$^{-/-}$ clones flattened the Wnt activity gradient (Fig. 5B), lowered the renewal rate to uniform across the clones (Fig. 5C), and resulted in a significant loss of long-term clonal expansion alongside substantial reduction of Wnt activity (Fig. 5, D to F). These observations confirmed that increased progenitor renewal and clonal expansion after p53 loss were dependent on activation of Wnt/β-catenin signaling.

Imaging of clones at different stages of expansion revealed that this robust Wnt activity gradient was maintained not only in well-separated clones (fig. S4, E and F) but also in densely populated regions where clonal fusion was occurring (fig. S4, G and H). Indeed, Wnt activity within the 0-to-50-μm and 50-to-100-μm bins from the clone border was comparable between clones seeded at low and normal densities as well as the areas of clonal fusion (fig. S4I), indicating a similar ability to establish a radial Wnt activity gradient. Thus, the Wnt activity gradient was robust across a broad range of clonal densities and was maintained in both isolated and fused mutant clones, suggesting that it represented an intrinsic property of p53-deficient epidermal expansion rather than a consequence of clone spacing or local clone geometry.

A Wnt activity gradient was associated with rapid epithelial clonal expansion

Ctnnb1 deletion reduced both the gradient and amplitude of Wnt activity (Fig. 5, B to F), leaving it unclear whether one or both of these features contribute to clonal expansion. Thus, we systemically tested association of radial gradient– versus plateau-shaped Wnt activation with clonal expansion. We hypothesized that, because Sfrp1 is a secreted Wnt antagonist (*47*–*50*), clones that maintained *Sfrp1* suppression (i.e., *Trp53*$^{-/-}$, sh*Sfrp1*, *Trp53*$^{-/-}$ *Usp22*-ORF, and *Trp53*$^{-/-}$ *Lrp1*-ORF) may induce distinct pattern of Wnt activation from clones with depletion of *Lrp1* (*51*–*53*) or *Usp22* (*59*) or from constitutive Wnt activation using

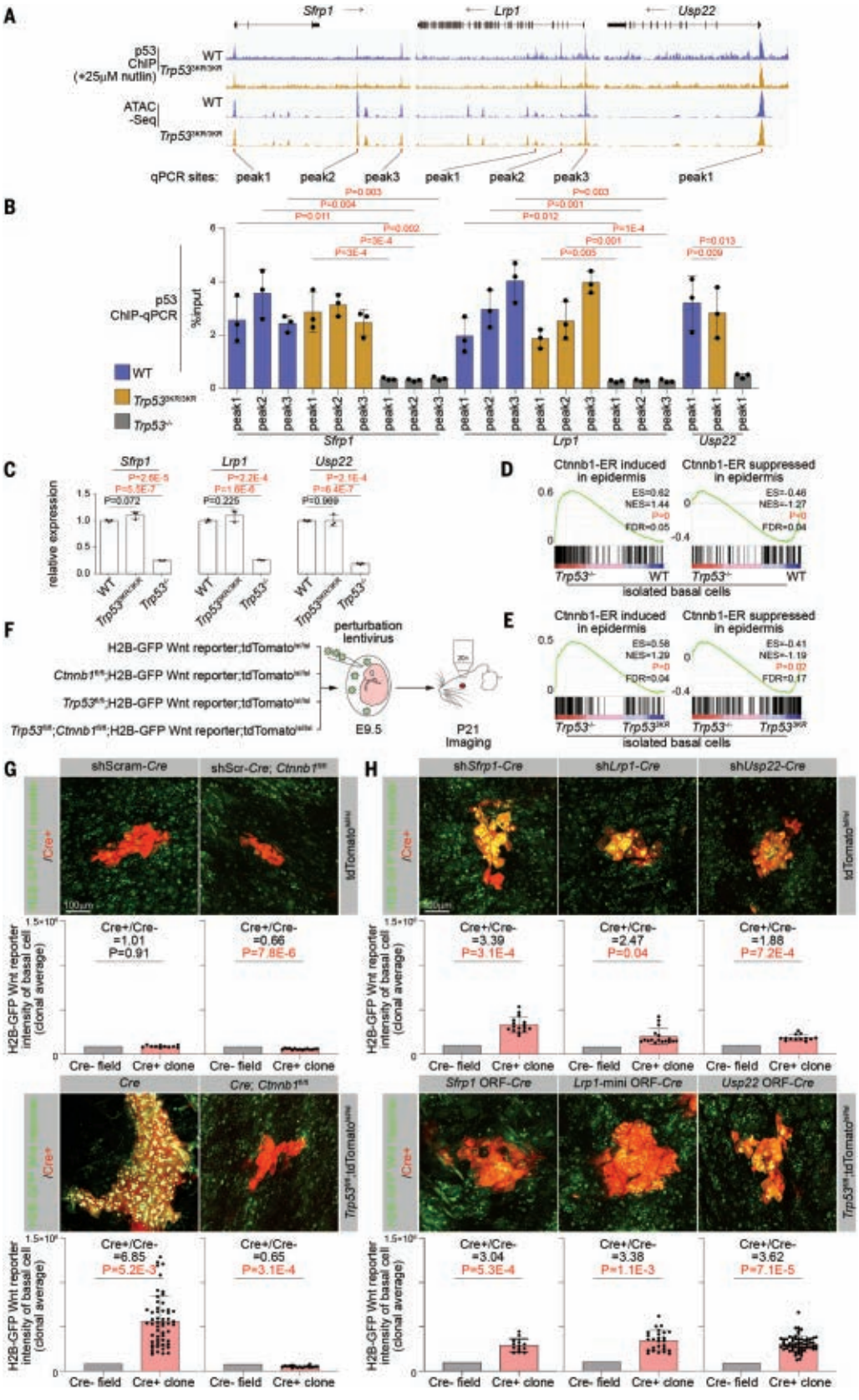

Fig. 4. p53 governed renewal and differentiation program controls Wnt activity in epidermis. (**A**) p53 ChIP-seq and ATAC-seq tracks at *Sfrp1*, *Lrp1*, and *Usp22* in WT and *Trp53*$^{3KR/3KR}$ basal progenitors. (**B**) ChIP-qPCR at p53-binding sites in untreated WT, *Trp53*$^{3KR/3KR}$, and *Trp53*$^{-/-}$ basal progenitors. $n = 3$ animals per genotype. (**C**) *Sfrp1*, *Lrp1*, and *Usp22* transcript abundance in the three genotypes. $n = 3$ biological replicates per genotype. (**D** and **E**) GSEA of published β-catenin–responsive transcriptional signatures (*54*) in *Trp53*$^{-/-}$ relative to WT (D) or *Trp53*$^{3KR/3KR}$ (E) basal progenitors. (**F**) Mouse strains, perturbations, and imaging strategy used to measure clonal Wnt activity.

(**G** and **H**) Representative maximum-intensity projections and mean basal H2B-GFP reporter intensity inside and outside tdTomato-positive clones. p53 loss increased Wnt activity, whereas *Ctnnb1* deletion suppressed it (G). Depletion of *Sfrp1*, *Lrp1*, or *Usp22* increased reporter activity; expression of the corresponding ORFs reduced reporter activity in *Trp53*$^{-/-}$ clones (H). Scale bars, 100 μm. Data are mean ± SD. GSEA statistics are shown in (D) and (E). Other comparisons used two-tailed Student's *t* tests; exact *P* values and sample sizes are shown.

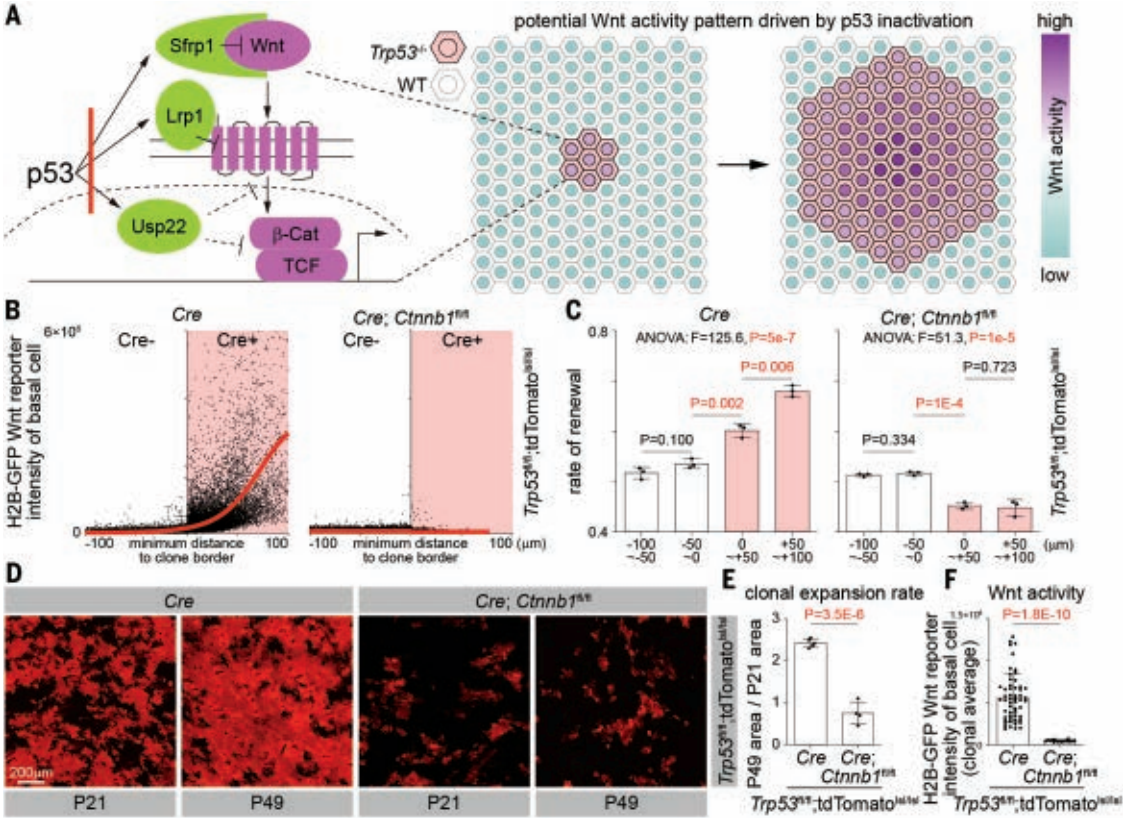

Fig. 5. p53 inactivation generated a radial gradient of Wnt activity. (**A**) Model for spatial Wnt regulation by the secreted antagonist *Sfrp1* and the cell-associated regulators *Lrp1* and *Usp22*. (**B**) Basal H2B-GFP reporter intensity plotted against distance from the clone boundary. Positive and negative distances indicate positions inside and outside the clone, respectively. Nonlinear regression (red) showed increasing Wnt activity from the edge toward the center of *Trp53*$^{-/-}$ clones. *Ctnnb1* deletion reduced reporter activity and abolished the gradient. (**C**) Progenitor renewal measured by EdU/BrdU pulse chase (*13*,*15*) across distance bins from the clone boundary. $n = 3$ animals per condition. (**D** to **F**) Representative clones (D), clonal expansion (E), and Wnt reporter activity (F) after *Ctnnb1* deletion in *Trp53*$^{-/-}$ clones. Scale bar, 200 μm. $n = 5$ *Trp53*$^{-/-}$ and 4 *Trp53*$^{-/-}$;*Ctnnb1*$^{-/-}$ animals. Data are mean ± SD. Curves in (B) were fitted by nonlinear regression; other comparisons used two-tailed Student's *t* tests. Exact *P* values are shown.

ΔN90 *Ctnnb1*-ORF (*61*). Indeed, nonlinear regression showed that depletion of *Sfrp1* was sufficient to generate a Wnt activity gradient in both WT and *Trp53*$^{3KR/3KR}$ backgrounds, whereas sh*Lrp1*, sh*Usp22*, and ΔN90 *Ctnnb1*-ORF resulted in a leveled Wnt activation across the clones (Fig. 6A and fig. S5, A to C). Specific to the gradient pattern of Wnt activity were the radial gradient of renewal (fig. S5D) and a significantly higher long-term expansion rate (Fig. 6, B and C). Notably, even in ΔN90 *Ctnnb1*-ORF clones (Fig. 6A and fig. S5, B and C), where the average Wnt activity was higher than is observed upon p53 loss and significantly higher than in *Sfrp1*-depleted clones (Fig. 6D), we recorded lower rates of progenitor renewal (fig. S5D) and clonal expansion (Fig. 6, B and C) relative to the Wnt gradient clones. Thus, the level of Wnt activity alone did not account for clonal growth. Measurements of clone roundness, border roughness, and circularity showed that gradient-forming clones exhibited more irregular shapes and rougher boundaries than did Wnt plateau clones, suggesting that the former may integrate more effectively with the surrounding tissue (Fig. 6, E and F, and fig. S5, E and F). Together, these analyses associated a radial Wnt activity gradient, rather than uniform elevation, with rapid expansion of p53-deficient clones.

Discussion

The connection between p53 and the renewal of stem and progenitor cells was first suggested three decades ago, with observation that p53 expression shifts the growth of mammary epithelial cells from geometric to linear, consistent with an increase in asymmetric pattern of division (*62*). The p53-dependent effect on stem cell divisions is also documented in daughter-cell doublets in both WT and *Erbb2* transgenic mammary glands (*63*) and in the planarian epidermis (*64*). Yet the underlying mechanism remained unclear. Studies of induced pluripotency provide another link: p53 depletion increases reprogramming efficiency (*65*–*69*). However, those reports suggest that p53 suppresses reprogramming via canonical cytostatic pathways and induction of p21, p16INK4a, and p14arf (*65*–*69*). Our unbiased identification of p53 targets in basal progenitors of the adult epidermis followed by an in vivo functional screen for key regulators of progenitor renewal among them led to the discovery of *Sfrp1*, *Lrp1*, and *Usp22* as direct targets of p53. These genes occupy upstream positions in a p53-governed Wnt-suppression network that restricts clonal expansion by maintaining a homeostatic rate of progenitor renewal.

Early ChIP-seq and expression profiling identify an array of Wnt ligands and receptors as transcriptionally activated by p53 in embryonic

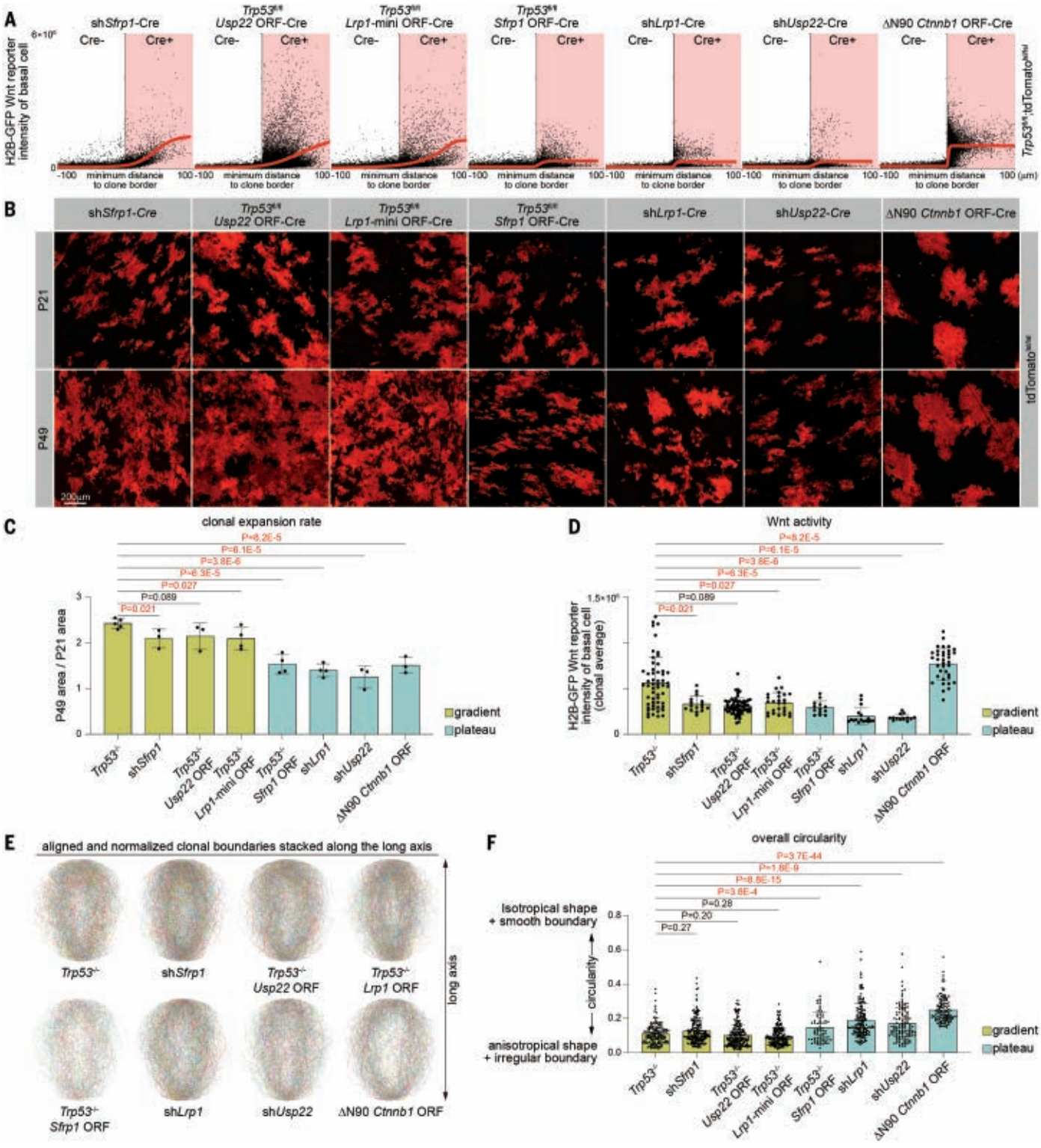

Fig. 6. A radial Wnt activity gradient was associated with rapid clonal expansion. (**A**) Basal H2B-GFP reporter intensity plotted against distance from the clone boundary. Positive and negative distances indicate positions inside and outside the clone, respectively. Nonlinear regression (red) identified gradient-shaped Wnt activation in sh*Sfrp1*, *Trp53*$^{-/-}$;*Usp22*-ORF, and *Trp53*$^{-/-}$;*Lrp1*-ORF clones and plateau-shaped activation in *Trp53*$^{-/-}$;*Sfrp1*-ORF, sh*Lrp1*, sh*Usp22*, and ΔN90-*Ctnnb1*-ORF clones. (**B** to **D**) Representative clones (B), clonal expansion (C), and mean Wnt reporter activity (D) under gradient- and plateau-forming conditions. Scale bar, 200 μm. (**E**) Clone boundaries were normalized for size, aligned along the major axis, and superimposed. (**F**) Clone circularity; higher values indicate more isotropic shapes with smoother borders. Green denotes gradient-forming conditions; blue denotes plateau-forming conditions. Individual points represent clones; bars show mean ± SD. Curves in (A) were fitted by nonlinear regression; comparisons in (C), (D), and (F) used two-tailed Student's *t* tests. Exact *P* values and sample sizes are shown.

stem cells (*70*). During embryonic development, the p53/63/73 family drives Wnt activation to promote mesendodermal development (*71*). In contrast, work in cultured fibroblasts showed that p53 can suppress both the protein level and the transcriptional activity of β-catenin through glycogen synthase kinase-3 beta–triggered proteasome degradation (*72*). However, further studies in fibroblasts and cancer cell lines show that p53 transactivates the miR-34 family of micro RNAs, which collectively target the 3′ untranslated regions of *Wnt1*/3, *Lrp6*, *Ctnnb1*, and *Lef1*, thereby acting as suppressors of Wnt activity (*73*). In this study, we did not observe significant p53 ChIP-seq peaks of previously reported p53 targets within the Wnt pathway. In addition, apart from down-regulation of *Fzd10* and *Wnt3a* transcripts in $Trp53^{3KR/3KR}$ and $Trp53^{-/-}$ basal progenitors, which does not correlate with increased Wnt activity measured by both transcriptional signature and in vivo reporter, we did not observe a significant change in the majority of reported p53 targets of the Wnt pathway.

Our findings suggest that p53 restricts clonal expansion in the adult epidermis, similar to its proposed role in preventing cheater clones during early embryogenesis (*74*). It does so through multimodal suppression of Wnt activity, and its loss generates a radial gradient over the normally uniform epidermal Wnt field (*57*, *75*). The importance of this spatial pattern is supported by the more restrained growth of clones with uniform Wnt activity (*21*, *22*, *76*), which may be vulnerable to intraclonal crowding or competition with neighboring cells (*21*, *22*, *76*, *77*). *Apc*-mutant clones in the intestinal epithelium achieve robust growth by coupling clonal Wnt activation with increased secretion of the Wnt inhibitor Notum, which suppresses Wnt activity in adjacent WT tissue without affecting the ligand-independent Wnt activity in the mutant clone (*78*, *79*). This non–cell-autonomous suppression may help Apc mutant clones overcome competition from neighboring cells but does not inform what may be happening during clonal expansion mediated by an intact Wnt pathway we document in $Trp53^{-/-}$ clones. It is tempting to speculate that a radial Wnt activity gradient may blur the boundary between mutant and WT fields, avoiding or minimizing competition. In this model, $Trp53^{-/-}$ clones sustain a higher rate of progenitor renewal at the center than at the edges and in surrounding WT cells, enabling sustained expansion over time. This mode of growth contrasts with that of $Hras^{G12V}$ mutant clones, which exhibit lower renewal at the center than at the edge (*15*) and undergo growth stabilization owing to intraclonal competition, an outcome not observed in $Trp53^{-/-}$ clones, which rapidly overtake the surrounding tissue. Together, these results establish a model wherein spatial architecture of a signaling pattern can determine how mutant clones expand within normal epithelium.

Materials and methods

Animals

We used approximately equal numbers of male and female animals throughout the study. $Trp53^{3KR/3KR}$ [donated by W. Gu (*29*, *35*)]; $Trp53^{fl/fl}$ (JAX008462); $Ctnnb1^{fl/fl}$ (JAX 004152); $R26^{mTmG}$ (JAX007576), $R26^{tdTomato}$ (JAX007914) Cre-reporter; and H2B-GFP Wnt-reporter mice (JAX 013752) were all on the C57BL/6 background. Mice were housed and cared for in AAALAC-accredited facilities at the Fred Hutchinson Cancer Center and Icahn School of Medicine at Mount Sinai. All animal experiments were conducted in accordance with Fred Hutchinson Cancer Center, Icahn School of Medicine at Mount Sinai, and ethical regulations and IACUC-approved protocols. Project license numbers: 50814 (Fred Hutchinson Cancer Center), IPROTO202100000070 (Icahn School of Medicine at Mount Sinai).

Lentiviral constructs

To achieve RNA interference–mediated gene depletion and minimize off-target effect in the p53-specific perturbation pool, we subcloned five to seven top scored and/or validated shRNAs from the TRCportal database of the Broad Institute (*80*) to target each candidate as previously described (*13*, *44*). In the 10×10 two-combination complementary test, ORF overexpression was achieved using pLX EF1 Barcode vector as we reported (*13*, *44*). To ensure equal lentivirus representation within lentiviral pools, plasmids were mixed in equimolar ratios. Expression of Cre recombinase in combination with either constitutive or tetracycline-inducible ORFs was achieved using the pLX-EF1α-Cre and pLKO-TRE-Cre vectors, respectively. Similarly, coexpression of Cre recombinase with constitutive shRNAs was accomplished using the pLKO-Cre vectors (*13*, *44*, *45*, *81*, *82*).

Lentivirus production and in utero injection

Large-scale production, concentration, and titration of lentivirus were performed as described before (*81*, *83*). Briefly, 120 ml viral supernatant produced by two T300 flasks of 293TN (Invitrogen) cells was concentrated using Centricon Plus-70 centrifugal filter unit (100 kDa, Millipore). Concentrated supernatant was subsequently pelleted through ultracentrifugation (Beckman Optima MAX-TL ultracentrifuge with MLS-50 rotor; 45,000 rpm, 1.5 hour). Lentiviral pellet was resuspended in 70 μl of viral resuspension buffer (VRB; 20 mM Tris pH 8.0, 250 mM NaCl, 10 mM $MgCl_2$, 5% sorbitol) as stock. Lentivirus stocks were serially diluted and titrated using genomic DNA from lentivirus expressing green fluorescent protein–infected epidermis as qPCR standard. Barcoded virus was diluted in VRB according to titration results to ensure in utero transduction of epidermis at an ~10% rate. Lentiviral transduction into primary keratinocyte cultures or in vivo through ultrasound-guided in utero microinjection was performed as previously reported (*45*, *81*, *83*). Briefly, developmental stage of mouse embryos was determined by Vevo 2100 or 3100 LT animal ultrasound imager using MS550S/D transducer (Fujifilm). One microliter of diluted virus was injected into amniotic cavity of E9.25–E9.75 embryos using glass capillary needle fitted on Celltram Vario micro injector (Eppendorf).

In vivo genetic screens

In vivo differentiation screens were performed as previously reported (*13*, *15*). Briefly, whole skin of mice at P1 was digested in 2 mg/ml dispase at 37°C for 1 hour to separate epidermis from dermis. Epidermal tissue was further digested into single cells with 0.25% trypsin for 30 min at 37°C. Epidermal cell suspension from 30 animals was pooled together to make one biological replicate of >50-fold coverage for each shRNA. To isolate basal progenitors from differentiated suprabasal cells, epidermal cells were stained with CD326/EpCAM-APC (G8.8, 1:50; BD Biosciences) and CD49f/α6-Integrin-PerCP/Cy5.5 (GoH3, 1:50; Biolegend), both cell populations were isolated using a BD FACSAria II machine (BD Biosciences). Genomic DNA from all samples was extracted using QIAamp DNA tissue mini kit (Qiagen). Barcode preamplification, sequencing, and data processing were performed as previously described (*13*, *15*, *44*, *45*, *84*). Briefly, PCR preamplification of the barcode region generated products that were made into sequencing libraries using NEBNext Ultra II DNA Library Prep Kit (NEB). Sequencing of barcode libraries was performed on Illumina Nextseq 550 at the Fred Hutchinson Cancer Center Genomic Core facility. Short reads were trimmed down to barcode region by FASTX-Toolkit (*85*) and mapped to predetermined barcode list using BWA (*86*). Relative enrichment of barcodes was calculated using Deseq2 program (*46*) from raw barcode counts of all replicates, and individual shRNA constructs were scored as significant with a P value cutoff at 0.01. To identify individual genes that affected differentiation, we used the following criteria: (i) More than two shRNAs targeting the same gene had a significant impact on differentiation and (ii) all significant shRNAs targeting the same gene affected differentiation in the same direction (e.g., pro-differentiation or pro-renewal).

Immunofluorescence and immunoblot analyses

The following primary antibodies were used: chicken anti-GFP (ab13970, 1:1000 for immunofluorescence (IF); Abcam); mouse anti-BrdU (MoBU-1, 1:100 for IF; Invitrogen); mouse anti-β-Actin (66009, 1:3000 for WB,

1:500 for IF; Proteintech); rabbit anti-Keratin10 (Poly19054, 1:1000 for IF; BioLegend); rabbit anti-mouse p53 (NCL-L-p53-CM5p, 1:500 for WB; Leica Biosystem); rabbit anti-active Caspase-3 (C92-605, 1:1000 for IF; BD Bioscience); mouse anti 4-HNE (HNEJ-2, 1:100 for IF; Abcam); mouse anti-β-Catenin (15B8, 1:2000 for WB, Sigma); rabbit anti-Lamin A/C (#2032, 1:1000 for WB, Cell Signaling); rabbit anti-Gapdh (14C10, 1:1000 for WB, Cell Signaling); rabbit anti-Sfrp1 (PA5119647, 1:200 for IF; Invitrogen); rabbit anti-Lrp1 (E2Q7S, 1:200 for IF, Cell Signaling); rabbit anti-Usp22 (55110-1-AP, 1:500 for IF, Proteintech). Tissues were processed for immunostaining as previously described (*13*, *44*, *45*, *81*) and mounted in ProLong Gold with or without 4′,6-diamidino-2-phenylindole (Life Technologies). EdU-BrdU pulse-chase assay was performed as we described (*13*, *15*), tissue sections were first processed for EdU on the basis of Click-iT chemistry before subsequent Keratin10 and GFP detection. Lastly, tissue sections were incubated in 2N HCl at 37°C for 30 min to denature DNA, quenched with 0.1M pH 8.5 sodium borate twice, and proceeded for BrdU immunofluorescence detection. Confocal images were taken on Zeiss LSM700 and Andor BC43 systems, using Plan-Apochromat 40×/NA1.4 oil objective (Zeiss) and Plan Apochromat Lambda 60×/NA1.42 oil objective (Nikon). Images were processed using Zeiss Zen and ImageJ software. Immunoblot was performed using Novex system (Invitrogen), and fluorescent signal was captured using Odyssey Fc system (LI-COR).

EdU-BrdU pulse-chase proliferation and differentiation assays and quantification

We used previously described assays (*13*, *15*) as follows: We first administered EdU to animals, followed by BrdU injection 2 hours later. Twenty-four hours later, we harvested epidermis and detected EdU and BrdU signal by IF. To calculate cell division interval as a direct measurement of proliferation, we defined the number of EdU^+-only cells as E. Because one cell division gives rise to two daughter cells, the number of cell divisions that give rise to EdU^+-only cells is E/2. Because only basal epidermal cells divide, the fraction of basal cells that divide within 2 hours to give rise to EdU^+-only cells is F = (E/2)/B (where B is total number of basal cells). We next extrapolate this measurement to the whole basal cell population using the following ratio: D/1 = 2 hours/F, where D stands for the average cell division interval (in hours). To directly quantify the rate of progenitor renewal, tissues were also stained for expression of Keratin10 to mark committed progeny. We calculated the rate of progenitor cell renewal of the EdU^+-only cells (E) using the following equation: rate of renewal = (number of $Keratin10^-$ E cells) / (total number of E cells).

ChIP-seq, ChIP-qPCR, and ATAC-seq analysis

For ChIP-seq analysis of mouse p53, 1×10^7 basal progenitor cells were isolated from P21 WT or *Trp53*$^{3KR/3KR}$ animals using CD326/EpCAM-APC (G8.8, 1:50; BD Biosciences) and CD49f/α6-Integrin-PerCP/Cy5.5 (GoH3, 1:50; Biolegend) staining as described (*13*, *15*). Nutlin-3a (25 μM; Sigma Aldrich) was added to digestion medium to stabilize p53 for 1 hour. For ChIP-qPCR, 1×10^7 basal progenitor cells were isolated from P21 WT, *Trp53*$^{3KR/3KR}$, and *Trp53*$^{-/-}$ animals without treatment. Cells were fixed and processed for ChIP using Pierce Magnetic ChIP Kit (Thermo Fisher) and rabbit anti-mouse p53 (NCL-L-p53-CM5p, 5ug/ChIP; Leica Biosystem). For ChIP-seq, purified ChIP DNA bound by nutlin-stabilized p53 was prepped into sequencing library with NEBNext Ultra II DNA Library Prep Kit (NEB). For ChIP-qPCR, purified ChIP DNA bound by native level of p53 was subjected for qPCR with the following primer pairs detecting p53 peaks identified by ChIP-seq: *Sfrp1* peak1 (TTACAGCGTCCAACTCCGAC, CGGCCAGAAGGATCGGTTTA), *Sfrp1* peak2 (TCGCAAATCGGAAAGGCATC, TCGGCTAGCATGATCCTCCA), *Sfrp1* peak3 (ATACCGGAAGGTAAGCCCCT, CACACAGGCAAACTGCTGAC), *Lrp1* peak1 (TCTTCTTGCCTTCACGTCCC, TGTGGCCCTTCATGGCTTAG), *Lrp1* peak2 (ACCAGCGACCCAATTAAGGG, AAAAGACAGCGAAGGCGGTA), *Lrp1* peak3 (TCTCAGTCGGTTTGGGAACG, CTTCAAACGTCGCCCACAAG), and *Usp22* peak1 (CTGTCCGCGGGAAAGGATAA, GCCTTCCAAGGAGAGACTCG). ATAC-seq was performed on 5×10^5 basal progenitor cells using Illumina Tagment DNA TDE1 Enzyme and Buffer Kit. Both ChIP and ATAC libraries were sequenced on Illumina Nextseq 550. Short reads were mapped via BWA (*86*); ChIP and ATAC peaks were called using MACS2 (*87*); peak annotation, peak distribution analysis, and motif enrichment were performed using ChIP-seeker (*42*) and Homer (*43*); heatmaps around transcription start sites were generated using seqMINER (*88*).

RNA-seq and gene set analyses

We separated epidermis from the stroma using collagenase-dispase digestion (*13*, *15*, *45*, *83*). Single cells were obtained after 15 min trypsin digestion, and stained for basal integrin expression (*13*, *15*, *45*, *83*). We FACS isolated α6-Integrin^{high} epidermal progenitors and extracted their RNA using QIAamp RNA mini kit (Qiagen). RNA quality was assessed using Agilent 2100 Bioanalyzer, with all samples passing the quality threshold of RNA integrity numbers > 8. Library was prepared using NEBNext mRNA sample preparation kit, and cDNA was sequenced on Illumina Nextseq 550. Reads were mapped to mm10 build of the mouse genome using STAR (*89*) and summarized with featureCounts (*90*) and normalized using Deseq2 (*46*) and cufflinks (*91*). Differential expression between multiple groups following distinct pattern were achieved with pattern match algorithm of the MeV package (*41*) with a cutoff at $P < 0.01$. GSEA (*92*, *93*) analyses were performed using normalized expression data with GSEA 4.3.2, and Gene Ontology (GO) enrichment analyses were performed using gProfiler (*94*).

Whole-mount imaging and image analysis

Freshly harvested epidermis was mounted in phosphate-buffered saline, sealed in chamber slides, and imaged using high-speed spinning disc confocal (BC-43, Andor/Oxford instrument) with 20×0.8 Plan Apo LD air objective (Nikon) to minimize potential signal attenuation. In mTmG Cre reporter animals, cells per clone were counted using Imaris (Oxford instrument), and the three mode of division types were assessed as we previously described (*13*, *15*). In H2B-GFP Wnt reporter;TdT Cre reporter animals, 3D image analysis was achieved using Imaris (Oxford instrument). Specifically, local suprabasal Z-planes were removed, individual basal H2B-GFP Wnt reporter nuclei were then identified, and 3D sum of intensity was quantified using spot detection function; clone border was determined using surface reconstruction function according to TdT Cre reporter expression; minimal distance between clone border and individual H2B-GFP Wnt reporter nuclei were calculated using distance transformation function (fig. S3B). Clonal boundaries were outlined as regions of interest and aligned by Image J (*95*). Clonal roundness was calculated as $(4\times A)/(\pi\times MA^2)$, where A represents clone area and MA represents major axis length; normalized radial variance as $[(1/N) \times \Sigma(ri - \bar{r})^2] / \bar{r}^2$, where N represents the total number of boundary points, ri represents the distance of the i-th boundary point from the clone centroid, and $\bar{r}$ represents the mean radial distance; circularity as $(4\pi\times A)/P^2$, where A represents clone area and P represents clonal perimeter.

Statistics and reproducibility

All quantitative data were expressed as mean ± SD. Differences between groups were assayed using two-tailed Student's *t* test. Significant differences were considered when $P < 0.05$. Nonlinear regression was performed using Prism 6 (GraphPad software). No statistical methods were used to predetermine sample sizes. For the in vivo genetic screens, sample sizes were selected to provide >50-fold coverage of each perturbation. All other quantitative in vivo experiments included at least three animals or biological replicates. Exact sample sizes are reported in the figure legends. Experiments were not randomized, and investigators were not blinded to genotype or experimental condition during data collection or outcome assessment. No animals or data

points were excluded from the analyses, and no predefined inclusion or exclusion criteria were applied.

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SUPPLEMENTARY MATERIALS

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RESEARCH ARTICLE SUMMARY

METABOLISM

Mitochondrial carrier SLC25A34 links clock, diet, and temperature control of adipocyte lipid cycling

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INTRODUCTION: Brown and beige thermogenic adipocytes (fat cells) have uniquely high energy expenditure capacity. This activity is critical for defending body temperature in small mammals and is linked to cardiometabolic health in humans. Energy expenditure from thermogenic fat cycles daily on a rhythm set by the circadian clock, rising after waking and dropping during sleep. At any point, however, this activity can be suddenly induced by exposure to cold temperature or by ingestion of a calorie-dense, high-fat meal. How thermogenic fat can seamlessly achieve both rhythmic continuity and acute responsiveness is a fundamental question of energy homeostasis that has remained unknown.

RATIONALE: We set out to understand how diverse cues from the body's clock, environmental temperature, and diet are molecularly integrated in adipocytes to facilitate both anticipatory and adaptive regulation. By combining unbiased analysis of DNA-binding patterns, gene expression, and protein levels in thermogenic fat from mice that were subject to different circadian, environmental, and dietary perturbations, we identified the orphan mitochondrial metabolite transporter SLC25A34 as the convergence point of all three physiological paradigms.

RESULTS: Similar to fat's thermogenic activity, the expression of *Slc25a34* mRNA exhibited a daily rhythm, which was mediated by the nuclear receptors REV-ERBα and REV-ERBβ. REV-ERBs are transcriptional repressors in the circadian clock that recruit histone deacetylase 3 to the *Slc25a34* promoter during sleep to shut off transcription. When animals wake and REV-ERB circadian repression is lifted, the peroxisome proliferator–activated receptors α and γ (PPARα and PPARγ) are able to bind the *Slc25a34* promoter and restore expression. However, at any time, when animals are suddenly confronted with an unanticipated need for adipocyte energy expenditure such as exposure to cold temperatures or eating a lipid-rich meal, REV-ERB repression is rapidly overridden and PPARs are activated by lipolytic signals to boost SLC25A34 levels. When the energy demand is met, the REV-ERB–mediated rhythmicity of *Slc25a34* is reestablished.

Our biochemical and metabolic data support a functional model in which SLC25A34 transports oxaloacetate into mitochondria. In doing so, SLC25A34 helps to maintain high tricarboxylic acid (TCA) cycle activity, which is required for powering energy expenditure in thermogenic adipocytes. The need for SLC25A34 during times of high energetic demand arises because thermogenic adipocytes not only oxidize fatty acids in the mitochondria but also simultaneously synthesize lipids in the cytosol, a process collectively known as lipid cycling. Lipid synthesis requires continuous production of acetyl–coenzyme A (acetyl-CoA) in the cytosol, and the two major pathways responsible for this acetyl-CoA pool also generate oxaloacetate as a by-product, which then needs to be transported back into the mitochondria. Increased cytosolic acetyl-CoA additionally promotes the transcription of genes linked to mitochondrial respiration. Thus, SLC25A34 supports both sides of this self-reinforcing cycle in which lipids are continually broken down and resynthesized.

CONCLUSION: These findings identify SLC25A34 as a molecular node that integrates circadian, temperature, and dietary cues and reveal how thermogenic fat activity can sustain a daily rhythm while remaining acutely responsive to sudden energy demands. This mechanism may offer a new entry point for therapeutically enhancing energy expenditure in metabolic disease. □

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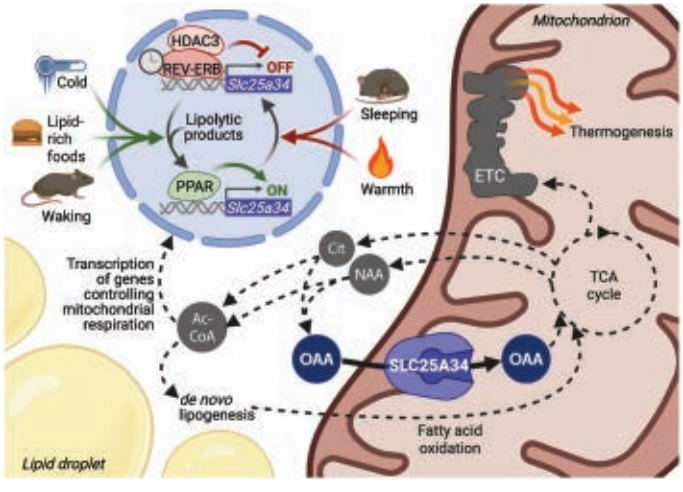

The circadian clock, diet, and temperature control lipid homeostasis through SLC25A34. During sleep, REV-ERBs repress *Slc25a34*. After waking, cold exposure, or lipid-rich meals, PPARs rapidly induce expression. We hypothesize that SLC25A34 imports oxaloacetate (OAA) into the mitochondria to support fatty acid oxidation and lipid synthesis and maximize TCA cycle and electron transport chain (ETC) activity. Cit, citrate; NAA, *N*-acetylaspartate. [Figure created with BioRender.com]

METABOLISM

Mitochondrial carrier SLC25A34 links clock, diet, and temperature control of adipocyte lipid cycling

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Adipocyte lipid metabolism is coordinated by circadian rhythms, diet, and environmental temperature, but how these diverse signals are molecularly integrated remains unknown. We showed that these cues converge on the orphan mitochondrial transporter SLC25A34 to orchestrate lipid cycling. During the sleep phase, the adipocyte clock suppresses *Slc25a34* expression through the REV-ERB transcriptional repressors. Entering the active phase, consuming lipid-rich diets, or exposure to cold abolishes REV-ERB repression, and lipolytic signals stimulate *Slc25a34* transcription through the peroxisome proliferator–activated receptors. SLC25A34 is proposed to import oxaloacetate into mitochondria, dually supporting the tricarboxylic acid cycle and cytosolic acetyl–coenzyme A (acetyl-CoA) production. Elevated cytosolic acetyl-CoA then fuels the synthesis of lipids and promotes the transcription of genes enhancing mitochondrial oxidation. Thus, SLC25A34 confers circadian, dietary, and temperature control of adipocyte lipid metabolism.

The circadian clock synchronizes physiological processes with environmental cues, integrating signals from sunlight, diet, physical activity, and temperature to optimize metabolic function across tissues (*1*–*3*). Circadian metabolic control must balance anticipated daily energetic needs with acute adaptation to unexpected perturbations. Adipose thermogenesis is an ideal paradigm in which to study this coordinated anticipatory and adaptive nature of the clock. In mice and humans, the energy-expending activity of thermogenic adipocytes is regulated by exposure to cold temperature (*4*–*6*), diet composition and quantity consumed (*7*, *8*), and the daily oscillations of the circadian clock (*9*–*11*). In rodents, these energy-dissipating cells are critical for defending body temperature against environmental cold exposure (*12*), whereas in humans, brown adipose tissue (BAT) appears to improve lipid and glucose homeostasis and is associated with cardiometabolic protection (*13*). In mice and humans, BAT metabolic activity begins to increase just before waking (*9*, *14*, *15*), peaks late in the waking period, and then declines to its lowest point during sleep. Given the energetic cost of BAT activity, clock repression of thermogenesis during sleep may have conferred an evolutionary advantage by reducing unnecessary energy expenditure.

Regardless of the time of day, acute exposure to cold temperature can override circadian control and rapidly stimulate adipose thermogenesis through the sympathetic nervous system (*4*). Although the specific metabolic pathways that mediate the interaction between the clock and temperature cues are unresolved, lipid utilization likely contributes because numerous elements of adipocyte lipid metabolism are regulated by both programs (*4*, *16*, *17*). A unique hallmark of BAT lipid metabolism is the ability to simultaneously increase lipid synthesis in the cytosol and fatty acid β-oxidation in the mitochondria (*18*–*20*). Such synthesis and breakdown fuel an energy-dissipating lipid cycle that promotes thermogenesis and represents a possible therapeutic strategy against obesity and metabolic disease (*21*, *22*). Both environmental cold (*18*–*20*) and the circadian clock (*23*) can influence this dual activation of anabolic and catabolic lipid pathways. However, how continuous circadian rhythmicity and acute cold responsiveness are molecularly integrated to coordinate thermogenic metabolism remains unknown.

Clock and temperature control of the orphan mitochondrial transporter *Slc25a34*

To understand how circadian and cold-regulated signaling pathways orchestrate thermogenic metabolism in adipocytes, we used a combination of proteomics, transcriptomics, and genomic binding to identify candidate factors in BAT (Fig. 1A and table S1). We searched for (i) proteins that increased in level in the BAT of mice exposed to 4°C cold exposure for 3 weeks versus thermoneutrality [fold change (FC) >2] (*24*); (ii) proteins that increased in level in the BAT of whole-body *Rev-erbα* knockout (KO) mice compared with controls at 29°C (FC >2) (table S2); (iii) genes with a *Rev-erbα*–binding site within 10 kilobases (kb) of the transcriptional start site that showed higher occupancy by REV-ERBα at thermoneutrality than in cold conditions (*9*); and (iv) genes that were regulated both by cold and REV-ERBα and REV-ERBβ at the transcriptional level [FC >2 in wild-type (WT) mice upon cold exposure and in BAT-specific *Rev-erbα* and β double KOs (*Rev-erbα*/β BKO) compared with WT mice, false discovery rate (FDR) <0.05 in all cases] (table S3). Only two factors satisfied all criteria: the well-established, canonical thermogenic effector uncoupling protein 1 (UCP1, also known as SLC25A7) and the uncharacterized, orphan mitochondrial metabolite transporter SLC25A34. Both proteins are members of the SLC25A family of mitochondrial solute carriers (*25*, *26*) highlighting the critical importance of metabolite and ion exchange across the inner mitochondrial membrane (IMM) for adipocyte function.

At thermoneutrality (29°C), *Slc25a34* expression was the highest in the heart, followed by slow-twitch skeletal muscle, kidney, and multiple brain regions (Fig. 1B). BAT was among the organs with the lowest amounts of *Slc25a34* mRNA. However, upon 24 hours of cold exposure (4°C), *Slc25a34* mRNA accumulated in BAT 90-fold over the levels at thermoneutrality, causing BAT to supplant heart as the most *Slc25a34*-enriched tissue (Fig. 1B). Transcription of *Slc25a34* increased in BAT after only 8 hours of cold exposure and remained elevated throughout 3 weeks of cold adaptation (Fig. 1C). At the protein level, the abundance of SLC25A34 was significantly increased after 3 days of cold exposure and then for the duration of cold acclimation (Fig. 1D). We next examined SLC25A34 protein localization and stability in thermogenically active adipocytes. Using the β_3-adrenergic receptor agonist CL-316,243 to mimic cold exposure in cultured brown adipocytes, we could confirm mitochondrial localization of an exogenously expressed

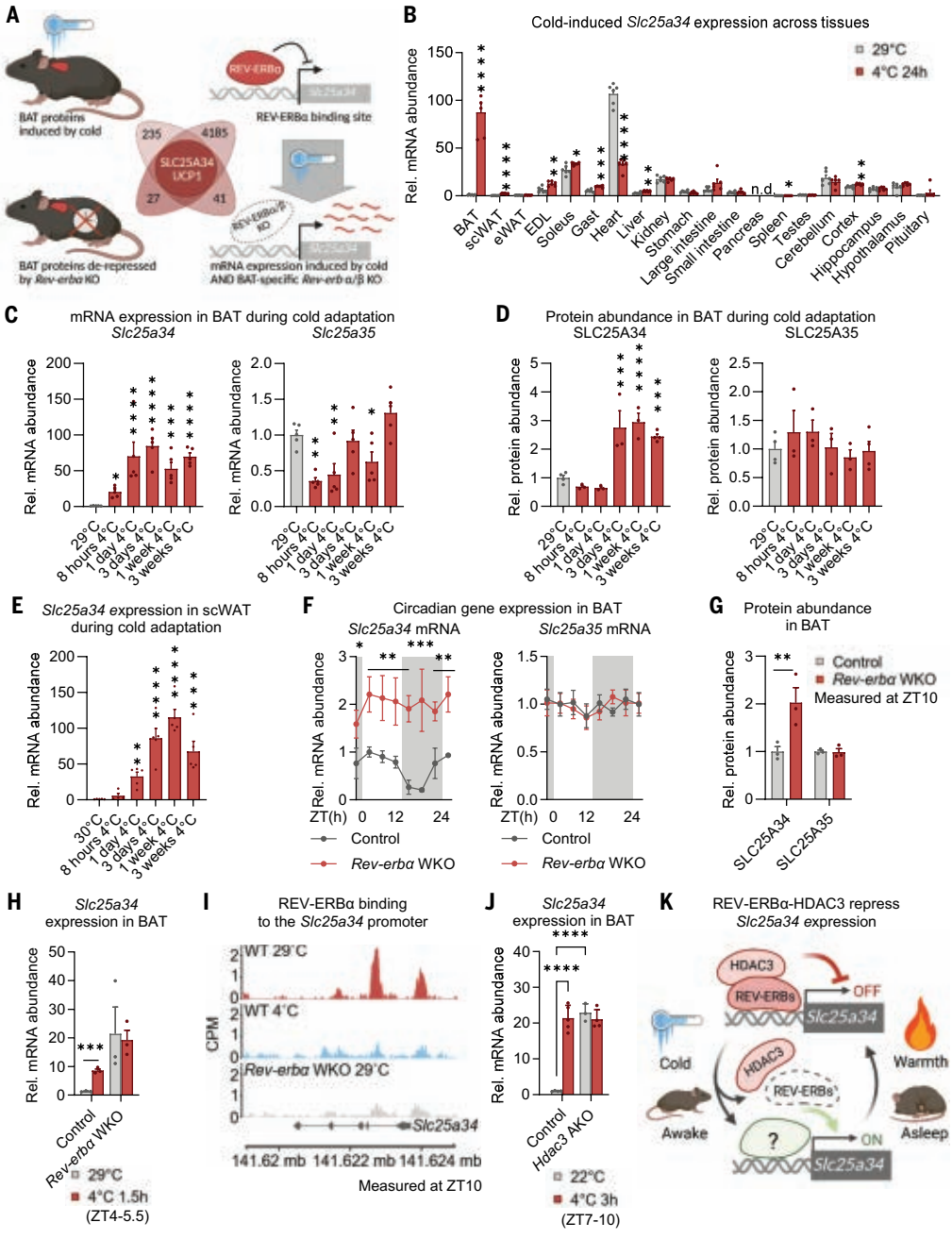

Fig. 1. Clock and temperature control of the orphan mitochondrial transporter *Slc25a34* through the REV-ERB circadian repressors. (**A**) Schematic of the multi-omic approach to discover *Slc25a34* comparing the proteome of BAT from (i) cold-exposed mice (top left, 3 weeks cold exposure versus thermoneutrality, FC > 2) and (ii) *Rev-erbα* WKO mice at thermoneutrality (bottom left, *Rev-erbα* WKO versus control, FC > 2), (iii) the BAT cistrome from control and Rev-erbα WKO mice (top right, REV-ERBα–binding sites located within 10 kb of the nearest gene that show higher occupancy in cold conditions than at thermoneutrality), and (iv) the BAT transcriptome from cold-exposed

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Rev-erbα/β BKO mice (bottom right, genes induced by cold in control mice (FC > 2, FDR < 0.05) and genes induced by double *Rev-erbα/β* KO in BAT at thermoneutrality (FC > 2, FDR < 0.05). (**B**) *Slc25a34* expression across tissues expressed as the fold change relative to the expression in the tissue at 29°C. (**C** and **D**) *Slc25a34* and *Slc25a35* mRNA expression (C) and protein abundance (D) in BAT during cold adaptation. Significance indicates effects of cold exposure compared with 29°C. (**E**) *Slc25a34* mRNA expression in scWAT during cold adaptation. Significance indicates effects of cold exposure compared with 29°C. (**F**) Circadian expression of *Slc25a34* and *Slc25a35* mRNA in control and *Rev-erbα* WKO mice acclimated to 29°C. (**G**) SLC25A34 and SLC25A35 protein abundances in BAT of control and *Rev-erbα* WKO mice at 22°C at ZT10. (**H**) *Slc25a34* mRNA abundance in BAT of control and *Rev-erbα* WKO mice at 29°C and after 1.5 hours of 4°C exposure (ZT4 to ZT5.5). (**I**) REV-ERBα occupancy in the proximal *Slc25a34* promoter in BAT at 29°C and after 6 hours (ZT4 to ZT10) of 4°C exposure of control and *Rev-erbα* WKO mice. (**J**) *Slc25a34* mRNA abundance in BAT from control and *Hdac3* AKO mice at 22°C and after 3 hours at 4°C (ZT7 to ZT10). (**K**) Schematic depicting the role of REV-ERBs in coordinating circadian and cold-mediated regulation of *Slc25a34* expression. For all panels, data are represented as mean ± SEM. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001, unpaired two-tailed multiple *t* tests with the Benjamini, Krieger, and Yekutieli two-stage linear step-up procedure for multiple-comparisons correction [(B) and (H)]; one-way ANOVA with the Benjamini, Krieger, and Yekutieli two-stage linear step-up procedure for multiple comparisons correction [(C) to (E)]; or two-way ANOVA with the Benjamini, Krieger, and Yekutieli two-stage linear step-up procedure for multiple comparisons correction [(F), (G), and (J)].

hemagglutinin (HA)–tagged SLC25A34 (fig. S1, A and B), consistent with the SLC25A carrier family. However, unlike other canonical adipocyte carriers, such as UCP1 and SLC25A1 (the citrate transporter), SLC25A34 was markedly less stable, as evidenced by its decreased abundance after inhibition of protein synthesis with cycloheximide (fig. S1, C and D).

Cold exposure did not cause accumulation of SLC25A35, the closest homolog of SLC25A34, which is proposed to have a similar transport function based on sequence conservation in key substrate-binding residues (*27*, *28*). In fact, the abundance of *Slc25a35* mRNA was acutely suppressed by cold exposure (fig. S1E and Fig. 1C), and protein concentrations remained unchanged throughout cold acclimation (Fig. 1D). In subcutaneous white adipose tissue (scWAT), which undergoes recruitment of energy-expending beige adipocytes during cold adaptation, both transporters accumulated after cold exposure but *Slc25a34* to a nearly 40-fold greater extent (Fig. 1E and fig. S1F). Thus, the abundance of SLC25A34 is strongly and sustainably increased in thermogenic adipose tissues after exposure to environmental cold temperature.

Circadian and cold control of *Slc25a34* by REV-ERBs and HDAC3-mediated transcriptional repression

In addition to regulation by environmental temperature, expression of *Slc25a34* in BAT also oscillates over 24 hours (Fig. 1F). Whole-body genetic KO of the circadian nuclear receptor *Rev-erbα* (hereby referred to as *Rev-erbα* WKO) not only abolished this rhythmicity but also resulted in constitutively increased *Slc25a34* expression compared with that in control littermates (Fig. 1F), demonstrating that REV-ERBα plays a critical role in clock-mediated repression of *Slc25a34* transcription. Consistent with this, the abundance of SLC25A34 protein was increased in BAT from *Rev-erbα* WKO mice (Fig. 1G and table S2). Conversely, *Slc25a35* expression was not circadian nor was its protein abundance changed upon *Rev-erbα* deletion (Fig. 1, F and G). Given that the circadian clock, through REV-ERBα, modulates temperature-dependent thermogenic control (*9*), we investigated the interaction between REV-ERBα suppression and cold enhancement of *Slc25a34* transcription. We measured *Slc25a34* and *Slc25a35* expression in BAT from thermoneutral-acclimated and cold-exposed *Rev-erbα* WKOs and control littermates. Cold exposure increased *Slc25a34* expression in the control mice. However, in *Rev-erbα* WKO BAT, the amounts of *Slc25a34* were already maximally increased at thermoneutrality and could not be further increased upon cold exposure (Fig. 1H), suggesting that derepression of REV-ERBα control is a critical step in the effects of cold on transporter abundance. Indeed, REV-ERBα binds to two sites in the proximal promoter and the first intron of *Slc25a34* under thermoneutral conditions, and this binding was abolished by cold temperature (Fig. 1I). In contrast to *Slc25a34*, *Slc25a35* was downregulated by cold to the same extent in both *Rev-erbα* WKOs and control littermates (fig. S1G) despite the presence of the REV-ERBα–binding site in its proximal promoter (fig. S1H).

Because of the global nature of the *Rev-erbα* WKO, we could not rule out that REV-ERBα–dependent regulation in the central nervous system or non-BAT peripheral tissues might indirectly influence BAT *Slc25a34* expression. Therefore, we conditionally deleted *Rev-erbα* and β in brown adipocytes (*Rev-erbα*/β BKOs) to determine whether the observed regulation of *Slc25a34* expression in BAT was cell autonomous. We depleted both REV-ERBs because they share the same consensus DNA-binding sites, and *Rev-erbβ* has been shown to compensate for the loss of *Rev-erbα* in other tissue-specific KOs (*29*). Moreover, although *Rev-erbα* was previously shown to be down-regulated by cold exposure (*9*), we found that cold similarly decreased *Rev-erbβ* in BAT (fig. S2, A and B), suggesting that both REV-ERBs might contribute to the cold regulation of *Slc25a34*. Indeed, BAT *Slc25a34* expression was increased in *Rev-erbα/β* BKO mice compared with littermate controls both at thermoneutrality and in the cold (fig. S2C and table S3). Conversely, *Slc25a35* expression was similar between control and *Rev-erbα/β* BKO mice and was significantly decreased by cold exposure in both genotypes (fig. S2D). Circadian rhythmicity of *Slc25a34* expression was additionally blunted in *Rev-erbα/β* BKO, along with that of *Ucp1* and core clock REV-ERB target brain and muscle ARNT-like 1 (*Bmal1*) (fig. S2, E to G). Collectively, these data show that REV-ERB transcriptional repressors cell autonomously control both cold and circadian regulation of *Slc25a34*.

REV-ERBs mediate transcriptional repression through the recruitment of histone deacetylase 3 (HDAC3) and subsequent histone deacetylation (*30*). To address whether HDAC3 contributes to cold regulation of *Slc25a34*, we measured its expression in the BAT of mice with adipose-specific genetic deletion of *Hdac3* (*Hdac3* AKO) and littermate controls at room temperature (RT, 22°C) and after acute exposure to cold. BAT *Slc25a34* expression in *Hdac3* AKO mice was increased at RT compared with that of control mice and was not further increased by cold exposure (Fig. 1J), indicating that HDAC3 acts as the REV-ERB–directed corepressor of *Slc25a34*. Conversely, BAT *Slc25a35* expression was decreased in the *Hdac3* AKO compared with control animals at RT and was further suppressed by cold in the *Hdac3* AKO mice (fig. S2H). In a previously published RNA-sequencing (RNA-seq) dataset of BAT from *Hdac3* AKO and control mice at thermoneutrality (*31*), *Slc25a34* was the third most up-regulated gene when *Hdac3* was deleted (fig. S2, I and J). Together, these data show that REV-ERBα, REV-ERBβ, and HDAC3 coordinate circadian and temperature-dependent repression of *Slc25a34* at thermoneutrality and during sleep, which is relieved by waking and acute cold exposure (Fig. 1K).

Cold- and diet-induced lipid metabolism stimulate *Slc25a34* expression through peroxisome proliferator-activated receptor α and γ

Although our findings establish REV-ERBα– and REV-ERBβ–dependent repression as a key event in the regulation of *Slc25a34* abundance, the cell-autonomous derepression of REV-ERBs could not explain the entirety of the transcriptional control by cold exposure. Therefore, we next sought to identify the transcription factor or factors contributing to the activation of *Slc25a34* expression. Given the numerous transcriptional regulators involved in thermogenic metabolism in adipocytes (*32*),

we sought to identify candidate activators by performing pathway enrichment analysis of the top 50 proteins that exhibited similar expression patterns as SLC25A34 throughout cold adaptation (*24*) (fig. S3A and table S4). We hypothesized that determining the pathways co-regulated with SLC25A34 might reveal the factors responsible for the transcription control. We found that nearly all proteins co-regulated with SLC25A34 were linked to biological processes related to lipid metabolism and biosynthesis (Fig. 2A). Therefore, we tested whether peroxisome proliferator-activated receptors (PPARs), the canonical regulators of adipocyte lipid metabolism (*33*), could be transcriptional activators of *Slc25a34*. Indeed, both PPARα and PPARγ bind to a regulatory region near *Slc25a34* that is enriched for active histone marks (*34*). Agonists for PPARα (GW7647) or PPARγ (rosiglitazone) cell autonomously increased *Slc25a34* expression in brown adipocytes alongside their canonical targets *Ucp1* and *Elovl3* (Fig. 2, B to E, and fig. S3, B and C). Small interfering RNAs (siRNAs) targeting either factor significantly decreased transporter mRNA abundance (Fig. 2, F and G). Thus, both PPARα and PPARγ appear to regulate *Slc25a34* expression in vitro; however, PPARγ can both control PPARα expression (*35*) and compensate for its loss (*36*), confounding assignment of specific PPAR dependency.

Therefore, we performed RNA sequencing of BAT from mice with adipose-specific genetic depletion of *Pparα* (*Pparα* AKO; targeting exon 5) and control littermates. Deletion of *Pparα* resulted in only 10 significantly down-regulated mRNAs (Fig. 2H, fig. S3D, and table S5). Among these, *Slc25a34* showed the greatest reduction (Figs. 2, H and I, and fig. S3D), revealing it to be one of the most robust and selective PPARα targets. By contrast, neither *Ucp1* nor *Slc25a35* expression was affected by *Pparα* deletion (Fig. 2J and fig. S3E). This is most likely because *Pparγ*, which was expressed similarly in BAT from *Pparα* KOs compared with control animals (fig. S3F), could compensate for some genes such as *Ucp1* but not *Slc25a34* in the absence of *Pparα*. We further confirmed an essential role for PPARα in regulating *Slc25a34* expression in vivo using a second adipose-specific KO model in which exon 4 was deleted (*37*) (fig. S3, G to J). To address the effect of acute *Pparα* depletion on physiological BAT *Slc25a34* regulation, we used a tamoxifen-inducible *Pparα* genetic depletion mouse model expressing CreERT2 under control of the *Ucp1* promoter (*Pparα* iBKO). *Pparα* iBKO and littermate control mice acclimated to RT were exposed to thermoneutrality or cold for 3 days concurrent with the tamoxifen-induced depletion of *Pparα*. *Pparα* expression was decreased in *Pparα* iBKO mice both at thermoneutrality and after cold exposure compared with that in control animals (fig. S3K). This acute ablation of *Pparα* in BAT significantly blunted the cold-induced accumulation of *Slc25a34*, demonstrating the importance of PPARα for physiological temperature–sensitive regulation of *Slc25a34* (Fig. 2K). Conversely, cold-induced *Ucp1* expression was unaffected by *Pparα* ablation, suggesting full compensation by PPARγ (Fig. 2L). Thus, whereas both PPARα and PPARγ can increase *Slc25a34* expression, PPARα appears to be the predominant physiological regulator of this transporter.

In addition to cold, the transcription-activating functions of PPARs can also be modulated by products of lipolysis, which act as PPAR agonists (*38*, *39*). Given that lipid catabolism is increased in BAT during cold exposure (*6*), we evaluated whether lipolysis influences *Slc25a34* expression. Brown adipocytes were treated with SR-3420, a small molecule that directly enhances adipose triglyceride lipase (ATGL)–dependent triglyceride lipolysis by disrupting the interaction between its coactivator αβ hydrolase domain-containing protein 5 (ABHD5) and Perilipin 1 (*40*, *41*). SR-3420 increased the expression of *Slc25a34* mRNA in the brown adipocytes (Fig. 2M). Furthermore, the lipolytic induction of *Slc25a34* expression was completely prevented by blocking fatty acid activation with the long-chain acyl–coenzyme A synthetase (ACSL) inhibitor triacsin C or by preventing the transport of fatty acids into the mitochondria with the carnitine palmitoyltransferase 1a (CPT1a) inhibitor etomoxir (Fig. 2M). These data suggest that endogenous lipid products from mitochondrial fatty acid β-oxidation activate *Slc25a34* expression in brown adipocytes, likely under PPARα-dependent control.

Exogenous dietary lipids also activate PPARs and BAT thermogenic capacity similarly to cold exposure (*42*). Therefore, we compared the effects of a 60% high-fat diet (HFD) and a ketogenic diet with 93% fat content with chow diet (12% fat) (Fig. 2N) on the expression of the transporter in BAT of mice acclimated to RT (22°C). A HFD modestly increased *Slc25a34* mRNA abundance, but the ketogenic diet increased mRNA abundance of the transporter to a much greater extent (Fig. 2O). Bulk RNA sequencing further revealed that *Slc25a34* was the second most induced mRNA in BAT after ketogenic diet feeding (Fig. 2P and table S6). Consistent with the ketogenic diet results, fasting also increased the expression of *Slc25a34* in BAT, skeletal muscle, and heart, but not liver (Fig. 2Q), whereas *Slc25a35* and *Ucp1* expression was decreased (fig. S3, L and M). Therefore, *Slc25a34* is dually controlled by circadian REV-ERB-HDAC3 repression and cold-induced, lipid-liganded PPARα/γ activation (Fig. 2R).

SLC25A34 controls mitochondrial respiration in thermogenic adipocytes in vitro and in vivo

To investigate the role of the transporter in mitochondrial function, *Slc25a34* was acutely depleted in immortalized murine brown adipocytes. Three different siRNAs against the transporter (fig. S4, A to D) significantly decreased respiratory capacity, particularly norepinephrine (NE)–stimulated and carbonyl cyanide *p*-trifluoromethoxyphenylhydrazone (FCCP)–dependent maximal respiration (Fig. 3A and fig. S4E). Conversely, continuous measurement of basal respiration over the course of siRNA knockdown (KD) revealed that the effects on basal respiration were periodic and not sustained (fig. S4F). Depletion of *Slc25a34* also reduced maximal respiratory capacity of the brown adipocytes not treated with NE (Fig. 3B) and NE-stimulated and maximal respiration in primary brown adipocytes (Fig. 3C). Despite the significant reduction in respiratory capacity, *Slc25a34* depletion did not affect adipocyte differentiation (fig. S4G). Conversely, depleting *Slc25a35* had little or no impact on NE-stimulated respiratory capacity (Fig. 3D and fig. S4H), highlighting a functional distinction between the two homologs.

SLC25 carriers transport a diverse array of metabolites across the IMM, including amino acids, fatty acids, nucleotides, and cofactors (*25*, *26*). Sequence analysis suggests that SLC25A34 may be a carboxylic acid transporter (*25*, *26*). Given that several carboxylic acids serve as mitochondrial fuel, we evaluated how mitochondrial respiration in brown adipocytes with or without SLC25A34 changed in response to different substrates. We permeabilized cultured brown adipocytes with a plasma membrane permeabilizer that serves to dilute out the cytosolic metabolites and endogenous respiratory substrates. We then supplemented the permeabilized cells with different substrates for respiratory complex I (malate, pyruvate, and glutamate), complex II (succinate), and donating electrons into the Q-pool (octanoylcarnitine, palmitoylcarnitine, and glycerol-3-phosphate) alongside adenosine 5′-diphosphate (ADP) and magnesium and measured their respiratory capacity (Fig. 3E). We observed lower coupled respiration in the cells depleted of *Slc25a34* on nearly all the tested substrates (Fig. 3F and fig. S4I), indicating that the mitochondrial deficiency was likely the result of a broader disruption of tricarboxylic acid (TCA) cycle activity or respiratory capacity. Thus, our findings indicate that SLC25A34 is required for the mitochondrial function of brown adipocytes.

To test the role of *Slc25a34* in vivo, we generated mice in which *Slc25a34* can be deleted in brown adipocytes in an inducible manner (*Slc25a34* iBKO). We directly assessed BAT thermogenic function using an experimental paradigm in which animals were anesthetized at thermoneutrality, injected with NE, and changes in oxygen consumption were measured in indirect calorimetric cages (*43*) (Fig. 3G). NE-induced respiration was significantly lower in *Slc25a34* iBKO mice compared with littermate controls (Fig. 3, H to J). This occurred despite unchanged *Ucp1* mRNA levels between genotypes (fig. S4J), indicating

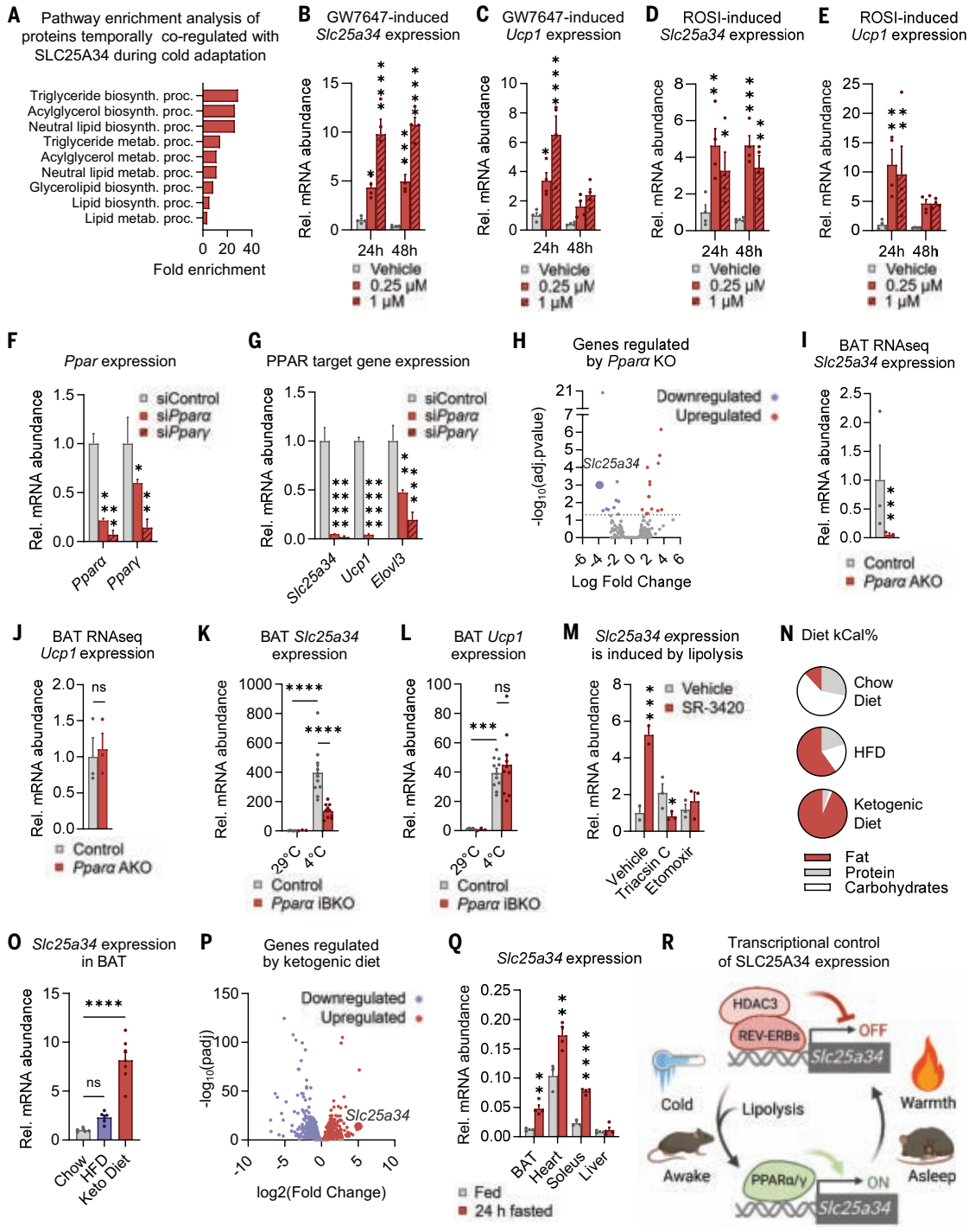

Fig. 2. Cold- and diet-induced lipid metabolism activate PPARα and PPARγ to mediate temperature control of *Slc25a34*. (**A**) Pathway enrichment of proteins temporally co-regulated with SLC25A34 in BAT during cold adaptation. (**B** to **E**) Regulation of *Slc25a34* and *Ucp1* mRNA by the PPARα agonist GW7647 [(B) and (C)] or the PPARγ agonist rosiglitazone [(D) and (E)]. (**F** and **G**) Gene expression in brown adipocytes after siRNA KD of *Pparα* and *Pparγ*. (**H**) Volcano plot comparing BAT transcriptomes of control and *Pparα* AKO (targeted exon 5) mice acclimated to 22°C. (**I** and **J**) Gene expression in BAT of control and *Pparα* AKO (targeted exon 5) mice quantified in RNA-seq analysis. (**K** and **L**) Gene expression in BAT from control and *Pparα* iBKO mice acclimated to 22°C and exposed to 29°C or 4°C for 3 days. (**M**) *Slc25a34* mRNA in brown adipocytes after treatment with the lipolytic activator SR-3420 and the ACSL inhibitor Triacsin C or the CPT1a inhibitor etomoxir. (**N** and **O**) Diet composition and effects of diet challenges on BAT *Slc25a34* mRNA (harvested between ZT4 and ZT6). (**P**) Volcano plot depicting genes from BAT RNA-seq that are regulated by ketogenic diet. (**Q**) *Slc25a34* mRNA expression analysis across tissues of WT mice fed or fasted for 24 hours. (**R**) Schematic representation of how PPARα coordinates *Slc25a34* expression with REV-ERBs in BAT. For all panels, data are represented as mean ± SEM. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$, two-way ANOVA with the Benjamini, Krieger, and Yekutieli two-stage linear step-up procedure for multiple-comparisons correction [(B) to (G) and (K) to (M)]. For (I) and (J), the *P* value was computed using Bioconductor software's package and corrected for multiple testing using the Benjamini-Hochberg mode of the R function *p.adjust* to compute a FDR. For (O), one-way ANOVA with the Benjamini, Krieger, and Yekutieli two-stage linear step-up procedure for multiple-comparisons correction was used. For (Q), unpaired two-tailed multiple *t* tests with the Benjamini, Krieger, and Yekutieli two-stage linear step-up procedure for multiple-comparisons correction was used.

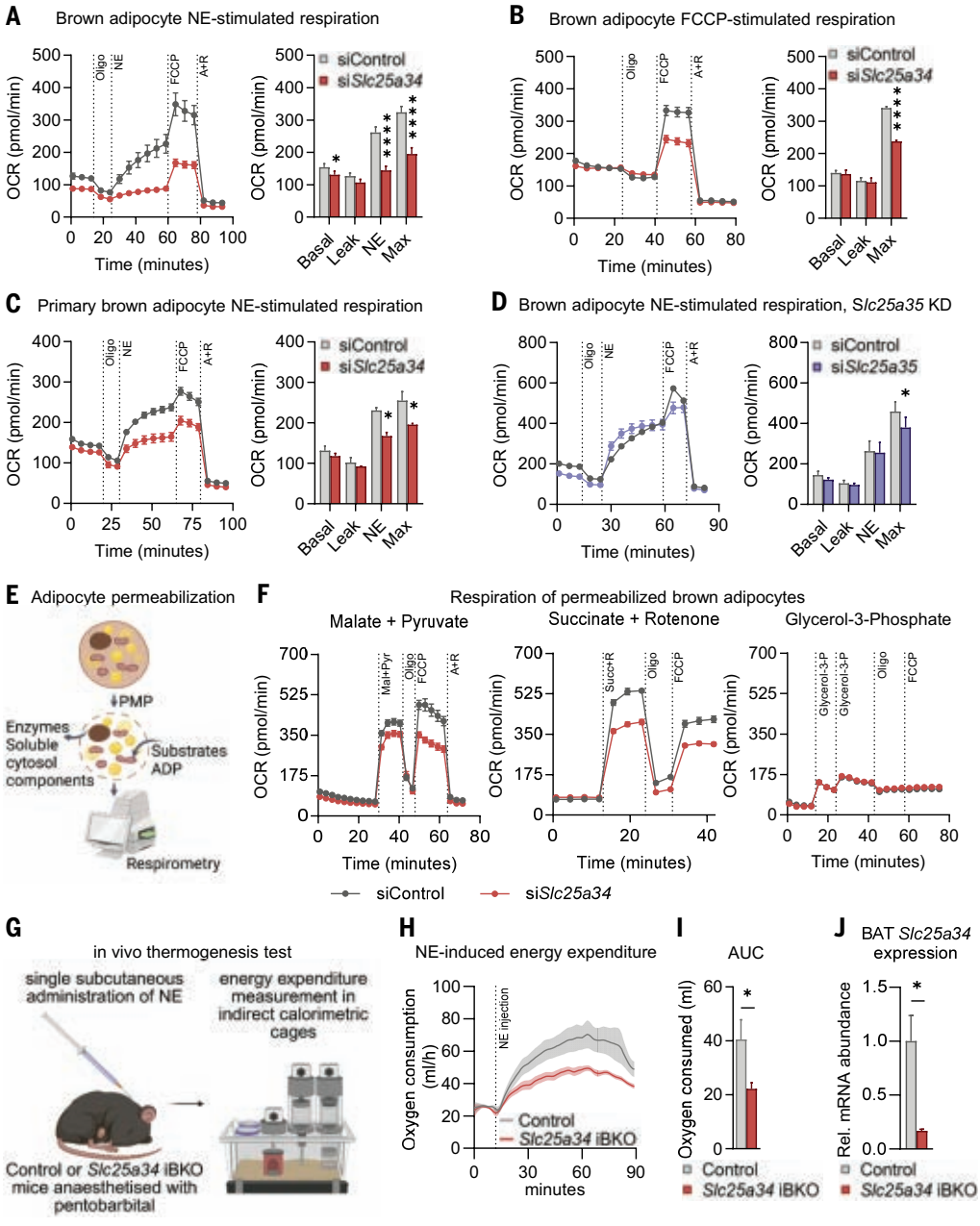

Fig. 3. SLC25A34 controls mitochondrial respiratory capacity in thermogenic adipocytes in vivo and in vitro. (**A** and **B**) NE-induced (A) and maximal (B) OCR of brown adipocytes with control or siRNA-mediated *Slc25a34* KD. Shown are a representative trace (left) and quantification of maximal OCR values reached during each treatment in $n = 9$ (A) and $n = 3$ (B) independent experiments (right). (**C**) OCR in primary brown adipocytes with control or siRNA-mediated *Slc25a34* KD. Shown are a representative trace (left) and quantification of maximal OCR values reached during each treatment in two independent experiments (right). (**D**) OCR in brown adipocytes with control or siRNA-mediated *Slc25a35* KD. Shown are a representative trace (left) and quantification of maximal values reached during each treatment in four independent experiments (right). (**E**) Schematic of experimental design for assessing respiration in permeabilized brown adipocytes. (**F**) Respiration of permeabilized brown adipocytes after 4 days of siRNA-mediated *Slc25a34* KD and addition of specific respiratory substrates (from left to right): pyruvate and malate, succinate with addition of rotenone, and glycerol-3-phosphate. (**G**) Schematic of experimental design for the in vivo NE-stimulated thermogenesis test in *Slc25a34* iBKO and control mice. (**H** and **I**) Oxygen consumption (H) and corresponding area under the curve (I) of anaesthetized mice during the NE-stimulated thermogenesis test. (**J**) *Slc25a34* mRNA expression in BAT mice from the NE-stimulated thermogenesis test. KD of *Slc25a34* in (A) to (C) and (F) was performed with siRNA #1; measurements were made after 4 days of KD induction. For all panels, data are represented as mean ± SEM. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$, two-way repeated-measures ANOVA with the Šidák correction for multiple comparisons [(A) to (D)] or unpaired two-tailed *t* test [(I) and (J)].

that SLC25A34 mediates adrenergic-induced mitochondrial respiratory activity in vitro and in vivo.

SLC25A34 defends mitochondrial oxaloacetate availability and supports cytosolic acetyl-CoA production

To obtain deeper insights into the potential function of SLC25A34, we profiled the metabolomes of brown adipocytes at steady state 4 days after siRNA-mediated depletion of the transporter. Of the 82 differentially regulated metabolites (table S7), the most significant change was the accumulation of *N*-acetylaspartate (Fig. 4, A and B). *N*-acetylaspartate is synthesized in mitochondria from oxaloacetate (OAA)–derived aspartate and exported to the cytosol, where sequential hydrolysis by aspartoacylase (ASPA) and acetyl–coenzyme A (acetyl-CoA) synthetase 2 (ACSS2) generates cytosolic acetyl-CoA (*44*). Consistent with disrupted utilization of *N*-acetylaspartate, whole-cell concentrations of acetyl-CoA, which largely reflect the cytosolic and nuclear concentrations (*45*), were significantly decreased after 4 days of *Slc25a34* depletion (Fig. 4C). Thus, SLC25A34 may help to maintain the acetyl-CoA pool in thermogenic adipocytes.

Given the large number of metabolites changed at 4 days, we next performed a time course with earlier time points to distinguish primary from secondary metabolite changes (Fig. 4, D and E). Even after just 1 day of depletion, before any changes in cellular respiratory capacity manifested, *N*-acetylaspartate was still the most dysregulated among only five significantly altered metabolites (Fig. 4, F to I, and tables S8 and S9), indicating that it may be directly affected by the loss of SLC25A34. These rapid metabolite changes after 1 day of siRNA are consistent with the relatively low protein stability that we observed for SLC25A34 (fig. S1, C and D). The other acute metabolite changes included dysregulation of aspartate, α-ketoglutarate, proline, and betaine (Fig. 4, I to M). Thus, ablation of *Slc25a34* may disrupt the malate-aspartate shuttle (MAS) and/or the TCA cycle. *N*-acetylaspartate metabolism by ASPA and ACSS2 is not the only route to generate cytosolic acetyl-CoA in adipocytes. Citrate exported from the mitochondria is metabolized to acetyl-CoA by ATP

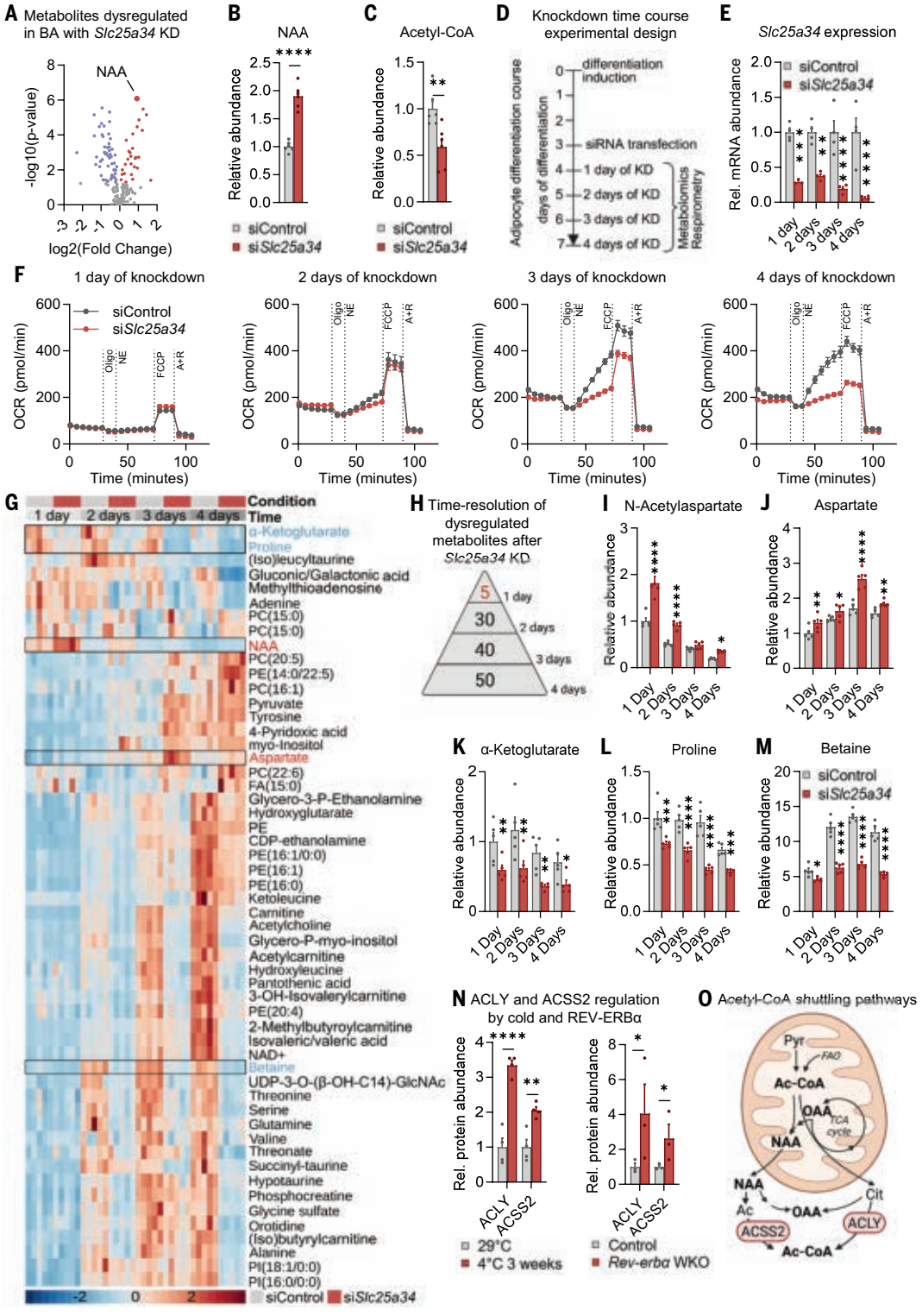

Fig. 4. SLC25A34 promotes cytosolic acetyl-CoA production. (**A**) Global differential metabolite changes between brown adipocytes with control or *Slc25a34* KD after 4 days of KD induction. BA, brown adipocytes. (**B** and **C**) Relative abundance of *N*-acetylaspartate (NAA) (B) and acetyl-CoA (C) in brown adipocytes with control or *Slc25a34* KD after 4 days of the KD induction. (**D**) Schematic of the experimental design. (**E**) *Slc25a34* expression. (**F**) Mitochondrial respiration. (**G**) Heatmap of metabolite changes. (**H**) Overview of metabolite changes. (**I** to **M**) Specific metabolite abundances from an *Slc25a34* siRNA-mediated KD time course in brown adipocytes. In (G), the metabolites outlined in black boxes represent the five significantly changed metabolites after 1 day of *Slc25a34* KD. *Slc25a34* KD on all panels was performed with siRNA #1. (**N**) Regulation of ACSS2 and ACLY protein abundance by cold in WT mice (left) and by REV-ERBα in *Rev-erbα* WKO mice (right). (**O**) Model showing how the generation of cytosolic acetyl-CoA by mitochondrial export of citrate (Cit) and NAA both lead to the production of OAA in the cytosol. For all panels, data are represented as mean ± SEM. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001, unpaired two-tailed Student's *t* test [(B) and (C)]; two-way ANOVA with the Benjamini, Krieger, and Yekutieli two-stage linear step-up procedure for multiple comparisons correction [(E), (I) to (M), and (N), left]; or Wilcoxon rank-sum test [(N), right].

citrate lyase (ACLY) (*46*, *47*). Both ACLY and ACSS2, the key enzymes in the two acetyl-CoA–generating pathways, are regulated by REV-ERBα and cold temperature in BAT, similarly to SLC25A34 (Fig. 4N), providing a potential link between these pathways. Moreover, the *N*-acetylaspartate and citrate routes of cytosolic acetyl-CoA production each result in the generation of OAA in the cytosol (Fig. 4O), which is the proposed substrate of the yeast homolog of SLC25A34 (*28*, *48*).

Therefore, we tested whether SLC25A34 played a role in balancing the cytosolic and mitochondrial pools of OAA. Because of the labile nature of OAA, we chemically derivatized samples with sodium borohydride, converting OAA to a more stable compound, deuterated malate, for quantification. We purified mitochondria with a rapid HA-tag–based protocol (*49*) from brown adipocytes with or without depletion of *Slc25a34* and performed targeted metabolomics analysis of purified mitochondria and of the mitochondria-depleted (cytosolic) fraction and total cell lysates (Fig. 5A and fig. S5A). In control cells, OAA levels were higher in the cytosol relative to the mitochondria, establishing a concentration gradient or ratio consistent with potential SLC25A34-mediated OAA import into mitochondria (Fig. 5B). Depletion of *Slc25a34* further increased the cytosolic-to-mitochondria OAA ratio by raising cytosolic levels and lowering the mitochondrial pool without altering total cellular OAA (Fig. 5B). We also observed dysregulation of the MAS metabolites aspartate, glutamate, and malate (fig. S5, B to D), consistent with the established role of OAA in this circuit. Collectively, these findings support a model in which SLC25A34 facilitates the import of OAA into mitochondria.

To further explore the biochemical feasibility of OAA as a substrate for SLC25A34, we performed docking simulations of OAA with human SLC25A34 in Autodock Vina with the SLC25A34 structure predicted by AlphaFold. These simulations were done with hydrated substrate and flexible residues in the conserved binding sites and matrix salt bridge network. OAA could dock on SLC25A34 with high affinity (–15.42 kcal/mol) in a binding pose in which it formed polar contacts between its carboxyl and carbonyl groups and two arginine residues (R176 and R277) in the conserved common substrate-binding site characteristic of SLC25 carriers (fig. S5E). This pose also permitted the formation of polar contacts between the carbonyl and carboxyl groups of OAA and the four residues that make up the matrix salt bridge network (K31, D228, E28, and K128), which braced SLC25A34 in a matrix closed conformation. Docking to the conserved binding site in a manner that can disrupt the matrix salt bridge network is necessary for substrate transport by SLC25A carriers (*50*).

Because a lower mitochondrial OAA pool could result from reduced glucose- or glutamine-dependent anaplerosis, we next performed multi-isotope tracing by incubating control and *Slc25a34*-depleted brown adipocytes with [$^{13}C_6$]-glucose and [$^{13}C_5$]-glutamine for 5, 30, and 60 min (see Fig. 5C for the 60-min time point and table S10 for all time points). We additionally included [$^{13}C_2$,$^{2}H_3$]-acetate, which enters the TCA cycle directly as acetyl-CoA, allowing us to determine OAA availability independently of anaplerotic flux. Consistent with our previous analysis, we found that depletion of *Slc25a34* significantly increased the total pool of *N*-acetylaspartate (Fig. 5D). However, in contrast to our earlier metabolomic analysis (Fig. 4J and fig. S5B), global aspartate abundance was decreased in the tracing dataset, potentially due to differences in media composition and analytical platforms (see the Materials and methods). Therefore, we prioritized analysis of the isotopologue enrichment to gain insight into carbon routing in the presence and absence of SLC25A34. [$^{13}C_6$]-glucose tracing revealed increased m+2 labeling in acetyl-CoA and increased m+2 and m+3 labeling in citrate, fumarate, malate, and aspartate in the absence of *Slc25a34* (Fig. 5, E and F, and fig. S6B), suggesting elevated glucose-derived anaplerosis. Glutamine additionally supplied more TCA carbons after loss of *Slc25a34*, as [$^{13}C_5$]-glutamine tracing showed a shift toward m+4 labeling in TCA intermediates and aspartate and increased glutamine-derived acetyl-CoA (Fig. 5, G and H, and fig. S6C). However, [$^{13}C_2$,$^{2}H_3$]-acetate labeling of acetyl-CoA, TCA intermediates, and *N*-acetylaspartate, which is dependent on the OAA pool, was broadly diminished by *Slc25a34* depletion (Fig. 5, I and J, and fig. S6D), consistent with the limiting OAA levels due to the absence of SLC25A34. Collectively, the isotope-tracing metabolomic data are consistent with the hypothesis that SLC25A34 maintains TCA flux by importing OAA from the cytosol to sustain its mitochondrial availability. When the transporter is depleted, glucose- and glutamine-driven anaplerosis are elevated to partially compensate for TCA carbon supply, whereas acetate entry into the TCA cycle is constrained. In this model, disrupted OAA transport leads to impaired *N*-acetylaspartate turnover and decreased total citrate, functionally linking SLC25A34 to two major pathways for generating cytosolic acetyl-CoA.

SLC25A34 couples lipid cycling with transcriptional control of mitochondrial oxidation

In adipocytes, one of the primary metabolic fates of cytosolic acetyl-CoA is to support de novo lipogenesis (DNL) (*51*). We assessed DNL through the incorporation of radiolabeled carbons from ^{14}C-glucose into triglycerides after control or siRNA-mediated depletion of *Slc25a34* (Fig. 6A). ^{14}C-glucose was used because it radiolabels cytosolic acetyl-CoA, and subsequently triglycerides, through TCA-derived citrate through ACLY (*52*) and *N*-acetylaspartate through ACSS2 (*53*), the two rate-limiting enzymes that we found to be cold and circadian regulated (Fig. 4N), similar to SLC25A34. Consistent with the acetyl-CoA measurements, loss of *Slc25a34* significantly diminished glucose-supported triglyceride production in brown adipocytes (Fig. 6B). In further support of the role for SLC25A34 in DNL, we found that insulin, one of the key lipogenic activators in adipocytes, also significantly induced *Slc25a34* expression similarly to known DNL mediators (fig. S7, A to D).

In addition to fueling DNL, cytosolic acetyl-CoA is also a critical mediator of transcriptional control of nuclear-encoded genes (*46*, *54*). To assess the potential impact of *Slc25a34*-dependent acetyl-CoA production on nuclear transcription, we globally profiled gene expression from brown adipocytes with or without *Slc25a34* depletion using an siRNA construct #1 as well as a siRNA construct pool (#2) (fig. S4A). When comparing the expression profiles of the two siRNAs, the shared changes revealed that transcription of 1489 genes was increased and that of 1685 genes was decreased (fig. S7E, Fig. 6C, and table S11). Despite causing a general decrease in the abundance of acetyl-CoA, *Slc25a34* ablation did not globally affect histone or total protein acetylation (fig. S7, F to H) but instead specifically affected the expression of particular gene networks. Pathway enrichment analysis revealed that the up-regulated genes were related to inflammatory and interferon pathways, underscoring the importance of SLC25A34 in maintaining cellular homeostasis (fig. S7I). To define the temporal sequence of these changes, we profiled a panel of inflammatory and interferon-responsive genes over a 4-day *Slc25a34* KD time course. *Il6*, *Cxcl1*, and *Icam1* were induced from day 1 and remained elevated, indicating an early and sustained nuclear factor κB (NF-κB)–driven proinflammatory state (fig. S7, J to L). *Il16*, a CD4-binding chemoattractant cytokine, was induced from day 2 onward, consistent with a transition to broader immune communication (fig. S7M). Classical interferon-stimulated genes (e.g., *Ifit1*, *Cxcl10*, *Isg15*, *Ifi44*, and *Gbp4*) showed coordinated up-regulation only at later time points, consistent with a delayed interferon response after prolonged mitochondrial dysfunction (fig. S7, N to R).

The genes down-regulated by *Slc25a34* depletion were the ones significantly linked to the mitochondrial TCA cycle and the respiratory chain (Fig. 6D). Consistent with this transcriptional down-regulation, we observed a significant decrease in citrate synthase (CS) activity upon loss of *Slc25a34* (Fig. 6E). However, this defect did not appear to reflect a reduction in mitochondrial number because the ratio of mitochondrial DNA to nuclear DNA remained unchanged (fig. S8A).

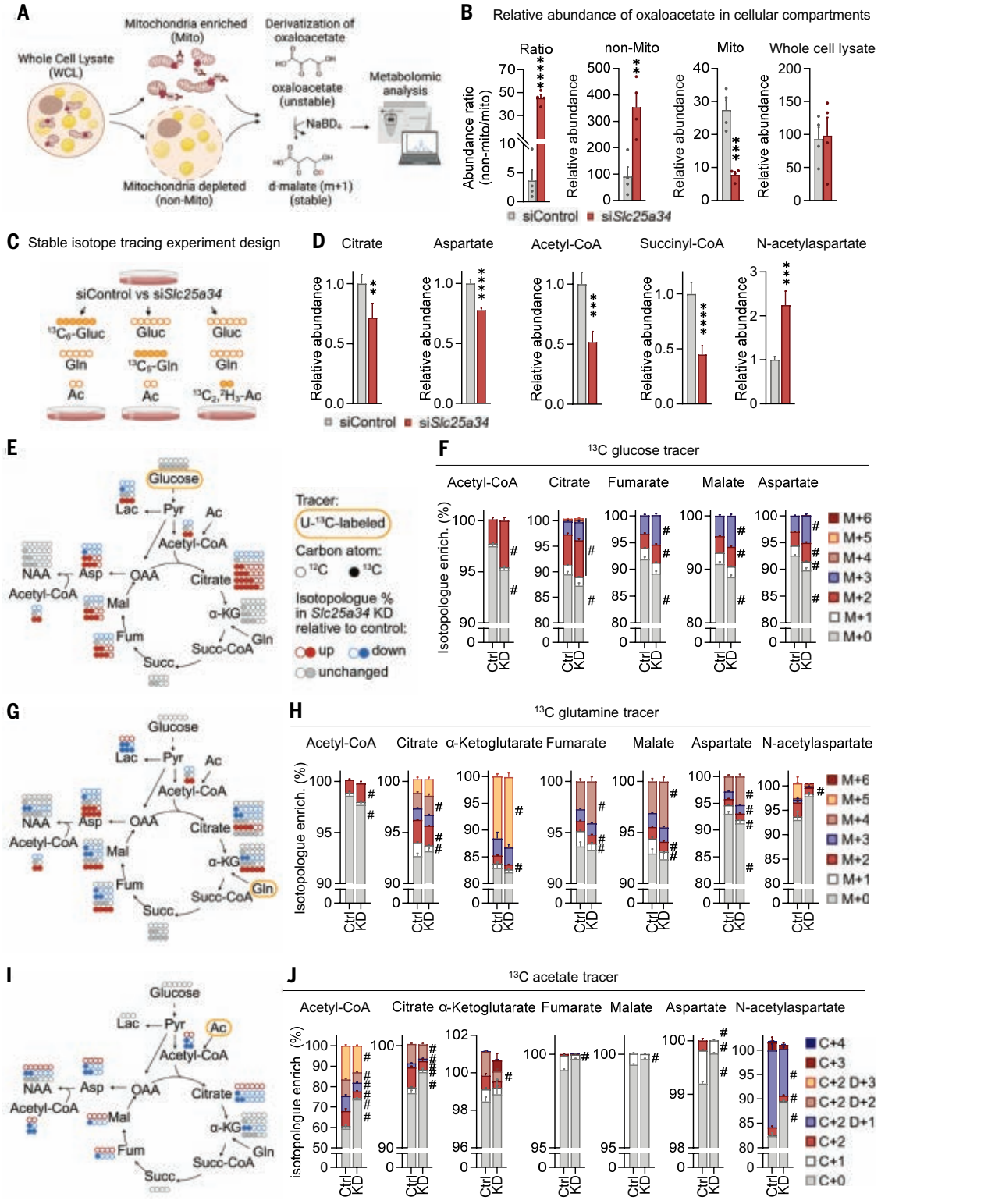

Fig. 5. Triple-isotope tracing implicates SLC25A34 in routing substrates into the TCA cycle and OAA-to-NAA cataplerosis. (**A** and **B**) Schematic for organelle-specific metabolomics (A) and abundance of OAA (B) across cellular compartments. Data shown are the average of four independent experiments. (**C**) Schematic of the experimental design of stable isotope tracing in brown adipocytes. (**D**) Relative abundance of unlabeled aspartate, citrate, acetyl-CoA, succinyl-CoA, and NAA in *Slc25a34* KD and control cells. (**E** and **F**) Schematic overview (E) and isotopologue enrichment percentage from [$^{13}C_6$]-glucose tracing (F) for different metabolites. (**G** and **H**) Schematic overview (G) and isotopologue enrichment percentage from [$^{13}C_5$]-glutamine tracing (H) for different metabolites. (**I** and **J**) Schematic overview (I) and isotopologue enrichment percentage from [$^{13}C_2$,$^{2}H_3$]-acetate tracing (J) for different metabolites. Ctrl, Control cells; KD, cells with *Slc25a34* KD. *Slc25a34* KD on all panels was performed with siRNA #1, and measurements are made after 4 days of KD induction. The tracing experiment was repeated three times, all measurements were combined, and the average values are shown. For all panels, data are represented as mean ± SEM. $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$, unpaired two-tailed Student's *t* test [(B) and (D)], $\#P < 0.05$ siControl versus si*Slc25a34*, multiple paired *t* test with the Benjamini, Krieger, and Yekutieli two-stage linear step-up procedure for multiple-comparisons correction [(F), (H), and (J)].

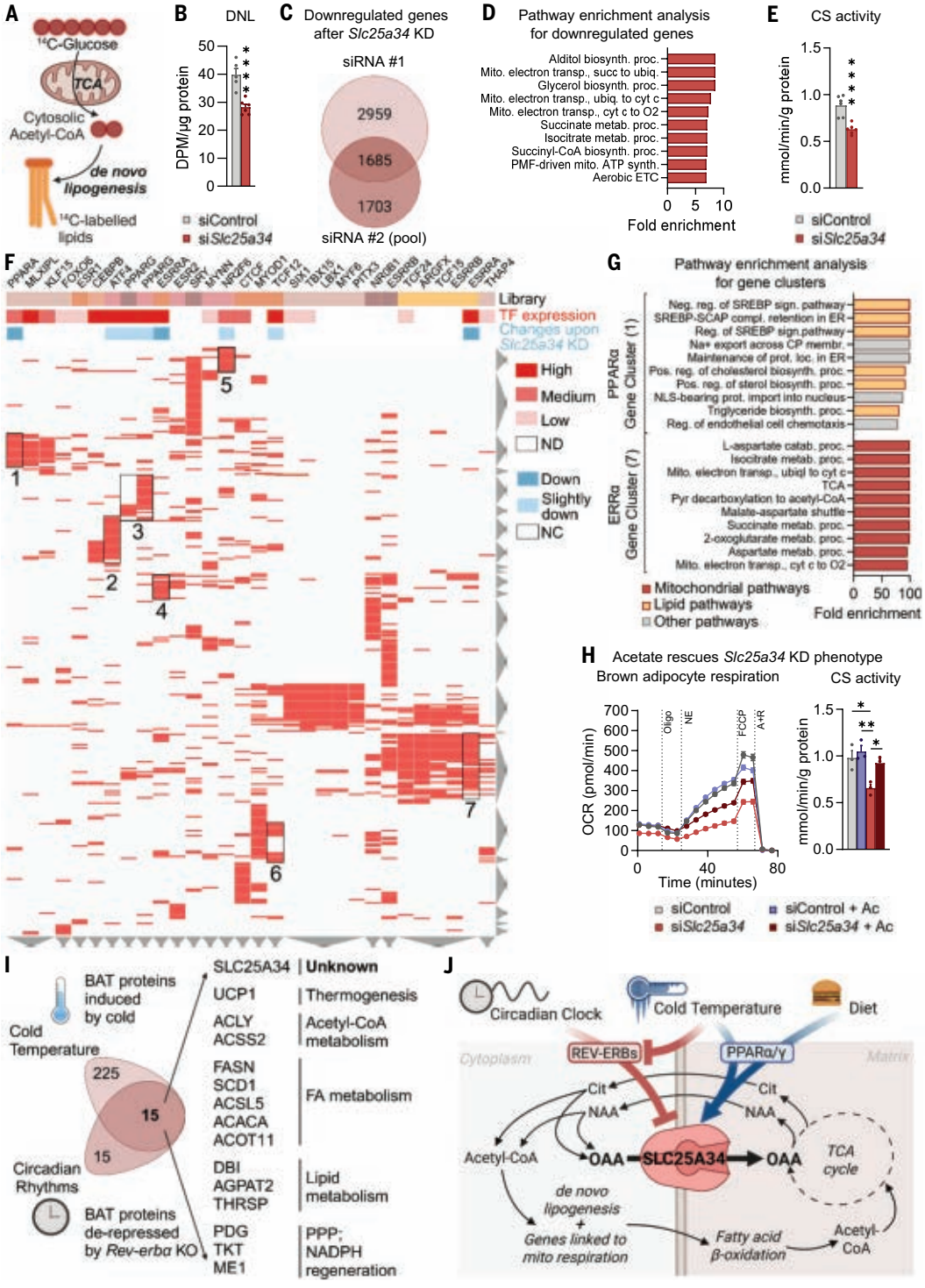

Fig. 6. SLC25A34 couples lipid cycling with transcriptional control of mitochondrial oxidation. (**A** and **B**) Schematic and assessment of radiolabeled glucose-derived DNL in brown adipocytes with control or siRNA-mediated *Slc25a34* KD after 4 days of KD. (**C**) Comparison of genes significantly down-regulated after *Slc25a34* KD using two different siRNAs (siRNA #1 and siRNA #2 (pool)). (**D**) Pathway enrichment for the conserved gene signature down-regulated by both siRNAs. (**E**) CS activity of brown adipocytes with control or *Slc25a34* KD after 4 days of KD. (**F** and **G**) Promoter motif analysis of the genes down-regulated by loss of *Slc25a34* for candidate transcription factors (F) and pathway enrichment of factor-specific gene clusters (G). (**H**) Effect of acetate supplementation on respiration and CS activity of brown adipocytes with siRNA-mediated *Slc25a34* KD. Shown are a representative seahorse trace (left) and CS activity (right). *Slc25a34* KD was performed with siRNA #1, measurements done after 4 days of KD induction. The experiment was repeated five times (see fig. S8G for quantification). (**I**) Classification of the protein cluster jointly regulated by cold and the circadian clock. (**J**) Schematic showing circadian and cold- and diet-dependent transcriptional control of *Slc25a34* by REV-ERBs and PPARα/γ and a proposed model by which the SLC25A34-mediated import of OAA into mitochondria facilitates cytosolic acetyl-CoA production and intracellular lipid cycling. For all panels, data are represented as mean ± SEM. **P* < 0.05, ****P* < 0.001, *****P* < 0.0001, unpaired two-tailed Student's *t* test [(B) and (E)] or one-way ANOVA with the Benjamini, Krieger, and Yekutieli two-stage linear step-up procedure for multiple-comparisons correction [(H), right].

Therefore, our findings demonstrate that SLC25A34 affects mitochondrial function at least in part through transcriptional control of nuclear genes linked to specific mitochondrial pathways.

As a transporter embedded in the IMM, SLC25A34 cannot directly regulate nuclear genes. To understand the downstream transcriptional control, we examined the promoters of SLC25A34-regulated genes for transcription factor motifs. To refine our focus to the most relevant candidates, we further assessed how highly expressed each transcriptional regulator was in brown adipocytes and whether its expression was changed after depletion of *Slc25a34*. Among the genes that showed increased expression after *Slc25a34* ablation, promoters in the cluster associated with genes implicated in inflammatory pathways were most enriched in the signal transducers and activators of transcription (STAT) and interferon regulatory factor (IRF) transcription factor motifs (fig. S8B and table S12). These factors were also highly expressed basally in brown adipocytes and were increased after *Slc25a34* depletion. Thus, STATs and IRFs appear to mediate the induction of inflammatory pathways after loss of the transporter (fig. S8C). Analysis of the genes that showed decreased mRNA expression after loss of *Slc25a34* revealed seven clusters linked to transcription factors that satisfied the criteria for brown adipocyte expression and SLC25A34 dependency (Fig. 6F and table S12). Pathway enrichment analysis found significant associations with three of the seven clusters. Clusters 1 and 3 represent targets of PPARα and PPARγ and comprise genes linked to lipid synthesis and catabolism (Fig. 6G and fig. S8D). This SLC25A34-dependent association with PPAR lipid metabolism genes may reveal a positive reinforcement mechanism, given that PPARα and PPARγ also drive *Slc25a34* expression. The genes in cluster 7 are related to mitochondrial respiratory function and were most strongly linked to estrogen-related receptor α (ERRα) control (Fig. 6G). Thus, the resulting decrease in cytosolic acetyl-CoA appears to act through specific transcription factor pathways.

We investigated whether the dysfunction caused by depleting *Slc25a34* could be alleviated by exogenously boosting cytosolic acetyl-CoA (fig. S8E). This was done by supplementing the cell media with acetate (fig. S8F), which is metabolized in the cytosol by ACSS2 to produce acetyl-CoA, theoretically circumventing the requirement for mitochondrial export of *N*-acetylaspartate or citrate. Indeed, culturing brown adipocytes with acetate mitigated the respiratory defects caused by *Slc25a34* KD (Fig. 6H and fig. S8G). Furthermore, acetate supplementation restored mitochondrial CS activity in *Slc25a34*-depleted cells (Fig. 6H). These findings support the idea that SLC25A34 influences mitochondrial function partly through cytosolic acetyl-CoA production.

Our data demonstrate that SLC25A34 has an essential role in both cytosolic DNL and mitochondrial respiration, thereby coordinating both anabolic and catabolic lipid processes across organelles. This role is further highlighted by mapping proteins co-regulated by cold and the circadian clock onto their metabolic pathways: Nearly all support lipid synthesis, shuttling, re-esterification, or mitochondrial thermogenesis and may be functionally coordinated by SLC25A34 (Fig. 6I and fig. S8H). Collectively, our findings are consistent with a model in which the circadian clock, environmental temperature, and diet leverage mitochondrial import of OAA through SLC25A34 to couple and accelerate the citrate and *N*-acetylaspartate pathways and generate cytosolic acetyl-CoA (Fig. 6J). This acetyl-CoA pool fuels DNL and the transcriptional control of mitochondrial respiratory capacity, both of which serve to increase mitochondrial lipid oxidation and thermogenesis and, ultimately, complete an SLC25A34-dependent circuit of lipid synthesis and breakdown.

To investigate the role of SLC25A34 in human biology, we performed siRNA KD of the transporter in primary in vitro differentiated brown adipocytes from four independent donors. Loss of *SLC25A34* reduced cyclic adenosine 3′,5′-monophosphate (cAMP)–inducible respiratory capacity in three of the four donor adipocytes (Fig. 7A) and transcription of genes linked to thermogenic function in adipocytes from all donors (Fig. 7B). In humans, SLC25A34 may contribute to adipose and whole-body energy homeostasis beyond thermogenic BAT (*13*). A meta-analysis of 24 clinical studies (https://adiposetissue.org/) (*55*–*78*) revealed that expression of *SLC25A34* in scWAT was negatively correlated with circulating C-reactive protein, triglycerides, and low-density lipoprotein cholesterol, Homeostasis Model Assessment of Insulin Resistance (HOMA-IR), body-mass index (BMI), waist circumference, and waist-to-hip ratio (WHR), and positively correlated with circulating high-density lipoprotein cholesterol and the ratio of stimulated versus basal lipolysis (Fig. 7C). We found no significant associations with *SLC25A34* expression in omental adipose tissue, possibly indicating that SLC25A34 may play a depot-specific beneficial role in promoting systemic metabolic health.

Discussion

The circadian clock has been evolutionarily fine-tuned over hundreds of millions of years to anticipate recurring daily metabolic needs and coordinate tissue-specific homeostatic programs accordingly (*79*). This rhythmic framework must also integrate rapid responses to acute environmental and dietary changes. Chronic modern-day perturbations, such as irregular eating patterns, high-calorie diets, and reduced exposure to temperature fluctuations, can overwhelm this capacity, contributing to neurologic, psychiatric, cardiometabolic, and immune disorders (*79*–*82*). Delineating the mechanisms that integrate circadian machinery with tissue-specific metabolic control is therefore critical for understanding evolutionary biology and identifying therapeutic targets.

Adipose thermogenesis exemplifies the integration of circadian, environmental, and dietary control of metabolism. Thermogenic function is suppressed by the clock during sleep and rises before awakening to peak during the waking period, thereby governing daily fluctuations in systemic lipid and glucose homeostasis (*9*, *14*, *16*). However, this rhythmicity must also be coordinated with rapid BAT responsiveness to cold (*4*, *83*), which itself influences glucose and lipid metabolism (*17*), whereas dietary composition and caloric density further shape thermogenic output (*84*). We discovered that the orphan mitochondrial metabolite carrier SLC25A34 represents a key regulatory node in BAT downstream of all three inputs by integrating circadian control through REV-ERBα and β-HDAC3 repression with cold temperature and dietary regulation through lipid-mediated PPARα and PPARγ activation. The individual contributions of REV-ERBα and REV-ERBβ remain unclear, but given their functional redundancy (*85*), shared DNA-binding sites (*29*), and parallel cold down-regulation in BAT, they likely play similarly important roles. The stronger effect observed in the *Rev-erbα* WKO probably reflects indirect contributions from other tissues such as the liver and WAT.

A key hallmark of BAT activation is the simultaneous engagement of mitochondrial fatty acid β-oxidation and cytosolic DNL, enabling lipid cycling (*86*). Fatty acid β-oxidation and glucose catabolism (*52*) feed the mitochondrial acetyl-CoA pool and TCA cycle, and the resulting metabolites citrate and *N*-acetylaspartate are exported to the cytosol. Their catabolism yields acetyl-CoA that primes DNL, the lipid products of which can be reincorporated into triglycerides or returned to mitochondria for β-oxidation, thus completing the cycle. Notably, generating cytosolic acetyl-CoA from citrate or *N*-acetylaspartate also produces OAA, the cytosolic fate of which has remained unresolved. In *Slc25a34*-depleted brown adipocytes, we observed *N*-acetylaspartate accumulation, dysregulated TCA intermediates, a shift of OAA from mitochondria to cytosol, and reduced [$^{13}C_2$,$^{2}H_3$]-acetate incorporation into citrate and downstream TCA intermediates, suggesting that mitochondrial OAA becomes limiting without SLC25A34. These data are consistent with the observed decrease in CS activity and may reflect substrate feedback regulation.

Therefore, we propose a working model in which SLC25A34 imports cytosolic OAA into mitochondria, supporting a shuttle that couples

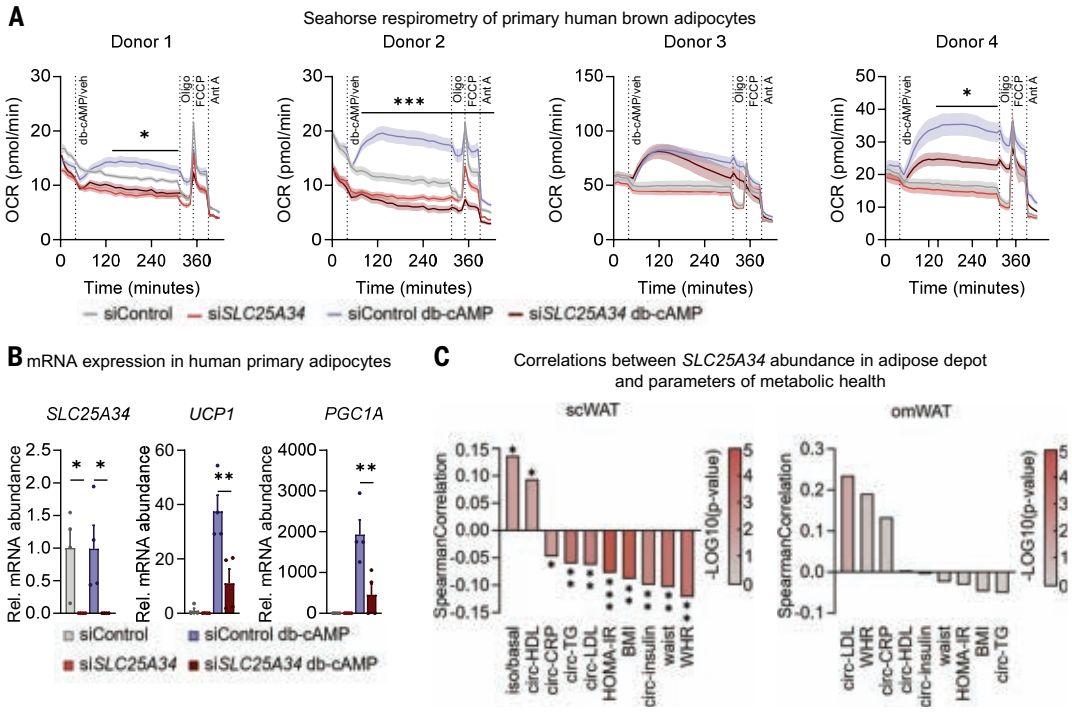

Fig. 7. SLC25A34 shapes human adipocyte respiration and is associated with cardiometabolic health. (**A** and **B**) Basal and cAMP-inducible OCR of primary human brown adipocytes from four donors with *SLC25A34* KD (A) and expression of thermogenesis-related genes (B). Each data point in (B) represents one of the four donors. db-cAMP, dibutyryl-cAMP. (**C**) Meta-analysis of 24 studies showing correlations between *SLC25A34* expression in subcutaneous WAT (scWAT) and omental WAT (omWAT) and metabolic health parameters in human patients. Shown are the ratio of stimulated versus basal lipolysis (iso/basal), circulating high-density lipoprotein (circ-HDL), circulating C-reactive protein (circ-CRP), circulating triglycerides (circ-TG), circulating low-density lipoprotein (circ-LDL), HOMA-IR, BMI, circulating insulin (circ-insulin), waist circumference (waist), and WHR. For all panels, data are represented as mean ± SEM. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, for siControl db-cAMP versus *siSLC25A34* db-cAMP, repeated-measures two-way ANOVA with the Geisser-Greenhouse correction with Šídák multiple-comparisons test (A) or one-way ANOVA with Tukey's multiple-comparisons test (B).

citrate and *N*-acetylaspartate export to cytosolic acetyl-CoA production. This would supply substrate for DNL and re-esterification while sustaining mitochondrial OAA for continued TCA cycle activity. SLC25A34 also appears to help maintain an anti-inflammatory state in brown adipocytes, because its loss triggers an early, sustained NF-κB–driven response followed by the induction of interferon-stimulated genes, although the mechanisms linking SLC25A34 to protection from mitochondrial stress and inflammation remain unknown. SLC25A34-mediated lipid cycling is also sensitive to systemic nutritional and hormonal cues: Fasting and a ketogenic diet, which favor lipolysis and β-oxidation, and insulin, which favors DNL, all increase *Slc25a34* expression in BAT. These data are consistent with a role for SLC25A34 in both catabolic and anabolic lipid metabolism, with the balance between these arms likely prioritized by biological context.

Members of the SLC25 family often transport multiple substrates across the IMM (*87*). In our work, the most significant metabolite change observed upon acute *Slc25a34* depletion was increased total *N*-acetylaspartate, raising the possibility that it is a substrate of SLC25A34, with its accumulation reflecting the loss of its counterexchange with OAA and entrapment within mitochondria. SLC25A34 may also transport, or be part of a circuit for, other related dicarboxylates such as malate, consistent with the substrate selectivity of its closest deorphanized homolog, the yeast OAA carrier OAC1 (*48*). Testing direct transport will ultimately require in vitro assays with recombinant protein in reconstituted liposomes, which may be challenging given the low stability of both OAA and SLC25A34.

Notably, both SLC25A34 and its paralog SLC25A35 share common ancestry with the yeast OAA carrier gene *OAC1* (*28*) and may have retained shared transport capacity while acquiring divergent regulation: *Slc25a35* is highly expressed at baseline in BAT, suppressed by cold, and not circadian regulated, suggesting a role in routine maintenance, whereas *Slc25a34* is cold induced and subject to strong circadian and dietary control, consistent with a role under specific energetic demands. Recent work showed that SLC25A35 mediates mitochondrial phosphoenolpyruvate (PEP) efflux to support glycerolipid synthesis (*88*), suggesting that both homologous transporters contribute to lipid metabolism through distinct mechanisms. Although both can partially compensate for Oac1 in yeast (*28*), only *Slc25a34* depletion disrupted respiration in brown adipocytes in our studies, indicating that the two carriers have become specialized for different cellular contexts within the mammalian mitochondrial carrier network.

SLC25A34 also influences the transcription of nuclear-encoded genes linked to lipid metabolism and mitochondrial respiration. Whether this occurs directly through cytosolic acetyl-CoA as a histone acetyltransferase substrate or through a lipid-derived signaling molecule remains unknown, although motif enrichment implicating PPARα and PPARγ in SLC25A34 target genes favors the latter. Such a mechanism would connect circadian, dietary, and temperature-sensing pathways to coordinate cytosolic lipid synthesis and mitochondrial lipid oxidation. Although we focused on adipose thermogenesis, REV-ERBα/β and PPARα/γ, the primary regulators of *Slc25a34*, have key roles in other tissues (*85*, *89*, *90*), suggesting that SLC25A34's impact likely extends beyond adipocytes. For example, mitochondrial OAA transport has been observed in heart, kidney, and liver (*91*–*96*), with the heart having the highest *Slc25a34* expression. Furthermore, *N*-acetylaspartate, the metabolite most affected by *Slc25a34* loss in BAT, is also the brain's second most abundant metabolite and supports myelination and neuronal integrity (*97*), raising the possibility of a

role in neuronal health. If and to what extent the metabolic signature of SLC25A34 in thermogenic adipocytes is conserved across these functionally distinct cell and tissue types remains to be determined.

Materials and methods

Animal models

The WT (C57Bl/6JBomTac) mouse studies [tissue panel (Fig. 1B and fig. S1A)], ketogenic diet study (Fig. 2, N and O), cold acclimation study (Fig. 1, C to E, and fig. S1B), and primary adipocyte culture from mice were performed with approved protocols from The Danish Animal Experiments Inspectorate (permit number: 2018-15-0201-01441) and the University of Copenhagen (project numbers P18-312 and P19-374). Mice were housed in an enriched environment with ad libitum access to a standard chow diet (Altromin, 1310) and water. Light in the facility was set to a 12 hour light/dark cycle (light: 6:00 a.m. to 6:00 p.m. and dark: 6:00 p.m. to 6:00 a.m.) at 22 ± 2°C. Studies were performed in male and female mice between 8 and 20 weeks of age. Mice were housed at RT (22 ± 2°C) unless otherwise stated. Housing temperatures used in the studies: thermoneutrality, 29°C; RT, 22° ± 2°C; and cold, 4° to 5°C.

Details of the housing conditions and studies performed on *Rev-erbα* WKO, *Rev-erbα/β* BKO, *Hdac3* AKO, *Pparα* AKO, *Pparα* iBKO, and *Slc25a34* iBKO mice are specified below.

Tissue panel: The tissue panel experiment was performed as described in (*98*). Briefly, 20-week-old C57Bl/6JBomTac (Taconic Biosciences) mice were acclimatized to thermoneutrality for 3 weeks before the 24 hours cold challenge at 4°C. Tissues were harvested at the end of cold challenge between 9:00 a.m. [Zeitgeber time 3 (ZT3)] and 1:00 p.m. (ZT7).

Cold acclimation study: This study is published in (*24*). For cold exposure experiments, 12- to 16-week-old mice were individually housed and placed into climate-controlled rodent incubators (Memmert HPP750Life) set to 29°C to acclimate to thermoneutrality for 2 weeks. Cold exposure mice were then moved to an incubator set to 4°C and kept there until dissection. Tissues were harvested between 9:00 a.m. (ZT3) and 1:00 p.m. (ZT7). BAT was used for proteomics analysis and gene expression analysis with quantitative polymerase chain reaction (qPCR).

High-fat diet and ketogenic diet study (Fig. 2O): Adult (8 weeks) male C57BL/6J mice were purchased from Janvier Breeding Centre (France), and either placed on a standard chow diet (Brogaarden, Denmark) or a 60% high-fat diet (HFD) (Research Diets) for 24 weeks. The mice were group-housed with up to 8 mice per cage at 22°C on a 12:12 hour light-dark cycle. The mice had free access to food and water. Before the ketogenic diet study all mice were singly housed with 2 weeks for acclimatization. For the consecutive 3 weeks the mice from the chow diet either continued receiving chow diet (5 mice) or were given a ketogenic diet (BioServ F3666); fresh ketogenic diet was given to the mice every second day. HFD group (6 mice) continued receiving HFD for the 3 weeks. All mice were fed ad libitum and therefore the food intake was not isocaloric. After 27 weeks since the beginning of the diet intervention the mice were sacrificed, and BAT was used to measure gene expression.

Ketogenic diet study (Fig. 2P): Adult (14-weeks old) C57Bl/6N male mice acclimated to 22°C and fed with standard chow diet (Brogaarden, Denmark) were singly housed for 2 weeks before the experiment. For 3 weeks, the mice were given either a chow diet (4 mice) or a ketogenic diet from BioServ (F3666, NJ, USA) (4 mice) ad libitum. Fresh ketogenic diet was given to the mice every second day. The food intake between the two groups was not isocaloric. After 3 weeks of diet intervention, the mice were sacrificed and tissues harvested.

Fasting study (Fig. 2Q and fig. S2, L and M): For the fasting study, 2- to 16-week-old mice were individually housed at RT. Mice were housed on a 12:12-hours light-dark cycle (lights on at 6:00 a.m., lights off at 6:00 p.m.). For the fasting, the food was removed from the fasting group at 9:00 a.m. (ZT 3) and both fasted and control groups were sacrificed and tissues harvested in 24 hours at 9:00 a.m. (ZT 10).

***Rev-erbα* WKO:** Studies with the *Rev-erbα* whole body KO mice (*Rev-erbα* WKO) were performed with an approved protocol from the University of Pennsylvania Perelman School of Medicine Institutional Animal Care and Use Committee as described in (*9*). The *Rev-erbα* WKO mice were obtained from B. Vennström and backcrossed seven or more generations with C57Bl/6 mice. Mice were housed on a 12:12-hours light-dark cycle (lights on at 7:00 a.m., lights off at 7:00 p.m.). Gene expression and protein analysis were performed on 12- to 16-week-old male *Rev-erbα* WKO mice and WT littermates.

For the circadian gene expression experiment (Fig. 1F) mice were singly housed at 29°C before the experiment. Tissues were harvested at ZT-1, ZT3, ZT7, ZT11, ZT15, ZT19, ZT23, and ZT27 (2 to 4 animals per group per time point). For the proteomics study (Fig. 1G), mice were acclimated to thermoneutrality at 29°C, and tissues were harvested at 5:00 p.m. (ZT 10) (table S2). The cold exposure study (Fig. 1H), was performed in climate-controlled rodent incubators. All WT and *Rev-erbα* WKO mice used in the studies were first placed in individual cages with access to food and water and allowed to acclimate to 29°C for 2 weeks before cold challenge. For the cold challenge mice were exposed to cold at 4°C at 11:00 a.m. (ZT 4) for 1.5 hours and tissues were harvested at ZT5.5.

***Rev-erbα/β* BKO:** All animal work was approved by the University of Pennsylvania Perelman School of Medicine Institutional Animal Care and Use Committee (IACUC protocol number 804747 issued to Mitchell Lazar). Male *Rev-erbα*$^{fl/fl}$/*Rev-erbβ*$^{fl/fl}$ (*99*) mice were maintained on a C57Bl6/J background (Jackson Labs, Stock 008661). To produce *Rev-erbα/β* BKO, *Rev-erbα*$^{fl/fl}$/*Rev-erbβ*$^{fl/fl}$ mice were bred to Ucp1-Cre mice maintained on a C57BL/6J background (Jackson Labs Technologies, Inc., B6.FVB-Tg[Ucp1-Cre]1Evdr/J, Stock 024670) (*23*). Mice were bred and maintained at 20° to 22°C on a 12:12 hours light:dark cycle with ad libitum access to food and water unless otherwise noted. The normal chow diet (NCD, LabDiet, 5010) consisted of 12.7% kcal fat, 28.7% kcal protein, and 58.5% kcal carbohydrates. Gene expression was performed on 11–16-week-old male *Rev-erbα*/β BKO mice and WT littermates. Cold exposure experiments were performed in climate-controlled rodent incubators set to 29° and 4°C. Mice from the thermoneutrality acclimated group (5 WT and 5 KO mice) were housed at 29°C for 2 weeks and the tissues were harvested at 5:00 p.m. (ZT10). Mice from the cold challenge group (5 WT and 6 KO mice) were singly housed at 29°C for 4 days and acutely exposed to 4°C cold in prechilled cages at 4°C with bedding and free access to standard chow, without a cotton nestlet or access to water, and the cage lid partly open. Cold challenge lasted for 6 hours from 11:00 a.m. (ZT 4) to 5:00 p.m. (ZT 10). Tissues were harvested at 5:00 p.m. (ZT10) and BAT was used for RNA sequencing (table S3).

For the circadian study, mice were maintained under thermoneutral conditions (29°C) for 14 days before tissue collection. To assess circadian gene expression, BAT was harvested at six time points, 4 hours apart over 24 hours (ZT2, ZT6, ZT10, ZT14, ZT18, and ZT22) from $n = 3$ mice per group per genotype. BAT RNA was used for RNA sequencing. The accession number for the data is GEO no. GSE331054.

***Hdac3* AKO:** Studies with *Hdac3* AKO mice were performed according to ethical regulations and protocols approved by the Institutional Animal Care and Use Committee (IACUC) of the Perelman School of Medicine at the University of Pennsylvania and the Children's Hospital of Philadelphia as described in (*31*). Mice were group housed with enrichment in a temperature and humidity controlled, specific-pathogen free animal facility at 22°C under a 12:12-hour light-dark cycle with free access to standard chow (LabDiet, 5010) and water. To generate the

mouse strain with the adipose-specific *Hdac3* KO (*Hdac3* AKO), *Hdac3*$^{f/f}$ mice maintained on a C57BL/6 background were bred to B6;FVB-Tg(Adipoq-Cre)1Evdr/J mice from the Jackson Laboratory (stock no.: 010803) (*31*). All experiments used male age-matched littermates group housed at 4–5 animals per cage. Gene expression studies were performed on 10- to 12-week-old male mice at ZT 10. Cold-exposure experiments were performed in climate-controlled rodent incubators (Powers Scientific) maintained at 22°C or 4° to 5°C, with interior temperature monitoring by digital thermometers. For the cold exposure experiment, mice were allowed to acclimate to 22°C for 2 weeks. Cold exposure mice were placed in pre-chilled cages at 4° to 5°C with bedding, a cotton nestlet, free access to standard chow and water, and the cage lid partly open at 2:00 p.m. (ZT7). Mice were sacrificed and tissues harvested after 3 hours of cold exposure 5:00 p.m. (ZT10). Animal numbers were 4 WT and 3 KO in the RT group, 5 WT and 6 KO in the cold group).

***Pparα* AKO (exon 5 targeted):** All experiments with Pparα AKO mice with targeted exon 5 were performed according to procedures approved by the University of Pennsylvania's Institutional Animal Care and Use Committee as described in (*34*). Male WT C57BL/6J (stock 000664) mice, as well as F1 intercrossed progeny with 129S1/SvlmJ (B6129SF1/J, stock 101043), were purchased from Jackson Laboratory. *Pparα*$^{flox/flox}$ mice were kindly provided by F.J. Gonzales (National Institutes of Health, Bethesda) and have been described previously (*100*). To generate *Pparα* AKO mice, we crossed Adiponectin-Cre mice with *Pparα*$^{flox/flox}$ mice. All mice were housed, 5 per cage, in a temperature-controlled, specific pathogen–free facility with a 12-hour light / 12-hour dark cycle and ad libitum access to water and food at RT (22 ± 2°C). 1Twelve-week-old male mice were sacrificed between ZT3 and ZT4, and BAT was used for RNA sequencing (3 WT and 3 KO mice).

***Pparα* AKO (exon 4 targeted):** All experimental procedures and protocols for *Pparα* AKO (exon 4) mice complied with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and were approved by the Institutional Animal Care and Use Committee of the University of Mississippi Medical Center. The *Pparα* AKO (exon 4) mice were previously generated and described in (*37*). In short, homozygous *Pparα*$^{flox/flox}$ (from Dr. Walter Wahli, first published in (*101*)) and heterozygous *Adipoq*-Cre mice were crossed to create *Pparα* AKO (exon 4) mice and *Pparα*$^{flox/flox}$ littermate controls. Male and female mice were housed under standard conditions at RT with full access to water until 8 weeks of age. Mice were then switched to a normal chow (NCD) consisting of a 17% fat diet (Teklad 22/5 rodent diet, #860, Harland Laboratories, Inc., Indianapolis, IN, USA) for 30 weeks and had access to food and water ad libitum. After 30 weeks, mice were euthanized by overdose of isoflurane anesthesia at ZT3 to ZT5, and tissues were immediately collected and frozen in liquid nitrogen. Tissue samples were stored at −80°C until use.

***Pparα* iBKO:** The study with *Pparα* iBKO mice were approved by the Danish Animal Experiment Inspectorate (approval #2018-15-0201-01459) and complied with the ARRIVE guidelines. Male mice of 7-8 weeks old were housed at 22°C, under a 12:12-hours light-dark cycle, with free access to food and drinking water. To generate mice with brown adipose-specific *Pparα* KO (*Pparα* iBKO), *Pparα*$^{f/f}$ mice (kindly provided by F.J. Gonzales, National Institutes of Health, Bethesda) (*100*) were crossed with the tamoxifen-inducible B6-Tg(Ucp1-cre/ERT2)426Biat mouse line (kindly provided by Prof. Christian Wolfrum) (*102*). For the cold exposure experiment, RT acclimated mice were given 2 mg of tamoxifen (Sigma-Aldrich T5648) in 100 µl of corn oil (Sigma-Aldrich C8267) per mouse by oral gavage once per day for 2 consecutive days. Then the mice were split into 2 groups and transferred to 29° or 4°C for 3 days, after which the tissue was harvested. One more tamoxifen gavage was given to all mice on the first day of the temperature challenge. Control mice were *Pparα*$^{f/f}$ without Ucp1-Cre/ERT2 and were given an identical dosage of tamoxifen. Mice of all groups were sacrificed between ZT3-ZT5.

***Slc25a34* iBKO:** The study with *Slc25a34* iBKO mice was performed with approved protocols from The Danish Animal Experiments Inspectorate (permit number: A2025-15-0202-00236) and the University of Copenhagen (project number: P25-131).

Conditional *Slc25a34* KO mice (*Slc25a34*$^{f/f}$) on the C57BL/6N genetic background were generated by Genoway (France) by flanking exon 1, which contains the start codon, with *loxP* sites. To generate mice with brown adipose-specific *Slc25a34* KO (*Slc25a34* iBKO), *Slc25a34*$^{f/f}$ mice were crossed with the tamoxifen-inducible B6-Tg(Ucp1-cre/ERT2)426Biat mouse line (kindly provided by Christian Wolfrum) (*102*). To induce KO in this model, 2 mg of tamoxifen (Sigma-Aldrich T5648) in 100 µl of corn oil (Sigma-Aldrich C8267) per mouse was delivered by oral gavage once per day for 3 consecutive days. Control mice were *Slc25a34*$^{f/f}$ without Ucp1-Cre/ERT2 and were given an identical dosage of tamoxifen.

In vivo thermogenesis test: The NE challenge presented in Fig. 3, G and I, was performed on 14- to 18-week-old *Slc25a34* iBKO mice and littermate control female mice (8+8) maintained on a chow diet. Mice were acclimated to RT. The experiment was performed 2 weeks after tamoxifen treatment to induce the KO. Food was removed from the cages 2 hours before the NE test. NE response was assessed in anesthetized mice (intraperitoneal injection of 75 mg/kg BW pentobarbital with 7,5 mg/kg lidocaine) after an intraperitoneal injection of 1 mg/kg BW NE (Sigma-Aldrich, A9512). Oxygen consumption and carbon dioxide production were recorded using the Promethion core system (Sable Systems International) with data processing using OneClickMacro and Macrointerpretersetup. The NE challenge was performed at 33°C, and mice were sacrificed after the study. NE was injected at 10:30 a.m. (ZT 4.5) and mice were sacrificed at 12:30 p.m. (ZT6.5).

Cell models

Immortalized brown adipocyte (BMC) culture: Murine brown preadipocytes immortalized with SV40 large T antigen (referred to as BMC) were kindly provided by Patrick Seale (*103*). Preadipocytes were cultured in basal DMEM (Gibco, 31966) containing 10% FBS (Sigma-Aldrich, F7524) and 1% penicillin/streptomycin. Cells were passaged when they reached 60-70% of confluence. Media was changed every 2nd day. On the day of 100% confluency, cell differentiation was induced by supplementing of standard media with the differentiation cocktail (insulin (20 nM) (Sigma-Aldrich, I9278), dexamethasone (1 mM) (Sigma-Aldrich, D1756), rosiglitazone (0.5 mM) (Cayman Chemicals, 71740), T3 (1 nM) (Sigma-Aldrich, T6397), and 3-isobutyl-1-methylxanthine (IBMX) (250 µM) (Sigma-Aldrich, I5879)). On day 2 of differentiation, media was switched to maintenance media containing insulin and T3. Maintenance media was changed every 2nd day. Cells were harvested or assayed on day 7 of differentiation unless stated otherwise in the figure legends. The cells were maintained at 37°C in a humidified atmosphere with 5% CO2.

Generation of an HA-SLC25A34 overexpressing stable cell line: HA-SLC25A34 overexpressing BMCs were generated by retroviral transduction (pMMLV; VectorBuilder). Retrovirus was generated using the Plat-E packaging cell line (*104*). Cells were transfected with 10 µg of DNA using Lipofectamine 2000 (Invitrogen, #52887) after the manufacturer's instructions. Media was changed after 12-16 hours, and after 48 hours, media was collected and filtered using a 0.45 µm filter. 1 ml of the viral supernatant was added drop wise to a 10cm dish containing 5 ml of DMEM with 10% FBS and 4 µg/µl polybrene. Media was changed after 24 hours, and transduced cells were selected with media containing 2 µg/ml puromycin.

Primary murine brown adipocyte culture: Primary brown preadipocytes were isolated from WT C57BL/6J male and female mice as described

in (*98*). Male and female tissues were pooled for the generation of primary cell culture. BATs were harvested from 6-8 weeks old mice and put into ice-cold DMEM (Gibco, 31966) media. Tissues were minced and subsequently digested with DMEM (Gibco, 31966) supplemented with 2% BSA (Sigma-Aldrich, A7030) and 0.2% collagenase type I (Worthington Biochemical Corp., LS004197) at 37°C for 40 min. The digests were centrifuged (5 min, 400 g) to obtain the stromal vascular fraction (SVF) and pellets were resuspended in DMEM with 10% FBS (Sigma-Aldrich, F7524) and 1% penicillin/streptomycin. The cell suspensions were passed through a 70 μm strainer and distributed onto the plate-format of interest. On day 1 after cell harvest, the isolated preadipocytes were rinsed with DMEM containing 10% FBS and 1% penicillin/streptomycin and thereafter, media was changed every 2nd day. The cells were passaged up to one time before differentiation. On the day of 100% confluency, cell differentiation was induced by supplementing of standard media with the differentiation cocktail (insulin (86 nM) (Sigma-Aldrich, I9278), dexamethasone (0.1 μM) (Sigma-Aldrich, D4902), rosiglitazone (0.5 mM) (Cayman Chemicals, 71740), T3 (1 nM) (Sigma-Aldrich, T6397), and 3-isobutyl-1-methylxanthine (IBMX) (250 μM) (Sigma-Aldrich, I5879)). On day 2 of differentiation, media was switched to maintenance media containing insulin and T3. Maintenance media was changed every second day. Cells were harvested or assayed on day 7 of differentiation. The cells were maintained at 37°C in a humidified atmosphere with 5% CO2.

Primary human brown adipocyte cell culture: Deep cervical adipose tissue collection was approved by the Medical Research Council of Hungary (20571-2/2017/EKU) followed by the EU Member States' Directive 2004/23/EC on presumed consent practice for tissue collection. All experiments were performed in accordance with the guidelines of the Helsinki Declaration. Written informed consent was obtained from all participants before the thyroid surgery. Patients with known diabetes, malignant tumor, or with abnormal thyroid hormone levels at the time of surgery were excluded. SVF was isolated from each biopsy by digestion in collagenase from *Clostridium histolyticum* (120 U/ml) (Sigma-Aldrich, C0773), filtering (pore size 140 μm), and centrifugation at 330 *g* for 7 min. Pellets were resuspended in Dulbecco's modified Eagle's medium/nutrient F-12 Ham (DMEM-F12, Sigma-Aldrich, T2252) medium supplemented with 10% FBS (Thermo Fisher Scientific, A5256701SA), penicillin/streptomycin (100 U/ml) (Roche, 11074440001), biotin (33 μM) (Sigma-Aldrich, B4639), and pantothenic acid (17 μM) (Sigma-Aldrich, P5155). After 24 hours, nonadherent cells were removed by washing with PBS, and adherent cells were cultured further until reaching 90% confluency. The absence of Mycoplasma was verified using a PCR Mycoplasma Test Kit I/C (Promocell, catalog no. PK-CA91).

Primary human brown adipocyte differentiation from SVF was initiated for 3 days with DMEM/F-12 HAM medium containing penicillin/streptomycin (100 U/ml), biotin (33 μM), pantothenic acid (17 μM), human apo-transferrin (10 μg/ml) (Sigma-Aldrich, T2252), 20 nM human insulin (Sigma-Aldrich, I9278), 100 nM hydrocortisone (Sigma-Aldrich, H0888), 0.2 nM T3 (Sigma-Aldrich, T6397), rosiglitazone (2 μM) (Cayman Chemicals, 71740), dexamethasone (25 nM) (Sigma-Aldrich, D1756), and 500 μM IBMX (Sigma-Aldrich, I5879). After the initial 3-day induction period, rosiglitazone, dexamethasone, and IBMX were omitted, and then the media was changed every 2nd day. The cells were maintained at 37°C in a humidified atmosphere with 5% CO_2 (*105*).

siRNA mediated gene KD in murine brown adipocytes: Primary and immortalized murine brown preadipocytes were differentiated as stated above. On day 3 of differentiation, cells were reverse transfected according to (*106*). In brief, siRNA targeting the gene of interest or relevant control siRNA were diluted in Opti-MEM (Gibco, 51985) to a final concentration of 300 nM. RNAiMAX (Invitrogen, 13778-150) was added to a final concentration of 30 μl/ml. The siRNA mix was added to the bottom of the wells in the plate-format of interest and allowed to incubate at RT for 30 min. In the meantime, cells were trypsinized, counted, and resuspended in culture media. Finally, the cell suspension was distributed on top of the siRNA mix to a final siRNA concentration of 50 nM. Media was changed 2 days after transfection and cells were harvested or assayed on day 7 of the differentiation unless otherwise is stated in the figure legends. siRNA constructs used in the study are listed in table S14.

siRNA-mediated knock-down of *SLC25A34* in human brown adipocytes: Transfection and gene silencing experiments were performed using DharmaFECT1 transfection reagent (Horizon Discovery, T-2001) on day 12 of differentiation by following the manufacturer's instructions. Mature brown adipocytes were incubated with a mixture of DharmaFECT1 and SLC25A34-targeting siRNA (100 nM) (Dharmacon ON-TARGETplus SMARTpool Human SLC25A34 siRNA, Horizon Discovery, L-032041-02) or nontargeting (scrambled) siRNA (100 nM) (Dharmacon plus nontargeting control pool, Horizon Discovery, D-001810-10) for 48 hours (qPCR) or 96 hours (respirometry) at 37°C in a humidified atmosphere with 5% CO_2. After the incubation time, adipocytes were activated with dibutyryl-cAMP (db-cAMP, 500 μM) (Sigma-Aldrich, D0260) for 8 hours in DMEM/F-12 HAM medium supplemented with penicillin/streptomycin (100 U/ml), biotin (33 μM), and pantothenic acid (17 μM) (*107*).

Acetate rescue study: For the acetate rescue study, immortalized brown adipocytes were grown on 10 cm cell culture dishes. On the day of confluency, the culture medium was exchanged to differentiation medium for the control dish and to differentiation media supplemented with 10 mM sodium acetate for the acetate-treated dish. The differentiation protocol was standard for both groups. On day 3 of differentiation, siRNA transfection was performed as described above. The acetate-treated group always received culture media freshly supplemented with 10 mM sodium acetate during transfection and regular media changes. Seahorse respirometry was performed on day 7 in seahorse media containing standard respiratory substrates, without sodium acetate supplementation. Cells were harvested on day 7 for CS activity assay and mtDNA/nDNA quantification. Experiment was repeated 5 times.

Insulin treatment study: BMC brown adipocytes were differentiated according to the protocol described above. On day 6 of differentiation, insulin was omitted from the medium for 24 hours. On day 7, cells were treated with 2 nM or 20 nM insulin or left untreated as controls. Cells were harvested after 4 and 24 hours of treatment. The experiment was repeated once.

In vitro assays

Immunofluorescence assay: Cells were reseeded on day 3 of differentiation onto Matrigel-coated glass coverslips. On day 10, cells were serum-starved in DMEM containing 0.2% BSA for 2 hours and stimulated with 1 nM CL-316,243 for 1 hour. Cells were washed with PBS and fixed with 4% paraformaldehyde for 15 min, followed by quenching with 50 μM glycine. Cells were blocked and permeabilised in 5% normal swine serum (NSS) containing 0.1% saponin in PBS for 30 min, before 1 hour incubation with primary antibodies (TOMM20, ProteinTech, #11802-1-AP; HA, BioLegend, #901515) diluted in 2% NSS containing 0.1% saponin for 1 hour at RT. After PBS washes, cells were incubated with appropriate secondary antibodies for 1 hour at RT. Coverslips were washed in PBS and distilled water before mounting using ProLong Gold antifade mounting medium (Invitrogen, #P36930). Images were acquired on a Leica SP8 laser scanning confocal microscope with a 63×, NA 1.4 Plan-Apochromat oil immersion objective. Acquired images were median de noised using Leica Application Suite software (v3.5.7.2). Fluorescence intensity across indicated ROIs were plotted as line profiles.

Subcellular fractionation for isolation of mitochondria from adipocytes: Mitochondria were isolated from cultured brown adipocytes following

the differential centrifugation protocol described by Frezza *et al.* (*108*). Cells were washed with ice-cold PBS and scraped in ice-cold homogenisation buffer supplemented with protease inhibitors. Cells were pelleted by centrifugation at 600*g* for 10 min at 4°C and resuspended in homogenisation buffer. Cell disruption was performed using a glass teflon douncer (18 strokes). The homogenate was centrifuged at 600*g* for 10 min twice at 4°C to remove unlysed cells and debris. The resulting supernatant was centrifuged at 7000*g* for 10 min at 4°C to pellet mitochondria. A post-mitochondrial supernatant was collected. The mitochondrial pellet was washed with isolation buffer and re-centrifuged at 7000*g* for 10 min at 4°C before final resuspension in 100-150 µl buffer. Protein concentration was determined using BCA assay followed by immunoblotting of the indicated proteins. Antibodies used for immunoblotting: PMP70 (Abcam, #ab3421); calnexin, CXN (Abcam, #ab22595); total OXPHOS rodent WB antibody cocktail, OXPHOS (Abcam, ab110413); vinculin, VCL (Cell Signaling Technology, #13901).

Cycloheximide chase assay: Cells were reseeded into 24-well plates on day 3 of differentiation. Cells were pretreated with 10 nM CL-316,243 for 24 hours before the start of the experiment. On day 10, protein synthesis was inhibited by addition of cycloheximide (50 µg/ml), and cells were harvested at 0, 3, 6, 9, and 24 hours after treatment. Cells were washed twice with ice-cold PBS and lysed in RIPA buffer (50 mM Tris pH 7.5, 150 mM NaCl, 1% Triton X-100, 0.5% sodium deoxycholate, 0.1% SDS, 1 mM EDTA) supplemented with protease and phosphatase inhibitors (cOmplete Mini, Roche; PhosSTOP, Roche). Lysates were sonicated and centrifuged at 16,000*g* for 20 min at 4°C. Protein concentration was determined using BCA assay.

Immunoblotting: (fig. S1, B and C) Samples were denatured in Laemmli buffer containing TCEP and heated at 65°C for 10 min. Samples were resolved by SDS-PAGE on 4-20% gradient gels (Bio-Rad) and transferred to nitrocellulose membranes using a Trans-Blot Turbo system (Bio-Rad). Membranes were blocked using 5% milk in TBS-T (0.1% Tween-20) for 1 hour at RT and incubated with primary antibodies in 5% BSA TBS-T overnight at 4°C. Samples were analyzed by immunoblotting using antibodies against HA (BioLegend, #901515, 1:2000), SLC25A1 (ProteinTech, #15235-1-AP, 1:1000), UCP1 (ProteinTech, #83870-1-RR, 1:1000) and TOMM20 (ProteinTech, #11802-1-AP, 1:1000). GAPDH (Cell Signaling Technology, #97166S) was used as a loading control. After washing, membranes were incubated with HRP- or Alexa488-conjugated secondary antibodies (1:2500) in 5% milk in TBS-T for 1 hour at RT. Signal was detected using enhanced chemiluminescence (Thermo Fisher) or fluorescence on a ChemiDoc system (Bio-Rad). (figs. S4, C and D, and S7, F to H)

Cells were lysed in ice-cold RIPA buffer containing 0.1% SDS, protease inhibitors (Merck, #11836153001). Total protein concentration was measured using the Pierce Bicinchoninic Acid (BCA) Assay Kit (Thermo Scientific, #23225), with BSA as the standard. Proteins were separated by SDS-PAGE and transferred to a polyvinylidene fluoride (PVDF) membrane. For total protein quantification, the membranes were stained with Revert 700 Total Protein Stain for Western Blot Normalization (Li-Cor, #926-11011) and developed using Odyssey system (700 channel). Then the membranes were destained with Tris-buffered saline containing 0.1% Tween 20 (TBS-T) until clear. Membranes were blocked with 5% BSA or nonfat dry milk in TBS-T and incubated with primary antibodies, diluted in TBS-T with 5% nonfat dry milk, overnight at 4°C. Afterward, horseradish peroxidase-conjugated secondary antibodies were applied. Protein bands were detected after chemiluminescent substrate (Cytiva, RPN2232) addition using an Odyssey XF Imaging System. Protein band intensities were quantified with ImageStudio Lite Version 5.2 software. Specific signals with subtracted background signals for each blot were normalized to the housekeeping protein histone H3 (for S4C, S4D, S7F, S7H-right) or to total protein intensity for the protein pan-acetylation blot (S7G, S7H-left). Relative protein abundances were calculated by dividing the measured signals by their respective normalization factors, enabling relative quantitative comparisons. Antibodies used in the assay: SLC25A34 (Sigma-Aldrich, #HPA069806), H3 (Cell Signaling Technology, #9715), H3K27Ac (Cell Signaling Technology, #4353), pan Ac-Lys (Cell Signaling Technology, #9441).

DNA extraction and qPCR for mtDNA/nDNA quantification: DNA was extracted from cultured murine brown adipocytes (BMC) using AllPrep DNA/RNA/Protein Mini Kit (QIAGEN, 80204) according to the manufacturer's protocol. DNA concentration as measured with a NanoDrop spectrophotometer (Thermo Fisher Scientific). Gene expression was determined based on real-time qPCR using SYBR green (PP00259, Primerdesign) using LightCycler 480II (Roche) according to the manufacturer's protocol. The data was analyzed with the ΔΔCT method. mtDNA was quantified by the expression of *Nd1,* and nDNA by the expression of *Hk2*. All primers are listed in table S13.

DNL assay in cultured mouse brown adipocytes: Radiolabeling with [^{14}C]-glucose: Murine brown adipocytes were cultured and siRNA mediated KD was performed as described above in 12 well plates. On day 7 of differentiation, culture media was exchanged to 0.5 ml per well of standard DMEM media supplemented with T3 and Insulin, containing ^{14}C-labeled glucose (D-[^{14}C(U)] Revvity Health Sciences Inc NEC042V250UC, Lot 3106342; 250 µCi (9.25 MBq); 265 mCi/mmol (9.805 GBq/mmol); 1.25 ml ethanol:water (3:97); 0.5 µCi per 0.6 ml of media. A few wells were left untreated as a negative control. Cells were cultivated in the presence of ^{14}C-labeled glucose for 6 hours. After 6 hours, cells were rinsed twice with ice-cold PBS and frozen on plates at –70°C until subsequent analysis.Lipid extraction and scintillation counting: Lipid extraction was performed as described in (*109*) with modifications. 0.4 ml of H_2O was added to each well and cells were harvested using a rubber policeman into 15 ml conical Falcon tubes. 0.5 ml chloroform and 1.0 ml methanol were added to each sample, followed by vortexing. Next, 0.5 ml chloroform and 0.5 ml of 0.2 N HCl were added, and samples were vortexed again. Phase separation was achieved by centrifugation at 2500 rpm for 5 min at RT. The lower (organic) phase was transferred to a clean 15 ml tube and washed once with a pre-prepared "blank upper phase" (generated by mixing 1 ml chloroform, 1.0 ml methanol, 0.5 ml of 0.2 N HCl, and 0.4 ml water per sample). After a second centrifugation, the cleaned lower phase was transferred to a 2 ml glass vial and organic solvents were evaporated to dryness under nitrogen flow. Dried lipids were resuspended in 75 µl chloroform:methanol (2:1, v/v), and 60 µl was transferred to scintillation vials containing 5 ml scintillation fluid (Ultima Gold, Revvity Health Sciences Inc 6013329). Scintillation was counted using Hidex 600 SLe Automatic TDCR Liquid Scintillation Counter.

CS activity assay: The mitochondrial function of cultured brown adipocytes (BMC) was estimated by measuring the activity of CS in the cellular extracts. The assay is based on the protocol from (*110*) and was performed as described in (*111*) with modifications. Briefly, the cells were grown in 6 well plates (1000000 cells per well) and harvested by adding 400 µl of the lysis buffer (50 mmol/l Hepes, 137 mmol/l NaCl, 10 mmol/l Na4P2O7, 20 mmol/l NaF, 5 mmol/l EDTA, 1 mmol/l MgCl2, 1 mmol/l CaCl2, 2 mmol/l Na3VO4, 5 mmol/l nicotinamide, 10 µmol/l trichostatin A, Halt protease inhibitor cocktail (Thermo Scientific, Waltham, MA, USA), 1% (v/v) NP-40, 10% (v/v) glycerol, demineralized water). The extracts were mixed with a reaction mix containing 0.1 m Tris-HCl, pH 8.1, 0.4 mm acetyl-CoA sodium salt (Sigma-Aldrich) and 1 µm 5,5′-dithiobis(2-nitrobenzoic acid) (Sigma-Aldrich), and absorbance at 412 nm was measured every 30 s for 5 min (basal slope). Oxaloacetic acid at 10 mm was used as the reaction substrate, and the change in absorbance after its addition to the mix was measured every 30 s for another 5 min (reaction slope). CS activity was then calculated from the delta of the reaction and basal slopes and presented in µmol/min/g protein. Protein content was measured in a separate set

of samples produced in the same experiment using Bradford assay (Quick Start Bradford 1x Dye Reagent #5000205m Bio-Rad) according to manufacturer protocol.

Proteomics

Proteomics was performed exactly as described in (*24*). BAT proteomics data from the cold adaptation study is published and provided as Supplementary Material Data S1 in (*24*). BAT proteomics from *Rev-erbα* WKO mice is provided in table S1. BATs were mechanically lysed with a homogenizer with 2 ml SDS lysis buffer containing 2.0% SDS w/v, 150 mM NaCl, PhosStop (Roche, Madison, WI) phosphatase inhibitors, EDTA free protease inhibitor cocktail (Promega, Madison, WI) and 50 mM HEPES, pH 8.5. Lysates were reduced with 5 mM DTT and cysteine residues were alkylated at RT with iodoacetamide (14 mM) in the dark as previously described (*112*). Protein content was purified by methanol/chloroform extraction. Protein disks were resuspended in 8 M Urea containing 50 mM HEPES (pH 8.5) and concentrations were measured by BCA assay before protease digestion. 1 mg of protein lysates were diluted to 4 M urea and digested with LysC (Wako, Japan) in a 1/100 enzyme/protein ratio overnight. Protein extracts were diluted further to a 1.0 M urea concentration and trypsin (Promega, Madison, WI) was added to a final 1/200 enzyme/protein ratio for 6 hours at 37°C. Digests were acidified with 20 µl of 20% formic acid (FA) to a pH ~ 2 and subjected to C18 solid-phase extraction (SPE) (Sep-Pak, Waters, Milford, MA).

Isobaric labeling of peptides was performed using 10-plex tandem-mass tag (TMT) reagents (Thermo Fisher Scientific, Rockford, IL). TMT reagents (0.8 mg) were dissolved in 42µl dry acetonitrile (ACN) and 10 µl was added to 100 µg of peptides dissolved in 100 µl of 200mM EPPS, pH 8.0. After 1hr (RT), the reaction was quenched by adding 4 µl of 5% hydroxylamine. Labeled peptides were combined, acidified with FA (pH ~2) and diluted to a final ~5% ACN concentration before C18 SPE on Sep-Pak cartridges (50 mg).

Basic pH reversed-phase HPLC (bpHrp) TMT labeled peptides were subjected to orthogonal bpHrp fractionation. Labeled peptides were solubilized in buffer A (5% ACN 10 mM ammonium bicarbonate, pH 8.0) and separated by an Agilent 300 Extend C18 column (5 µm particles, 4.6 mm ID and 220 mm in length). Using an Agilent 1100 binary pump equipped with a degasser and a photodiode array (PDA) detector (Thermo Scientific, San Jose, CA), a 50 min linear gradient from 12% to 45% acetonitrile in 10 mM ammonium bicarbonate pH 8 (flow rate of 0.8 ml/min) separated the peptide mixtures into a total of 96 fractions. 96 Fractions were consolidation into 12 samples, acidified with 10 µl of 20% formic acid and vacuum dried. Each sample was re-dissolved in 5% formic acid, desalted using StageTips, dried by vacuum centrifugation, and reconstituted for LC-MS/MS analysis.

Mass spectrometry analysis: All bpHrp fractions were subjected to LC-MS/MS analyses onto a LTQ Orbitrap Velos Fusion (Thermo Scientific San José, CA) instrument equipped with a Famos autosampler (LC Packings, Sunnyvale, CA) and an Agilent 1100 binary HPLC pump (Agilent Technologies, Santa Clara, CA). Peptides were separated onto a 100 µm I.D. microcapillary column packed first with ~1 cm of Magic C4 resin (5µm, 100 Å, Michrom Bioresources, Auburn, CA) followed by ~25 cm of Maccel C18AQ resin (1.8 µm, 200 Å, Nest Group, Southborough, MA). Peptides were separated by applying a gradient from 5 to 35% ACN in 0.5% FA over 180 min at ~200 nl/min. Electrospray ionization implemented through applying a voltage of 1.86 kV using an inert gold electrode through a PEEK junction at the end of the microcapillary column. The LTQ Orbitrap Velos Fusion was operated in data-dependent manner for the MS methods. The MS survey scan was performed in the Orbitrap in the range of 400-1300 *m/z* at a resolution of 3x104, followed by the selection of the ten most intense ions (TOP 10) for CID-MS2 fragmentation in the ion trap using a precursor isolation width window of 2 *m/z*, AGC setting of 2000, and a maximum ion accumulation of 150 ms. Normalized collision energy was set to 35% and an activation time of 20ms. Ions within a 10 ppm *m/z* window around ions selected for MS2 were excluded from further selection for fragmentation for 120s. Directly after each MS2 event, 6-10 of most intense fragment ion in an *m/z* range between 110-160% of the precursor *m/z* was selected for HCD-MS3 (*113*, *114*). The fragment ion isolation width was set to 4 *m/z*, AGC was set to 20,000, the maximum ion time was 250ms, normalized collision energy was set to 60% and an activation time of 50ms for each MS3 scan. For all MS3 scans, Orbitrap resolving power was set to 30,000 (@ 400 *m/z*).

Data processing: A compilation of in-house software was used to convert mass spectrometric data (Raw file) to a mzXML format, as well as to correct monoisotopic *m/z* measurements and erroneous peptide charge state assignments. Assignment of MS/MS spectra was performed using the Sequest algorithm by searching the data against a protein sequence database including all entries the Mouse Uniprot database (download date June, 2014) containing known contaminants such as human keratins and its reverse decoy components (*115*). Sequest searches were performed using a 20 ppm precursor ion tolerance and requiring each peptides N-/C- termini to have trypsin protease specificity while allowing up to three missed cleavages. TMT tags on peptide N termini/lysine residues (+229.162932 Da) and carbamidomethylation of cysteine residues (+57.02146 Da) were set as static modifications, and methionine oxidation (+15.99492 Da) was set as variable modification. A MS2 spectra assignment FDR of <1% was achieved by applying the target-decoy database search strategy (*115*). Filtering was performed using an in-house linear discrimination analysis algorithm to create one combined filter parameter from the following peptide ion and MS2 spectra metrics: Sequest parameters XCorr and ΔCn, peptide ion mass accuracy and charge state, peptide length and mis-cleavages. Linear discrimination scores were used to assign probabilities to each MS2 spectrum for being assigned correctly and these probabilities were further used to filter the dataset to a 1% protein-level FDR (*112*).

TMT reporter ion and quantitative data analysis: For quantification, a 0.003 *m/z* window centered on the theoretical *m/z* value of each ten reporter ions and the closest signal intensity from the theoretical *m/z* value was recorded. Reporter ion intensities were further de-normalized based on their ion accumulation time for each MS3 spectrum and adjusted based on the overlap of isotopic envelopes of all reporter ions (manufacturer specifications). Total signal to noise values for all peptides were summed for each TMT channel, and all values were adjusted to account for variance in sample handling. For each peptide, a total minimum signal to noise value of 200 was required (*113*, *114*).

Identification of proteins temporarily co-regulated with SLC25A34: To identify proteins with expression dynamics similar to SLC25A34 during cold adaptation, we performed hierarchical clustering using the Perseus software platform [version 1.5.5.5, (*116*)]. The cold adaptation BAT proteomics dataset was analyzed by calculating the Pearson correlation coefficient between each protein and the expression profile of SLC25A34. Proteins were then ranked according to their correlation values, and the top 50 most similarly temporally regulated proteins were selected for downstream pathway enrichment analysis (table S4).

Gene expression analysis

RNA extraction and qPCR from murine tissues and cells: Tissues were isolated and immediately snap-frozen in liquid nitrogen. Cells were immediately lysed using TRI reagent (T9424, Sigma-Aldrich). Total RNA was extracted from the tissues and cells using TRI reagent

(T9424, Sigma-Aldrich) followed by isolation using RNeasy Mini Kit (74106, Qiagen). Reverse transcription was performed on 1000 ng RNA using the High-Capacity cDNA Reverse Transcription kit (4368814, Applied Biosystems). Gene expression was determined based on real-time qPCR using SYBR green (PP00259, Primerdesign) using LightCycler 480II (Roche) according to manufacturer's protocol. The data was analyzed with the ΔΔCT method and normalized to the housekeeping gene 36b4. All primers are listed in table S13.

RNA extraction and qPCR from human brown adipocytes: Human adipocytes were immediately lysed using TRI reagent (T9424, Sigma-Aldrich). Total RNA was extracted by chloroform–isopropanol phase separation. RNA concentration and purity were assessed with a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific). Reverse transcription was performed on 1000 ng RNA using the High-Capacity cDNA Reverse Transcription kit (4368814, Applied Biosystems). Gene expression was determined based on real-time qPCR using LightCycler 480II (Roche) according to the manufacturer's protocol. Validated TaqMan assays (*ACTB*: Hs01060665_g1, *GAPDH*: Hs99999905_m1, *PPARGC1A*: Hs01016719_m1, *UCP1*: Hs00222453_m1, *SLC25A34*: Hs01371979_m1) used in qPCR were designed and supplied by Thermo Fisher Scientific. The data was analyzed with the ΔΔCT method and normalized to the geometric mean of *ACTB* and *GAPDH* Ct values.

ChIP-sequencing: The REV-ERBα ChIP-seq experiment was performed as described in (*9*). In particular, the study was done on BAT samples from three mice acclimated to 29°C harvested at 5:00 p.m. (ZT10) with or without a 6 hours cold challenge, exactly as previously described in (*9*). The data are published GSE79167. Murine BAT was harvested immediately after euthanasia. It was quickly minced and cross-linked in 1% formaldehyde for 20 min, followed by quenching with 1/20 volume of 2.5 M glycine solution and two washes with ice-cold PBS. Chromatin fragmentation was performed by sonication in ChIP SDS lysis buffer (50 mM HEPES, 1% SDS, 10 mM EDTA at pH 7.5) using probe sonication. Proteins were immunoprecipitated in ChIP dilution buffer (50 mM HEPES, 155 mM NaCl, 1.1% Triton X-100, 0.11% Na-deoxycholate, Complete protease inhibitor tablet at pH 7.5) using the Cell Signaling Technology antibody (#2124). Cross-linking was reversed overnight at 65°C in elution buffer (50 mM Tris-HCL, 10 mM EDTA, 1% SDS at pH 8), and DNA was isolated using phenol/chloroform/isoamyl alcohol. Deep sequencing was performed by the Functional Genomics Core (J. Schug and K. Kaestner) of the Penn Institute for Diabetes, Obesity, and Metabolism using the Illumina Genome Analyzer IIx and Illumina HiSeq 2000 and sequences were obtained using the Solexa Analysis Pipeline.

RNA-seq and analysis of *Pparα* AKO (exon 5 targeted) BAT: RNA extraction, sequencing and data analysis of BAT from *Pparα* AKO mice was performed exactly as described in the corresponding section of the method details in (*34*). A list of differentially regulated genes is provided in table S5 and the data are available on Zenodo (*117*).

RNA-seq and analysis of BAT from ketogenic diet fed mice: The RNA sequencing and data analysis were performed as described in (*98*). In particular, for the generation of RNA-seq libraries, polyadenylated mRNA was isolated from 1 μg of total RNA by incubation with oligo-dT beads and prepared according to manufacturer's protocol (TrueSeq 2, Illumina). Samples were sequenced on the Illumina HiSeq 1500 platform. Sequencing reads were mapped to the mouse or human reference genome (version mm9 or hg19) using STAR (*118*). Tag directories were generated using HOMER (*119*) and exon reads were counted using iRNA-seq (*120*). Normalization and identification of differentially expressed genes was performed using DESeq2 (*121*). A list of differentially regulated genes is provided in table S6 and the accession number for data is GEO no. GSE348259.

RNA-seq and analysis of BAT from Rev-erbα/β BKO mice at 29° and 4°C, 6h and cultured brown adipocytes: RNA extraction was performed as stated above. Isolated RNA was treated with DNase I (NEB-M0303L) by mixing 250 ng of RNA, 8 Units of DNase I and DNase buffer in 10 μl reaction volume per sample and incubation the reaction mix for 15 min at 37°C. Reaction was quenched by thorough vortexing of the samples and incubation at 85°C for 5 min.

Total RNA integrity (RIN) was assessed using the TapeStation RNA High Sensitivity (HS) kit (Agilent Technologies). mRNA-seq libraries were prepared using the Universal Plus mRNA-seq library preparation kit with NuQuant (TECAN, Männedorf, Switzerland, Cat. #M01485 V8) according to the manufacturer's instructions. Library concentrations were quantified using Qubit with the NuQuant quantification method and the 1X dsDNA High Sensitivity Assay Kit (Thermo Fisher Scientific, catalog no. Q33231). Library quality was evaluated using the TapeStation DNA D1000 HS kit (Agilent Technologies). Sequencing was performed on an Illumina NovaSeq 6000 platform using SP 100-cycle v1.5 reagents, generating 52-bp paired-end reads. Libraries were sequenced at a molarity of 2 nM with a 1% PhiX spike-in control.

Pre-processing of sequencing data was performed by the Single-Cell Omics platform at the Novo Nordisk Foundation Center for Basic Metabolic Research. In short, sample-demultiplexed FASTQ files were generated from BCL files using the bcl2fastq software (v. 2.20.0.422), and gene-level count matrices were then generated by the nf-core RNA-seq pipeline [v. 3.5, (*122*)]. Specifically, this pipeline first trims the reads using the Trim Galore (v. 0.6.7) software, deduplicates the reads based on UMIs through UMI-tools [v. 1.1.2, (*123*)] before finally mapping the reads to the GRCm38 genome using STAR [v. 2.6.1d, (*118*)] and quantifying the gene counts using the salmon software [v. 1.5.2, (*124*)].

Differential gene expression analysis was similarly performed by the Single-Cell Omics platform at the Novo Nordisk Foundation Center for Basic Metabolic Research. Here, all samples were analyzed together using the EdgeR [v. 3.38.0, (*125*)] and limma [v. 3.52.0, (*126*)] R-packages based on the specified contrasts. Specifically, the differential gene expression analyses were performed using the glmQLFit-function with robust=TRUE. A list of differentially regulated genes is provided in table S11. The data for BAT from *Rev-erbα/β* BKO mice at 29° and 4°C for 6 hours are available at ArrayExpress (https://www.ebi.ac.uk/biostudies/arrayexpress) under no. E-MTAB-17580. The data for siRNA-mediated *Slc25a34* KD in brown adipocytes are available at ArrayExpress under no. E-MTAB-17577.

Pathway enrichment analysis: All pathway enrichment analyses were performed using Gene Ontology (GO) enrichment analysis, type: PANTHER Overrepresentation Test (released 7 August 2024). Genes expressed in mouse BAT were used as a reference list. The presented pathways are the top 10 enriched pathways sorted based on the fold enrichment score with FDR < 0.05 (test type: Fisher's exact correction: FDR).

Transcription factor prediction based on differential gene expression: To identify potential transcriptional regulators in the dataset, we calculated an average log fold change (logFC) for each gene based on two independent siRNA KD datasets (Sigma and Dharmacon) for all genes significantly dysregulated in both datasets (FDR < 0.05). Two groups of genes were selected: down-regulated genes (logFC ≤ −0.5, $n = 542$) and up-regulated genes (logFC ≥ 0.5, $n = 676$). These gene groups were used to predict candidate transcription factors using ChEA3 (*127*). The top predicted transcription factors were identified for each group. To contextualize these findings, each transcription factor was annotated based on its expression level in brown adipocytes as high, medium, low, or not detected [ND]), and whether its own mRNA was significantly down-regulated after the *Slc25a34* KD. The relationship between the transcription factors and their target genes was visualized using clustergrams. Genes forming the clusters indicated in Fig. 6F and fig. S8B are shown in table S12.

Metabolomic analyses

Derivatization of extracted metabolites and untargeted steady-state metabolomics analysis (Fig. 4, A and B): siRNA-mediated KD in murine brown preadipocytes (BMC) was performed as described above. siRNA transfection was performed on day 3 of differentiation and cells were harvested on day 7 of differentiation. Cells were washed briefly with ice-cold 1× PBS, and residual PBS was removed by aspiration. Metabolites were extracted by adding 200 µl of ice-cold extraction solvent (methanol:acetonitrile:H_2O 50:30:20) directly to the cells, followed by scraping and transfer into prelabeled, chilled 1.5-ml Eppendorf tubes. Samples were vortexed vigorously for 2 min at 4°C and centrifuged at 1000*g* for 2 min at 4°C. The resulting supernatant was transferred to new prelabeled tubes and kept on an ice bath until further processing. For 3-NPH derivatization, samples (20 µl) were derivatized with 50 mM 3-NPH (in 50% acetonitrile) in the presence of 30 mM EDC (in 50% acetonitrile and 6% pyridine) for 45 min at 4°C with gentle shaking. Appropriate blanks, quality control (QC, pool of all samples) and controls were included, including pure standards and samples without extract (solvent only). Derivatized metabolites were analyzed by LC-MS as follows. Samples of 3 µl were injected into a Vanquish Horizon UPLC system (Thermo Fisher Scientific) equipped with a Zorbax Eclipse Plus C18 guard column (2.1 × 50 mm, 1.8 µm; Agilent Technologies) and an analytical column (2.1 × 150 mm, 1.8 µm; Agilent Technologies), maintained at 40°C. Analytes were eluted at a flow rate of 400 µl/min using a gradient of eluent A (0.1% formic acid in water) and eluent B (0.1% formic acid in acetonitrile): 0 to 1 min, 10% B; 1 to 1.5 min, 10 to 40% B; 1.5 to 5 min, 40 to 95% B; 5 to 7.6 min, 95% B; 7.6 to 8 min, 95 to 10% B. The system was equilibrated for 3.5 min under initial conditions. The eluent flow was coupled to a TimsTOF Pro (Bruker) for mass spectrometric analysis in negative ion mode. Data were processed using Metaboscape (Bruker, v. 2025) with the In-Silico Derivatization workflow, which includes automated library structure derivatization, CCS prediction, and in-silico fragmentation for automated annotation.

Underivatized samples were extracted as described above. Instead of derivatization, the supernatant was lyophilized and resuspended in 25 µl of 0.1% formic acid in water just before analysis. From each sample, 5 µl was transferred to a pooled QC sample. From each sample, 2 µl was injected into a Vanquish Horizon UPLC system (Thermo Fisher Scientific) equipped with a Zorbax Eclipse Plus C18 guard column (2.1 × 50 mm, 1.8 µm; Agilent Technologies) and an analytical column (2.1 × 150 mm, 1.8 µm; Agilent Technologies), maintained at 40°C. Analytes were eluted at a flow rate of 400 µl/min using a gradient of eluent A (0.1% formic acid in water) and eluent B (0.1% formic acid in acetonitrile): 0 to 1 min, 3% B; 1 to 3 min, 3 to 40% B; 3 to 5 min, 40 to 95% B; 5 to 7.6 min, 95% B; 7.6 to 8 min, 95 to 3% B. The system was equilibrated for 3.5 min under initial conditions. The eluent flow was coupled to a TimsTOF Pro (Bruker) for mass spectrometric analysis in positive and negative ion modes. Data were processed using Metaboscape (Bruker, v. 2025), with features annotated using the NIST2020 and MetaboBase libraries, integrated as add-ons within Metaboscape. Data provided in table S7.

Targeted steady-state metabolomic measurement of acetyl-CoA in cultured brown adipocytes (Fig. 4C): siRNA-mediated KD in murine brown preadipocytes (BMCs) was performed as stated above. siRNA transfection was performed on day 3 of differentiation and cells were harvested on day 7 of differentiation. Cells were washed briefly with ice-cold 1× PBS, and residual PBS was removed by aspiration. Metabolites were extracted by adding 200 µl of ice-cold extraction solvent (methanol:acetonitrile:H_2O 50:30:20) directly to the cells, followed by scraping and transfer into prelabeled, chilled 1.5-ml Eppendorf tubes. Samples were vortexed vigorously for 20 min at 4°C and centrifuged at 1000*g* for 2 min at 4°C. The supernatant was transferred to a new prelabeled tube and lyophilized by vacuum centrifugation. Samples were resuspended in 25 µl of which 3 µl were injected into a Vanquish Horizon UPLC system (Thermo Fisher Scientific) equipped with a Zorbax Eclipse Plus C18 guard column (2.1 × 50 mm, 1.8 µm; Agilent Technologies) and an analytical column (2.1 × 150 mm, 1.8 µm; Agilent Technologies), maintained at 40°C. Analytes were eluted at a flow rate of 500 µl/min using a gradient of eluent A (50 mM ammonium acetate in water) and eluent B (100% methanol): 0 to 0.5 min, 10% B; 0.5 to 7.2 min, 10 to 99% B; 7.2 to 8 min, 99% B; 8 to 8.1 min, 99 to 10% B; 8.1 to 12 min, 10% B. The eluent flow was coupled to a Q Exactive HF (Thermo Fisher Scientific) for mass spectrometric analysis in negative ion mode. Data were processed using MZmine 2.53 (*128*) with the Targeted Feature Detection module, targeting CoA esters annotated using in-house standards. Data from the above analyses were processed in Metabolink (*129*) to perform blank subtraction, QC-based filtering, drift correction, data merging, and probabilistic quotient normalization.

Untargeted steady-state metabolomic analysis of cultured brown adipocytes with *Slc25a34* KD time course (Fig. 4, D and F to M): For cell culture and transfection, BMC cells were differentiated as described above. On day 3 of differentiation, reverse transfection was performed as described above. Cells were harvested at 4 time points: on day 4 of differentiation (referred to as 1 day of KD), day 5 of differentiation (referred to as 2 days of KD), day 6 of differentiation (referred to as 3 days of KD), and day 7 of differentiation (referred to as 4 days of KD). For all time points, 1 hour prior the harvest, the medium was changed to DMEM without glucose, L-glutamine, phenol red, sodium pyruvate and sodium bicarbonate (Sigma-Aldrich, D5030) supplemented with 143 mM NaCl, 25 mM glucose, and 1 mM pyruvate.

For metabolomic sample preparation, metabolites were extracted using a method drafted from (*130*). Briefly, media were aspirated and cells were extracted with 0.7 ml of ice cold 50% aqueous liquid chromatography-mass spectrometry (LCMS) grade acetonitrile (extraction and re-suspension solvent). The plate was then placed on top of wet ice and allowed to rest for 10 min. Cells were scraped in the extraction solvent and transferred to 1.5 ml centrifuge tubes. Samples were centrifuged at top speed for 10 min in a centrifuge cooled at 0°C. Blanks were composed by treating an empty well as well as a cell culture well containing media sans cells in the same way as samples. Supernatants were transferred to fresh 1.5 ml centrifuge tubes and dried by speed vacuum. Dried residues were re-suspended in 0.2 ml of 50% aqueous LCMS grade acetonitrile. For the time course study, samples were re-suspended in 0.1 ml of 50% aqueous LCMS grade acetonitrile. All samples were centrifuged for 5 min at 20,238*g* without temperature control. 50 µl of supernatant were transferred to fresh autosampler vials. 5 µl of each sample were combined to form a QC pool sample, which was injected 6 times in the beginning of the queue, twice in the middle of the queue and twice at the end of the queue to serve as a performance control. In addition to the above blanks, a solvent blank was composed with the re-suspension solvent. Samples, blanks, and QC pool were placed in a pre-chilled (6°C) Thermo Scientific Dionex 3000 autosampler and injected as described below. In all analyses, samples were injected in a random order.

For liquid chromatography–mass spectrometry (LCMS), in the non–time course study, samples were injected at 5 µl in negative ion mode. For positive ion mode analysis, samples were diluted 1/10 and injected at 2.5 µl. In the time course study, samples were injected at 5 µl in both modes. Analytes were separated using a gradient at a flow rate of 0.4 ml/min between 0.006% (v/v) LCMS grade ammonium hydroxide in LCMS grade water (mobile phase B) and 0.006% (v/v) LCMS grade ammonium hydroxide (mobile phase A) in LCMS grade acetonitrile over a heated (25°C) Phenomenex Amide (2 × 100 mm, particle size 3 µm, pore size 100 Å) column fitted with a Phenomenex SecurityGuard guard cartridge containing amide resin. The column was equilibrated at 10% mobile phase B, which was then increased to 70% B over 10 min.

The column was washed over 5 min at 100% B, then re-equilibrated at 10% B for 10 min. Analytes were electrospray ionized and detected in both negative and positive ion mode using a Bruker Impact II Q-TOF mass spectrometer operated in full scan mode (*m/z*: 50 to 1000). The instrument was mass calibrated before use on each day using an infusion of 4.5 mM sodium acetate. The calibrant was also infused at the beginning of each analysis and used to mass calibrate each sample. The instrument parameters are as follows: cone voltage, 30 kV; collision energy, 7 eV; nebulizer gas flow, 10 L/min; nebulizer gas temperature, 220°C; nebulizer pressure, 2 bar; and scan rate, 2 Hz. Ions were assigned identities based upon MS/MS fragmentation and matching to Metlin (*131*) and HMDB (*132*) mass spectral libraries.

For analysis, raw data was converted to .cdf format using Bruker DataAnalysis software (version 4.3). Peaks were detected, aligned, and integrated using XCMS online (*133*). Ions eluting before 1 min and isotopes were removed from further analysis as well as ions with an intensity below 10,000 counts. MetaboAnalyst (version 4.0) was used for all statistical analyses and heatmap generation (*134*). Outliers were based upon interquartile range. Data were normalized to sum intensity, generalized log-transformed (*135*) and Pareto scaled. All multiple comparisons were FDR < 0.05 corrected. Data are shown in table S8 (statistical analysis of the 54 dysregulated metabolites) and table S9 (shows relative intensities for detected annotated metabolites that are significantly dysregulated by *Slc25a34* KD at either time point). Some metabolites are duplicated because they were detected in both ion modes.

Mitochondrial immunoprecipitation and targeted steady-state metabolomics analysis of cultured brown adipocytes (Fig. 5, A and B, and fig. S5, B to D): For cell culture and transfection, cells were seeded on 15 cm tissue culture dishes, grown to 100% confluency, and induced to differentiate. On day 3 of differentiation, reverse transfection was performed following the protocol from (*106*). siRNA targeting *Slc25a34* (si*Slc25a34*) or a nontargeting control siRNA (si*Control*) were diluted in Opti-MEM (Gibco, catalog no. 51985) to a final siRNA concentration of 300 nM. Lipofectamine RNAiMAX (Invitrogen, catalog no. 13778-150) was added at a final concentration of 30 μl/ml. The siRNA-transfection mix was incubated at RT for 30 min. In parallel, cells were trypsinized, counted, and resuspended in complete culture medium. A total of 1.5×10^6 cells per well were seeded on top of the siRNA mixture, resulting in a final siRNA concentration of 50 nM. In addition, adenovirus encoding 3XHA-EGFP-OMP25 (for rapid mitochondrial purification (*49*)) was added dropwise to each well. The media was refreshed 2 days after transfection. On day 7 of differentiation, cells were harvested for anti-HA bead pulldown as described in (*136*).

For rapid mitochondria isolation (HA pulldown), cells were washed twice with PBS and once with 1 ml KPBS. Homogenization was performed using a Dounce homogenizer with 25 strokes. A 50 μl aliquot of the homogenate was collected for Western blotting analysis. The remaining homogenate was transferred to a 2 ml microcentrifuge tube and centrifuged at 1000 g for 2 min. The supernatant was transferred to a fresh 2 ml tube, and 60 μl of pre-washed anti-HA magnetic bead slurry (Thermo Scientific, catalog no. 88836) was added. Samples were rotated for 3 min at 4°C. Beads were magnetically separated, and the flowthrough was collected as the post-mitochondrial (cytosolic) fraction. Beads were then resuspended in 1 ml KPBS and divided into two 500 μl aliquots for downstream metabolomic and Western blotting analyses.

For metabolite extraction and derivatization, metabolites were extracted by eluting from beads in 380 μl of 50% methanol. Samples were centrifuged at 17,000*g* for 10 min, and the supernatants were collected. Supernatants were vortexed for 10 s, followed by the addition of 220 μl acetonitrile and another 10-s vortex. Samples were bead-beaten for 6 min at 20 Hz using 1.4 mm ceramic beads. Next, 600 μl dichloromethane and 300 μl water were added, and samples were vortexed for 1 min before incubating on ice for 10 min. Samples were centrifuged for 10 min at 1500*g* at 1°C, and the aqueous phase was transferred to a new 1.5-ml microcentrifuge tube. To quantify unstable α and β-keto acids (such as OAA), a reduction was performed before sample drying and derivatization as described previously (*137*). The pH of each aqueous sample was adjusted to 10 to 12 using 2 M NaOH, verified with pH paper. While on ice, 15 μl of freshly prepared $NaBD_4$ solution (5 mg/ml in 50 mM NaOH) was added. The reaction was allowed to proceed at RT for 1 hour. Samples were returned to ice, and 10 μl of cold 1 M HCl was slowly added to quench the reaction. Samples were dried using a SpeedVac concentrator.

Dried samples were reconstituted in 30 μl of pyridine and then centrifuged at RT at 13,000 g for 2 min. The resulting supernatants were transferred to GC/MS autosampler vials containing 70 μl of *N*-tert-butyldimethylsilyl-*N*-methyltrifluoroacetamide (MTBSTFA) for derivatization, and then were incubated at 70°C for 1 hour. One microliter of each derivatized sample was injected for GC/MS analysis.

GC/MS analyses were performed using an Agilent 7890A gas chromatograph coupled to a 5975C mass selective detector, equipped with an Agilent 7693 autosampler. Chromatographic separation was achieved using a DB-5ms column (30m length with 10m guard length, 0.25mm inner diameter, 0.25μm film thickness). The GC oven program was initiated with a 1-min hold at 120°C, followed by a temperature ramp of 10°C per minute to 300°C. A final bake-out step was held at 320°C for 10 min. The injector and MS interface temperatures were maintained at 285°C. Helium was used as the carrier gas at a constant flow rate of 1.5 ml/min. The mass spectrometer operated in electron ionization mode at 70 eV. Data acquisition was performed in selected ion monitoring (SIM) mode and full scan mode (mass range: 50 to 700 Da). Data analysis was performed using Agilent GCMS MassHunter. Metabolite relative abundance was normalized to protein from each corresponding sample. Data provided in table S8.

Isotope tracing analysis (Fig. 5, C to J, and fig. S6, A to D): Cell culture, transfection, and stable isotope supplementation: BMC cells were differentiated as described above. Reverse transfection was performed on day 3. On day 7, 2 hours before tracer supplementation, the medium was replaced with base medium consisting of DMEM without glucose, glutamine, or phenol red (ThermoFisher Scientific, A1443001), supplemented with 10% FBS, 1% penicillin-streptomycin, 5 mM glucose, 0.5 mM glutamine, and 1 mM acetate.

After 2 hours, cells were divided into four groups and the medium was replaced with the same base medium, modified as follows: (i) glucose tracing: 5 mM [$^{13}C_6$]-glucose (Cambridge Isotope Laboratories, CLM-1396-5) instead of unlabeled glucose; (ii) glutamine tracing: 0.5 mM [$^{13}C_5$]-glutamine (Cambridge Isotope Laboratories, CLM-1822-H) instead of unlabeled glutamine; (iii) acetate tracing: 1 mM [$^{13}C_2$,$^{2}H_3$]-acetate (Sigma-Aldrich, S8750) instead of unlabeled acetate; and (iv) nonlabeled group for steady-state metabolomics: no labeled substrates (base medium).

Cells from tracing groups were harvested 5, 30, and 60 min after tracer addition. Samples for steady-state metabolomics were harvested 60 min after the second medium change.

Liquid chromatography-mass spectrometry for isotope tracing: Acyl-CoA quantification and isotope tracing was conducted by liquid chromatography-high resolution mass spectrometry as previously published (*138*, *139*). Samples were lysed by sonication at 5x 0.5 s pulses at 50% intensity, and lysate centrifuged at 17,000*g* for 10 min at 4°C. Oasis HLB SPE columns was equilibrated with 1 ml methanol followed by 1 ml of water, supernatant applied, desalted with 1 ml water, and eluted with 1 ml methanol with 25 mM ammonium acetate. Extracts were dried under nitrogen and resuspended in 50 μl of 5% (w/v) 5-sulfosalicylic acid in water. 5 μl was injected for analysis into an Ultimate 3000 Quaternary Ultra High-Pressure Liquid Chromatograph coupled to a QExactive Plus (Thermo Fisher) mass spectrometer in

positive ESI mode as described previously (*139*). Chromatography conditions were as follows: buffer A was 5 mM ammonium acetate; buffer B was 95% acetonitrile (v/v) with 5 mM ammonium acetate and buffer C was 80% acetonitrile with 0.1% formic acid, with a gradient of 98% A and 2% B for 1.5 min, 80% A and 20% B at 5 min, 100% B at 12 min with a flow rate of 0.2 ml/min; followed by 100% B at 16 min at 0.3 ml/min; 100% from 17-21 min and 98% A and 2% B from 22-28 min at 0.2 ml/min. Mass spectrometry conditions were: auxiliary gas 10 arbitrary units (arb), sheath gas 35 arb, sweep gas 2 arb, spray voltage 4.5 kV, capillary temperature 425°C, S-lens RF-level 50, auxiliary gas heater temperature 400°C, and in-source CID 5 eV. Scan parameters were alternating full scan from 760 to 1800 *m/z* at 140,000 resolution and data-independent acquisition (DIA) looped three times with all fragment ions multiplexed at a normalized collision energy (NCE) of 20 at a resolution of 280,000. An isolation width of 7 *m/z* with an offset of 3 *m/z* was used to capture all relevant isotopologues for targeted acyl-CoA thioesters. Data integration was performed in Tracefinder v4.1 (Thermo Fisher), and isotopic enrichment determined using FluxFix by comparing to unlabeled samples (*140*).

Non–acyl-CoA metabolites were measured by liquid chromatography-high resolution mass spectrometry adapted from Petrova *et al.* (*141*). Samples were quenched in 1 ml 80% methanol/water (v/v) pre-chilled to -80°C and spiked with 50 µl 1 ng/µl d3-propionylcarntine pre-diluted in 80% methanol (v/v). Samples were vortexed for 1 min and chilled to -80°C for 30 min, centrifuged at 18000 g for 10 min at 4°C. The supernatant was transferred to a deep well 96-well plate and evaporated to dryness under nitrogen gas. Samples were reconstituted in 100 µl and 2 µl of the sample was injected from a 4°C autosampler into a ZIC-pHILIC 150 × 2.1 mm 5 µm particle size column (EMD Millipore) with a ZIC-pHILIC 20 × 2.1 guard column in a Vanquish Duo UHPLC System (Thermo Fisher) at 25°C. Chromatography conditions were as follows: buffer A was acetonitrile; buffer B was 20 mM ammonium carbonate, 0.1% (v/v) ammonium hydroxide in water without pH adjustment, with a gradient of 0.5 min at 20% A followed by a linear gradient from 20% to 80% B; 20 to 20.5 min: from 80% to 20% B; 20.5 to 28 min: hold at 20% B at a 0.150 ml/min flow rate. Column elute was introduced to a Q Exactive Plus with a HESI II probe operating in polarity switching mode with full scans from 70 to 1000 *m/z* with an insource fragmentation energy of 1. Instruments were controlled with XCalibur 4.1 and data was analyzed on Tracefinder 5.1 using a 5 ppm window from the predominant negative ion of each isotopologue. Area under the curves for each analyte was normalized to the matched internal standard or the nearest surrogate internal standard. For isotope tracing, isotopologue enrichment was calculated using FluxFix (*140*). Statistical analysis was performed on the values obtained from FluxFix. For plotting purposes, negative values were assigned a value of 0.

Respirometry

Seahorse XF96 respirometry with intact brown adipocytes: BMC cells were differentiated and transfected on day 3 as described above. During the reverse transfection, cells were replated to Seahorse XF96 Cell Culture Microplates (Agilent Technologies) at a density of 20,000 cells/well. Cell culture medium was changed 1 hour before the first measurement to DMEM (Sigma-Aldrich, D5030, supplemented with 143 mM NaCl, pH 7.45) supplemented with 25 mM glucose, 1 mM pyruvate (Gibco, 11360-070), and 4 mM glutamine. Real-time oxygen consumption rate (OCR) was measured under basal conditions and after injections of oligomycin A (1 µM) (Leak) (Cayman Chemicals, 11342), NE (1 µM) (Sigma-Aldrich, A9512) (NE), FCCP (0.5 µM) (Cayman Chemicals, 15218) (MAX), antimycin A/rotenone (1 µM each) (Cayman Chemicals, N/A) (Cayman Chemicals, 13995) (nonmitochondrial oxygen consumption). OCR was assessed using a Seahorse XFe96 Extracellular Flux Analyzer (Agilent Technologies). The presented values have not been normalized. Each dot represents an independent Seahorse experiment. Bar values show the OCR at the final measurement point of each injection phase, except for NE-stimulated respiration, which is quantified as the maximal OCR after NE injection.

Seahorse XF96 respirometry with human brown adipocytes: BMC cells were differentiated and transfected on day 3 as described above. During the reverse transfection, cells were replated to Seahorse XFe96/XF Pro cell culture microplates (Agilent Technologies) at a density of 15000 cells/well. Cell culture medium was changed 1 hour before the first measurement to Seahorse XF DMEM (Agilent Technologies, 103575-100, pH 7.4) supplemented with 5.55 mM glucose. Real-time OCR was measured under basal conditions and after injections of dibutyryl-cAMP (500 µM) (Sigma-Aldrich, D0260), oligomycin (2 µM) (Leak) (Merk-Millipore, 495455), FCCP (1 µM) (Sigma-Aldrich, C2920) (MAX), antimycin A (10 µM) (Sigma-Aldrich, A8674) (nonmitochondrial oxygen consumption). OCR was assessed using a Seahorse XF Pro Extracellular Flux Analyzer (Agilent Technologies). The presented values have not been normalized.

Seahorse XF96 respirometry with permeabilized brown adipocytes: Cells were cultured on Seahorse XF96 Cell Culture Microplates (Agilent Technologies) at a density of 20000 cells/well. Before the experiment, the cells were rinsed with the MiR05 media (Oroboros. Instruments, product 60101-01) once and supplemented with MiR05 media containing the Seahorse XF Plasma Membrane Permeabilizer (Agilent Technologies) in 1 nM concentration, 2 mM ADP (Jena Bioscience: NU-1198) and 1 mM $MgCl_2$, immediately after the measurements were started. The compounds were injected in the following concentrations: malate (2 mM) (Sigma-Aldrich: M1000), pyruvate (5 mM) (Sigma-Aldrich: P2256), glutamate (10 mM), succinate (10 mM) (Sigma-Aldrich: S2378), octanoylcarnitine (0.5 mM) (APExBIO Technology: B6371-50), palmitoylcarnitine (0.04 mM) (Sigma-Aldrich: P4509), glycerophosphate (10mM) (Santa Cruz Biotechnology: sc-215789), oligomycin A (1 µM) (Cayman Chemicals, 11342), FCCP (1 µM) (Cayman Chemicals, 15218), antimycin A/rotenone (1 µM) (Cayman Chemicals, N/A) (Cayman Chemicals, 13995). The presented values represent the raw measurement values and have not been normalized. The quantification for statistical analysis represents the maximal OCR values achieved after the respiratory substrate injection without FCCP.

Resipher respirometry: BMC cells were differentiated and transfected on day 3 as described above. During the reverse transfection, cells were replated to Corning 96-well Clear Flat Bottom Polystyrene TC-treated Microplates (CORNING, 3599) at a density of 60,000 cells/well. Total medium volume per well was 200 µl. Cells were allowed to attach to the plate for 2 hours. Afterward, oxygen concentration and oxygen flux were continuously measured using Resipher (Lucid Scientific) at 37°C, 10% CO_2 for 4 days throughout the course of siRNA-mediated *Slc25a34* depletion. Culture medium was changed once on day 5 of differentiation according to the differentiation protocol (~48 hours after transfection). Resipher oxygen sensing lid contains micro probes with optical oxygen sensors that scan between 0.55 to 0.95 mm above the cells to measure the oxygen concentration gradient within the medium column. For the oxygen flux measurements with 4-hour resolution are shown. For relative flux quantification, 12-hour average fluxes from day 1 (24 to 36 hours), day 2 (55 to 67 hours), day 3 (76 to 88 hours), and day 4 (96 to 108 hours) were normalized to control group fluxes measured at 12 to 24 hours across four independent experiments.

Molecular docking

The hydrated and flexible docking simulations were performed in Python using Autodock Vina (*142*, *143*). AlphaFold PDB file containing the receptor coordinates for Slc25a34 (AF-Q6PIV7-F1-v4) was first modified with the reduce script (*144*) ($ reduce -BUILD receptor.pdb), adding all missing hydrogen atoms. PDBQT file and grid parameters

of the receptor were then generated using the python package Meeko ($ mk_prepare_receptor.py) with the following grid coordinates (x, y, z; box size), 0 0 0; 30 30 30 (*145*). E28, K31, K128, R176, D228, and R277 were specified to be flexible residues. Affinity maps and water maps were finally computed with autogrid4 and the python Vina package mapwater.py, respectively. Ligand coordinates were obtained through PubChem (sdf files, 3D structures) and hydrogen atoms were first added with the python package Molscrub ($ scrub.py -I ligand.sdf -0 ligandH.sdf). The meeko script ($ mk_prepare_ligand.py -i ligandH.sdf -o ligand.pdbqt -w) was then used to add explicit water molecules to the ligands and generate the PDBQT files (*145*). The Docking affinities were calculated using Autodock Vina with Autodock 4 forcefield and the exhaustiveness set at 32, following Vina hydrated docking protocol (https://autodock-vina.readthedocs.io/). The computed docking was visualized in the software PyMOL (Schrödinger), showing residues implicated in the matrix salt bridges of the transporter, as well as interacting with the ligands.

Quantification and statistical analysis

All statistical tests were performed using GraphPad Prism software. Data are presented as means ± SEM. All statistical tests are stated in the figure legends. For the two-way ANOVA tests, corrections for multiple comparisons were made by controlling the FDR using the two-stage step-up method of Benjamini, Krieger, and Yekutieli unless otherwise indicated in the figure legends.

Cell studies were performed at least three times independently. unless otherwise noted. Cell studies for steady-state metabolomic analyses shown in Fig. 4, A to C and D to N, were performed once.

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E-MTAB-17580; the accession number for data presented in fig. S2, E to G (circadian time course, BAT from *Rev-erbα/β* BKO mice) is GEO no. GSE331054; the data presented in Fig. 2, H to J, and fig. S3, D to F (BAT from *Pparα* AKO mice) are available on Zenodo (*117*); the accession number for data presented in Fig. 2P (BAT from mice fed with ketogenic diet) is GEO no. GSE348259; the data presented in Fig. 6, C, D, F, and G, and fig. S6, E and I (siRNA-mediated *Slc25a34* KD in brown adipocytes) are available at ArrayExpress under no. E-MTAB-17577. **License information:** Copyright © 2026 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

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RESEARCH ARTICLE SUMMARY

ARTHRITIS

Hypertension fuels osteoarthritis through neuroendocrine signaling

Su-Jin Kim†, Hyo-Eun Kim†, Ki-Ho Chung, Gyuseok Lee, Ka Hyon Park, Young-Gwon Kim, Mangeun Kim, Weon-Young Choi, Soo-Chang Joo, Hae Jin Kee, Seung-Hee Kwon, Jong-Keun Seon, Gun-Woo Lee, Yoonkyung Won, Jeong-Tae Koh, Jin Han, Jang-Soo Chun*, Je-Hwang Ryu*

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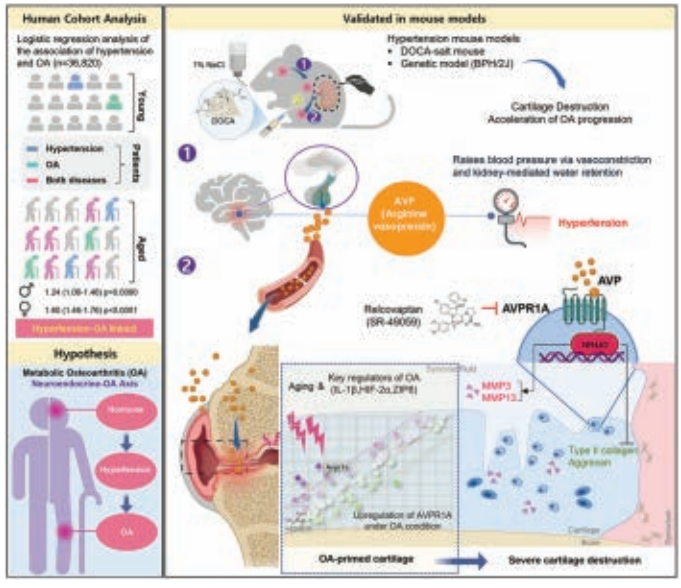
INTRODUCTION: Osteoarthritis (OA), the leading cause of disability among older adults, has long been considered a disease driven primarily by mechanical wear and tear. However, accumulating evidence indicates that OA is also influenced by systemic metabolic, endocrine, and vascular factors. Hypertension, which is highly prevalent in aging populations, is associated with an increased OA risk and greater severity. However, the mechanisms linking elevated blood pressure to impaired cartilage homeostasis remain undefined.

RATIONALE: Transcriptomic analyses in mice identified *Avpr1a*, which encodes the receptor for the neuroendocrine hormone arginine vasopressin (AVP), as one of the most strongly up-regulated genes in OA chondrocytes. Further screening revealed that *Avpr1a* is the only AVP receptor gene that is up-regulated in articular cartilage. Because the AVP-AVPR1A axis plays a central role in blood pressure regulation, its dysregulation within cartilage suggests a direct molecular link between hypertension and OA. Given that circulating AVP concentration increases under hypertensive conditions, we hypothesized that hypertension-induced elevations in AVP act on OA-primed chondrocytes—sensitized through mechanical injury or aging—that up-regulate AVPR1A, thereby activating catabolic signaling pathways.

RESULTS: We combined human cohort analyses with two hypertensive mouse models to define the mechanistic relationship between hypertension and OA. Logistic regression of population-scale data demonstrated that hypertension was independently associated with OA prevalence, with progressively stronger associations observed in advanced structural disease. In both mouse models, hypertension markedly exacerbated experimental OA, increasing cartilage erosion and subchondral bone plate thickening without affecting synovitis or osteophyte maturity. We further demonstrated that AVP is not synthesized locally within joint tissues but instead reaches the synovial fluid through the systemic circulation. AVPR1A expression was increased in OA cartilage but remained low in healthy, young cartilage. Both mechanical wear and tear and natural aging acted as priming factors by up-regulating basal AVPR1A expression. Consequently, this priming, rather than hypertension alone, was required to activate the AVP-AVPR1A signaling axis. Activation of this pathway promoted catabolic signaling while suppressing anabolic signaling without inducing cellular senescence. Cartilage destruction required concurrent elevation of both AVP and AVPR1A. Genetic deletion or local intra-articular pharmacological antagonism of AVPR1A prevented hypertension-induced cartilage degeneration, demonstrating that hypertension-driven OA progression depends specifically on AVP-AVPR1A signaling. Finally, transcriptomic and functional analyses identified NR4A3 as the principal transcriptional effector linking hypertensive endocrine signaling to chondrocyte catabolism, independent of the renin-angiotensin-aldosterone system or oxytocin receptors.

CONCLUSION: These findings establish the AVP-AVPR1A axis as an endocrine-skeletal signaling pathway through which systemic hypertension accelerates OA progression. Hypertension elevates circulating AVP, OA primes chondrocytes to express AVPR1A, and their combined activation drives cartilage matrix breakdown and subchondral bone remodeling. Targeting AVPR1A or its downstream mediators offers a therapeutic strategy to slow OA progression in hypertensive individuals experiencing joint injury or advanced age. □



Hypertension accelerates OA through AVP-AVPR1A signaling. Elevated circulating AVP in hypertension exacerbates cartilage degeneration in mechanically injured or aged joints. Under OA conditions, chondrocytes selectively up-regulate AVP receptor AVPR1A, rendering articular cartilage highly responsive to systemic AVP. Activation of this signaling axis drives downstream NR4A3 transcription, increasing catabolic enzyme expression while suppressing anabolic cartilage matrix synthesis to promote disease progression. DOCA, deoxycorticosterone acetate; HIF-2α, hypoxia-inducible factor 2α; IL-1β, interleukin-1β; ZIP8, zinc importer ZIP8. [Figure created with BioRender.com]

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ARTHRITIS

Hypertension fuels osteoarthritis through neuroendocrine signaling

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Osteoarthritis (OA) is a degenerative joint disease, the progression of which is accelerated by systemic metabolic stress. Here, we identified a neuroendocrine pathway through which hypertension accelerates OA pathogenesis in joints predisposed by mechanical injury or aging. Nationwide cohort analyses and hypertensive mouse models demonstrated that elevated blood pressure exacerbates OA pathogenesis. OA-primed chondrocytes up-regulated arginine vasopressin (AVP) receptor 1A (AVPR1A), rendering them responsive to hypertension-associated circulating AVP. AVP-AVPR1A signaling enhanced catabolic signaling while suppressing anabolic regulators, thereby promoting cartilage catabolism. Genetic deletion or pharmacological inhibition of AVPR1A protected against hypertension-driven OA acceleration. Transcriptomic profiling and gene-silencing analyses identified NR4A3 as the principal downstream transcriptional mediator. Collectively, these findings establish the AVP-AVPR1A axis as a mechanistic link between systemic hypertensive stress and joint degeneration.

Osteoarthritis (OA) is the most prevalent chronic joint disorder and a leading cause of disability among older adults (typically ≥60 years old). It is characterized by the progressive and irreversible deterioration of articular cartilage, accompanied by subchondral bone remodeling, synovial inflammation, and chronic pain that limits daily activity (*1*). Despite its high prevalence, no existing therapy can halt or reverse disease progression; clinical management remains largely focused on symptom relief. The socioeconomic burden of OA continues to rise, with 595 million individuals affected globally in 2020, corresponding to 7.6% of the global population and a 132.2% increase in total cases since 1990 (*2*). These challenges highlight the urgent need to elucidate the biological mechanisms that initiate and exacerbate cartilage degeneration, with the ultimate goal of identifying new disease-modifying therapeutic targets.

OA has traditionally been considered a degenerative disorder arising from mechanical wear and aging. However, accumulating evidence now supports OA as a multifactorial disease driven in part by systemic metabolic disturbances rather than mechanical stress alone. Clinical and experimental studies have shown that OA shares key pathological features with metabolic syndrome (MetS), in which obesity, insulin resistance, dyslipidemia, and hypertension collectively promote low-grade inflammation, oxidative stress, and aberrant subchondral bone remodeling that disrupt joint homeostasis (*3*–*5*). Adipokines and pro-inflammatory cytokines released from visceral adipose tissue accelerate chondrocyte catabolism and matrix degradation (*6*, *7*), whereas hyperglycemia and lipid peroxidation impair extracellular matrix turnover, thereby linking systemic metabolic dysregulation to local cartilage degeneration (*8*, *9*). Consistent with this concept, our previous work demonstrated that dysregulated cholesterol metabolism in chondrocytes contributes to OA pathogenesis (*10*).

Among the components of MetS, hypertension has emerged as a particularly consistent and independent risk factor for OA. Epidemiological cohorts, including the Framingham and Rotterdam studies, have shown that individuals with hypertension have a higher incidence of knee and hand OA, even after adjustment for age, body-mass index, and physical activity (*11*–*14*). A meta-analysis further reported that hypertensive individuals have nearly twice the risk of developing radiographic OA and an ~1.5-fold higher risk of symptomatic OA compared with normotensive controls (*15*). Longitudinal analyses have also demonstrated that elevated blood pressure predicts accelerated joint space narrowing and subchondral bone sclerosis, particularly among postmenopausal women (*16*, *17*).

Mechanistically, hypertension promotes vascular remodeling and microvascular rarefaction within the subchondral bone, resulting in local ischemia, nutrient deprivation, and subsequent cartilage degeneration (*18*, *19*). Endothelial dysfunction and oxidative stress further impair subchondral perfusion, establishing a proinflammatory microenvironment within joint tissues (*20*). Hypertensive states also increase circulating concentrations of vasoactive hormones such as angiotensin II and arginine vasopressin (AVP), both of which can influence chondrocyte metabolism and extracellular matrix turnover through receptor-mediated pathways (*21*–*23*). Despite these epidemiological and mechanistic links between hypertension and OA, the molecular mediators that convert elevated blood pressure into disrupted cartilage homeostasis remain poorly defined, highlighting the need to identify the biochemical axis that connects cardiovascular stress with skeletal tissue degeneration.

Here, we identified an endocrine-skeletal signaling pathway that explains how hypertension accelerates OA pathogenesis. We demonstrate that elevated blood pressure promotes OA development through activation of the AVP-AVPR1A axis. Under hypertensive conditions, increased circulating AVP stimulates up-regulation of its receptor AVPR1A in chondrocytes, which is induced by mechanical injury or natural aging, thereby triggering the expression of matrix-degrading enzymes that erode the extracellular cartilage matrix. Genetic ablation of *Avpr1a* or local pharmacological inhibition of its activity within the joint markedly attenuated hypertension-driven acceleration of cartilage destruction and disease severity. These findings establish the AVP-AVPR1A axis as a molecular link between hypertension and OA, revealing a mechanistic pathway through which systemic hypertensive stress acts as a neuroendocrine amplifier of joint degeneration.

Results

Epidemiological and experimental evidence linking hypertension to OA pathogenesis

We first examined the association between hypertension and OA using population-based data from the Korea National Health and Nutrition Examination Survey (KNHANES; 2007–2021), an ongoing, nationally representative cross-sectional survey conducted by the Korea Disease Control and Prevention Agency. To evaluate age-dependent patterns in middle-aged and older adults, participants aged 45 years or older (n = 36,820) were stratified into 10-year age categories. Logistic regression models were adjusted for potential confounders, including sex, education level, household income, smoking status, alcohol consumption, diabetes, and body weight. In this initial population-based

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analysis, OA was defined as self-reported physician-diagnosed OA. Our analysis of the KNHANES datasets revealed an association between hypertension and OA, with the prevalence of hypertension among individuals with OA peaking in the 65- to 74-year age group and declining thereafter in participants aged ≥75 years (Fig. 1A and fig. S1A). This association remained in both males and females when the entire cohort of participants aged 45 years or older was analyzed (Fig. 1B and fig. S1B). To further investigate the association between hypertension and radiographic OA, we analyzed a subset of the KNHANES database (2009–2013, $n = 13{,}717$ participants) consisting of participants aged 50 years or older who underwent knee radiography with Kellgren-Lawrence (KL) grading. In this subset, radiographic OA was defined as a KL grade ≥2. Hypertension was associated with radiographic knee OA severity, with the strength of the association increasing progressively across advancing KL grades and reaching its highest level in participants with KL grade 4 disease (fig. S2, A to C). Collectively, these findings provide epidemiological evidence linking hypertension to OA pathogenesis.

To experimentally validate the relationship between hypertension and OA, we used two hypertensive mouse models (Fig. 1, C and D): the nephrectomy and deoxycorticosterone acetate (DOCA)–salt–induced hypertension model (*24*) and the genetically hypertensive BPH/2J mouse strain (*25*). OA was induced in both models by destabilization of the medial meniscus (DMM) (*26*). Compared with normotensive controls, DOCA-salt–treated hypertensive mice exhibited elevated systolic blood pressure under both sham-operated and DMM conditions (Fig. 1E). Under DMM conditions, hypertension exacerbated cartilage erosion and subchondral bone plate thickening, a hallmark of OA (*27*), whereas synovitis and osteophyte maturity were not affected (Fig. 1, F and G). These degenerative changes progressed further between 6 and 8 weeks after surgery (fig. S3, A to C). Similarly, genetically hypertensive BPH/2J mice exhibited higher systolic blood pressure than normotensive BPN/3J mice under both sham- and DMM-operated conditions (Fig. 1H). Under DMM conditions, hypertensive BPH/2J mice also displayed more severe cartilage erosion and greater subchondral bone plate thickening than normotensive BPN/3J mice, whereas synovitis and osteophyte maturity remained unchanged (Fig. 1, I and J). Collectively, these findings provide complementary epidemiological and experimental evidence that hypertension accelerates OA progression.

Identification of AVPR1A as a molecular linker between hypertension and OA

To identify molecular mediators linking hypertension and OA, we analyzed microarray (Fig. 2, A and B) and RNA sequencing (RNA-seq) (Fig. 2, C and D) datasets obtained from primary mouse chondrocytes. To mimic the OA phenotype, primary chondrocytes were stimulated with interleukin-1β (IL-1β) or infected with adenoviruses overexpressing OA-associated catabolic factors, including hypoxia-inducible factor-2α (Ad-HIF-2α) or the zinc importer ZIP8 (Ad-ZIP8) (*10*, *28*, *29*). Analysis of the microarray datasets, focusing on hypertension-related genes, identified *Avpr1a*, which encodes AVPR1A, as one of the most

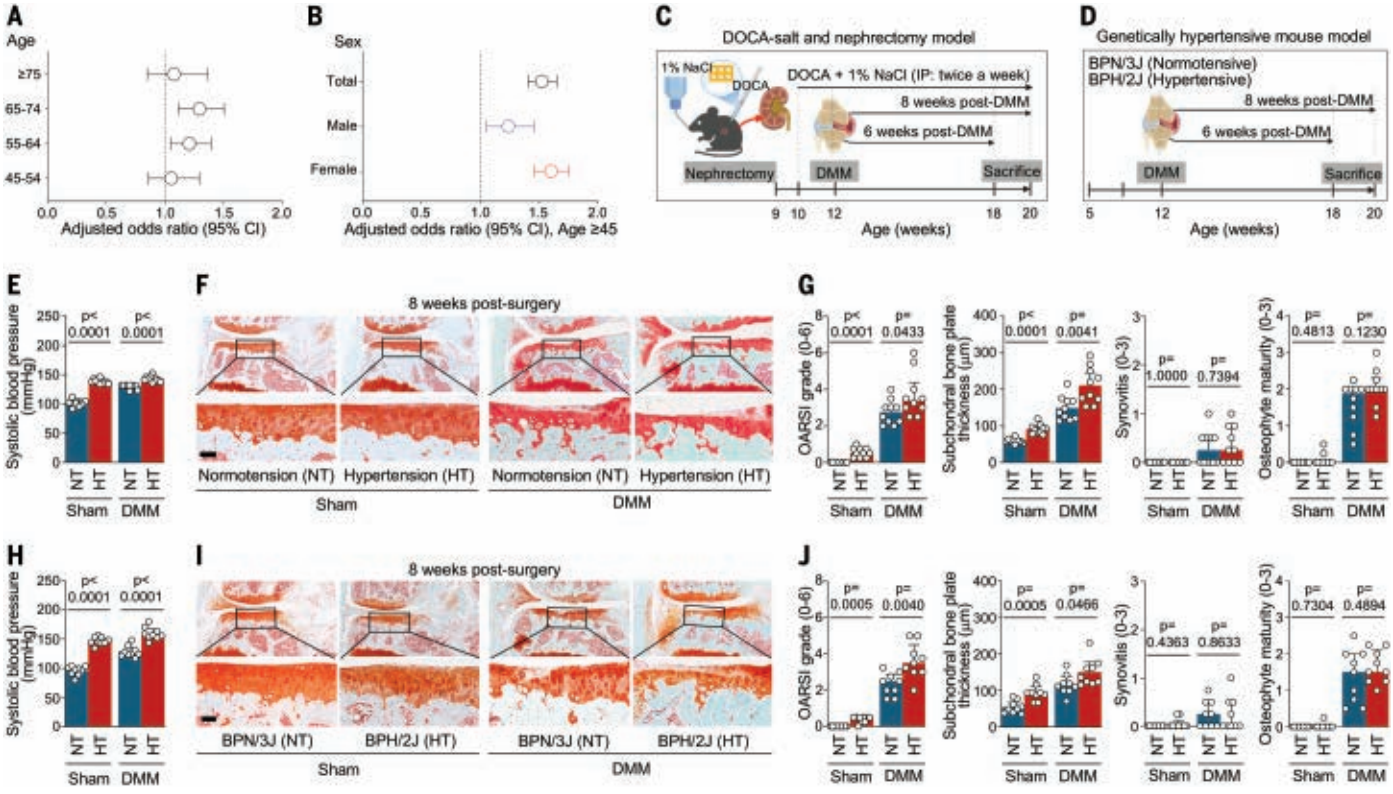

Fig. 1. Hypertension is associated with accelerated OA pathogenesis. (**A** and **B**) Logistic regression analysis of the association between hypertension and OA using KNHANES data grouped by age (A) and sex (B) (2007–2021; $n = 36{,}820$ participants aged ≥45 years). (**C** and **D**) Experimental schematics for DOCA-salt–induced (C) and genetically hypertensive BPH/2J (D) mouse models. IP, intraperitoneally. (**E** to **G**) Normotensive (NT) and DOCA-salt–induced hypertensive (HT) mice subjected to sham or DMM surgery ($n = 10$ mice per group). (**H** to **J**) Genetically hypertensive BPH/2J (HT) and normotensive BPN/3J (NT) mice subjected to sham or DMM surgery ($n = 9$ mice per group). Shown are systolic blood pressure [(E) and (H)], representative images of Safranin O–stained knee joint sections [(F) and (I)], and quantification of OARSI score, subchondral bone plate thickness, synovitis score, and osteophyte maturity [(G) and (J)]. OARSI grade, synovitis score, and osteophyte maturity are shown as median (IQR) and were compared by two-tailed Mann-Whitney *U* tests [(G) and (J)]. Systolic blood pressure and subchondral bone plate thickness are shown as mean with 95% CI and were compared by two-tailed Student's *t* tests [(G), (H), and (J)] or a Mann-Whitney *U* test for non-normally distributed data (E). Scale bars in (F) and (I) are 50 μm.

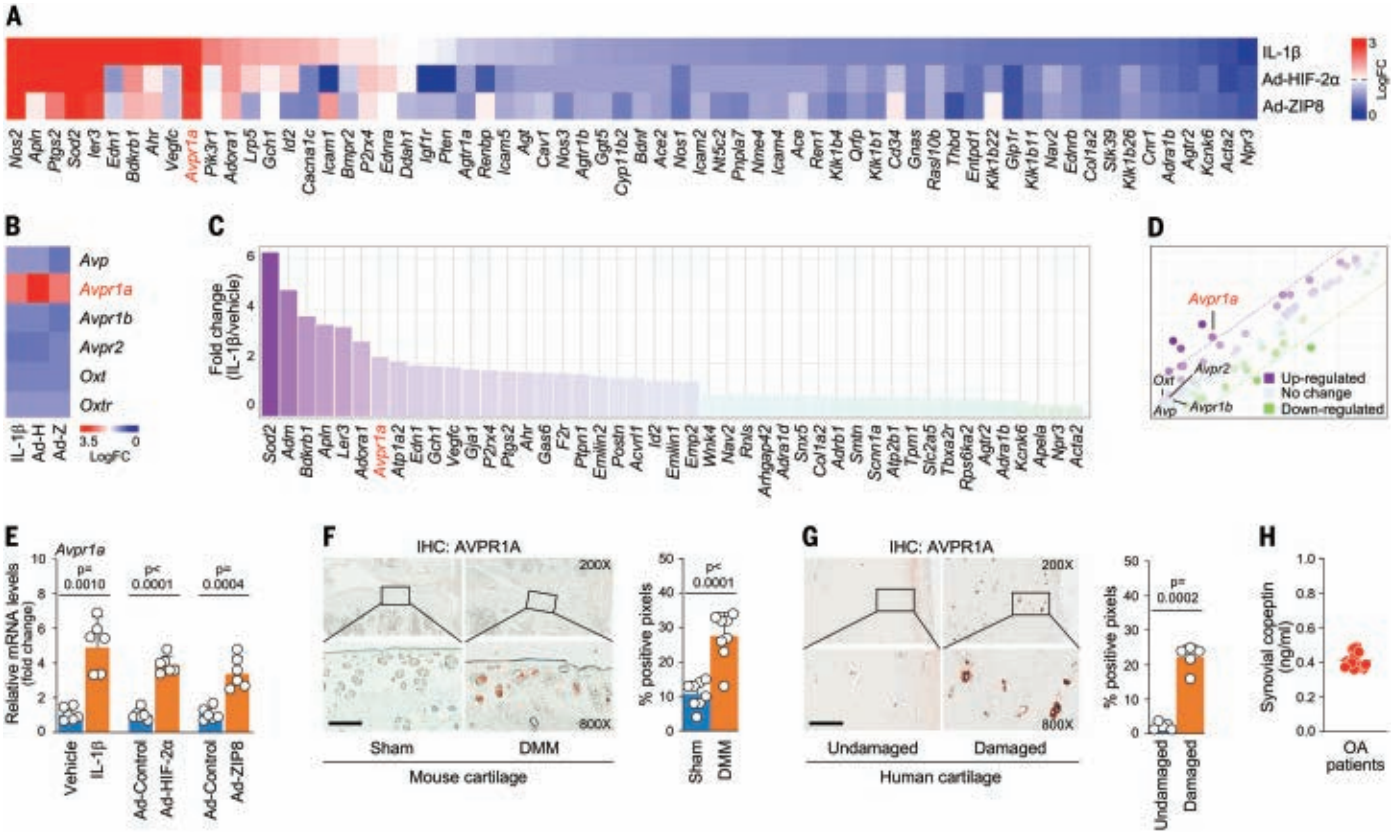

Fig. 2. AVPR1A links hypertension to OA. (**A** and **B**) Heatmaps of microarray analysis showing hypertension-related genes (A) and vasopressin or oxytocin signaling components (B) in primary mouse chondrocytes treated with IL-1β (1 ng/ml, 36 hours) or infected with Ad-HIF-2α or Ad-ZIP8 (800 MOI, 24 hours) ($n = 4$ biological replicates). FC, fold change. (**C** and **D**) RNA-seq analysis of IL-1β–treated primary mouse chondrocytes, presented as a bar plot (C) and scatter plot (D). (**E**) *Avpr1a* mRNA expression in primary mouse chondrocytes treated with IL-1β or infected with Ad-Control, Ad-HIF-2α, or Ad-ZIP8 ($n = 6$ biological replicates). (**F** and **G**) Representative immunohistochemistry (IHC) images and semiquantitative analysis of AVPR1A expression in articular cartilage from sham- and DMM-operated mice ($n = 8$ mice per group) (F) and paired damaged and adjacent undamaged human OA cartilage ($n = 5$ patients) (G). (**H**) Synovial fluid copeptin concentrations in patients with radiographic OA (KL grades 3 and 4; $n = 16$ patients). Data are presented as geometric mean with geometric SD factor and were analyzed by a two-tailed paired *t* test (E). Data are presented as mean ± SD and were analyzed by a two-tailed Student's *t* test (F) or a two-tailed paired *t* test (G). Scale bars in (F) and (G) are 50 μm.

robustly up-regulated transcripts in OA-like chondrocytes (Fig. 2A). By contrast, the expression of other genes associated with AVPR1A signaling, including *Avp*, *Avpr1b*, *Avpr2*, *Oxt* (oxytocin), and *Oxtr* (oxytocin receptor), remained unchanged (Fig. 2B). Consistent with these findings, RNA-seq analysis of IL-1β–treated chondrocytes also revealed up-regulation of *Avpr1a* (Fig. 2, C and D). The increased expression of *Avpr1a* in OA-like chondrocytes was further validated by quantitative reverse transcription PCR (qRT-PCR) (Fig. 2E). Up-regulation of AVPR1A in chondrocytes was also observed in the articular cartilage of DMM-operated mice by immunohistochemical staining. Whereas cartilage from sham-operated mice exhibited minimal AVPR1A staining, strong immunoreactivity was detected in DMM-operated joints (Fig. 2F). The specific up-regulation of *Avpr1a* in DMM cartilage was further confirmed by qRT-PCR using laser-microdissected mouse articular cartilage (fig. S4, A and B). The clinical relevance of AVPR1A up-regulation was further supported by the analysis of human OA cartilage. Both immunohistochemical and qRT-PCR analyses demonstrated higher AVPR1A protein and mRNA expression in damaged regions than in paired undamaged areas (Fig. 2G and fig. S4, C and D).

AVP must be present within the joint microenvironment to activate AVPR1A signaling. Indeed, substantial amounts of AVP, quantified using its stable surrogate marker copeptin (*30*), were detected in the synovial fluid of patients with advanced OA (Fig. 2H and table S1). AVP is synthesized in the hypothalamic neurons and released into the systemic circulation through the posterior pituitary. Consistent with this established biology, qRT-PCR analysis demonstrated that *Avp* mRNA was undetectable in joint-resident cells, including chondrocytes and fibroblast-like synoviocytes (FLSs), under both basal and IL-1β–stimulated conditions (fig. S5A). Moreover, AVP protein was not detected in joint tissues, including articular cartilage and synovium, from DMM-operated mice (fig. S5B). These findings suggest that AVP present in the synovial fluid is derived from the systemic circulation rather than produced locally within the joint. Together, the OA-associated up-regulation of AVPR1A in joint tissues, together with the presence of circulating AVP within the joint microenvironment, provides a molecular basis linking hypertension to OA pathogenesis.

The AVP-AVPR1A axis mediates hypertension-accelerated OA pathogenesis

Next, we investigated the biological role of the AVP-AVPR1A axis in hypertension-accelerated OA progression. To independently evaluate the effects of AVP and AVPR1A, mice received either continuous AVP infusion via osmotic minipumps (fig. S6, A and B) or intra-articular (IA) injection of an adenovirus expressing AVPR1A (Ad-AVPR1A) to induce AVPR1A overexpression in the joint (fig. S6B). Neither 4 weeks of AVP infusion nor IA-mediated AVPR1A overexpression alone, or in combination, induced OA-like morphological changes in sham-operated joints (fig. S6C). Furthermore, prolonged AVP infusion for 12 weeks

also failed to induce OA-like joint pathology (fig. S6, D to F). By contrast, when combined with DMM surgery, AVP infusion increased systolic blood pressure (Fig. 3A) and exacerbated DMM-induced cartilage destruction and subchondral bone plate thickening (Fig. 3, B and C), whereas synovitis and osteophyte maturity remained unchanged (fig. S6G). These findings indicate that AVP-AVPR1A signaling is insufficient to initiate severe, fully fledged joint degeneration in the absence of OA-inducing stimuli and instead functions as a potentiator, rather than an initiator, of OA progression. To evaluate this pathway under hypertensive conditions, AVPR1A was overexpressed in the joints of DOCA-salt–induced hypertensive mice. IA overexpression of AVPR1A did not alter systolic blood pressure (Fig. 3D) but markedly exacerbated cartilage degeneration and subchondral bone plate thickening (Fig. 3, E and F). Moreover, combined AVP infusion and IA-mediated AVPR1A overexpression markedly accelerated OA progression in DMM-operated mice, whereas sham-operated joints remained unaffected (Fig. 3, G to I). The lack of pathology in sham-operated joints further reinforces that mechanical

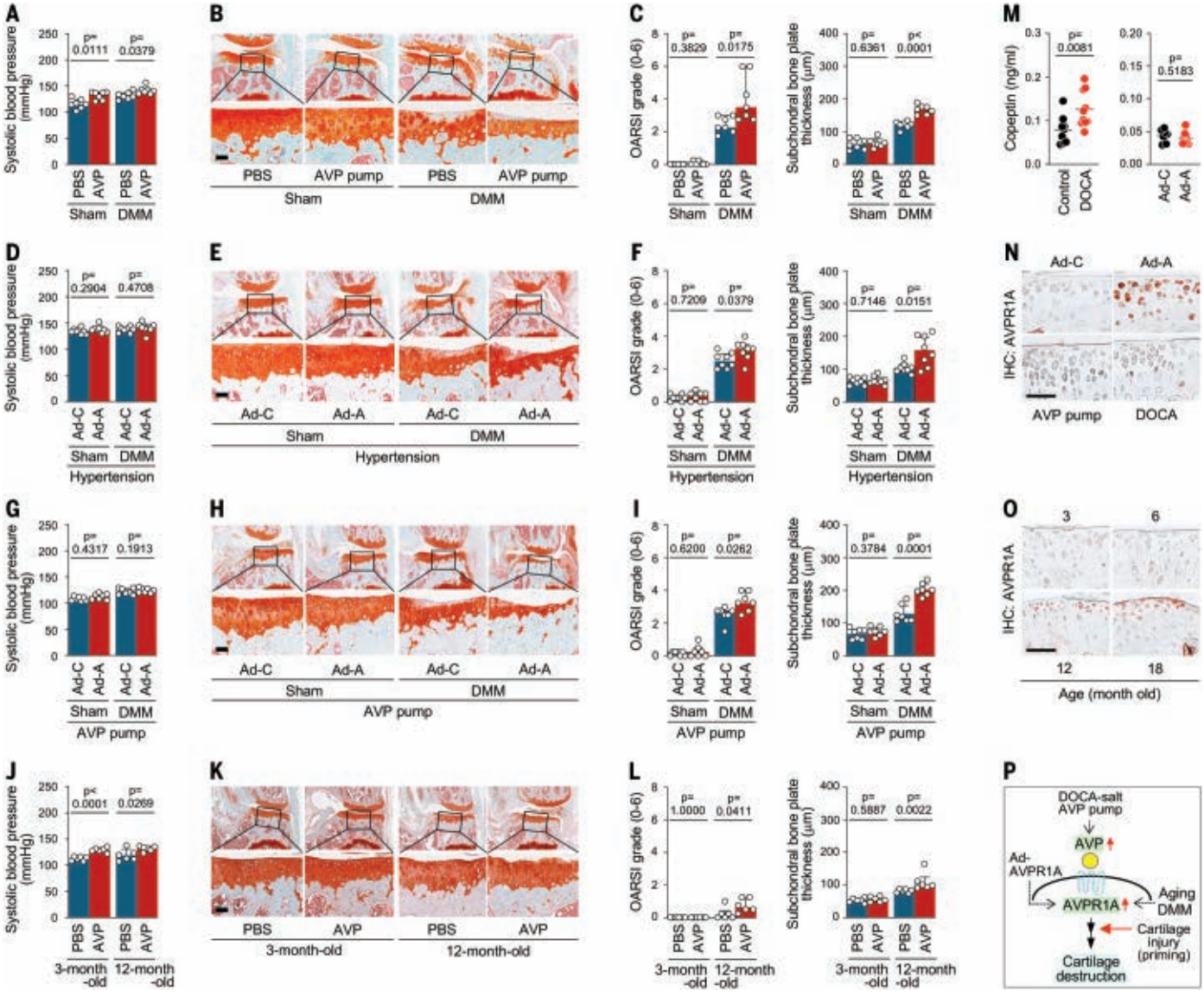

Fig. 3. The AVP-AVPR1A axis is required for hypertension-induced acceleration of OA. (**A** to **C**) Effects of systemic AVP infusion (4 weeks) in sham- or DMM-operated mice ($n = 7$ mice per group). Systolic blood pressure (A), Safranin O staining (B), and quantification of OARSI score and subchondral bone thickness (C) are shown. (**D** to **F**) Effects of AVPR1A overexpression in DOCA-salt hypertensive mice receiving IA injections of Ad-Control (Ad-C) or Ad-AVPR1A (Ad-A) ($n = 8$ mice per group). Systolic blood pressure (D), Safranin O staining (E), and quantification of OARSI score and subchondral bone thickness (F) are shown. (**G** to **I**) Combined effects of AVP infusion and IA injection of Ad-C or Ad-A in sham- or DMM-operated mice ($n = 7$ mice per group). Systolic blood pressure (G), Safranin O staining (H), and quantification of OARSI score and subchondral bone thickness (I) are shown. (**J** to **L**) Effects of AVP infusion in young (3-month-old) and aged (12-month-old) mice ($n = 6$ mice per group). Systolic blood pressure (J), Safranin O staining (K), and quantification of OARSI score and subchondral bone thickness (L) are shown. (**M**) Serum copeptin levels in DOCA-salt ($n = 10$ mice per group) and IA injection of Ad-C or Ad-A mice ($n = 8$ mice per group). (**N** and **O**) AVPR1A immunostaining in cartilage (N) and in aging mice (3 to 18 months; $n = 5$ mice per group) (O). (**P**) Proposed model. Data for OARSI grades are presented as median (IQR) and were compared using two-tailed Mann-Whitney *U* tests [(C), (F), (I), and (L)]. Systolic blood pressure and subchondral bone thickness are presented as mean with 95% CI (normally distributed) or median (IQR) (non-normally distributed) and were compared using two-tailed Student's *t* tests [(C), (D), (F), (G), (I), and (J)] or Mann-Whitney *U* tests [(A) and (L)]. Copeptin levels (M) are presented as mean ± SD and were compared using a two-tailed Student's *t* test. Scale bars in (B), (E), (H), (K), (N) and (O) are 50 μm.

or inflammatory priming is required for AVP-AVPR1A signaling to drive OA progression.

Given that OA is predominantly an age-related joint disease, we next investigated whether natural aging could serve as a physiological priming event analogous to injury-induced priming. Systemic AVP infusion increased systolic blood pressure in both 3- and 12-month-old mice (Fig. 3J). However, AVP infusion did not induce morphological changes in the joints of young 3-month-old mice, whereas it caused cartilage erosion and subchondral bone plate thickening in aged 12-month-old mice (Fig. 3, K and L). These findings indicate that natural aging, similar to DMM-induced joint injury, establishes a permissive state that enables pathological responses to AVP. To further define this mechanism, we examined the systemic and local components of the AVP-AVPR1A axis. Whereas circulating copeptin concentrations were elevated in DOCA-salt–induced hypertensive mice (Fig. 3M), AVPR1A expression was locally up-regulated in injured joints and joints receiving Ad-AVPR1A injection (Fig. 3N), demonstrating a compartmentalized signaling axis in which the ligand is supplied systemically while the receptor is induced locally. Notably, this local induction of AVPR1A was not restricted to mechanically injured joints. Immunohistochemical analysis of articular cartilage from 3-, 6-, 12-, and 18-month-old mice revealed a progressive age-dependent increase in AVPR1A expression under basal conditions (Fig. 3O and fig. S7, A and B). Consistently, 18-month-old genetically hypertensive BPH/2J mice exhibited more severe spontaneous cartilage damage than normotensive BPN/3J controls (fig. S8, A and B). Collectively, these findings demonstrate that the coexistence of systemic AVP and elevated local AVPR1A expression, established either by mechanical joint injury or by physiological aging, is critical for susceptibility to accelerated OA progression (Fig. 3P).

Genetic deletion or pharmacologic inhibition of AVPR1A mitigates hypertension-accelerated OA progression

To determine whether AVPR1A is required for OA pathogenesis, we used mice with genetic deletion (knockout) of *Avpr1a* (*Avpr1a*$^{-/-}$), which exhibit normal skeletal development but do not express *Avpr1a* (Fig. 4A and fig. S9, A and B). Immunohistochemical analysis confirmed the absence of AVPR1A protein in the articular cartilage of *Avpr1a*$^{-/-}$ mice (fig. S9C). At 6 weeks after DMM surgery, protective effects of *Avpr1a* deletion were not detectable (fig. S9, D and E). However, at 8 weeks, *Avpr1a*$^{-/-}$ mice exhibited markedly reduced cartilage erosion and subchondral bone plate thickening compared with wild-type (WT) mice (Fig. 4, B and C), suggesting that the absence of AVPR1A suppresses

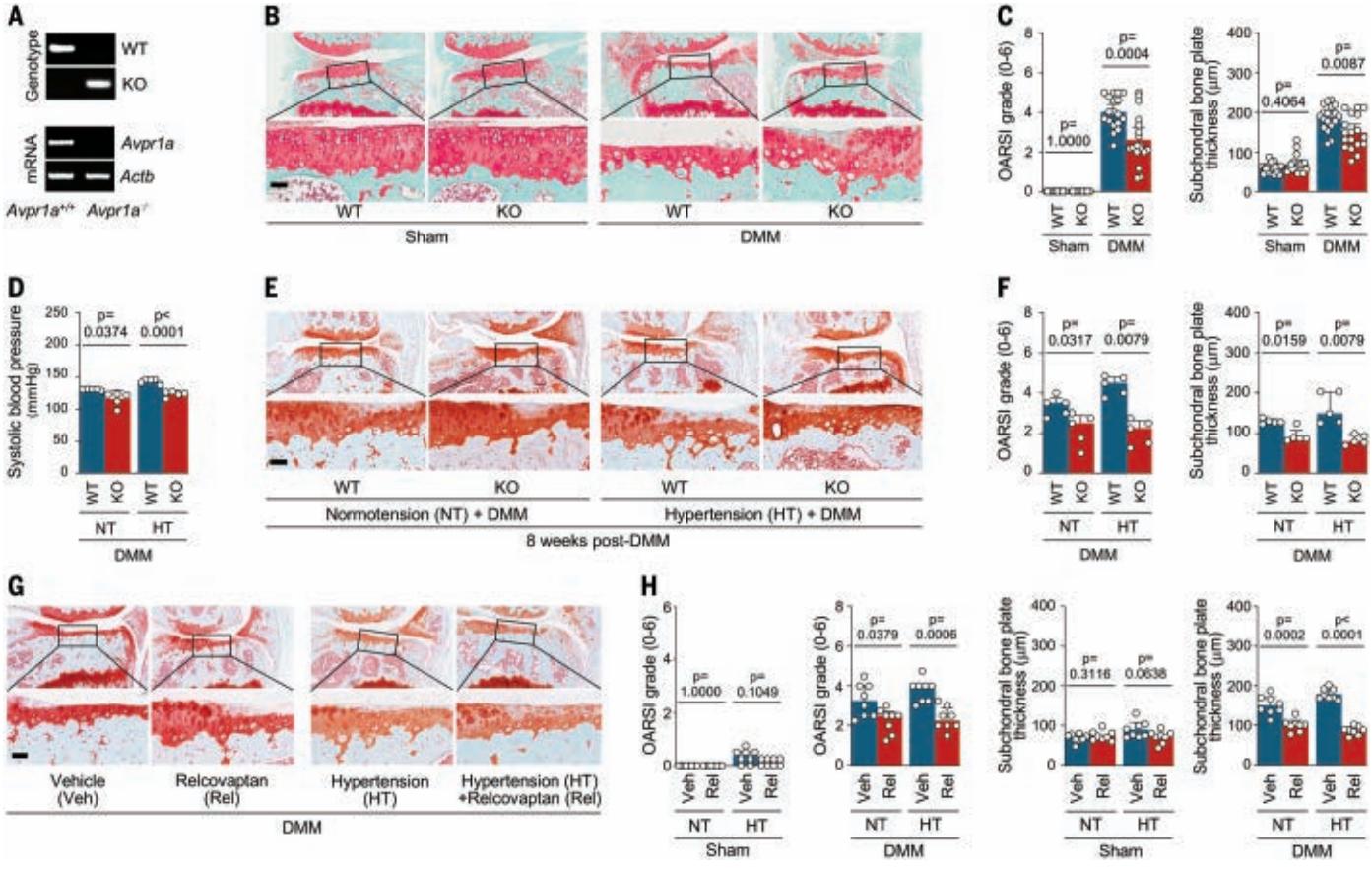

Fig. 4. Genetic deletion or pharmacological inhibition of AVPR1A mitigates OA development. (**A** to **C**) Genotyping (PCR) and *Avpr1a* mRNA expression (RT-PCR) in primary chondrocytes isolated from WT and *Avpr1a*$^{-/-}$ knockout (KO) mice (A). Shown are representative Safranin O–stained joint sections (B) and quantification of OARSI score and subchondral bone plate thickness (C) in sham- and DMM-operated WT and *Avpr1a*$^{-/-}$ mice 8 weeks after surgery ($n = 18$ mice per group). (**D** to **F**) Effects of hypertension in *Avpr1a*-deficient mice. Normotensive and hypertensive WT and *Avpr1a*$^{-/-}$ mice were subjected to sham or DMM surgery. Systolic blood pressure (D), representative Safranin O–stained joint sections (E), and quantification (F) ($n = 5$ mice per group) are shown. (**G** and **H**) Pharmacological inhibition of AVPR1A. Normotensive and hypertensive WT mice received weekly IA injections of vehicle or relcovaptan (10 mM, 10 μl per joint). Representative Safranin O–stained joint sections (G) and quantification (H) in sham- and DMM-operated mice ($n = 8$ mice per group) are shown. OARSI grades are presented as median (IQR) and were compared using two-tailed Mann-Whitney *U* tests [(C), (F), and (H)]. Systolic blood pressure is presented as mean with 95% CI and was compared using a two-tailed Student's *t* test (D). Subchondral bone plate thickness data are presented as mean with 95% CI (normally distributed) or median (IQR) (non-normally distributed) and were compared using two-tailed Student's *t* tests [(C) and (H)] or a Mann-Whitney *U* test (F). Scale bars in (B), (E), and (G) are 50 μm.

progressive joint destruction rather than preventing early damage. Systolic blood pressure in DMM-operated *Avpr1a*$^{-/-}$ mice was lower than that in WT mice, with a greater reduction observed under DOCA-salt–induced hypertensive conditions than under normotensive conditions (Fig. 4D). Moreover, the inhibitory effects of *Avpr1a* deletion on cartilage destruction and subchondral bone plate thickening under normotensive conditions were even more pronounced when hypertension was present (Fig. 4, E and F). Similar reductions in systolic blood pressure and preservation of joint homeostasis were also observed in sham-operated *Avpr1a*$^{-/-}$ mice (fig. S9, F and G).

Because whole-body deletion of *Avpr1a* may affect systemic physiology, we next investigated the local role of AVPR1A within the joint using relcovaptan, a selective AVPR1A antagonist (*31*). IA administration of relcovaptan attenuated DMM-induced cartilage destruction and subchondral bone plate thickening, with more pronounced protective effects under hypertensive conditions (Fig. 4, G and H). Consistent with these findings, immunohistochemical analysis revealed that AVPR1A was expressed predominantly in articular cartilage, with only minimal expression in the synovium of DMM-operated joints (fig. S10, A and B). These results suggest that the protective effects of IA-administered relcovaptan are mediated primarily through inhibition of AVPR1A signaling in articular cartilage. Relcovaptan also attenuated the mild cartilage degeneration observed in sham-operated hypertensive mice (Fig. 4, G and H, and fig. S11). To further evaluate its therapeutic efficacy, relcovaptan was administered intra-articularly at 5, 10, and 20 mM. All three doses reduced cartilage degeneration and subchondral bone plate thickening in DMM-operated hypertensive mice (fig. S12, A to D). Although the magnitude of protection did not increase in a strictly dose-dependent manner, each dose reduced OA severity in relcovaptan-treated mice compared with vehicle-treated controls (fig. S12, A to D).

In addition to the AVP-AVPR1A axis, we investigated whether the renin-angiotensin-aldosterone system (RAAS) contributes to hypertension-accelerated OA progression. To directly evaluate the role of RAAS signaling within the joint, DOCA-salt–induced hypertensive mice were treated with losartan, an angiotensin II type 1 receptor antagonist (*32*), either systemically by oral administration or locally via IA injection. In contrast to AVPR1A blockade, losartan failed to attenuate cartilage degeneration or subchondral bone plate thickening when administered by either route (fig. S13, A and B). Collectively, these findings identify the AVP-AVPR1A axis as a key mediator of hypertension-accelerated OA progression and demonstrate that both genetic deletion and pharmacological inhibition of AVPR1A confer robust protection against OA, particularly under hypertensive conditions.

The AVP-AVPR1A axis promotes catabolic remodeling in chondrocytes

To elucidate how AVP-AVPR1A signaling contributes to cartilage degeneration, we examined the expression of catabolic and anabolic markers in primary chondrocytes. Among the molecules analyzed, the matrix-degrading enzymes MMP3 and MMP13, key mediators of cartilage destruction in OA (*33*, *34*), were modestly up-regulated by either AVP treatment or AVPR1A overexpression alone (fig. S14, A and B). By contrast, combined treatment with AVP and Ad-AVPR1A markedly increased the mRNA and protein expression of MMP3 and MMP13 while concomitantly suppressing the anabolic markers *Sox9*, *Col2a1*, and *Acan* (Fig. 5, A and B, and fig. S15A). AVP also potentiated IL-1β–induced expression of MMP3 and MMP13 (Fig. 5, C and D, and fig. S15B), an effect that was dose-dependently suppressed by the AVPR1A antagonist relcovaptan (Fig. 5, E and F, and fig. S15C). Consistent with these in vitro findings, immunohistochemical analysis of cartilage from DMM-operated mice demonstrated that relcovaptan markedly attenuated the AVP-induced MMP3 and MMP13 up-regulation in vivo (fig. S15D).

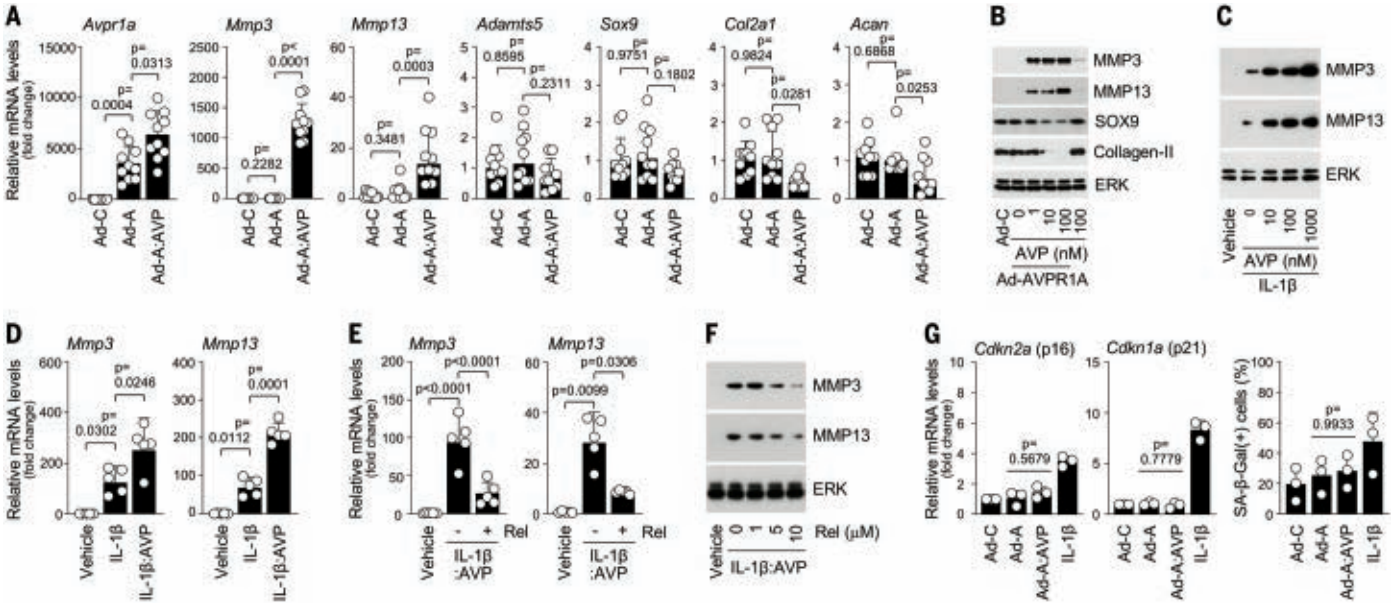

Fig. 5. AVP and AVPR1A cooperatively regulate catabolic and anabolic programs in chondrocytes. (**A** and **B**) Relative mRNA expression [(A); $n = 10$ biological replicates] and representative immunoblots (B) of the indicated molecules in primary mouse chondrocytes infected with Ad-Control (Ad-C) or Ad-AVPR1A (Ad-A) (400 MOI, 24 hours) followed by AVP treatment (100 nM, 24 hours). (**C** and **D**) Representative immunoblots (C) and relative mRNA expression (D) of the indicated molecules in chondrocytes treated with vehicle, IL-1β (1 ng/ml), or IL-1β with AVP (indicated concentrations or 1 μM) for 24 hours ($n = 5$ biological replicates). (**E** and **F**) Relative mRNA expression (E) and representative immunoblots (F) in chondrocytes treated with IL-1β (1 ng/ml) and AVP (100 nM) with increasing concentrations of relcovaptan for 24 hours ($n = 5$ biological replicates). (**G**) *Cdkn2a* and *Cdkn1a* mRNA levels and quantification of SA-β-gal–positive cells in chondrocytes infected with Ad-C or Ad-A (400 MOI) or treated with AVP (100 nM, 36 hours); IL-1β (1 ng/ml) served as a positive control ($n = 3$ biological replicates). Data are presented as geometric mean with geometric SD factor (or median with IQR for non-normally distributed data) for (A), (D), and (E) and as mean ± SD for (G). Statistical significance across groups [(A), (D), (E), and (G)] was evaluated by standard one-way ANOVA with Tukey's post-hoc test (equal variances), Welch's ANOVA with Games-Howell test (unequal variances), or Kruskal-Wallis test with Dunn's test [(A); non-normally distributed].

Because relcovaptan may exhibit off-target activity toward the oxytocin receptor (OTR) at high concentrations (*35*), we evaluated the potential contribution of OTR signaling. Under IL-1β–induced OA-mimicking conditions, *Oxtr* mRNA expression was undetectable in both primary chondrocytes and FLSs (fig. S16A), whereas *Avpr1a* expression was markedly up-regulated (Fig. 2). Relcovaptan effectively inhibited downstream catabolic markers (Fig. 5, E and F) without altering *Avpr1a* expression, and *Oxtr* remained undetectable (fig. S16, B and C). Furthermore, the highly selective OTR antagonist SSR126768 (*36*) failed to suppress IL-1β– and AVP-induced *Mmp3* expression or affect *Avpr1a* expression (fig. S16, D and E). These findings demonstrate that AVP-induced catabolic signaling in chondrocytes is mediated specifically through AVPR1A rather than OTR.

To exclude potential contributions from other systemic pathways, including the RAAS, we examined the expression of angiotensin II receptors and their potential involvement in primary chondrocytes. Under OA-mimicking conditions induced by IL-1β, Ad-HIF-2α, or Ad-ZIP8, the mRNA expression of *Agtr1a*, *Agtr1b*, and *Agtr2* was either undetectable or unchanged (fig. S17, A to C). Moreover, recombinant angiotensin II failed to induce the expression of *Avpr1a* or *Mmp3* (fig. S17D). To further verify whether these catabolic responses were linked to cellular senescence, costimulation with AVP and AVPR1A overexpression was evaluated. It did not alter the expression of the senescence markers *Cdkn2a* (p16) and *Cdkn1a* (p21), nor did it increase the number of senescence-associated β-galactosidase (SA-β-gal)–positive cells (Fig. 5G), indicating that the AVP-AVPR1A axis primarily promotes catabolic remodeling rather than cellular senescence. Collectively, these findings suggest that OA-associated AVPR1A up-regulation cooperates with circulating AVP to amplify cartilage-degrading enzymes and suppress anabolic gene expression in chondrocytes, independently of off-target OTR signaling, angiotensin II signaling, or cellular senescence.

Transcriptional landscape regulated by AVP-AVPR1A signaling

To elucidate the transcriptional mechanisms downstream of AVP-AVPR1A signaling, we performed transcriptomic analyses to identify the transcription factors that mediate this pathway. RNA-seq revealed distinct transcriptional profiles among the Ad-Control, Ad-AVPR1A, and Ad-AVPR1A with AVP groups, as illustrated by the heatmap (Fig. 6A) and volcano plots (Fig. 6B) of differentially expressed genes (DEGs). Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis indicated activation of inflammatory, extracellular matrix–degrading, and oxidative stress–related pathways (Fig. 6C). Radar plot analysis further showed that AVP stimulation in AVPR1A-overexpressing chondrocytes enhanced the expression of catabolic genes such as *Mmp3* and *Mmp13* while suppressing anabolic genes, including *Col2a1* and *Acan* (Fig. 6D). Transcription factor profiling identified *Nr4a3* (nuclear receptor 4A3) and *Hand2* (heart- and neural crest derivatives-expressed protein 2) as two of the most prominently up-regulated regulators under combined AVP and Ad-AVPR1A stimulation (Fig. 6E). Their induction was validated by qRT-PCR in primary chondrocytes (Fig. 6F) and further confirmed by immunohistochemistry in the articular cartilage of adult mice administered AVP and Ad-AVPR1A (Fig. 6G).

To determine the functional roles of NR4A3 and HAND2, we performed small interfering RNA (siRNA)–mediated knockdown (targeted gene silencing) of *Nr4a3* or *Hand2* in primary chondrocytes costimulated with AVP and Ad-AVPR1A. Silencing *Hand2* did not alter the expression of catabolic or anabolic markers (fig. S18A). By contrast, *Nr4a3* knockdown effectively reversed the AVP-induced catabolic phenotype by reducing *Mmp3* and *Mmp13* expression while restoring *Col2a1* expression; *Acan* expression, however, remained unchanged (Fig. 6H). We next overexpressed NR4A3 in primary chondrocytes using an adenoviral vector (Ad-NR4A3). NR4A3 overexpression markedly increased *Mmp13* expression while robustly suppressing *Col2a1* and *Acan*, whereas *Mmp3* expression remained unchanged (fig. S18B). These gain- and loss-of-function studies identified NR4A3 as the principal downstream transcriptional effector that mediates AVP-AVPR1A–induced extracellular matrix remodeling in chondrocytes. Collectively, these findings demonstrate that AVP-AVPR1A signaling orchestrates a catabolic transcriptional program through NR4A3, thereby promoting extracellular matrix degradation and disrupting cartilage homeostasis during OA progression.

Discussion

This study establishes the AVP-AVPR1A signaling axis as a mechanistic link between hypertension and accelerated OA progression through integrated analyses spanning population-based epidemiology, hypertensive mouse models, and molecular investigations. Our findings provide molecular evidence that systemic hypertension can directly accelerate articular cartilage degeneration through endocrine AVP-AVPR1A signaling, thereby uncovering a pathophysiological connection between two highly prevalent age-related disorders.

We first established hypertension as a risk factor for OA. Analysis of the KNHANES dataset revealed an epidemiological association between hypertension and OA, consistent with previous clinical studies and large-scale meta-analyses (*13*, *37*, *38*). This association was most pronounced in the knee joint and in individuals with advanced structural damage (KL grade 4), suggesting that hypertension is more strongly linked to disease progression than to OA onset. Although the association increased progressively with age up to age 74 years, it weakened in individuals aged ≥75 years, likely reflecting survivorship bias and competing clinical risks, as reported in national health care statistics (*39*). This association extends to both radiographic and symptomatic OA (*13*, *15*). Recent evidence further indicates that hypertension influences knee OA pain in an age-dependent manner (*40*), consistent with studies linking blood pressure to knee-pain severity and pain sensitivity (*41*, *42*). Using two independent hypertensive mouse models (*24*, *25*), we further confirmed that elevated blood pressure causally exacerbates OA pathology. Thus, the epidemiological observations are supported by experimental evidence, collectively indicating that hypertension functions primarily as an accelerator of established OA. These findings align with recent studies implicating hypertension-associated vascular dysfunction and hemodynamic stress in joint degeneration (*20*, *43*). Moreover, DMM surgery itself elevated baseline systolic blood pressure, suggesting a bidirectional relationship in which chronic joint pain and surgical stress trigger sympathetic activation, promoting secondary hypertension (*44*).

We identified the AVP-AVPR1A axis as a molecular mediator linking hypertension to cartilage catabolism. AVPR1A, a vasoconstrictive G protein–coupled receptor (*45*), was markedly up-regulated in OA-mimicking chondrocytes and in murine and human OA cartilage, whereas genetic deletion and local pharmacological inhibition of AVPR1A attenuated OA progression. In parallel, substantial amounts of circulating AVP, measured using its stable surrogate marker copeptin (*30*), were detected in the synovial fluid of OA patients, supporting the concept that endocrine AVP signaling engages locally induced AVPR1A to accelerate OA. One of the key findings of this study is that the AVP-AVPR1A axis functions as a context-dependent amplifier of OA rather than as a primary initiator. We therefore propose a two-hit model for hypertension-driven OA acceleration. The first hit is local joint priming. Young, healthy joints express little or no AVPR1A, rendering them largely unresponsive to circulating AVP. By contrast, mechanical injury or physiological aging up-regulates AVPR1A expression, thereby priming the joint. We further speculate that an intact extracellular matrix limits AVP penetration, whereas its degradation during early OA facilitates ligand access to AVPR1A-expressing chondrocytes. The second hit is the systemic endocrine signal generated by hypertension, in which circulating AVP accesses the joint microenvironment and activates the locally up-regulated AVPR1A. Although chronic hypertensive stress induces mild baseline vulnerability, severe cartilage destruction emerges only when systemic AVP is coupled with locally induced AVPR1A.

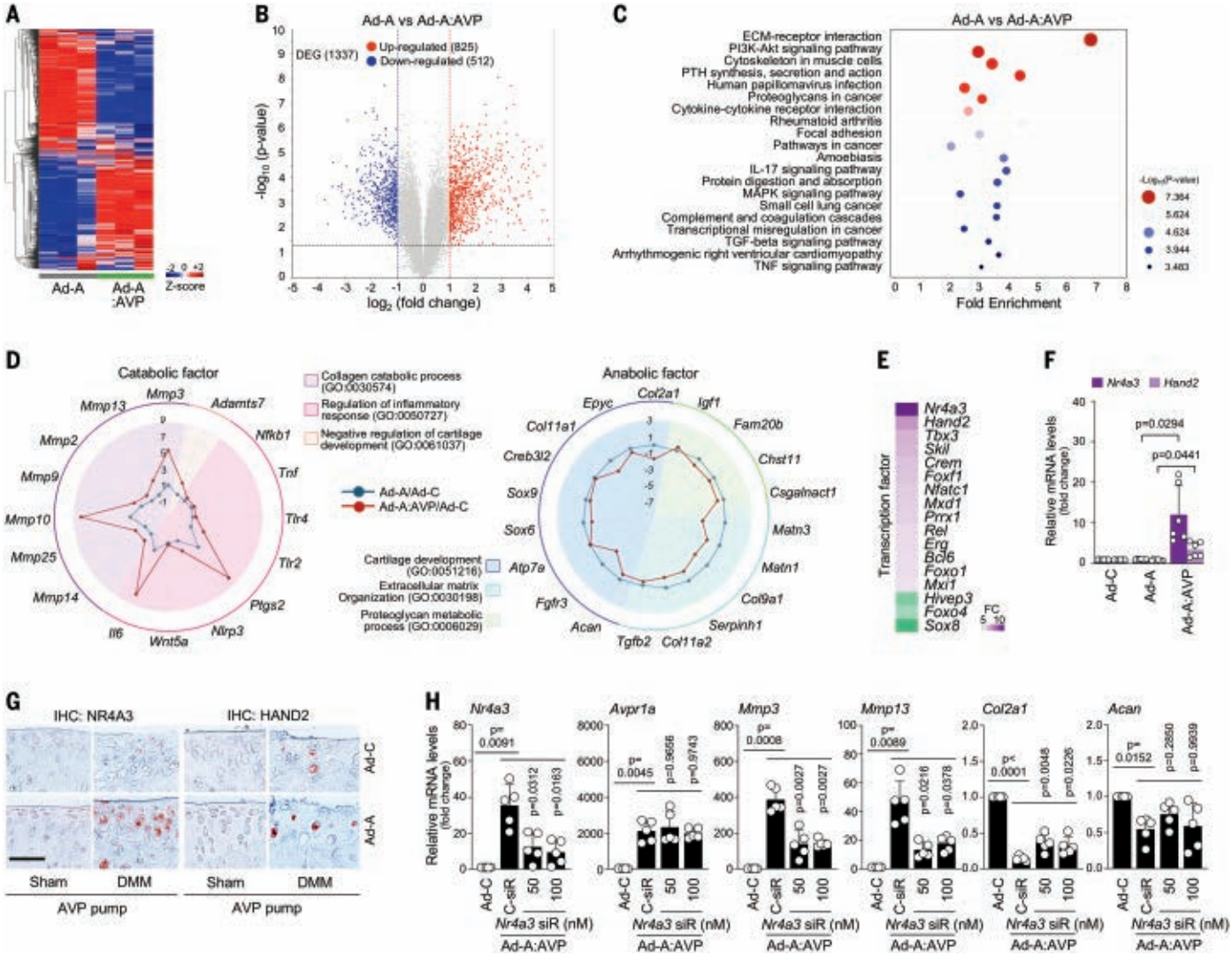

Fig. 6. Transcriptional landscape regulated by the AVP-AVPR1A axis in OA chondrocytes. (**A** to **C**) RNA-seq analysis of primary chondrocytes infected with Ad-Control (Ad-C), Ad-AVPR1A (Ad-A), or Ad-A with AVP (100 nM) ($n = 3$ biological replicates). Shown are a heatmap of DEGs (A), a volcano plot comparing Ad-A versus Ad-A:AVP (B), and a KEGG pathway enrichment analysis of DEGs (C). (**D**) Radar plot of representative catabolic and anabolic genes across gene ontology categories. The radial axis shows the log2 fold change. (**E**) Heatmap of transcription factors significantly up-regulated in Ad-A:AVP versus Ad-A alone. (**F**) qRT-PCR analysis of *Nr4a3* and *Hand2* expression in primary mouse chondrocytes infected with Ad-C or Ad-A (400 MOI), with or without AVP (100 nM) treatment for 24 hours ($n = 6$ biological replicates). (**G**) Representative immunostaining images of NR4A3 and HAND2 in articular cartilage from mice subjected to sham or DMM surgery, systemic AVP infusion, and IA injection of Ad-C or Ad-A ($n = 3$ mice per group). (**H**) qRT-PCR analysis in chondrocytes treated with 100 nM control siRNA (C-siR) or the indicated concentrations of *Nr4a3* siRNA for 6 hours, infected with Ad-C or Ad-A (400 MOI, 2 hours), and treated with AVP (100 nM) for 36 hours ($n = 5$ biological replicates). Data are presented as mean ± SD. Statistical significance was determined by Welch's ANOVA with Games-Howell test (F) or standard one-way ANOVA with Tukey's test (equal variances) or Welch's ANOVA with Games-Howell test (unequal variances) (H). Statistical significance for RNA-seq data [(A) to (E)] was defined as adjusted $P < 0.05$. Scale bars in (G) are 50 μm.

Our findings extend this model by demonstrating that chronological aging (3, 6, 12, and 18 months) up-regulates *Avpr1a* expression, rendering the aging joint vulnerable even in the absence of mechanical trauma. Consistently, systemic AVP infusion or spontaneous hypertension (18-month-old BPH/2J mice) markedly exacerbated cartilage degeneration in aged, but not in young, mice. Indeed, prolonged 12-week AVP infusion failed to induce joint pathology in young mice, confirming that this lack of response reflects unprimed receptors rather than insufficient ligand exposure. Thus, local AVPR1A induction precedes and permits the pathological actions of circulating AVP, which closely parallels our epidemiological observations where the hypertension-OA association became progressively stronger with advancing age.

Within our two-hit model, the systemic endocrine signal is mediated by AVP. The robust detection of copeptin in OA synovial fluid, together with the absence of AVP synthesis in joint-resident cells, indicates that synovial AVP is derived from the systemic circulation. Increased synovial vascular permeability associated with chronic synovitis and pathological angiogenesis in OA likely facilitates AVP entry into the joint (*46*, *47*). Once inside the joint, circulating AVP activates locally up-regulated AVPR1A, amplifying cartilage catabolism. This increased permeability allows even basal circulating AVP to access the joint, explaining why AVPR1A blockade attenuates cartilage destruction even under normotensive conditions. The clinical relevance is further supported by publicly available human transcriptomic data from the National Center for Biotechnology Information

(NCBI) Gene Expression Omnibus (GEO; accession no. GSE215039). Primary human chondrocytes stimulated with IL-1β and IL-6 lacked detectable expression of *AVP*, *AVPR1B*, or *AVPR2*, whereas *AVPR1A* was constitutively expressed and further up-regulated under catabolic conditions. Together with our ex vivo demonstration of increased AVPR1A expression in damaged human OA cartilage, these findings identify enhanced AVPR1A expression as a characteristic feature of the OA cartilage phenotype.

Hypertension is a heterogeneous disorder with both essential and secondary forms (*48*). Although recent studies suggest that cellular senescence contributes to OA-like phenotypes in spontaneously hypertensive rats (*49*) and DOCA-salt–induced hypertensive rats (*50*), mechanistic interpretation remains challenging. The spontaneous cartilage degeneration and impaired subchondral perfusion observed in spontaneously hypertensive rats (*49*), together with its baseline skeletal abnormalities (*51*), may confound the specific contribution of hypertension to OA progression. To overcome this limitation, we used complementary mouse models (DOCA-salt and genetic BPH/2J) in combination with DMM-induced OA, *Avpr1a* genetic targeting, and pharmacological interventions. Despite their distinct etiologies—secondary, low-renin hypertension (*52*) versus spontaneous, essential-like hypertension (*25*)—both models similarly accelerated OA progression, identifying systemic AVP as a shared neuroendocrine driver. Consistently, pharmacological inhibition of AVPR1A with relcovaptan (*31*) attenuated OA progression, whereas blockade of the RAAS with the angiotensin II type 1 receptor antagonist losartan (*32*) failed to provide comparable protection. Because angiotensin II neither induced *Avpr1a* expression nor activated catabolic gene programs in chondrocytes, the previously reported joint-protective effects of RAAS inhibitors (*53*) are likely attributable to systemic blood pressure reduction or downstream cross-talk rather than direct modulation of chondrocyte responses.

Mechanistically, we identified NR4A3 as a key downstream transcriptional effector linking AVP-AVPR1A signaling to chondrocyte catabolism. NR4A3 is an immediate-early transcription factor induced by G protein–coupled receptors (*54*) that regulates inflammatory responses and matrix metalloproteinase expression (*55*). This regulatory role is supported by our in silico identification of conserved NR4A-binding motifs within the *Mmp3*, *Mmp13*, and *Col2a1* promoters, together with recent evidence demonstrating NR4A3 occupancy at the *Mmp3* locus (*55*). Our functional analyses reveal a promoter-specific regulatory hierarchy. Although *Nr4a3* deficiency markedly suppresses the overall AVP-driven catabolic program, NR4A3 overexpression alone sufficiently transactivates *Mmp13* and strongly represses anabolic markers (*Col2a1* and *Acan*). By contrast, *Mmp3* requires additional AVP-dependent cofactors and is not activated by NR4A3 overexpression alone. Furthermore, HAND2 loss-of-function analysis further supports NR4A3 as a central regulator of this catabolic transition. Although our in vitro findings indicate that AVP-AVPR1A signaling primarily drives catabolic remodeling rather than directly inducing cellular senescence, physiological aging—an intrinsic process associated with cellular senescence and tissue remodeling (*56*–*58*)—progressively increases AVPR1A expression. This age-dependent induction primes the joint, creating a permissive microenvironment in which systemic hypertensive signaling can accelerate cartilage degeneration.

Several limitations of this study should be acknowledged. First, assessment of cross-sectional associations between fluid (synovial or serum) copeptin and radiographic OA severity was limited by the demographic homogeneity of our end-stage cohort (KL grade 3 or 4) and the potential confounding effects of antihypertensive medications. Although our KNHANES analysis robustly links systemic hypertension with advanced structural degeneration, longitudinal studies with rigorous medication control will be required to define neuroendocrine fluid dynamics across OA stages. Second, the KNHANES dataset lacks information on prior joint injury; however, this epidemiological limitation further highlights the importance of a rigorously controlled in vivo DMM model. Third, although human data showed a stronger association between hypertension and OA in females, our in vivo validation was performed in male mice to minimize sex-related variability in both DMM-induced OA (*59*) and DOCA-salt hypertensive models (*60*). Future studies incorporating estrogen-controlled conditions or female cohorts are warranted. Fourth, with respect to age-dependent joint priming, we could not fully distinguish the effects of chronological aging from cumulative mechanical loading. Fifth, existing available V1A receptor antagonists either lack AVPR1A specificity (for example, tolvaptan) (*61*) or are limited to short-term clinical use (for example, conivaptan) (*31*), underscoring the need for future trials that evaluate long-term therapeutic outcomes. Finally, our global knockout models cannot exclude contributions from other joint-resident cell types. Nevertheless, the protective effects of IA relcovaptan and the preferential up-regulation of AVPR1A in chondrocytes strongly support a chondrocyte-autonomous mechanism, which should be definitively tested using conditional knockout approaches.

Our integrated epidemiological, genetic, pharmacological, and mechanistic analyses identify the AVP-AVPR1A axis as a molecular link between hypertension and accelerated OA progression. By demonstrating that systemic AVP signaling cooperates with locally induced AVPR1A in mechanically injured or aging joints, this study provides a mechanistic framework that explains how chronic hypertension accelerates cartilage degeneration. Our findings highlight the AVP-AVPR1A axis as a promising therapeutic target for preventing hypertension-accelerated OA progression and preserving joint integrity in vulnerable populations.

Materials and methods

Logistic regression analysis of KNHANES

The KNHANES is an ongoing, nationwide, cross-sectional survey designed to monitor the health and nutritional status of the Korean population since 1998. The survey is conducted annually by the Korea Disease Control and Prevention Agency. All study procedures were approved by the KDCA Institutional Review Board (approval nos. 2007-02CON-04-P to 2018-01-03-5C-A), and written informed consent was obtained from all participants. To evaluate the association between hypertension and OA, we analyzed data from the fourth through eighth KNHANES cycles (2007–2021). Approximately 150,000 individuals were sampled during the study period, of whom approximately 105,000 completed the survey. After excluding participants with missing data for variables of interest, 36,820 individuals aged ≥45 years were included in the final analysis. Hypertension was defined as a systolic blood pressure ≥140 mmHg, a diastolic blood pressure ≥90 mmHg, or current use of antihypertensive medication (*62*). OA was defined as a self-reported physician diagnosis. To evaluate age-specific patterns among middle-aged and older adults, participants aged ≥45 years were stratified into four age groups (45 to 54, 55 to 64, 65 to 74, and ≥75 years). Logistic regression models were adjusted for sex, education level, household income, smoking status, alcohol consumption, diabetes, and body weight. Age was not included as a covariate because the analyses were performed separately within predefined age strata. Information regarding previous joint injury or surgery was not included because these variables were not consistently available in the KNHANES database. To further evaluate the association between hypertension and radiographic knee OA severity, we analyzed a KNHANES subgroup from 2009 to 2013, during which knee radiographs were obtained from participants aged ≥50 years. In this subgroup (n = 13,717), radiographic OA was defined as a KL grade ≥2. Because the proportional odds assumption required for ordinal logistic regression was not satisfied, multinomial logistic regression and grade-specific binary logistic regression analyses were performed to estimate stage-specific odds ratios across KL grades.

Hypertensive mouse models

To investigate the impact of systemic hypertension on OA progression, we used two independent hypertensive mouse models: the

DOCA-salt–induced hypertension model (*24*) and a genetic hypertension model using BPH/2J mice (*25*). For the DOCA-salt model, unilateral nephrectomy was performed in 9-week-old male mice (Fig. 1C). Beginning at 10 weeks of age, mice received intraperitoneal injections of DOCA (D7000, Sigma-Aldrich; 20 mg/kg dissolved in olive oil) twice weekly and were provided with 1% NaCl in their drinking water for the indicated experimental period. DMM or sham surgery was performed at 12 weeks of age (Fig. 1C). Sham-operated control mice received vehicle (olive oil) injections without DOCA treatment. For the genetic hypertension model, spontaneously hypertensive male BPH/2J mice and normotensive male BPN/3J control mice (003005, 003004, The Jackson Laboratory, Bar Harbor, ME, USA) were used without pharmacological intervention. DMM or sham surgery was performed at 12 weeks of age, and mice were euthanized 6 or 8 weeks later for subsequent analyses. For aging-related analyses, 18-month-old BPH/2J and BPN/3J mice were used to evaluate spontaneous cartilage degeneration under hypertensive and normotensive conditions, respectively.

Human OA cartilage and synovial fluid samples

Human OA cartilage and synovial fluid samples were obtained from patients undergoing total knee arthroplasty at Chonnam National University Hospital (Gwangju, Republic of Korea) after written informed consent was obtained from all participants. The study was approved by the Institutional Review Board of Chonnam National University Hospital (CNUHH-2023-122 and CNUHH-2026-025). Clinical characteristics of the study participants, including age, sex, body weight, and other relevant demographic information, are summarized in table S1. To evaluate localized gene expression in human OA cartilage, cartilage specimens from each patient were macroscopically separated into undamaged regions, characterized by an intact and smooth articular surface, and damaged regions exhibiting erosion and fibrillation. Total RNA was extracted from each region, and qRT-PCR was performed to analyze gene expression. Protein expression in damaged and corresponding undamaged cartilage regions was evaluated by immunohistochemistry. Synovial fluid samples were collected from patients with radiographic OA (KL grades 3 and 4), centrifuged at 3000*g* for 10 min to remove cellular debris, aliquoted, and stored at −80°C until analysis. Age-matched healthy control synovial fluid samples were not included because obtaining synovial fluid from individuals of comparable age without underlying joint degeneration is ethically and practically challenging. Detailed demographic information, including the age and sex of patients included in the synovial fluid analysis, is provided in table S1.

Animal models of experimental OA

All C57BL/6 mice were purchased from Damool Science (Daejeon, Republic of Korea) and maintained under specific pathogen–free conditions with free access to food and water. All animal experiments were approved by the Institutional Animal Care and Use Committee of Chonnam National University (CNU IACUC-YB-2023-96) and Gwangju Institute of Science and Technology (GIST-2021-069). Experimental OA was induced in 12-week-old mice, unless otherwise indicated, by DMM in the right knee, whereas sham surgery was performed on the contralateral left knee as an internal control. The overall experimental design and the number of mice allocated to each experimental group are summarized in fig. S19. Male mice were used exclusively to minimize biological variability arising from sex hormones. This experimental design was chosen because endogenous estrogen protects against cartilage degeneration in the DMM model (*59*), and female mice exhibit relative resistance to DOCA-salt–induced hypertension (*60*). To investigate the contribution of hypertension to OA progression, we used the DMM model because it provides a well-defined onset and reproducible progression of OA (*26*). Spontaneous age-related OA models were not used for these mechanistic experiments because aging is accompanied by systemic low-grade inflammation ("inflammaging"), hormonal alterations, and metabolic dysregulation, all of which independently contribute to OA pathogenesis and could confound the specific effects of hypertension and AVP-AVPR1A signaling (*4*, *63*). The role of AVPR1A in hypertension-associated OA was investigated using *Avpr1a*$^{-/-}$ mice (046134, The Jackson Laboratory), which are viable and exhibit normal skeletal development. Skeletal preparations from embryonic day 18.5 (E18.5) mice were stained with Alcian blue and Alizarin red to visualize cartilage and mineralized bone, respectively, as previously described (*4*, *64*). Genotyping of *Avpr1a*$^{-/-}$ mice and assessment of *Avpr1a* mRNA expression were performed by RT-PCR, as described below. To examine the influence of physiological aging on AVPR1A expression and OA susceptibility, wild-type mice aged 3, 6, 12, and 18 months were analyzed for age-dependent molecular and pathological changes in articular cartilage.

Pharmacological intervention of experimental OA

For systemic AVP administration, osmotic minipumps (ALZET model 1004, DURECT Corporation, CA, USA) were filled with AVP (113-79-1, Bachem AG, Bubendorf, Switzerland) and implanted subcutaneously in the dorsal region 2 weeks after sham or DMM surgery for the indicated durations. The pumps continuously delivered AVP at a rate of 24 ng/hour (0.11 μl/hour). At the designated endpoints, mice were euthanized and knee joints were harvested for subsequent analyses. To evaluate the effects of prolonged ligand exposure, a separate cohort of mice received continuous systemic AVP infusion for 12 weeks. Osmotic minipumps were surgically replaced every 4 weeks under light anesthesia to maintain uninterrupted AVP delivery. To investigate the effects of systemic AVP on the aging joint, 3-month-old (young) and 12-month-old (aged) WT mice were implanted subcutaneously with osmotic minipumps delivering either AVP (24 ng/hour) or phosphate-buffered saline (PBS) for 28 days. At the end of the treatment period, mice were euthanized and knee joints were collected for subsequent analyses.

To modulate AVP-AVPR1A signaling in vivo, sham- or DMM-operated mice received IA injections of an adenovirus expressing mouse AVPR1A (Ad-AVPR1A) or an empty control adenovirus (Ad-Control) (5×10^8 plaque-forming units in 10 μl per joint). Adenoviral vectors (Vector Biolabs, Malvern, PA, USA) were administered into both knee joints once weekly for three consecutive weeks. For pharmacological inhibition of AVPR1A, the selective antagonist relcovaptan (*31*) (SR49059; Axon Medchem, Groningen, Netherlands) was administered by IA injection (10 μl per joint) once weekly for 6 weeks. IA injections were initiated 2 weeks after sham or DMM surgery, and mice were euthanized 6 weeks after the first injection (8 weeks after DMM surgery) for histological analyses. Because the IA pharmacokinetics of relcovaptan have not been established, this dosing regimen was selected to (i) minimize injection-related artifacts associated with frequent IA administration, (ii) maintain adequate local drug exposure, and (iii) inhibit AVPR1A signaling during the early phase of OA progression, when pathological remodeling is initiated. Losartan, an angiotensin II type 1 receptor antagonist (*32*), and SSR126768, an OTR antagonist (*36*), were also administered using a similar experimental protocol.

Assessment of OA parameters in experimental mice

For histological analyses, knee joints were fixed in 4% paraformaldehyde, decalcified in 0.5 M EDTA, embedded in paraffin, and processed as previously described (*10*). Serial frontal sections (5 μm thick) were stained with Safranin O/Fast Green and scanned using an Aperio CS2 slide scanner (Leica Biosystems, Vista, CA, USA). Cartilage degeneration was evaluated using three sections per joint collected at 100-μm intervals by four independent observers blinded to the experimental groups. OA severity was graded according to the Osteoarthritis Research Society International (OARSI) scoring system (score range, 0 to 6), and the highest score among the medial femoral condyle, medial tibial plateau, lateral femoral condyle, and lateral tibial plateau was recorded for each joint (*10*, *29*, *65*). Subchondral bone plate thickness, a histological indicator of subchondral bone sclerosis, was measured using

Aperio ImageScope software (Leica Biosystems) (*10*, *29*, *66*). The subchondral bone plate was defined as the compact bone immediately beneath the calcified cartilage. Thickness was measured as the perpendicular distance from the cartilage-bone interface to the boundary of the bone marrow cavity. Multiple measurements were obtained across the weight-bearing region of the medial tibial plateau in each section and averaged to generate a single value per section. Representative values for each mouse were calculated by averaging measurements from three serial sections collected at equal intervals. Synovitis was evaluated in joint sections stained with Safranin O and hematoxylin according to the method previously described (*10*, *29*, *67*). Synovial inflammation was graded on a scale of 0 to 3 based on the degree of inflammatory cell infiltration (0, none; 1, mild; 2, moderate; and 3, severe). The most severely affected synovial regions adjacent to the femur and tibia were scored. Three serial sections collected at equal intervals were analyzed, and the mean score was used as the representative value for each mouse. Osteophytes were identified in Safranin O–stained sections, and osteophyte maturity was assessed as previously described (*10*, *29*). Three serial sections per mouse were analyzed, and the average score was used as the representative value.

Measurement of AVP

Because circulating AVP is highly unstable, AVP amounts were assessed indirectly by measuring copeptin, a stable surrogate marker of AVP secretion (*30*). Serum samples were collected from sham- or DMM-operated mice at 8 weeks after surgery, aliquoted, and stored at −80°C until analysis. Serum copeptin concentrations were quantified using a mouse copeptin enzyme-linked immunosorbent assay (ELISA) kit (CEA365Mu, Cloud-Clone Corp., Katy, TX, USA) according to the manufacturer's instructions. Copeptin concentrations in synovial fluid were measured using a human copeptin ELISA kit following a similar protocol. Synovial fluid samples (50 µl per well) were analyzed in duplicate. The assay had a detection range of 12.35 to 1000 pg/ml, and all measured values fell within the linear range of the standard curve.

Determination of blood pressure in mice

Systolic blood pressure was measured weekly in conscious mice using a noninvasive tail-cuff system (BP-2000, Visitech Systems Inc., Apex, NC, USA) (*68*). To minimize stress and improve measurement reliability, mice were acclimated to the procedure for three consecutive days before the initial recording and prewarmed at 37°C for 10 min before each measurement. For each mouse, 10 successful systolic blood pressure recordings were obtained and averaged for analysis. Both DOCA-salt–treated mice and genetically hypertensive BPH/2J mice exhibited stable hypertension (systolic blood pressure >150 mmHg) before DMM surgery, confirming successful establishment of the hypertensive phenotype.

Immunohistochemistry

AVPR1A expression was evaluated by immunohistochemistry in mouse articular cartilage and in paired damaged and adjacent undamaged regions of human OA cartilage using a commercially available anti-AVPR1A antibody (bs-11598R, Bioss Antibodies, Woburn, MA, USA). Tissue sections were incubated overnight at 4°C with a rabbit anti-AVPR1A primary antibody, followed by detection using the EnVision+ horseradish peroxidase (HRP) system (K5007, Agilent Technologies, Santa Clara, CA, USA) and visualization with 3-amino-9-ethylcarbazole (AEC) substrate (SK-4200, Vector Laboratories, Newark, CA, USA). The specificity of the anti-AVPR1A antibody was validated using cartilage sections from *Avpr1a*$^{-/-}$ mice, which exhibited no detectable immunoreactivity. For negative controls, the anti-AVPR1A antibody was replaced with normal mouse immunoglobulin G (IgG) (sc-2025; Santa Cruz Biotechnology, Dallas, TX, USA) at the same concentration. Additional antibodies used for immunostaining included rabbit anti-AVP antibody (ab213708), rabbit anti-HAND2 antibody (ab200040), and rabbit anti-MMP13 antibody (ab39012) from Abcam (Cambridge, UK); rabbit anti-NR4A3 antibody (NBP2-46246) from Novus Biologicals (Centennial, CO, USA); and rabbit anti-p16 antibody (MA5-17142) from Invitrogen (Carlsbad, CA, USA).

Primary chondrocyte and FLS cultures

Primary mouse articular chondrocytes were isolated and cultured as previously described (*69*). Articular cartilage was harvested from the knee joints of 4- to 6-day-old ICR mice, and immature chondrocytes were isolated by digestion with 0.2% collagenase type II. Neonatal mice were used because isolation of adult articular chondrocytes is technically difficult and adult chondrocytes rapidly undergo dedifferentiation during in vitro culture (*69*). Isolated chondrocytes were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin at 37°C in a humidified atmosphere containing 5% CO_2. To generate OA-like chondrocytes, cells were treated on day 3 of culture with IL-1β (1 ng/ml) for 36 hours or infected for 24 hours with adenoviruses encoding hypoxia-inducible factor-2α (Ad-HIF-2α) or the zinc transporter ZIP8 (Ad-ZIP8), both established mediators of OA pathogenesis (*10*, *28*, *29*). Adenoviral infection was performed at a multiplicity of infection (MOI) of 400. To investigate the direct effects of the RAAS, chondrocytes were treated with angiotensin II (A9525, Sigma-Aldrich, St. Louis, MO, USA) at 10 to 1000 nM, alone or in combination with IL-1β (1 ng/ml), for 36 hours. Chondrocyte senescence was evaluated by measuring the expression of the senescence markers p16 and p21 and by SA-β-gal staining (*70*). Additional experimental treatments are described in the corresponding figure legends or relevant sections. Primary mouse FLSs were isolated and cultured as previously described (*10*, *29*). Cells from passages 4 to 8 were used for all experiments. FLS purity was confirmed by flow cytometry using antibodies against CD90 and CD14, demonstrating >90% CD90-positive cells and <1% CD14-positive cells (Abcam). FLSs were maintained in DMEM supplemented with 10% FBS and 1% penicillin-streptomycin and were treated with the indicated reagents or infected with the appropriate adenoviral vectors as described for each experiment.

Biochemical and molecular analyses

mRNA levels of the indicated genes were quantified by qRT-PCR. Total RNA was extracted from cultured cells using TRI reagent (TR118, Molecular Research Center, Cincinnati, OH, USA). Total RNA was also extracted from human cartilage macroscopically separated into "undamaged" (intact, smooth surface) and "damaged" (eroded, fibrillated lesion) areas. Cartilage tissue of sham- and DMM-operated mice was isolated from samples obtained using a Leica LMD6 laser microdissection system (Leica Microsystems) (*71*). Total RNA was subsequently extracted from the microdissected tissues using the RNeasy FFPE kit (73504, Qiagen, Hilden, Germany) according to the manufacturer's instructions. qRT-PCR was performed using SYBR Premix Ex Taq (RR420, Takara, Shiga, Japan) on a QuantStudio 1 system (Thermo Fisher Scientific, Waltham, MA, USA). Gene expression was normalized to *Gapdh* and analyzed using the $2^{-\Delta\Delta Ct}$ method. Primer sequences are listed in table S2. Whole-cell lysates were subjected to Western blotting using antibodies against MMP3 (ab52915, Abcam), MMP13 (18165-AP, Proteintech, Rosemont, IL, USA), collagen II (MAB8887, Chemicon, Temecula, CA, USA), SOX9 (ab185966, Abcam), and ERK (610408, BD Biosciences, San Jose, CA, USA). Protein signals were visualized using enhanced chemiluminescence (ECL, RPN2235, Cytiva, Marlborough, MA, USA) and quantified with Image J software.

For RNA-seq, primary chondrocytes were incubated for 24 hours with IL-1β (1 ng/ml), Ad-AVPR1A, or a combination of Ad-AVPR1A and AVP. Total RNA with an RNA integrity number >8.0 was extracted using TRIzol reagent and subjected to paired-end sequencing by Ebiogen (Seoul, Republic of Korea) on the Illumina NovaSeq 6000 platform. DEGs were identified using a threshold of an absolute fold change > 2 and

$P < 0.05$. Functional enrichment and KEGG pathway analyses were performed using ExDEGA software.

Data visualization, schematic illustration, and statistical analysis

All quantitative data were analyzed and visualized as dot plots using GraphPad Prism (version 10; GraphPad Software, San Diego, CA, USA). Experimental schematics and mechanistic illustrations were created using BioRender.com. All data were generated from independent biological replicates. For quantitative analyses, data from all independent experiments or animal cohorts were pooled for statistical analysis. Representative histological and biochemical images were selected from experiments showing results consistent with those obtained across independent biological replicates. The exact sample size (n) for each experiment is provided in the corresponding figure legends and fig. S19. Statistical analyses were performed using IBM SPSS Statistics (version 26; IBM Corp., Armonk, NY, USA). Data normality and homogeneity of variance were assessed using the Shapiro-Wilk and Levene tests, respectively. qRT-PCR data with a log-normal distribution were log-transformed before analysis, with normality assessed on log-transformed values, and are presented as geometric mean with geometric SD factor; the remaining qRT-PCR data are presented as arithmetic mean ± SD. Ordinal variables (including OARSI grade) and unpaired non-normally distributed data are presented as median with interquartile range (IQR) and were analyzed using two-tailed Mann-Whitney U tests for two-group comparisons or Kruskal-Wallis tests with Dunn's post-hoc test for multiple-group comparisons. Paired non-parametric data were analyzed using Wilcoxon signed-rank tests. Normally distributed continuous variables, including systolic blood pressure and subchondral bone plate thickness, are presented as mean with 95% confidence intervals (CIs) and were analyzed using two-tailed unpaired or paired Student's t tests for two-group comparisons or standard one-way analysis of variance (ANOVA) followed by Tukey's post-hoc test (equal variances) or Welch's ANOVA followed by Games-Howell post-hoc test (unequal variances) for multiple-group comparisons. For population-based epidemiological analyses, multivariable, ordinal, and multinomial logistic regression analyses were performed to calculate adjusted odds ratios (ORs) with 95% CIs. For high-throughput RNA-seq data, statistical significance was defined as adjusted $P < 0.05$. All other tests were two-tailed, and statistical significance was defined as $P < 0.05$.

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SUPPLEMENTARY MATERIALS

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Figs. S1 to S19; Tables S1 and S2; MDAR Reproducibility Checklist; Data S1

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BIOMATERIALS

Hierarchical crystalline organic-inorganic framework enabling high-modulus toughening in nacre

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Enhancing toughness in nanocomposites often involves viscoelastic organic phases, but this typically reduces stiffness. We describe a high-modulus toughening mechanism in mature *Cristaria plicata* nacre arising from a crystalline organic-inorganic framework composed of aragonite nanorods alternately integrated with intratabular biomolecular crystallites. Due to the critical slip-length effect, these crystalline units approach the theoretical limits of modulus and strength. The nanoconfined biomolecular crystallites function as nanopins at interlamellar interphases, alleviating stress concentrations and enabling cross-scale synchronized deformation, thereby conferring high fracture toughness. Inspired by this mechanism, reconstituted nacre fabricated from regenerated aragonite flakes simultaneously achieved high stiffness and crack resistance. These insights offer a promising pathway toward reliable, high-performance structural materials.

Nacre has long attracted intense interest for its outstanding strength, stiffness, and toughness (*1*, *2*). Its nanostructure has been recognized for more than six decades, and the neatly staggered arrangement of inorganic crystallites has been identified as the primary source of its exceptional mechanical performance. This structural concept has served as a guide for the design of nacre-inspired materials. Fabrication strategies, including layer-by-layer deposition (*3*, *4*), casting (*5*), and ice templating (*6*, *7*), have been developed to assemble oriented inorganic crystallites into strong and tough nanocomposites. Gao *et al.* (*8*) demonstrated theoretically that when inorganic crystallites operate at their critical length, they can approach theoretical strength and modulus. Further, the organic phase is often viewed as a homogeneous adhesive that serves as a ductile, energy-dissipating matrix (*9*, *10*). However, these soft phases typically hinder load transfer between inorganic crystallites, creating an inherent trade-off: improved toughness at the expense of stiffness, which is a critical limitation for load-bearing applications. Recent studies showed that nanoconfined polymer crystallites can enhance a material's toughness without compromising its stiffness (*11*, *12*). Therefore, we investigated whether similar biomolecular crystallization exists in natural nacre and, if so, what its contribution to nacre's toughening mechanisms might be.

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We describe a high-modulus toughening mechanism in the mature nacre of *Cristaria plicata* (*13*, *14*) driven by a crystalline organic-inorganic framework (COIF). This framework maintains the classical brick-and-mortar architecture while incorporating a hierarchical structure within the tablets, where 29 ± 3.9 nm aragonite nanorods are aligned perpendicular to the tablet plane and separated by 4.9 ± 0.4 nm gaps filled with intratabular biomolecular crystallites. By harnessing the critical slip-length effect, these crystalline units approach the theoretical limits of modulus and strength. The nanoconfined biomolecular crystallites function as nanopins at interlamellar interphases, alleviating stress concentrations and enabling synchronized deformation across multiple length scales. Guided by this principle, we fabricated reconstituted nacre from regenerated aragonite tablets and resins.

Structural analysis

C. plicata shells consist of three distinct layers: the periostracum, the prismatic calcite, and the nacreous layer (Fig. 1A). After mechanical polishing, the nacreous layer was isolated (Fig. 1B). This consists of a transitional layer (TL) near the inner surface and a mature layer (ML) adjacent to the prismatic layer (fig. S1). The nacre functions as the primary load-bearing component, conferring impact resistance to the shell (*15*, *16*). Although mainly composed of aragonite (~95 wt %), it also contains a minor organic fraction (~5 wt %) interspersed within the tablets, which is essential for toughness (*17*). Although the nacre's toughness is grounded in this organic-inorganic composition, we observed substantial differences in fracture toughness between ML and TL despite their similar organic content. Mechanical testing results showed that the ML's fracture toughness was nearly three times that of the TL (K_{Ic}: 3.2 ± 0.2 versus 1.1 ± 0.1 MPa·m$^{1/2}$), indicating substantially greater resistance to crack initiation (Fig. 1C). This correlation between structural complementarity and fracture toughness motivated us to conduct detailed cross-scale investigations into the organizational arrangement of organic-inorganic components within each nacre sublayer, aiming to uncover the underlying mechanism responsible for their divergent mechanical performances.

The evolution of the organic-inorganic architecture during nacre growth was first investigated by scanning electron microscopy (SEM). Both the TL and ML exhibited the characteristic brick-and-mortar structure of alternating aragonite tablets and interlamellar organics (Fig. 1, D and F). By contrast, high-resolution transmission electron microscopy (HRTEM) revealed substantial differences in their secondary structures. In the TL, aragonite tablets are disordered and loosely packed (Fig. 1G$_1$), with the organic matrix remaining largely amorphous both between (Fig. 1G$_2$) and within (Fig. 1G$_3$) the tablets, showing no detectable crystalline features (figs. S2 and S3). Conversely, the ML that develops from the TL during shell growth exhibits a refined hierarchical structure (Fig. 1H$_2$ and figs. S4 and S5). Crystalline biomolecular domains are present in both the interlamellar and intratabular organics (figs. S6 and S7). Interlamellar biomolecular crystallites are distributed within the ~10-nm interlamellar gaps and display a crystallinity of ~18.7% (Fig. 1H$_2$ and fig. S8, A and B). Instead of homogeneous $CaCO_3$ nanoparticles inside the tablets, we found oriented aragonite rods aligned perpendicular to the tablet direction. The aragonite nanorods with a diameter of 29 ± 3.9 nm form the array and are separated by 4.9 ± 0.4 nm gaps that are filled with nanoconfined intratabular biomolecular crystallites (IT-BCs), which exhibit a substantially higher crystallinity of ~80.3% (Fig. 1H$_3$ and figs. S8, C and D, and S9). Because of the low intrinsic contrast between interlamellar biomolecular crystallites and IT-BCs, fast Fourier transform analysis of HRTEM images was used to enhance periodic signals, revealing strand spacings of ~8 Å in interlamellar biomolecular crystallites versus ~4 Å in IT-BCs, consistent with molecular-scale ordering within the biomolecular crystallites (*18*). The smaller spacings observed in IT-BCs are suggestive of more ordered, silk-like, β-sheet–rich nanocrystallites (fig. S10), potentially resembling previously reported monoclinic-like

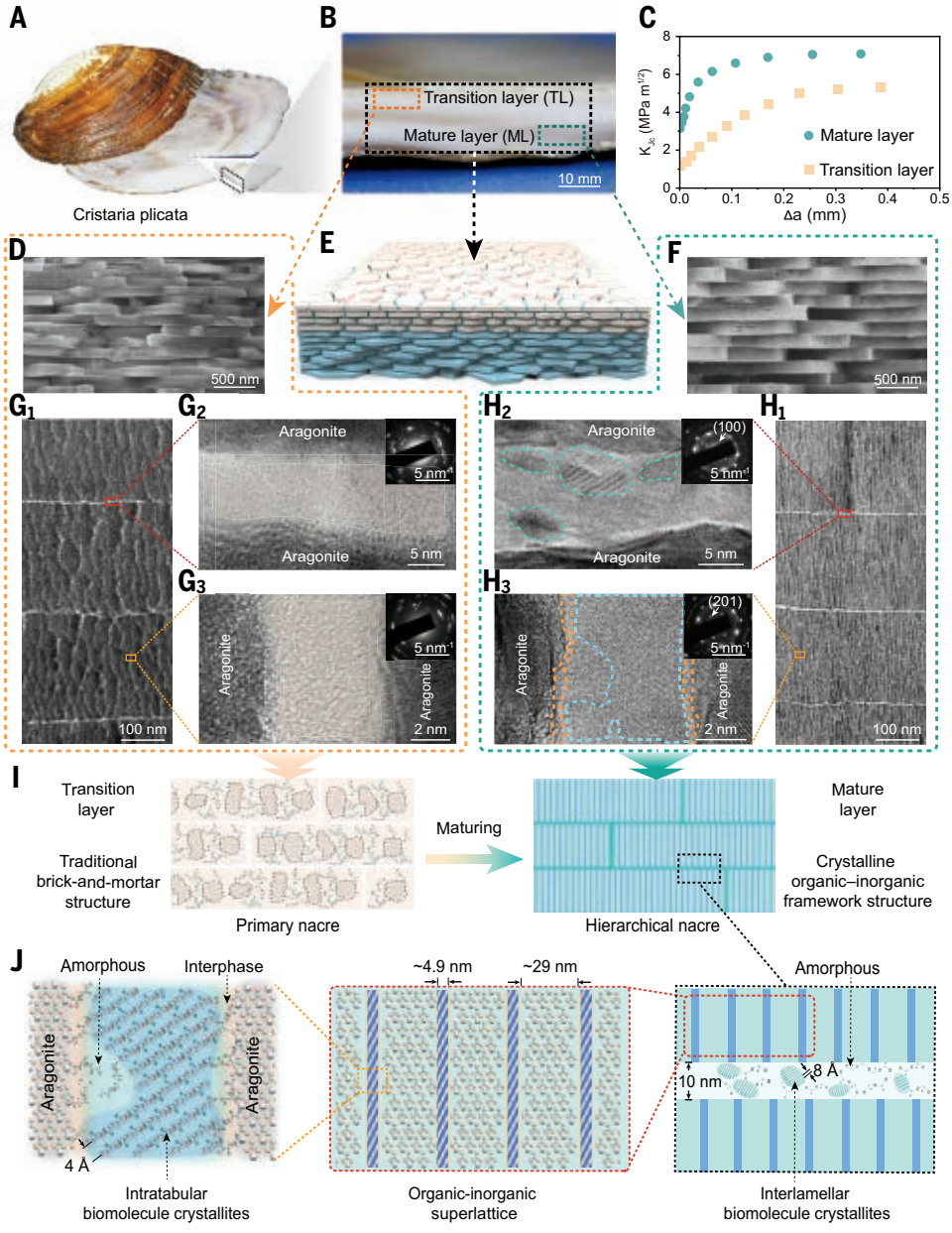


Fig. 1. Structural comparison between the ML and TL in *C. plicata* nacre. (**A**) Optical photo of *C. plicata* nacre. (**B**) Optical photograph of a *C. plicata* nacre bulk sample harvested from the lower half of the shell showing the TL at the top and the ML at the bottom. (**C**) Rising crack-propagation resistance curves of the ML and TL evaluated by the steady-state fracture toughness (K_{Jc}). (**D**) Cross-sectional SEM image of the TL. (**E**) Schematic diagram of *C. plicata* nacre in cross-section. (**F**) Cross-sectional SEM image of the ML. (**G_1**) Cross-sectional TEM image of the TL. (**G_2**) Enlarged HRTEM and electron diffraction images from the red box in G_1. (**G_3**) Enlarged HRTEM and electron diffraction images from the orange box in G_1. (**H_1**) Cross-sectional TEM image of the ML. (**H_2**) HRTEM and electron diffraction images from the red box in H_1. The cyan area is the interlamellar biomolecular crystallites. (**H_3**) HRTEM and electron diffraction images from the orange box in H_1. The blue area indicates the IT-BCs. (**I**) The TL nacre, composed of a brick-and-mortar structure formed by interlamellar organic matrix and aragonite grains, transforms into the ML nacre with COIF structure, which consists of a biomolecular framework and aragonite nanorods. (**J**) Schematic illustration of the ML showing its hierarchical structure.

antiparallel β-sheet packing in silk II structures. Furthermore, the interlamellar diffraction features exhibit broader profiles and slightly larger d-spacings, indicating that the confined interlamellar organic network forms a coupled protein-chitin hybrid assembly (*19*). This transition underscores the formation of crystallization in the organic phase and its integration with aragonite nanorods during nacre maturation (fig. S11, A and B).

Thermogravimetric analysis confirmed that the two layers have essentially equivalent inorganic-to-organic ratios (fig. S11, C and D). However, comparison with the continuous organic framework of decalcified nacre revealed a nonuniform spatial distribution of organics within ML: ~48 vol % resides in the interlamellar regions (fig. S11E), whereas ~52 vol % is embedded within intratabular domains (figs. S12 and S13). These findings indicate that the ML deviates from the classical brick-and-mortar description (Fig. 1I). Although structural features such as mineral bridges (*20*), nanoasperities (*21*), inorganic nanograins (*22*), and interlamellar organics (*9*) have been identified in previous studies, they mostly represent localized features associated with the inorganic phase, with the organic phase traditionally viewed as a passive amorphous adhesive. We therefore propose that the ML is more appropriately described as an interconnected hierarchical COIF, where oriented aragonite nanorods interlace with nanoconfined IT-BCs to form a continuous three-dimensional (3D) network (Fig. 1J). Within this framework, organics serve not merely as passive adhesives but also as active structural elements anchored within aragonite tablets to seamlessly integrate into a unified, hierarchical composite.

Mechanical behavior characterization

The mechanical properties of the ML and TL were systematically compared. In both tensile (Fig. 2A) and bending tests (Fig. 2B), the ML exhibited higher modulus ($E_{tension}$: 62 ± 3.6 versus 53 ± 3.2 GPa; $E_{bending}$: 58 ± 4.7 versus 45 ± 4.2 GPa) and strength ($\sigma_{tension}$: 220 ± 9.3 versus 170 ± 8.9 MPa; $\sigma_{bending}$: 180 ± 6.1 versus 140 ± 7.6 MPa) than the TL. Nanoindentation further confirmed that the ML has a higher surface modulus than the TL ($E_{surface}$: 108 ± 8.8 versus 89 ± 6.5 GPa) (Fig. 2C). Although unnotched samples showed only modest increases in strength with maturation, a slope change in the initial stage of bending tests was observed. In addition, tensile tests indicated a slightly higher toughness in the TL ($T_{tension}$: 1.4 ± 0.08 MJ/m^3) compared with the ML ($T_{tension}$: 1.2 ± 0.06 MJ/m^3). By contrast, fracture mechanics assessments revealed a pronounced enhancement in fracture toughness. Single-edge notched beam tests demonstrated that the ML's maximum fracture toughness (K_{Jc}) reached 7.1 ± 0.2 MPa·m$^{1/2}$, which is 1.4 times that of TL. Its fracture energy release rate (G_{Ic}) and plastic zone radius (r_p) were nearly three times and five times, respectively, that of the TL (G_{Ic}: 176.2 versus 23.4 J/m^2, r_p: 51.1 versus 9.3 μm), indicating substantially greater resistance to crack propagation (fig. S14) (*23*). Under cyclic loading, the ML endured >10^5 cycles at 80% of its maximum bending load while retaining 91% of

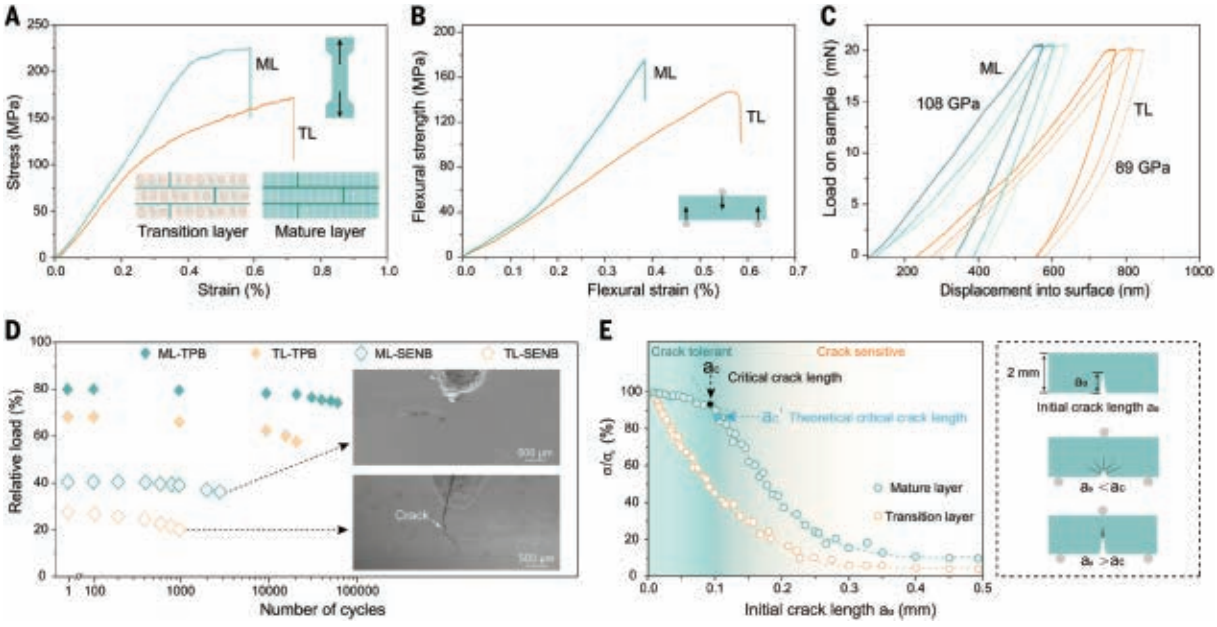


Fig. 2. Comparison of mechanical properties between the ML and TL in *C. plicata* nacre. (**A**) Tensile stress-strain curves of the ML and TL. (**B**) Flexural stress-strain curves of the ML and TL. (**C**) Depth-dependent nanomechanical response of the ML and TL by nanoindentation. (**D**) Relative load versus cycle number curves for $\varphi_{Max}/\varphi_0 = 0.8$ (φ_0, three-point bending maximum load, $R_0 = 0.25$) in three-point bending test and single-edge notched beam test mode (φ_0, single-edge notched beam maximum load, $R = 0.25$). (**E**) Crack tolerance in the ML and TL, expressed as normalized strength (σ/σ_0) versus initial crack length (a_0), is quantified by the critical crack length (a_c), the threshold below which cracks do not degrade material strength.

its strength, which was five times higher than that of the TL (Fig. 2D and fig. S15) (*24*). SEM observations confirmed these findings: Cracks propagated substantially from notches in the TL after ~200 cycles, whereas the ML's notch remained structurally intact until ultimate failure at 3000 cycles. Fatigue crack growth analysis further confirmed the superior fatigue resistance of the ML, with a higher threshold stress intensity factor range (K_{th}: 2.1 ± 0.2 versus 1.2 ± 0.1 MPa·m$^{1/2}$) (fig. S16).

To elucidate the underlying basis of the observed mechanical difference, we examined the crack-tolerance characteristics of the two layers (*25*). In brittle materials, fracture occurs catastrophically when the initial crack length (a_0) exceeds the critical crack length (a_c), at which point the stress intensity factor (K_I) surpasses the critical stress intensity factor (K_{Ic}), resulting in rapid crack propagation at stresses well below the ultimate tensile strength (σ_{uts}). Conversely, when the initial crack length (a_0) is below a_c, the stress concentration is insufficient to drive crack growth, allowing the material to withstand stresses approaching its σ_{uts}. Thus, a_c serves as a quantitative measure of crack tolerance, marking the transition from strength-controlled failure to brittle fracture. Based on the σ_{uts} and K_{Ic}, the theoretical critical crack length (a_c', calculated in eq. E19 in the supplementary materials) value is ~101 μm for the ML and ~10 μm for the TL, representing a 10-fold difference. Despite comprising only ~5% organic content, the ML attains a critical crack length comparable to that of polymeric materials, underpinning its exceptional crack tolerance (*26*).

Experimental tests on specimens with varying initial crack lengths validated these predictions (Fig. 2E). In the ML, fracture strength exhibited two distinct regimes: for small cracks, the strength decreased only modestly with increasing a_0; once a_0 surpassed the transition point (a_c), the strength dropped sharply. The experimentally determined a_c closely matched the theoretical a_c', confirming the model's accuracy. By contrast, the TL showed no clear strength-controlled regime, consistent with its much lower a_c. Collectively, these findings demonstrate that despite sharing similar macroscopic architecture and overall toughness, the ML and TL have markedly different mechanical behaviors.

Cross-scale synchronized deformation by the shear-pinning model

To investigate the distinct deformation behaviors of two layers, we combined digital image correlation (DIC) and in situ ultra-small-angle x-ray scattering (USAXS) (fig. S17). In the TL, DIC revealed pronounced displacement localization beneath the notch during bending (Fig. 3A), consistent with concentrated crack-tip deformation and nonuniform strain distribution across the thickness (Fig. 3B). By contrast, the ML exhibited a more spatially distributed displacement field with cross-scale synchronized deformation spanning multiple structural layers (Fig. 3, C and D). The upper layers underwent tension, whereas the lower layers experienced compression, with both stress states more evenly distributed across the thickness. This suggests that the COIF architecture effectively redistributes deformation and alleviates crack-tip stress concentration. Complementary USAXS analysis at the nanoscale supported these findings in the TL (Fig. 3E): Interlamellar organics experienced shear deformation, resulting in an ~20% increase in tablet spacing under strain (Fig. 3F). Conversely, in the ML (Fig. 3G), peak positions and spacing (~190 nm) remained unchanged (Fig. 3H), demonstrating that the COIF structure constrains localized deformation and facilitates efficient stress transfer.

Building on these observations, we developed finite-element models to analyze the deformation behavior of TL and ML structures (fig. S18). In the models, aragonite tablets were represented as assemblies of rods and organics with varying aspect ratios, with the lowest ratio corresponding to ML and the highest to TL. In the TL model (Fig. 3I), deformation was primarily governed by interfacial shear within the interlamellar organics, resulting in highly localized strain and pronounced displacement concentration near the crack tip, consistent with DIC observations (*27*). As the aspect ratio decreased, the displacement gradients at the crack tip progressively diminished (Fig. 3J), with the ML structure exhibiting up to 73% reduction in displacement gradients compared with the TL (fig. S19). Finite-element model simulations also provided estimates of the elastic moduli for the organic phases: The interlamellar organics displayed an effective modulus of 7 GPa, whereas

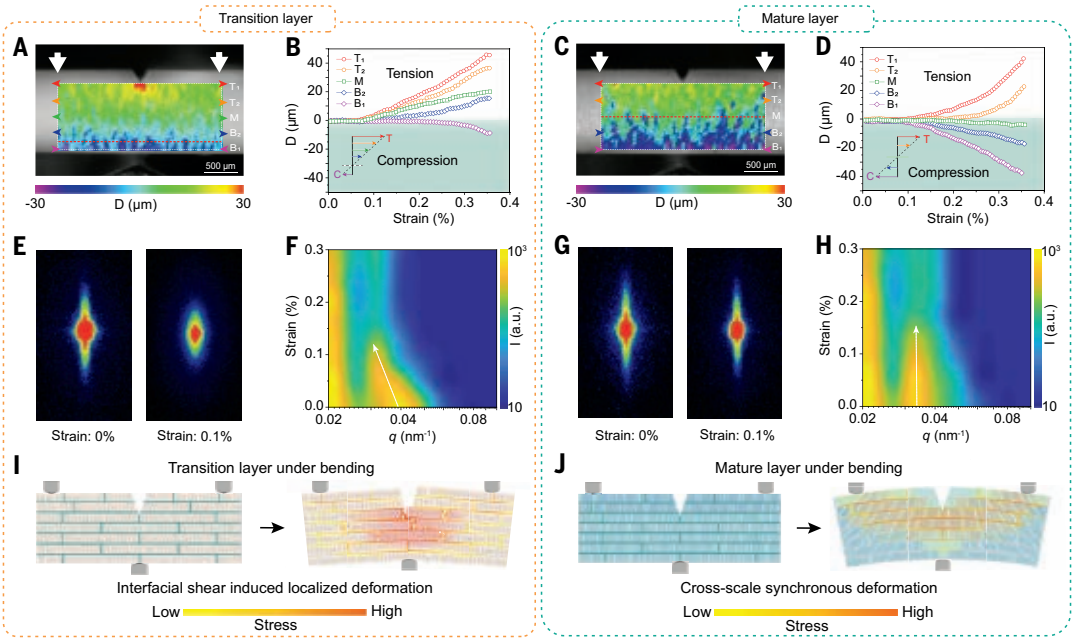


Fig. 3. Deformation mechanisms in the TL and ML of *C. plicata* nacre. (**A**) DIC results for a notched TL sample, showing substantial stress concentration below the notch. (**B**) Displacement versus bending strain along the length for T_1, T_2, M, B_2, and B_1 indicated in the boxed region of (A). (**C**) DIC results for a notched ML nacre sample revealing no obvious stress concentration beneath the notch. (**D**) Displacement versus bending strain along the sample length at positions T_1, T_2, M, B_2, and B_1 indicated in the boxed region of (C). (**E**) Evolution of the 2D USAXS pattern during bending of the TL. Inset values denote applied flexural strains. (**F**) Contour maps of 1D-integrated USAXS profiles for the TL. The region below the notch exhibits increased interlayer spacing due to localized slip, causing the scattering peak to shift to a lower vector (q) and its stabilization at ~0.032 nm^{-1} for bending strains exceeding 0.1%. (**G**) Evolution of the 2D USAXS pattern during the bending of the ML. Inset values correspond to applied flexural strains. (**H**) Contour maps of 1D-integrated USAXS profiles for the ML. The interlayer spacing beneath the notch remains constant due to cross-scale synchronized deformation, with the scattering vector (q) persisting at ~0.033 nm^{-1}. (**I**) Schematic illustration of interfacial shear deformation within the organic matrix during bending of the TL, leading to stress concentration at the notch tip. (**J**) Schematic illustration of cross-scale synchronized deformation of the organic phase within the COIF architecture during ML bending, resulting in the redistribution of stress away from the notch tip and a substantial reduction in peak stress. a.u., arbitrary units.

the intratabular organics reached 20 GPa, with these values substantially higher than those measured from recrystallized organic extracted from nacre (fig. S20). This enhanced stiffness arises from nanoconfinement effects, in which biomolecular chains confined within the nanometer scale interlamellar and intratabular spaces experience restricted conformational freedom and dynamic rearrangements, thereby increasing their apparent modulus (*11*, *28*). Consequently, nacre attains a high overall modulus despite its substantial organic content.

On the basis of these finite-element model insights, we propose a shear-pinning model of the COIF architecture to explain the ML's superior crack tolerance relative to the TL. In the TL, localized deformation primarily arises from shear stress transmitted through the interlamellar organics, where the inherently weak organic–inorganic interphases limit further resistance to crack propagation (fig. S21). Such deformation behavior is consistent with previously proposed nacre-toughening mechanisms that primarily operate through localized interfacial energy dissipation during crack growth (*1*, *21*). By contrast, in the ML, the nanoconfined IT-BCs act as nanoscale pins that anchor adjacent tablets and enable cross-scale synchronized deformation, thereby redistributing stress concentration ahead of the crack, thus suppressing crack initiation and delaying crack instability (fig. S22). This cross-scale synchronized deformation mechanism underlies the ML's substantially higher crack tolerance relative to the TL. Notably, this deformation behavior is also consistent with the transition observed in the initial strain regime of the ML-bending tests shown in Fig. 2B, suggesting that cross-scale synchronized deformation becomes increasingly pronounced during bending, leading to a nonlinear increase in apparent stiffness.

High-modulus toughening at critical slip length

Gao *et al.* (*8*, *29*) proposed that in nacre, the inorganic crystallites must withstand large tensile stresses without fracturing. They developed a simple model, based on the Griffith criterion, to define a critical length scale for these crystallites above which fracture strength becomes size dependent and failure is governed by crack-tip stress concentration, whereas below this scale, the presence of cracks no longer compromises the material's strength. Their estimate suggests an inorganic platelet thickness of ~30 nm. However, experimental confirmation of such a nanoscale structure in nacre had remained elusive. We quantified the characteristic dimensions of the intratubular units using HRTEM, revealing that the aragonite nanorods have a characteristic scale of 29 ± 3.9 nm, consistent with Gao *et al.*'s predicted critical length (Fig. 4A). Furthermore, nanoconfined IT-BCs act as high-modulus biomolecular crystallites fulfilling the Griffith criterion. Calculations based on this criterion indicate that it has a characteristic size of 4.9 ± 0.4 nm, consistent with our observations (Fig. 4B and eq. E21 in the supplementary materials). Therefore, we propose a critical slip length effect in which both the organic and inorganic crystallites, having reached their critical scales, exhibit near-theoretical modulus and strengths. When the relative slip between the organic and inorganic crystallites reaches its limit, cracks are unable to penetrate the crystallites and instead remain confined to their interphases.

Under these conditions, the interphases between aragonite nanorods and IT-BC domains constitute the weakest link and govern subsequent crack behavior (fig. S23). To investigate this, we used a modified Orowan model to analyze crack propagation in the ML (*8*, *30*). Crack extension requires elastic strain energy (W_e) to overcome the surface energy cost (W_s) associated with the creation of new interphases (*31*, *32*). According to the Griffith and Orowan criteria, the critical stress (σ_c) of the amorphous interphase organics is lower than that of the crystalline domains, rendering the interphase the preferential site for crack initiation (Fig. 4C). However, further crack propagation necessitates additional energy dissipation through shear-induced plastic deformation and disruption of molecular chains within the amorphous interphase. This process generates a

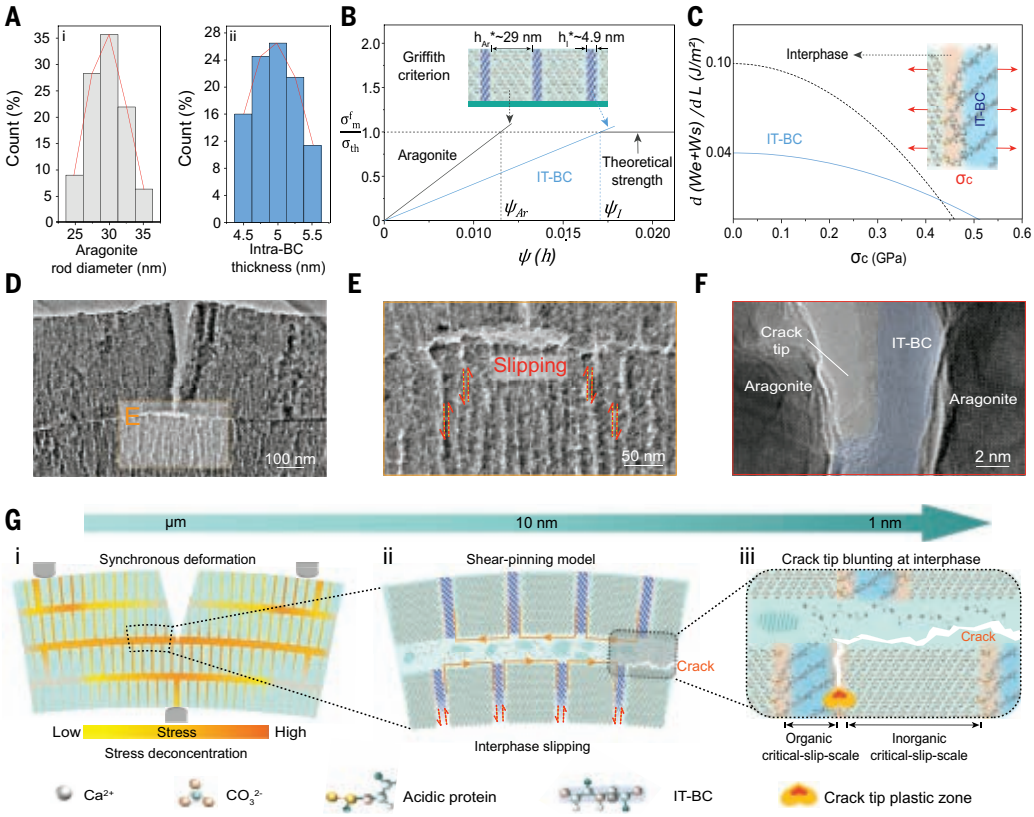
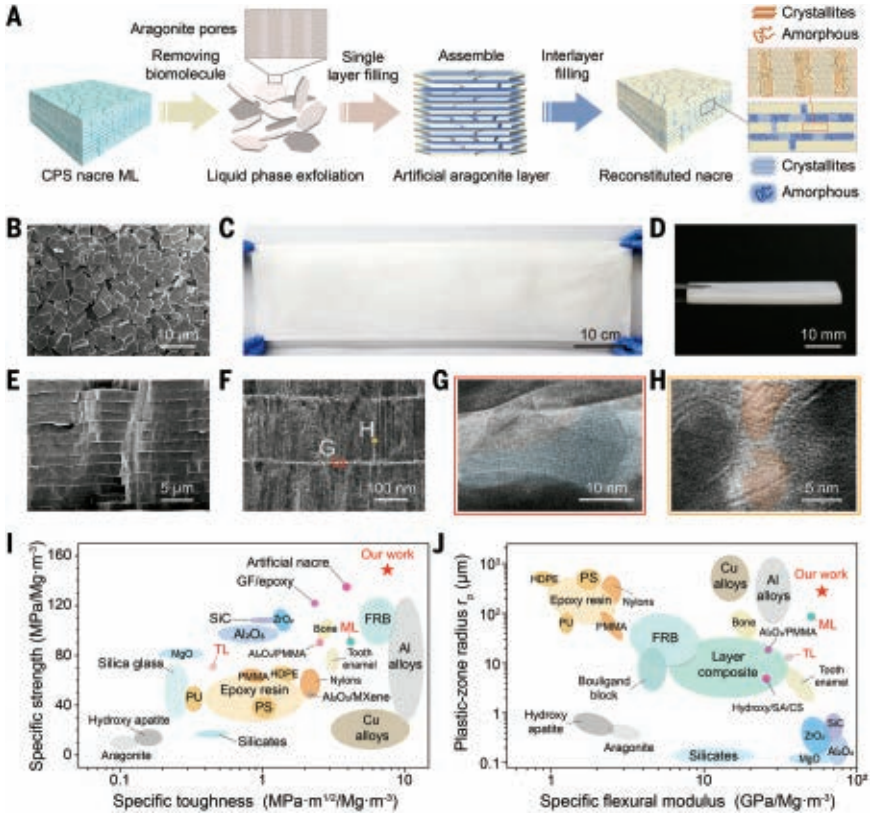


Fig. 4. High-modulus toughening mechanisms in the ML. (**A**) Size distribution of inorganic aragonite nanorod diameter and the IT-BC thickness. (**B**) The aragonite nanorod diameter and IT-BC thickness are consistent with the Griffith criterion. (**C**) Critical stresses of the IT-BCs and the intratablet interphase were calculated based on the Griffith and Orowan criteria, respectively. (**D**) HRTEM image of cross-scale synchronized deformation in the ML. (**E**) Enlarged HRTEM image from the orange box in (D). The aragonite rod exhibited interphase slip during the synchronized deformation bending process. (**F**) HRTEM characterization showing that the interphase-propagating crack tip undergoes substantial blunting due to nanoconfined strain energy dissipation. (**G**) Schematic illustration of the ML showing its deformation behavior across different scales.

Fig. 5. Fabrication and characterization of reconstituted nacre. (**A**) Schematic illustration of the assembly process for bulk reconstituted nacre. (**B**) SEM image of the regenerated aragonite tablets. (**C**) Large-area composite films fabricated by layer-by-layer assembly of the regenerated aragonite tablets and a crystalline polymer. (**D** and **E**) Optical photo (D) and cross-sectional SEM image (E) of the reconstituted nacre. (**F**) TEM image revealing the COIF structure in the reconstituted nacre. (**G** and **H**) Resin between (G) and inside (H) the regenerated aragonite tablets. (**I** and **J**) Ashby-type plots of specific strength versus specific toughness (I) and plastic zone radius r_p versus specific flexural modulus (J) for representative materials, including metal alloys, glasses, engineering composites, and the reconstituted nacre composites developed in this work. Source data for (I) and (J) are listed in table S6 in the supplementary materials.



plastic zone at the crack tip, which blunts the crack front. In addition, cross-scale synchronized deformation effectively reduces stress concentration, thereby expanding the plastic zone and impeding crack propagation. HRTEM observations confirmed this mechanism, revealing plastic zones at crack tips (Fig. 4, D and E). Under applied stress, interfacial slipping between aragonite nanorods and IT-BC domains dissipates energy through shear deformation, whereas microcracks remain confined within individual tablets. This efficient redistribution of strain energy promotes crack-tip blunting and leads to rapid crack arrest (Fig. 4F and fig. S24).

Collectively, these findings reveal a high-modulus toughening mechanism in the ML driven by a hierarchical COIF. This architecture exploits the critical slip-length effect to approach the theoretical limits of stiffness and strength, whereas nanoconfined biomolecular crystallites function as nanopins that reduce stress localization and enable cross-scale synchronized deformation (Fig. 4G). Ultimately, the crack formed by the external load becomes blunted at the interphase. It thereby endows the mature layer with exceptional crack tolerance, as well as an unprecedented combination of high elastic modulus and fracture toughness.

Bioinspired reconstruction of the COIF in materials engineering

The potential of the COIF architecture in materials engineering is further advanced by a bioinspired strategy for fabricating nacre-mimetic composites (Fig. 5A). Using natural nacre as a template, biomacromolecules were removed through thermal treatment (260°C), followed by chemical exfoliation in *N*-methyl-2-pyrrolidone to produce single-layer aragonite platelets with lateral dimensions of ~5 μm (Fig. 5B). A low-viscosity, highly crystalline copolymer (polystearic-co-lauryl methacrylate) (PSMA-PLMA) was then infiltrated to refill nanopores and reconstruct the COIF. The monomer infiltrated uniformly into the aragonite nanopores and polymerized through a spreading-assisted in situ process, yielding large-area films (Fig. 5C). These films were stacked with biphenyl epoxy resin as the interlamellar adhesive and consolidated by hot pressing at 230°C (Fig. 5D and fig. S25). SEM imaging revealed dense cross-sections (Fig. 5E), whereas TEM confirmed uniform occupancy of PSMA-PLMA within the aragonite rods (Fig. 5H and fig. S26). Electron diffraction exhibited crystalline rings characteristic of both epoxy and PSMA polymers (figs. S27 and S28), indicating the successful construction of the COIF architecture (figs. S29 and S30).

The reconstituted nacre demonstrated exceptional mechanical performance, retaining 91% of its strength after 100,000 bending cycles (figs. S31 and S32). Variations in the crystalline-amorphous polymer ratio revealed that reduced crystallinity substantially shortened fatigue life: The low-modulus amorphous phases functioned more as adhesives, increasing crack sensitivity (figs. S33 and S34). This demonstrates synergistic toughening by a crack-bridging mechanism. Compared with samples without PLMA infiltration or with low crystallinity, the reconstituted nacre achieved a bending strength of 290 ± 5.1 MPa and a modulus of 43 ± 2.1 GPa, representing 2.5- and 2.0-fold increases, respectively, which highlights the COIF structure as the origin of these superior mechanical properties (fig. S35). In SLE1 (SMA-LMA-Epoxy-1), the addition of single-walled carbon nanotubes further enhanced the ultimate strain; its bending strength reached 293 ± 5.8 MPa and its fracture toughness K_{Ic} reached 7.2 ± 0.2 MPa·m$^{1/2}$, with a rising resistance curve K_{Jc} up to 12.8 ± 0.4 MPa·m$^{1/2}$ (fig. S36). Benchmarking against representative composites and metals, the reconstituted nacre outperformed most high-performance materials in both specific strength and toughness (Fig. 5I and fig. S37). At a modulus of 43 ± 4.4 GPa, the energy release rate (G_{Jc}) reached 1190 J/m^2, and the radius of the plastic zone (r_p) during bending reached 112 μm, three orders of magnitude larger than ceramics of similar stiffness (Fig. 5J). In addition, the a_c of the reconstituted nacre increased to ~200 μm, demonstrating exceptional crack tolerance.

In summary, we reveal a high-modulus toughening mechanism in the mature nacre of *C. plicata* driven by a COIF. This architecture leverages the critical slip-length effect to approach the theoretical limits of stiffness and strength, whereas nanoconfined biomolecular crystallites function as nanopins that reduce stress localization and enable cross-scale synchronized deformation, culminating in exceptional fracture toughness. When applied to reconstituted nacre, this strategy achieves outstanding performance (E ~43 GPa, K_{Jc} ~12.8 MPa·m$^{1/2}$, a_c ~0.2 mm, r_p ~112 μm), effectively decoupling stiffness from toughness. These findings establish a broader COIF-inspired design principle for robust bioinspired materials that is extendable to various nanomaterials with confined nanoscale architectures.

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SUPPLEMENTARY MATERIALS

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POLAR WANDER

Quadrupolar sea level fluctuations reveal episodes of rapid polar wander

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True polar wander (TPW) is presently occurring at a rate of ~10 centimeters per year, exceeding the mean rate of differential plate motion. Whether this rate is transient and negligible on geological timescales or whether Earth has experienced rapid TPW lasting millions of years is unclear. Paleomagnetic evidence has been used to propose fast episodes of TPW, but limitations of the paleomagnetic record have precluded definitive results. Rapid TPW will induce global sea level fluctuations associated with relative changes in centrifugal potential, offering an independent test. We evaluated this prediction using continental flooding reconstructions and identified multiple robust quadrupolar patterns since 320 million years ago, confirming that protracted rapid TPW has occurred on Earth and that couplings among Earth's rotational dynamics, mantle processes, and surface environments can be strongly episodic.

Earth tends to keep the axis of its maximum moment of inertia aligned with the spin axis to minimize rotational energy at constant angular momentum. When surface or mantle dynamics redistribute mass on the planet, the inertia tensor is perturbed and the entire solid Earth (the crust and mantle together) reorients to restore this alignment (*1*). To an external observer, true polar wander (TPW) would appear as a coherent rotation of the solid Earth relative to the fixed angular momentum vector. Because Earth's core and zonal climate belts are locked to the spin axis, TPW causes the solid Earth to rotate between them (*2*). Such events may thus have profound impacts on climate, the biosphere, and the geodynamo (*3*–*6*), depending on their rate and amplitude. A key question is whether Earth is susceptible to brief episodes of rapid, high-amplitude TPW, as has been controversially proposed for the Cretaceous (*7*–*9*), Jurassic (*10*–*12*), various intervals of the Paleozoic (*13*–*15*), and the Precambrian (*16*–*19*).

Because Earth's surface is continuously reshaped by differential tectonic motions, distinguishing TPW from other kinematic signals is inherently challenging. On geologic timescales, TPW can be inferred by comparing paleomagnetic and hotspot reference frames: Paleomagnetic data record both TPW and relative plate motion, whereas hotspot reference frames ideally capture only the latter (*20*, *21*). However, there is substantial evidence that hotspots move independently, complicating the isolation of TPW signals by this approach (*22*–*25*). Additionally, the oldest robust hotspot track of likely deep-mantle origin can only be traced back to ~130 million years ago (Ma) (*26*), precluding the use of hotspot reference frames in deeper time. TPW can alternatively be estimated from the residual of integrated global plate motions derived from kinematic models (*27*, *28*), but the extant fraction of lithosphere shrinks markedly backward in time. Before 200 Ma, only continental lithosphere remains intact, and for intervals where this lithosphere is bound up in large coherent assemblies (e.g., Pangea), it becomes intractable to distinguish TPW from meridional components of plate motion (*29*). In the absence of robust reference frames, a third, simplified approach infers TPW from abrupt shifts between successive paleomagnetic poles (*9*, *10*). Although all these methods rely directly on the paleomagnetic record, they have yielded divergent interpretations of TPW in the Cenozoic and Mesozoic—as being entirely negligible (*24*, *27*, *30*), substantial but persistently slow (*31*), or episodic and characterized by rapid, large amplitude shifts (*9*, *10*).

We explored a fundamentally different approach to detecting episodes of rapid TPW that exploits records of sea level change. Owing to Earth's rotation, the reference geoid has an approximately oblate form, as reflected by the equatorial bulge (Fig. 1). During a TPW event, the solid Earth rotates relative to the reference geoid and seeks to relax into alignment with the new equilibrium form. However, the viscosity of the mantle and elastic stresses in the lithosphere delay this relaxation, whereas the hydrosphere responds immediately (*32*, *33*). Consequently, regions of the solid Earth that are displaced away from the rotational equator experience transient regression, whereas regions that are displaced toward it experience transgression (*32*, *34*, *35*). Globally, this results in a pattern of sea level change described by a degree-2, order-1 spherical harmonic (Y_{21}; a tesseral quadrupole) with two nodal planes: one lying along the rotational equator and the other perpendicular to the first and framed by the planetary spin and TPW axes (Fig. 1). Sea level changes will be maximized in the areas furthest from both planes (i.e., at mid-latitudes and longitudes 90° from the TPW axis), with their sign antisymmetric across both planes.

Signals of TPW in sea level dynamics

Building on earlier conceptual work (*34*), Mound *et al.* (*36*) first proposed to exploit these predicted sea level changes to test hypothesized episodes of rapid TPW by comparing the outcome of numerical forward models of Earth's viscoelastic response to TPW forcing against stratigraphic estimates of sea level. However, the amplitude of modeled sea level change is sensitive to the specified mantle viscosity and lithosphere thickness (*32*, *36*), and the quantification of sea level from stratigraphic records is challenging and generally only feasible in a few well-studied sections that become increasingly sparse as we go backward in time. By contrast, we exploited the amplitude-insensitive but spatially extensive records of sea level change captured by continental flooding maps. These maps document the time-dependent distribution of exposed and inundated land, and the differencing of sequential maps yields binary maps of continental transgressions and regressions. Continental flooding maps are available at a "global" (pan-continental) scale and a temporal resolution of 10 million years (Myr) back to 540 Ma (*37*), making it feasible to interrogate them for signals of rapid TPW throughout the Phanerozoic.

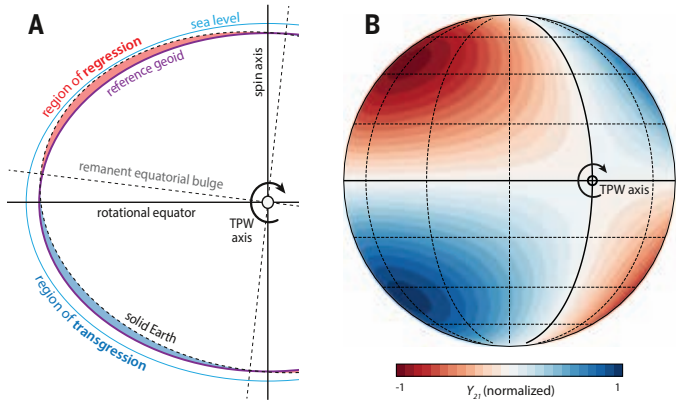

Fig. 1. Patterns of sea level change induced by TPW. (**A**) Cross section of Earth (with exaggerated oblate form) showing displacement of the solid Earth's transient equatorial bulge (with respect to the reference geoid) during a TPW event. (**B**) Normalized response of degree-2, order-1 spherical harmonic (Y_{21}); nodal planes are highlighted in black.

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For a given time, we tested whether the spatial distribution of coeval transgressions and regressions observed on the continents is consistent with the expected quadrupolar pattern induced by a TPW event. To this end, we used logistic regression in an inverse modeling approach to recover the best-fitting potential TPW axis and determine the significance of the TPW contribution relative to the null hypothesis that there is no TPW signal (see materials and methods). This approach does not recover the absolute rate of TPW directly but enables detection of intervals of enhanced TPW without requiring a priori knowledge of the event. Furthermore, forward modeling results (*35*) indicate that excitation of the Y_{21} patterns resolvable by the flooding maps used here requires TPW rates $\geq 0.6°$ Myr^{-1}, which we designate "rapid." The detection of Y_{21} patterns requires reliable information on the time-dependent positions of the observation sites. For latitudes, absolute positions are necessary, but for longitudes, relative positions are sufficient: The use of relative longitudes simply means that the recovered TPW axis will be referenced to the continents rather than fixed to the mantle. Relative longitudes are best constrained after the formation of Pangea, so we restrict our analysis to the last 320 Myr, and adopt the "zero-longitude" Africa assumption (*29*) for convenience.

Our assessment of the continental flooding dynamics every 10 Myr back to 320 Ma reveals four intervals during which a TPW contribution is significant at the 5% level (see materials and methods): the Oligocene to Miocene (30 to 20 Ma), the mid-Cretaceous (100 to 90 Ma), the Late Jurassic to Early Cretaceous (150 to 140 Ma), and the Early Jurassic (200 to 190 Ma) (Fig. 2). Two of these intervals correspond closely with intervals when especially rapid TPW has been proposed in the Mesozoic.

Episodic rapid TPW since 320 Ma

TPW in the Cenozoic is generally estimated to be muted [if detectable (*27*, *38*)] but associated with persistent clockwise rotations about axes located around 90°E (*21*, *31*). Notably, the only significant TPW inversion that we found for the Cenozoic, resulting from inversion of the 30-to-20-Ma difference map, corresponds to a clockwise rotation about a TPW axis located at 93°E (Fig. 3). This could reflect a brief episode of as-yet-unrecognized rapid TPW at this time, but among the four significant results observed in the last 320 Myr, this model is associated with the poorest fit to the observations and the smallest ratio between the modeled TPW and eustatic terms (Fig. 2). Throughout the remainder of the Cenozoic, if significant TPW occurred, it did so at moderate rates that did not noticeably affect sea level at the temporal resolution considered here.

In contrast to the muted activity of the Cenozoic, several episodes of extremely fast TPW ($\geq 2.5°$ Myr^{-1}) have been proposed in the Mesozoic on the basis of time-varying paleomagnetic pole comparisons. One main episode is hypothesized to have occurred in the Late Cretaceous, around 84 Ma (*7*, *8*), possibly as an oscillatory couple (*9*). An earlier episode is hypothesized to have occurred in the Late Jurassic between 160 and 148 Ma (*10*–*12*), with a retromotion phase perhaps extending into the earliest Cretaceous (*39*, *40*) (Fig. 2). However, the interpretations of the paleomagnetic data supporting these various proposed TPW episodes remain debated (*41*–*43*), and a preliminary analysis of sea level data was found incompatible with numerical models of a very fast ($\geq 3.2°$ Myr^{-1}) Late Cretaceous event (*44*). Integrated plate motion analyses (*28*, *29*) have also recognized episodes of enhanced TPW in the

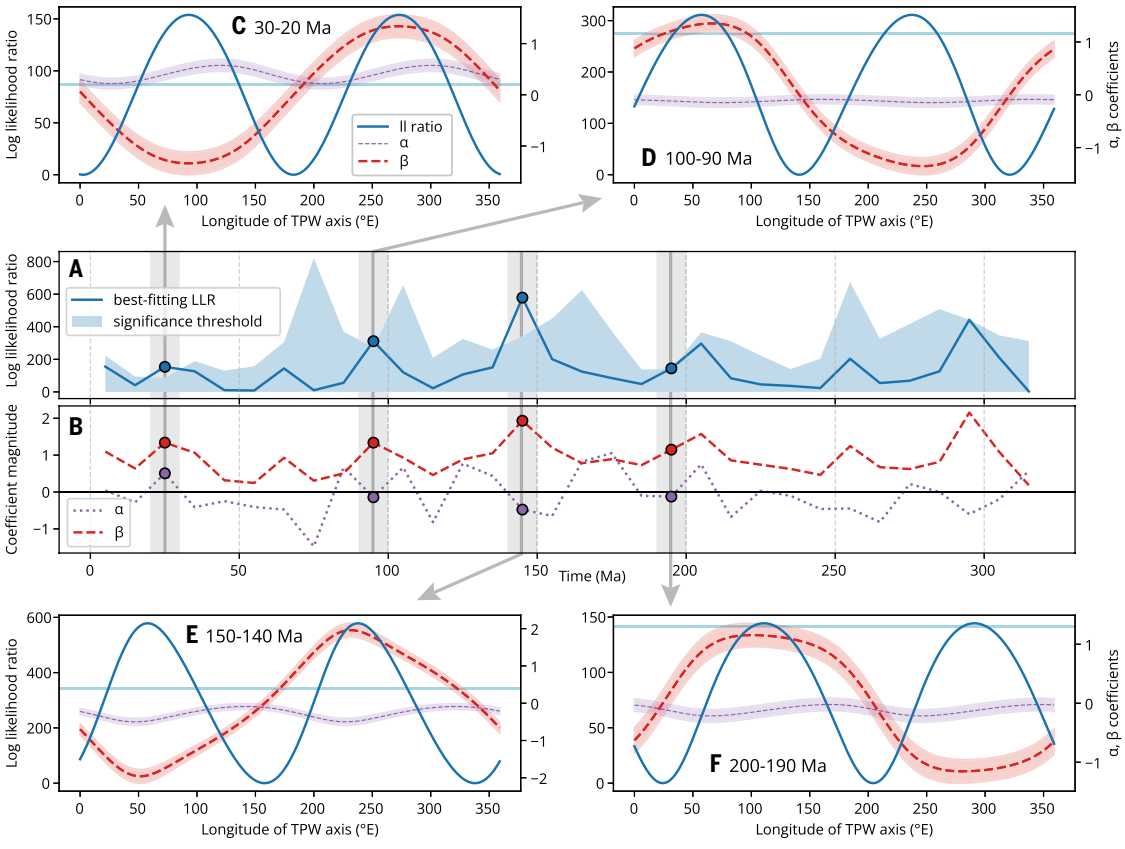

Fig. 2. Estimated contributions of TPW to sea level change since 320 Ma. (**A**) Log-likelihood ratio of best-fitting TPW model per time, compared against the significance threshold determined from permutation tests (see materials and methods). Intervals with significant results are highlighted by circles and gray vertical bands. (**B**) Specific magnitude of eustatic (α) and TPW (β) contributions to observed sea level changes. (**C** to **F**) Full inversion results for the four intervals where a significant TPW contribution is detected. Each panel shows the estimated parameters (α, β) and the log-likelihood ratio for each candidate TPW longitude. Light blue lines show the 95th percentile of the permutation tests that provide the critical statistic for assessing significance.

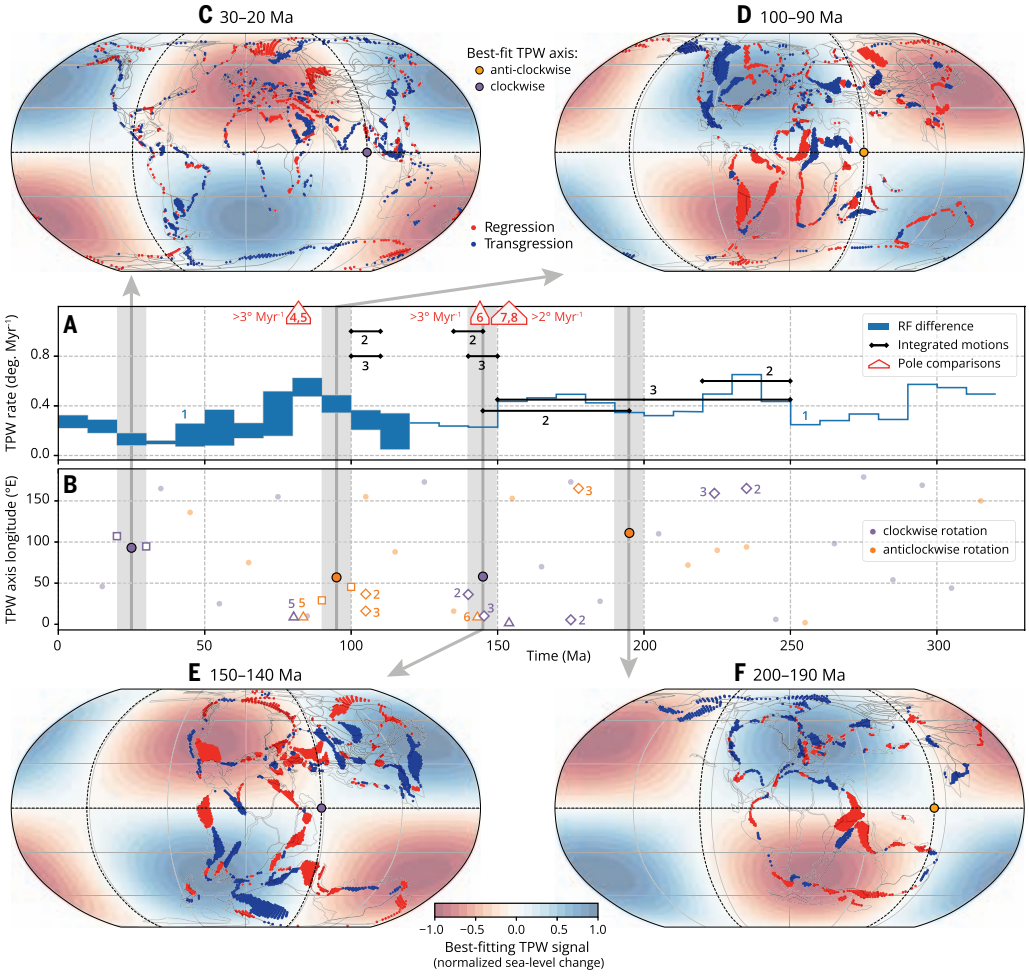

Fig. 3. TPW inferences since 320 Ma. (**A**) Estimated rates of TPW from reference frame (RF) differencing methods, the residual of integrated global motions from kinematic plate models, and comparisons of paleomagnetic poles. Numbers denote references: 1: (*31*), 2: (*28*), 3: (*29*), 4: (*8*), 5: (*9*), 6: (*39*), 7: (*10*), 8: (*12*). RF difference estimates of 0 to 120 Ma depict the range of results reported by (*31*); the earlier curve is based on the mantle frame of (*45*) in (*31*). (**B**) Estimated longitudes of TPW axis, colored by rotation sense. Large (small) circles highlight results that are (are not) significant at the 5% level. Open symbols denote other estimates from RF differencing (squares), integrated global plate motions (diamonds), and paleomagnetic pole comparisons (triangles) with numbers as above. The RF differencing estimates are with respect to mantle reference frames (see text). (**C** to **F**) Difference map reconstructions overlain on predicted Y_{21} values and nodal planes (dashed lines) with respect to the best-fitting TPW axis location.

mid-Cretaceous (110 to 100 Ma) and the Late Jurassic to Early Cretaceous (150 to 135 Ma), but with more moderate (although still rapid) estimated rates of 0.8° to 1.0° Myr^{-1}. By contrast, estimates from the differencing of paleomagnetic and hotspot reference frames back to 120 Ma (*31*) suggest only a slight elevation of TPW activity in the mid-to-Late Cretaceous (≤0.6° Myr^{-1}), with the exception of Doubrovine (*25*), who reported a burst of rapid TPW (~1° Myr^{-1}) between 110 and 100 Ma. A recent extension of the reference frame differencing approach back to 320 Ma [adopting a "tectonic rules"–based mantle reference frame (*45*) in lieu of hotspot tracks before 120 Ma] found no evidence of substantially faster rates of TPW during the Jurassic, with rates remaining ≤0.5° Myr^{-1} (*31*).

Our inversions corroborate the occurrence of rapid TPW events in the mid-Cretaceous and near the Late Jurassic–Early Cretaceous boundary. The 100-to-90-Ma flooding difference map exhibits widespread regression across South America, west and southern Africa, and eastern Asia, whereas it shows that North America and Europe experienced transgression (Fig. 3). These dynamics reflect minimal eustatic change (as reflected by low α values) and a dominant TPW contribution associated with an anticlockwise rotation about an axis at ~53°E. This is in close agreement with the estimated axis positions of mid-to-Late Cretaceous TPW inferred from integrated plate motions (*28*, *29*) and reference frame differencing (*31*) (the latter are reported in a mantle reference frame; restoring these to a paleomagnetic reference frame would move them ~5° to 20° eastward, closer to our estimate). Temporally, however, this event falls between the rapid TPW episodes inferred from integrated plate motions (110 to 100 Ma) and that proposed from pole comparisons at around 84 Ma (*8*, *9*). We propose that the broad consensus among these datasets—and in particular the strong substantiation from our independent sea level observations—establishes the veracity of a mid-Cretaceous rapid TPW event but that existing age uncertainties have aliased the signal among them. Notably, this finding circumvents uncertainties tied to hotspot motion and the limited global coverage of paleomagnetic data, which have impeded validation of this TPW event by existing methods (*22*, *24*, *42*). An important caveat is that a short-period (<10 Myr) oscillatory event, such as that proposed by Mitchell *et al.* (*9*), may induce sea level responses that largely self-cancel across adjacent 10-Myr time steps, reducing their detectability. Discrepancies in numerical calibration of the geological timescale near the Santonian-Campanian (*46*) boundary may also obscure our ability to resolve a TPW event at that time. Nevertheless, our results constrain the intervals over which such events

could plausibly have occurred and which could be further interrogated with detailed flooding reconstructions of finer temporal resolution.

The 150-to-140-Ma flooding difference map reveals transgressions in South America, Antarctica and eastern Asia, whereas it indicates that southwestern North America, western Europe, and eastern Australia experienced regression (Fig. 3). This distribution is approximately the inverse of that observed between 100 and 90 Ma and is associated with a significant TPW contribution from a clockwise rotation about an axis at ~58°E. This TPW contribution is the strongest of any observed since 320 Ma. The estimated TPW axis location again corresponds well with those estimated from integrated plate motions (*28*, *29*), and in this case the timing is also identical (Fig. 2). This timing also overlaps with a proposed retromotion phase associated with the so-called monster polar shift (*39*, *40*), but the sense of that proposed retromotion is the opposite of that assessed here. Again, we contend that the correspondence between the largely independent sea level dynamics and paleomagnetic constraints substantiates the occurrence of a Late Jurassic–Early Cretaceous rapid TPW event.

The final significant result, observed at 200 to 190 Ma, is associated with a regression in the southern part of a fragmenting Pangea while a transgression was experienced in its north (Fig. 3). The best-fitting TPW contribution is an anticlockwise rotation about an axis at ~111°E. TPW estimates from integrated plate motions (*28*, *29*) and the extended reference frame differencing approach (*31*) have identified moderate TPW activity (~0.4° to 0.6° Myr^{-1}) from 250 to 150 Ma. Both paleomagnetic approaches estimate those 250-to-150-Ma TPW episodes to have occurred about axes closer to 0°E longitude, but they also infer that a major reversal in the rotation sense occurred at ~200 Ma, coincident with the timing of our significant result. This coincidence in timing and discrepancy in axis orientation is notable and warrants further scrutiny; but as the inversion is only marginally significant and the model fit to the observations is poor, we prefer to interpret this result as possibly a chance outcome.

The absence of any significant TPW signals among our late Carboniferous and Permian inversions is particularly notable because estimations from integrated plate motions during this time span are limited by the fact that nearly all continental lithosphere was bound into the supercontinent Pangea, which exhibited strong meridional motion from 320 to 250 Ma. Consequently, TPW is not discriminable from meridional plate motion for these times [unless the TPW axis location is assumed a priori (*29*)], leading to critical unanswered questions about the nature of Pangea's northward migration (*47*). Given the size of the landmass, the lack of evidence of fringing subduction along its northern perimeter, and its particularly rapid northward pace during the late Carboniferous (>1° Myr^{-1} from 320 to 310 Ma), it has been argued that TPW was primarily responsible for the observed northward motion of the supercontinent (*47*, *48*). Our results challenge this hypothesis because there is no indication of a significant contribution to the sea level dynamics during this time and because the best-fitting TPW axis locations are far from an east-west orientation (Fig. 2), where they would otherwise be expected had the northward motion of Pangea been due predominantly to TPW. The result with the strongest log-likelihood ratio (300 to 290 Ma), which falls just beneath the significance threshold, is associated with a TPW axis near 0°E longitude, perpendicular to the axis describing the main (meridional) motion of the supercontinent.

These findings refute the view of TPW as negligible or persistently slow and highlight the need to consider TPW as an episodic control on sea level change (*32*, *35*) and likely other global environmental and biological dynamics (*3*–*6*). They further imply that similar episodes of fast TPW are likely to have punctuated earlier intervals of Earth history.

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SUPPLEMENTARY MATERIALS

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ELECTROCHEMISTRY

Dynamic multimodal cation-carbon interaction for efficient adiponitrile electrosynthesis

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Adiponitrile electrosynthesis from acrylonitrile hydrodimerization is critical for the nylon industry. However, the key step of single-electron acrylonitrile reduction demands toxic heavy-metal electrodes and leads to low energy efficiency. We developed a dynamic multimodal cation-electrode interaction strategy to enable efficient synthesis with environmentally friendly carbon electrodes. Optimal benzyltrihexylammonium cations create a water-depleted interface in aqueous electrolytes by squeezing the electric double layer through multimodal noncovalent interactions between the benzyl moieties of the cations and carbon electrodes. This strategy achieves selectivity comparable to the standard lead-based system but at substantially lower overpotentials, as demonstrated from the laboratory scale to a 159–square centimeter electrolyzer. This work underscores the importance of tailoring cation-centered microenvironments for sustainable, industrial organic electrosynthesis in more conductive aqueous systems.

Adiponitrile (ADN) is a central organic chemical for producing nylon 6,6, a prominent polymer used in automobiles, engineering plastics, and textiles (*1*, *2*). It also has emerging applications in producing hexamethylene diisocyanate (*3*, *4*), a critical component for high-performance polyurethane (*5*–*7*), and electrolyte additives for lithium-ion batteries (*8*–*10*). Accordingly, the global market has been valued at more than $10 billion and is anticipated to grow at a compound annual growth rate of 8.2% (*11*). ADN production often relies on the hydrocyanation of butadiene using the toxic hydrogen cyanide (HCN) (*12*, *13*). Emergingly stringent environmental requirements make the safe and sustainable use of HCN more challenging (*14*).

An alternative route is the electrochemical hydrodimerization of acrylonitrile (AN) (*15*, *16*). Since its early discovery in the 1950s and 1960s (*17*–*19*), it remains a viable option for ADN production and accounts for one of the largest applications of organic electrosynthesis (*20*–*24*). ADN electrosynthesis proceeds through cathodic AN hydrodimerization coupled with anodic water oxidation. AN first receives an electron to form the AN anion radical, which further couples with another AN toward ADN. Alternatively, quenching of the AN anion radical by water with further reduction forms propionitrile (PN) as the major side-product, whereas condensation with multiple ANs forms undesired trimer or oligomer side-products (*25*, *26*). The prerequisite of AN anion radical formation necessitates a high overpotential, which unavoidably promotes the competing hydrogen evolution reaction (HER) (*27*–*29*). Consequently, electrode materials with extremely high HER overpotentials, such as Pb and Cd, are necessary (*30*, *31*), but these metals suffer from emerging environmental concerns. Despite the maturity of plant-scale toxic metal sequestration, metal leaching into the electrolyte during operation generates hazardous wastewater requiring costly treatment. The inertness of these electrodes also causes large overpotentials and low energy efficiencies.

A promising candidate is a C-based electrode because of its low cost, high environmental friendliness, and high conductivity (*32*–*34*), but generally it is not considered attractive because of the parasitic HER and AN hydrogenation. To drive efficient ADN electrosynthesis and improve selectivity on C-based electrodes, it is necessary to precisely control the interfacial structure near the electrode (*35*–*38*).

Comparative ADN electrosynthesis performance on C versus Pb electrodes with various cations

In this study, we proposed a dynamic multimodal cation-carbon interaction strategy for tuning the electrode-electrolyte interfacial structure and enabling efficient ADN electrosynthesis on C electrodes (Fig. 1A). In the conventional ADN electrosynthesis system, quaternary ammonium salt (QAS) is an essential component (*39*, *40*). The potential-driven accumulation of QAS cations in the electric double layer can spontaneously enrich AN substrates for subsequent hydrodimerization. For instance, on Pb electrodes, the tetrabutylammonium cation (TBA^+) produced 73.2% ADN selectivity, whereas the tetramethylammonium cation (TMA^+) produced no ADN (Fig. 1B and fig. S1). However, monotonically increasing the alkyl lengths decreases ADN selectivity beyond butyls. This volcanic trend was consistent on C electrodes but with lower ADN selectivity than that observed on Pb electrodes. This selectivity markedly reduced over increasing overpotentials or current densities (Fig. 1C, leftmost bar for Bu_4N^+), making C-based electrodes unviable under industrial operating conditions. Nevertheless, a critical advantage of the C-based electrode was its lower overpotential by ~139 mV compared with the Pb system (Fig. 1D).

When we substituted one alkyl chain of the QAS cation with a benzyl group, we observed a substantial improvement in ADN selectivity. Under different alkyl lengths, all benzyl-bearing QASs exhibited higher ADN selectivities than their tetraalkyl counterparts (Fig. 1B). More relevantly, the selectivity peak under the volcanic dependence shifted toward longer alkyl chains, reaching >80% under $Bn(Hexyl)_3N^+$, which outperforms TBA^+. This improved selectivity could be generalized to various C electrodes with high graphitization degrees (fig. S2). C fiber paper shows the highest selectivity, with the graphite powder electrode showing comparable performance. The slightly lower ADN selectivities for other C electrodes are attributed to their higher HER activities (figs. S3 and S4).

Only when the alkyl chain length was extended to ≥8 did the ADN selectivity decline, with both higher trimer selectivity and expedited HER kinetics (Fig. 1B and fig. S5). Substituting benzyl with naphthylmethyl [$Nap(Hexyl)_3N^+$] also decreased the ADN selectivity (Fig. 1B and fig. S1, B8), which suggests that the appropriate cation-carbon interaction is necessary for the desired chemoreactivity. Furthermore, $Bn(Hexyl)_3N^+$ not only enhances the ADN selectivity but further reduces the overpotential by 133 mV, achieving higher current densities and yields at lower overpotentials (Fig. 1D). At −0.88 V versus the reversible hydrogen electrode (RHE), the ADN yield under $Bn(Hexyl)_3N^+$ reached ~56.9%, which was higher than the yields of 16.5 and 14.2% obtained with $Bn(Bu)_3N^+$ and Bu_4N^+, respectively (Fig. 1E). High ADN selectivities under benzyl-based QASs could be retained at higher overpotentials or higher current densities, which holds great promise for industrial applications.

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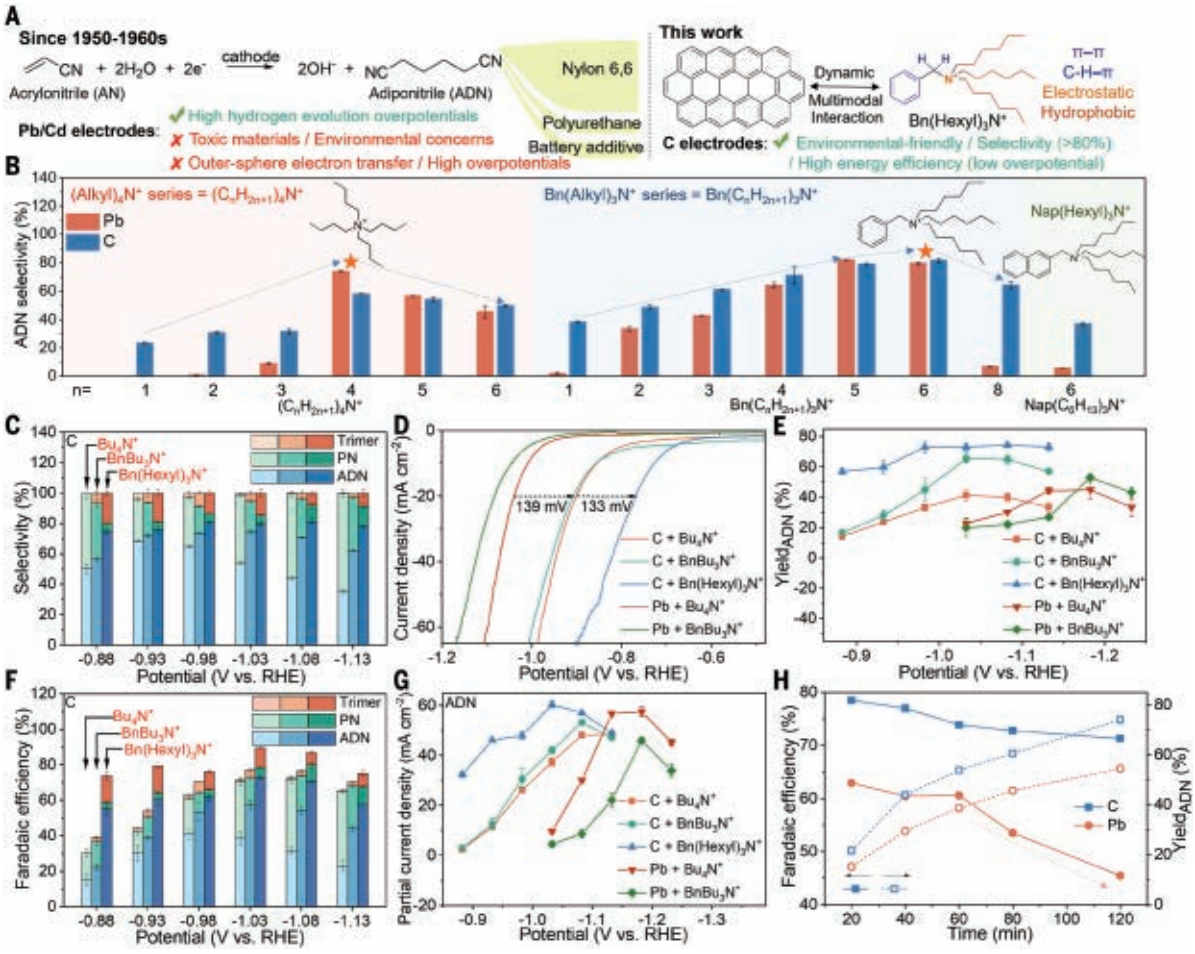

Fig. 1. Selective and efficient ADN electrosynthesis facilitated by multimodal cation-electrode interaction on C electrodes. (**A**) Schemes of the conventional ADN electrosynthesis and the dynamic multimodal cation-electrode interaction strategy for improving ADN electrosynthesis. (**B**) ADN selectivity on the Pb and C electrodes with different QAS cations. Electrolysis was performed at −1.03 V versus RHE on the C electrode for 2 hours and at −1.13 V versus RHE on the Pb electrode for 2 hours to ensure similar current densities. (**C**) Product selectivities on the C electrode with different QAS cations under various applied potentials. The electrolysis duration is 2 hours. (**D**) Polarization curves obtained with different combinations of electrodes and cations in the electrolyte with 0.3 M AN at a scan rate of 2 mV s^{-1}. (**E**) Yield of ADN over different potentials on the Pb or C electrode with different QAS cations. (**F**) FEs on the C electrode with different QAS cations under various applied potentials. (**G**) Partial current densities of ADN versus potential curves on the C and Pb electrodes with different QAS cations. (**H**) Time-dependent FEs and yield of ADN for the C/$Bn(Hexyl)_3N^+$ and Pb/Bu_4N^+ systems at −1.03 V versus RHE and −1.13 V versus RHE, respectively.

Faradaic efficiency (FE) is another crucial criterion. Under typical Bu_4N^+, the FEs on C-based electrodes were generally lower than those on Pb electrodes, especially under higher overpotentials (Fig. 1F and figs. S6 to S8). However, installing the benzyl moieties tremendously increased the FEs. The optimal FE on $Bn(Hexyl)_3N^+$ at −1.03 V versus RHE could reach 72.9%, which is higher than the conventional Pb system at this potential (58.8%) and under identical current densities (44.8%) (fig. S7). By integrating the FEs and current, the derived partial ADN current densities reached 59.5 mA cm^{-2} only at −1.03 V versus RHE (Fig. 1G). Notably, the peak ADN current densities were similar on C and Pb electrodes, which confirms their similar hydrodynamic properties.

Another indispensable advantage of the C/$Bn(Hexyl)_3N^+$ system is its high selectivity and FE under high conversions (Fig. 1H). Traditionally, ADN electrosynthesis requires a high AN concentration (>0.4 M), whereas decreasing the AN concentration markedly lowered the selectivity and FE (table S1). This inevitably limits the overall yield in a batch reactor. As the reaction proceeds with the consumption of AN, the Pb system showed much decreased FEs because of the lower AN concentration in late stages. Comparatively, the C/$Bn(Hexyl)_3N^+$ system almost maintained the ADN selectivity and FE over reaction progression.

Interfacial double-layer reconstruction induced by cation-carbon interaction

Previous understanding about the QAS function was restricted to creating hydrophobic interfaces by electrostatic enrichment for coadsorbing AN (*41*). However, in our system, the presence of possible cation-carbon interaction complicates the interfaces and warrants further mechanistic investigation. Differential capacitance was first analyzed by alternating-current cyclic voltammetry (ACV) and electrochemical impedance spectroscopy (EIS). Without AN, tetraalkyl QASs obviously decreased the capacitance, and further installing benzyl or naphthylmethyl could minimize the capacitance, likely because of a more compact double layer (Fig. 2A and fig. S9), but $Nap(Hexyl)_3N^+$ showed negligible potential-dependent responses, corresponding to a rigid interface that relates to its inferior performance. With AN, the capacitance increased by two orders of magnitude, attributed to AN anion radical formation (Fig. 2B and fig. S10). QASs with hexyl or octyl groups exhibited lower onset potentials compared with butyl counterparts, demonstrating a major contribution of alkyl chains for AN accumulation. These trends could also be observed in the EIS spectra on C electrodes (figs. S11 and S12) but were less obvious on Pb electrodes (fig. S13).

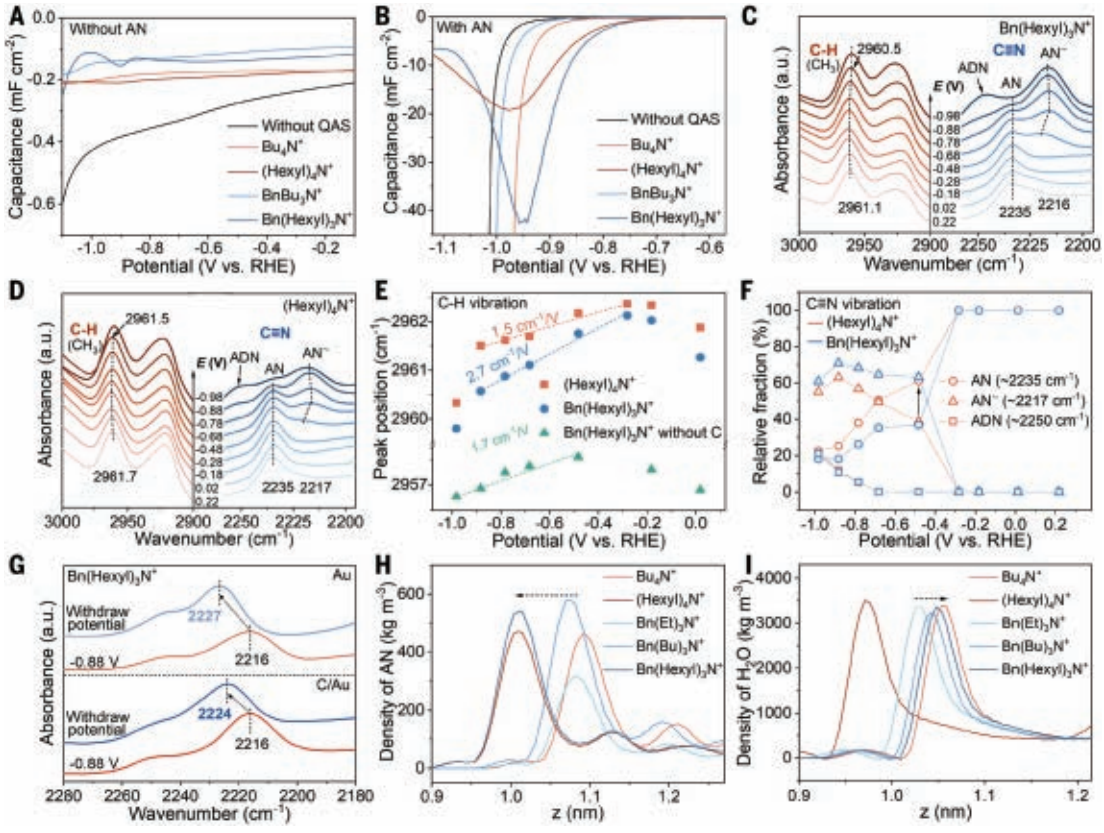

Fig. 2. Interfacial analysis under the cation-electrode interaction for ADN electrosynthesis. (**A** and **B**) ACV curves of different QAS cations on the C electrode without AN (A) and with AN (B). (**C** and **D**) In situ ATR-SEIRAS spectra on the graphene/Au electrode for $Bn(Hexyl)_3N^+$ (C) and $(Hexyl)_4N^+$ (D) under different potentials. a.u., arbitrary units. (**E**) Measured peak positions of the C-H vibration for different combinations of cations and electrodes under different potentials. (**F**) Calculated relative fraction of different CN vibrational peaks for different combinations of cations and electrodes under different potentials. (**G**) Comparison of the ATR-SEIRAS spectra of the Au and graphene/Au electrodes on withdrawing the potential from −0.88 V versus RHE. The graphene/Au electrode showed a smaller spectral shift. (**H** and **I**) MD-simulated distribution of AN (H) and H_2O (I) molecules under different QAS systems.

To elucidate the cation-electrode interaction, we performed in situ attenuated total reflection–surface-enhanced infrared absorption spectroscopy (ATR-SEIRAS) (Fig. 2, C to G). Because hexyl-based QASs are most effective in accumulating AN, we mainly compared $(Hexyl)_4N^+$ on graphene-coated Au, $Bn(Hexyl)_3N^+$ on graphene-coated Au, and $Bn(Hexyl)_3N^+$ on Au. All spectra showed the QAS C-H vibration within 2850 to 3000 cm^{-1}, and extrapolating the peak positions at ~2955 to 2965 cm^{-1} (terminal CH_3 asymmetric stretch) showed potential-dependent behavior (*42*). The peaks first blue-shifted from 0.12 to −0.28 V versus RHE and then red-shifted beyond −0.28 V (Fig. 2E). Such a Stark shift is an indicator of the electric field strength exerted on QASs (*43*). Benzyl incorporation and graphene coating increase the Stark tuning rate (Fig. 2, C to E, and fig. S14), suggesting an increased local electric field from a more compact cation layer and more uniform dipole alignment by benzyl-carbon interaction. The vibrational shift was highly reversible, excluding any irreversible reactions (fig. S15). Moreover, switching to $Nap(Hexyl)_3N^+$ almost eliminated C-H vibrational mode changes in the presence of graphene, suggesting its possible multilayer stacking to form a more rigid interface (fig. S16).

For CN vibration, free solvated AN peaked at ~2234 cm^{-1} (fig. S17). On applying potentials below −0.68 V, a peak at ~2216 cm^{-1} appeared and intensified with more negative potentials (Fig. 2, C and D, and fig. S18). According to its consistency with capacitance surging, this peak was assigned to the AN anion radical. On withdrawing the potential, this peak immediately shifted to ~2224 cm^{-1} on C and ~2227 cm^{-1} on Au, demonstrating its stabilization on C-based electrodes (Fig. 2G). The evolution of this peak was accompanied by a subtle shoulder above 2250 cm^{-1} belonging to reaction products (mainly ADN; fig. S17). By deconvoluting the AN, AN anion radical, and ADN peaks, we found that the benzyl groups led to faster generation of AN anion radicals with larger proportions at the interface (Fig. 2F), which explains the promoted kinetics and selectivity observed on $Bn(Hexyl)_3N^+$ and C-based electrodes. The larger Stark tuning rate and faster anion radical generation were observed on other C electrodes, pointing to the universal cation-carbon interaction (figs. S19 to S21).

Dynamic multimodal cation-carbon interaction for displacement of interfacial water on C

Because the AN anion radical reacts either with AN toward ADN or with H_2O toward unwanted PN, the interfacial distribution of AN and H_2O is essential for ADN selectivity. We further carried out molecular dynamics (MD) simulation and vibrational sum-frequency generation (VSFG) spectroscopy (figs. S22 and S23). MD simulation revealed AN accumulation at ~1.0 nm away from the electrode (z = 0 nm) using $Bn(Hexyl)_3N^+$ (Fig. 2H). Decreasing the alkyl lengths exhibited inferior enrichment, with distribution peaks farther away and lower intensities. Comparatively, $(Hexyl)_4N^+$ also demonstrated similar AN distribution, implying the dominant role of alkyls in enriching AN. This is consistent with the simulated trends of AN-QAS interaction in bulk, with the impact of alkyl chains overwhelming that of benzyl groups (fig. S24).

For H_2O, $Bn(Hexyl)_3N^+$ showed the first H_2O distribution peak only beyond 1.0 nm, where AN distribution started to decline (Fig. 2I), which suggests that there is spatial separation between AN-enriched and

H_2O-enriched regions. Decreasing alkyl lengths only slightly shifted H_2O distribution toward the electrode. By contrast, $(Hexyl)_4N^+$ created a substantially closer H_2O peak within 1.0 nm, indicating the colocation of H_2O and AN at the interface that substantially enhances PN formation. This difference reveals that the benzyl-carbon interaction prevents the inclusion of H_2O-containing spaces by hexyl groups, resulting in a more hydrophobic interface and spatial separation between AN and H_2O.

This insight is further supported by VSFG spectroscopy—an interfacial-sensitive molecular technique (*44*). The VSFG signal is a second-order nonlinear signal generated by focusing infrared and visible pulses onto the sample (Fig. 3A). Because of the selection rule, the signal from the centrosymmetric region is prohibited, which allows us to probe the liquid-solid interfaces.

We first focused on the OH stretch region of graphene-water interfaces (Fig. 3B). The VSFG spectrum of the graphene/$NaHCO_3$ aqueous solution shows two OH stretch bands at ~3200 and ~3600 cm^{-1} (*45*). The 3600 cm^{-1} band is assigned to the dangling OH group of water interacting with multilayer graphene, whereas the 3200 cm^{-1} band is the OH group of H-bonded water. After adding $Bn(Hexyl)_3N^+$, the 3600 cm^{-1} band is unchanged, whereas the 3200 cm^{-1} band exhibits a marked decrease. This indicates that by interacting with graphene, $Bn(Hexyl)_3N^+$ substantially disrupted the H-bond networks at the graphene-water interfaces. Although the presence of a dangling OH peak suggests that water molecules are not completely repelled away from the Stern layer, the reduction of the H-bonded peak indicates that the interfacial H-bond network is broken apart by $Bn(Hexyl)_3N^+$, likely by reducing the density of interfacial water molecules in the first few layers through replacement.

To gain insights into the molecular interactions between $Bn(Hexyl)_3N^+$ and the C electrode, we investigated the cation spectra (Fig. 3, C and D). We assigned the VSFG peaks as follows: terminal methyl symmetric (CH_3r^+) at ~2878 cm^{-1}, methyl asymmetric (CH_3r^-) at 2956 cm^{-1}, and Fermi resonance (CH_3FR) at ~2922 cm^{-1} (*46*–*50*). First, by comparing spectra with and without graphene, we resolved an extra band at ~2986 cm^{-1} on the graphene/Au electrode, assigned to the benzyl C-H stretch modes (*51*). The appearance of the aromatic C-H mode is because of the reorientation of $Bn(Hexyl)_3N^+$ coming closer to the interface in the presence of C, supported by a sign change in the heterodyne-detected VSFG signal of the aromatic C-H (fig. S25).

Notably, the spectral features of $Bn(Hexyl)_3N^+$ exhibited qualitatively similar potential dependence on Au and C/Au electrode surfaces—its VSFG peaks decrease when sweeping the potential negatively from −0.1 to −0.8 V (Fig. 3, C and D, and fig. S26). Thus, $Bn(Hexyl)_3N^+$ was very responsive to the potential bias, and graphene did not add additional screening effects. By contrast, the spectra of $(Hexyl)_4N^+$ presented quite differently (Fig. 3, E and F). The overall VSFG signals were much weaker compared with those of $Bn(Hexyl)_3N^+$ (Fig. 3G) because of the higher symmetry of this cation (fig. S27). More relevantly, although the signal had a weak response to the potential bias on

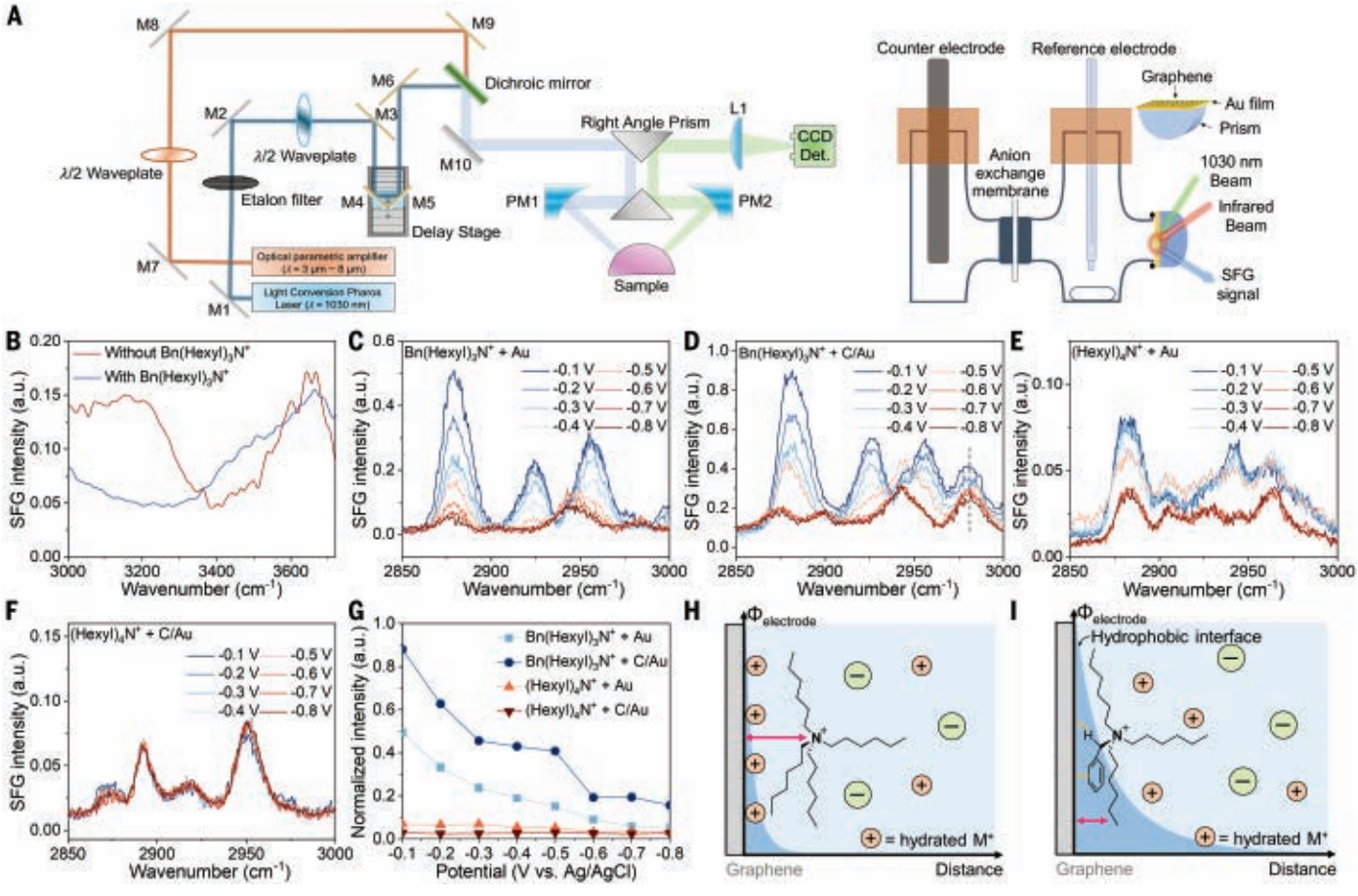

Fig. 3. VSFG spectroscopic analysis of the multimodal cation-carbon interaction. (**A**) Schematic diagram of the VSFG setup and the electrochemical cell and electrode. CCD, charge-coupled device. (**B**) VSFG spectra of the graphene-water interface with and without $Bn(Hexyl)_3N^+$. (**C** and **D**) VSFG spectra of $Bn(Hexyl)_3N^+$ on the Au and graphene/Au electrodes, respectively. (**E** and **F**) VSFG spectra of $(Hexyl)_4N^+$ on the Au and graphene/Au electrodes, respectively. (**G**) Comparison of CH_3r^+ peak intensities in the potential-dependent VSFG spectra of $Bn(Hexyl)_3N^+$ and $(Hexyl)_4N^+$. The electrolyte is a 0.5 M $NaHCO_3$ aqueous solution, and the potential range is −0.1 to −0.8 V (versus Ag/AgCl). (**H** and **I**) Schematics of the quaternary ammonium cations for $(Hexyl)_4N^+$ (H) and $Bn(Hexyl)_3N^+$ (I) on the graphene-water interface under the applied cathodic potential.

Au, such a response was completely absent on the graphene/Au electrode (Fig. 3, E and F). It was notable that a single benzyl group substitution made the QAS cation more responsive to external potentials.

Such a response could be quantitatively seen by plotting CH_3r^+ peak intensity versus applied potentials (Fig. 3G). For $Bn(Hexyl)_3N^+$, the intensity decreased as the potential shifted negatively (*52*). Thus, considering the noticeable spectral changes in the CH_3r^- peaks and the attractive force from negative potentials, we assign the decreasing signal to the cancellation of coherent signals from oppositely oriented $Bn(Hexyl)_3N^+$ in the bilayer (*53*). By contrast, the peak intensity was nearly constant for $(Hexyl)_4N^+$. The responsiveness to external potential indicated that $Bn(Hexyl)_3N^+$ experiences dominant electrostatic attraction from the electrode, whereas such interaction was screened out for $(Hexyl)_4N^+$. The screening should come from the metal cations in the electrolyte, sandwiched between the electrode and $(Hexyl)_4N^+$ (Fig. 3H). Thus, for $(Hexyl)_4N^+$, the cation experiences less surface potential. By contrast, for $Bn(Hexyl)_3N^+$, a large signal intensity and potential dependence indicates closer distances to the electrode because of the additional cation-electrode interaction (Fig. 3I).

Toward the identity of such interaction, π-π interaction between benzyl and graphene was plausible, as evidenced by lower ADN selectivity when replacing the benzene ring with cyclohexane (fig. S28). However, electrolysis using the $Bn(Hexyl)_3N^+$-based QAS containing a bulky *tert*-butyl group in the para position demonstrated similar ADN selectivity (fig. S28). This discrepancy pointed to an additional noncovalent force that compensates for the suboptimal geometry. A likely candidate is the C-H–π interaction originating from the benzylic CH_2 group: The high polarity of this C-H dipole, amplified by conjugated benzene and adjacent ammonium, renders the bond relatively soft to form such an interaction. This interaction is further supported by the benzylic C-H stretch observed in VSFG (Fig. 3D). Introducing an extra methylene unit slightly lowered ADN selectivity while retaining activity, which suggests that although individual C-H–π contacts may weaken, their cumulative effect from multiple methylene units still effectively constructs a hydrophobic interface that tunes the orientation of both the benzene ring and alkyl chains (figs. S28 and S29). Beyond the type of interaction, its dynamic nature proved equally essential. Rigid interfacial structures, arising from stronger interactions as seen with $Nap(Hexyl)_3N^+$ and $Bn_2(Hexyl)_2N^+$, substantially reduced ADN selectivity (Fig. 1B and fig. S28).

Building on this mechanistic understanding, the physical picture becomes clear: The additional π-π and C-H–π interactions between benzyl and C electrodes effectively displace water molecules from the first few interfacial layers, substantially reducing the density of H-bonded water networks and their availability for undesirable byproduct formation (Fig. 3, H and I). This hydrophobic interface also

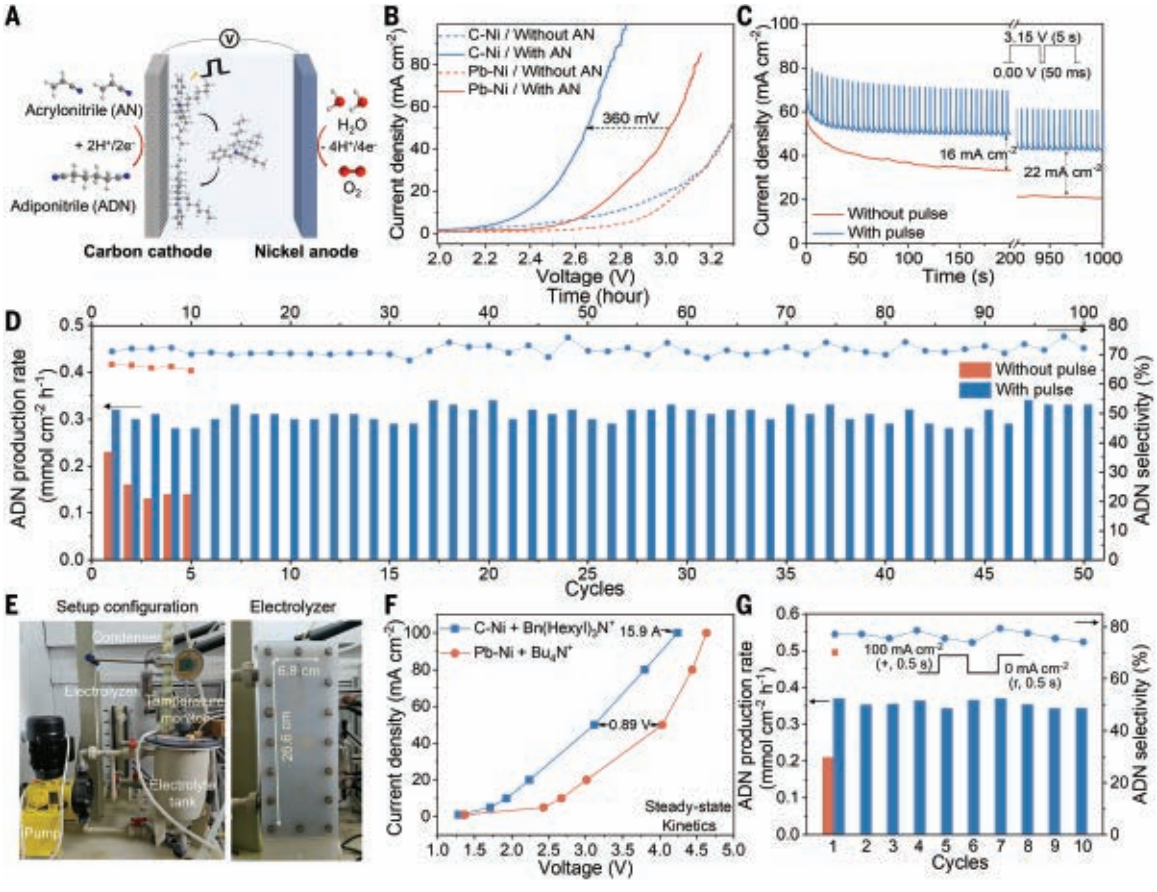

Fig. 4. Two-electrode laboratory-scale to scaled-up electrolyzers using the dynamic multimodal cation-electrode interaction with C electrodes. (**A**) Schematic diagram of the pulsing strategy for controlling the dynamic cation-electrode interface toward the two-electrode ADN electrosynthesis device. (**B**) Linear sweep voltammetry curves for different combinations of electrodes with and without AN. (**C**) Chronoamperometric curves of the C-Ni electrolyzer with the $Bn(Hexyl)_3N^+$ cation without and with the pulsing strategy. (**D**) ADN production rates and selectivities over repetitive runs of the C-Ni electrolyzer with the $Bn(Hexyl)_3N^+$ cation without and with the pulsing strategy. (**E**) Photos of the scaled-up reactor and its affiliated apparatus. (**F**) Steady-state kinetics of the system of C-Ni electrodes with $Bn(Hexyl)_3N^+$ compared with the standard system of Pb-Ni electrodes with Bu_4N^+. The kinetics curve is measured by the steady-state potentials at different values of constant current densities. (**G**) ADN production rates and selectivities over repetitive runs of the scaled-up C-Ni electrolyzer with the $Bn(Hexyl)_3N^+$ cation without and with the pulsing strategy. Each cycle contains 2-hour electrolysis at 100 mA cm^{-2} and a 2-hour resting period, for a total of 4 hours.

facilitates local accumulation of AN, thereby enhancing ADN electrosynthesis. Notably, this interaction motif is not limited to C electrodes; enhanced performances on five different metal electrodes under $Bn(Hexyl)_3N^+$ confirm its broader applicability (fig. S30).

Scaled-up demonstration and pulsed operation for efficient C-based ADN electrosynthesis

To illustrate the practical applicability, we constructed a two-electrode ADN electrosynthesis system using a Ni foam anode and a C cathode under $Bn(Hexyl)_3N^+$ (Fig. 4). A standard electrolyzer featuring a Pb cathode and TBA^+ was compared side by side. Without AN, both electrolyzers only exhibited slowly increasing current after 2.8 V, which suggests that the HER was suppressed by QASs (Fig. 4B). After AN addition, polarization curves substantially shifted to lower voltages. Under 50 mA cm^{-2}, the voltage difference reached 360 mV, accounting for ~12% energy efficiency. A voltage difference of 150 mV could still be maintained when compared with a Pb-Ni electrolyzer using $Bn(Hexyl)_3N^+$, which demonstrates the contributions of both the C electrode and cation (fig. S31).

The product selectivity with $Bn(Hexyl)_3N^+$ and C-Ni electrodes was higher than that of the conventional systems. However, over long-term operation, we resolved the declining current and ADN production rate alongside the formation of oily species from AN oligomerization (Fig. 4, C and D). Postmortem electrode analysis confirmed the residual N content but no evidence of corrosion (figs. S32 and S33). Considering the advantage of dynamic potential controls in electrochemical systems and the noncovalent cation-electrode interactions (*25*), we applied square-wave pulses to dynamically let $Bn(Hexyl)_3N^+$ undergo adsorption-desorption cycles to prevent the formation of a persistent hydrophobic interface (Fig. 4A). Notably, under the three-electrode system, the ADN selectivity further improved to 85%, with side-product selectivity tailored using pulses (fig. S34). Switching to the two-electrode practical system, we obtained steadier current over long-term electrolysis, and the ADN production rate and selectivity could be maintained over at least 50 cycles without decay. The slight decline in ADN selectivity from 80 down to 70 to 75% in this two-electrode electrolyzer originated from the membraneless system, in which the anodic dioxygen (O_2) quenched AN anion radicals near the cathode.

We further scaled up the reactor to 159 cm^2 and increased the current density to 100 mA cm^{-2} with a total of 15.9 A (Fig. 4, E to G, and fig. S35). The optimized condition of 0.5 s of electrolysis at 100 mA cm^{-2} and 0.5 s of rest at 0 mA cm^{-2} produced the highest ADN selectivity of 77.3% (fig. S36). Cycling under this condition produced 75 to 80% ADN selectivity and a similar areal production rate to that produced by the small-scale electrolyzer (Fig. 4F). Notably, the synergized C and $Bn(Hexyl)_3N^+$ showed a much lower voltage required compared with the standard Pb system. The largest voltage difference was 0.89 V at 50 mA cm^{-2}, corresponding to 22% energy saving. This sustained ADN production with high selectivity over repetitive cycles (Fig. 4G) clearly demonstrates the possibility of substituting Cd/Pb cathodes with low-cost and environmentally friendly C cathodes with higher energy efficiencies toward real applications of ADN electrosynthesis. It also emphasizes the importance of engineering cation-based interfaces for realizing efficient organic electrosynthesis systems in conducting aqueous electrolytes toward industrial applications.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Figs. S1 to S44; Table S1; References (*54*–*64*)

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PLANETARY ACCRETION

Incorporation of atmospheric components in planetary magma oceans through bubble–metal compounds

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Atmophile elements (carbon, hydrogen, and nitrogen) were thought to accrete late during planetary formation, yet cosmochemical and astrophysical evidence suggests that they were already present while small bodies, such as Vesta, were forming. High-temperature magmatic events on small bodies, however, should have outgassed these ultravolatile elements because gravity would have been insufficient to retain them. In this study, experiments reveal that gas bubbles strongly attach to metal droplets in magma ocean–like mixtures, forming compounds that resist segregation. Early flotation of such compounds would have enhanced mantle siderophile endowment, eliminating the need for late accretion. On small bodies, low gravity would have enabled millimeter- to centimeter-sized compounds to sink, trapping atmophiles in their interiors. This mechanism links core formation to the early entrapment of atmophile elements and provides a framework for reconciling volatile delivery across planetary bodies and planetary accretion models.

Ninety years ago, Victor Goldschmidt (*1*) classified the chemical elements as a function of their properties to explain their distribution on planet Earth. Siderophile (iron affinity) elements are preferentially partitioned into the core; chalcophile (sulfide affinity) elements are ore-forming; lithophile (silicate affinity) elements are partitioned into the mantle and the crust; and atmophile designates volatile elements such as carbon (C), hydrogen (H), sulfur (S), and nitrogen (N), which constitute the atmospheres around rocky planets (*1*–*5*). Because atmophile elements are also the building blocks of life (*2*), understanding their origin, provenance, and distribution among the various planetary reservoirs constitutes seminal and hotly debated topics (*3*, *6*).

Isotopic data for Earth's C, H, and N are indistinguishable from those of most chondrites, suggesting delivery during planetary formation by a mixture of carbonaceous and noncarbonaceous chondritic materials (*6*–*10*). However, the timing of this delivery remains debated (*6*, *11*–*13*). One view proposes that planets initially formed dry (*13*) from inner solar system materials, with atmophile elements added later. The so-called "late veneer" (*14*), or the late influx of chondrites invoked to replenish siderophile elements in Earth's mantle after core formation (*15*), likely contributed part of the atmophile inventory (*6*, *13*). However, mass-balance calculations indicate that a late veneer cannot account for Earth's H, C, and N abundances, implying that much of the atmophile budget was already present during the main stages of accretion (*5*–*12*).

Several observations support the early presence of atmophile elements in the inner solar system [for example, (*6*–*12*, *16*)]. High-water contents and deuterium (D)/H ratios measured in samples from Vesta (*17*), a differentiated asteroid ~300 km in diameter, indicate that volatile elements were present during its formation. Iron meteorites, interpreted as fragments of planetesimal (small differentiated body) cores, contain elevated C and N abundances, which are inconsistent with extensive "open to space" magmatic outgassing during differentiation (*18*). Some also contain noble gases (*19*), which are ultra-atmophile elements and thus unexpected for small, molten bodies exposed to degassing in space. Bubbles and voids in angrite meteorites (*20*) may further record volatile-rich gas phases trapped during magmatic events. Last, solar-like noble gases [helium (He) and neon (Ne)] preserved in Earth's deep mantle (*21*, *22*) suggest that gas of nebular composition was captured and incorporated into planetary interiors in the first ~4 million years of solar system history (*23*–*27*).

The challenge of retention of atmophile elements in small bodies

The conundrum here is that for atmophile geochemical signatures to be conveyed in planetary mantles and cores, a viable mechanism enabling their incorporation, despite low gravity in the case of planetesimals, is required. Because the solubility of atmophiles is higher in molten than in solid rocks (*3*), dissolution in early magma oceans, which prevailed during planetary accretion, seems a suitable mechanism. Yet the gas pressure at the surface of a magma ocean must have been sufficiently high to enable dissolution of ultra-atmophile elements (*4*, *5*, *23*, *24*, *26*, *27*). For a planetary body, the surface gas pressure is given by

$$P = \frac{g \times m^{gas} \times M^{tot}}{4 \times \pi \times r^2} \quad (1)$$

where g is gravitational acceleration, M^{tot} and r are the total mass and radius of the body, and m^{gas} is the mass fraction of gas at the surface of the planet. The solubility S_i of a gas component i into a molten silicate can be simplified and related to the equilibrium partial pressure of the gas P_i as

$$S_i = k_i \times P_i^n \quad (2)$$

where k_i stands for the thermodynamic constant, which varies for each gas component i, and the exponent n is a constant depending on the stoichiometry of the gas dissolution reaction (*3*, *28*). Water (H_2O) is the most soluble component in silicate melts, followed by S, chlorine (Cl), N, carbon dioxide (CO_2), and carbon monoxide (CO) (*3*, *4*, *29*). The noble gases are less soluble components (*30*). Broadly speaking, P values become significant in terms of atmophile solubilities if $r > 1000$ km (*23*, *24*, *26*, *27*, *31*). Therefore, this implies that dissolving atmophile elements in a molten planetary surface layer requires sufficiently large planetary bodies, as indicated in the compilation of McCubbin and Barnes (*12*).

Standard models (*11*, *16*, *32*) propose that planets formed through collisions between planetesimal and "embryo" ($r \sim 100$ to 1000 km) bodies. Therefore, incorporating atmophile elements into planetary mantles requires these elements to have already been present in small bodies, despite their low mass and weak gravity. In addition, large planetary bodies formed relatively late in the evolution of the inner solar system, likely after dissipation of the solar nebula, making gravitational capture of solar noble gases improbable (*23*, *24*, *26*). This conundrum favors alternative formation scenarios [for example, (*26*, *33*)] involving rapid planetary growth and early capture of a nebular hydrogen (H_2)–He atmosphere (*26*, *27*).

In this work, we report on experiments, supported by attachment theory, that demonstrate how gas bubbles formed upon melting during planetary accretion could have been systematically attached to metal droplets. This process links planetary core formation to the accretionary capture of atmophile elements. This mechanism for the entrapment of ultravolatile elements breaks some of Goldschmidt's seminal rules with the following two major implications: It relaxes the restrictive size–age conditions thought to be required for building

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volatile-bearing planetesimals and planetary embryos and also provides an explanation for the excess siderophile elements observed in many planetary mantles, without invoking a late veneer.

Experimental evidence for gas bubble attachment on metal droplets in magma oceans

We conducted moderately high-pressure experiments to expand on the experimental observations long collected by metallurgists at atmospheric pressure (*34*). The tests were run in internally heated pressure vessels and piston cylinders at Institut des Sciences de la Terre d'Orléans (ISTO), Orléans, France (*35*, *36*). All experiments produced a silicate magma (glass) coexisting with fluid-bearing bubbles and metal droplets [molten iron (Fe)–rich alloys] (Fig. 1 and figs. S1 to S3). Starting materials comprised basaltic glasses (table S1), Fe-rich metal, sulfide, nitride, and graphite under H_2-rich reducing conditions (*35*) representative of magma oceans on small planetary bodies (50 to 1000 km diameter). Runs were performed at 1275° to 1400°C and 40 to 1050 MPa (table S2) then rapidly quenched and characterized by using scanning electron microscopy and three-dimensional (3D) microtomography imaging (Figs. 1 and 2 and figs. S1 to S3). Full experimental procedures, starting compositions, and analytical methods are provided in the supplementary materials.

Regardless of experimental duration (5 min to 4 hours), gas bubbles were systematically attached to metal droplets after quenching, indicating very sluggish gas–silicate–metal segregation. 3D x-ray tomography images (Fig. 1) show bubble–metal attachment and slow segregation controlled by the competing buoyancies of bubble–droplet compounds, depending on their relative sizes. In the shortest runs, these assemblages are homogeneously distributed throughout the sample, with multiple metal droplets commonly attached to a single bubble (Fig. 1A and figs. S1 and S2). In longer runs, bubble–droplet attachment became predominantly one-to-one (Fig. 1, B and C) with the development of a density-driven segregation: Low-density assemblages rose, and dense assemblages sank, although some reached near-neutral buoyancy and remained suspended in the sample interior. Droplet–bubble pairs increased in size during experiments, indicating coalescence during buoyancy-driven migration, which is consistent with previous observations in magmatic systems (*36*). In some cases, metal droplets are transported to the top of the capsule by attachment to large bubbles, where they become trapped in gas pockets formed by coalescence at the platinum (Pt) interface. Conversely, large metal droplets accumulate at the bottom, whereas some sink while still attached to small bubbles (Fig. 1C).

In Fig. 2, we display an experiment conducted on S-rich starting material, which produced biphasic Fe molten alloys (C-rich versus S-rich). A full 3D scan from the top to the bottom is available in fig. S4. This sample shows bubbles significantly larger than S-free droplets, with the pronounced coalescence of an especially large central cavity (~500 μm) decorated with numerous metallic droplets. The bubble–droplet attachment is characterized by the S-rich portion of the molten alloy systematically wetting the silicate melt and the bubble, whereas the C-rich portion is encapsulated in the S-rich Fe alloy. These experiments were used to evaluate the silicate–bubble–metal interface and to obtain information on the three-phase (metal–silicate–gas) contact line, which can be described as a balance of the contact angles between the three phases. The measured angles are relatively high, and their distribution—shown as a histogram—for different bubble–metal pairs, as well as the average values, is presented in Fig. 2. The dispersion of the angle values is high. Possible volume changes of gas bubbles during fast quenching of the experiments were ignored.

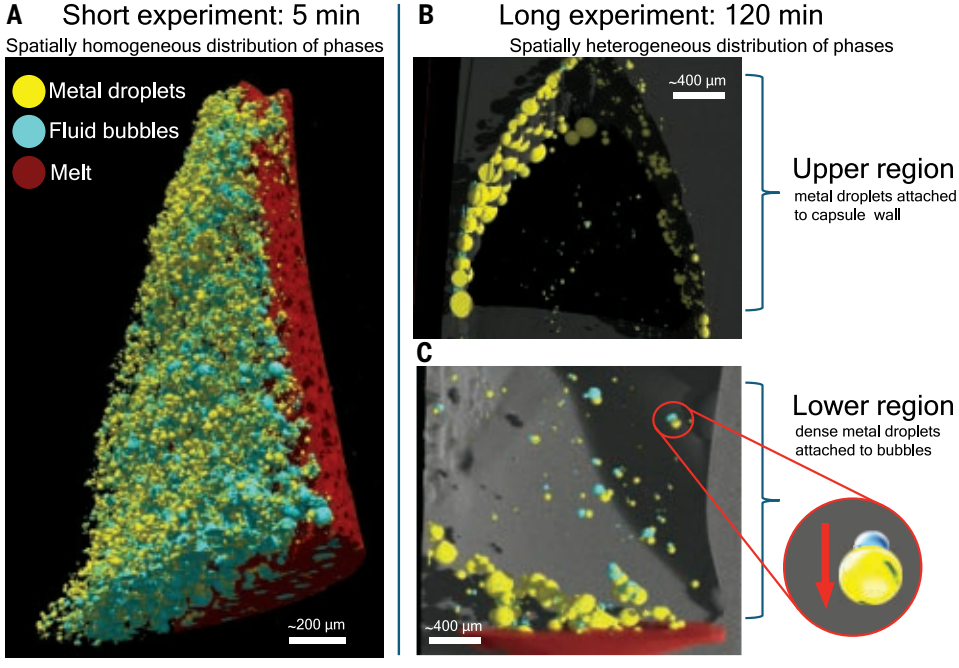

Fig. 1. 3D image of bubble–droplet compounds. (**A** to **C**) X-ray tomography images of (A) short (5 min; METABUL-1) and [(B) and (C)] long (120 min; METABUL-3) experiments (table S2). Fluid bubbles are displayed in cyan, and metal droplets are displayed in yellow. The silicate glass matrix is in red when displayed (otherwise transparent). All images show the strong droplet–bubble attachment. (B) Positive buoyancy regime in which metal droplets, which were attached to large bubbles, were plated atop the Pt capsule. (C) Negative buoyancy regime, with large droplets that had already sunk and intermediate droplet–bubble pairs falling. Shown is different scaling: In the short experiment scenario, fluid bubbles have diameters of 50 to 80 μm, whereas in long experiment conditions, sinking large metal droplets have diameters in the range of 120 to 230 μm.

Modeling bubble attachments

The nearly systematic attachment of bubbles to dense particles in silicate melt has long been reported by metallurgists for metal–bubble compounds at 1 bar (*34*), as well as by geologists studying oxide–bubble (*37*) and sulfide–bubble compounds in a pressure range of 1 to 1000 bar (*36*, *38*, *39*). Our experiments, together with literature data, show that metal–bubble attachment occurs from 10^5 Pa (1 bar) to 10^5 MPa (10 kbar), corresponding to fluid density ranging from ca. 0.001 to 1 g/cm^3.

In this study, we framed a thermodynamic description of attachment that is consistent with previous treatments of metal–oxide mineral–bubble systems (*37*), in which the attachment energy (ΔG) is defined as the energy of the attached configuration minus the combined energies of the isolated and noninteracting compounds. In this energy change between the preattachment and postattachment, we accounted for the interfacial areas involved as well as their associated surface energies. This balance can be expressed as

$$\Delta G = \Delta A_B \sigma_{BL} + \Delta A_M \sigma_{ML} + A_C \sigma_{BM} \quad (3)$$

where ΔA_B and ΔA_M are the changes in the surface area of the bubble and the metal droplet, respectively, between the separated and joint scenarios. A_C is the surface

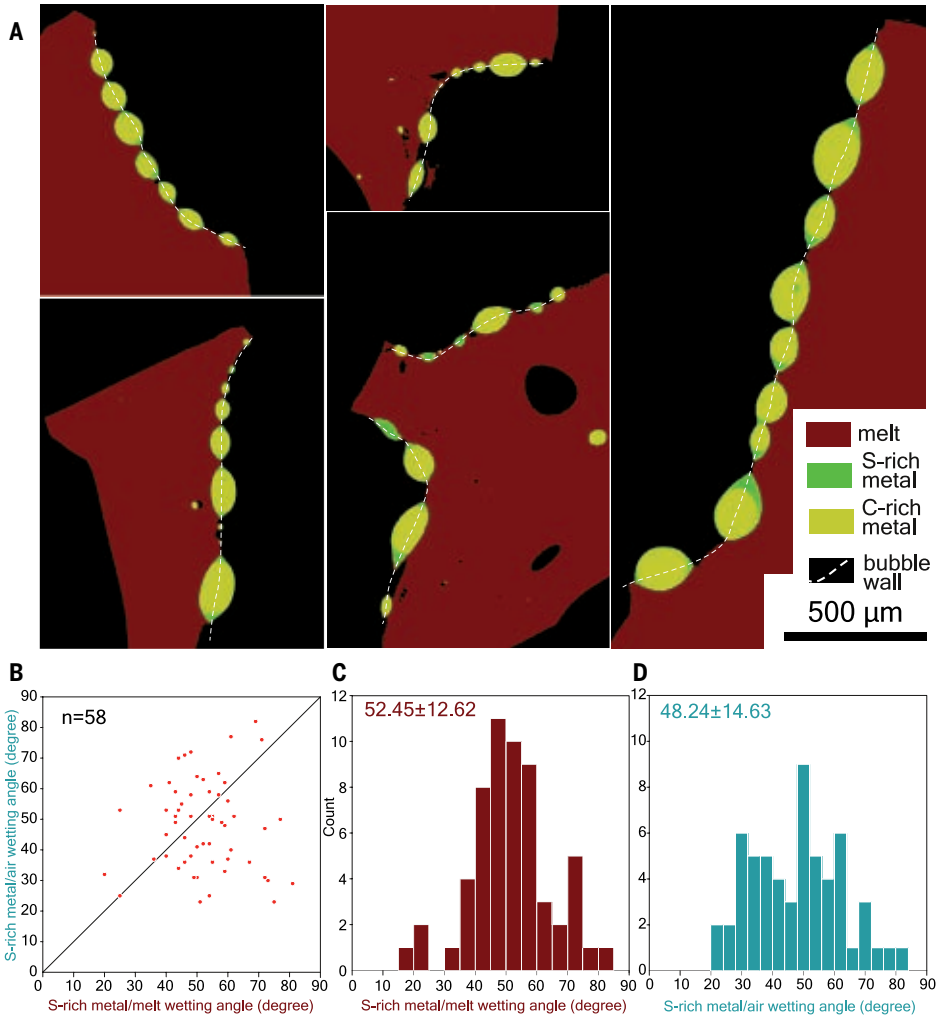

Fig. 2. Metal–gas–silicate wetting angle measurements. (**A**) 2D images selected from 3D x-ray tomography on a sample that is fully detailed in fig. S4 (table S2, run MetalBul 5). Different droplets are decorating a very large bubble atop the sample. The biphasic molten metal alloys take the form of eyes with a C-rich core encapsulated in a S-rich rim that is deformed by the gas–melt–metal interfacial energies. (**B**) Plot of the metal–melt versus metal–gas angles. (**C**) Histogram of the angle distribution between metal and silicate melt. (**D**) Histogram of the angle distribution between metal and gas. The large bubble on which the angles have been measured probably resulted from a late coalescence event during the last step of the experiment, just before the quenching.

area of the interface formed between them, and σ_{BM}, σ_{BL}, and σ_{ML} are the surface tensions of the involved phases, where B, M, and L denote bubble, metal, and (silicate) liquid, respectively. Before attachment, the bubble and the metal droplet are considered here as spheres. After attachment, they are spherical caps, connected by a spherical interface. By equating the volumes before and after attachment, one can calculate the new radii of the bubble, the metal drop, and the interface, and in turn, their surface areas. In the present approach, both the bubble and the metal droplet are treated as deformable entities with conserved volumes, similarly to that by Yao and Mungall (*39*) and in contrast to previous studies in which one of the phases was assumed to be solid and nondeformable (*37*, *40*). Another outcome of this calculation is Δl, which represents the length by which the bubble and the droplet have interpenetrated each other during attachment (the decrease in distance between their center points) and from which we define the force of attachment as

$$\Delta F_{att} = \frac{\Delta G}{\Delta l} \quad (4)$$

Assuming that the densities of the molten metal in the droplet (m), surrounding liquid (l), and fluid in the bubble (b) follow the following relationships, $\rho_m > \rho_l > \rho_b$, the buoyancy force on the bubble is positive and defined as

$$F_b^{buo} = V_b(\rho_l - \rho_b)g \quad (5)$$

The buoyancy force on the droplet is negative and defined as

$$F_m^{buo} = V_m(\rho_l - \rho_m)g \quad (6)$$

where V_b and V_m are the volumes of the bubble and metal droplet, respectively, and g is gravitational acceleration. The difference between these two forces is the force that tends to separate (detach) the bubble and the droplet

$$F_{det} = F_b^{buo} - F_m^{buo} \quad (7)$$

and detachment takes place when

$$F_{det} > F_{att} \quad (8)$$

The attachment force correlates positively with bubble size, whereas negative buoyancy correlates with droplet size, and positive buoyancy correlates with bubble size. It thus appears that the attachment versus detachment bifurcation is essentially a function of the relative sizes of bubble versus droplet (and the density of each phase).

The simultaneous use of the above equations (*41*) allowed us to construct a universal regime diagram describing bubble–metal droplet attachment and their resulting buoyancy behavior (Fig. 3). The diagram highlights the region where attachment is not expected because of excessive individual buoyancies of the low- and high-density phases. This regime is favored for large particle sizes [for example, radius (Rb) > 1 cm] and is less pronounced under low-gravity conditions, such as those relevant to asteroids and Vesta (Fig. 3). A broad domain, with objects that are millimeters to centimeters in radius, exists in which attachment is possible. This domain is further divided into the following two subregimes: one in which the bubble–droplet compound ascends because of a net upward buoyancy, and another in which the aggregate sinks under a net downward buoyancy. This model is consistent with the size distribution of negatively buoyant bubble–droplets observed in our experiments and shown in Fig. 1C (fig. S5). Because buoyancy scales directly with gravity, these attachment fields expand substantially under low-gravity conditions. As a result, for small rocky bodies, such as an asteroid of 50 km radius and Vesta, the attachment regimes—both ascending and descending—extend to larger particle sizes (Rb up to centimeter scale). Millimeter- to centimeter-radius metal droplets are comparable with metallic nuggets found in chondrites. Also, voids with similar size and attached to metal and sulfide grains are described in some achondrites (the Angrites) (*20*), which may correspond to gas bubbles formed during incorporation of atmophile elements and subsequent magmatic degassing events on small bodies.

A mechanism to enhance the incorporation of atmophile elements in molten small bodies

Millimeter-sized metal and sulfide grains commonly occur in accretionary materials, whether carbonaceous or noncarbonaceous chondrites, which also contain substantial amounts of volatiles (*5*, *8*–*10*). During accretion, molten chondritic material is therefore expected to have generated both gas bubbles and metal droplets simultaneously. In magma oceans, small metal droplets would have attached to gas bubbles, forming composite bubble–droplet pairs.

If these assemblages reached neutral buoyancy, as observed in some experiments (Fig. 1), they would have enhanced the retention of both siderophile and atmophile elements in the magma ocean. Under appropriate size conditions—small bubbles coupled with relatively larger metal droplets—these compounds are predicted to sink (Fig. 3). Bubble–droplet attachment thus provides an efficient mechanism to sequester ultravolatile elements by transporting them into the interior of molten bodies.

This process has two main consequences: (i) Bubbles, as expressions of fluid saturation, could have been stabilized at depth in magma oceans; and (ii) their descent would have generated elevated internal fluid pressures (~tens of megapascals, depending on gravity and burial depth). Increasing pressure enhances the solubility of volatile elements in both silicate and metallic melts. This mechanism would enable the physical burial of bubbles and chemical dissolution of volatiles in small bodies, even under high-temperature, low-gravity, and low-surface-pressure conditions (Fig. 4).

Sinking bubbles shrink because of compression, decreasing their positive buoyancy while having a moderate effect on the attachment forces, as calculated in fig. S6. The sinking–compression–solubilization mechanism is thus a runaway process. Among the various relevant volatile species, H_2O, the most soluble in silicate melt (*28*), is thus moderately affected by the enhanced solubility of the bubble–droplet compounds. Less soluble components such as CO_2 (*28*), N_2 (*35*), and noble gases (*29*) require high gas pressure to dissolve in the silicate melt (*27*). Depending on how deep the bubble–droplet compounds sink (Fig. 4), this can greatly affect the solubilization of C, N, and noble gases in the magma ocean on small bodies despite the very low surface pressure. To be comprehensive, volatile solubility in the molten metal is also linked to gas pressure (*3*, *25*, *42*), with C and N being the most soluble (*3*, *18*), and H only significantly affected if $P > 1$ GPa (*3*). Noble gas incorporation into molten metals is also expected to be affected by this process (*25*).

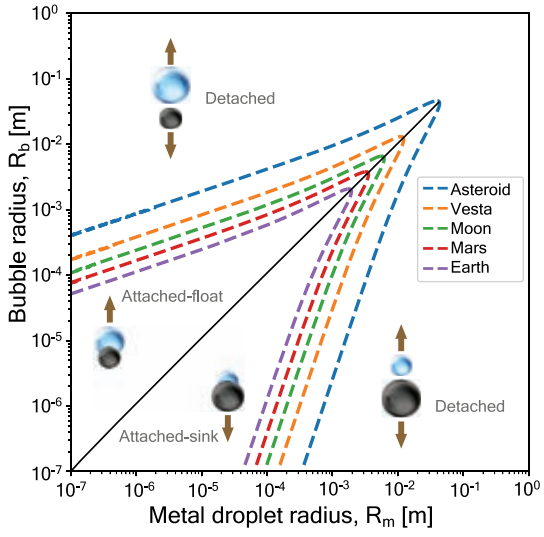

Fig. 3. Universal regime diagram illustrating the dynamics of bubble–droplet compounds in magma oceans under different gravity conditions. Attachment and buoyancy calculations for bubble–droplet compounds were performed for near-surface initial conditions, corresponding to a pressure of 1 bar and bubbles at a depth of ~1 m in the magma ocean. Under sinking conditions, silicate melt and metal densities remained effectively constant, whereas the fluid density increased. The effect of such a fluid density increase is shown in the supplementary materials (fig. S6) to favor attachment.

The runaway sinking–compression–solubilization mechanism may thus lead to the physical disappearance of the bubble at a certain depth. However, it is expected that some bubble–droplet compounds could survive sinking and remain attached upon reaching the core–magma ocean interface (Fig. 4). Bubble–droplet compounds thus constitute an attractive solution to chemically and physically store ultravolatile elements in the magma ocean and core of planetary bodies. Attachment is described here for Fe-rich molten alloys but also applies to planetesimals producing sulfide melts [for example, (*43*)].

A mechanism acting against metal–silicate segregation

So far, we have focused our discussion on the description of a mechanism to incorporate atmophile elements in the molten rocky bodies that are too small to hold these volatiles by gravity. Further implications of our findings are for the timing and effectiveness of metal–silicate segregation. Bubble–droplet bonding delays this segregation and must thus be considered when model ages for metal–silicate separation in small and large bodies are established on the basis of radioactive decay rates of lithophile versus siderophile elements (such as hafnium-tungsten) (*44*).

Neutrally buoyant bubble–droplet compounds may have prevented the complete removal of ultrasiderophile elements (such as the platinum group) from planetary mantles. The mantles of Earth and other solar system bodies (*14*, *45*) show an excess of these elements above numerical estimates: On Earth, concentrations are ~100 times greater than predicted from core–magma ocean partitioning (*14*, *15*, *46*). This enrichment is classically attributed to a chondritic late veneer (*11*, *13*, *14*, *32*, *45*). From this study, bubble–droplet flotation provides an alternative, physically grounded mechanism for retaining amounts of metal in magma oceans that are sufficient to account for mantle ultrasiderophile budgets. Using existing partition coefficients and mantle abundances (*46*), we estimate that the flotation of only 0.1 to 0.3 wt % of Fe-rich metal would reproduce upper mantle platinum group element concentrations (supplementary materials). This mechanism could operate broadly across planetary systems without requiring a late veneer. Because these elements are extremely siderophile, their relative proportions in metallic droplets would remain chondritic, thus mimicking a late-veneer signature. In addition, flotation and redistribution of metal droplets may have influenced magma ocean redox conditions, challenging the concept of a strongly oxidized shallow magma ocean (*47*). Instead, floating metal droplets should allow for more reduced conditions, even in the shallowest region of magma oceans, with implications for the composition of outgassed atmospheres (*5*). Only upon magma ocean solidification, the Fe and nickel (Ni) of such droplets would get oxidized and incorporated into the silicates, enabling the formation of minute amounts of nuggets of platinum group elements.

Overall, bubble–droplet attachment provides a unified framework that links the behavior of ultrasiderophile and ultra-atmophile elements, which otherwise display contrasting geochemical affinities. We have established a mechanism to inspire future work focused on magma ocean simulations [for example, (*48*, *49*)]. In particular, a more quantitative assessment of the amount of atmophile elements that could be stored by this mechanism in the planetary core and deep magma ocean is needed. This would require atmophile solubility laws in metal and molten silicate to be coupled with a model of magma ocean mechanics. Assessments of the evolution of the fluid–melt–metal surface tensions at high pressure are needed to evaluate the importance of such a process on large molten planetary bodies, including exoplanetary systems

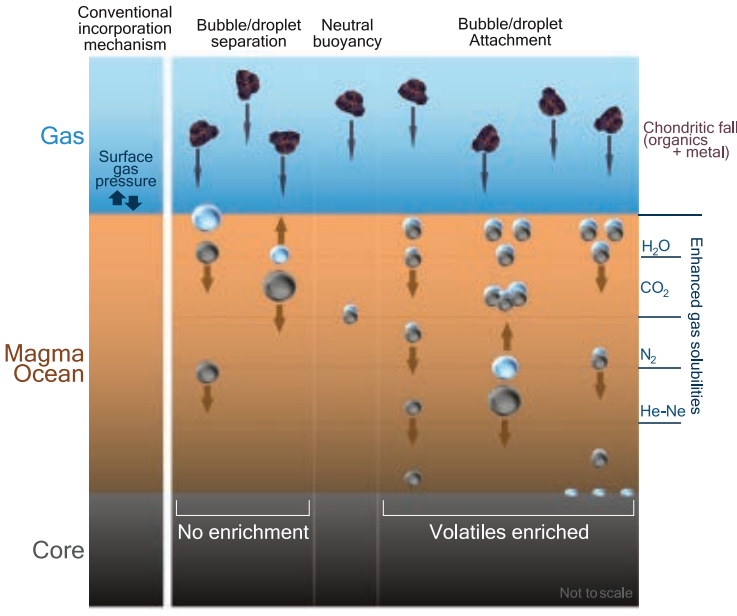

Fig. 4. Schematic diagram illustrating the dynamics of bubble–droplet association during sinking in core-forming magma oceans. The magma ocean is shown here topped by a gas phase that is expected to be low in pressure (≪1 bar), whereas at the base of the magma ocean, a molten metallic core is subjected to higher pressure (tens to thousands of bar for body sizes of 50 to 1000 km). (Left) The conventional dissolution mechanism of volatile elements requiring high gas surface pressure and thus restricted to large planetary bodies. (Right) The various types of bubble–droplet compounds, graphically illustrating the atmophile elements affected by the sinking of these compounds. The deeper the compounds sink, the more low-solubility elements are affected.

in which interactions between the fluid envelope and silicate melt are predicted to produce metal (*50*). Upon magma ocean solidification (*48*), bubble attachment to crystals and sulfides could also occur, changing both chemical and mechanical dynamics, as observed and modeled in volcanic rocks [for example, (*36*–*39*)]. Last, the resilience of bubble–droplet attachment to dilatation–compression–dissolution of the fluid in response to the convective movements of magma oceans and the associated shear processes must be addressed.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Figs. S1 to S6; Tables S1 to S3; References (*51*–*56*)

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BACTERIAL IMMUNITY

Phage proteases activate CBASS antiphage immunity

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Cyclic oligonucleotide–based antiphage signaling systems (CBASS) are bacterial immune pathways evolutionarily related to cGAS-STING (cyclic GMP-AMP synthase–stimulator of interferon genes) in humans. In CBASS, cGAS/DncV-like nucleotidyltransferase (CD-NTase) enzymes sense phage infection and synthesize nucleotide signals to initiate antiviral defense. Here, we reveal phage prohead protease activity as a widespread mechanism of CD-NTase activation. We reconstitute CBASS recognition of phage T4 in vitro and identify proteolytic cleavage of a CD-NTase activation loop as a trigger of immune signal synthesis. Phage prohead proteases directly activate CBASS in vivo, explaining how immunity is initiated late during infection. Structure-based phylogenetic analyses identify activation loops in clade A, D, and G CD-NTases and define residues controlling recognition of distinct phage families. Our results establish CBASS recognition of phage protease activity as a widespread antiviral defense mechanism.

Animal, plant, and bacterial cells each synthesize specialized nucleotide signals to respond to viral infection and initiate antiviral defense (*1*–*4*). In animals, cyclic GMP-AMP synthase (cGAS)–like receptors sense infection through recognition of viral DNA or RNA and synthesize a cyclic dinucleotide signal to activate the receptor STING (stimulator of interferon genes) (*5*–*7*). Bacteria encode an evolutionarily related pathway, named CBASS (cyclic oligonucleotide–based antiphage signaling systems), in which cGAS/DncV-like nucleotidyltransferase (CD-NTase) enzymes, which are ancestrally related to human cGAS, sense phage infection and synthesize cyclic di- and trinucleotide signals to activate downstream effector proteins and antiviral defense (*8*–*10*). CBASS operons that consist solely of a CD-NTase and an effector protein are designated as type I operons and are the most common form of CBASS. Other CBASS operons (types II to IV) encode additional CD-NTase–associated proteins (Cap) that regulate CD-NTase activation through poorly understood mechanisms (*11*–*15*). Activation of CBASS immunity causes an abortive infection response, whereby bacteria become dormant or commit to cell death to restrict phage replication and prevent viral spread to neighboring cells (*8*, *16*, *17*). As a result of this substantial fitness cost, CBASS is activated only late during the phage replication cycle and functions as a last line of antiviral defense. Consistent with this role in antiviral immunity, recent studies have identified CBASS-resistant escape mutations in phage genes, including the major capsid, scaffolding proteins, and late viral noncoding RNAs associated with the final steps of virion formation (*18*–*20*), but the specific molecular factors recognized by most CD-NTase enzymes are unknown.

A phage factor expressed late during infection activates CBASS signaling

To define the factors sensed by CBASS antiphage defense, we sought to biochemically reconstitute the key step of virally induced CD-NTase activation in vitro using the model virus *Escherichia coli* phage T4. Phage T4 and other Tevenvirinae family phages encode a set of anti-CBASS (Acb) proteins dedicated to disrupting the host nucleotide signals that control CBASS antiphage defense (*18*, *21*, *22*). We confirmed that an engineered phage T4 variant lacking functional *acb1* and *acb2* (T4 Δacb1/Δacb2) is restricted by an *E. coli* type II CBASS operon (*23*), demonstrating that Acb proteins are required to mask the effect of an unknown factor from phage T4 that activates CBASS (Fig. 1, A and B). Inactivating mutations to the *E. coli* CD-NTase CdnG (*Ec*CdnG), regulatory Cap2/Cap3, and DNase effector *Ec*Cap5 proteins each restored phage replication, supporting that phage T4 Δacb1/Δacb2 restriction occurs through canonical CD-NTase activation and downstream CBASS signaling (Fig. 1C and fig. S1A) (*15*, *16*, *22*, *24*). To biochemically reconstitute CD-NTase activation, we prepared a soluble lysate from cells infected with phage T4 Δacb1/Δacb2 and mixed this sample with a separate lysate prepared from cells expressing an effector-dead variant of CBASS (*Ec*Cap5 H56A). We analyzed CD-NTase activation in the resulting reconstitution reactions by developing an in vitro *Ec*Cap5 biosensor assay that induces cleavage of an indicator double-stranded DNA (dsDNA) plasmid in response to the *Ec*CdnG nucleotide signal 3′2′-cGAMP (Fig. 1D and fig. S1, B and C) (*25*, *26*). Using this reconstituted system, we observed that CD-NTase–dependent 3′2′-cGAMP synthesis occurs only in the presence of lysates from cells infected with phage T4 Δacb1/Δacb2 and not from uninfected cells

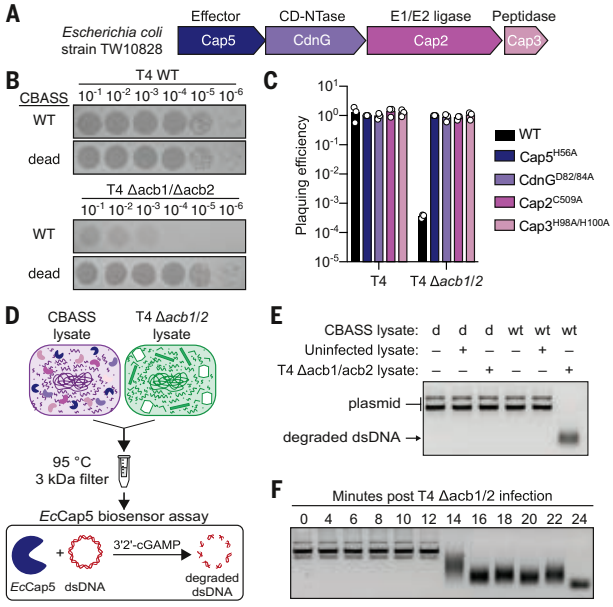

Fig. 1. A phage factor expressed late during infection activates CBASS signaling. (**A**) Schematic of the type II CBASS operon from *E. coli* strain TW10828 used in this study. (**B**) Representative plaque assay depicting serial dilutions of WT phage T4 or phage T4 Δacb1/Δacb2 plated on cells expressing either WT or dead *Ec*CdnG D82/84A CBASS operons. (**C**) Quantification of the data in (B) and fig. S1A. (**D**) Experimental schematic depicting biochemical reconstitution of CBASS sensing T4 Δacb1/Δacb2 phage infection by mixing CBASS and phage-infected lysates. Activation of CBASS signaling in the reconstitution reactions is then analyzed using the *Ec*Cap5-based 3′2′-cGAMP biosensor assay. (**E**) Agarose gel depicting production of 3′2′-cGAMP and cleavage of indicator dsDNA by *Ec*Cap5 when CBASS lysates are mixed with T4 Δacb1/Δacb2 infected lysates. For CBASS lysates, "d" indicates cells expressing operons encoding an inactive *Ec*CdnG (D82/84A) and "wt" indicates cells expressing an operon encoding WT *Ec*CdnG and an inactive *Ec*Cap5 (H56A). An en dash indicates buffer was added to the reaction and a plus sign indicates that lysate was added to the reaction. (**F**) Agarose gel depicting production of 3′2′-cGAMP and cleavage of indicator dsDNA when CBASS *Ec*Cap5 H56A lysates are mixed with T4 Δacb1/Δacb2 phage lysates harvested 14 min after infection or later.

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(Fig. 1E). We tested a timecourse of phage T4 Δacb1/Δacb2 infection and observed CD-NTase activation in all lysates prepared 14 min after infection or later (Fig. 1F). These results demonstrate that CBASS activation specifically occurs in response to a soluble factor produced during late stages of phage T4 infection.

CBASS signaling is activated by proteolytic activity

Phage T4 infection in *E. coli* occurs through a well-defined sequence of molecular events that include early shutoff of host gene expression, replication of the phage genome during the middle phase, and formation of new virions in the late stages of infection. T4 virion formation is a tightly regulated process in which capsid subunits assemble around a protein scaffold to form a structure called the prohead, which is then processed by the phage-encoded prohead protease gp21 to form the mature capsid that is competent for genome packaging (*27*). The timing of CBASS immune activation observed in our reconstitution experiments occurs immediately after the formation of phage T4 proheads, approximately 12 min after infection (Fig. 1F) (*27*). We therefore used a genome fragment approach to screen phage T4 genes that are expressed late during infection (*28*) and observed that the viral prohead protease protein gp21 was the only gene sufficient to trigger CBASS-dependent toxicity in vivo (Fig. 2, A and B, fig. S2, A to G, and table S1). In cells expressing inactive CBASS, expression of wild-type (WT) T4 gp21 did not substantially reduce cell numbers relative to catalytically dead gp21, but did alter colony size and morphology, indicating that gp21 activity causes low levels of toxicity in *E. coli* independent of CBASS signaling (Fig. 2, A and B, and fig. S2, E to G). However, in cells expressing WT CBASS, expression of active gp21 reduced cell viability by ~2.5 log relative to inactive gp21, suggesting that *Ec*CdnG activation may occur in response to proteolytic activity and not simply complex formation with a specific viral factor (Fig. 2, A and B, and fig. S2H). To extend our findings beyond the model phage T4, we tested prohead protease genes from divergent phages that encode Acb proteins and are therefore likely natural targets of CBASS immunity. The prohead proteases from *E. coli* phages RB32, RB69, *Aeromonas* phage Aeh1, and *Pseudomonas* phage phiKZ were sufficient to cause CBASS-dependent toxicity (Fig. 2, A and B, and fig. S2H). Diverse phage proteases cleaved *Ec*CdnG to an ~28-kDa species (Fig. 2C), suggesting that phage proteases act directly on *Ec*CdnG to initiate CBASS signaling. Notably, the prohead protease from phiKZ was sufficient to cause CBASS-dependent toxicity but did not directly cleave *Ec*CdnG, raising the possibility of limited cleavage or that alternative mechanisms of *Ec*CdnG activation may exist (Fig. 2, A to C). Together, these data demonstrate that phage proteases directly cleave CD-NTase enzymes and are sufficient to trigger CBASS immune signaling in vivo.

Phage prohead proteases are known to be restrained by autoproteolysis that rapidly terminates enzymatic function (*29*, *30*). To test the ability of *Ec*CdnG to detect proteolysis, we therefore modeled prohead protease activity in vitro by incubating purified *Ec*CdnG with a panel of foreign proteases of varying specificity and monitored activation using the *Ec*Cap5 biosensor assay. *Ec*CdnG potently responded to foreign protease activity in vitro with the model proteases proteinase K, trypsin, Glu-C, and chymotrypsin, each triggering production of 3′2′-cGAMP (Fig. 2D and fig. S2I). Previous analyses have demonstrated that CD-NTase activation can be induced in vitro using elevated CD-NTase enzyme concentrations, high pH (>8.5), and/or manganese divalent cations (*9*, *16*, *25*, *31*–*33*). Notably, we observed that protease-dependent activation of *Ec*CdnG was 50 times more potent than previously developed artificial reaction conditions (Fig. 2E and fig. S2J). Together, these results demonstrate that foreign protease activity is sufficient to activate *Ec*CdnG to catalyze nucleotide immune signal synthesis.

Cleavage of an unstructured loop activates *Ec*CdnG

To define how foreign proteases induce CD-NTase activation, we monitored *Ec*CdnG protein stability in vitro during activation of 3′2′-cGAMP synthesis. In the presence of trypsin, activation of *Ec*CdnG 3′2′-cGAMP synthesis cooccurred with cleavage of *Ec*CdnG into a ~28-kDa protein species (Fig. 3, A to C), similar to what we observed during in vivo activation by phage prohead proteases (Fig. 2C). We compared the AlphaFold3-predicted structure of *Ec*CdnG with experimentally determined structures of CD-NTase enzymes (*9*, *31*, *32*, *34*–*36*) and identified a surface-exposed,

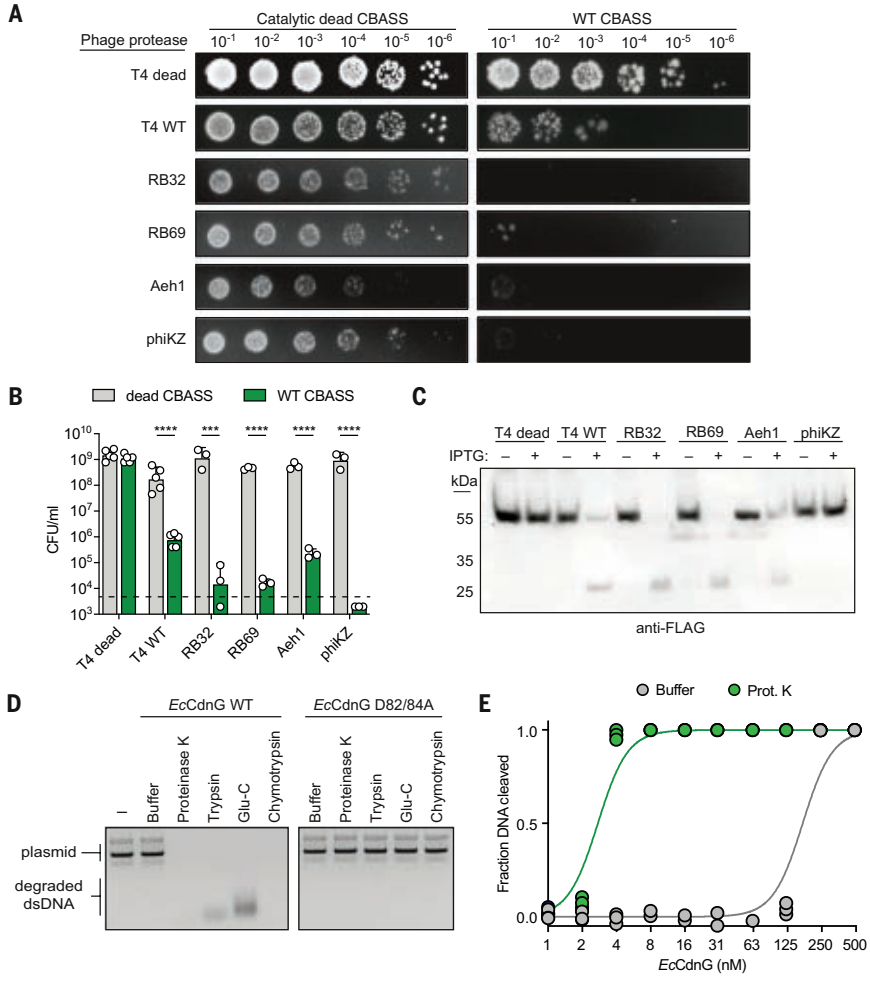

Fig. 2. CBASS signaling is activated by proteases. (**A**) Representative toxicity assay depicting serial dilutions of *E. coli* cotransformed with the indicated CBASS operon and prohead protease gene and plated on LB supplemented with arabinose and IPTG. For CBASS, "catalytic dead" refers to *Ec*CdnG D82/84A mutations and "T4 dead" refers to H85A/S140A mutations. (**B**) Quantification of the data in (A). The dashed line indicates the limit of detection. Data are presented as mean ± SD from $n = 3$ to 5 independent experiments. *** indicates $P < 0.001$, **** indicates $P < 0.0001$. (**C**) Representative anti-FLAG immunoblot of 3×FLAG-*Ec*CdnG cleavage when coexpressed with the indicated phage prohead proteases. (**D**) Agarose gel depicting *Ec*Cap5 activation and cleavage of indicator dsDNA when either WT or catalytically inactive *Ec*CdnG is treated with the indicated protease. (**E**) Quantification of *Ec*Cap5 cleavage of indicator dsDNA when titrating amounts of *Ec*CdnG are treated with proteinase K.

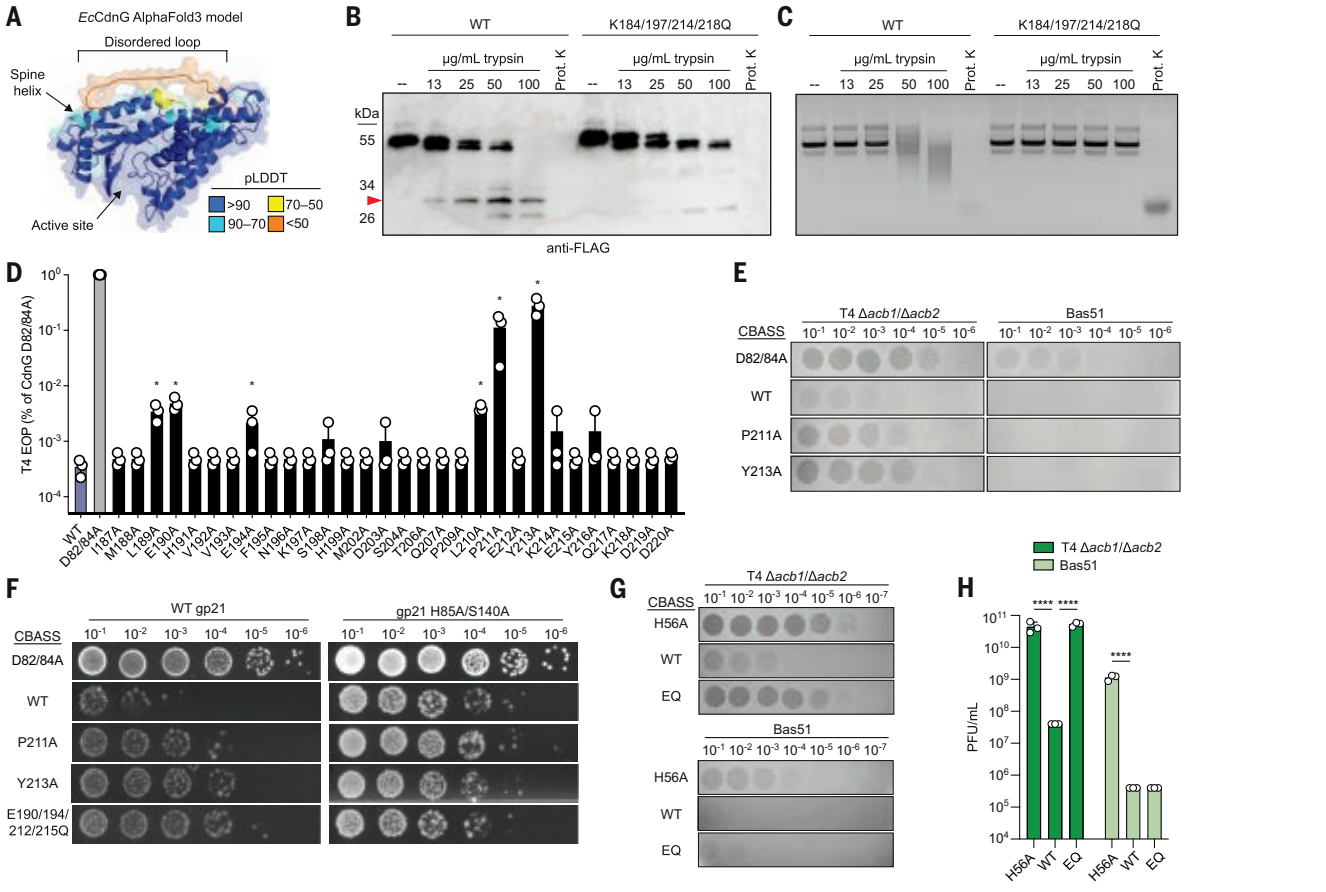

Fig. 3. Cleavage of an unstructured loop activates *Ec*CdnG. (**A**) AlphaFold3 model of *Ec*CdnG highlighting the unstructured loop and canonical features of CD-NTase proteins. Model is colored by pLDDT confidence score. The flexible C-terminal tail of *Ec*CdnG was removed for clarity. (**B**) Representative anti-FLAG immunoblot of either WT or trypsin-resistant 3×FLAG-*Ec*CdnG treated with titrating amounts of trypsin or proteinase K. Red arrow indicates ~28 kDa species that cooccurs with 3′2′-cGAMP synthesis. (**C**) Agarose gel depicting *Ec*Cap5 activation and cleavage of indicator dsDNA when either WT or trypsin-resistant 3×FLAG-*Ec*CdnG is treated with titrating amounts of trypsin or proteinase K. (**D**) Efficiency of plaquing (EOP) of T4 Δacb1/Δacb2 phage plated on cells encoding CBASS operons with the indicated *Ec*CdnG mutation. Data are normalized to the *Ec*CdnG D82/84A dead operon and are presented as mean ± SD from $n = 3$ independent experiments. * indicates $P < 0.05$ compared with the wild type. (**E**) Representative plaque assays of serial dilutions of T4 Δacb1/Δacb2 phage or Bas51 phage plated on cells encoding the indicated *Ec*CdnG point mutations. (**F**) Representative toxicity assay depicting serial dilutions of *E. coli* cotransformed with CBASS operons encoding the indicated *Ec*CdnG mutations and either WT or catalytically dead T4 prohead protease plated on LB supplemented with arabinose and IPTG. (**G**) Representative plaque assays of T4 Δacb1/Δacb2 phage or Bas51 phage plated on cells encoding either inactive *Ec*Cap5 (H56A), WT, or *Ec*CdnG E190Q, E194Q, E212Q, E215Q (EQ) CBASS operons. (**H**) Quantification of the data in (G). Data are presented as mean ± SD from $n = 3$ independent experiments. **** indicates $P < 0.0001$.

disordered loop consisting of ~30-amino acids in the center of the canonical CD-NTase bi-lobed enzyme architecture, where cleavage would result in production of a ~28-kDa product (Fig. 3A). Substitution of all four Arg/Lys trypsin cleavage sites to glutamine within this disordered *Ec*CdnG loop (K184Q, K197Q, K214Q, K218Q) abolished the ability of *Ec*CdnG to produce nucleotide immune signals in response to trypsin protease activity (Fig. 3, A to C). Notably, the trypsin-resistant mutant *Ec*CdnG enzyme retained the ability to sense proteinase K activity and restricted phage T4 Δacb1/Δacb2 in vivo (Fig. 3C and fig. S3, C and D), demonstrating that the mutated *Ec*CdnG enzyme is functional and could still respond to proteases with alternative cleavage site specificities.

The phage T4 gp21 protease is known to preferentially cleave target substrates C-terminal of glutamate residues within the amino acid motif I/LxE (*30*, *37*). The *Ec*CdnG disordered loop contains the residue LPE, consistent with the ability to detect gp21 protease activity during phage T4 infection. We used alanine-scanning mutagenesis to disrupt the *Ec*CdnG disordered loop in vivo and observed that mutations to residue P211 or the adjacent residue Y213 in the LPE motif attenuated the ability of CBASS by more than 100 times to defend against phage T4 Δacb1/Δacb2 infection (Fig. 3, D and E, and fig. S3A). To further define the ability of this disordered loop to control the specificity of *Ec*CdnG phage recognition, we screened the BASEL phage collection for additional susceptible phages and identified that the *Ec*CdnG CBASS operon successfully defends against phages Bas48 to 59 of the *Vequintavirinae* family (fig. S3B) (*38*). *Ec*CdnG mutations P211A and Y213A partially disrupted CBASS defense against phage T4 but not against phage Bas51, suggesting that these mutations alter the specificity of CBASS phage recognition and do not compromise overall immune function (Fig. 3E and fig. S3A). Expression of the *Ec*CdnG P211A or Y213A mutant operons resulted in similar levels of toxicity as that of the WT operon in the absence of gp21 activity, further supporting the functionality of these mutant operons (Fig. 3F and fig. S3, E and F). However, when coexpressed with active T4 gp21, we observed a ~2.5-log reduction in cell viability with the WT operon but only a ~1-log reduction with the P211A and Y213A operons, further suggesting that *Ec*CdnG loop mutations do not abolish CBASS signaling generally but instead affect gp21-dependent activation of CBASS in vivo (Fig. 3F and fig. S3, E and F). Notably, mutation of the *Ec*CdnG residue E212 that resides within the putative LPE motif was not

sufficient to abolish defense (Fig. 3D and fig. S3A). However, the phage T4 gp21 protease is known to more weakly target noncanonical xxE motifs (*39*). We observed that an *Ec*CdnG variant with all glutamate residues in the disordered loop mutated to glutamine (E190Q, E194Q, E212Q, E215Q) was no longer able to respond to gp21 expression or defend against phage T4 replication, but retained the ability to restrict phage Bas51 (Fig. 3, F to H, and fig. S3, E and F). Together, these data demonstrate that *Ec*CdnG contains a surface-exposed activation loop that functions as a molecular tripwire to sense phage protease function.

CD-NTase proteins use cleavage of activation loops as a widespread mechanism of viral recognition

CD-NTases comprise a diverse family of thousands of signaling enzymes in CBASS immunity that segregate into distinct clades designated A to H (*9*, *11*). Members of distinct clades often share less than 20% sequence identity and there are no clear structural features that distinguish each clade, but all CD-NTase enzymes share a catalytic core capable of synthesizing cyclic nucleotide signals. To determine whether sensing protease activity is common throughout CBASS immunity, we clustered CD-NTase protein sequences from each clade by identity and combined analysis of experimental and AlphaFold3-predicted structures to search each cluster for conserved patterns of potential activation loops. These analyses revealed that ~24% of all CBASS operons contain a CD-NTase with a large surface-exposed activation loop in the same three-dimensional location as the activation loop of *Ec*CdnG (Fig. 4A). Activation loops are present in nearly all CD-NTase sequences in clades A and G and ~18% of the sequences in clade D, suggesting that these CBASS operons may share a unified mechanism of immune activation (Fig. 4A). Structure-guided alignments of diverse representative CD-NTases reveal that within each clade, the activation loops contain limited sequence similarity but are framed by the same conserved secondary structure elements. To determine whether diverse CD-NTases with putative activation loops also respond to foreign protease activity, we purified 12 proteins representing each major cluster in clades A, D, and G, and quantified the ability of these enzymes to signal in response to proteinase K (Fig. 4B, fig. S4A, and table S2). Notably, for the CD-NTase clade clusters predicted to contain activation loops, seven of the nine representative enzymes catalyzed nucleotide immune signal synthesis in response to proteinase K treatment (Fig. 4B). Notably, the enzymes that contained putative activation loops but did not respond to proteinase K treatment were either poorly expressed in *E. coli* or purified with an N-terminal MBP fusion, suggesting that these CD-NTases may in fact respond to protease activity under more optimal experimental conditions (fig. S4A). CD-NTase enzymes responsive to protease function notably include the major model operons used to define the function of CBASS immunity *Vibrio cholerae* DncV and *Enterobacter cloacae* CdnD (*8*, *9*, *15*, *16*, *40*). By contrast, three of three CD-NTases from the clade D clusters lacking predicted activation loops failed to respond to protease treatment as expected (Fig. 4B). Together, these findings indicate that proteolytic

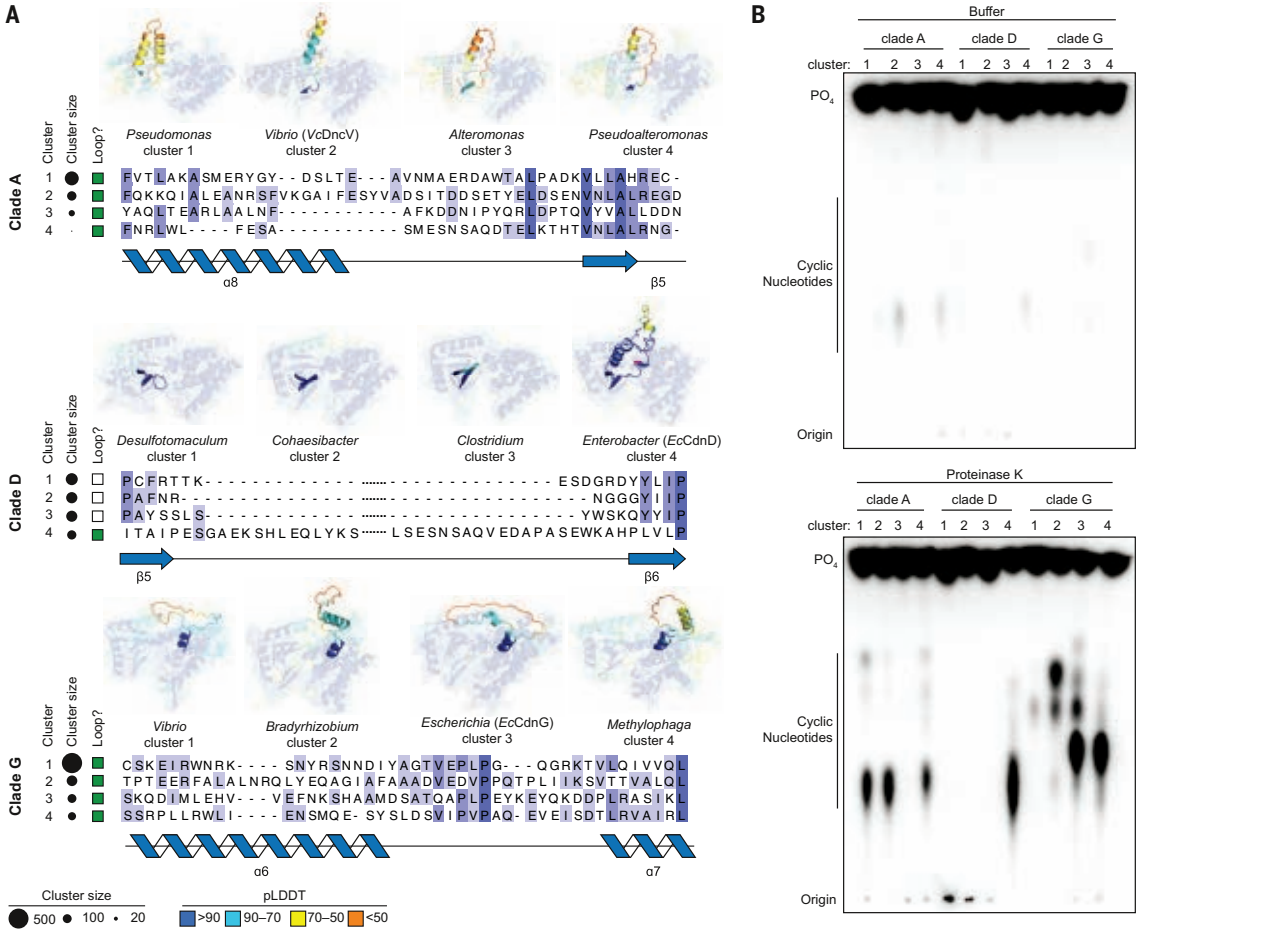

Fig. 4. CD-NTase proteins use cleavage of activation loops as a widespread mechanism of viral recognition. (**A**) AlphaFold3 models and sequence alignments of representatives from the four largest clusters of CD-NTases in clades A, D, or G. The strength of shading in the sequence alignments indicates the degree of residue conservation. (**B**) Representative TLC assays depicting proteinase K-dependent activation of nucleotide immune signal synthesis by the CD-NTase cluster representatives shown in (A).

cleavage of the CD-NTase activation loops is a widespread mechanism of activation shared across diverse bacteria, and also suggest that detection of phage proteases is a major trigger controlling CBASS immune signaling.

Discussion

This study identifies viral protease activity as a major trigger of CBASS antiphage defense. We demonstrate that phage prohead proteases cleave *Ec*CdnG to activate CBASS signaling in vivo and show that CD-NTase enzymes sensitively respond to proteolytic cleavage as a direct biochemical trigger in vitro. Mechanistically, our data support a model in which a disordered CD-NTase activation loop acts as a protease-sensitive regulatory element that, once cleaved, enables nucleotide immune signal synthesis (fig. S4B). We further show that activation loop architectures are conserved across >1300 CD-NTase enzymes in multiple clades, suggesting that sensing of phage maturation through detection of protease activity is a widely conserved activating signal in CBASS immunity. A diverse set of proteases with distinct substrate specificities were each sufficient to activate *Ec*CdnG, demonstrating that cleavage at distinct positions within the activation loop can trigger enzyme activation. This finding raises the possibility that additional phage or host-encoded proteases could contribute to CBASS activation in other settings, but evolution of CD-NTase activation loops is likely restrained by host protease activity to prevent inadvertent activation and autoimmunity. Mechanistically, CD-NTase activation loop cleavage may facilitate enzyme oligomerization or release of inhibitory molecules such as folate metabolites known to restrain CBASS activation (*35*, *41*, *42*). Defining the kinetics, spatial constraints, and structural transitions linking activation loop cleavage to CD-NTase catalytic function during phage infection will be an important goal of future research.

Protease-dependent activation is consistent with the observation that CBASS functions at a late stage of infection and suggests that previously reported CBASS escape mutations in phage maturation factors may alter the regulation of prohead protease activity (*8*, *18–20*). Notably, the type II CBASS components Cap2 and Cap3 are dispensable for protease-dependent activation of *Ec*CdnG in vitro but are required for phage restriction in cells. Cap2 and Cap3 are known to regulate ubiquitin-like conjugation of CD-NTases to the host protein phage shock protein A (pspA) and other unidentified host proteins, although their specific role in CD-NTase regulation during phage infection remains unclear (*14*, *15*, *22*). Our data suggest that these CBASS accessory proteins may function in localization or scaffolding roles to position CD-NTase enzymes at the inner membrane, where phage protease activity and the initial stages of virion maturation occur. Notably, a type I CBASS operon from *Staphyloccocus aureus* that encodes a clade E CD-NTase enzyme is activated by structured phage RNAs and does not contain an activation loop, suggesting that mechanisms of CD-NTase activation may be specific to CD-NTase clades or CBASS operon types (*20*). The specific roles of Cap2 and Cap3 in CBASS activation during phage infection will be an important area of future investigation. Other nucleotide-based antiphage systems such as Pycsar also encode Cap2 and Cap3 homologs, suggesting that sensing of phage protease activity could regulate additional immune signaling pathways (*15*). Finally, our results parallel mechanisms of pathogen detection by animal and plant immune pathways including NLRP1 and CARD8 inflammasomes in mammals and disease-resistance proteins in plants, which are directly cleaved and activated by viral and bacterial proteases (*43–47*). Collectively, these findings reveal a major mechanism of viral sensing in CBASS immunity and define pathogen protease activity as a conserved foreign signal that activates immunity in multiple kingdoms of life.

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SUPPLEMENTARY MATERIALS

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MATERIALS SCIENCE

Synergistic polar states by selective atomic gradients

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Materials that combine distinct properties within a single phase are of fundamental and technological interest. Here, we present a broadly applicable strategy to achieve strong bulk polarity that not only coexists with its originally incompatible properties, such as metallicity, but also synergistically enables exceptional functionalities. A combination of thin-film synthesis, atomic-scale imaging, and theoretical calculations reveals that A-site selective atomic gradients induce strong polar states in otherwise centrosymmetric perovskite oxides ABO_3. These polar states unconventionally coexist with various preexisting properties, leading to bulk polar metallicity with tunable nonreciprocal transport; high-κ, low-loss dielectricity with an equivalent oxide thickness below 0.1 nanometer; and giant pyroelectricity. This work may enable the development of new multifunctional materials with unusual coexisting and synergistic properties.

The polar state, characterized by a permanent electric dipole moment, has served as the basis for key functionalities, including nonlinear optical, pyroelectric, and piezoelectric properties (*1*–*3*). Beyond these traditional applications, the polar state offers even more exciting prospects for new functional phenomena when combined with other electronic properties (*4*–*7*). However, a bulk polar state typically excludes several important electronic properties like metallicity, magnetism, and superconductivity (*8*, *9*). In particular, macroscopic electric polarization is well defined only in insulators, not in metals (*10*), and electrostatic screening by itinerant electrons in metals hinders long-range dipole ordering and polar state (*11*). Furthermore, polar states are limited to specific crystal point groups, rather than abundant crystalline structures such as orthorhombic *Pnma* (*12*). This restriction reduces potential combinations of the polar state with other structural distortions (e.g., oxygen octahedral rotation in perovskite oxides) that could enhance various functionalities (*13*). Therefore, developing a universal method to achieve a strong bulk polar state that is compatible and synergistic with existing properties is of great interest.

To this end, we focus on a transition metal perovskite oxide, ABO_3, which is an important class of materials owing to its wide range of properties and potential applications (*14*). Transition metal atoms (i.e., B in ABO_3) and their d electrons mainly determine the electronic properties of ABO_3 (*15*). By contrast, non–transition metal atoms (i.e., A in ABO_3) usually determine the lattice size of ABO_3. On the basis of these distinct roles of atoms, we engineer a "selective atomic gradient" by controlling the composition ratio of only A-site atoms (i.e., A and A′ atoms) in each atomic layer of $(A,A')BO_3$ films (Fig. 1A). The difference in ionic radii of A and A′ generates a large strain gradient in epitaxial films of $(A,A')BO_3$. The strain gradient can, in turn, create strong bulk polarity, characterized by the off-center displacement (δ_B) of B-site atoms, through the phenomenon of flexoelectricity (*16*, *17*). Notably, this applies to any class of materials, including metals (*6*, *18*, *19*). Unlike traditional compositional gradients widely explored in perovskite oxides (*20*–*22*), our layer-by-layer A-site selective gradient generates a giant strain gradient, inducing bulk polarity via flexoelectricity. It provides a universal pathway to polarity—challenging for traditional gradients, especially in metals where conduction electrons screen purely compositional effects. Furthermore, the flexoelectrically induced bulk polarity could synergize with other properties, such as oxygen octahedral rotations, primarily stemming from B-site or oxygen atoms.

Here, we demonstrate such a synergistic bulk polar state by the A-site selective atomic gradients using two distinct $(A,A')BO_3$, i.e., perovskite iridate $(Sr,Ca)IrO_3$ and titanate $(Ba,Ca)TiO_3$. Perovskite iridate is an intriguing metallic system with strong spin-orbit coupling (*23*). Our atomic-gradient $(Sr,Ca)IrO_3$ films realize a bulk Rashba metal with the synergistic coexistence of strong bulk polarity, metallicity, spin-orbit coupling, and IrO_6 octahedral rotation. By contrast, perovskite titanate is a representative dielectric (*24*), but ultrathin films suffer from large leakage caused by chemical complexity and point defects (*25*). In atomic-gradient $(Ba,Ca)TiO_3$, the coexistence of large polarization and TiO_6 octahedral rotation generates a divergent polar structure that suppresses leakage in the ultrathin limit while preserving the intrinsic dielectric response. Consequently, our atomic-gradient $(Ba,Ca)TiO_3$ systems offer high-κ, low-loss ultrathin oxides with an equivalent oxide thickness below 0.1 nm and also exhibit giant pyroelectricity. These results indicate that synergistic polar states by selective atomic gradients can be created in various material systems, engendering unusual coexisting properties and new functions.

Polar metallic iridates by atomic gradients

We synthesize metallic $(Sr,Ca)IrO_3$ epitaxial films with A-site selective atomic gradients, using a pulsed laser deposition technique, on a (001) $SrTiO_3$ substrate (*26*). By sequential deposition, we finely control the deposited amounts of $SrIrO_3$ and $CaIrO_3$ within a single atomic layer. By monitoring the reflection high-energy electron diffraction intensity, we precisely control both the two-material sequential deposition and single-atomic-layer deposition (fig. S1). This allows us to accurately fabricate an atomic layer with the desired compositional ratio of Sr and Ca and realize films with A-site atomic gradients, where the Sr/(Sr + Ca) ratio systematically changes for each atomic layer. We refer to the atomic-gradient films, which start with $SrIrO_3$ ($CaIrO_3$) at the bottom and end with $CaIrO_3$ ($SrIrO_3$) at the top, as $(Sr_b,Ca_t)IrO_3$ [or $(Ca_b,Sr_t)IrO_3$]. These films exhibit the well-defined A-site atomic gradients (Fig. 1B), while preserving good crystallinity and ordinary metallic behavior (figs. S2 and S3).

The difference in ionic radii between Sr and Ca naturally generates substantial strain gradients in atomic-gradient epitaxial films. To detect these strain gradients, we image and analyze the in-plane and out-of-plane lattice parameters of local unit cells (u.c.) by scanning transmission electron microscopy (STEM) (fig. S4). Figure 1C unveils a clear lattice gradient, where the out-of-plane lattice parameter is higher (lower) in the Sr-rich (Ca-rich) region of $(Sr_b,Ca_t)IrO_3$. Near the bottom interface this trend is less clear, which might arise from the chemical mixing of $SrIrO_3$ and $SrTiO_3$ (*27*), or from octahedral rotation–related lattice-volume changes. Comparing the measured lattice parameters with bulk values yields large transverse strain gradients of $\partial u_{11}/\partial x_3 = 6.59 \times 10^6\ m^{-1}$ in $(Sr_b,Ca_t)IrO_3$ films (fig. S4),

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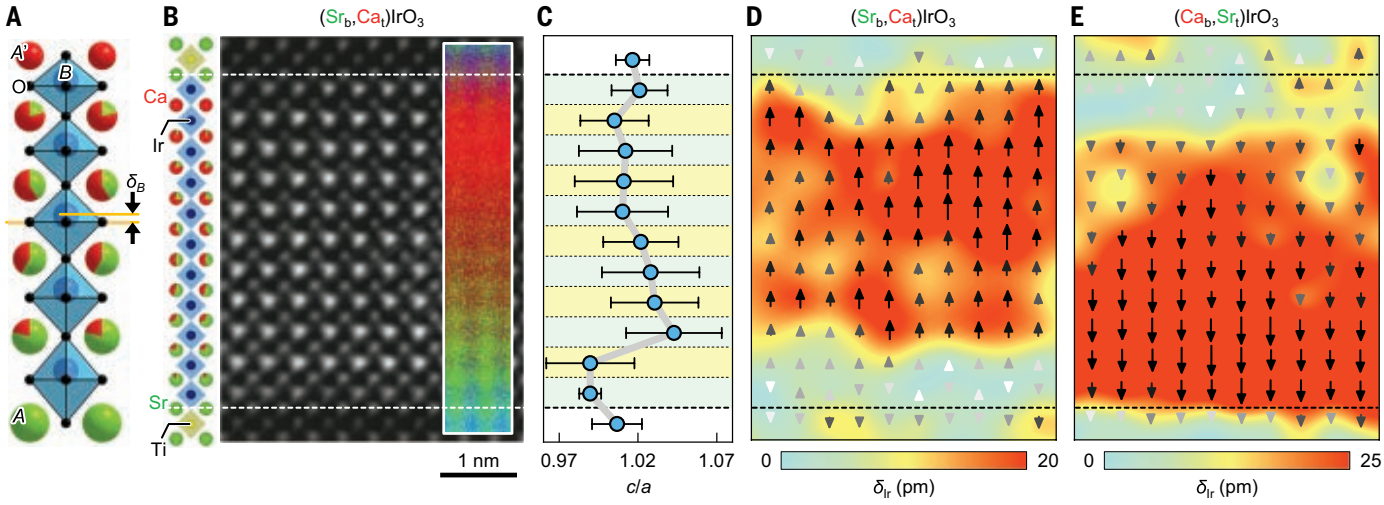

Fig. 1. Polar metals by selective atomic gradients. (**A**) A schematic of perovskite oxides (A,A′)BO_3 with A-site selective atomic gradients. δ_B represents the out-of-plane off-centering of B-site atoms. (**B**) High-angle annular dark-field (HAADF)–STEM image of the 11-u.c.-thick (Sr$_b$,Ca$_t$)IrO_3 film, taken in the [100] zone axis of the $SrTiO_3$ substrate. A schematic on the left illustrates 11-u.c.-thick (Sr$_b$,Ca$_t$)IrO_3, sandwiched by $SrTiO_3$. The inset is the energy-dispersive spectroscopy (EDS) mapping, showing the A-site atomic gradient from Sr (green) to Ca (red). (**C**) Average tetragonality, c/a, of each atomic layer in (B). The error bar indicates the standard deviation. (**D**) Mapping of δ_{Ir} in (B). (**E**) Mapping of δ_{Ir} in 11-u.c.-thick (Ca$_b$,Sr$_t$)IrO_3.

consistent with geometric phase analysis (fig. S5). Therefore, A-site selective atomic gradients in ABO_3 can effectively generate giant strain gradients without compromising the original properties, potentially inducing strong bulk polarity.

Figure 1D presents the Ir off-centering (δ_{Ir}) in (Sr$_b$,Ca$_t$)IrO_3 films, measured with reference to the center of the four neighboring A-site atoms. (Sr$_b$,Ca$_t$)IrO_3 films clearly show δ_{Ir}, which reaches 10 pm on average and even 30 pm locally, and whose detailed profile follows the evolution of local strain gradients. By contrast, solid-solution (Sr$_{0.5}$Ca$_{0.5}$)IrO_3 films without atomic gradients show no noticeable δ_{Ir} (fig. S6), despite their chemical composition and strain being on average identical to those of atomic-gradient films. Furthermore, (Ca$_b$,Sr$_t$)IrO_3 films with the reversed atomic gradients [i.e., with Ca-rich (Sr-rich) layers near the bottom (top) interface] exhibit Ir off-centering with a direction opposite to that of (Sr$_b$,Ca$_t$)IrO_3 (Fig. 1E). Our optical second-harmonic generation (SHG) (fig. S7) and density functional theory (DFT) calculations (fig. S8) also consistently manifest the bulk polarity and obvious δ_{Ir} in atomic-gradient iridate films, respectively. These results provide evidence for the critical role of atomic gradients in inducing bulk polarity in otherwise centrosymmetric metals.

Tunable nonreciprocal transport

Polar metallic iridates containing heavy elements (i.e., Ir) satisfy the key ingredients of Rashba interaction, that is, broken inversion symmetry and spin-orbit coupling (*28*). Furthermore, our STEM studies reveal that large Ir off-centering in our films could even coexist with substantial IrO_6 octahedral rotation (fig. S9 and fig. S10), which typically conflicts with and suppresses polar distortions (*12*). The coexisting octahedral rotation could intensify the mixing of d-orbitals through spin-orbit coupling, leading to an increased Berry curvature (fig. S11) and a consequently larger Rashba interaction in our polar metals (*13*). The general form of the Rashba interaction is

$$H_R = \alpha_R(\boldsymbol{\sigma} \times \mathbf{k}) \cdot \hat{\mathbf{z}} \quad (1)$$

where the Rashba parameter, α_R depends on the strength of spin-orbit coupling and inversion asymmetry and $\mathbf{k}$ denotes the wave vector, $\boldsymbol{\sigma}$ the Pauli matrices, and $\hat{\mathbf{z}}$ the direction of the polar axis. H_R splits the electronic band into the two oppositely spin-polarized bands, and DFT calculations find a large Rashba splitting (Fig. 2A and fig. S12) and the presence of spin texture on the states on the Fermi surface (Fig. 2B) (*26*) in our systems. In addition, the spin projection along the Γ-Y direction (Fig. 2C and fig. S13) manifests the Rashba-type spin-splitting around the Fermi energy owing to the Ir off-center displacement.

When an in-plane magnetic field is applied to Rashba conductors, the additional Zeeman effect distorts the Fermi surfaces asymmetrically, making the electrical resistance depend on the direction of the current. This so-called nonreciprocal resistance can be written as (*29*)

$$R(I) - R(-I) = R_0\mathbf{B} \cdot (\mathbf{I} \times \mathbf{P}) = R_0\gamma BI\cos\theta \quad (2)$$

where the electric current $\mathbf{I}$ flows along $\hat{\mathbf{x}}$, the magnetic field $\mathbf{B}$ is applied along the in-plane direction, the bulk polarity $\mathbf{P}$ is along $\hat{\mathbf{z}}$, the angle θ is between $\hat{\mathbf{y}}$ and $\mathbf{B}$ (inset in Fig. 2D), R_0 is the resistance at zero magnetic field, and γ is the nonreciprocal coefficient. Thus, the nonreciprocal resistance could serve as an experimental probe for exploring Rashba conductors.

We detect nonreciprocal resistance as the second-harmonic voltage signal in lock-in measurements (*29*). For an ac input current [$I^{\omega}(t) = I_0\sin\omega t$], the nonlinear voltage caused by the nonreciprocal transport can be expressed as

$$V^{2\omega}(t) = \frac{1}{2}\gamma R_0 BI_0^2\cos\theta\left\{1+\sin(2\omega t-\frac{\pi}{2})\right\} \quad (3)$$

Measuring the second-harmonic signal with a $\pi/2$ phase shift yields the nonreciprocal magnetoresistance, defined as $\Delta R^{2\omega} \equiv V^{2\omega}/I_0 = (\gamma R_0 BI_0\cos\theta)/2$. We consider the B-antisymmetric components (i.e., $[\Delta R^{2\omega}(B) - \Delta R^{2\omega}(-B)]/2$) to eliminate background signals, e.g., arising from thermal effects. (Sr$_b$,Ca$_t$)IrO_3 films clearly exhibit a finite $\Delta R^{2\omega}$ at $\theta = 0°$ and $B \neq 0$, but a negligible $\Delta R^{2\omega}$ at $\theta = 90°$ (Fig. 2D); the angular dependence follows cosθ (Fig. 2E). Furthermore, $\Delta R^{2\omega}$ is tunable by controlling the atomic gradients: (Ca$_b$,Sr$_t$)IrO_3 films exhibit $\Delta R^{2\omega}$ with the sign opposite to that of (Sr$_b$,Ca$_t$)IrO_3 (Fig. 2F), whereas (Sr$_{0.5}$Ca$_{0.5}$)IrO_3 films without atomic gradients present no $\Delta R^{2\omega}$ (fig. S14). According to a phenomenological theory (*30*), the coefficient γ is proportional to $\alpha_R\tau^2/\varepsilon_F$, where τ is the spin relaxation time and ε_F is the Fermi energy. Although our metallic iridates generally have a high carrier concentration of $>10^{21}$ cm^{-3} (i.e., large ε_F), the measured γ value is as large as 2.9×10^{-3} $A^{-1}T^{-1}$ and even comparable to that of some Rashba semiconductors [e.g., α-GeTe (*31*)]. This implies that large Ir off-centering, coexisting with spin-orbit coupling and

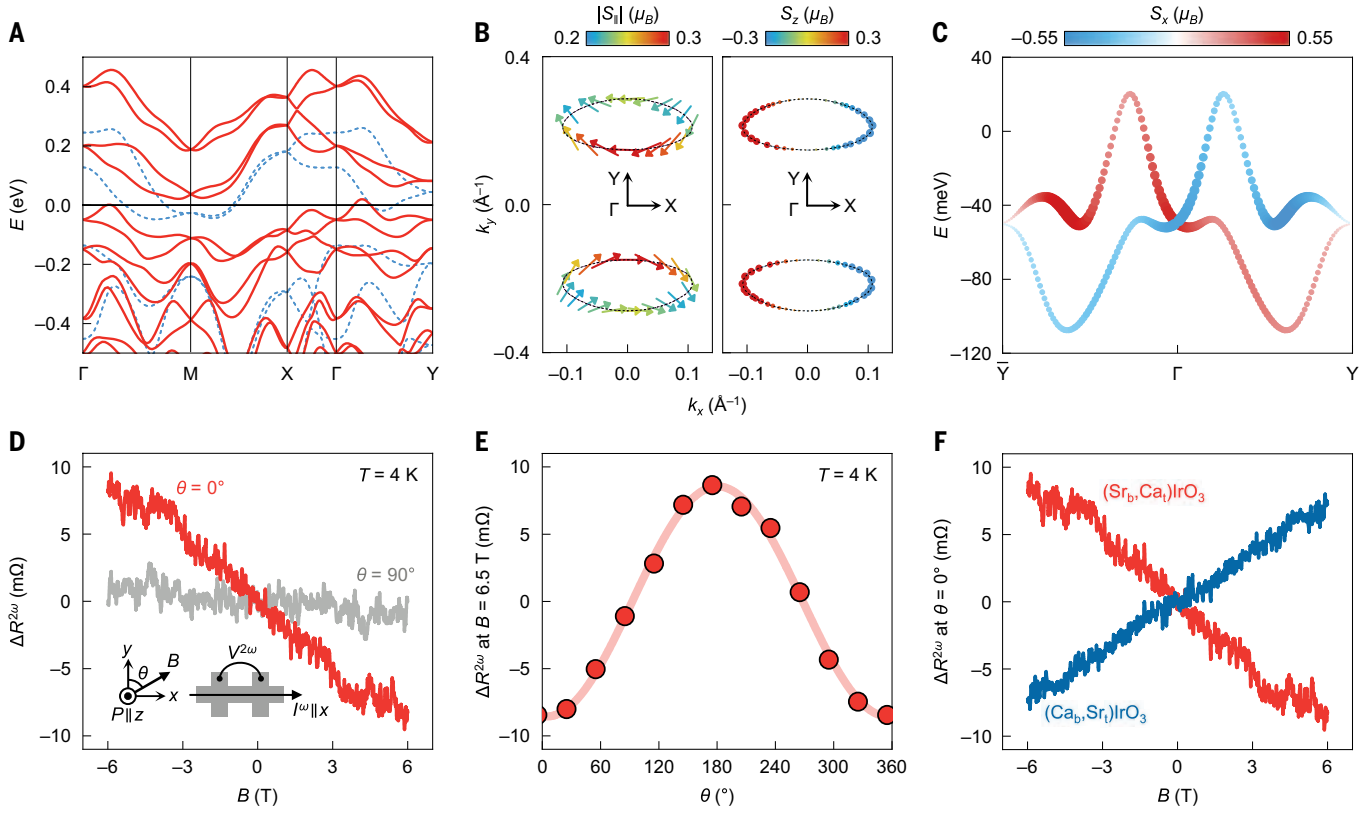

Fig. 2. Rashba-type band structures and nonreciprocal transport in polar metallic iridates. (**A**) Calculated band structures of bulk $SrIrO_3$ with (red solid lines) and without Ir off-center displacement (blue dashed lines). A large Rashba splitting of ~100 meV in the bands around the Fermi energy is observed. The Fermi energy is defined as zero. (**B**) Calculated spin texture of the states on the Fermi surface. The left panel shows the in-plane spin components on the Fermi surface in the (110) plane, where the length and color of the arrows represent the spin's magnitude. The right panel plots the out-of-plane spin component along the [110] direction. The color and size of the dots represent the direction and magnitude of these out-of-plane moments. (**C**) Calculated spin-projected band structures along the $\bar{Y} - \Gamma - Y$ high-symmetry line of the bands in (A), showing Rashba spin texture. (**D**) Magnetic field dependence of the measured nonreciprocal magnetoresistance $\Delta R^{2\omega}$ at $T = 4$ K of 11-u.c.-thick $(Sr_b,Ca_t)IrO_3$. $\Delta R^{2\omega}$ is antisymmetrized with respect to B to eliminate background signals, e.g., arising from thermal effects. (**E**) Angular dependence of $\Delta R^{2\omega}$ at $T = 4$ K and $B = 6.5$ T of $(Sr_b,Ca_t)IrO_3$. The experimental $\Delta R^{2\omega}$ data (red circles) are well fitted by a cosθ curve (red line). (**F**) Magnetic field dependence of $\Delta R^{2\omega}$ at $T = 4$ K and $\theta = 0°$ of $(Sr_b,Ca_t)IrO_3$ (red line) and $(Ca_b,Sr_t)IrO_3$ (blue line).

octahedral rotation, leads to sizable spin band splitting (i.e., a large α_R) in polar metallic iridates, in line with DFT results (Fig. 2A).

Giant polar divergence in atomic-gradient titanates

We further extend our approach to dielectric perovskite titanate $ATiO_3$ to demonstrate the synergistic combination of inherent polarization response, strong bulk polarity, and octahedral rotation. We synthesize (Ba,Ca)TiO_3 epitaxial films with A-site selective atomic gradients on $SrRuO_3$-buffered (001) $SrTiO_3$ substrate (Fig. 3A), by controlling the deposited amount of $BaTiO_3$ and $CaTiO_3$ within a single atomic layer (fig. S15). We refer to the atomic-gradient films, which start with $BaTiO_3$ ($CaTiO_3$) at the bottom and end with $CaTiO_3$ ($BaTiO_3$) at the top, as $(Ba_b,Ca_t)TiO_3$ [or $(Ca_b,Ba_t)TiO_3$]. $(Ba_b,Ca_t)TiO_3$ films demonstrate a well-defined A-site atomic gradient (Fig. 3A), while preserving good crystallinity (figs. S2, S3, and S16) and consequent lattice gradient (Fig. 3B and fig. S17). The local lattice measurements determine the strain gradient as 1.1×10^7 m^{-1} (figs. S18 and S19), large enough to flexoelectrically polarize the films. Figure 3C presents polarization vectors for each unit cell of $(Ba_b,Ca_t)TiO_3$ films, estimated on the basis of the measured positions of A-site, B-site, and oxygen atoms (*26*). $(Ba_b,Ca_t)TiO_3$ films show a large downward polarization of up to 40 μC cm^{-2} on average, reaching 60 μC cm^{-2} locally in the Ba-rich layers (fig. S20). $(Ca_b,Ba_t)TiO_3$ films with reversed atomic gradients exhibit opposite directed (i.e., upward) polarization (fig. S18), but $(Ba_{0.5}Ca_{0.5})TiO_3$ films without atomic gradients show no noticeable polarization (fig. S21). These results show the critical role of atomic gradients in inducing large electric polarization in otherwise centrosymmetric (Ba,Ca)TiO_3, consistent with SHG and DFT results (figs. S22 and S23).

Although the local strain gradient is nearly uniform throughout the $(Ba_b,Ca_t)TiO_3$ films (Fig. 3B and fig. S18B), the local polarization varies greatly in magnitude (Fig. 3D). The A-site atomic gradient from Ba to Ca induces a crystal-structure gradient from tetragonal to orthorhombic *Pnma* (Fig. 3E), in addition to the strain gradient. Whereas BO_6 octahedral rotation in the *Pnma* structure tends to compete with bulk polarity (*12*), $(Ba_b,Ca_t)TiO_3$ films show the interplay between polarity and TiO_6 octahedral rotation, as well as their coexistence. First, the polarization remains as large as 30 μC cm^{-2}, even in the middle layers with $(Ba_{0.5}Ca_{0.5})TiO_3$ or $(Ba_{0.4}Ca_{0.6})TiO_3$ compositions that exhibit sizable TiO_6 octahedral rotation. Furthermore, as the Ca concentration increases, the enhanced TiO_6 octahedral rotation steeply suppresses the electric polarization, resulting in an extraordinarily large divergence of atomic polarization, $\nabla \cdot \mathbf{P}$, in atomic-gradient (Ba,Ca)TiO_3 (Fig. 3D). By contrast, DFT calculations reveal that atomic-gradient (Ba,Sr)TiO_3 without TiO_6 octahedral rotation exhibits a relatively uniform polarization with negligible divergence (fig. S24). This underscores that the large $\nabla \cdot \mathbf{P}$ in atomic-gradient (Ba,Ca)TiO_3 films originates from the coexistence of large flexoelectric polarization and TiO_6 octahedral rotation.

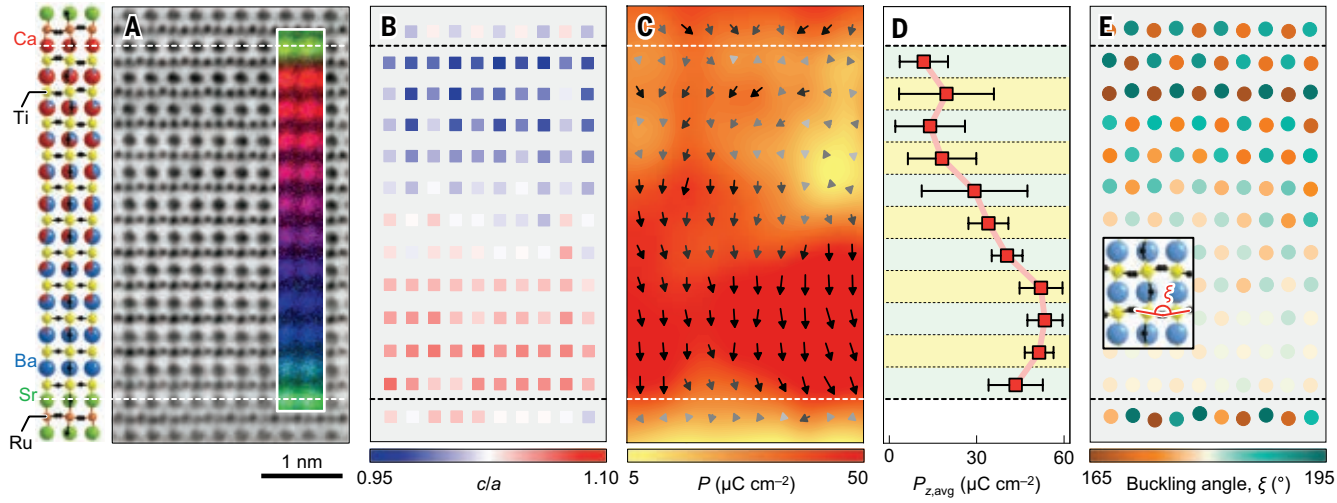

Fig. 3. Giant polar divergence in atomic-gradient titanates. (**A**) Annular bright-field (ABF)–STEM image of the 11-u.c.-thick $(Ba_b,Ca_t)TiO_3$ film, taken in the (pseudo)cubic [110] zone axis. A schematic on the left illustrates 11-u.c.-thick $(Ba_b,Ca_t)TiO_3$, sandwiched by $SrRuO_3$. The inset is the EDS mapping, showing the A-site atomic gradient from Ba (blue) to Ca (red). (**B**) Mapping of *c*/*a*, measured in (A). (**C**) Mapping of local atomic polarization, estimated in (A). (**D**) The red squares indicate average out-of-plane polarization component, $P_{z,avg}$, of each atomic layer in (C). The error bar indicates the standard deviation. (**E**) Mapping of the horizontal oxygen buckling angle, ξ, measured in (A). ξ = 180° near the bottom interface indicates a tetragonal structure, and the deviation of ξ from 180° serves as a direct measure of the TiO_6 octahedral rotation in the orthorhombic (*Pnma*) structure (fig. S18).

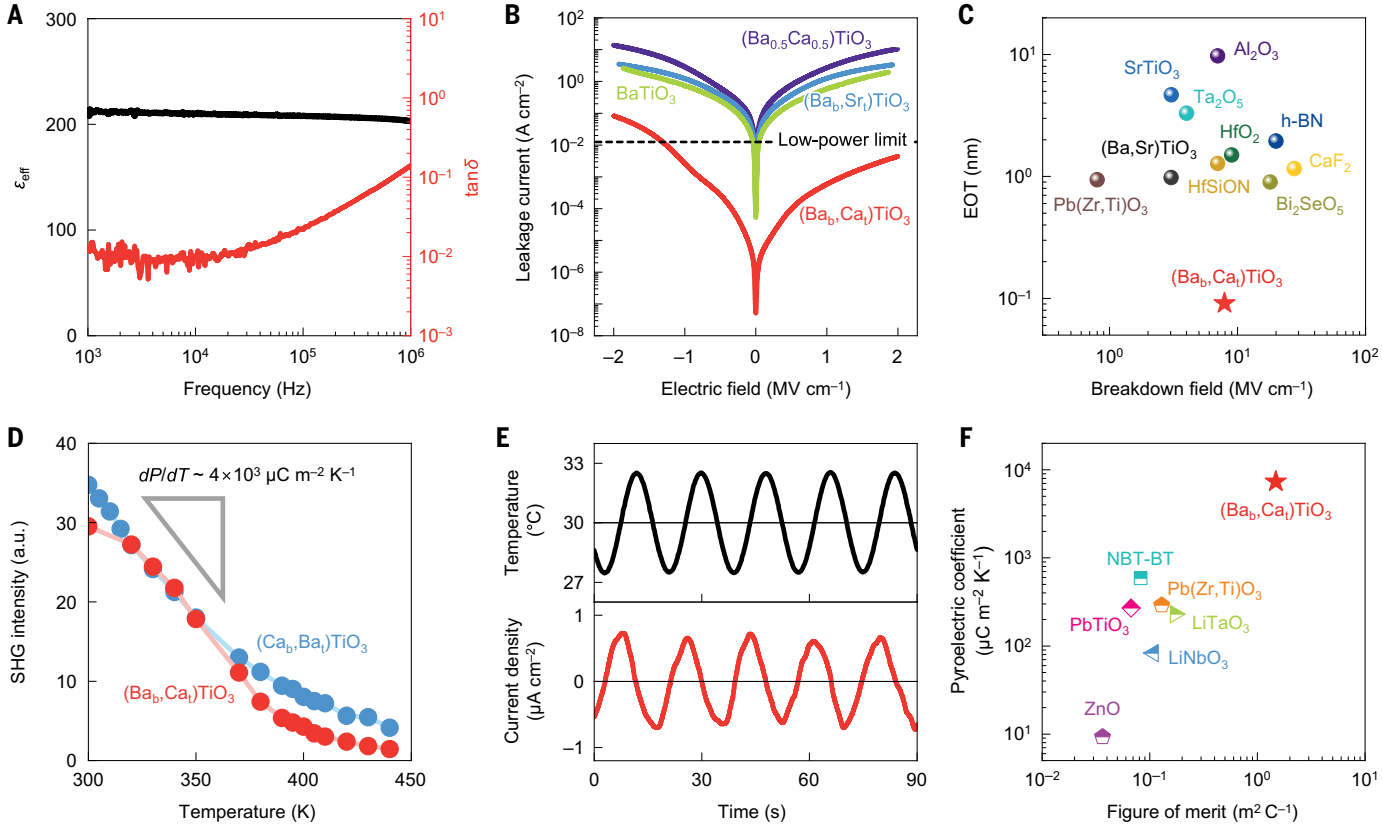

Fig. 4. Dielectric and pyroelectric properties of atomic-gradient titanates. (**A**) Dielectric constant (ε_{eff}) and loss tangent (tanδ) of 13-u.c.-thick $(Ba_b,Ca_t)TiO_3$ as a function of frequency. (**B**) Electric field–dependent leakage current density of $(Ba_b,Ca_t)TiO_3$ (red), $BaTiO_3$ (green), $(Ba_b,Sr_t)TiO_3$ (blue), and solid-solution $(Ba_{0.5}Ca_{0.5})TiO_3$ (purple) films. The thickness of all films used for the measurements is 13 u.c. (i.e., 5.1 nm). The black dashed line marks the low-power limit (*32*,*33*). (**C**) The equivalent oxide thickness (EOT) versus breakdown field of our $(Ba_b,Ca_t)TiO_3$ films and various dielectric materials (*34*–*38*). The EOT of $(Ba_b,Ca_t)TiO_3$ is estimated using the film thickness (t = 5.1 nm), which ensures low leakage currents below the low-power limit. (**D**) Temperature dependence of the optical second-harmonic generation (SHG) intensity. (**E**) Pyroelectric current density generated under applied sinusoidal temperature variation. (**F**) Comparison of the pyroelectric coefficients and the figures of merit of atomic-gradient $(Ba_b,Ca_t)TiO_3$ with those of conventional single crystalline and epitaxial pyroelectric materials (*2*,*22*,*47*–*49*).

Consequently, the divergence of electric polarization induces a large negative bulk bound charge ρ_b via simple electrostatics, $\rho_b = -\nabla \cdot \mathbf{P}$ (*11*, *21*). The bulk bound charge ρ_b is estimated to be as large as $\rho_b = -1 \times 10^8$ C m^{-3} (corresponding to ~10^{21} e cm^{-3}) and could substantially deplete the extrinsic carriers (i.e., electrons) in our dielectric oxide films (fig. S24). Thus, the polar divergence could largely suppress the leakage current in ultrathin films, while preserving good dielectric response of titanates.

High-κ, low-loss ultrathin titanates

Figure 4A shows the dielectric constant (ε_{eff}) and loss tangent (tanδ) as a function of frequency in ~5.1-nm-thick $(Ba_b,Ca_t)TiO_3$ films (*26*). Even at the ultrathin limit, our atomic-gradient films exhibit a low tanδ, close to 0.01, in the frequency range of 10^3 to 10^5 Hz, while preserving high $\varepsilon_{eff} > 200$. We measure the leakage current density in the atomic-gradient films to be below 10^{-2} A cm^{-2} at +2 MV cm^{-1} (i.e., at ~1 V; Fig. 4B), which meets the low-power limit (*32*, *33*). By contrast, solid-solution $(Ba_{0.5}Ca_{0.5})TiO_3$ films without atomic gradients exhibit substantial leakage, a common problem in ultrathin perovskite oxide dielectrics, likely caused by unavoidable point defects (*25*). Furthermore, single-layer $BaTiO_3$ and atomic-gradient $(Ba_b,Sr_t)TiO_3$ films, characterized by almost uniform electric polarization and minimal polar divergence (figs. S24 and S26), similarly show large leakage currents (Fig. 4B). These results highlight the critical role of giant polar divergence in suppressing leakage current in ultrathin geometry.

We compare the dielectric properties of $(Ba_b,Ca_t)TiO_3$ films and other common dielectrics. Figure 4C plots the equivalent oxide thicknesses (EOTs) and breakdown fields of various dielectrics, including HfO_2 and single-crystalline CaF_2 (*34*–*38*). The EOT is calculated as $3.9t/\varepsilon_{eff}$, where 3.9 is the dielectric constant of silicon oxide and t the film thickness. We find that the EOT of $(Ba_b,Ca_t)TiO_3$ films is as small as 0.09 nm and outperforms common dielectrics; their breakdown field is comparable to that of widely used dielectrics including HfO_2 and Al_2O_3 (fig. S27). This sub-0.1-nm-EOT dielectric, which also ensures low leakage currents below the low-power limit, holds promise for high-performance dielectric applications, such as field-effect transistors (*39*). Furthermore, advances in oxide membrane technology would facilitate the integration of our high-κ atomic-gradient films with various two-dimensional materials, paving the way for diverse studies in material properties and functionalities (*40*).

Giant pyroelectricity

We next investigate pyroelectricity, a crucial functionality of polar dielectrics. Despite their 5-nm thickness, our atomic-gradient (Ba,Ca)TiO_3 films exhibit notably low leakage currents, which is essential for reliable pyroelectric applications (*41*). The pyroelectric coefficient p of a flexoelectrically polarized material could be expressed as

$$p = \frac{\partial P_{flexo}}{\partial T} = \frac{\partial(\mu_{eff}\,\partial u_{11}/\partial x_3)}{\partial T} = \frac{\partial \mu_{eff}}{\partial T}\frac{\partial u_{11}}{\partial x_3} + \mu_{eff}\frac{\partial(\partial u_{11}/\partial x_3)}{\partial T} \quad (4)$$

where μ_{eff}, $\partial u_{11}/\partial x_3$, and T indicate flexoelectric coefficient, strain gradient, and temperature, respectively. Whereas $\partial u_{11}/\partial x_3$ is not expected to depend strongly on temperature, μ_{eff} could exhibit pronounced temperature dependence. Particularly in a paraelectric material possessing precursor polar regions, μ_{eff} can be exceptionally large and vary with highly negative $\partial\mu_{eff}/\partial T$ over a wide temperature range (*42*, *43*). We find that $(Ba_{0.5}Ca_{0.5})TiO_3$ without atomic gradients is in an overall paraelectric state at room temperature (figs. S21 to S23); however, precursor polar regions may emerge in (Ba,Ca)TiO_3 thanks to the competing interplay between polar instability and TiO_6 octahedral rotation (*44*), and/or from local chemical and structural inhomogeneities (*45*). This could not only result in a large μ_{eff}, consistent with the observed large flexoelectric polarization (Fig. 3C), but also lead to a large negative $\partial\mu_{eff}/\partial T$, implying giant pyroelectricity in atomic-gradient (Ba,Ca)TiO_3. As expected, our SHG measurements reveal a considerable reduction in bulk polarity across a wide temperature range as the temperature increases (Fig. 4D). On the basis of the room-temperature electric polarization measured in our films (Fig. 3), the slope of SHG intensity versus temperature indicates that giant pyroelectricity with p as high as 4×10^3 μC m^{-2} K^{-1} could be present.

We estimate pyroelectricity by dynamically modulating the temperature and measuring the generated current (*26*). Under sinusoidal temperature modulation, $(Ba_b,Ca_t)TiO_3$ and $(Ca_b,Ba_t)TiO_3$ films produce an alternating current with phase shifts of 90° and −90°, respectively, confirming the pyroelectric effect as the current-generation mechanism (Fig. 4E and fig. S28) (*46*). The amplitude of current density increases linearly with the amplitude of applied temperature variation, confirming once more its pyroelectric nature (fig. S28). The pyroelectric coefficient p measured in atomic-gradient (Ba,Ca)TiO_3 films reaches 7.34×10^3 μC m^{-2} K^{-1} at room temperature. This value is quantitatively consistent with the result from SHG measurements and is at least an order of magnitude higher than those of typical polar materials (Fig. 4F) (*2*, *22*, *47*–*49*). The substantial pyroelectric coefficient also allows for a large figure of merit $F_V = p / c_p\varepsilon_0\varepsilon_{eff}$, where c_p is the heat capacity.

Our work introduces flexoelectrically driven pyroelectricity, with effects surpassing those of conventional polar materials. This differs from recently reported giant pyroelectricity in free-standing nanomembranes (*50*), where reduced dimensionality alters lattice dynamics as the proposed mechanism—one that is inapplicable to our substrate-bonded epitaxial films (fig. S29). Crucially, we specifically achieve giant pyroelectricity in vertical ultrathin capacitor geometry, which is more relevant to applications but historically challenging owing to depolarization fields and leakage currents. The measured pyroelectric currents ideally track the applied dT/dt and exceed those of ferroelectric $BaTiO_3$ films of similar thickness by two orders of magnitude (fig. S29).

Discussion and outlook

Bulk polar states are typically incompatible with several technologically important electronic properties (*8*, *9*) and some functional structural distortions (*12*). Our study challenges this traditional notion by proposing that selective atomic gradients can universally create strong bulk polarity in otherwise centrosymmetric materials. Notably, this polar state is not only compatible but also synergistic with the existing properties, enabling emerging functionalities. To validate this concept, we have used perovskite iridates and titanates, demonstrating strong bulk polarity and associated functional properties in both materials. The successful demonstration across two distinct materials highlights the versatility and broad applicability of our approach. Further applying this method to systems, such as magnets and superconductors, could enable the design and discovery of new quantum and topological states (*7*, *51*). Thus, selective atomic gradients emerge as a promising avenue for breakthroughs in diverse material systems, advancing both fundamental research and practical applications involving polar symmetry.

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SUPPLEMENTARY MATERIALS

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QUANTUM SYSTEMS

Stationary entanglement of a levitated oscillator with an optical field

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Stationary entanglement between the motion of macroscopic objects and light is a long-standing goal of quantum optomechanics, with implications for both fundamental tests of quantum physics and emerging quantum technologies. We report the generation of quantum entanglement between the center-of-mass motion of a nanosphere levitated in an optical tweezer inside an optical cavity and the electromagnetic field. Using heterodyne detection, we reconstructed the full set of optomechanical correlations and observed a violation of separability bounds between the mechanical motion and a propagating optical mode, demonstrating the distribution of nonclassical correlations beyond the interaction region. The entanglement was generated at room temperature and remained robust over a broad range of parameters. Our results establish levitated optomechanical systems as a platform for continuous-variable quantum communication and for tests of macroscopic quantum physics.

Entanglement is one of the defining features of quantum mechanics (*1*, *2*), and demonstrating its control in macroscopic systems is a crucial goal with far-reaching implications for quantum technologies and for probing the foundations of quantum physics. Optomechanical platforms have enabled exquisite control over mechanical motion, leading to milestones such as the preparation of non-Gaussian states (*3*, *4*) and entangled mechanical oscillators (*5*–*9*). Furthermore, pulsed optomechanical entanglement between a drum oscillator and a microwave (*10*), and between nano-oscillators and single photons (*11*), has been demonstrated in ultracryogenic environments.

Optomechanical interaction can entangle distinct optical fields, even exploiting a classical mechanical system (*12*, *13*), but stationary entanglement between mechanical motion and propagating light has remained elusive, despite long-standing theoretical proposals (*14*–*19*). Such light-matter entanglement is of particular interest because it provides a natural interface between stationary and flying quantum degrees of freedom, enabling nonlocal correlations to be distributed across space. This capability is a key resource for quantum communication networks, where mechanical oscillators act as local quantum memories and optical fields distribute information between distant nodes (*20*, *21*).

Among optomechanical platforms, levitated systems have recently emerged as a particularly attractive approach (*22*), combining strong light-matter interactions (*23*, *24*) with exceptional isolation from the environment (*25*, *26*). The center-of-mass motion of optically trapped dielectric nanospheres has already reached the one-dimensional (1D) and 2D quantum regimes (*27*–*33*).

In this work, we achieved a further milestone by demonstrating stationary, room-temperature, optomechanical entanglement between the mechanical motion of a levitated nanosphere and a propagating optical field, evidenced by correlations violating separability bounds.

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Experimental procedure

We loaded a 100-nm-diameter silica sphere in an optical tweezer (*34*) created by two laser fields (denoted A and B) at 1064 nm, superposed in an optical fiber with identical linear polarization and a power ratio of 3:1. A slight ellipticity of the focused beam, with the major axis aligned along the polarization direction (see inset in Fig. 1B), split the transverse motion into two mechanical modes with eigenfrequencies $\Omega_x/2\pi$ = 110.6 kHz and $\Omega_y/2\pi$ = 98.4 kHz. The nanosphere was placed at the center of an optical cavity whose axis is almost orthogonal to that of the tweezer (Fig. 1A). Both trapping lasers were phase-locked to an auxiliary laser stabilized to a cavity resonance, which enabled precise control of their detunings $\Delta_{A,B}$ from two cavity modes separated by twice the free spectral range (FSR = 3.07 GHz). Light scattered from the tweezer fields populated the two cavity modes, thereby coupling the particle motion to the intracavity fields through coherent scattering (*23*, *35*–*37*). Details of the experimental setup are provided in the supplementary materials (*38*) and in a previous work (*24*).

When the nanosphere is located near a node of the cavity standing wave, the dominant optomechanical coupling involves motion along the cavity axis [the bright mechanical mode (*39*–*41*), with eigenfrequency $\Omega_b \simeq (\Omega_x + \Omega_y)/2$]. The red-detuned field A directly cools this bright mode. Because the linear polarization is oriented at ~45° relative to the cavity axis, both transverse motional degrees of freedom are cooled. As a consequence, the direction orthogonal to the cavity axis (dark mode), even if it is not directly coupled to the cavity field, undergoes efficient sympathetic cooling (*30*, *32*, *40*). Operating in the resolved-sidebands regime, where the cavity decay rate κ is smaller than Ω_b, optimal cooling is achieved for detuning $-\Delta_A \simeq \Omega_b$. In this regime, the optomechanical interaction is well described by a beam-splitter Hamiltonian $\hbar g_A\left(\hat{a}_A^\dagger \hat{b} + \hat{a}_A \hat{b}^\dagger\right)$ ($\hbar$ is the reduced Planck's constant, g_A is the coupling rate, and $\hat{a}_A$ and $\hat{b}$ are the annihilation operators of the optical and mechanical modes, respectively). The field outgoing the cavity for each optical mode $\alpha = (A, B)$ is

$$a_\alpha^{\mathrm{out}} = \sqrt{\eta}\left(e^{i\Phi_\alpha} a_\alpha^{\mathrm{in}} + i g_\alpha \sqrt{\kappa}\, \chi_\alpha x_b\right) + \sqrt{1-\eta}\; a_\alpha^{v} \qquad (1)$$

where a_α^{in} denotes the vacuum input field, Φ_α a frequency-dependent phase, χ_α the cavity susceptibility, and a_α^{v} the vacuum field associated with the overall propagation and detection losses, quantified by the efficiency η. This expression highlights that besides enabling cooling and stabilization, the outgoing field A also acts as a meter for the bright-mode displacement x_b (here normalized to its zero-point fluctuations).

The B field, blue-detuned with respect to the nearest cavity resonance, plays the central role in the experiment. When $\Delta_B \simeq \Omega_b$, its interaction with the nanosphere motion is well described by a two-mode squeezing Hamiltonian $\hbar g_B\left(\hat{a}_B \hat{b} + \hat{a}_B^\dagger \hat{b}^\dagger\right)$, which generates entanglement between the electromagnetic field and the bright mechanical mode. These quantum correlations are preserved in the field propagating out of the cavity. Inside the cavity, the electromagnetic field modes are uniquely determined by the boundary conditions. By contrast, propagating modes allow for a certain degree of arbitrariness in their definition. We may introduce a mode annihilation operator as

$$a_\xi = \int_{-\infty}^{\infty} \xi(-t) a^{\mathrm{out}}(t)\mathrm{d}t = \int_{-\infty}^{\infty} \xi(\omega) a^{\mathrm{out}}(\omega)\frac{\mathrm{d}\omega}{2\pi} \qquad (2)$$

where the envelope function $\xi(t)$ is normalized according to $\int_{-\infty}^{\infty} |\xi(t)|^2 \mathrm{d}t = 1$ (*17*).

It can be shown that a_ξ obeys the canonical bosonic commutation relations $\left[a_\xi, a_\xi^\dagger\right] = 1$. The corresponding mode quadratures are defined as $Q_\xi = a_\xi + a_\xi^\dagger$ and $P_\xi = i\left(a_\xi^\dagger - a_\xi\right)$. In the next sections, we consider propagating optical modes defined by a flat spectral window centered at $-\Omega_b$ with width Γ_ξ.

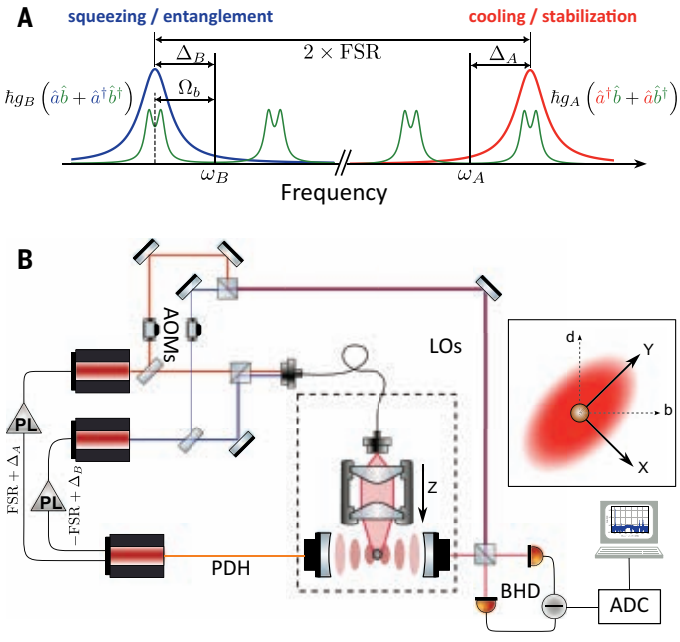


Fig. 1. Experimental configuration. The motion of a levitated nanoparticle is coupled to the field of an optical cavity through coherent scattering. The confining potential is obtained by overlapping two fields, *A* and *B*, in the optical tweezer, at different frequencies. This enables simultaneous driving of the cavity on opposite sides of its resonances, with detunings Δ_A (red) and Δ_B (blue): The former cools and stabilizes the motion, whereas the latter generates entanglement between the mechanical and optical subsystems. (**A**) The spectrum of the bright mode, represented by the green curves, has a double-peaked shape recalling the two resonances in 2D motion. The detuning of the *A* and *B* fields is set with respect to different longitudinal modes separated by two FSRs at a value close to the mechanical bright-mode frequency Ω_b. (**B**) This frequency configuration was achieved by phase-locking (PL) both trapping lasers to an auxiliary laser, which is frequency stabilized to a cavity resonance using a Pound-Drever-Hall (PDH) locking scheme. The cavity output fields were measured in a heterodyne configuration using the same balanced detection (BHD). We used two different local oscillator (LO) frequencies, detuned by 1.4 and 2.0 MHz with respect to the corresponding trapping fields by means of acousto-optical modulators (AOMs). The heterodyne signals were recorded after a high-resolution analog-to-digital converter (ADC). The inset depicts the beam shape of the tweezer (polarized along *Y*) in the transverse plane. The label "b" ("d") refers to "bright" ("dark") and corresponds to the direction along (orthogonal to) the cavity axis. Measurements were performed in an ultrahigh-vacuum environment, at a pressure of $\simeq 3.5\times10^{-8}$ mbar.

The assessment of optomechanical entanglement relies on the covariance matrix $\mathbf{V}$ describing the joint state of the bright mechanical mode and the propagating field mode *B*. Its elements are defined as $V_{ij}=0.5\langle\{u_i,u_j\}\rangle$, where $\mathbf{u}=(Q_\xi,P_\xi,x_b,p_b)$, x_b and p_b denote the position and momentum of the bright mode normalized to their respective zero-point fluctuations and $\{\cdot,\cdot\}$ indicates the anticommutator.

Data analysis and modeling

To compute the covariance matrix $\mathbf{V}$, we used two complementary approaches. The first infers $\mathbf{V}$ by fitting the heterodyne spectra within the full optomechanical model, which is then used, together with the fitted parameters, to reconstruct the covariance matrix. This approach is less sensitive to detection imperfections and provides a robust estimate of the system entanglement. The second uses field *A* as an interferometric probe of the nanosphere position, enabling the reconstruction of all relevant correlations directly from the measured output fields with minimal model assumptions.

Intracavity operators are not directly accessible, but the system state is encoded in the fields exiting the cavity, and we retrieved this information using heterodyne detection. To this purpose, two local oscillators, detuned by $\Omega^A_{LO}/2\pi=1.4$ MHz and $\Omega^B_{LO}/2\pi=2.0$ MHz, were combined and directed onto a single balanced detection that probes the light transmitted by the cavity. The heterodyne photocurrent operator is $i_{het}=\alpha^*_{LO}\,a^{out}\,e^{i\Omega_{LO}t}+\alpha_{LO}\,a^{out\,\dagger}\,e^{-i\Omega_{LO}t}$, where α_{LO} is the local oscillator field (*42*). i_{het} is Hermitian, and hence it satisfies $i_{het}(-\omega)=[i_{het}(\omega)]^\dagger$. From the heterodyne signal, we experimentally reconstructed the spectral correlation matrix $\mathbf{A}=\langle a_i\,a_j\rangle$ of the output field ladder operators $\mathbf{a}=\left(a^{out}_A,a^{out\,\dagger}_A,a^{out}_B,a^{out\,\dagger}_B\right)$, where $O^\dagger(\omega)=\mathcal{F}\left[O^\dagger(t)\right]$ (*18*, *38*).

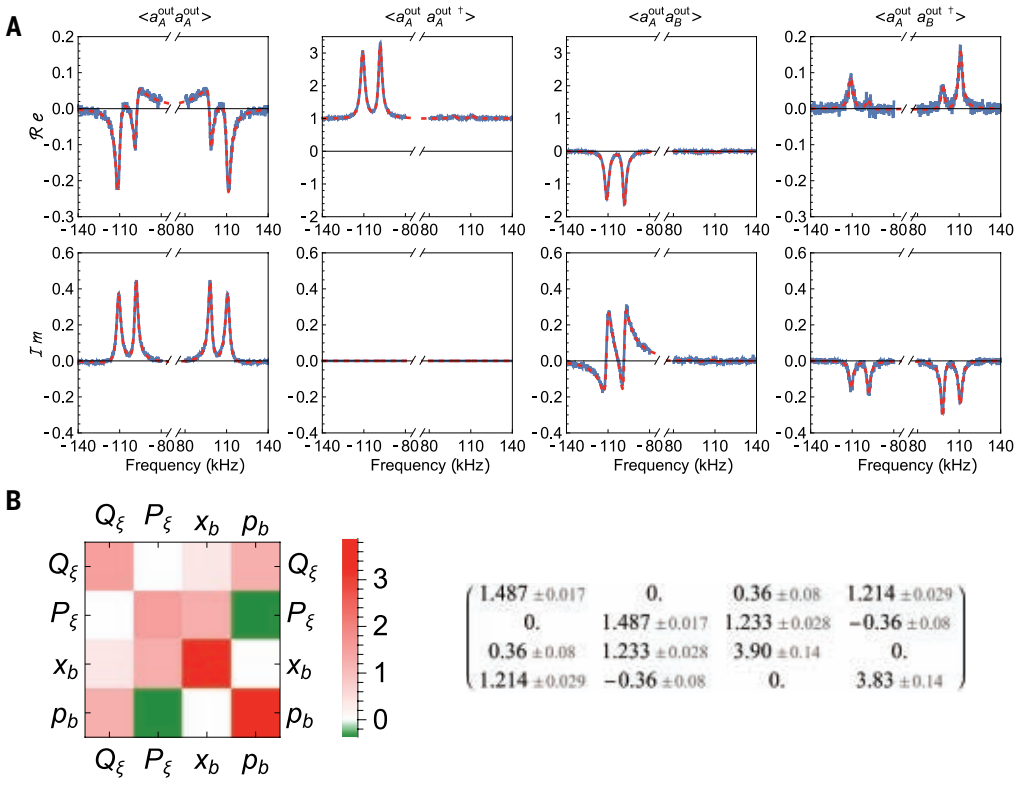
Because this requires a phase-sensitive analysis, the nanosphere was placed at a small but finite offset from the nodes of the *A* and *B* intracavity stationary waves. The resulting weak coherent contributions to a^{out}_α provided a stable phase reference, allowing the field quadratures to be determined consistently across the full dataset.

The heterodyne signal was continuously acquired and segmented into 10-ms-long intervals. For each interval, the Fourier transform of the heterodyne signal, $v(\omega)$, was computed and assembled into the 4D data vector $\mathbf{v}(\Omega)=\left(v^*\left(\Omega^A_{LO}-\Omega\right),v\left(\Omega^A_{LO}+\Omega\right),v^*\left(\Omega^B_{LO}-\Omega\right),v\left(\Omega^B_{LO}+\Omega\right)\right)$. From this vector, compensated for the frequency-dependent argument of the electronic response, we evaluated the spectral correlation matrix $\mathbf{C}$ defined as $C_{ij}=\langle v_i(-\Omega)v_j(\Omega)\rangle$, where $\langle\cdot\rangle$ denotes ensemble averaging. Calibration of $\mathbf{C}$ to the standard quantum noise level of fields *A* and *B* (obtained by blocking the signal path in the heterodyne detection) was followed by subtraction of electronic noise. This procedure transformed $\mathbf{C}$ into a new matrix $\mathbf{A}^\phi=\mathbf{RAR}^T$, where $\mathbf{R}$ is a constant matrix accounting for the phases, ϕ_A and ϕ_B, of the two detected coherent components with respect to the intracavity fields, which define the reference phase in the model.

The spectral matrix $\mathbf{A}$ contains all the information that can be extracted from the two outgoing fields. Owing to intrinsic symmetries, it comprises 16 different real-valued, frequency-dependent functions. The elements $A^\phi_{1,2}$ and $A^\phi_{3,4}$ are independent of ϕ_A and ϕ_B and represent the usual heterodyne power spectral density of the fields *A* and *B*, respectively. These are fitted simultaneously to the optomechanical model to extract the system parameters, namely, the pairs of mechanical bare frequencies, heating rates, and optomechanical coupling rates. Independent measurements provide $\kappa/2\pi=58\pm0.6$ kHz and $\eta=0.283\pm0.006$. The full $\mathbf{A}^\phi$ matrix is then fitted to the model to determine ϕ_A and ϕ_B, with all other parameters being fixed to the values obtained from the previous fitting procedure. Representative examples of elements of $\mathbf{A}$ are shown in Fig. 2A around the bright-mode sidebands, along with the respective fitting modeled curves.

In the first approach, fitting only the two heterodyne power spectral densities was sufficient to determine $\mathbf{V}$ relying on the full set of fitted parameters. We emphasize that the same parameters also reproduce all 14 remaining independent elements of the spectral correlation matrix $\mathbf{A}$, up to the phase factors ϕ_A and ϕ_B. This excellent agreement between experiment and theory therefore provides a stringent quantitative validation of the method.

In the second approach, all correlations were extracted directly from the output fields with minimal reliance on the model, exploiting the fact that field *A* provides a continuous interferometric measurement of the nanosphere position. This measurement can be calibrated by inverting Eq. 1. Explicitly, we define $x_{meas}=\frac{a^{out}_A}{i\sqrt{\eta\,\kappa}\,g_A\,\chi_A}$ and $p_{meas}=-i\frac{\omega}{\Omega_b}x_{meas}$, which can be used as measured estimators of the mechanical quadratures x_b and p_b when evaluating $\mathbf{V}$ (*38*). The calibration relies primarily on independently determined quantities (κ, η, and vacuum level). The only parameter taken from the fitted model is the optomechanical coupling rate, which is tightly constrained by the observed width of the mechanical peaks and therefore only weakly correlated with the remaining fit parameters. Importantly, the noise properties and correlations that quantify the quantum character of



$$\begin{pmatrix} 1.487 \pm 0.017 & 0. & 0.36 \pm 0.08 & 1.214 \pm 0.029 \\ 0. & 1.487 \pm 0.017 & 1.233 \pm 0.028 & -0.36 \pm 0.08 \\ 0.36 \pm 0.08 & 1.233 \pm 0.028 & 3.90 \pm 0.14 & 0. \\ 1.214 \pm 0.029 & -0.36 \pm 0.08 & 0. & 3.83 \pm 0.14 \end{pmatrix}$$

Fig. 2. Optomechanical correlations. (**A**) First row of the complex-valued spectral matrix **A**, shown around the bright-mode sidebands, along with the theoretical model (dashed-red line). Each column in the figure represents the real (top) and imaginary (bottom) part of **A**. Only the $A^{\phi}_{1,2}$ and $A^{\phi}_{3,4}$ elements, that is, the heterodyne power spectral densities, are fitted to obtain the theoretical curves. (**B**) 2D portrayal of the covariance matrix **V** fully characterizing the optical-mechanical Gaussian state and corresponding numerical values. The estimated dynamical parameters for this dataset are $\Omega_b/2\pi = 106 \pm 0.1$ kHz, $g_A/2\pi = 11.7 \pm 0.2$ kHz, and $g_B/2\pi = 6.3 \pm 0.1$ kHz. The heating rates are $\Gamma_x/2\pi = 3.00 \pm 0.05$ kHz and $\Gamma_y/2\pi = 2.67 \pm 0.05$ kHz. All quoted errors indicate one standard deviation of ~10 independent samples.

the system emerge directly from the processed data, instead of being deduced from the fit of the model. This second approach fully exploits the experimental matrix **A**, which provides a complete description of the correlations between the output fields A and B. The covariance matrix **V** is obtained from **A** by applying a suitable linear transformation, described in the supplementary materials, and integrating over frequency. A representative covariance matrix obtained with this method is shown in Fig. 2B.

Results and discussion

The entanglement between the optical and mechanical subsystems is quantified by the smallest symplectic eigenvalue, ν_-, of the partially transposed covariance matrix (*43*, *44*). With our normalization, $\nu_- = 1$ if the two oscillators are in uncorrelated vacuum states, and the presence of entanglement is witnessed by $\nu_- < 1$. Figure 3 reports the values of ν_- calculated from the covariance matrix **V** directly reconstructed through **A**, as a function of the width Γ_ξ of $\xi(\omega)$. Entanglement emerges when the bandwidth Γ_ξ of the optical mode becomes sufficiently large to encompass the full spectral response of the mechanical bright mode (*17*), characterized by two peaks at $\sim\Omega_x$ and $\sim\Omega_y$ broadened by optomechanical interaction. The minimum value of ν_-, corresponding to maximal entanglement, is achieved around $\Gamma_\xi/2\pi = 35$ kHz and reaches $\nu_- = 0.918 \pm 0.029$. Here, the quoted statistical uncertainty represents one standard deviation obtained from multiple measurements, whereas an additional systematic uncertainty of ± 0.02 arises from the independent estimates of η and κ. Increasing Γ_ξ beyond this point slowly dilutes the mechanical contribution to the optical mode, leading to a slow raise of ν_- while increasing the chance of including experimental imperfections. Entanglement is commonly quantified by the logarithmic negativity E_N, defined as $E_N = \max\{0, -\log_2(\nu_-)\}$ (*45*, *46*). At its maximum, we obtained $E_N = 0.12 \pm 0.04$, including both the statistical and the systematic uncertainty.

Figure 3 also reports the values of ν_- obtained by estimating **V** from the full optomechanical model using all the fitted parameters, including the heating rates. The model reproduces the overall trend of the values extracted directly from the experimental covariance matrix but consistently yields lower values of ν_-, with a minimum of $\nu_- = 0.884 \pm 0.011$ ($E_N = 0.177 \pm 0.019$). We attribute this discrepancy to slow fluctuations of the system parameters during the acquisition of the full time series, most notably of the relative phases ϕ_A and ϕ_B, which are not fully captured by the stationary description. To mitigate these effects, we restricted the data analysis to time intervals over which the fitted parameters remained approximately constant. In practice, each independent estimate of the correlation matrix **A,** and hence of the covariance matrix **V** and the symplectic eigenvalue ν_-, was obtained by averaging over, typically, 2000 consecutive time windows of 10 ms. Residual parameter fluctuations during this 20-s acquisition time nevertheless mix slightly different Gaussian states and reduce the observable entanglement. As a consequence, the entanglement actually present in the system is more faithfully captured by estimating **V** through the optomechanical model evaluated with the fitted parameters, whereas the values of ν_- directly inferred from the data should be regarded as a conservative estimate of the entanglement that can be effectively harnessed under the present level of system stability. We also note that the rectangular spectral filter used here was not optimal. Tailored mode shaping could increase the optomechanical fidelity and enhance the observable entanglement (*42*).

The results discussed so far were obtained for detunings satisfying $-\Delta_A \simeq \Delta_B \simeq \Omega_b$. This condition, however, is not essential, and a finite degree of entanglement is present even when it is relaxed. Indeed, we repeated the experiment while varying the detuning, keeping only the condition $-\Delta_A \simeq \Delta_B$. Figure 4 shows the symplectic eigenvalue ν_- derived from the covariance matrix **V** reconstructed directly from the measured correlation matrix **A**, together with the corresponding values obtained by computing **V** with the optomechanical model using parameters fitted to the same experimental data.

Entanglement persists over a detuning range exceeding 40 kHz, consistent with the expectation that it is maintained within a bandwidth on the order of the cavity linewidth κ. This robustness implies that stable optomechanical entanglement can be achieved without fine control of the detuning.

An important and nontrivial question is whether entanglement is also present inside the cavity, between the intracavity mode of field B and the mechanical oscillator. This is reasonable but not obvious,

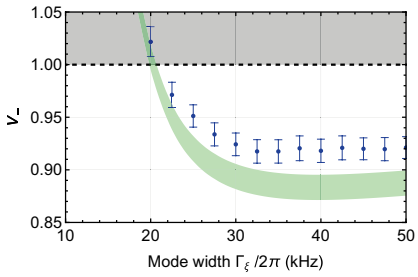


Fig. 3. Optimization of the optical mode width. The smallest symplectic eigenvalue ν_- of the partially transposed covariance matrix **V** is plotted as a function of the mode width Γ_ξ. The data points with error bars (SEM over about 10 independent data acquisitions) were obtained from **V** directly reconstructed from the experimental correlation matrix. The green-shaded area derives from **V** computed with our optomechanical model, using the parameters that best fit the heterodyne spectra. The width of this band reflects systematic uncertainties in the model parameters (mainly the efficiency η), with negligible contribution from statistical uncertainty. The gray-shaded region highlights the threshold above which the mechanical and optical subsystems become separable.

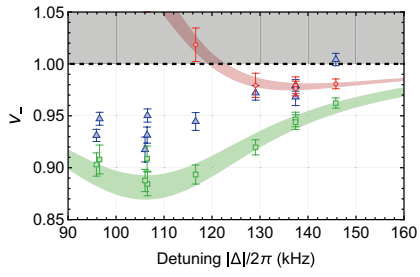


Fig. 4. Symplectic eigenvalue versus cavity detuning. Experimental data points shown by blue markers with error bars (SEM over about 10 independent data acquisitions) were obtained from the covariance matrix directly reconstructed from the measured correlations. Green markers were instead extracted from the same experimental data through the optomechanical model, using the parameters that best fit the heterodyne spectra. In this case, the uncertainty is dominated by the systematic error on the detection efficiency η. In these measurements, we set $-\Delta_A \simeq \Delta_B = |\Delta|$. Data points with similar detuning were acquired at different times and therefore with slightly different system parameters. Intracavity entanglement is inferred from the symplectic eigenvalues associated with the covariance matrix calculated, within the same model and parameter set, for the joint state of the motion along an optimized direction and the intracavity *B* field (red points, with error bars again set by the systematic uncertainty on η). The green- and red-shaded regions were obtained by taking the extremal model predictions generated by the full set of parameters extracted from all datasets (at all detunings). These bands therefore illustrate the overall detuning dependence of the entanglement in our system.

because the intracavity field mode is uniquely determined by the cavity boundary conditions and cannot be tailored as in the case of the propagating field. Although intracavity observables cannot be accessed directly, the same optomechanical model that quantitatively reproduces the measured output correlations allowed us to reconstruct the corresponding intracavity covariance matrix from the fitted parameters. A crucial issue is the motion along the tweezer axis (the z mode), whose mechanical frequency is $\Omega_z/2\pi = 19.3$ kHz. This motion is coupled to the cavity field owing to a small angle between the cavity axis and the tweezer transverse plane. As a consequence, the z mode also produces a peak in the spectrum of the bright mode through the cavity-mediated coupling, slightly increasing its variance. We have indeed added this contribution in the entries V_{33} and V_{44} before computing the values of ν_- here reported (*38*). The z motion has no effect on the propagating optical mode, because the filter ξ removes the spectral region around the z mode resonance. By contrast, the z mode peak is not negligible in the spectra of the cavity optical modes and, indeed, it destroys the entanglement with the transverse mechanical bright mode considered thus far. Nevertheless, optomechanical entanglement is partially recovered by considering a slightly tilted direction of motion, thereby including a projection of the tweezer axis. This enhances the optomechanical correlations in the spectral region of the z mode. The symplectic eigenvalue ν_- associated with the covariance matrix describing the cavity B mode and the optimized motion direction, shown in Fig. 4 as a function of the detuning, confirms the presence of intracavity entanglement, albeit at a reduced level compared with that of the propagating mode. The minimum inferred value of ν_- for the intracavity variables is $\nu_- = 0.976 \pm 0.007$, corresponding to a logarithmic negativity of $E_N = 0.035 \pm 0.010$, and occurs at a detuning of $|\Delta| / 2\pi = 137$ kHz.

Concluding remarks

Our demonstration of quantum entanglement between the center-of-mass motion of a levitated nanosphere and a propagating optical field establishes levitated optomechanical systems as a versatile platform for generating and distributing light-matter entanglement at room temperature, without ultracryogenic environments or stringent parameter tuning, providing a contribution to macroscopic quantum technologies. The demonstrated entanglement can be further enhanced by realistic improvements, for example, on η and Γ. Together with the long mechanical coherence time, this finding highlights levitated optomechanical systems as a promising platform for quantum state transfer and quantum-memory applications. Beyond center-of-mass motion, additional mechanical degrees of freedom, including rotational and librational modes (*47*–*50*), provide further opportunities for multimodal quantum control. Combined with the prospect of trapping and simultaneously cooling multiple levitated nanospheres in a single optical cavity (*51*, *52*), this architecture could enable multipartite entanglement mediated by a shared optical field (*53*). More broadly, the ability to prepare nonclassical states of increasingly massive objects may open the way to experimental tests of the possible quantum nature of gravity (*54*, *55*) and conjectured modifications of quantum mechanics at macroscopic scales.

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SUPPLEMENTARY MATERIALS

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PIEZOELECTRICS

Ultrahigh piezoelectricity by polaron-defect complexes

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Materials capable of delivering giant and rapid piezoelectric responses at kilohertz frequencies are pivotal for advanced electromechanical technologies. However, their development has been constrained by a long-standing magnitude–speed trade-off: Ferroelectrics with intrinsic electric dipoles offer fast response but limited polarization, whereas ionic defect motion generates highly polarizable defect dipoles but responds slowly, limiting kilohertz-rate applications. We overcame this fundamental limitation by devising 8 mole % K-doped $AgNbO_3$ ($Ag_{0.92}K_{0.08}NbO_3$) thin films engineered to host dynamically reconfigurable polaron-defect complexes through electro-elastic interactions between small electron polarons and lattice defects. Field-driven multi–unit-cell polaron hopping reconfigures these complexes, producing giant defect polarization, and orders-of-magnitude faster kinetics of polaron hopping (versus ionic migration) enables their rapid response. This synergy leads to an unprecedented field-induced effective piezoelectric coefficient of ~7170 picometers per volt with ~6.2% strain at kilohertz frequencies in engineered silver niobate–based films.

Piezoelectric materials enable bidirectional and efficient interconversion of mechanical and electrical energy, underpinning key technologies from kilohertz actuators to ultrasonic transducers for applications in consumer electronics, robotics, and beyond (*1*–*4*). To enable next-generation miniaturized microelectromechanical systems (MEMS), piezoelectric thin films need to deliver giant and rapid electromechanical responses at kilohertz frequencies and beyond (*5*–*11*). The longitudinal effective piezoelectric coefficient $d^*_{33,f}$ can be approximated as $d^*_{33,f} = 2QP\varepsilon_0\varepsilon_r$, where the electrostrictive coefficient (Q), polarization (P), and relative permittivity (ε_r) collectively determine the magnitude of piezoelectric response (*12*). Therefore, achieving a high $d^*_{33,f}$ requires not only large P, but also large ε_r, which determines the magnitude of polarization modulation (ΔP) induced by a small alternating electric field (E_{AC}).

Ferroelectrics, in which spontaneous polarization (P_s) arises from the collective alignment of intrinsic electric dipoles, exhibit outstanding piezoelectric coefficients, particularly in Pb-based bulk single crystals (*3*). In bulk ferroelectrics, strategies such as introducing morphotropic phase boundaries and local structural heterogeneity have been developed to lower the polarization rotation barrier and enhance ε_r, improving $d^*_{33,f}$ (*13*–*15*). In thin films, however, substrate clamping severely suppresses polarization rotation and ferroelectric domain-wall motion, limiting ε_r and reducing $d^*_{33,f}$ from 2000 pm V^{-1} in Pb-based bulk single crystals to only 20 to 80 pm V^{-1} in their thin-film counterparts (*16*–*18*). The intrinsic lattice response, governed by picometer-scale ionic displacement around equilibrium lattice sites, also limits the achievable modulation of spontaneous polarization (ΔP_s) under moderate electric fields (<500 kV cm^{-1}). Consequently, most ferroelectric films exhibit piezoelectric strains below 1% despite their rapid intrinsic electromechanical response (*17*, *19*–*22*).

Beyond conventional ferroelectric thin films, recent studies have demonstrated large field-induced effective piezoelectric responses in nonclassical electromechanical films, including CeO_2- and HfO_2-based fluorite-structured oxides (*5*, *23*). Within this broader class, a representative defect-mediated polarization strategy exploits mobile ionic defects and their interactions with charged dopant defects to establish extrinsic ionic defect dipoles with large defect polarization (P_d), offering an alternative route to overcome the limited polarization magnitude of conventional ferroelectrics (*2*, *5*, *24*–*28*). This strategy was realized in centrosymmetric Gd-doped CeO_2 (CGO) films containing mobile oxygen vacancies ($V_O^{\cdot\cdot}$), where a direct current electric field (E_{DC}) drives oxygen-vacancy migration over multiple unit cells (nanometer-scale distances), generating enormous P_d and ε_r and enabling giant field-induced piezoelectricity (*5*, *26*, *29*). However, the slow kinetics of ionic-defect migration restricts this enhancement to the low-frequency regime (<10 Hz) (*25*, *30*). In practice, the E_{DC}-induced $d^*_{33,f}$ in CGO films decreases from ~1000 pm V^{-1} at ~10 Hz to only ~140 pm V^{-1} at 1 kHz (*5*). Thus, achieving large field-induced polarization with fast dynamics at kilohertz frequencies remains a major challenge.

Design paradigm for ultrahigh kilohertz effective piezoelectric response

To overcome this unresolved magnitude–speed trade-off in effective piezoelectric response, we engineered dynamically reconfigurable electromechanical polaron-defect (PD) complexes enabled by fast small-polaron hopping kinetics (Fig. 1A). Small polarons—localized electrons or holes accompanied by lattice distortion (*31*)—have been reported to induce electromechanical responses in fluorite oxides through charge localization–driven structural distortions (*32*–*34*). Inspired by ionic defect dipoles, we exploited this lattice-coupling capability by integrating defect-bound polarons with charged lattice defects (*35*–*37*). This PD-coupled state combines the large defect polarization with the rapid electronic hopping kinetics of polarons, overcoming the sluggish kinetics of ionic-defect migration.

In this specific design, relatively immobile positively charged lattice defects introduce local lattice distortions and associated elastic dipoles (*38*–*40*), creating electrostatic and elastic trapping potentials that stabilize nearby electronic polarons and form electromechanically active PD complexes. In the unpoled state, these PD complexes are randomly oriented, resulting in negligible net defect polarization. Under a direct current (DC) poling field, localized polarons hop between neighboring lattice sites near the defect centers, triggering reconfiguration of the PD complexes. This field-driven reconfiguration induces large defect polarization (P_d) through charge redistribution and promotes local elastic dipole alignment, enhancing distortion anisotropy and electrostrictive

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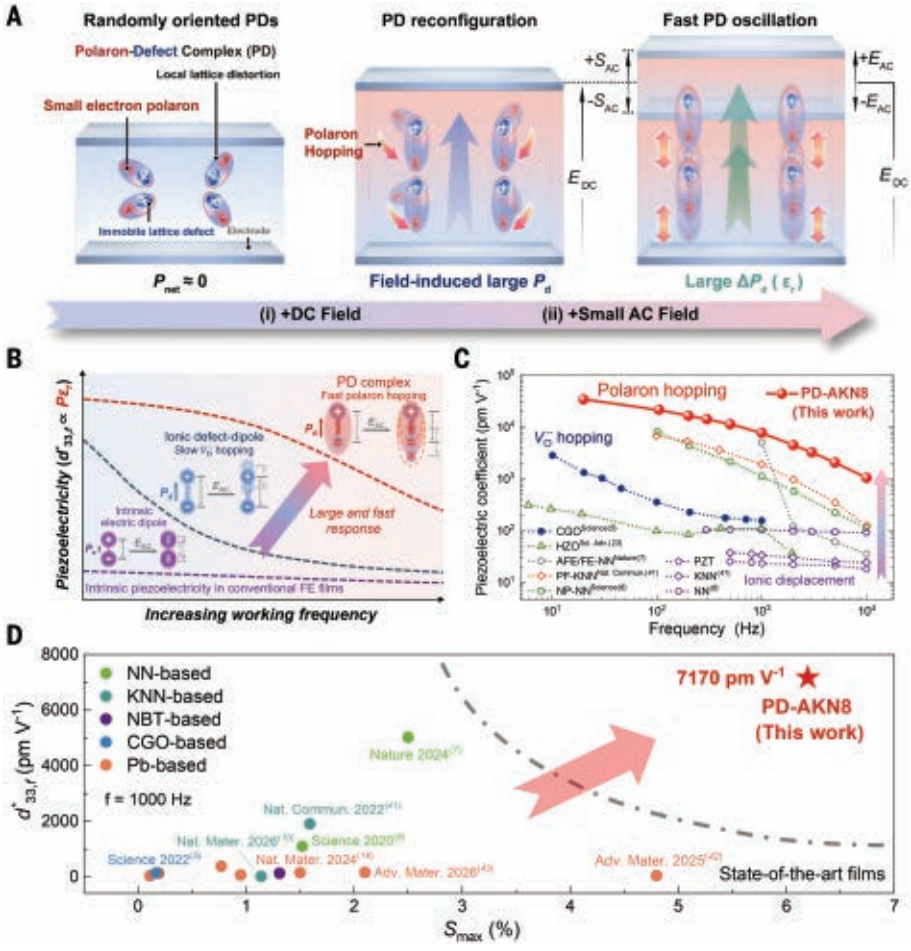

Fig. 1. PD-complex paradigm toward ultrahigh-kilohertz piezoelectricity. (**A**) The schematic diagram of the roadmap for achieving effective piezoelectric responses through PD-complex dynamics under DC and AC electric fields. (**B**) Frequency-dependent effective piezoelectric responses enabled by three distinct mechanisms: intrinsic electric dipoles mediated by ion displacement (purple), ionic defect dipoles mediated by oxygen vacancy hopping (blue), and polaron-defect complexes mediated by small polaron hopping (red). The spring indicates the electro-elastic coupling between mobile species (oxygen vacancies or small polarons) and lattice defects. (**C**) Frequency-dependent effective piezoelectric coefficients for PD-AKN8, benchmarked against representative thin-film systems. The PZT data were measured in this work by using the same "DC + AC" protocol used for PD-AKN8 (fig. S2C), whereas the reported HZO data correspond to in-plane d_{xx} measured with a lateral-electrode geometry and are included for mechanistic context. (**D**) Comparison of $d^*_{33,f}$ and maximum strain (S_{max}) for state-of-the-art films at 1000 Hz. The dominant electromechanical mechanisms and electrical measurement protocols are categorized and summarized in table S1. NN-based, $NaNbO_3$-based films; KNN-based, (K,Na)NbO_3-based films; NBT-based, (Na,Bi)TiO_3-based films; and CGO-based, Gd-doped CeO_2-based films. "Pb-based" includes PZT-based, Pb(Mg,Nb)O_3-$PbTiO_3$-based (relaxor) ferroelectric, and $PbZrO_3$-based antiferroelectric films.

coupling. Superimposing a small alternating current (AC) field drives oscillatory reconfiguration of the activated PD complexes through polaron hopping, producing a large modulation of defect polarization (ΔP_d) and consequently a large effective piezoelectric strain (strain $\propto E_{AC}$) (Fig. 1A). Because the field-responsive species is an electronic polaron rather than a migrating ion, a small E_{AC} can generate a large ΔP_d and thus enhanced permittivity (ε_r) at kilohertz frequencies (Fig. 1B). Through the simultaneous enhancement of P_d and ε_r, PD-complex engineering unlocked exceptional field-induced effective piezoelectric performance, extending well into the kilohertz regime (Fig. 1B).

We realized this design in 8 mol % K-doped $AgNbO_3$ ($Ag_{0.92}K_{0.08}NbO_3$) thin films engineered to host dynamically reconfigurable PD complexes (PD-AKN8). These films combined large field-induced defect polarization with rapid response dynamics. As a result, PD-AKN8 operated in a distinct kilohertz electromechanical regime relative to both conventional and polar structure–engineered ferroic films and emerging nonclassical field-induced electromechanics, including phase instability–mediated $Hf_{0.5}Zr_{0.5}O_2$ (HZO) and oxygen vacancy redistribution–mediated CGO films (Fig. 1C and table S1) (*5*–*7*, *23*, *41*). Under an E_{DC} bias of 375 kV cm^{-1}, PD-AKN8 achieved a field-induced kilohertz $d^*_{33,f}$ of 7170 pm V^{-1} at 1 kHz, which is approximately 50 times greater than that of CGO films at the same frequency (fig. S1) (*5*). At 5 and 10 kHz, $d^*_{33,f}$ remained as high as 2080 pm V^{-1} and 1090 pm V^{-1}, respectively, exceeding advanced lead-free benchmarks by 6 to 10 times and lead zirconate titanate (PZT) thin films by 10 to 20 times (figs. S1 and S2) (*5*–*7*, *41*). The films also delivered a giant maximum piezoelectric strain (S_{max}) of 6.2% under a moderate E_{AC} at 1 kHz, which is more than double that of the lead-free benchmark (~2.5% at 1 kHz) (*7*). The simultaneous realization of both ultrahigh effective piezoelectric coefficients and giant strains at kilohertz rates has not been demonstrated in previously reported electromechanical film systems (Fig. 1D and table S1) (*5*–*7*, *10*, *16*, *18*, *20*–*22*, *41*–*43*).

Atomically resolved electromechanical PD complexes

The PD-AKN8 thin films (~200 nm thick) were grown on Nb-doped $SrTiO_3$ (011) by means of chemical solution deposition (CSD) (supplementary materials, materials and methods) and characterized with high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) (fig. S3). HAADF-STEM imaging revealed a high density of square-shaped Nb-antisite defect (Nb^{4+}_{Ag}) units, some of which further aggregated into defect-unit clusters (Fig. 2A and figs. S4 and S5). These regions exhibited darker HAADF contrast than that of the surrounding matrix, which is consistent with Nb enrichment because Nb has a lower atomic number than that of Ag. By contrast, reference $AgNbO_3$ (AN) films prepared by means of pulsed laser deposition showed no pronounced local compositional fluctuations or Nb-antisite–related defect units (fig. S4).

This difference was closely related to the actual film composition and the CSD process. Quantitative energy-dispersive x-ray spectroscopy (EDS) analysis showed that PD-AKN8 deviated from the nominal stoichiometry and exhibited pronounced local Ag-site deficiency and compositional fluctuations (fig. S6). During organic pyrolysis, partial reduction of Ag^+ to metallic Ag further promoted local Ag depletion and Nb-rich regions (fig. S7) (*44*), favoring the formation and clustering of Nb-antisite defect units (*45*–*47*). These coherently embedded units acted as localized positively charged defect centers, and their clustering further strengthened the local positive electrostatic field (Fig. 2B and fig. S8). For charge compensation, nearby excess electrons were localized on adjacent Nb sites as small electron polarons (Nb′), reducing Nb^{5+} to Nb^{4+} states; their electrostatic and elastic coupling with relatively immobile Nb^{4+}_{Ag} stabilized the PD complexes (Fig. 2, B and C, and supplementary text section 1).

Defocus-series imaging further resolved a three-dimensional (3D) architecture of the defect network. Defect units exhibited clear extension

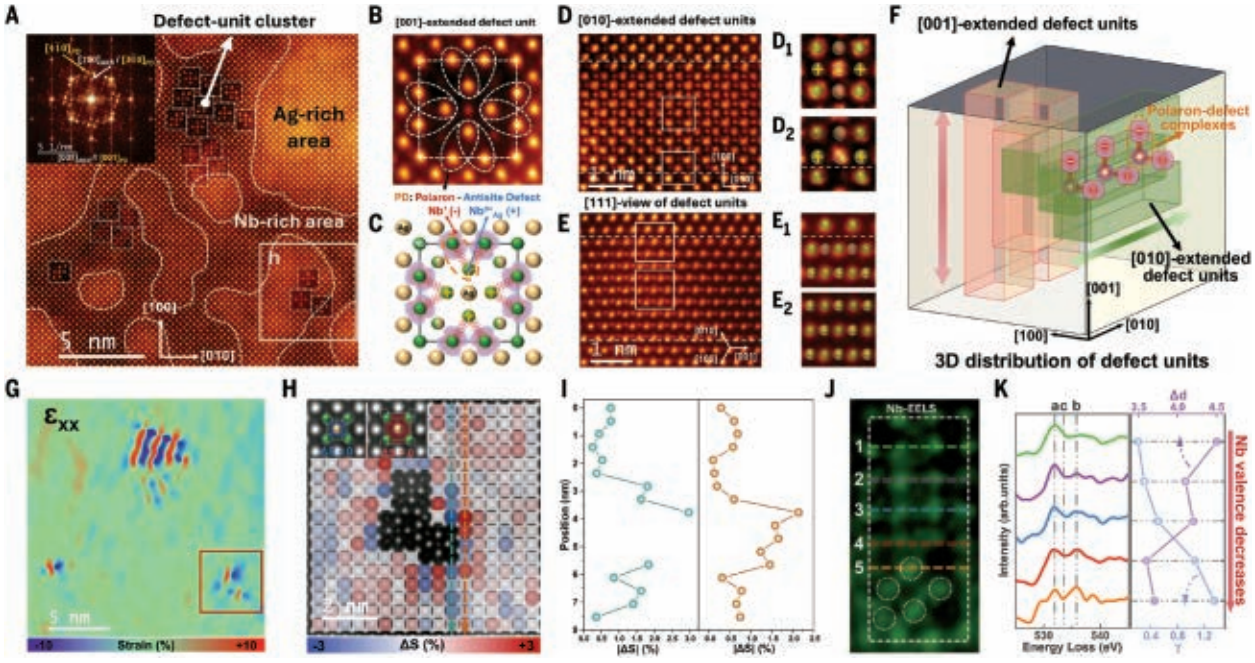

Fig. 2. Atomic-scale structure characterization of PD complexes in PD-AKN8. (**A**) HAADF-STEM image of PD-AKN8 viewed along the [001] zone axis. This processed and annotated image is derived from the same underlying raw HAADF-STEM acquisition shown in fig. S4A$_2$. (Inset) Corresponding FFT pattern, in which the yellow arrow indicates diffraction spots arising from defect ordering. (**B**) Enlarged atomic projection of a [001]-extended defect unit. White dashed ellipses indicate representative local motifs containing Nb_{Ag}^{4+} antisite defects and neighboring polaron-associated regions. (**C**) Structural model derived from the motifs in (B), illustrating the electromechanically coupled PD complex. Yellow and green spheres indicate Ag and Nb, respectively, and green spheres with plus signs indicate Nb_{Ag}^{4+}. Pink translucent spindle-shaped regions indicate localized excess electrons on neighboring Nb sites, corresponding to small electron polarons (Nb′). Springs indicate the electromechanical coupling between Nb′ and immobile Nb_{Ag}^{4+} antisite defects. Orange dashed ellipses indicate the resulting PD complexes. (**D**) HAADF-STEM images of [010]-extended defect units (dashed outlines). (D_1 and D_2) Magnified images of boxed regions in (D). (**E**) View of defect units along the [111] zone axis (dashed outlines). (E_1 and E_2) Magnified images of boxed regions in (E). (**F**) Proposed 3D structural model of a defect-unit network hosting PD complexes. For clarity, possible [100]-extended defect units are not shown. (**G**) GPA strain map (ε_{xx}) derived from (A). (**H**) Mapping of local Nb-sublattice strain around defect units derived from the red box in (G) [corresponding to the white-boxed region in (A)]. (Inset) A schematic of Nb-sublattice volume expansion ($\Delta S > 0$; red) and contraction ($\Delta S < 0$; blue). (**I**) Profiles of absolute lattice distortion magnitude along the blue and yellow paths marked in (H). (**J**) Atomic-resolution Nb EELS map centered on a defect unit. The dashed circles indicate Nb_{Ag}^{4+} antisite defects. Colored lines indicate spectral acquisition paths. (**K**) Spectral analysis of the defect unit. (Left) EELS spectra from the regions defined in (J). (Right) Quantitative analysis of O K-edge variations. Parameters γ and Δd quantify spectral changes associated with the defect unit. $\gamma = (I_b\text{-}I_c)/(I_a\text{-}I_c)$, where I_a, I_b, and I_c are peak intensities. Δd is the energy difference (in electron volts) between peaks a and b.

along the [001] zone axis, appearing as through-type or half-embedded features with extension lengths of more than 10 nm, accompanied by measurable lattice strain (figs. S9 to S11). These defect units also extended along the [010] and [100] directions and formed clusters with widths of 1 to 5 nm, as shown in the [001] view (Fig. 2D and fig. S12). Imaging along the [111] projection further confirmed the existence of such spatially extended defect-unit clusters (Fig. 2E and figs. S13 and S14). Consistently, quantitative HAADF-STEM analysis confirmed the substantial projected area of Nb-antisite–affected regions (fig. S5). Together, these observations support a structural model in which defect-unit clusters form a dense, interconnected 3D defect network that hosted a high density of PD complexes throughout the entire film volume (Fig. 2F).

To quantitatively probe the electromechanical coupling within the complexes, we performed correlated strain and electronic structure analysis. Geometric phase analysis (GPA) revealed strong, heterogeneous lattice distortion localized around the defect units (Fig. 2G). Quantitative Nb-sublattice strain mapping showed that strain fluctuation was highly confined and decayed to background levels within approximately three unit cells from the defect center (Fig. 2, H and I). Atomic-resolution electron energy-loss spectroscopy (EELS) provided complementary electronic evidence. Fine-structure analysis of the O K-edge revealed a systematic shift as the probe moved from the matrix toward a defect unit, indicating a gradual reduction of the Nb valence state that is consistent with small electron polaron formation (Fig. 2, J and K) (*48*). This valence reduction decayed over the same spatial scale (approximately three unit cells) as that of the strain field. The coincident spatial decays of the strain and the electronic perturbations provided quantitative evidence supporting the stabilization of PD complexes through elastic and electrostatic coupling between Nb_{Ag}^{4+} defects and Nb′ polarons, a key factor in enhancing the local strain anisotropy that underlies the macroscopic piezoelectric enhancement.

Investigation of electromechanical responses in PD-AKN8

We quantified the electromechanical responses of PD-AKN8 under superimposed AC/DC fields using laser scanning vibrometry (LSV) (supplementary materials, materials and methods, fig. S15A, and supplementary text section 2). Contributions from substrate bending were systematically eliminated to ensure the measured response reflected the film response (supplementary text section 3 and fig. S15B) (*6*, *18*, *20*). Within the electrode area, the surface displacement profile was highly uniform and robust, whereas signals were strongly suppressed outside this area (Fig. 3A), which ruled out artifacts from local breakdown. The films exhibited $d^*_{33,f}$ >7000 pm V^{-1} at 1 kHz and >1000 pm V^{-1} at 10 kHz, together with industrial-grade fatigue resistance (>40 million cycles at 10 kHz without time-dependent drift) (fig. S15C). The corresponding measured vibration profiles were directly visualized in movie S1.

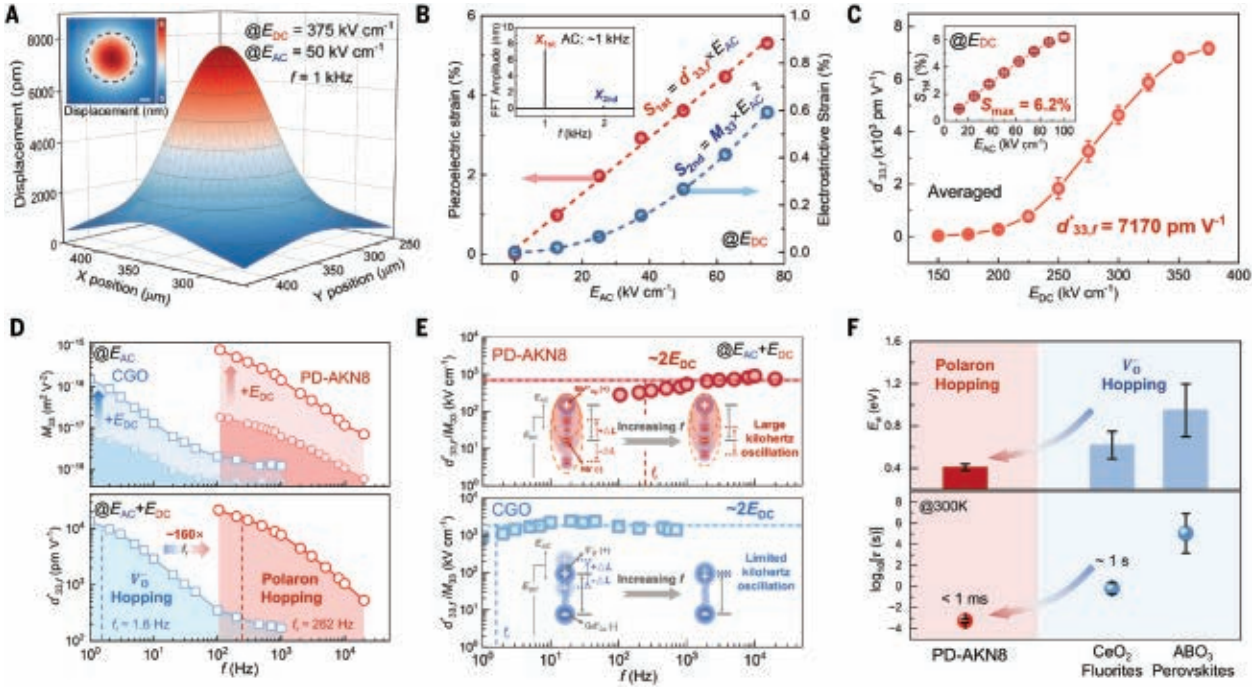

Fig. 3. Electromechanical responses and relaxation dynamics in PD-AKN8. (**A**) 3D surface displacement profile of PD-AKN8 under E_{DC} = 375 kV cm^{-1} and E_{AC} = 50 kV cm^{-1} at 1 kHz. (Inset) Corresponding 2D displacement profile. The dashed circle indicates the top electrode boundary. (**B**) Linear and quadratic electromechanical strains as a function of E_{AC} at ~1 kHz under E_{DC} of 375 kV cm^{-1}. (Inset) FFT amplitude spectrum of the harmonic displacements as a function of frequency under E_{DC} of 375 kV cm^{-1} and E_{AC} of 50 kV cm^{-1} at ~1 kHz. (**C**) E_{DC}-dependent averaged $d^*_{33,f}$ of PD-AKN8 at 1 kHz measured across six independent electrodes. (Inset) Averaged first-harmonic electromechanical strain (S_{1st}) as a function of E_{AC} up to 100 kV cm^{-1} at 1 kHz across six independent electrodes. (**D**) Comparison of frequency-dependent (top) electrostrictive coefficients (M_{33}) and (bottom) effective piezoelectric coefficients ($d^*_{33,f}$) for PD-AKN8 (red) and CGO (blue) films. (Top) Frequency-dependent M_{33} values of PD-AKN8 (at E_{AC} = 50 kV cm^{-1}) and CGO (at E_{AC} = 500 kV cm^{-1}) before and after E_{DC} poling. The applied E_{DC} values for PD-AKN8 and CGO are 375 and 1000 kV cm^{-1}, respectively. (Bottom) Frequency-dependent $d^*_{33,f}$ values under superimposed E_{DC} and E_{AC}, in which the dashed lines indicate the fitted relaxation frequencies (f_r) extracted from $d^*_{33,f}$ spectra for both films. (**E**) Frequency-dependent $d^*_{33,f}/M_{33}$ ratios of (top) PD-AKN8 and (bottom) CGO films under E_{DC}. (Insets) Schematic illustrations of the frequency-dependent dynamics of (top) a PD complex in PD-AKN8 and (bottom) an ionic defect dipole in CGO. (**F**) (Top) Activation energy and (bottom) relaxation time of PD-AKN8 compared with oxygen-vacancy–dominated systems at room temperature. (Top) Activation energy (E_a) for small-polaron hopping in PD-AKN8 (this work) and oxygen-vacancy migration in reported CeO_2-based fluorites and ABO_3-type perovskites. (Bottom) Relaxation time [$\tau = 1/(2\pi f_r)$] extracted from impedance spectra and electromechanical measurements at room temperature for PD-AKN8 and CGO films. For KNN-based perovskites, τ was extrapolated from high-temperature (>400 °C) impedance data, in which oxygen-vacancy migration dominates.

Multiple mutually reinforcing experiments supported that the giant electromechanical response in PD-AKN8 originated from field-induced defect polarization (P_d) rather than intrinsic ferroelectric polarization (P_s). To establish the defect-mediated origin, we applied the electrostrictive framework established in P_d-dominated CGO films, in which DC bias–induced effective piezoelectric coefficients follow $d^*_{33,f} \approx 2M_{33}E_{DC}$ with a large field-related electrostrictive coefficient M_{33} (typically >10^{-17} m^2 V^{-2}) (supplementary text section 4) (*5*). Under an E_{DC} bias with a superimposed ~1 kHz E_{AC}, fast Fourier transform (FFT) harmonic analysis enabled separate determination of $d^*_{33,f}$ and M_{33} from the first- and second-harmonic components (X_{1st} at ~1 kHz and X_{2nd} at ~2 kHz), respectively (supplementary text section 2 and Fig. 3B, inset). The first-harmonic strain (S_{1st}) obtained from X_{1st} exhibited a linear dependence on E_{AC}, whereas the second-harmonic strain (S_{2nd}) obtained from X_{2nd} showed a quadratic dependence, confirming piezoelectric and electrostrictive contributions, respectively (Fig. 3B). This enabled extraction of $d^*_{33,f}$ >7000 pm V^{-1} and $M_{33} \approx 1.05 \times 10^{-16}$ m^2 V^{-2}, satisfying the electrostrictive relation $d^*_{33,f} \approx 2M_{33}E_{DC}$. This M_{33} value exceeded those of conventional bulk ferroelectrics by two to three orders of magnitude and approached that reported for DC-biased CGO films (~10^{-16} m^2 V^{-2} at 1 Hz) (*5*, *49*), despite being achieved at kilohertz frequencies. The defect-polarization origin was further corroborated by a drastic increase in apparent relative permittivity ($|\varepsilon_r|$) with E_{DC} at kilohertz frequencies in PD-AKN8, which is opposite to the trend observed in ferroelectrics (*43*) and the reference AN, but consistent with defect-mediated systems (such as CGO) (figs. S16 and S17). Furthermore, the field- and frequency-dependent polarization, current peaks, and dielectric dispersion ruled out ferroelectric switching as the main origin of polarizability enhancement in PD-AKN8 (figs. S18 to S21).

DC-bias–dependent measurements further indicated that the Nb-antisite–related small-polaron hopping process in PD-AKN8 dominates the piezoelectricity enhancement. Below ~200 kV cm^{-1}, the $d^*_{33,f}$ increased gradually with E_{DC}; above this threshold, it exhibited a sharp rise (Fig. 3C). Similar threshold activation behavior was observed in a DC-bias–dependent current-density measurement in PD-AKN8, which exhibited defect-related localized electron conduction behavior (fig. S21). We ruled out the possible electromechanical contribution of current-driven effects, such as free-carrier transport, electrothermal, or electrochemical effects (figs. S22 to S24 and supplementary text section 5). The averaged $d^*_{33,f}$ reached 7170 ± 230 pm V^{-1} at an E_{DC} of 375 kV cm^{-1} (fig. S25), representing a 350-fold enhancement over the field and frequency-insensitive response of defect-free reference AN films (~20 pm V^{-1}) (fig. S26). Concurrently, the induced piezoelectric strain reached 6.2% under E_{AC} of 100 kV cm^{-1} at 1 kHz (Fig. 3C, inset, and fig. S27), which is orders of magnitude greater than the electromechanical strain (<0.2%) of conventional ferroelectric thin films under comparable E_{AC} (fig. S28). Additionally, $d^*_{33,f}$ increased with increasing K^+ doping up

to 8 mol % in PD-AKN but declined at 12 mol % (fig. S29), a trend that correlated qualitatively with the concentration of antisite defects and localized electrons from reciprocal space mapping (RSM) and x-ray absorption near-edge structure (XANES) spectral analysis (figs. S30 to S32). To exclude the possible effect of K^+ doping on ferroelectric or other properties, we introduced Mn into PD-AKN8 to suppress the antisite-defect and localized-electron concentration. This suppression was accompanied by a similar drop in $d^*_{33,f}$, supporting a direct correlation between the piezoelectric response and PD-complex concentration (fig. S33).

The markedly greater $|\varepsilon_r|$ (~10^6) in PD-AKN8 compared with that of CGO (~10^3) at 1 kHz under DC bias (fig. S16) already suggested superior defect-hopping dynamics at kilohertz frequencies (*5*). Frequency-dependent relaxation of electromechanical response provided decisive evidence that the mobile species were small polarons, not oxygen vacancies. Under E_{AC} drive only, M_{33} exhibited a nonideal Debye relaxation with a characteristic frequency of ~200 Hz (Fig. 3D, top, and fig. S34), matching the small polaron hopping regime (10^1 to 10^3 Hz) (*50*); this frequency was two orders of magnitude greater than short-range $V_O^{\cdot\cdot}$ hopping in CGO films (~0.7 Hz) and four orders of magnitude above long-range $V_O^{\cdot\cdot}$ migration (~0.02 to 0.05 Hz) (*5*, *25*), excluding $V_O^{\cdot\cdot}$ as the dominant mobile species. Under combined E_{AC} and E_{DC}, M_{33} increased more than 10-fold and was accompanied by giant $d^*_{33,f}$ with a relaxation frequency (f_r) of ~260 Hz, which is again more than two orders of magnitude (~160-fold) greater than that of CGO films (~1.6 Hz) (Fig. 3D, bottom, and figs. S35 and S36) (*5*). Crucially, the relation $d^*_{33,f}/M_{33} \approx 2E_{DC}$ held above their respective relaxation frequencies (Fig. 3E), which supports an electrostrictive origin of the field-induced piezoelectricity through polaron-hopping–modulated defect polarization.

Independent electrical transport measurements and broadband dielectric spectroscopy yielded a consistent activation energy (E_a) of 0.41 ± 0.03 eV (figs. S37 and S38), which is in excellent agreement with reported values for small polaron hopping in oxides but notably less than reported barriers for oxygen-vacancy migration in CeO_2-based fluorites (0.5 to 0.75 eV) (*29*, *30*, *51*) and in related ABO_3 perovskites (0.7 to 1.2 eV) (*52*, *53*), as well as for the migration of other relevant ionic defects possibly involved in our system (Fig. 3F, top, and tables S2 and S3) (*54*). This lower barrier exponentially enhanced defect hopping kinetics and shortened the relaxation time (derived from room-temperature dielectric spectroscopy) in PD-AKN8 to 0.4 to 0.8 ms, which is orders of magnitude faster than those in oxygen-vacancy–dominated CeO_2-based fluorites (0.2–2.2 s) (*5*, *30*) and (K,Na)NbO_3 (KNN)–based perovskite ceramics (10^3 to 10^6 s) (*52*, *55*), enabling large-amplitude, high-speed electromechanical responses and overcoming the kinetic limitations of ionic migration (Fig. 3F, bottom).

Field-induced PD complex reconfiguration drives ultrahigh piezoelectricity

The ultrahigh piezoelectric response arose from electric field–driven reconfiguration of PD complexes, which generated giant defect polarization (P_d). To directly probe this field-induced P_d and local symmetry change, we performed electric field–induced second-harmonic generation (SHG) microscopy, a sensitive probe of local noncentrosymmetry, for which the SHG intensity generally scales as $I^{2\omega} \propto |P|^2$ (*56*–*58*).

Under normal (90°) incidence (sensitive to in-plane polarization), PD-AKN8 showed localized SHG "hotspots" (areas with relatively stronger intensity) at zero field (Fig. 4A). Because the SHG spatial resolution is much larger than individual Nb-antisite–related defect units, these hotspots are attributed to regions with statistically higher defect-unit density and stronger local noncentrosymmetry, rather than individual atomic-scale defects (*59*). The darker background also contains field-responsive defect units, but with a lower statistical density, which is consistent with the multiscale chemical and defect-density fluctuations revealed with EDS and HAADF-STEM (figs. S6 and S39). Upon applying a DC bias, the SHG intensity increased sharply in both hotspots and the background, which is consistent with a ~240% enhancement in the spatially averaged SHG response (Fig. 4B and fig. S40), suggesting a common PD-complex–mediated response mechanism. By contrast, the reference AN displayed homogeneous SHG signals that were an order of magnitude weaker at zero field and showed negligible field-induced modulation, suggesting a minimal contribution from intrinsic electric dipoles (Fig. 4B and fig. S41). The frequency dependence of the SHG intensity mirrored that of $d^*_{33,f}$ (Fig. 4B, inset), directly linking the polarization dynamics to polaron-mediated PD-complex reconfiguration. At 45° incidence (sensitive to both in-plane and out-of-plane polarization) (Fig. 4C, inset), the overall SHG modulation depth reached an extraordinary ~250% V^{-1} at 8 V (Fig. 4C), which is approximately two orders of magnitude greater than that reported for classical ferroelectrics (<5% V^{-1}) (*60*). The giant, frequency-dispersive SHG enhancement further supports that the large polarization originates from field-induced PD-complex reconfiguration rather than conventional ferroelectric switching or slow ionic migration mechanisms.

Complementarily, operando 4D-STEM lattice-parameter mapping provided nanoscale structural evidence for field-induced PD-complex reconfiguration (Fig. 4D and fig. S42). The Nb-antisite–rich hotspot regions exhibiting enlarged out-of-plane lattice parameters became more pronounced and more numerous with increasing DC bias, whereas low-defect-concentration regions changed only slightly. Consistently, operando microfocused synchrotron 2θ–ω x-ray diffraction (XRD) showed low-angle shifting and intensity enhancement of the defect region–related shoulder peak under electric fields (fig. S43), indicating that field-enhanced defect polarization is coupled to anisotropic out-of-plane lattice expansion in PD-complex–rich regions.

We performed density functional theory (DFT) calculations on an AN supercell containing a Nb_{Ag}^{4+} antisite defect to elucidate the polaron-defect coupling and its field-driven reconfiguration (Fig. 4E). In Fig. 4E, left and middle left, charge-compensating electrons localized unevenly as small polarons near the antisite defect, as also confirmed from spin density differences (fig. S44), forming electromechanically active PD complexes. For comparison, the dashed outlines marked in Fig. 4E, middle left, indicate the antisite-only configuration (without polarons) (figs. S44 and S45), revealing the polaron localization–induced changes in adjacent Nb–Nb distance (fig. S46). The resulting Nb-sublattice distortion reflects the electro-elastic polaron-defect coupling. Under an applied electric field (Fig. 4E, middle right and right), simulated field-driven charge redistribution, associated with polaron delocalization from the Nb_{Ag}^{4+} antisite defect, redistributed defect polarization, drove pronounced Nb–Nb bond elongation, and thereby enhanced the local coupled distortion anisotropy. These observations were consistent with the operando SHG, 4D-STEM, and microfocused XRD (Fig. 4, A to D, and fig. S43). A reference AN without such coupling showed negligible distortion (fig. S47), highlighting the essential role of such coupling in the large electromechanical response.

To track the microscopic reconfiguration pathway, we performed electric field–dependent SHG polarimetry at 45° incidence (Fig. 4F). At low fields (0 to 4 V), SHG patterns showed an increase in the I_0/I_{90} ratio in the *p*-out measurement condition (SHG polarized parallel to the plane of incidence illustrated in Fig. 4C, inset) and a transition from a weak twofold to a pronounced fourfold pattern in the *s*-out condition (SHG polarized perpendicular to the plane of incidence). These changes were consistent with an evolution from centrosymmetric-like (*C*) to tetragonal-like (*T*) symmetry, indicating the local symmetry breaking through initial PD-complex alignment.

At higher fields (4 to 7 V), where $d^*_{33,f}$ rose sharply, the small polarons (Nb′) hopped among neighboring Nb sites, reconfiguring the local defect polarization and the accompanying lattice distortion while preserving local tetragonal symmetry, indicating progressive polarization extension within the PD complexes. This stage was reflected in a consistently enhanced fourfold *s*-out pattern and a plateau in the I_0/I_{90} ratio in *p*-out

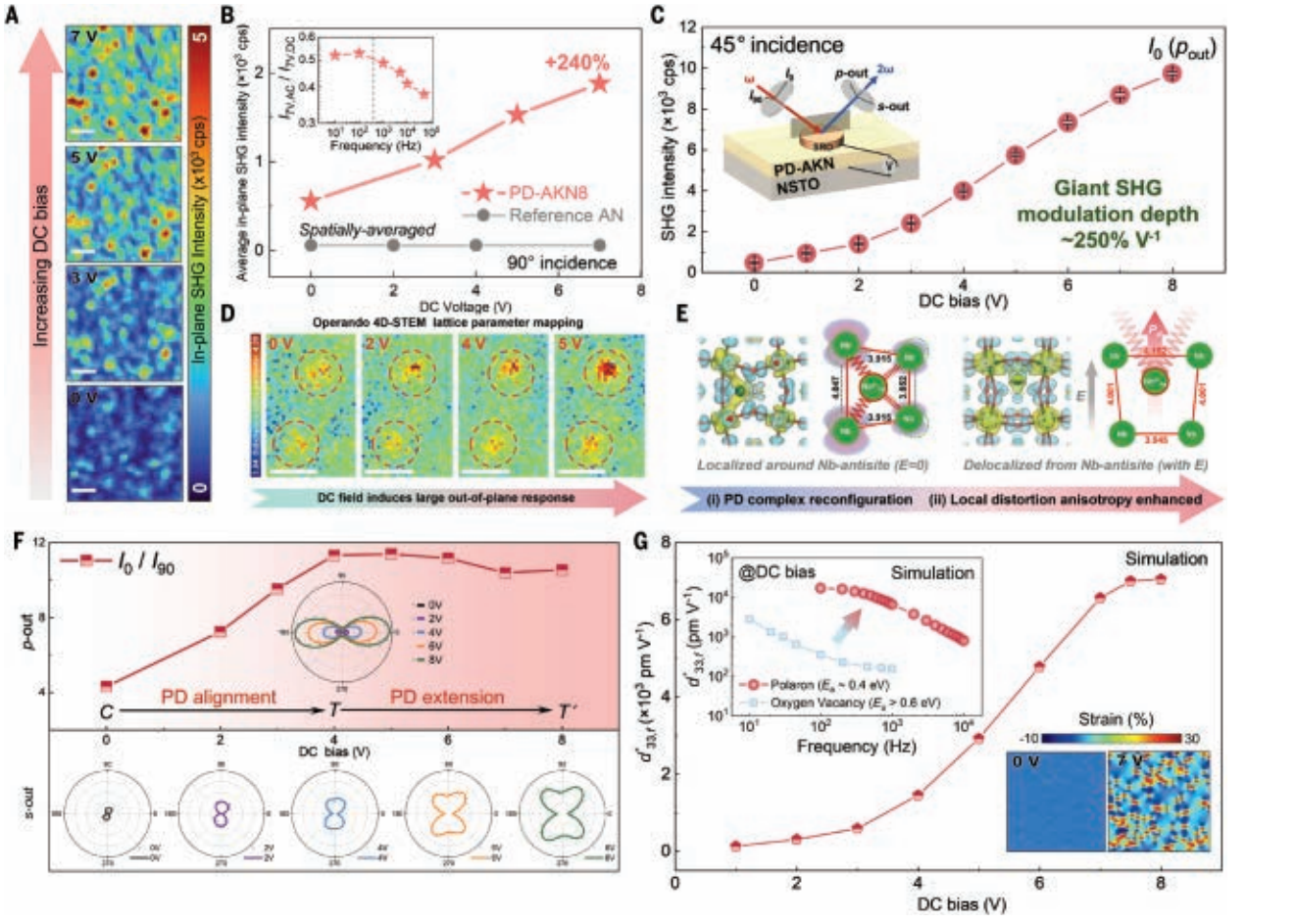

Fig. 4. Operando structural and polarization-state evolution under electric fields with theoretical modeling. (**A**) Operando SHG microscopy of PD-AKN8 under a 90° incidence geometry with increasing DC bias from 0 to 7 V. Scale bar, 2 μm. (**B**) Evolution of the averaged in-plane SHG intensity as a function of DC bias in PD-AKN8 compared with reference AN. (Inset) Frequency-dependent, normalized, averaged in-plane SHG intensity of PD-AKN8 under AC modulation (0 to 7 V sinusoidal waveform), with a relaxation feature indicated with the dashed line. The SHG intensity was background-corrected by subtracting the 0 V DC signal and normalized to the net response at 7 V DC. (**C**) SHG intensity as a function of DC bias measured under a 45° incidence geometry in PD-AKN8. (Inset) Schematic illustration of the 45° incidence measurement geometry. (**D**) Operando 4D-STEM out-of-plane lattice-parameter maps of PD-AKN8 under increasing DC bias from 0 to 5 V. The color bar is in angstroms. Defect-rich hotspot regions exhibit progressively increased out-of-plane lattice expansion. Red dashed circles indicate representative hotspots. Scale bar, 20 nm. (**E**) DFT-calculated structures, charge-density-difference isosurfaces, and Nb-sublattice schematics of an AN supercell containing an Ag-site Nb antisite (left) without an electric field and (right) under an applied electric field (representative structures were derived from calculations in figs. S44 and S45). Dashed lines and circles indicate the antisite-only (without polarons) structural motifs. Pink isosurfaces, springs, and labels indicate localized electrons (small polarons), PD coupling, and Nb–Nb distances (angstroms), respectively. (**F**) SHG intensity as a function of input light polarization angle under (top) *p*-out (SHG polarization parallel to the plane of incidence) and (bottom) *s*-out (SHG polarization perpendicular to the plane of incidence) conditions in PD-AKN8. (Top) I_0/I_{90} ratio as a function of DC bias under *p*-out polarization, with polar plots shown in the inset. (Bottom) Polar plots under *s*-out polarization at different DC biases. Solid curves indicate fits to the experimental data. (**G**) Phase-field simulations of $d^*_{33,f}$ as a function of DC bias in PD-AKN. (Top left inset) Simulated frequency-dependent $d^*_{33,f}$ under 7 V DC bias compared with oxygen-vacancy-dominated scenarios. (Bottom right inset) Simulated strain maps before and after applying a 7-V DC bias to PD-AKN.

detection. Above ~7 V, both the SHG intensity and $d^*_{33,f}$ gradually saturated, which is consistent with the finite coupling length (approximately three unit cells) observed in STEM (Fig. 2, I and K). The entire process corresponded to a continuous symmetry lowering toward a locally more distorted tetragonal phase, quantified by an increase in the SHG tensor ratio $\chi_{33}/\chi_{33(DC=0)}$ (fig. S48) (*58*), confirming a two-stage PD-complex reconfiguration that progressively enhanced local distortion anisotropy.

Phase-field simulations, constrained by the experimentally determined activation energy (E_a) and SHG observations, provided a quantitative framework connecting PD-complex reconfiguration dynamics with macroscopic piezoelectric response. The simulations reproduced both the magnitude and the two-stage evolution of $d^*_{33,f}$ under combined DC and AC fields (Fig. 4G and fig. S49A), suggesting a kinetically constrained, barrier-mediated reconfiguration process. The pronounced field-induced electromechanical response was attributed to PD-complex reconfiguration, which generated large defect polarization and induced strong local strain anisotropy through electrostrictive coupling (Fig. 4G, bottom inset, and fig. S50), contributing to the macroscopic strain. Owing to the moderate activation barrier of small-polaron hopping ($E_a \approx 0.41$ eV) and associated rapid relaxation dynamics (<1 ms), large piezoelectric outputs (>1000 pm V^{-1}) were sustained up to ~10 kHz under DC-bias activation (fig. S49B). By contrast, oxygen-vacancy–dominated systems with higher migration barriers ($E_a > 0.6$ eV) were kinetically limited to ~10 Hz (Fig. 4G, top inset), underscoring the kinetic advantage of the PD-complex–mediated mechanism.

Discussion and generalizability

The superior electromechanical performance originates from an optimized coupling between small electron polarons and the electrostatic and elastic fields associated with charged antisite defects, enabled by a moderate activation barrier for polaron hopping. This intermediate coupling places small polarons in an advantageous regime between two extremes: (i) weakly coupled large-polaron systems or delocalized free carriers with a low transport barrier (<0.1 eV) (*61*), in which insufficient lattice coupling limits field-induced structural deformation (fig. S22), and (ii) oxygen vacancy–mediated migration process with strong ionic-defect-lattice coupling and high activation barriers (>0.6 eV), resulting in electromechanical responses that are typically restricted to low frequencies and high fields (*5*).

Because this polaron-defect design paradigm does not rely on ferroelectric polarization, it holds broad generalizability across nonferroelectric material systems. We successfully extended this strategy to centrosymmetric, nonperovskite Nb-doped TiO_{2-x} thin films, in which PD complexes were deliberately engineered (*37*, *62*, *63*). These films exhibited robust field-induced piezoelectric responses ($d^*_{33,f}$ > 1000 pm V^{-1}) at 1 kHz under moderate fields, accompanied by characteristic polaron-type relaxation and the same electrostrictive relationship $d^*_{33,f} \approx 2M_{33}E_{DC}$ (fig. S51). To further evaluate scalability and device relevance, PD-AKN8 films were uniformly deposited on commercial 4-inch Si wafers by using the same CSD route and patterned into millimeter-scale electrodes. Despite statistical averaging effects and mechanical boundary constraints associated with enlarged electrode areas, the films retained piezoelectric responses exceeding those of commercial PZT films, supporting the scalability of PD-complex engineering and its compatibility with MEMS-relevant device geometries (fig. S52).

We overcame the long-standing magnitude–speed trade-off in high-performance electromechanical materials through a PD-complex design paradigm. By designing dynamically reconfigurable electromechanical PD complexes, we achieved ultrahigh-kilohertz-rate performance in silver niobate–based films, with a record-breaking effective piezoelectric coefficient $d^*_{33,f}$ of ~7170 pm V^{-1} and a strain of ~6.2% at kilohertz frequencies. This polaron-piezoelectric effect originates from dynamic PD-complex reconfiguration driven by fast small-polaron hopping, which enables giant defect polarization modulation at kilohertz frequencies. The combination of giant electromechanical response, high-speed operation, scalable processing, and retained performance under enlarged electrode geometries positions PD-AKN8 as a sustainable, high-performance candidate for MEMS applications.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S52; Tables S1 to S3; References (*65*–*77*); Movie S1

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DYNAMIC POLYMERS

Precise length and charge segmentation of billion-dalton supramolecular polymers

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Supramolecular polymers with precise micrometer-scale segments of opposite charge are challenging to synthesize as discrete assemblies in water. These systems could be used to pattern charged surfaces, interact with the membranes of cells, or form hierarchical structures. We report a living polymerization of fibrillar peptide assemblies with a central positively charged segment flanked by negative ones that can have a collective molar mass on the order of billions of daltons. The self-capping supramolecular polymerization required an untwisted β sheet core persisting along the entire length of the assembly as well as dormant ends in the absence of a monomer source. The positive domains of the segmented structures were anchored to the membranes of neurons, leaving the connected negative segments in a highly dynamic state near the cells through charge repulsion. The segmented architecture generated extensive neurite growth and synaptic activity.

Chemically distinct components within and around cells activate growth, movement, development of specialized functions, and tissue architectures. One-dimensional (1D) assemblies are particularly abundant in living systems, and analogous synthetic structures with spatially organized components and precisely defined dimensions can be realized through several synthetic approaches. For example, 1D architectures that exhibit chemical segmentation along their lengths have been generated by crystallization-driven self-assembly, where block copolymers form cylindrical micelles with crystalline cores (*1*–*3*). A similar strategy has been developed with small molecules to create segmented tubular structures (*4*). Other approaches have involved directional hydrogen bonding and π–π stacking to achieve living supramolecular polymerization of linearized assemblies of molecules (*5*–*7*). Also, self-sorting within 1D assemblies has been investigated to produce segmented structures using partially incompatible polymer micelles or small molecules (*8*, *9*). Folded peptides have been connected by click chemistry to construct segmented 1D assemblies (*10*). Peptide amphiphiles (PAs; peptides conjugated to lipophilic segments) have been extensively investigated to create nanoscale filaments for regeneration, robotic materials, and electroactive systems (*11*–*14*).

In supramolecular systems, it has been extremely difficult to introduce length and segmentation precision. Because electrostatic charge is a dominant feature of biological interfaces, controlling the spatial arrangement of charged domains in supramolecular assemblies would provide a powerful mechanism to direct cell-material interactions. In this work, we synthesized in water 1D supramolecular polymer fibrils with controlled length and also precise segmentation with oppositely charged domains. The fibrils and their segments reached micrometer-scale lengths, and unexpectedly, the living supramolecular polymerization was fast at one end and slow at the other. This differential reaction rate allowed us to produce asymmetric fibrils with short and long terminal segments of defined chemical structure. We grew these 1D assemblies at millimolar concentration under physiological conditions that might be functionally important for the use of the structures in biological systems. In previous work, it has been common to investigate the response of cells and tissues to the chemistry of covalent polymers and surface patterning (*15*–*17*). It is still critical to understand how more dynamic environments, such as supramolecular structures, could promote bioactivity without fixing cell directionality or tissue structure. Here, we investigated, as an example, how precisely segmented and dynamic structures interact with neurons, which require plasticity in neurite growth and synaptogenesis. Furthermore, our work allowed us to establish some of the key design principles to synthesize these chemical systems.

Precise length control of supramolecular polymers

We synthesized a PA with the cationic peptide sequence $VVAAG(RGG)_3$, lipidated with a 16-carbon tail and conjugated to a terminal poly(ethylene glycol) (PEG) chain (R-PA; Fig. 1A and table S1). The VVAA sequence was utilized given its strong tendency to create stable β sheet secondary structure, whereas the flexible RGG sequence was inspired by its presence in biological systems to bind negatively charged biopolymers (*18*). The oligomeric PEG domain was integrated into the structure to minimize uncontrolled aggregation of supramolecular assemblies. After annealing at 99°C in phosphate-buffered saline (PBS), R-PA formed long fibrils that we visualized by transmission electron microscopy (TEM) (Fig. 1B and fig. S1) and (after labeling with a fluorescent dye) confocal laser scanning microscopy (CLSM) (fig. S2). The assemblies melted into spherical micelles at 68°C and reassembled into fibrils during cooling at 53°C (figs. S3 to S6 and tables S2 and S3). Unexpectedly, the length of the fibrils could be precisely controlled by simply varying the annealing temperature, with each condition resulting in assemblies exhibiting very low dispersity ($Đ < 1.1$) in number average length (L_N) (Fig. 1, C and E, and fig. S7). Annealing at 80°C (heating at 1°C/min to 80°C, holding for 1 hour, then cooling at 1°C/min to 25°C) yielded 2.6-μm-long fibrils, whereas annealing at 85°C resulted in a nearly 10-fold increase in length ($L_N = 20.3$ μm). Longer fibrils at the same monomer concentration made the solutions more viscous, increasing from 1.2 to 2.2 mPa·s at a shear rate of 1000 s^{-1} (fig. S8), demonstrating the importance of precise length control on properties.

The narrow length distribution of the polymers was consistent with a nucleation-growth supramolecular assembly mechanism. A discrete number of nuclei would begin to grow at the same time and at the same rate, so differences in the number of nuclei should control the precise fibril length. All samples were bath sonicated before assembly to generate very short fibrous seed particles (figs. S9 and S10). As the annealing temperature was increased, we expected the number of nonmelted particles that could act as nuclei upon cooling to decrease, lengthening the resultant fibrils (Fig. 1D). In this way, the system is an example of self-seeding (*19*, *20*), a mechanism not previously reported for supramolecular polymers in aqueous solution.

We directly tested the nucleation-growth mechanism by manually seeding the system to generate length-controlled assemblies. Preassembled fibrils were first probe sonicated to produce seeds (fig. S11). A metastable solution of micelles was separately prepared by heating the R-PA to 99°C then cooling to 65°C (just below the melting temperature but above the temperature of reassembly). We then added the seeds to the micelles, which acted as the nuclei upon cooling (fig. S12). Varying the micelle to seed ratio allowed us to achieve linear control over fibril lengths, again with very low dispersity ($Đ < 1.1$).

Living supramolecular polymerization of segmented assemblies

We hypothesized that the observed supramolecular polymerization of fibrils had "living" character and was occurring by addition of monomer

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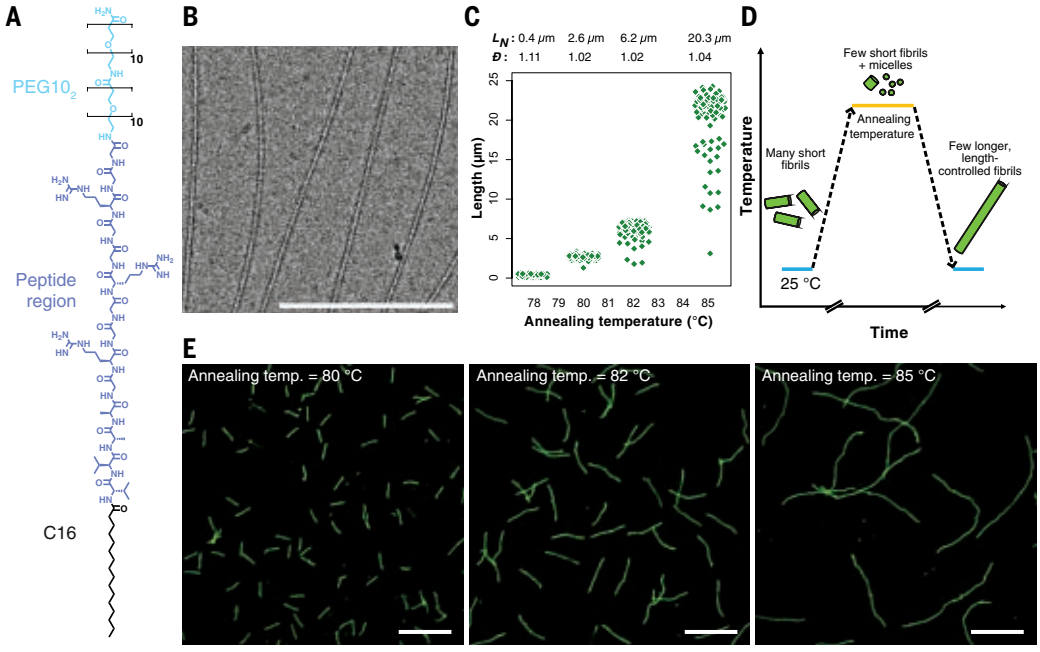

Fig. 1. Length-controlled self-seeded assembly of a PA (C16-VVAAGRGGRGGRGG-PEG10- PEG10). (**A**) Molecular structure of **R-PA**. (**B**) Cryo-EM of R-PA. Scale bar, 250 nm. (**C**) Plot of length versus annealing temperature. $N > 100$ fibrils for each sample, with 100 randomly chosen data points displayed. For annealing temperature 78°C, length was measured by atomic force microscopy. For annealing temperatures 80°, 82°, and 85°C, length was measured by CLSM. (**D**) Schematic of the self-seeding mechanism in which the annealing temperature controls the length of the resultant fibrils. (**E**) Representative CLSM images of length-controlled PA fibrils following annealing at the indicated temperature. Scale bars, 10 μm. Single-letter abbreviations for the amino acid residues referenced throughout this paper and figure are as follows: V, Val; A, Ala; G, Gly; R, Arg; E, Glu.

from micellar aggregates to growing ends. We prepared short fibrillar segments of 0.4 μm in length using the self-seeded annealing approach described above but incorporating a small percentage of Cy5-labeled PA. These short segments were added to a solution of melted Cy3-labeled PA micelle at 65°C, and upon cooling, we observed the formation of segmented supramolecular polymers with distinct dye-labeled domains. Using aqueous media with a high PA concentration, we generated three-segment assemblies at high yields with controllable lengths that depended on the relative number of short seed segments added to the micelles (Fig. 2, A and B, and figs. S13 and S14). Notably, we observed that all fibrils contained a short and a long segment of the Cy3 monomers at opposite ends of the seed. With an 11:1 ratio of micelles to short seed segments, the long Cy3 ends of the assemblies were on average 3.1 times the length of the short ones, and both segments were effectively monodisperse (Fig. 2C). We synthesized very long, segmented, asymmetric fibrils with total lengths >30 μm and very low dispersity using this technique (Fig. 2D and fig. S15). These experiments provided strong evidence for living supramolecular polymerization at both ends of the fibril.

The observed asymmetric segmentation suggested that the two ends of the assemblies were different. To test this hypothesis, we added three-segment fibrils to Cy5-labeled micelles to generate five-segment supramolecular polymers (Fig. 2E and fig. S16). The resulting five-segment assemblies consistently revealed growth of a new short end next to the preexisting short segment and a new long end next to the preexisting long segment (Fig. 2F). This correlated asymmetric growth at each end established that the supramolecular assemblies had an inherent directionality. Wide-angle x-ray scattering (WAXS) (fig. S17) and attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) (fig. S18) experiments showed evidence of ordered β sheet structures within the fibrils. Chirality from the amino acid residues in the PA molecules, combined with ordered β sheet packing, likely created a directionality that ultimately determined the molecular nature of the faces at the ends of the assemblies (Fig. 2G). Dissimilar monomer addition kinetics at each end would result from differences in the exposed faces at the living ends, thus resulting in asymmetric growth rates. A similar asymmetry has been reported during the crystallization of tubular peptide crystals (*21*) and elongation of naturally occurring amyloid fibrils (*22*). Slower cooling rates further increased the asymmetry of ends, whereas faster cooling decreased it, indicating that the directional growth is thermodynamically controlled (figs. S19 and S20).

We next investigated supramolecular dynamics by mixing long Cy3-labeled and short Cy5-labeled supramolecular polymers of R-PA. In these systems, we observed an increase in Förster resonance energy transfer (FRET) efficiency after about 32 hours, which stabilized at a level corresponding to 10% molecular exchange (figs. S21 to S24). After 6 days, CLSM still showed two distinct populations of fibrils with little visible mixing and no change in overall length. We hypothesized that molecules on the edges of the assemblies engaged in this exchange rather than those on the ends. In the segmented fibrils, we did not observe any further changes by CLSM in length or mixing over a period of 6 weeks (fig. S25).

Supramolecular polymers with precise oppositely charged segments

Based on our results above, we next considered whether we could create segmented assemblies with well-defined domains of opposite charge. These systems could become highly functional, taking advantage of electrostatic interactions centered on specific segments to attach nanoparticles or macromolecules, anchor the assemblies in controlled configurations to charged surfaces, or even create hierarchical structures (*23*). We designed a negatively charged monomer (E-PA) with the sequence $VVAAG(EGG)_3$, lipidated and PEGylated in the same manner as the positively charged analog (Fig. 3A and table S1). When annealed (fig. S26), E-PA formed long supramolecular fibrils that resembled those formed by the original positively charged analog. We observed two prominent fibril orientations using cryo–electron microscopy (cryo-EM), a thinner, electron-dense one and a wider, more diffuse orientation, likely caused by a thin ribbon structure lying on its edge or face, respectively (Fig. 3B).

Class-averaged high-resolution cryo-EM was used to observe the nanometer-scale structure of E-PA, revealing a zipper-like arrangement of electron density within the assembly core, consistent with β sheet signatures observed by FTIR (Fig. 3C and figs. S27 to S30). The width of the highly organized core was narrow compared with the total fibril width, which we attributed primarily to the PEG chains and disordered dynamic peripheral regions whose presence was suggested by FRET experiments. 3D reconstruction was unsuccessful, as

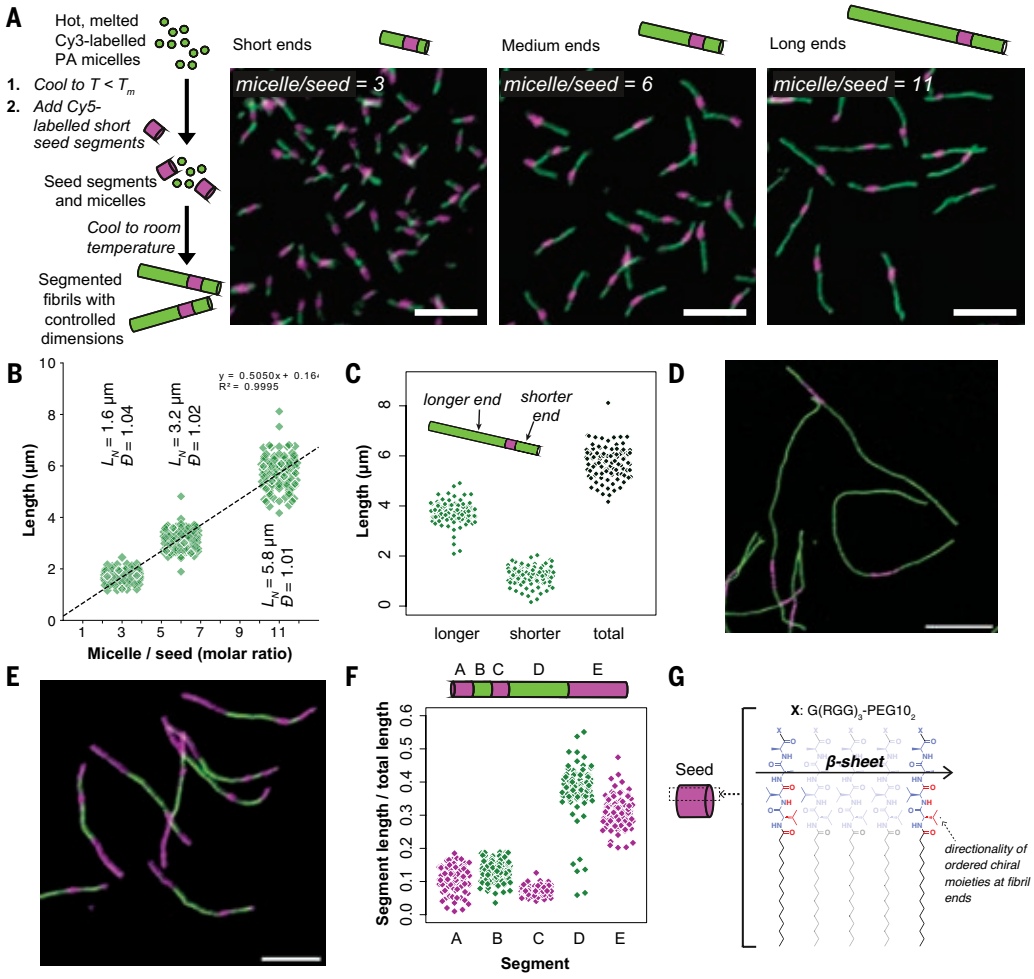

Fig. 2. Asymmetric living polymerization of segmented R-PA supramolecular polymers. (**A**) Schematic of the synthesis of segmented supramolecular polymers (left), and three CLSM images of segmented assemblies with short, medium, and long asymmetric end lengths grown using molar ratios of micelles to short seed segments equal to 3, 6, and 11, respectively. Scale bars, 5 µm. (**B**) Total lengths of fibrils versus the micelle/seed ratio. (**C**) Lengths of the short and long ends and the total lengths of the fibrils with micelle/seed = 11. $L_{N,\,long} = 3.7$ µm, $Đ_{long} = 1.01$; $L_{N,\,short} = 1.2$ µm, $Đ_{short} = 1.09$; $L_{N,\,total} = 5.8$ µm, $Đ_{total} = 1.01$. (**D**) CLSM images of very long segmented fibrils. Scale bar, 10 µm. $L_{N,\,total} = 34$ µm, $Đ_{total} = 1.04$. (**E**) CLSM images of fibrils with five discrete segments, generated through multiple rounds of living supramolecular polymerization. Scale bar , 5 µm. (**F**) Segment lengths of five-segment assemblies, labeled sequentially from A to E, where A is the end segment with the shortest length. (**G**) Schematic of the β sheet directionality and asymmetric ends. For length measurements in panels (B) to (D) and (F), $N > 100$ fibrils; length measured from CLSM images.

the fibrils did not display a helical twist and adopted only two preferred orientations on the cryo-EM grid, together preventing the angular sampling that is required (*24*). For R-PA, diffuse electron density around the fibrils prevented the acquisition of well-resolved class-averaged images, most likely the result of arginine-phosphate interactions (figs. S31 and S32). However, small-angle x-ray scattering (SAXS) and WAXS experiments showed near-identical profiles for R-PA and E-PA, providing evidence that the identity of the charged groups did not appreciably impact their respective structures (Fig. 3D). Annealing Cy3-labeled E-PA (table S1) at specific temperatures also produced length-controlled assemblies through the self-seeding mechanism that we described above for R-PA (figs. S33 and 34).

We next attempted the synthesis of supramolecular polymers with oppositely charged segments along their lengths. First, we confirmed that E-PA generated precise segmented structures by using monomers labeled with two different dyes (Cy3 and Cy5, figs. S35-36). Then, we used short Cy5-labeled R-PA seed fibrils and added them to a melted solution of Cy3-labeled E-PA. This process yielded precisely segmented assemblies bearing both positively and negatively charged domains (Fig. 3E, figs. S37-38). Furthermore, we observed again the asymmetric growth of the segments flanking the seed. We were surprised to find that even in concentrated solutions, massive aggregation between domains of opposite charge was not observed. The network of precisely segmented fibrils shown in Fig. 3E had spatially defined colors, rather than amorphous aggregates with less defined contrast or color. We propose that the presence of PEGylated termini on the PA molecules suppressed aggregation. The possibility of creating precise positive and negative domains must originate in the critical role played by the highly ordered β-sheet core present in assemblies of both monomers.

Design rules and mechanism for precisely controlled PA assembly

We carried out a number of experiments to understand the possible mechanisms operating in the precision supramolecular polymerization investigated here. We synthesized a version of E-PA without PEG that terminated in the peptide sequence EGGE (table S1). Imaging of this system by TEM immediately after probe sonication revealed the formation of relatively short fibrils which grew substantially further after 24 hours (Fig. 4A). However, the PEGylated analog E-PA after similar sonication formed extremely short structures that remained stable (Fig. 4A). The use of PEG10 units on the terminus of the PAs allowed us to systematically vary the total length of PEG through solid-phase peptide synthesis. Despite observing similar internal structures and thin ribbon fibril morphologies regardless of PEG length (figs. S39 to S42), we found that a minimum of two PEG10 units was required to stabilize fibril lengths and allow precise synthesis (figs. S43 and S44). In a modified R-PA without PEG, we also observed extensive fibril-fibril aggregation, which interfered with precision supramolecular polymerization. Furthermore, when PEG with a comparable molecular weight to PEG10-PEG10 was present free in solution with the non-PEGylated variant of E-PA, partial end stabilization was observed but only at very high (10 wt%) PEG concentration (figs. S45 and S46).

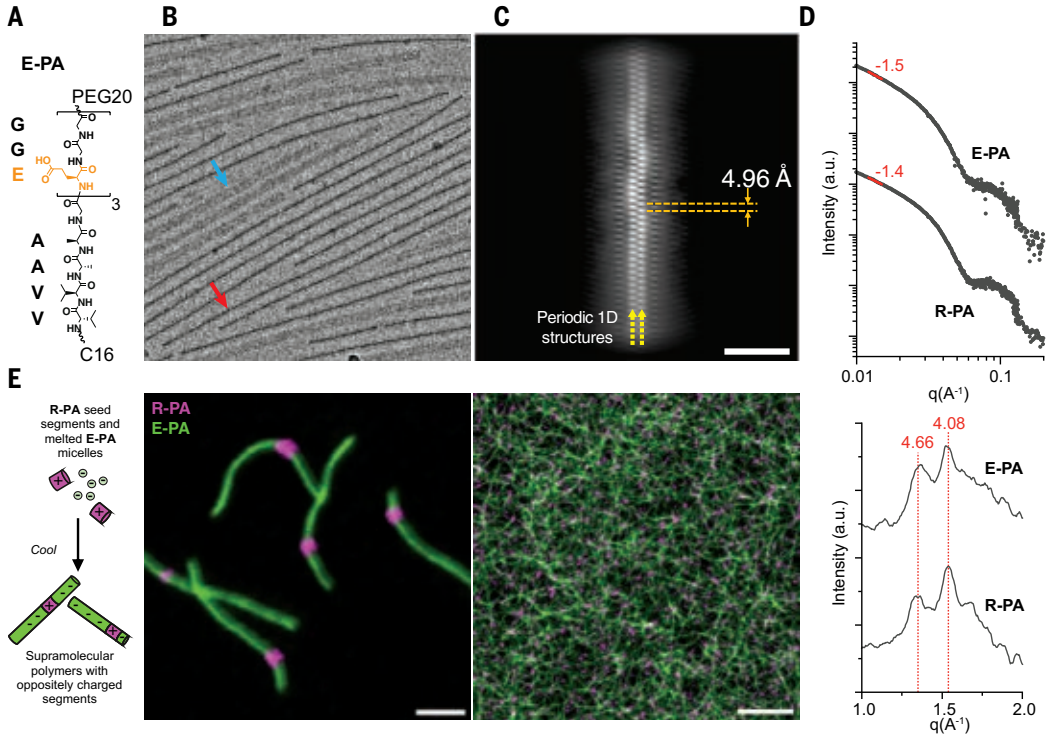

Fig. 3. Synthesis of segmented supramolecular polymers with adjacent regions of opposite charge. (**A**) Molecular structure of **E-PA**. (**B**) Cryo-EM of **E-PA**. Blue and red arrows indicate two different orientations of fibrils. Scale bar, 250 nm. (**C**) High-resolution image based on the averaged cryo-EM 2D class of E-PA showing 4.96-Å 1D periodic spacing. Scale bar, 5 nm. (**D**) SAXS (top) and WAXS (bottom) of E-PA and R-PA. (**E**) (Left) Schematic of the synthesis of segmented supramolecular polymers with domains of opposite charge. (Middle) CLSM images of segmented fibrils with a positively charged R-PA center (magenta) and negatively charged E-PA ends (green) obtained at a low concentration (0.25 μM) and (right) at a higher concentration (16 μM). Scale bar left panel = 2 μm; right panel = 10 μm. a.u., arbitrary units.

Based on the observations above, we hypothesized that all assemblies contain a "cap" structure formed by interactions between the end surfaces of the fibrils and the long PEG chains, which are locally concentrated through their conjugation to the PA. This cap structure prevents fusion of fibril ends or Ostwald ripening mechanisms, thus giving rise to stable length precision. The protective interactions of the PEG segments with the fibril ends cannot be permanent as this would inhibit further addition of monomer in a living manner. We therefore propose that elongation of the fibrils occurs through the continuous diffusion of monomer molecules into a transient cap at high temperatures. Controlled growth from both ends was observed when seeding at 68° or 65°C; however, at 62.5°C, the dispersity of the short ends increased, and their elongation became uncontrolled at 60°C (Đ = 2.25), with many failing to grow at all (Fig. 4B and fig. S47). This asymmetry in dispersity of the end segments cannot be explained by the asymmetric growth rates alone. Instead, this result provides support for our hypothesis that indeed there is a cap that controls the precision of the supramolecular polymerization. Furthermore, we speculate that the interactions between PEG and the two fibril ends are distinct, resulting in caps that exhibit different behaviors (Fig. 4C). At 60°C, the cap at the end of the short segment has lost the rapid dynamic behavior required for controlled supramolecular polymerization, thus leading to nonsynchronous growth. By contrast, the cap at the end of the long segment remains dynamic enough to sustain synchronous growth. Similarly to the asymmetric growth rates described earlier, we hypothesize that the distinct interactions between the cap oligomers and each of the two fibril ends is also a consequence of the directionality of the β-sheet. This would be caused by differences in the chemical moieties exposed at each end of the fibrils.

We then investigated the specificity of PEG as a hydrophilic oligomer in this cap-mediated mechanism for precise supramolecular assembly. When PEG on E-PA was completely replaced with poly(sarcosine) of comparable length, we observed that the fibril ends were indeed stabilized following sonication (fig. S48). Similar to PEG, poly(sarcosine) exhibits high solubility and flexibility (*25*), allowing the formation of a dynamic cap. We also replaced the amide end group on the PEGylated E-PA with a carboxylic acid, a methyl ester, or a hydroxyl group and found that none of these changes disrupted the fibril end stabilization (figs. S49 to S51). Additionally, the sarcosine E-PA and all end group–modified PEGylated E-PAs exhibited similar morphology and β-sheet internal structures (fig. S52) and were able to be seeded with PEGylated R-PA to give fibrils of precise length (Fig. 4D and fig. S53). These results demonstrate the generality of the mechanism discovered, which requires interactions between the hydrophilic oligomer and the peptide sequence to create the cap structure. As to what the interactions between the hydrophilic oligomer and the fibril end may be, it is noteworthy that both PEG and poly(sarcosine) contain many hydrogen bond acceptors. The amide groups and the side chains of the peptide at the fibril end can act as hydrogen bond donors in a weak interaction with the polymer backbones in the cap. In the case of arginine, there could be other interactions beyond hydrogen bonding involving the π electrons (*26*).

To explore the effect of peptide sequence on precision of length and segmentation, we investigated a constitutional isomer of R-PA in which VVAA was mutated to AAVV. This change resulted in the formation of twisted fibrils with a broad distribution in pitch lengths (Fig. 4E), in contrast to the highly uniform, untwisted structures formed by R-PA (Fig. 1B). Using the isomeric PA as the monomer during supramolecular polymerization, we could not control segment length, and the correlation between the lengths of the initial seeds and their terminal segments was inconsistent (Fig. 4F and figs. S54 to S56). The primary WAXS peak linked to β sheet secondary structure at 4.66 Å remained unchanged in the isomeric assemblies, but a second peak at a shorter *d*-spacing (4.06 Å) became broader (fig. S57). The isomeric assemblies also exhibited a broader melting transition (fig. S58) and weaker hydrogen bonding based on FTIR spectra (fig. S59). These observations suggested the existence of a population of fibrils with a heterogeneous internal structure in their cores, which in turn resulted in nonuniform growth rates and thus lack of precision. Notably, melted AAVV isomer could not be elongated from R-PA seed segments, which highlights the specificity of the hydrogen bonding that controls PA supramolecular polymerization (fig. S60). We interpret the variability observed as the result of a dynamic distribution of configurations with similar

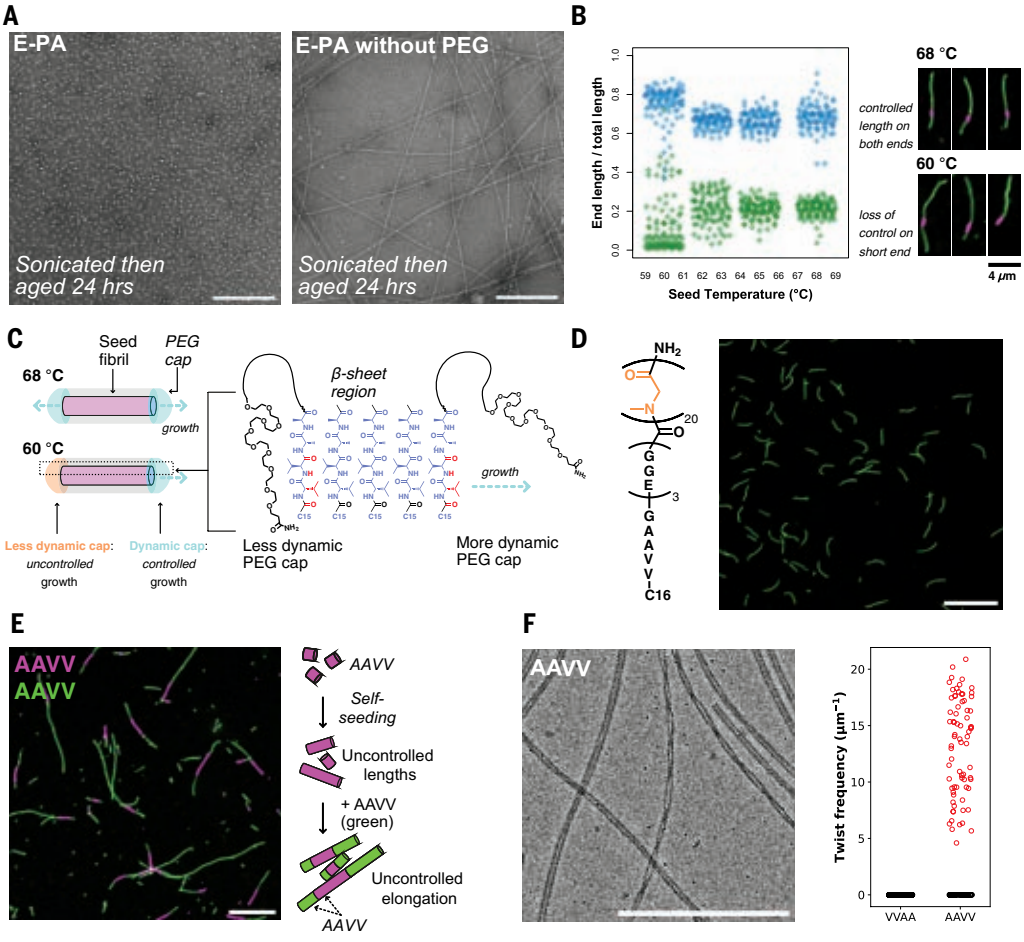

Fig. 4. Design rules and mechanism for precisely controlled PA assembly. (**A**) TEM micrographs of E-PA with and without PEG that have been probe sonicated and then aged for 24 hours. (**B**) (Left) Ratio of end length to total fibril length of the short (green) and long (blue) end segments of R-PA fibrils following seeding at different temperatures and (right) representative CLSM micrographs of the resultant assemblies at 68° and 60°C. (**C**) Schematic of the proposed mechanism for fibril end stabilization by PEG caps at 68° and 60°C. (**D**) (Left) Chemical structure of PA in which the PEG oligomer is replaced by poly(sarcosine) and (right) a CLSM image of the precise-length fibrils formed with this monomer. Scale bar, 10 μm. (**E**) (Left) CLSM image of supramolecular assemblies obtained using the AAVV isomer of **R-PA** (scale bar, 10 μm.) and (right) schematic of the outcome based on the image shown. (**F**) (Left) Representative cryo-TEM micrograph of the AAVV isomer of R-PA (scale bar, 250 nm) and (right) fibril twist frequency measurements of VVAA and AAVV.

energies, in which the pitch of individual assemblies could fluctuate. This highly dynamic state would lead to nonuniform growth rates and thus prevent the formation of precise supramolecular assemblies. It is also possible that a sequence change results in caps unable to sustain controlled growth.

Enhanced neuron-fibril interactions using segmented charged architectures

Although positively charged nanoparticles can be cytotoxic in high concentration (*27*), we anticipated that the positively charged domains would anchor the segmented assemblies to the negative surfaces of cells. The anchor would keep the negative segments in proximity to the cells and at the same time allow them to remain dynamic to capture important extracellular matrix (ECM) components. The combined action of many assemblies acting in this way could create supramolecular scaffolds that serve as both adhesive sites and recruitment centers for growth factors. We chose neurons to test our architectures for two reasons. First, their cell membranes are more negatively charged than those of other cells as they have a higher concentration of gangliosides (*28*), providing a strong electrostatic attraction (*29*) with the positively charged anchors in our segmented structures. Second, neurons are a critical target for regenerative medicine owing to their limited capacity to regrow after injury or in diseased states of the adult central nervous system (*30*).

We tested the well-defined segmented structures with oppositely charged domains on mouse cortical neurons. We used a negatively charged PA in which one glutamic residue was removed from E-PA to increase its melting point, which improved stability at physiological temperatures without impacting fibril structure (figs. S61 to 63 and table S1). Neurons were allowed to adhere to a commonly used positively charged poly-D-lysine (PDL) surface for 24 hours before supramolecular fibrils 10 to 12 μm in length with various charge architectures were added to the surrounding media at a concentration of 45 μM (figs. S64 and S65). As expected, negatively charged fibrils revealed very weak interactions with the neurons, which could originate in their electrostatic repulsion by the plasma membranes (Fig. 5A). By contrast, segmented fibrils containing a small positively charged central domain (just 12.5 mol % of the total assembly) and long negatively charged ends were efficiently captured by neurons (Fig. 5A and figs. S66 to S69). Some interaction is also observed between the fibrils and the PDL surface, but they are concentrated around the neurites.

In agreement with our earlier hypothesis, live-cell microscopy confirmed that the fibrils exhibited intense movement, yet they still remained strongly anchored near the cell surfaces (movies S1 and S2). We clearly observed Brownian motion of the negatively charged ends of the segmented assemblies, revealing movements on the order of microns per second (Fig. 5B), whereas the positively charged anchors remained effectively static (fig. S70). Entirely positively charged fibrils interacted very strongly with neuron membranes to form a dense coating around the cells and did not reveal any evidence of movement (figs. S68 and S69 and movies S3 and S4). Also, assemblies created by supramolecular polymerization of mixed positive and negative monomers, which were much more random than segmented ones, interacted very weakly with cells (figs. S66 and S67).

We investigated the morphological response of neurons to the presence of charge-segmented assemblies in the media and observed a greater increase in neurite length over time (Fig. 5C) relative to negatively or positively charged assemblies (figs. S71 and S72). Also,

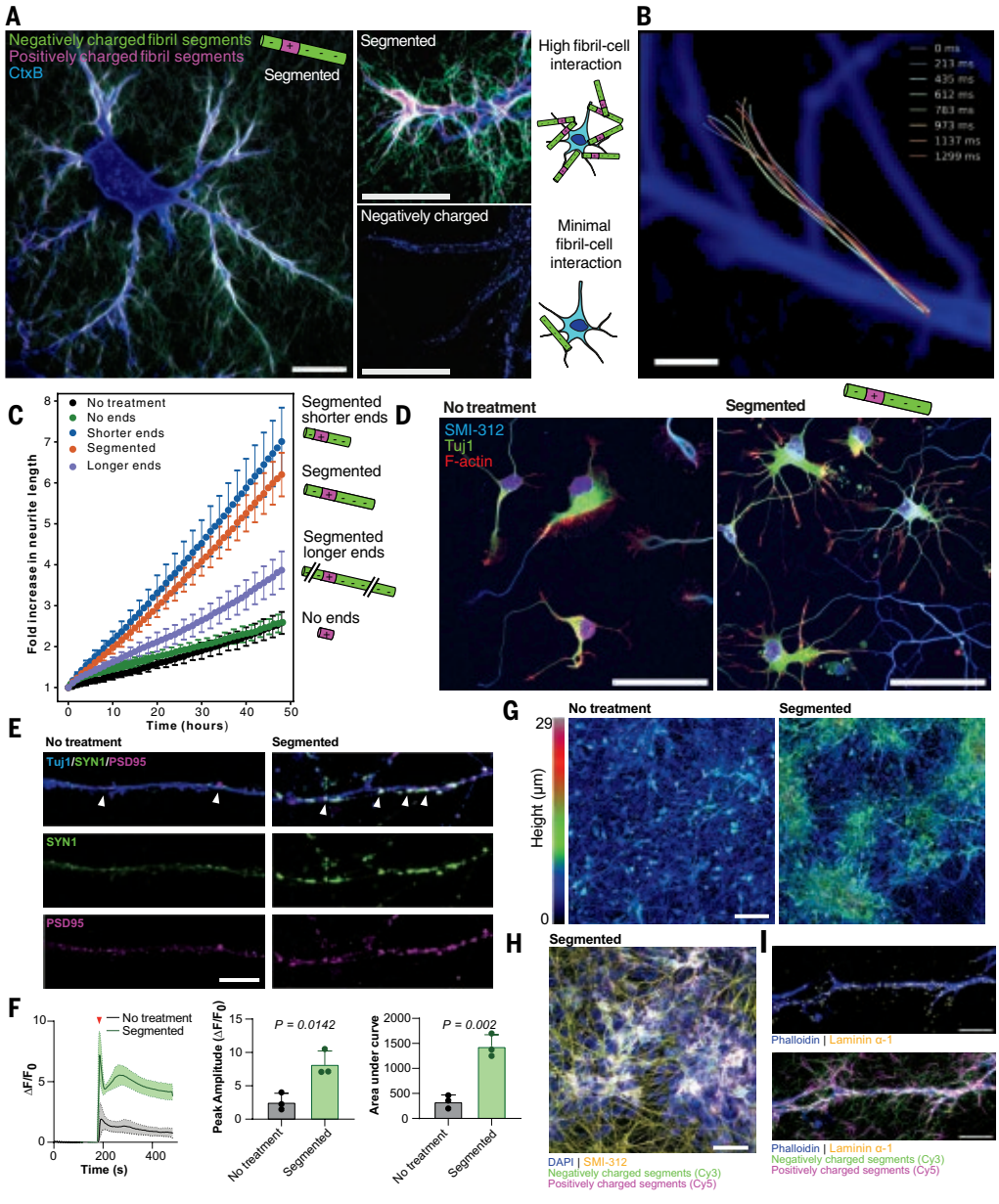

Fig. 5. Neuron growth and activity are enhanced by charge-segmented fibrils. (**A**) (Left) Representative single-plane CLSM images of live primary mouse cortical neurons interacting with dye-labeled segmented or negatively charged fibrils 24 hours after treatment and (right) corresponding schematics of this interaction. Scale bars, 10 μm. (**B**) CLSM image of a neurite and a traced overlay of a negatively charged domain within a segmented fibril showing its position at the times indicated [each tracked frame is indicated by a different color, and the neuronal membrane is stained with cholera toxin B (blue)]. Scale bar, 2 μm (**C**) Fold increase in mean neurite length over 48 hours following treatment with different supramolecular assemblies with varying charged domain architectures (shown schematically on the right). The amount of positive segment was kept constant in all segmented treatments (equivalent to a concentration of 5.6 μM of R-PA), whereas the lengths of the negative ends were varied (equivalent to 20, 39.4, or 118 μM of negatively charged PA for the "segmented shorter ends," "segmented," and "segmented longer ends" architectures, respectively). Error bars represent the SEM. Neurite length was measured in real-time using 28 to 42 images across three wells per treatment per time point. (**D**) Representative CLSM images of fixed primary mouse cortical neurons either untreated (left) or treated with segmented fibrils (right) for 48 hours. SMI-312 indicates axons (blue), Tuj1 indicates neurons (green), and F-actin is labeled by phalloidin (red). Scale bar, 50 μm. (**E**) Representative CLSM maximum-intensity z-projections of synapse formation on neurites of primary mouse cortical neurons either (left) untreated or (right) treated with charge-segmented fibrils for 14 days. SYN1 indicates presynaptic terminals (green), PSD95 indicates postsynaptic terminals (magenta), and Tuj1 indicates neurons (blue). SYN1+ and PSD95+ puncta are indicated by white arrows. Scale bar, 10 μm. (**F**) (Left) Ca^{2+} response of primary mouse cortical neurons either untreated or treated with segmented fibrils for 7 days. Averaged intracellular Ca^{2+} traces were measured by relative change in fluorescence ($\Delta F/F_0$) over time. Baseline fluorescence was recorded for 3 min prior to KCl stimulation, indicated by the red arrow. (Middle) Peak response amplitude and (right) duration of response calculated by AUC are shown as well. Soma fluorescence was measured from 11 to 37 cells per well across $n = 3$ independent wells per condition. For each well, single-cell responses were averaged, and data are presented as mean ± SEM (Welch's unpaired t test). (**G**) Representative height-coded maximum intensity projections of human NPCs either (left) untreated or (right) treated with segmented fibrils for 7 days. Projections were generated from combined MAP2, SMI-312, and 4′,6-diamidino-2-phenylindole (DAPI) fluorescence channels (labeling dendrites, axons, and nuclei, respectively) and pseudocolored according to z-position. Scale bar, 100 μm. (**H**) Representative CLSM maximum intensity projection of human NPCs treated with dye-labeled segmented fibrils for 7 days. DAPI indicates nuclei (blue), SMI-312 indicates axons (yellow), negatively charged segments are indicated in green, and positively charged segments are indicated in magenta. Scale bar, 50 μm. (**I**) Representative CLSM image of a neurite from primary mouse cortical neurons treated with segmented fibrils for 24 hours. Phalloidin indicates F-actin in neurites (blue), and laminin α-1 indicates secreted laminin (yellow) on segmented fibrils in which negatively and positively charged segments are shown in green and magenta, respectively. The top panel shows the isolated neurite and laminin signals from the bottom panel. Scale bar, 5 μm.

neurons grown on laminin or fibronectin substrates did not reveal the level of neurite extension observed in the presence of segmented assemblies (fig. S71). Mixed assemblies with randomly distributed positive and negative monomers had no effect on neurite outgrowth despite having the same overall molecular composition as the segmented ones (figs. S71 and S72). Furthermore, positively charged assemblies were cytotoxic, as established by live-dead assays (figs. S73 and 74), which was not the case for negative or segmented ones. In addition to increased neurite length, the segmented assemblies generated neurons with a high degree of arborization, as evidenced by fluorescence microscopy and Sholl analysis (figs. S75 to S77). This morphological effect is associated with greater neuronal maturation (*31*).

Varying the lengths of the positive and negatively charged regions in the segmented fibrils (fig. S78) impacted the extent of neurite

outgrowth (Fig. 5C and figs. S79 and S80). Compared with the charge-segmented assemblies described above, halving the lengths of the negatively charged ends while keeping the size and concentration of the positive anchor constant produced a similar increase in neurite growth (Fig. 5C, "shorter ends"). By contrast, when the ends were very long, the bioactive effect was diminished (Fig. 5C, "longer ends"). We observed less fibril-cell interaction for the long fibrils, indicating that the long negative segments inhibited anchoring to cell membranes (figs. S81 and S82). When the positive-to-negative charge ratio was kept constant and the total fibril length varied, we again observed a reduced fibril-cell interaction for assemblies that were ~30 μm in length (despite their longer anchors compared with the "longer ends" treatment) and no enhancement of neurite growth (figs. S80 to S82). Short positively charged fibrils without any negative ends (treated at the same concentration of positive domains within the segmented structures) also did not enhance neurite growth (Fig. 5C, "no ends," and fig. S72). Thus, our experiments demonstrate that negatively charged segments close to the cells are essential for bioactivity and the segmented assemblies seem to overcome the expected repulsion. The shortest charge-segmented fibrils exhibited the highest fibril-cell interaction, but their impact on neurite growth was not enhanced compared with that of medium-length assemblies (figs. S79 to S82). Together, these results suggest a balance between anchor strength and repulsion from negative end segments is beneficial.

We investigated the influence of segmented fibrils on synaptogenesis and functional maturation. Neurons treated with charge-segmented fibrils of medium lengths for 14 days formed more synapses than those without treatment or neurons grown on laminin or Matrigel. This was shown by higher colocalization of pre- and postsynaptic markers, synapsin-1 (SYN1) and postsynaptic density (PSD95), respectively (Fig. 5E and fig. S83). To determine whether this increase in synapses corresponded to enhanced neuronal network activity, we measured Ca^{2+} influx using a Fluo-4 dye. Neurons treated with segmented fibrils for 7 days exhibited intense calcium responses, characterized by a threefold increase in peak calcium response (as measured by relative fluorescence, defined as $\Delta F/F_0$) and a fourfold increase in cumulative calcium activity, calculated by area under the curve (AUC), when stimulated with KCl compared with controls that received no treatment (Fig. 5F and fig. S84). Notably, on the PDL substrate without any synthetic assemblies added to the media, the neuron somas exhibited aggregation with neurite fasciculation (fig. S85), indicative of poor cell-substrate adhesion most likely because ECM components were absent (*32*). By great contrast, in the presence of segmented assemblies, we did not observe such aggregation, and the complex mature morphologies of the neurons were preserved with greater neurite dispersion. These data demonstrate the ability of segmented fibrils to greatly enhance synaptic maturation and calcium activity in primary cortical neurons. The potential importance of this observation relative to controls and other common extracellular substrates is the ability of the anchored supramolecular assemblies to contribute to neuronal function.

As mentioned above, the negative domains around the cells were highly dynamic, which may allow the localized assemblies to be more reconfigurable and able to adapt to the complex morphology of maturing neurons (*33*). It has been shown that adhesive interactions that are too weak or too strong can inhibit neurite outgrowth by preventing growth cone attachment or advancement, respectively (*34*). The oppositely charged domains and fibril dynamics in the localized segmented assemblies may help balance these adhesive and repulsive forces. Indeed, much greater neurite extension and retraction dynamics were observed when neurons were treated with the segmented rather than the positively charged assemblies (movies S5 and S6 and fig. S86), which is likely related to changes in cytoskeletal remodeling (*35*). This exploratory behavior is important for neurons to find their targets both during development and in diseased states.

We previously found highly dynamic supramolecular assemblies that display the laminin-mimetic signal IKVAV exhibit neural bioactivity (*36*). The charge-segmented assemblies in this work behave as systems that we could describe as "superbioactive" based on our observations of combined neurite growth, maturation, and synaptic activity without the presentation of specific biological signals that are covalently integrated in the primary chemical structure. Also, in differentiating human neural progenitor cells (NPCs), we observed unexpected 3D structures of neurons in the presence of the segmented assemblies. The neurons organized into dense raised domains containing mostly somas and dendrites colocalized with the supramolecular assemblies and connected by axons outside these clusters (Fig. 5, G and H, and figs. S87 to S89). Furthermore, these raised domains were enriched in laminin relative to nonsegmented assemblies or those without treatment (fig. S90). Immunostaining of mouse cortical neurons treated with fluorescently labeled segmented assemblies also revealed endogenous laminin α-1 bound to the fibrils concentrated around the neurites (Fig. 5I and figs. S91 to S96). Laminins are important ECM proteins in nervous tissue, as they are involved in cell-ECM adhesion and signaling (*37*). Both neurotrophic factors and laminin are known to bind to negatively charged glycosaminoglycans in the ECM through their cationic heparin-binding domains (*38*–*40*), and thus, we expect that the anchored negatively charged segments play a similar role (*41*, *42*). In our view, another axis of the high bioactivity is the intense dynamics visualized by confocal microscopy in the negatively charged segments that intimately surround the entire cell and adapt to neuronal growth. We also speculate that the apparent "tapping" motion of negatively charged fibril ends on the cell membrane (Fig. 5B and movie S1) could enhance signaling between any fibril-bound proteins and cell-surface receptors. The known supramolecular dynamics at timescales much shorter than this motion could further contribute to the observed bioactivity, as reported previously by our laboratory (*11*, *43*).

Discussion

We created length-controlled peptide assemblies with precisely segmented domains of opposite charge and dimensions comparable to the diameter of cells. Based on imaging, we estimate that the longest of these assemblies exceeds a collective molar mass of 1 billion Da (when assuming ~10 PA molecules per fibril cross-section). The precise control over the supramolecular assembly of these systems was enabled primarily by hydrophilic oligomers conjugated to the peptide. These oligomers gave rise to the formation of dynamic cap structures formed by the monomers themselves, which open to allow the addition of monomer at high temperature and close again to stabilize the ends when monomer is consumed. We refer to this mechanism as self-capping supramolecular polymerization (SCSP). This mechanism worked in synergy with the presence of a highly ordered core in the assemblies that provides a strong driving force for further supramolecular growth. The controlled growth required for precision was undermined by structural heterogeneity among the assemblies. Specifically, this precision was no longer achieved when changes in peptide sequence yield filament morphologies that are not flat but instead twist with variable pitch. The assemblies had an inherent directionality in their structure that manifested as asymmetric growth from the two ends of a fibril. Future work could investigate the design of other molecules that are not necessarily PAs conjugated to hydrophilic oligomers that meet the requirements for SCSP.

Using neurons, we discovered that precise supramolecular polymers with tuned length and charge segmentation greatly enhanced growth and synaptic activity without having to use specific bioactive signals. This inherent bioactivity was also revealed by the 3D organization of human neurons from 2D cultures. By contrast, negatively charged assemblies were not highly bioactive, whereas positively charged ones were cytotoxic. The simple inclusion of a shorter positively charged segment between two negatively charged ones acted as an electrostatic anchor to form a localized, dynamic scaffold that captured ECM proteins. In contrast to micropatterning- and microtexture-based approaches that typically use materials with predetermined structures, the solution-deposited

supramolecular polymers responded to the environment in which they were applied, diffusing to and associating with neurites. This approach offers several advantages over static materials, such as the versatility to apply them to nonplanar, 3D growing tissues as a regenerative therapy. Our results serve as proof-of-principle that precise arrangement of electrostatic charge in cell-scale supramolecular constructs can not only augment bioactivity as a scalable therapy but also uncover unknown roles of the ECM.

The structures investigated in this work point out the great functional advantages that supramolecular polymers provide, namely the opportunity to integrate dynamic behavior synergistically with molecular order. This combination was critical in our discovery of SCSP as a strategy to create precise supramolecular assemblies. Moving forward, precision in these systems could be exploited in multiple ways, for example, by using the chemical nature of the segments to build larger scale hierarchical or folded structures. It may even be possible to covalently attach assemblies to surfaces at specific sites or to mineralize segments to create devices or macroscale functional networks.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Figs. S1 to S96; Tables S1 to S5; References (*45*–*54*); Movies S1 to S6

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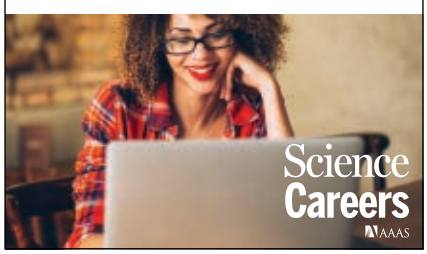

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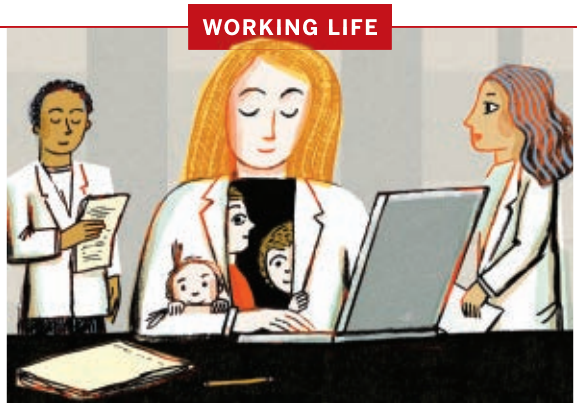

Making the invisible visible

Alexandra Piotrowski-Daspit

I was rushing out of the office to pick up my two young children from day care when I ran into a senior colleague in the hallway. They wanted to chat about a proposal I was working on and how I was settling into my new tenure-track position. I was grateful for their interest and collegiality, and I didn't want them to think I wasn't giving the job my full attention. So, I acted as if I had all the time in the world. I didn't mention where I was headed, or that I had a kids' pickup deadline. I barely made it. That night after dinner and bedtime I caught up on emails, never letting on that I was working so late because of my family responsibilities. I was convinced I had to manage my demands invisibly.

To the outside world, I was building a lab, mentoring students, teaching, writing grants, and pursuing the milestones expected of an early-career faculty member. But outside the margins of my CV, I was also coordinating child care, navigating day care illnesses and closures, recovering from pregnancies, breastfeeding, pumping, and trying to think clearly through antepartum and postpartum brain fog.

For a while, I thought the solution was to become more efficient. If I planned carefully enough, woke up early enough, answered enough emails after bedtime, maybe I could make the chaotic parts of motherhood invisible in my professional life. I could be the same kind of early-career scientist I had imagined before children, just with less sleep.

This approach almost worked, until it didn't. Day care would call and a sick child would erase a carefully planned day. School closures would rearrange an entire week. Travel anywhere involved heroic efforts to organize small lives in addition to mine. No amount of optimization gave me full control and I was simply exhausted. There was almost always something I felt I was not doing as well as I wished I could.

Then, when I was a little over 1 year into my faculty position and pregnant with my third child, an email landed in my inbox inviting me to join a gathering for pretenure mothers at my institution. A few assistant professors had bonded over the ups and downs of juggling motherhood with the responsibilities of a tenure-track faculty member, and they wanted to connect with others doing the same. The invitation offered something I had not realized I needed: a space where my identities as a mother and scientist did not have to remain separate. I filled out the scheduling poll.

At our lunches, I kept hearing people say aloud the things I had been facing silently: the guilt of leaving campus early, the exhaustion of writing after bedtime, the pressure of keeping up with peers whose lives might look very different from ours. We shared strategies, offered reassurance, and considered what types of support we actually needed. Those conversations did not magically make working life easier. But they made it less lonely.

I began to see that my struggle was not a personal failure to become efficient enough. The many demands in my life were real, even if academic culture kept them out of view.

I became more willing to acknowledge my limitations, accept help, and resist blaming every disruption on poor organization. If I need to move a meeting because of child care pickup or a sick child at home, I say so plainly. I am careful not to make my trainees responsible for my family logistics, but I also do not pretend those logistics do not exist. I want my trainees to see that being a good scientist does not require pretending to have a life untouched by caregiving, illness, rest, or limits.

This shift has also changed how I think about my life outside the lab. Recently I was volunteering at my oldest son's kindergarten Aloha Day party. Another mother arrived with a watermelon carved to look like Stitch from *Lilo & Stitch*. It was spectacular compared with my store-bought popsicles. The old me might have felt a familiar pang: Should I be doing more? Does my contribution betray my divided attention? Then I looked at the delighted children and felt grateful. This parent had offered something I could not, but I had shown up in the best way I could. I did not need to disguise the constraints of my career to prove I was a devoted mother, just as I do not need to disguise the demands of motherhood to prove I am a serious scientist. □

Alexandra Piotrowski-Daspit is an assistant professor at the University of Michigan. Do you have an interesting career story to share? See our author guidelines at https://scim.ag/WorkingLife.

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